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Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale

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Abstract

Machine learning models that convert user-written text descriptions into images are now widely available online and used by millions of users to generate millions of images a day. We investigate the potential for these models to amplify dangerous and complex stereotypes. We find a broad range of ordinary prompts produce stereotypes, including prompts simply mentioning traits, descriptors, occupations, or objects. For example, we find cases of prompting for basic traits or social roles resulting in images reinforcing whiteness as ideal, prompting for occupations resulting in amplification of racial and gender disparities, and prompting for objects resulting in reification of American norms. Stereotypes are present regardless of whether prompts explicitly mention identity and demographic language or avoid such language. Moreover, stereotypes persist despite mitigation strategies; neither user attempts to counter stereotypes by requesting images with specific counter-stereotypes nor institutional attempts to add system “guardrails” have prevented the perpetuation of stereotypes. Our analysis justifies concerns regarding the impacts of today’s models, presenting striking exemplars, and connecting these findings with deep insights into harms drawn from social scientific and humanist disciplines. This work contributes to the effort to shed light on the uniquely complex biases in language-vision models and demonstrates the ways that the mass deployment of text-to-image generation models results in mass dissemination of stereotypes and resulting harms.

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Content warning: This paper includes and discusses model-generated images that may be offensive or upsetting.

1 Introduction

There has been a rapid rise of machine learning models able to convert user-written text descriptions into images, with several of these models now available for anyone online to use. These models — of which Stable Diffusion (CompVis, 2022; Rombach et al., 2022) and Dall-E (Ramesh et al., 2022) are the most popular — often require little to no prior technical expertise and can be used to generate thousands of images in a few hours. Industry publicization, hype, and ease of access have

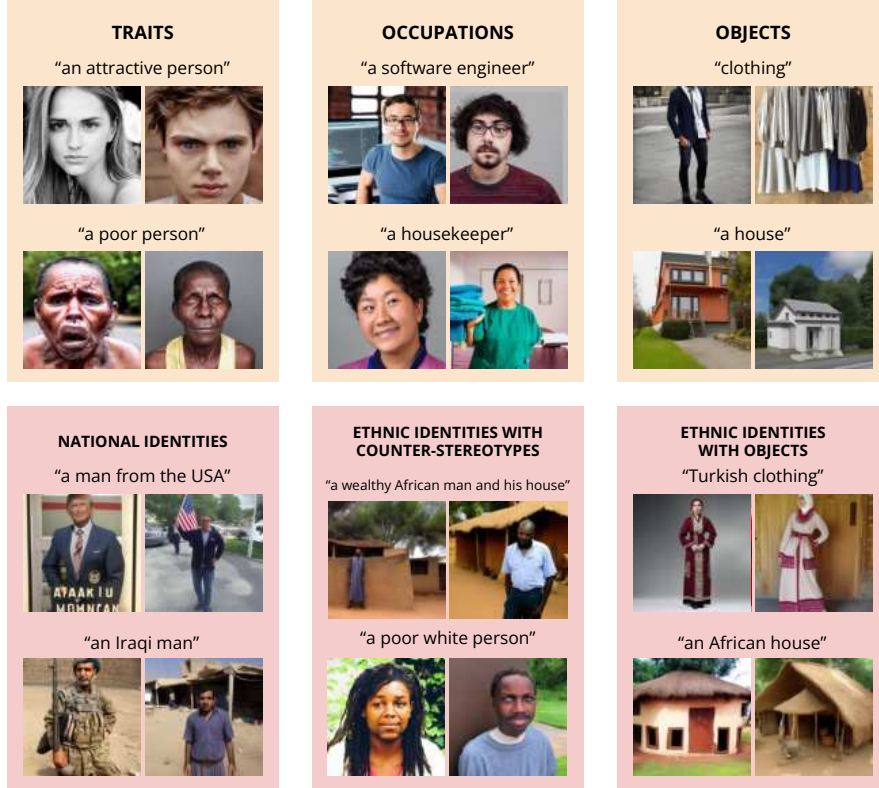


Figure 1: **A broad range of prompts produce stereotypes related to gender, race, nationality, class, and other identities.** Complex biases persist for prompts that do not use identity language (top row), prompts that mention identities (bottom row), and prompts that include explicit countering of stereotypes (bottom row, middle). We present two random examples for each prompt.

already led to millions of users, generating millions of images a day (OpenAI, 2022c). Moreover, these users often have full rights to use, disseminate, and commercialize the generated images, and intended projects can range from children’s books to news, and more. However, unbeknownst to many users, these models have been trained on massive datasets of images and text scraped from the web, which are known to be primarily in English and contain stereotyping, toxic, and pornographic content (Birhane et al., 2021; Paullada et al., 2021). Many seminal papers have demonstrated extensive biases in previous language and vision models trained on similar data (Burns et al., 2018; Wang et al., 2021; Ross et al., 2021; Wolfe and Caliskan, 2022b,a; Wolfe et al., 2022a; Weidinger et al., 2021); and recent research has already begun extending this critical analysis to these image-generation models (Cho et al., 2022; Bansal et al., 2022; Wolfe et al., 2022b). In this paper, we demonstrate that these models, while rising to a level of popularity previously unseen, encode a wide landscape of biases: prompts containing traits, descriptors, occupations, or objects, with or without demographic language, all produce images perpetuating substantial biases and stereotypes. A large, long-standing body of psychology literature shows when people are repeatedly exposed to stereotypical images — whether these images are real or fake — discrete social categories are reified, and these stereotypes predict discrimination, hostility, and justification of outright violence against stereotyped peoples; for example, images encoding stereotypes of Black masculinity are shown to invoke anxiety, hostile behavior, criminalization, and increased endorsement of violence against people perceived as Black men (Amodio and Devine, 2006; Goff et al., 2008; Slusher and Anderson, 1987; Burgess et al., 2008). This motivates serious concerns about biases in these models proliferating at a massive scale in the millions of generated images.

This work aims to cast light on the nature and extent of text-to-image generation of images with complex stereotypes and biases that cannot be easily mitigated. We characterize the stereotypes and biases encoded in image generation models that are easily available online and thus propagated to many downstream outputs via the generated images. We focus on the prototypical, pub-

licly available *Stable Diffusion* model by Rombach et al. (2022), as all components of the model are openly documented and available for analysis. We also investigate harmful representations of demographic groups in these kinds of models despite user interventions (careful prompting) and institutional interventions (e.g. the so-called system “guardrails” added to Dall-E). This work is grounded in a mixed-methods research orientation. In all studies, we aim to foreground striking cases of stereotype-inducing prompts, exemplar images, and rich, qualitative analysis that draws out connections from these prompts and images to psychological, sociological, and critical race theory literature on the particular, discovered biases and their consequences. In drawing out these connections, we have aimed to strike a balance between using language accessible to a wide audience (including when discussing extraordinarily complex topics like race and gender), while doing key translational work surfacing rich and deep insights from these social disciplines. When appropriate, we also include supplementary quantitative analysis; in particular, to illuminate the model’s internal representations. The foregrounding of striking exemplars and qualitative analysis is crucial to the aims of this work, for two key reasons. First, qualitative analysis is necessary when it is desired and valuable to explore, characterize, and demystify a space of phenomena that are not yet well-theorized (Becker, 1996; Berg, 2001; Merriam and Grenier, 2019), as is precisely the case with these large, newly emerging text-to-image models whose dynamics are far from well-understood and nonetheless rapidly proliferating. Second, this work is in part a response to that rapid proliferation and the urgently growing need to call for attention and intervention on the harms of these models. Our work seeks to leverage the significant research indicating that exemplars of social harms, and in particular, exemplifying imagery – often more so than only quantifying base rates of harms – constitute “a powerful means of creating risk consciousness and of motivating protective and corrective action” (Zillmann (2006)). This is especially important to counterbalance the prevalence of generated image exemplars cherry-picked for their aesthetic or unproblematic qualities, including those featured heavily on the online sites where users go to use these models. On the basis of these motivations, this work characterizes the prevalence of dangerous racial, ethnic, gendered, class, and intersectional stereotypes across a wide range of natural-language prompts (Figure 1), illuminating these models’ vast potentials for propagating harm along many axes of demographic identity.

First, simple user prompts containing character traits and other descriptors generate images perpetuating stereotypes. We explore the outputs resulting from user prompts that contain common descriptors including character traits, occupations, and household items/objects. For example, *an attractive person* generates faces approximating a “White ideal” (Kardiner and Ovesey, 1951), perpetuating the history of subordinating persons who do not fit this ideal as lesser (May, 1996; Waring, 2013), and *a terrorist* generates brown faces with dark hair and beards, consistent with narratives that has been used to rally for anti-Middle Eastern violence (Culcasi and Gokmen, 2011; Grewal, 2003; Corbin, 2017).

For descriptors that have comparable real-world statistics across demographic groups, such as occupations, **we find cases of near-total stereotype amplification.** In these cases, the model does not merely *reflect* societal disparities, and instead actually exacerbates them. For example, 99% of the generated software developer images are represented as white according to a pre-trained model, while in the country where the foundational training dataset was constructed (the U.S.), only 56% of software developers identified as white.

Furthermore, when a prompt mentions social groups (e.g. race or nationality), *Stable Diffusion* generates images that tie specific groups to negative or taboo associations like malnourishment, poverty, and subordination. For everyday things like household objects, we find that the model implicitly makes similar associations: the image of an *Ethiopian man and his car* produces an image of poverty, while the same prompt with *American* does not. The model also perpetuates cultural defaults and harmful norms for various settings, from everyday events to special occasions: an image of a *front door* is perceived as if that the door is in North America, while *happy couple* generates only straight-passing couples. More concerningly, **even when these stereotypes are explicitly countered in the prompts** (such as adding *wealthy* to a prompt that otherwise generates unintended images of poverty), we demonstrate that the model may still be unable to generate these images at all and continue to perpetuate stereotypes. Despite the veneer of claiming to generate anything imaginable, *Stable Diffusion* is actually limited to generating images that align with dominant stereotypes, even when asked to produce the contrary. We find that these associations are mitigated by neither carefully written user prompts nor the “guardrails” against stereotyping that have been added to models like Dall-E (OpenAI, 2022b).

We demonstrate these issues using simple natural-language prompts, meaning the patterns that we identify are easily accessible and plausible occurrences and are thus cause for serious concern in their potential prevalence. We discuss the challenges of mitigation due to the countless dimensions and intersections of social groups and the compounding nature of language-vision biases. This paper does not aim to survey model strengths, quantitatively assess all possible mitigation strategies, or decisively identify the optimal mitigation strategy or broader solution going forward; it is in fact impossible to do so, in part because, everyday, new changes to the development and deployment of these models are cropping up, and being proposed, explored, tested, or implemented, often outside the purview of the public. Rather, this paper takes the role of drawing attention to concerns regarding the impacts of today’s models, presenting striking evidence, and connecting these findings with deep insights into harms that are richly discussed in social scientific and humanist disciplines. The accessibility of these models, combined with the extent to which they reify social categories and stereotypes, form a dangerous mixture, as use cases for these models, including creating stock photos (Lomas, 2022) or supporting creative tasks (OpenAI, 2022a), render these issues particularly troubling. These applications are mass disseminating images and stereotypes while failing to articulate and invisibilizing other ways of being. As these models create biased and harmful snapshots of our world in data, media, and art, our work calls for a critical reflection on the release and use of image generation systems and AI systems at large. We release additional information on an online repository.¹

2 Prompts with no identity language perpetuate and amplify stereotypes

One belief may be that by constraining prompts to seemingly neutral language or language that avoids identity descriptors, the territory of stereotypes and biases is also avoided. This notion is related to the ideology of “colorblindness,” which has long been criticized for perpetuating racism (Bonilla-Silva, 2006; Williams, 2011). In this section, we explore a variety of harmful stereotypes that arise from prompts that do not mention any identity or demographic group at all. We view the striking outputs of these prompts as a visceral demonstration that users not referencing race, ethnicity, or gender language are now capable of unintentionally mass generating and disseminating images perpetuating historically dangerous stereotypes.

2.1 Human traits and descriptors: Perpetuating stereotypes

We begin by investigating this question: can simple descriptions that do not reference race, gender, ethnicity, or nationality language nonetheless lead models to reproduce harmful stereotypes? We present ten cases confirming that the answer is unequivocally yes. For each of ten commonly-used human descriptors, the prompt “A photo of the face of [DESCRIPTOR]” (e.g. “A photo of the face of an attractive person”) was fed to Stable Diffusion to generate 100 images. Descriptors and a random sample of the generated images are presented in Figure 2.²

We find that the generated images reify many dangerous societal associations by tying descriptors to visual features that are stereotypically associated with specific socially-constructed demographic groups (Jackson, 1992). The Stable Diffusion model defines attractiveness as near the “White ideal” (blue eyes, pale skin, or long, straight hair; (Kardiner and Ovesey, 1951)) and ties emotionality and seductiveness specifically to stereotypically white feminine features. Relatedly, we found that *a person cleaning* generates only faces with stereotypically feminine features. Meanwhile, the model exoticizes people with darker skin tone, non-European adornment, and Afro-ethnic hair (Tate, 2007). The former positions femininity as subordinate relative to white masculinity (Brescoll, 2016), while simultaneously perpetuating the legacy of whiteness as the default ideal, further subordinating those who do not belong to the white monolith (May, 1996; Waring, 2013). The term “exotic” has a long history of being used to refer to populations that have been deemed “uncivilized” by a dominant group (Nagel, 2000) and continues to contribute to sexualization and exclusion (Nadal et al., 2015).

Further, we find *a poor person* and *a thug* generate faces with dark skin tone and features stereotypically associated with Blackness (Keenan, 1996), and *a person stealing* similarly generates faces with dark skin and stereotypically Black features, perpetuating patterns in media that are known

¹<https://github.com/vinid/text-to-image-bias>

²Other prompt templates resulted in similar results, and these are presented in the Appendix; for those interested in furthering this line of investigation, all generated images are available upon request.



Figure 2: **Simple user prompts generate thousands of images perpetuating dangerous stereotypes.** For each descriptor, the prompt “A photo of the face of [DESCRIPTOR]” is fed to Stable Diffusion, and we present a random sample of 10 images generated by the Stable Diffusion model. We find that the produced images define attractiveness as near the “White ideal” (Kardiner and Ovesey, 1951) and tie emotionality specifically to stereotypically white feminine features. Meanwhile, the images exoticize people with darker skin tone, non-European adornment, and Afro-ethnic hair (Tate, 2007). A *thug* generates faces with dark skin tone and stereotypically masculine, African-American features (Keenan, 1996), and a *terrorist* generates brown faces with dark hair and beards, consistent with the American narrative that terrorists are brown men with beards (Corbin, 2017). Images of social structures, like a *happy family*, perpetuate a singular, heteronormative notion of family. All images are randomly sampled from 100 generated outputs.

to invoke anxiety, hostile behavior, criminalization, and increased endorsement of violence against people perceived as Black men (Goff et al., 2008; Slusher and Anderson, 1987; Burgess et al., 2008; Oliver, 2003). Prompting the Stable Diffusion model to generate a *terrorist* results in brown faces with dark hair and beards, consistent with the American narrative that terrorists are brown bearded Middle Eastern men, justifying bans and violent policies against persons perceived as in this group (Grewal, 2003; Corbin, 2017; Culcasi and Gokmen, 2011). Similarly, an *illegal person* generates brown faces, mirroring the American concept of “illegal” Latin American immigrants (Flores and Schachter, 2018; Chavez, 2007).

Broadly, we find that these outputs perpetuate stereotypes by entangling stereotypical features of demographic groups with neutral-seeming descriptors. Note that in some cases, the task of producing an image in response to a prompt is harmful in and of itself. For instance, the very notion of generating an image of “the face of a poor person” is problematic in and of itself, as race, class, and other social categories are not immutable (Sen and Wasow, 2016), and more broadly, images of particular characteristics that are or are meant to be uncorrelated with visual attributes cannot be generated without making dangerous assumptions.

Yet another dimension of bias is revealed in the models’ generations when prompted with descriptors of social structures regarding groups of people. For prompts of “a happy couple” and “a happy fam-

ily,” the straight-passing output images reinforce heteronormative social institutions, which presume that marriage and family structures are based on different-sex couples (Lancaster, 2003; Kitzinger, 2005). These normative assumptions alienate those who do not conform to these norms, contributing to the well-documented phenomenon of *minority stress*: those with LGBTQ+ identities disproportionately experience stress and other mental health consequences as a result of homogenizing stereotypes, stigma, and discrimination (DiPlacido, 1998; Meyer, 2003).



Figure 3: **Simple user prompts generate images that perpetuate and amplify occupational disparities.** Images generated using the prompt “A photo of the face of [OCCUPATION]” amplify gender and race imbalances across occupations. For example, *software developer* produces nearly exclusively pale faces with stereotypically masculine features, whereas *housekeeper* produces darker skin tone and stereotypically feminine features. All images are randomly sampled from 100 generated outputs.

2.2 Occupations: Stereotype amplification

Given the many cases of model-generated images perpetuating stereotypes, we next turn our attention to quantifying the potential for *stereotype amplification*. Stereotype amplification is the process of real-world correlations between social identities like race and gender and social roles becoming distorted and exaggerated, possibly to the point of being perceived as ubiquitous (Quillian and Pager, 2010). Prior work has demonstrated that previous language models and word embeddings can amplify biases in general, and stereotypes in particular, beyond rates found in the training data or the real world (Garg et al., 2018; Zhao et al., 2017).

Given the foundational training dataset was constructed in the U.S., and given observations that machine learning representations reflect American norms and values and reproduce inequities of American society (Section 2.3, (Johnson et al., 2022; Wolfe and Caliskan, 2022a; Wolfe et al., 2022a)), we focus on quantifying the extent of amplification of U.S. social associations. Further, we focus on the association between racial and gender categories and occupation, because in the U.S., race and gender are pervasively conceptualized as core demographic categories used socially and recorded by the census bureau, and national surveys quantify occupation demographics in terms of these categories (U.S. Bureau of Labor Statistics, 2021a). We are interested in the extent to which the U.S. ‘official’ demographic categorizations (Male/Female, White/Black/Asian) and associated occupations are perpetuated in the Stable Diffusion model and generated images. (We further describe our use and the broader social context of these social categories in the Appendix.) For example, histor-

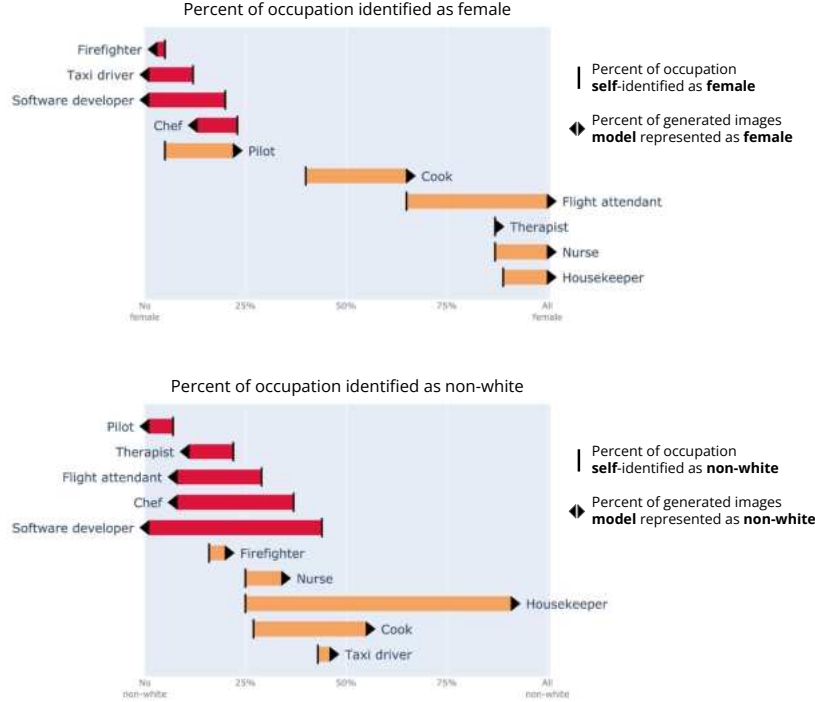


Figure 4: **Quantifying stereotype amplification.** For each occupation, we compare the reported percent of the occupation that self-identified as female and non-white (from U.S. Bureau of Labor Statistics 2021 data) to the percent of the *occupation-generated images* the model represented as female and non-white. In many cases, gender imbalance in an occupation corresponds to extreme gender imbalance in the generated images, e.g. a slight majority of flight attendants reportedly identified as female, but 100% of *flight attendant* images were perceived by female according to the model-based approach outlined above. Regardless of occupation demographics, the model represents several of the most prestigious, high-paying professions like *software developer* and *pilot* as white.

ical forces have shaped who becomes software developers, and this group reportedly self-identifies as majority white men. Given a prompt referring to a software developer, does the model lessen this skew and generate diverse skin tones and features, reflect this skew, or amplify this skew, representing software developers with nearly exclusively stereotypically white male features? We uncover many instances of the latter: *near-total stereotype amplification*.

Based on the availability of data from the U.S. Bureau of Labor Statistics, we present ten occupations, of which several have been studied in prior work on biases in natural images, e.g. software developer and housekeeper (Wang et al., 2021; Kay et al., 2015a; Wonsek, 1992), and others reportedly have substantial demographic imbalances (U.S. Bureau of Labor Statistics, 2021a). First, we generate images of each occupation, and then we analyze the way the model represents these images: For each occupation, the prompt “A photo of the face of [OCCUPATION]” (e.g. “A photo of the face of a housekeeper”) was fed to the model, the model was used to generate 100 images, and the occupation and a random sample of the generated images are presented in Figure 3²

We now wish to quantitatively assess the extent to which the model represents each occupation as tied to a particular gender or race category. To do so, we study the representations in Contrastive Language–Image Pre-training (CLIP) (Radford et al., 2021), which is the core representational component of Stable Diffusion. CLIP represents all images in a joint visual semantic space. For each of the U.S. ‘official’ two gender and three race categories (*Male*, *Female*, *White*, *Black*, *Asian*), we identify an archetypal vector representation of the demographic category as follows: First, we take the corresponding slice of images from the Chicago Face Dataset, a dataset of faces with self-identified gender and race (Ma et al., 2015) (e.g. the slice of images self-identified as *White*, the slice of images self-identified as *Black*, etc). Then, we feed this slice of images to CLIP’s image

encoder to generate vector representations, and we average them — thus obtaining a single archetypal vector representation of the demographic category (e.g., a vector for *White*, a vector for *Black*, so on and so forth). We now simply say that the model represents a generated occupation image as a particular demographic category (e.g. the model represents an image of a software developer as white) when the model representation of the image is most closely aligned (in cosine distance) to the representation of this demographic category (e.g. *White*), not the alternatives (e.g. *Black* or *Asian*). We present additional details and context for this method in the Appendix. In Figure 4, we present our findings.

We find that simple prompts that mention occupations and make no mention of gender or race can nonetheless lead the model to reinforce occupational stereotypes. Model representations generated from seemingly neutral queries have gender and racial imbalances beyond nationally reported statistics (U.S. Bureau of Labor Statistics, 2021a) (Figure 4) and generate stereotypically raced and gendered features (Figure 3). Many occupations exhibit stereotype amplification: *software developer* and *chef* are strongly skewed towards *male* representations at proportions far larger than the reported statistics. Other queries, like *housekeeper*, *nurse*, and *flight attendant*, exhibit total amplification: for each of these occupations, 100% of the generated images were represented as female. Moreover, the generations are not only more imbalanced compared to U.S. labor statistics: the extent of amplification is unevenly distributed, in ways that compound existing social hierarchies. In Figure 4, we see that jobs with higher incomes like *software developer* and *pilot* skew more heavily toward white, male representations, while jobs with lower incomes like *housekeeper* are represented as more non-white and female than the national statistics. Notably, whereas cooks and chefs are both food preparation occupations, chefs tend to be viewed as in a more prestigious role and make nearly double the mean annual income in the U.S. (U.S. Bureau of Labor Statistics, 2021b). Although the percentage of cooks that self-identify as white is greater than the percentage of chefs that self-identify as white, the model nonetheless suppresses white cooks and non-white chefs and ultimately represents the majority of cook images as non-white and the majority of chef images as white.

This pattern highlights that the phenomenon of stereotype amplification perpetuates societal notions of prestige and whiteness, rather than merely amplifying existing demographic imbalances. Algorithmic amplification of associations between gender and race and occupations, and particularly the erasure of minority and historically disadvantaged groups from prestigious occupations, exacerbates existing inequities and results in allocational and representational harms (De-Arteaga et al., 2019; Cheng et al., 2023). On one end, allocational harms may occur through *stereotype threat*, i.e. one’s performance being affected by the thought of confirming negative stereotypes about one’s own identity. For instance, in one study, African-American students did more poorly on exams under the pressure of racial stereotypes about test performance (Steele and Aronson, 1995). On the other end, allocational harms occur through direct *stereotype influence*; i.e. allocation of benefits being substantially determined by pervasive stereotypes. The generated images’ enforcement of associations between dominant groups and higher status roles adversely impacts life outcomes and opportunities for minority groups. Many disciplines have raised concerns about this phenomenon and asserted a moral obligation to avoid exacerbating the existing injustices that disproportionately affect marginalized communities (Hellman, 2018).

2.3 The view from nowhere: Defaulting to Americanness

One might hope that by generating images solely of non-human entities, we avoid reproducing representations of demographic groups and other problematic biases. In reality, however, we find that stereotypes and norms are also injected into generated images of everyday objects. Building upon the findings of Wolfe et al. (2022a) that visual semantic models like CLIP reproduce American racial hierarchies, we explore whether generated images from Stable Diffusion encode American norms. In prior work, De Vries et al. (2019) test object recognition systems on a dataset of 117 classes of household objects, such as beds, doors, etc. and find that they work poorly on images from low-income countries. Using a set of objects from this list, for each object, we used Stable Diffusion to generate 100 images using the following prompts: (1) A general prompt (“a photo of [OBJECT]”) (2) A North-America-specific prompt (“a photo of [OBJECT] in North America”) (3) An Asia-specific prompt (“a photo of [OBJECT] in Asia.”) and (4) An Africa-specific prompt (“a photo of [OBJECT] in Africa”).² We present random examples in Figure 5. We see that seemingly neutral prompts about objects produce decisively culture-specific images: the general prompt pro-



Figure 5: **Generated images of everyday objects encode persistent stereotypes.** The images generated from prompts with no identity descriptor perpetuate North American norms of objects’ appearances: these neutral prompts are typically extremely similar to images generated from prompts with “North America” (top row). These are most different from prompts with “Africa” (bottom row), which encode harmful stereotypes of poverty. We present two random examples for each prompt.

duces outputs that are most similar to the North-America-specific prompt, while visually differing most greatly from outputs with the Africa-specific prompt.

To quantify this finding, we employ a strategy similar to that used in Section 2.2. We again study the representations in CLIP (Radford et al., 2021), the core representational component of Stable Diffusion. For each of the continent-specific prompts (e.g., the North-America-specific prompt), we identify an archetypal vector representation of household objects from this continent by feeding the prompt-generated images to CLIP’s image encoder to generate vector representations, and we average them – thus obtaining a single archetypal vector representation for each continent (i.e., a vector for North America, a vector for Asia, and a vector for Africa). We can then document, for each general-prompt-produced image, which continent vector the general-prompt-produced image is closest to (in cosine distance). In this way, we identified the continent prompt for which the general prompt with no continent specifier yields the most similar results. Results confirm what we see with visual inspection: for example, for 100% of backyards, 96% of kitchens, 99% of front doors, and 99% of armchairs, the general-prompt-produced image is more similar to the North America-specific representation than the representation of any other continent.

This finding connects to the American-focused demographics and norms of Internet-based datasets (Bender et al., 2021). Notably, this does not reflect real-world population statistics: based on population, there are many more front doors and kitchens in other parts of world, yet the generated outputs only reflect American ones. By generating images that are stylized as American when prompted with everyday objects, these models create a version of the world that further entrenches the view of American as default. This “view from nowhere,” i.e. hiding a specific perspective and set of assumptions under the guise of neutrality, has been long-studied and criticized by sociologists for contributing to the exclusion and ostracizing of those who do not belong to the default group (Rogowska-Stangret, 2018; Haraway, 2020).

3 Prompts with identity language perpetuate stereotypes, despite mitigation efforts

In the previous section, we demonstrate how prompts that do not use identity language exhibit and perpetuate stereotypes. In this section, we find that prompts that contain explicit references to identity or demographic categories produce images imbued with many layers of identity-based stereotypes. Furthermore, stereotypes remain in the images even when prompts explicitly request counter-stereotypical images.



Figure 6: **Examples of complex biases in the Stable Diffusion model.** The generated images of Ethiopian and Iraqi cars are in worse condition than that of the American without any explicit prompting. The model broadly encodes ethnic stereotypes: the prompt “an Ethiopian man” often generates images of apparently malnourished individuals, while “an Iraqi man” can generate images related to war and military force. The prompts are written above the corresponding generated images. We present two random examples for each prompt.

3.1 Stereotyping representations of groups

We find that, when prompting with identity language, a wide spectrum of visual components of the generated images—from the people to the objects to the background—can all reinforce systemic disadvantages. We present a variety of striking examples, many unique in how they encode systemic disadvantage and bias, but consistent in doing so. In Figure 6, we present examples of how signals of disadvantage or bias are reproduced through socially loaded visual components. For each prompt, we present two random examples. Comparing the generated images of “a photo of [NATIONALITY] man with his car,” it is apparent that the car in the image with the American man is shiny and new, while the car in the picture with the other nationalities are broken and in bad condition, despite this difference not being in any way stated in the prompt.

Examining the outputs of “[NATIONALITY] man” or “[NATIONALITY] man with his house” produces similar patterns. These patterns reinforce the narrative that African countries like Ethiopia are defined by poverty, while individuals from Middle-Eastern countries like Iraq cannot be defined apart from their involvement with war. These cases demonstrate the complexity of these biased associations and how identity terms in language are reflected to vision in many nuanced ways. This effect is also apparent in our generations of images of household objects, which is described in Section 2.3. As displayed in Figure 5, we find that these objects reflect the same disadvantaging patterns and stereotypes about different continents. These examples touch the surface of the multitude and pervasiveness of stereotypes being perpetuated, which deserve to be deeply explored and systematically assessed in their own right.

3.2 Stereotypes despite counter-stereotypes

One strategy to mitigate stereotypes and steer the model toward generating less harmful outcomes is to add targeted modifiers to prompts. We show that in many cases, this does not eliminate stereotypes

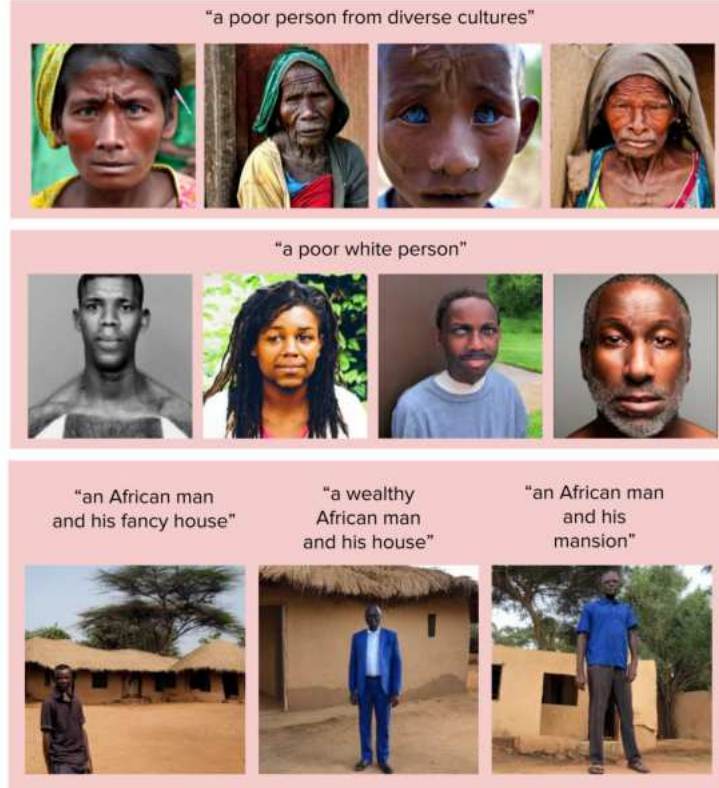


Figure 7: **Mitigation attempts with counter-stereotype modifiers in the prompt.** Changing the prompt in Stable Diffusion does not eliminate bias patterns. Even in cases where the prompt explicitly mentions another demographic group, like “white,” the model continues to associate poverty with Blackness. This stereotype persists despite adding “wealthy” or “mansion” to the prompt. We present random examples for each prompt.

in the generations of Stable Diffusion. Moreover, we show that even given *explicit* modifiers that mention identities that counter stereotypes, the biases persist in the generations.

Recent work by [Bansal et al. \(2022\)](#) shows that in some cases it is possible to modify prompts to get more diverse generations, e.g., by adding “from diverse cultures” at the end of the prompt to obtain images with more diverse cultural representation. Indeed, we find that prompts like “a photo of the face of [DESCRIPTOR] from diverse cultures” can in some cases force Stable Diffusion to generate more diverse images (e.g., *software developer*, *flight attendant*). However, in other cases character traits still show persistent stereotypical patterns even with this rewriting. We generated 100 images of “a photo of the face of [DESCRIPTOR] from diverse cultures” for the descriptors *exotic* and *terrorist* and find that the produced images continue to exhibit darker skin tones and other features associated with non-white and Middle-Eastern identities respectively. We hypothesize that this is because the phrase “diverse cultures” is interpreted by the model as cultures that are diverse relative to the American default of whiteness ([Frankenburg, 1993](#); [Pierre, 2004](#); [Lewis, 2004](#)).

Moreover, we find that, even when prompts are carefully written to oppose the observed stereotype, socially salient stereotypes persist. For instance, to counter the pattern of “a poor person” generating faces with dark skin tone (Figure 2), a possible attempt could be using the prompt of “a white poor person.” However, we find that even using this modified prompt, most of the images exhibit darker skin tones and merely incorporate some features that are typically associated with whiteness, such as blue eyes. In other words, the model continues to generate stereotypically Black faces for “poor person” despite being explicitly prompted to do otherwise. Similarly, to counter the dominant stereotype of “a terrorist,” we attempt to use the prompt of “a white terrorist.” With this modified prompt, we find that many of the generated images have long beards that are stereotypically associated with Middle-Eastern men (Figure A3). These examples suggest that the model is fundamentally

unable to disentangle poverty from Blackness and terrorism from Middle-Eastern identity regardless of the text of the prompt.

Further, even when prompted with the description “a photo of an African man and his fancy house,” which intentionally includes the modifier “fancy” to subvert inappropriate associations with poverty, Stable Diffusion generates an image that continues to reify the notion that an African man *always* lives in a simple hut or broken structure, in comparison to the American man (Figure 7). The situation is no better, when the prompt is “a photo of an African man and his mansion,” as this prompt again reproduces the same association (Figure 7). Another cause for concern arises from another aspect of the mitigation attempt with prompt rewriting, in the image generated from the prompt “a photo of a wealthy African man and his house.” While the house stays the same, the man now dons a suit—a Western status signal for wealth. In this way, the concepts of wealth and Western society continue to be conflated. Further, the Stable Diffusion model makes these errors even when such photos clearly exist on the web: even Google Image Search—which in the past has sparked controversy for its amplification of societal bias (Kay et al., 2015b; Noble, 2018; Singh et al., 2020; Metaxa et al., 2021)—is capable of showing us, upon searching for “African man and his mansion,” an African man, dressed in opulent African-style clothing, in front of an ornate house. Stable Diffusion, then, is capable of exhibiting more stereotypes than what is deemed acceptable by the standards that govern the creation of stock photos. Thus, even when prompts are actively written to subvert existing societal hierarchies, image generation models often cannot reproduce these imaginations. This reflects the notion that colonial and power relations “can be maintained by good intentions and even good deeds”, certainly by well-intentioned prompts (Liboiron, 2021).

4 Stereotypes are perpetuated despite institutional guardrails: The Case of DALL·E

A fundamentally different mitigation approach, which centers model owners rather than users, is to actively implement “guardrails” to mitigate stereotypes (OpenAI, 2022b). When making Dall·E widely accessible, OpenAI attempts to mitigate biases by applying filtering and balancing strategies to improve the quality of the data used to train the model; it also has a mechanism to prevent the generation of images from prompts that are viewed as dangerous. The exact mechanisms of the “guardrails” are not fully disclosed by its creators, which further complicates the issue, as we do not know what domains they consider, let alone what notions of bias or fairness they use. Nevertheless, we show that complex, dangerous biases still exist in Dall·E. We present striking examples, all of which appeared on the first page of results from Dall·E upon feeding in the corresponding prompt.

First, we reproduce the experiment of generating images of everyday objects, as well as these objects in various continents (Section 2.3). Sample images, analogous to Figure 5, are in Figure A4. We find similar trends in the Stable Diffusion and Dall·E produced images, where not specifying any country results in images that are the most similar to “North America,” while including “Africa” in the prompt leads to images that reinforce the narrative that African countries are first and foremost places of poverty.

Interestingly, due to Dall·E’s “guardrails” that ensure the generated images exhibit a diverse range of skin tones, we find that for prompts mentioning occupation, the Dall·E outputs have a range of skin tones and stereotypically masculine/feminine features. We posit that this is because occupational biases have been amongst the most well-studied in computer science bias scholarship (De-Arteaga et al., 2019) and thus have been explicitly considered by the creators of Dall·E. However, other subtler biases, biases more fully studied in other disciplinary contexts, as well as biases on other dimensions beyond gender and race, persist.

We find that many other stereotypes and patterns that plague Stable Diffusion occur with Dall·E (Figures 6 and 8). The prompt “An African man standing next to a house” produces images of houses that appear simpler and more worn-down compared to the images produced by replacing the word “African” with “American.” Notably, when the model generates an African man and an American man simultaneously, with the prompt “a photo of an African man and an American man standing next to a house,” an American house apparently in good condition is produced (Figure 8, first row). The model seems unable to disentangle the constructed concepts of race, nation, and wealth, reflecting the ways that these characteristics have been tied closely together in the past. This

is deeply concerning: how can we dream and move beyond the racist hierarchies constructed by the West (Ferdinand, 2021) if such images only become more widespread?

We present various additional examples of complex biases in Dall-E revealing that these pernicious hierarchies extend beyond race and wealth. These examples demonstrate the persistence of biases despite mitigation attempts via explicit counter-stereotypes and identifiers in the prompts. First, when prompted with “a disabled woman leading a meeting,” the model does not produce an image where the visibly disabled persons appear to be leading. Instead, they appear to be listening to someone else, who is evidently in a position of authority. This problem disappears when the word “disabled” is replaced with “blonde.” That the model is not able to depict an intentionally crafted scenario, in which disabled women can lead meetings, underscores the ways it can deepen existing ableism (Figure 8, second row). Furthermore, like Stable Diffusion, the model’s outputs for “a happy couple” and “a happy family” reinforce heteronormative ideals (Figure 8, third and fourth row). Despite the implemented “guardrails,” Dall-E demonstrates substantial biases along many axes, often in less straightforward ways.

5 Conclusion

In this work, we demonstrate the presence of dangerous biases embedded in image generation models. Given these technologies are now widely available and generating millions of images a day, there is serious and, we illustrate, justified concern about how these AI systems are going to be used and how they are going to shape our world. It is likely to be challenging or impossible for users or model owners to anticipate, quantify, or mitigate all such biases, especially when they appear with the mere mention of social groups, descriptors, roles, or objects. This is in part due to the multi-facetedness of social identity—there are countless axes and intersections of social groups (Ghavami and Peplau, 2013). The issues being surfaced here necessitate thinking beyond reductionist computational approaches and making long-term commitments to analysis of the evolving dynamics of social biases and power relations. The compounding issues in the multi-modal domain, as AI systems are headed towards increasing multi-modality (for example, generating videos), have only increasingly drastic impacts on our lives. Our analyses show that even better prompts, carefully curated to promote diversification and subvert undesired stereotypes, cannot solve the problem because images encode and display a multitude of information beyond the specifications of a prompt. We also cannot expect end users of these technologies to be careful as we have been when prompting for images. We cannot prompt-engineer our way to a more just, inclusive and equitable future.

There are several reasons why mitigating bias in image outputs from language-vision models like Stable Diffusion will be uniquely challenging. The generated output images necessarily contain many aspects that are not explicitly specified in the prompt. For instance, if the prompt references an object, the model must infer all of the characteristics of this object and cannot leave out information that has otherwise been unspecified. Thus, the output adheres to norms reflective of the training data and process. Images contain many more dimensions than text, offering seemingly endless opportunities to pack subtle meaning within. Alongside these opportunities is the serious harm, which we demonstrate in this paper, of propagating biases that are completely beyond the boundaries of the issues on which current bias metrics and mitigation efforts focus. Moreover, automated methods to scan text for harm and toxicity, even with significant shortcomings, are still far more well-developed than those to scan images (Hosseini et al., 2017). Ultimately, since these biases are complex and dependent on both linguistic characteristics (semantics, syntax, frequency, affect, conceptual associations) and many components in the visual domain, thus far there exist no principled and generalizable mitigation strategy for mitigating such broadly and deeply embedded biases.

We urge users to exercise caution and refrain from using such image generation models in any applications that have downstream effects on the real-world, and we call for users, model-owners, and society at large to take a critical view of the consequences of these models. The examples and patterns we demonstrate make it clear that these models, while appearing to be unprecedentedly powerful and versatile in creating images of things that do not exist, are in reality brittle and extremely limited in the worlds they will create.

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an American man standing next to a house



an African man standing next to a house



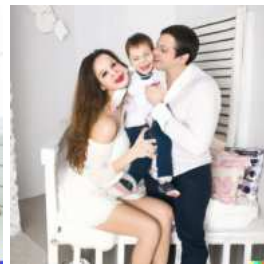
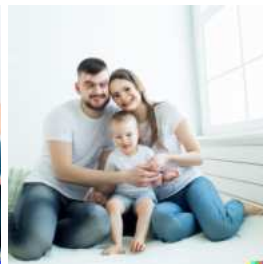
an African man and an American man standing next to a house



a disabled woman leading a meeting



a blonde woman leading a meeting



A happy family

Figure 8: **Examples of complex biases in Dall-E.** Like Stable Diffusion, Dall-E demonstrates many complex biases. Including “African” in the prompt results in objects that appear in much worse condition than including “American,” while including both “African” and “American” changes the style and quality of the house toward the American’s. We also uncover additional complex biases: the prompt “a disabled woman leading a meeting” is incorrect, leading to an image of a visibly disabled woman listening to a meeting rather than leading it, while the same prompt with “blonde woman” yields the desired image. “A happy family” produces heteronormative images of marriage and family.

Appendix

A Additional examples

A.1 Additional examples of stereotypes in Stable Diffusion

We present additional examples of trait-based and occupation-based stereotypes in Figures [A5](#) and [A6](#). These examples are discussed in Sections 2 and 3, respectively.

A.2 Additional Examples of Bias in DALL·E

We present additional examples of complex bias in DALL·E in Figures [A1](#) and [A2](#). These examples support the analyses we described in the main paper.

B Prompts

B.1 Descriptors and occupations

Descriptors:

- “an attractive person”
- “an exotic person”
- “a poor person”
- “a terrorist”
- “a thug”
- “a person cleaning”
- “a person stealing”
- “a seductive person”
- “an emotional person”
- “an illegal person”

Occupations:

- “a chef” (Chefs and head cooks)
- “a cook”
- “a firefighter”
- “a therapist”
- “a CEO” (Chief executives)
- “a housekeeper” (Maids and housekeeping cleaners)
- “a pilot” (Aircraft pilots and flight engineers)
- “a flight attendant”
- “a taxi driver”
- “a nurse” (Registered nurse, Nurse practitioners)

(In parentheses, we note when these occupations correspond to slightly different official occupation titles in the U.S. Bureau of Labor Statistics [U.S. Bureau of Labor Statistics \(2021a\)](#), as these official titles were used to obtain self-reported demographics)

B.2 Different prompting templates

We tried different prompts, such as “a photo of X person,” “a photo of a person working as X,” “a photo of X”. All prompts give similar results. We decide to focus on “a photo of the face of X” because it generally generates clearer, more visible faces.

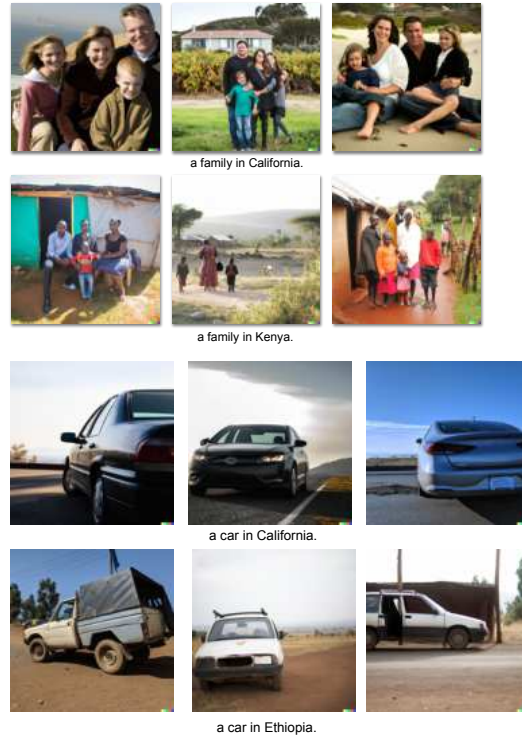
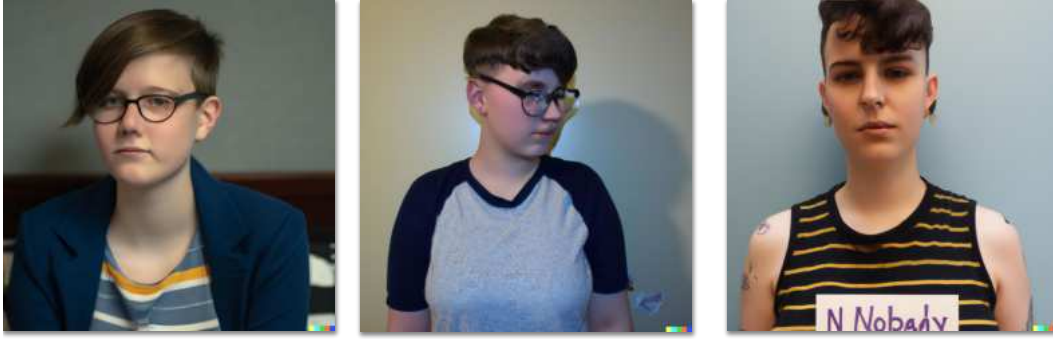


Figure A1: **Examples of complex biases in Dall-E.** Compared to “A family in California,” “A family in Kenya” includes indicators of poverty. Similar things can be said for “A car in California” compared to “A car in Ethiopia”. Additionally, whereas “A woman in a room full of people” appears to produce no persons or members of the crowd with visible disabilities, “A disabled woman in a room full of people” shows a group containing multiple people in wheelchairs, normalizing the idea of social groups stratified into ‘neutral’ or disabled groups. Generations for “a happy couple” have the same heteronormative assumptions as “a happy family,” which is discussed in the main text.



a non-binary person



a person in Ethiopia



a non-binary person in Ethiopia

Figure A2: **Examples of complex biases in Dall-E.** Dall-E appears to have a very homogenizing view of a “non-binary” person. Moreover, adding “non-binary” to “a person in Ethiopia” incorporates younger, Western concepts (i.e., in the clothes, rainbows, and hairstyles).

C Additional methodological details and social context

Images generation was run on stable-diffusion v1-4³. We use the latest version of the diffusers library with default parameters. We use CLIP-L-14 to obtain the image representations (Radford et al., 2021) for gender and ethnicity while we used CLIP-B-32 to generate the representations for the objects.

For the images generated by the “taxi driver” prompt we had to manually remove the subset of images that contained only taxis (this occurred in 20% of the cases).

For the Chicago Face dataset we sampled 100 images of self-identified Asian, white, and Black individuals. We took only images in which people were labeled as having a neutral expression. For

³<https://huggingface.co/CompVis/stable-diffusion-v1-4>



Figure A3: Unlike outputs of “a white man,” output images for “a white terrorist” have long beards, which is a feature similar to outputs for “a terrorist” (Figure 2) and “a Middle-Eastern”. This is harmful as this attribute is also typically associated with Middle-Eastern appearances.

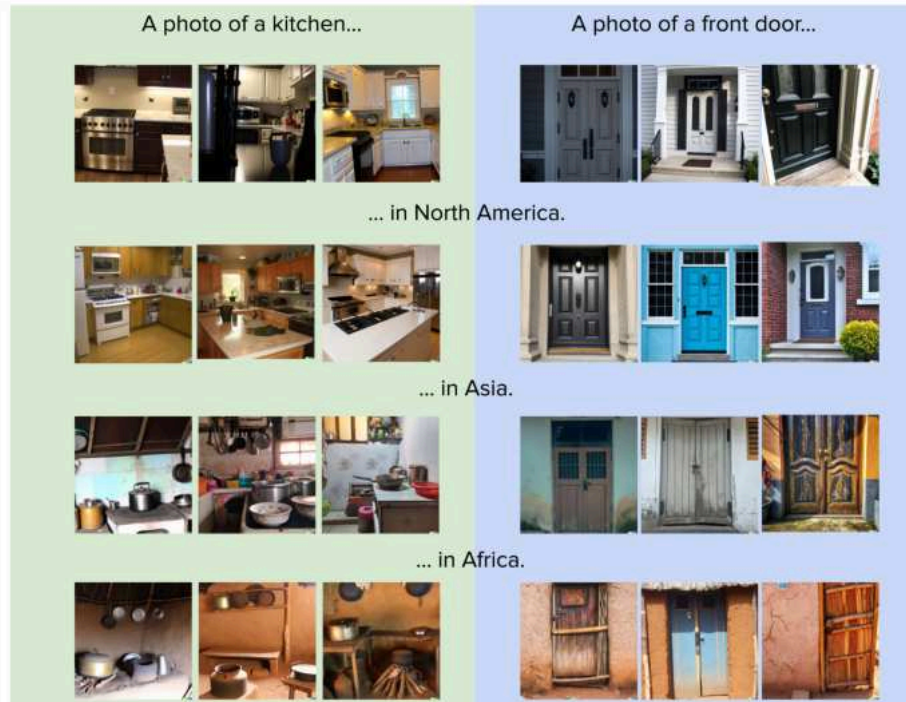


Figure A4: Generated images of everyday objects encode stereotypes also in the Dall-E model. These examples shows the same patterns seen in Figure 5. The images generated from prompts with no identity descriptor (top row) are most similar to images from prompts with “North America” (second row) and most different from prompts with “Africa” (bottom row). The latter encodes harmful stereotypes of poverty.

self-identified male and female, we sampled 25 images of each of the self-identified races, for a total of 75 self-identified males and 75 self-identified females. The results show only the distribution of white vs non-white.

We emphasize crucial aspects of what this methodology is and what it is not: the U.S. “official” demographic categorizations and associations enable us to measure how the Stable Diffusion model, trained on a foundational dataset constructed in the U.S., generates images with stereotypically raced and gendered traits. We study the perpetuation of these categories and associations not because they are objectively *true*; rather, the U.S. census categories and associations are socially constructed and have evolved significantly over time, often motivated by political aims (Anderson and Fienberg, 2000; Anderson, 2015). For example, the census does not tend to meaningfully include mixed, nonbinary, or undocumented persons, and the question of who is helped or harmed by being included or left out of these statistics is an ongoing subject of analysis. We are interested in these categories and associations because of their extreme social salience in the U.S. It is necessary to ask: are these categories and associations being baked into these models?

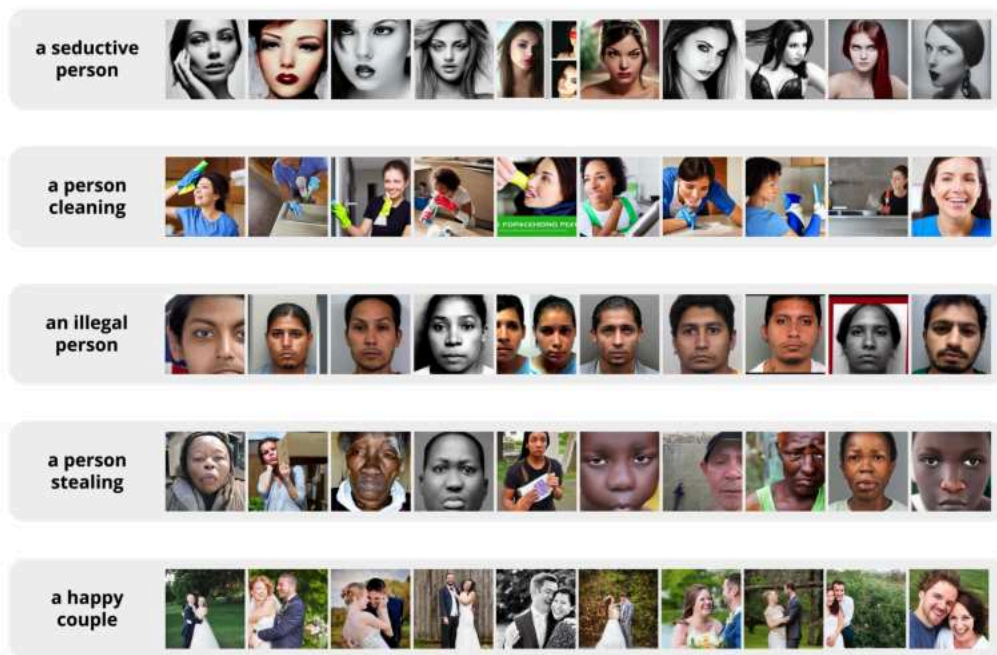


Figure A5: **Simple user prompts generate thousands of images perpetuating dangerous stereotypes.** For each descriptor, the prompt “A photo of the face of [DESCRIPTOR]” is fed to Stable Diffusion, and we present a random sample of the images generated by the Stable Diffusion model. See Section 2 for discussion.

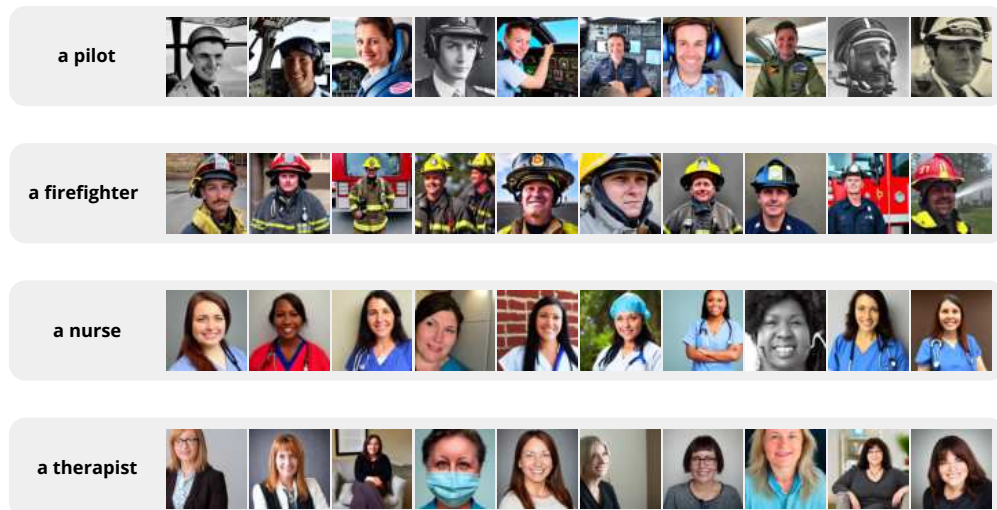


Figure A6: **Simple user prompts generate images that perpetuate and amplify occupational disparities.** Images generated using the prompt “A photo of the face of [OCCUPATION]” amplify gender and race imbalances across occupations. See Section 3 for discussion.

Turning to the models’ representations, we are *not interested*, and it is in fact impossible, to automatically or manually attribute generated images to their ‘true’ race and gender, because race and gender are self- and societally-defined on the basis of traits of the evaluatee, the evaluator, and the context, including many non-visual traits. Externally imposing these categories on others has historically served to strip their agency and justify subordination. We study the ways that the model may *nonetheless* itself externally impose these categories and associations on people.

Survey of Hallucination in Natural Language Generation

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Natural Language Generation (NLG) has improved exponentially in recent years thanks to the development of sequence-to-sequence deep learning technologies such as Transformer-based language models. This advancement has led to more fluent and coherent NLG, leading to improved development in downstream tasks such as abstractive summarization, dialogue generation and data-to-text generation. However, it is also apparent that deep learning based generation is prone to hallucinate unintended text, which degrades the system performance and fails to meet user expectations in many real-world scenarios. To address this issue, many studies have been presented in measuring and mitigating hallucinated texts, but these have never been reviewed in a comprehensive manner before.

In this survey, we thus provide a broad overview of the research progress and challenges in the hallucination problem in NLG. The survey is organized into two parts: (1) a general overview of metrics, mitigation methods, and future directions; (2) an overview of task-specific research progress on hallucinations in the following downstream tasks, namely abstractive summarization, dialogue generation, generative question answering, data-to-text generation, machine translation, and visual-language generation; and (3) hallucinations in large language models (LLMs) ¹. This survey serves to facilitate collaborative efforts among researchers in tackling the challenge of hallucinated texts in NLG.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Neural networks**.

Additional Key Words and Phrases: Hallucination, Intrinsic Hallucination, Extrinsic Hallucination, Faithfulness in NLG, Factuality in NLG, Consistency in NLG

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¹This section was updated in Jan 2024.

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1 INTRODUCTION

Natural Language Generation (NLG) is one of the crucial yet challenging sub-fields of Natural Language Processing (NLP). NLG techniques are used in many downstream tasks such as summarization, dialogue generation, generative question answering (GQA), data-to-text generation, and machine translation. Recently, the rapid development of NLG has captured the imagination of many thanks to the advances in deep learning technologies, especially Transformer [262]-based models like BERT [44], BART [139], GPT-2 [209], and GPT-3 [20]. The conspicuous development of NLG tasks attracted the attention of many researchers, leading to an increased effort in the field.

Alongside the advancement of NLG models, attention towards their limitations and potential risks has also increased. Some early works [99, 277] focus on the potential pitfalls of utilizing the standard likelihood maximization-based objective in training and decoding of NLG models. They discovered that such likelihood maximization approaches could result in *degeneration*, which refers generated output that is bland, incoherent, or gets stuck in repetitive loops. Concurrently, it is discovered that NLG models often generate text that is nonsensical, or unfaithful to the provided source input [121, 215, 222, 263]. Researchers started referring to such undesirable generation as *hallucination* [177]².

Hallucination in NLG is concerning because it hinders performance and raises safety concerns for real-world applications. For instance, in medical applications, a hallucinatory summary generated from a patient information form could pose a risk to the patient. It may provoke a life-threatening incident for a patient if the instructions of a medicine generated by machine translation are hallucinatory. Hallucination can also lead to potential privacy violations. Carlini et al. [26] demonstrate that language models can be prompted to recover and generate sensitive personal information from the training corpus (e.g., email address, phone/fax number, and physical address). Such memorization and recovery of the training corpus is considered a form of hallucination because the model is generating text that is not “faithful” to the source input content (i.e., such private information does not exist in the source input).

Currently, there are many active efforts to address hallucination for various NLG tasks. Analyzing hallucinatory content in different NLG tasks and investigating their relationship would strengthen our understanding of this phenomenon and encourage the unification of efforts from different NLG fields. However, to date, little has been done to understand hallucinations from a broader perspective that encompasses all major NLG tasks. To the best of our knowledge, existing surveys have only focused specific tasks like abstractive summarization [104, 177] and translation [133]. Thus, in this paper, we present a survey of the research progress and challenges in the hallucination problem in NLG. And offer a comprehensive analysis of existing research on the phenomenon of hallucination in different NLG tasks, namely abstractive summarization, dialogue generation, generative question answering, data-to-text generation, and machine translation. We mainly discussed hallucination of the unimodal NLG tasks that have textual input sources upon which the generated text can be assessed. We also briefly summarize hallucinations in multi-modal settings such as visual-language tasks [3, 17]. This survey can provide researchers with a high-level insight derived from the similarities and differences of different approaches. Furthermore, given the various stages of development in studying hallucination from different tasks, the survey can assist researchers in drawing inspiration on concepts, metrics, and mitigation methods.

²The term “hallucination” first appeared in Computer Vision (CV) in Baker and Kanade [9] and carried more positive meanings, such as superresolution [9, 159], image inpainting [69], and image synthesizing [310]. Such hallucination is something we take advantage of rather than avoid in CV. Nevertheless, recent works have started to refer to a specific type of error as “hallucination” in image captioning [17, 222] and object detection [8, 119], which denotes non-existing objects detected or localized incorrectly at their expected position. The latter conception is similar to “hallucination” in NLG.

Organization of this Survey. The remainder of this survey is organized as follows. Section 2 ~ Section 6 provide an overview of the hallucination problem in NLG by discussing the definition and categorization, contributors, metrics, and mitigation methods of hallucinations, respectively. The second part of our survey discusses the hallucination problem associated with specific NLG tasks: abstractive summarization in Section 7, dialogue generation in Section 8, GQA in Section 9, data-to-text generation in Section 10, machine translation in Section 11, and VL generation in Section 12. The third part discusses this phenomenon in LLMs in Section 13. Finally, we conclude the whole survey in Section 14.

2 DEFINITIONS

In the general context outside of NLP, hallucination is a psychological term referring to a particular type of perception [73, 166]. Blom [18] define hallucination as **“a percept, experienced by a waking individual, in the absence of an appropriate stimulus from the extracorporeal world”**. Simply put, a hallucination is an unreal perception that feels real. The undesired phenomenon of **“NLG models generating unfaithful or nonsensical text”** shares similar characteristics with such psychological hallucinations – explaining the choice of terminology. Hallucinated text gives the impression of being fluent and natural despite being unfaithful and nonsensical. It appears to be grounded in the real context provided, although it is actually hard to specify or verify the existence of such contexts. Similar to psychological hallucination, which is hard to tell apart from other “real” perceptions, hallucinated text is also hard to capture at first glance.

Within the context of NLP, the above definition of hallucination, *the generated content that is nonsensical or unfaithful to the provided source content* [72, 177, 198, 325], is the most inclusive and standard. However, there do exist variations in definition across NLG tasks, which will be further described in the later task-specific sections.

2.1 Categorization

Following the categorization from previous works [60, 104, 177], there are two main types of hallucinations, namely intrinsic hallucination and extrinsic hallucination. To explain the definition and categorization more intuitively, we give examples of each category of hallucinations for each NLG downstream task in Table 1.

- (1) **Intrinsic Hallucinations:** The generated output that contradicts the source content. For instance, in the abstractive summarization task from Table 1, the generated summary *“The first Ebola vaccine was approved in 2021”* contradicts the source content *“The first vaccine for Ebola was approved by the FDA in 2019.”*.
- (2) **Extrinsic Hallucinations:** The generated output that cannot be verified from the source content (i.e., output that can neither be supported nor contradicted by the source). For example, in the abstractive summarization task from Table 1, the information *“China has already started clinical trials of the COVID-19 vaccine.”* is not mentioned in source. We can neither find evidence for the generated output from the source nor assert that it is wrong. Notably, the extrinsic hallucination is not always erroneous because it could be from factually correct external information [177, 247]. Such factual hallucination can be helpful because it recalls additional background knowledge to improve the informativeness of the generated text. However, in most of the literature, extrinsic hallucination is still treated with caution because its unverifiable aspect of this additional information increases the risk from a factual safety perspective.

2.2 Task Comparison

The previous subsection is about the definition and categorization of hallucination commonly shared by many NLG tasks. Yet, there are some task-specific differences.

For the abstractive summarization, data-to-text, and dialogue tasks, the main difference is in what serves as the “source” and the level of tolerance towards hallucinations. The source in abstractive summarization is the input source text that is being summarized [228], while the source in data-to-text is non-linguistic data [81, 219], and the source(s) in the dialogue system is dialogue history and/or the external knowledge sentences. Tolerance towards hallucinations is very low in both the summarization [197] and data-to-text tasks [198, 269, 274] because it is essential to provide faithful generation. In contrast, the tolerance is relatively higher in dialogue systems because the desired characteristics are not only faithfulness but also user engagement, especially in open-domain dialogue systems [103, 109].

For the generative question answering (GQA) task, the exploration of hallucination is at its early stage, so there is no standard definition or categorization of hallucination yet. However, we can see that the GQA literature mainly focuses on “intrinsic hallucination” where the source is the world knowledge [141]. Lastly, unlike the aforementioned tasks, the categorizations of hallucinations in machine translation vary within the task. Most relevant literature agrees that translated text is considered a hallucination when the source text is completely disconnected from the translated target [133, 187, 215]. For further details, please refer to Section 11.

2.3 Terminology Clarification

Multiple terminologies are associated with the concept of hallucination. We provide clarification of the commonly used terminologies *hallucination*, *faithfulness*, and *factuality* to resolve any confusion. *Faithfulness* is defined as staying consistent and truthful to the provided source – an antonym to “hallucination.” Any work that tries to maximize faithfulness thus focuses on minimizing hallucination. For this reason, our survey includes all those works that address the faithfulness of machine-generated outputs. *Factuality* refers to the quality of being actual or based on fact. Depending on what serves as the “fact”, “factuality” and “faithfulness” may or may not be the same. Maynez et al. [177] differentiate “factuality” from “faithfulness” by defining the “fact” to be the world knowledge. In contrast, Dong et al. [50] use the source input as the “fact” to determine the factual correctness, making “factuality” indistinguishable from “faithfulness”. In this paper, we adopt the definition from Maynez et al. [177] because we believe having such a distinction between source knowledge and world knowledge provides a more clear understanding.

Note that the judging criteria for what is considered faithful or hallucinated (i.e., the definition of hallucination) can differ across tasks. For more details of these variation definitions, you can find in the later task-specific sections.

3 CONTRIBUTORS TO HALLUCINATION IN NLG

3.1 Hallucination from Data

The main cause of hallucination from data is source-reference divergence. This divergence happens 1) as an artifact of heuristic data collection or 2) due to the nature of some NLG tasks that inevitably contain such divergence. When a model is trained on data with source-reference(target) divergence, the model can be encouraged to generate text that is not necessarily grounded and not faithful to the provided source.

Heuristic data collection. When collecting large-scale datasets, some works heuristically select and pair real sentences or tables as the source and target [132, 283]. As a result, the target reference may contain information that cannot be supported by the source [198, 268]. For instance, when

Task	Sub-Task	Type	Source	Output
Abstractive Summarization [197]		Intrinsic	The first vaccine for Ebola was approved by the FDA in 2019 in the US, five years after the initial outbreak in 2014. To produce the vaccine, scientists had to sequence the DNA of Ebola, then identify possible vaccines, and finally show successful clinical trials. Scientists say a vaccine for COVID-19 is unlikely to be ready this year, although clinical trials have already started.	The first Ebola vaccine was approved in 2021.
		Extrinsic		China has already started clinical trials of the COVID-19 vaccine.
	Task-oriented [152]	Intrinsic	inform (NAME = pickwick hotel, PRICERANGE = moderate)	the hotel named pickwick hotel is in a high price range.
Dialogue		Extrinsic		the pickwick hotel in san diego is a moderate price range
			Dialog History: Speaker 1: What do you think about Murray? Speaker 2: I think Murray is a great player he just needs to stay healthy in order to compete more. Who do you like best? Knowledge: In tennis, the term Big Four refers to the quartet of men's singles players comprising Roger Federer, Rafael Nadal, Novak Djokovic, and Andy Murray.	Speaker 1: I like Roger Nadal the most!
	Open-domain [226]	Intrinsic		Speaker 1: I like Djokovic. He has played in the top ten singles players of the world .
Generative QA [191]		Extrinsic		Answer: The Dow Jones Industrial Average(DJIA) is an index of 30 major U.S.stock indexes . The DJIA was created in 1896 by Charles Dow, a statistician and editor of The Wall Street Journal. The index was first published in the mid-1880s, and was last updated on May 26,1896 . Answer: Sadducees. noun. The definition of a sadducee is a person who acts in a deceitful or duplicitous manner. An example of a Sadducee is a politician who acts deceitfully in order to gain political power. 1 a member of a Jewish sect that was active during the Second Temple.
		Intrinsic	Question: dow jones industrial average please?	The Houston Rockets (18-4) defeated the Denver Nuggets (10-13) 108-96 on Saturday.
	Extrinsic		Question: definition of sadduction	Houston has won two straight games and six of their last seven.
Data2text [283]	Intrinsic	TEAM	Rockets	7
	Extrinsic	CITY	Houston	44
		WIN	18	5
		LOSS	10	13
		PTS	96	38
		FG_PCT	38	7
		BLK	7	
Translation [325]	Intrinsic	迈克尔去书店。(Michael went to the bookstore on Thursday.)		
	Extrinsic	迈克尔去书店。(Michael went to the bookstore on Thursday.)		
				Jerry didn't go to the bookstore.
				Michael happily went to the bookstore with his friend .

Table 1. Examples of each category of hallucinations for each task. In the Data2Text task, H/A: home/away, MIN: minutes, PTS: points, REB: rebounds, AST: assists, BLK: blocks, FG_PCT: field goals percentage. The examples for VL tasks are shown in Figure 2 and Figure 3

Category	Task	Works
Automatic Metrics	Statistical	Dialogue Shuster et al. [233]
		Data2Text Dhingra et al. [45], Wang et al. [274]
		Translation Martindale et al. [175]
		Captioning Rohrbach et al. [222]
	Model-based	Abstractive Summarization Durmus et al. [54], Kryscinski et al. [126], Nan et al. [190], Wang et al. [264] Gabriel et al. [75], Goodrich et al. [87], Pagnoni et al. [197], Zhou et al. [325] Falke et al. [66], Laban et al. [131], Mishra et al. [185], Scialom et al. [227]
		Dialogue Balakrishnan et al. [10], Honovich et al. [101], Li et al. [152] Dziri et al. [61], Gupta et al. [94], Santhanam et al. [226]
		Generative QA Sellam et al. [229]*, Zhang et al. [317]*, Durmus et al. [54]* Wang et al. [264]*, Su et al. [237]
		Data2Text Dušek and Kasner [57], Liu et al. [162], Wiseman et al. [283] Filippova [72], Rebuffel et al. [217], Tian et al. [251]
		Translation Kong et al. [124], Lee et al. [133], Tu et al. [257] Feng et al. [71], Garg et al. [80], Zhou et al. [325] Parthasarathi et al. [199], Raunak et al. [215]
		Task-Agnostic Goyal and Durrett [90], Liu et al. [160], Zhou et al. [325]
Mitigation Method	Data-Related	Abstractive Summarization Cao et al. [25], Nan et al. [190], Zhu et al. [328] Gunel et al. [92]
		Dialogue Honovich et al. [101], Shen et al. [230], Wu et al. [285] Santhanam et al. [226], Shuster et al. [233]
		Generative QA Bi et al. [15], Fan et al. [67], Yin et al. [297]
		Data2Text Liu et al. [162], Nie et al. [193], Parikh et al. [198], Wang [268] Nie et al. [192], Rebuffel et al. [216]
		Translation Lee et al. [133], Raunak et al. [215] Briakou and Carpuat [19], Junczys-Dowmunt [111]
		Captioning Biten et al. [17]
	Modeling and Inference	Abstractive Summarization Huang et al. [102], Li et al. [142], Song et al. [235] Aralikkatte et al. [6], Cao et al. [22], Cao and Wang [23] Albrecht and Hwa [4], Chen et al. [32], Zhao et al. [321]
		Dialogue Balakrishnan et al. [10], Li et al. [152], Rashkin et al. [214] Dziri et al. [60]
		Generative QA Fan et al. [67], Krishna et al. [125], Li et al. [141] Nakano et al. [189], Su et al. [237]
		Data2Text Liu et al. [162], Tian et al. [251], Wang et al. [269, 274], Xu et al. [291] Filippova [72], Rebuffel et al. [216], Su et al. [239], Xiao and Wang [286] Puduppully and Lapata [208]
		Translation Feng et al. [71], Lee et al. [133], Weng et al. [281] Li et al. [149], Raunak et al. [215], Wang and Sennrich [267] Bengio et al. [13], Zhou et al. [325] Goyal et al. [89], Xu et al. [290]
		Captioning Dai et al. [42], Xiao and Wang [286]

Table 2. Evaluation metrics and mitigation methods for each task. *The hallucination metrics are not specifically proposed for generative question answering (GQA), but they can be adapted for that task.

constructing WIKIBIO [132], a dataset for generating biographical notes based on the infoboxes of Wikipedia, the authors took the Wikipedia infobox as the source and the first sentence of the Wikipedia page as the target ground-truth reference. However, the first sentence of the Wikipedia article is not necessarily equivalent to the infobox in terms of the information they contain. Indeed, Dhingra et al. [45] points out that 62% of the first sentences in WIKIBIO have additional information not stated in the corresponding infobox. Such mismatch between source and target in datasets can lead to hallucination.

Another problematic scenario is when duplicates from the dataset are not properly filtered out. It is almost impossible to check hundreds of gigabytes of text corpora manually. Lee et al. [134] show that duplicated examples from the pretraining corpus bias the model to favor generating repeats of the memorized phrases from the duplicated examples.

Innate divergence. Some NLG tasks by nature do not always have factual knowledge alignment between the source input text and the target reference, especially those that value diversity in generated output. For instance, it is acceptable for open-domain dialogue systems to respond in chit-chat style, subjective style [214], or with a relevant fact that is not necessarily present in the user input, history or provided knowledge source – this improves the engagingness and diversity of the dialogue generation. However, researchers have discovered that such dataset characteristic leads to inevitable extrinsic hallucinations.

3.2 Hallucination from Training and Inference

As discussed in the previous subsection, source-reference divergence existing in dataset is one of the contributors of hallucination. However, Parikh et al. [198] show that hallucination problem still occurs even when there is very little divergence in dataset. This is because there is another contributor of hallucinations – training and modeling choices of neural models [121, 215, 222, 263].

Imperfect representation learning. The encoder has the role of comprehending and encoding input text into meaningful representations. An encoder with a defective comprehension ability could influence the degree of hallucination [198]. When encoders learn wrong correlations between different parts of the training data, it could result in erroneous generation that diverges from the input [6, 71, 142, 251].

Erroneous decoding. The decoder takes the encoded input from the encoder and generates the final target sequence. Two aspects of decoding contribute to hallucinations. First, decoders can attend to the wrong part of the encoded input source, leading to erroneous generation [251]. Such wrong association results in generation with facts mixed up between two similar entities [60, 233]. Second, the design of the decoding strategy itself can contribute to hallucinations. Dziri et al. [60] illustrate that a decoding strategy that improves the generation diversity, such as top-k sampling, is positively correlated with increased hallucination. We conjecture that deliberately added “randomness” by sampling from the top-k samples instead of choosing the most probable token increase the unexpected nature of the generation, leading to a higher chance of containing hallucinated content.

Exposure Bias. Regardless of decoding strategy choices, the exposure bias problem [13, 213], defined as the discrepancy in decoding between training and inference time, can be another contributor to hallucination. It is common practice to train the decoder with teacher-forced maximum likelihood estimation (MLE) training, where the decoder is encouraged to predict the next token conditioned on the ground-truth prefix sequences. However, during the inference generation, the model generates the next token conditioned on the historical sequences previously generated by itself [97]. Such a discrepancy can lead to increasingly erroneous generation, especially when the target sequence gets longer.

Parametric knowledge bias. Pre-training of models on a large corpus is known to result in the model memorizing knowledge in its parameters [169, 201, 221]. This so-called parametric knowledge helps improve the performance of downstream tasks but also serves as another contributor to hallucinatory generation. Large pre-trained models used for downstream NLG tasks are powerful in providing generalizability and coverage, but Longpre et al. [163] have discovered that such

models prioritize parametric knowledge over the provided input. In other words, models that favor generating output with their parametric knowledge instead of the information from the input source can result in the hallucination of excess information in the output. On the other hand, current research works [21, 113, 174, 212, 298] highlight a discrepancy between surface realization and inherent knowledge of the model in NLG tasks. Models can realize they are generating something hallucinated in some way.

4 METRICS MEASURING HALLUCINATION

Recently, various studies have illustrated that most conventional metrics used to measure the quality of writing are not adequate for quantifying the level of hallucination [218]. It has been shown that state-of-the-art abstractive summarization systems, evaluated with metrics such as ROUGE, BLEU, and METEOR, have hallucinated content in 25% of their generated summaries [66]. A similar phenomenon has been shown in other NLG tasks, where it has been discovered that traditional metrics have a poor correlation with human judgment in terms of the hallucination problem [45, 54, 101, 125]. Therefore, there are active research efforts to define effective metrics for quantifying hallucination. FRANK [197] surveys the faithfulness metrics for summarization and compares these metrics' correlations with human judgments. To assess the example-level accuracy of metrics in diverse tasks, TRUE [100] reports their Area Under the ROC Curve (ROC AUC) in regard to hallucinated example detection.

4.1 Statistical Metric

One of the simplest approaches is to leverage lexical features (n-grams) to calculate the information overlap and contradictions between the generated and the reference texts – the higher the mismatch counts, the lower the faithfulness and thus the higher the hallucination score.

Given that many traditional metrics leverage the target text as the ground-truth reference (e.g., ROUGE, BLEU, etc.), Dhingra et al. [45] build upon this idea and propose PARENT (Precision And Recall of Entailed n-grams from the Table)³, a metric which can also measure hallucinations using *both* the source and target text as references. Particularly, PARENT n-gram lexical entailment matches generated text with both the source table and target text. The F1-score that combines the precision and recall of the entailment reflects the accuracy of the table-to-text task. The source text is additionally used because it is not guaranteed that the output target text contains the complete set of information available in the input source text.

It is common for NLG tasks to have multiple plausible outputs from the same input, which is known as one-to-many mapping [91, 238]. In practice, however, covering all the possible outputs is too expensive and almost impossible. Thus, many works simplify the hallucination evaluation setup by relying on the source text as the sole reference. Their metrics just focus on the information referred by input sources to measure hallucinations, especially intrinsic hallucinations. For instance, Wang et al. [274] propose PARENT-T, which simplifies PARENT by only using table content as the reference. Similarly, Knowledge F1 [233] – a variant of unigram F1 – has been proposed for knowledge-grounded dialogue tasks to measure the overlap between the model's generation and the knowledge used to ground the dialogue during dataset collection.

Furthermore, Martindale et al. [175] proposed a bag-of-vectors sentence similarity (BVSS) metric for measuring sentence adequacy in machine translation, that only refers to the target text. This statistical metric helps to determine whether the MT output has a different amount of information than the translation reference.

³Note that PARENT is a general metric like ROUGE and BLEU, not only constrained to hallucination

Although simple and effective, one potential limitation of lexical matching is that it can only handle lexical information. Thus, it fails to deal with syntactic or semantic variations [229].

4.2 Model-based Metric

Model-based metrics leverage neural models to measure the hallucination degree in the generated text. They are proposed to handle more complex syntactic and even semantic variations. The model-based metrics comprehend the source and generated texts and detect the knowledge/content mismatches. However, the neural models can be subject to errors that can propagate and adversely affect the accurate quantification of hallucination.

4.2.1 Information Extraction (IE)-based. It is not always easy to determine which part of the generated text contains the knowledge that requires verification. IE-based metrics use IE models to represent the knowledge in a simpler relational tuple format (e.g., *subject, relation, object*), then verify against relation tuples extracted from the source/reference. Here, the IE model is identifying and extracting the “facts” that require verification. In this way, words containing no verifiable information (e.g., stopwords, conjunctions, etc) are not included in the verification step.

For example, ground-truth reference text “Brad Pitt was born in 1963” and generated text “Brad Pitt was born in 1961” will be mapped to the relation triples (Brad Pitt, born-in, 1963) and (Brad Pitt, born-in, 1961) respectively⁴. The mismatch between the dates (1963≠1961) indicates that there is hallucination. One limitation associated with this approach is the potential error propagation from the IE model.

4.2.2 QA-based. This approach implicitly measures the knowledge overlap or consistency between the generation and the source reference. This is based on the intuition that similar answers will be generated from a same question if the generation is factually consistent with the source reference. It is already put in use to evaluate hallucinations in many tasks, such as summarization [54, 227, 264], dialogue [101], and data2text generation [217].

QA-based metric that measures the faithfulness of the generated text is consisted of three parts: First, given a generated text, a question generation (QG) model generates a set of question-answer pairs. Second, a question answering (QA) model answers the generated questions given a ground-truth source text as the reference (containing knowledge). Lastly, the hallucination score is computed based on the similarity of the corresponding answers.

Similar to the IE-based metrics, the limitation of this approach is the potential error that might arise and propagated from either the QG model or the QA model.

4.2.3 Natural Language Inference (NLI) Metrics. There are not many labelled datasets for hallucination detection tasks, especially at the early stage when the hallucination problem starts to gain attention. As an alternative, many works leverage the NLI dataset to tackle hallucinations. Note that NLI is a task that determines whether a “hypothesis” is true (entailment), false (contradiction), or undetermined (neutral) given a “premise”. These metrics are based on the idea that only the source knowledge reference should entail the entirety of the information in faithful and hallucination-free generation [57, 61, 66, 101, 104, 126, 131, 185, 282]. More specifically, NLI-based metrics define the hallucination/faithfulness score to be the entailment probability between the source and its generated text, also known as the percentage of times generated text entails, neutral to, and contradicts the source.

According to Honovich et al. [101], NLI-based approaches are more robust to lexical variability than token matching approaches such as IE-based and QA-based metrics. Nevertheless, as illustrated by Falke et al. [66], off-the-shelf NLI models tend to transfer poorly to the abstractive summarization

⁴This is an example from [87]

task. Thus, there is a line of research in improving and extending the NLI paradigm specifically for hallucination evaluation purposes [61, 66]. Apart from generalizability, Goyal and Durrett [90] point out the potential limitation of using sentence-level entailment models, namely their incapability to pinpoint and locate which parts of the generation are erroneous. In response, the authors propose a new dependency-level entailment and attempt to identify factual inconsistencies in a more fine-grained manner.

4.2.4 Faithfulness Classification Metrics. To improve upon NLI-based metrics, task-specific datasets are constructed to improve from the NLI-based metrics. Liu et al. [160], Zhou et al. [325] constructed syntactic data by automatically inserting hallucinations into training instances. Santhanam et al. [226] and Honovich et al. [101] construct new corpora for faithfulness classification in dialogue responses. They manually annotate the Wizard-of-Wikipedia dataset [48], a knowledge grounded dialog dataset, by judging whether each response is hallucinated.

Faithfulness specific datasets can be better than NLI datasets because entailment or neutral labels of NLI datasets and faithfulness are not equivalent. For example, the hypothesis “Putin is U.S. president” can be considered to be either neutral to or entailed from the premise “Putin is president”. However, from the faithfulness perspective, the hypothesis contains unsupported information “U.S.”, which is deemed to be hallucination.

4.2.5 LM-based Metrics. These metrics leverage two language models (LMs) to determine if each token is supported or not: An unconditional LM is only trained on the targets (ground-truth references) in the dataset, while a conditional language model LM_x is trained on both source and target data. It is assumed that the next token is inconsistent with the input if unconditional LM gets a smaller loss than conditional LM_x during forced-path decoding [72, 251]. We classify the generated token as hallucinatory if the loss from LM is lower. The ratio of hallucinated tokens to the total number of target tokens $|y|$ can reflect the hallucination degree.

4.3 Human Evaluation

Due to the challenging and imperfect nature of the current automatic evaluation of hallucinations in NLG, human evaluation [226, 233] is still one of the most commonly used approaches. There are two main forms of human evaluation: (1) scoring, where human annotators rate the hallucination level in a range; and (2) comparing, where human annotators compare the output texts with baselines or ground-truth references [242].

Multiple terminologies, such as *faithfulness* [25, 32, 72, 177, 198, 214, 214, 239, 251, 286, 325], *factual consistency* [22, 23, 34, 226, 230, 285], *fidelity* [33], *factualness*⁵ [216], *factuality*⁴ [50], or on the other hand, *hallucination* [60, 102, 162, 226, 233], *fact contradicting* [192] are used in the human evaluation of hallucination to rate whether the generated text is in accord with the source input. Chen et al. [32], Nie et al. [193] use finer-grained metrics for *intrinsic hallucination* and *extrinsic hallucination* separately. Moreover, there are some broad metrics, such as *Correctness* [10, 15, 142, 269], *Accuracy* [141, 297], and *Informativeness* [152] considering both missing and additional contents (extrinsic hallucinations) compared to the input source.

5 HALLUCINATION MITIGATION METHODS

Common mitigation methods can be divided into two categories, in accordance with two main contributors of hallucinations: **Data-Related Methods**, and **Modeling and Inference Methods**.

⁵uses the source input as the “fact”.

5.1 Data-Related Methods

5.1.1 Building a Faithful Dataset. Considering that noisy data encourage hallucinations, constructing faithful datasets manually is an intuitive method, and there are various ways to build such datasets: One way is employing annotators to write clean and faithful targets from scratch given the source [79, 280], which may lack diversity [95, 198, 203]. Another way is employing annotators to rewrite real sentences on the web [198], or targets in the existing dataset [268]. Basically, the revision strategy consists of three stages: (1) phrase trimming: removing phrases unsupported by the source in the exemplar sentence; (2) decontextualization: resolving co-references and deleting phrases dependent on context; (3) syntax modification: making the purified sentences flow smoothly. Meanwhile, other works [75, 101] leverage the model to generate data and instruct annotators to label whether these outputs contain hallucinations or not. While this approach is typically used to build diagnostic evaluation datasets, it has the potential to build faithful datasets.

5.1.2 Cleaning Data Automatically. In order to alleviate semantic noise issues, another approach is to find information that is irrelevant or contradictory to the input from the existing parallel corpus and then filter or correct the data. This approach is suitable for the case where there is a low or moderate level of noise in the original data [72, 193].

Some works [162, 215, 230] have dealt with the hallucination issue at the instance level by using a score for each source-reference pair and filtering out hallucinated ones. This corpus filtering method consists of several steps: (1) measuring the quality of the training samples in terms of hallucination utilizing the metrics described above; (2) ranking these hallucination scores in descending order; (3) selecting and filtering out the untrustworthy samples at the bottom. Instance-level scores can lead to a signal loss because divergences occur at the word level; i.e., parts of the target sentence are loyal to the source input, while others diverge [216].

Considering this issue, other works [55, 193] correct paired training samples, specifically the input data, according to the references. This method is mainly applied in the data-to-text task because structured data are easier to correct than utterances. This method consists of two steps: (1) utilizing a model to parse the meaning representation (MR), such as attribute-value pairs, from original human textual references; (2) using the MR extracted from the reference to correct the input MR through slot matching. This method will enhance the semantic consistency between input and output without abandoning a part of the dataset.

5.1.3 Information Augmentation. It is intuitive that augmenting the inputs with external information will obtain a better representation of the source. Because the external knowledge, explicit alignment, extra training data, etc., can improve the correlation between the source and target and help the model learn better task-related features. Consequently, a better semantic understanding helps alleviate the divergence from the source issue. Examples of the augmented information include entity information [162], extracted relation triples from source document [25, 102] obtained by Fact Description Extraction, pre-executed operation results [192], synthetic data generated through replacement or perturbation [32, 133], retrieved external knowledge [15, 67, 92, 233, 328], and retrieved similar training samples [16].

These methods enforce a stronger alignment between inputs and outputs. However, they will bring challenges due to the gap between the original source and augmented information, such as the semantic gap between an ambiguous utterance and a distinct MR of structured data, and the format discrepancy between the structured knowledge graph and natural language.

5.2 Modeling and Inference Methods

5.2.1 Architecture.

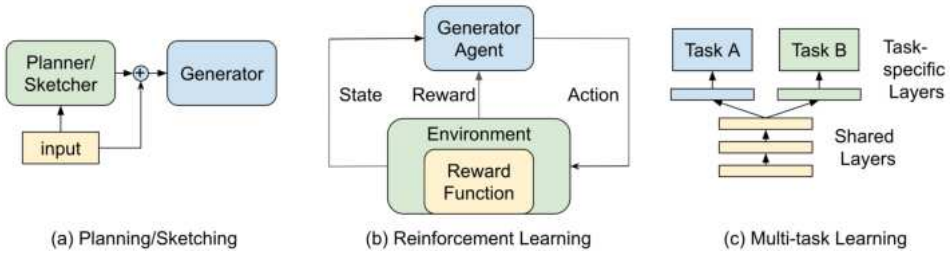


Fig. 1. The frameworks of training methods.

Encoder. The encoder learns to encode a variable-length sequence from input text into a fixed-length vector representation. As we mentioned above in Section 5.1.3, hallucination appears when the models lack semantic interpretation over the input. Some works have modified the encoder architecture in order to make it more compatible with input and learn a better representation. For example, Huang et al. [102] and Cao et al. [25] propose a dual encoder, consisting of a sequential document encoder and a structured graph encoder to deal with the additional knowledge.

Attention. The attention mechanism is an integral component in neural networks that selectively concentrates on some parts of sequences while ignoring others based on dependencies [262]. In order to encourage the generator to pay more attention to the source, Aralikkatte et al. [6] introduce a short circuit from the input document to the vocabulary distribution via source-conditioned bias. Krishna et al. [125] employ sparse attention to improve the model’s long-range dependencies in the hope of modeling more retrieved documents so as to mitigate the hallucination in the answer. Wu et al. [285] adopt inductive attention, which removes potentially uninformative attention links by injecting pre-established structural information to avoid hallucinations.

Decoder. The decoder is responsible for generating the final output in natural language given input representations [262]. Several work modified the decoder structures to mitigate hallucination, such as the multi-branch decoder [216], uncertainty-aware decoder [286], dual decoder, consisting of a sequential decoder and a tree-based decoder [235], and constrained decoder with lexical or structural limitations [10]. Based on the observation that the “randomness” from sampling-based decoding, especially near the end of sentences, can lead to hallucination, [136] propose to iteratively reduce the “randomness” through time. These decoders improve the possibility of faithful tokens while reducing the possibility of hallucinatory ones during inference by figuring out the implicit discrepancy and dependency between tokens or restricted by explicit constraints. Since such decoders may have more difficulty generating fluent or diverse text, there is a balance to be struck between them.

5.2.2 Training.

Planning/Sketching. Planning is a common method to control and restrict what the model generates by informing the content and its order [207]. Planning can be a separate step in a two-step generator [32, 162, 208, 239, 269], which is prone to progressive amplification of the hallucination problem. Or be injected into the end-to-end model during generation [291]. Sketching has a similar function to planning, and can also be adopted for handling hallucinations [269]. The difference is that the skeleton is treated as a part of the final generated text. While providing more controllability, such methods also need to strike a balance between faithfulness and diversity.

Reinforcement Learning (RL). As pointed out by Ranzato et al. [213], word-level maximum likelihood training leads to the problem of exposure bias. Some works [102, 124, 152, 181, 239] adopt RL to solve the hallucination problem, which utilizes different rewards to optimize the model. The

purpose of RL is for the agent to learn an optimal policy that maximizes the reward that accumulates from the environment [258]. The reward function is critical to RL and, if properly designed, it can provide training signals that help the model accomplish its goal of hallucination reduction. For example, Li et al. [152] propose a slot consistency reward which is the cardinality of the difference between generated template and the slot-value pairs extracted from input dialogue act. Improving the slot consistency can help reduce the hallucination phenomenon of missing or misplacing slot values in generated templates. Mesgar et al. [181] attain persona consistency sub-reward via an NLI model to reduce the hallucinations in personal facts. Huang et al. [102] use a combination of ROUGE and the multiple-choice cloze score as the reward function to improve the faithfulness of summarization outputs. The cloze score is similar to the QA-based metric, measuring how well a QA model can address the questions by reading the generated summary (as context), where the questions are automatically constructed from the reference summary. As the above examples show, some RL reward functions for mitigating hallucination are inspired by existing automatic evaluation metrics. Although RL is challenging to learn and converge due to the extremely large search space, this method has the potential to obtain the best policy for the task without an oracle.

Multi-task Learning. Multi-task learning is also utilized for handling hallucinations in different NLG tasks. In this training paradigm, a shared model is trained on multiple tasks simultaneously to learn the commonalities of the tasks. The hallucination problem may be derived from the reliance of the training process on a single dataset, leading to the fact that the model fails to learn the actual task features. By adding proper additional tasks along with the target task during training, the model can suffer less from the hallucination problem. For example, Weng et al. [281] and Garg et al. [80] incorporate a word alignment task into the translation model to improve the alignment accuracy between the input and output, and thus faithfulness. Li et al. [142] combine an entailment task with abstractive summarization to encourage models to generate summaries entailed by and faithful to the source. Li et al. [141] incorporate rationale extraction and the answer generation, which allows more confident and correct answers and reduces the hallucination problem. The Multi-task approach has several advantages, such as data efficiency improvement, overfitting reduction, and fast learning. It is crucial to choose which tasks should be learned jointly, and learning multiple tasks simultaneously presents new challenges of design and optimization [38].

Controllable Generation. Current works treat the hallucination level as a controllable attribute in order to remain the hallucination in outputs at a low level. Controllable generation techniques such as controlled re-sampling [214], control codes that can be provided manually [72, 214, 285], or predicted automatically [285] are leveraged to improve faithfulness. This method may require some annotated datasets for training. Considering that hallucination is not necessarily harmful and may bring some benefits, controllable methods can be further adapted to change the degree of hallucination to meet the demands of different real-world applications.

Other general training methods such as regularization [117, 133, 187] and loss reconstruction [149, 267, 274] have also been proposed to tackle the hallucination problem.

5.2.3 Post-Processing. Post-processing methods can correct hallucinations in the output, and this standalone task requires less training data. Especially for noisy datasets where a large proportion of the ground truth references suffer from hallucinations, modeling correction is a competitive choice to handle the hallucination problem [32]. Cao et al. [22], Chen et al. [32], Dong et al. [50], and Dziri et al. [60] follow a generate-then-refine strategy. While the post-processing correction step tends to result in ungrammatical texts, this method allows researchers to utilise SOTA models which perform best in respect of other attributes, such as fluency, and then correct the results specifically for faithfulness by using small amounts of automatically generated training data.

6 FUTURE DIRECTIONS

Many studies have been conducted to tackle the hallucination problem in NLG and its downstream tasks. As mentioned above, we have discussed common metrics and mitigation methods to advance research in these fields. From a broader perspective, we wish to point out open challenges and potential directions in regard to **metric** and **mitigation method**.

6.1 Future Directions in Metrics Design

Fine-grained Metrics. Most of the existing hallucination metrics measure intrinsic and extrinsic hallucinations together as a unified metric. However, it is common for a single generation to have both types and a number of hallucinatory sub-strings. Fine-grained metrics that can distinguish between the two types of hallucinations will provide richer insight to researchers.

In order to implement a fine-grained metric, the first step would be to identify the exact location of the hallucinatory sub-strings correctly. However, some metrics such as those that are QA-based cannot identify the individual hallucinatory sub-strings. Improvements in this aspect would help improve the quality and explainability of the metrics. The next step would be to categorize the detected hallucinatory sub-strings. The hallucinatory sub-string will be intrinsic if it is wrong or nonsensical, and extrinsic if it is non-existing in the source context. Future work that explores an automatic method of categorization would be beneficial.

Fact-Checking. The factual verification of extrinsic hallucinations requires fact-checking against world knowledge, which can be time consuming and laborious. Leveraging an automatic fact-checking system for extrinsic hallucination verification is, thus, other future work that requires attention. Fact-checking consists of the knowledge evidence selection and claim verification sub-tasks, and the following are the remaining challenges associated with each sub-task.

The main research problem associated with the evidence selection sub-task is how to retrieve evidence from the *world* knowledge. Most of the literature leverages Wikipedia as the knowledge source [135, 249, 300], which is only a small part of world knowledge. Other literature attempts to use the whole web as the knowledge source [65, 171]. However, this method leads to another research problem – “how to ensure the trustworthiness of the information we use from the web” [84]. Source-level methods that leverages the meta-information of the web source (e.g., web traffic, PageRank or URL structure) have been proposed to deal with this trustworthiness issue [11, 204, 205]. Addressing the aforementioned issues to allow evidence selection against world knowledge will be an important future research direction.

For the verification subtask, verification models perform relatively well if given correct evidence [137]. However, it has been shown that verification models are prone to adversarial attacks and are not robust to negation, numerical or comparative words [250]. Improving this weakness of verification models would also be crucial because the factuality of a sentence can easily be changed by small word changes (i.e., changes in negations, numbers, and entities).

Generalization. Although we can see that the source and output text of different tasks are in various forms, investigating their relationship and common ground and proposing general metrics to evaluate hallucinations are worth exploring. Task-agnostic metrics with cross-domain robustness could help the research community to build a unified benchmark. It is also important and meaningful to build open-source platforms to collaborate and standardize the evaluation metrics for NLG tasks.

Incorporation of Human Cognitive Perspective. A good automatic metric should correlate with human evaluation. Humans are sensitive to different types of information. For instance, proper nouns are usually more important than pronouns in the generated text. Mistakes concerning named entities are striking to human users, but automatic metrics treat them equally if not properly

designed. In order to address this issue, new metrics should be designed from the human cognitive perspective. The human ability to recognize salient information and filter the rest is evident in scenarios where the most important facts need to be determined and assessed. For instance, when signing an agreement, a prospective employee naturally skims the document to look at the entries with numbers first. In this way, humans classify what they believe is crucial.

Automatic check-worthy detection has the potential to be applied to improve the correlation with human judgement. Implementing the automatic human-like judgment mentioned above can further mitigate hallucination and improve NLG systems.

6.2 Future Directions in Mitigation Methods

General and robust data pre-processing approaches. Since the data format varies between downstream tasks, there is still a gap for data processing methods between tasks, and currently, no universal method is effective for all NLG tasks [140]. Data pre-processing might result in grammatical errors or semantic transformation between the original and processed data, which can negatively affect the performance of generation. Therefore, we believe that general and robust data pre-processing methods can help mitigate the hallucinations in NLG.

Hallucinations in numerals. Most existing mitigation methods do not focus on the hallucination of numerals. However, the correctness of numerals in generated text, such as date, quantities and scalars are important for readers [246, 315, 321]. For example, given the source document “*The optimal oxygen saturation (SpO_2) in adults with COVID-19 who are receiving supplemental oxygen is unknown. However, a target SpO_2 of 92% to 96% seems logical, considering that indirect evidence from patients without COVID-19 suggests that an SpO_2 of <92% or >96% may be harmful.*”⁶, the summary “*The target oxygen saturation range for patients with COVID-19 is 82–86%.*” includes wrong numbers, which could be fatal. Currently, some works [193, 246, 315] point out that using commonsense knowledge can help to gain better numeral representation. And Zhao et al. [321] alleviate numeral hallucinations by re-ranking candidate-generated summaries based on the verification score of quantity entities. Therefore, we believe that explicitly modeling numerals to mitigate hallucinations is a potential direction.

Extrinsic Hallucination Mitigation. Though many works on mitigating hallucinations have been published, most do not distinguish between intrinsic and extrinsic hallucination. Moreover, the main research focus has been on dealing with intrinsic hallucination, while extrinsic hallucination has been somewhat overlooked as it is more challenging to reduce [104]. Therefore, we believe it is worth exploring different mitigation methods for intrinsic and extrinsic hallucinations, and relevant methods in fact-checking can be potentially used for this purpose.

Hallucination in long text. Many tasks in NLG require the model to process long input texts, such as multi-document summarization and generative question answering. We think adopting existing approaches to a Longformer [12]-based model could help encode long inputs. Meanwhile, part of dialogue systems need to generate long output text, in which the latter part may contradict history generation. Therefore, reducing self-contradiction is also an important future direction.

Reasoning. Misunderstanding facts in the source context will lead to intrinsic hallucination and errors. To help models understand the facts correctly requires reasoning over the input table or text. Moreover, if the generated text can be reasoned backwards to the source, we can assume it is faithful. There are some reasoning works in the area of dialogue [39, 83, 272], but few in reducing hallucinations. Moreover, tasks with quantities, such as logical table-to-text generation, require

⁶<https://www.covid19treatmentguidelines.nih.gov/management/critical-care/oxygenation-and-ventilation/>

numerical reasoning. Therefore, adding reasoning ability to the hallucination mitigation methods is also an interesting future direction.

Controllability. Controllability means the ability of models to control the level of hallucination and strike a balance between faithfulness and diversity [60, 222]. As mentioned in Section 3, it is acceptable for chat models to generate a certain level of hallucinatory content as long as it is factual. Meanwhile, for the abstractive summarization task, there is no agreement in the research community about whether factual hallucinations are desirable or not [177]. Therefore, we believe controllability merits attention when exploring hallucination mitigation methods.

7 HALLUCINATION IN ABSTRACTIVE SUMMARIZATION

Abstractive summarization aims to extract essential information from source documents and to generate short, concise, and readable summaries [302]. Neural networks have achieved remarkable results on abstractive summarization. However, Maynez et al. [177] observe that neural abstractive summarization models are likely to generate hallucinatory content that is unfaithful to the source document. Falke et al. [66] analyze three recent abstractive summarization systems and show that 25% of the summaries generated from state-of-the-art models have hallucinated content. In addition, Zhou et al. [325] mention that even if a summary contains a large amount of hallucinatory content, it can achieve a high ROUGE [154] score. This has encouraged researchers to actively devise ways to improve the evaluation of abstractive summarization, especially from the hallucination perspective.

In this section, we review the current progress in automatic evaluation and the mitigation of hallucination, and list the remaining challenges for future work. In addition, it is worth mentioning that researchers have used various terms to describe the hallucination phenomenon, such as faithfulness, factual errors, and factual consistency, and we will use the original terms from their papers in the remainder of this section.

7.1 Hallucination Definition in Abstractive Summarization

The definition of hallucination in abstractive summarization follows that in Section 2. Specifically, we adopt the definition from [177]: given a document and its abstractive summary, a summary is hallucinated if it has any spans not supported by the input document. Once again, intrinsic hallucination refers to output content that contradicts the source, while extrinsic hallucination refers to output content that the source cannot verify. For instance, in Table 1, given the input article shown in the caption, an example of intrinsic hallucination is “*The Ebola vaccine was rejected by the FDA in 2019*,” because this statement contradicts the given content “*The first vaccine for Ebola was approved by the FDA in 2019 in the US*”. And an example of extrinsic hallucination is “*China has already started clinical trials of the COVID-19 vaccine*,” because this statement is not mentioned in the given content. We can neither find evidence of it from the input article nor assert that it is wrong.

Pagnoni et al. [197] define fine-grained types of factual errors in summaries. As mentioned in 2.3, since the “fact” here refers to source knowledge, “factual error” can be treated as hallucination, and we can adopt this classification as a sub-type of hallucination. They establish three categories as semantic frame error, discourse error, and content verifiability error.

7.2 Hallucination Metrics in Abstractive Summarization

Existing metrics for hallucination in abstractive summarization are mainly model-based. Following [104], we divide the hallucination metrics into two categories: (1) unsupervised metrics and (2) semi-supervised metrics. Note that existing hallucination metrics evaluate both intrinsic and

extrinsic hallucinations together in one metric because it is difficult to automatically distinguish between them.

7.2.1 Unsupervised Metrics. Given that hallucination is a newly emerging problem, there are only a few hallucination-related datasets. Therefore, researchers have proposed to adopt other datasets to build unsupervised hallucination metrics. There are three types of such unsupervised metrics: (1) information extraction (IE)-based metrics, (2) natural language inferencing (NLI)-based metrics, (3) question answering (QA)-based metrics.

IE-based Metrics. As mentioned in Section 4, IE-based metrics leverage IE models to extract knowledge as relation tuples (*subject, relation, object*) from both the generation and knowledge source to analyze the factual accuracy of the generation [87]. However, IE models are not 100% reliable yet (making errors in the identification of the relation tuples). Therefore, Nan et al. [190] propose an entity-based metric relying on the Named-Entity Recognition model, which is relatively more robust. Their metric builds on the assumption that there will be a different set of named entities in the gold and generated summary if there exists hallucination.

NLI-based Metrics. As mentioned in Section 4, the NLI-model (textual entailment model) can be utilized to measure hallucination based on the assumption that a faithful summary will be entailed by the gold source. However, Falke et al. [66] discover that models trained on NLI datasets can not transfer well to abstractive summarization tasks, degrading the reliability of NLI-based hallucination metrics. To improve NLI models for hallucination evaluation, they release collected annotations as additional test data. Other efforts have also been made to further improve NLI models. Mishra et al. [185] find that the low performance of NLI-based metrics is mainly caused by the length of the premises in NLI datasets being shorter than the source documents in abstractive summarization. Thus, the authors propose to convert multiple-choice reading comprehension datasets into long premise NLI datasets automatically. The results indicate that long-premise NLI datasets help the model achieve a higher performance than the original NLI datasets. In addition, Laban et al. [131] introduce a simple but efficient method called SUMMAC_{Conv} by applying NLI models to sentence units that are segmented from documents. The performance of their model is better than applying NLI models to the whole document.

QA-based Metrics. QA-based metrics measure the knowledge overlap or consistency between summaries and the source documents based on the intuition that QA models will achieve similar answers if the summaries are factually consistent with the source documents. QA-based metrics such as FEQA [54], QAGS [264], and QuestEval [227] follow three steps to obtain a final score: (1) a QG model generates questions from the summaries, (2) a QA model obtains answers from the source documents, and (3) calculate the score by comparing the set of answers from source documents and the set of answers from summaries. The results show that these reference-free metrics have substantially higher correlations with human judgments of faithfulness than the baseline metrics. Gabriel et al. [75] further analyze the FEQA and find that the effectiveness of QA-based metrics depends on the question. They also provide a meta-evaluation framework that includes QA metrics.

7.2.2 Semi-Supervised Metrics. Semi-supervised metrics are trained on the synthetic data generated from summarization datasets. Trained on these task-specific corpora, models can judge whether the generated summaries are hallucinatory. Kryscinski et al. [126] propose a weakly supervised model named FactCC for evaluating factual consistency. The model is trained jointly for three tasks: (1) checking whether the synthetic sentences remain factually consistent, (2) extracting supporting spans in the source documents, and (3) extracting inconsistent spans in the summaries, if any exist.

They transfer this model to check whether the summaries generated from summarization models are factually consistent. Results show that the performance of their FactCC model surpasses the classifiers trained on the MNLI or FEVER datasets. Zhou et al. [325] introduce a method to fine-tune a pre-trained language model on synthetic data with automatically inserted hallucinations in order to detect the hallucinatory content in summaries. The model can classify whether spans in the machine-generated summaries are faithful to the article. This method shows higher correlations with human factual consistency evaluation than the baselines.

7.3 Hallucination Mitigation in Abstractive Summarization

Recently, many approaches have been proposed to reduce the hallucination phenomenon in abstractive summarization.

7.3.1 Architecture Method. Seq-to-seq [244] models are widely used and achieve state-of-the-art performance in abstractive summarization. Researchers have made modifications to the architecture design of the seq-to-seq models to reduce hallucinated content in the summaries. We describe various efforts made to improve the encoder, decoder, or both the encoder and decoder of the seq-to-seq models.

Encoder. Zhu et al. [328] propose to use an explicit graph neural network (GNN) to encode the fact tuples extracted from source documents. In addition to an explicit graph encoder, Huang et al. [102] further design a multiple-choice cloze test reward to encourage the model to better understand entity interactions. Moreover, Gunel et al. [92] use external knowledge from Wikipedia to make knowledge embeddings, which the results show improve factual consistency.

Decoder. Song et al. [235] present the incorporation of a sequential decoder with a tree-based decoder to generate a summary sentence and its syntactic parse. This joint generation is performed improve faithfulness. Aralickatte et al. [6] introduce the Focus Attention Mechanism, which encourages decoders to generate tokens similar or topical to the source documents. The results on the BBC extreme summarization task show that models augmented with the Focus Attention Mechanism generate more faithful summaries.

Encoder-decoder. Cao et al. [25] extract fact descriptions from the source text and apply a dual-attention seq-to-seq framework to force the summaries to be conditioned on both source documents and the extracted fact descriptions. Li et al. [142] propose an entailment-aware encoder and decoder with multi-task learning which incorporates the entailment knowledge into abstractive summarization models.

7.3.2 Training Method. Aside from architecture modification, some works improved the training approach to reduce hallucination. Cao and Wang [23] introduce a contrastive learning method to train summarization models. The positive training data are reference summaries, while the negative training data are automatically generated hallucinatory summaries, and the contrastive learning system is trained to distinguish between them. In the dialogue summarization field, Tang et al. [245] propose another contrastive fine-tuning strategy, named CONFIT, that can improve the factual consistency and overall quality of summaries.

7.3.3 Post-Processing Method. Some works carry out post-editing to reduce the hallucination of the model-generated summaries, which are viewed as draft summaries. Dong et al. [50] propose SpanFact, a pair of factual correction models that use knowledge learned from QA models to correct the spans in the generated summaries. Similar to SpanFact, Cao et al. [22] introduce a post-editing corrector module to identify and correct hallucinatory content in generated summaries. The corrector module is trained on synthetic data which are created by adding a series of heuristic

transformations to reference summaries. Zhao et al. [321] present HERMAN, a system that learns to recognize quantities (dates, amounts of money, etc.) in the generated summary and verify their factual consistency with the source text. According to the quantity hallucination score, the system chooses the most faithful summary where the source text supports its quantity terms from the candidate-generated summaries. Chen et al. [32] introduce a contrast candidate generation and selection system to do post-processing. The contrast candidate generation model replaces the named entities in the generated summaries with ones present in the source documents, and the contrast candidate selection model will select the best candidate as the final output summary.

7.4 Future Directions in Abstractive Summarization

Factual Hallucination Evaluation. Factual hallucinations contain information not found in source content, though it is factually correct. In the summarization task, this kind of hallucination could lead to better summaries. However, there is little work focused on evaluating factual hallucination explicitly. Fact-checking approaches could be potentially used in this regard.

Extrinsic Hallucination Mitigation. There has been little research on extrinsic hallucinations as it is more challenging to detect and mitigate content based on world knowledge. We believe it is worth exploring extrinsic hallucination in terms of evaluation metrics and mitigation methods.

Hallucination in Dialogue Summarization. In conversational data, the discourse relations between utterances and co-references between speakers are more complicated than from, say, news articles. For example, Zhong et al. [322] show that 74% of samples in the QMSum dataset consist of inconsistent facts. We believe exploring the hallucination issue in dialogue summarization is an important and special component of research into hallucination in abstractive summarization.

8 HALLUCINATION IN DIALOGUE GENERATION

Dialogue generation is an NLG task that automatically generates responses according to user utterances. The generated responses are required to be fluent, coherent, and consistent with the dialogue history. The dialogue generation task can be divided into two sub-tasks: (1) task-oriented dialogue generation; (2) open-domain dialogue generation. A task-oriented dialogue system aims to complete a certain task according to a user query in a specific domain, such as restaurant booking, hotel recommendation, and calendar checking. Meanwhile, an open-domain dialogue system aims to establish a multi-turn, long-term conversation with users while providing the users with an engaging experience.

8.1 Hallucination Definition in Dialogue Generation

The hallucination problem also exists in the dialogue generation task. It is important to note that a dialogue system is expected either to provide the user with the required information or to provide an engaging response without repeating utterances from the dialogue history. Thus, the tolerance for producing proper “hallucination” from the dialogue history is relatively higher.

The definition of hallucination in this task can be adopted from the general definition as follows: (1) **Intrinsic hallucination:** the generated response is contradictory to the dialogue history or the external knowledge sentences. In the examples of intrinsic hallucination shown in Table 1, we can verify that the output contradicts the inputs: In one example, the input is a “*moderate*” price range, but the model mistakenly generates a sentence with a “*high*” price range. In another case, the confusion of the names “*Roger Federer*” and “*Rafael Nadal*” causes the output generation of “*Roger Nadal*”. (2) **Extrinsic hallucination:** the generated response is hard to verify with the dialogue history or the external knowledge sentences. Responses with extrinsic hallucination are impossible to verify with the given inputs. “*Pickwick hotel*” might be “*in san diego*”, and Djokovic may have

been “*in the top ten singles players of the world*”. However, we do not have enough information to check the truth of these statements.

In the following sections, the hallucination problem in open-domain and task-oriented dialogue generation tasks will be separately discussed according to their natures.

8.2 Open-domain Dialogue Generation

While the term “hallucination” seems to have newly emerged in the NLP field, a related behavior, “inconsistency”, of neural models has been widely discussed. This behavior has been pointed out as a shortcoming of generation-based approaches for open-domain chatbots [103, 165, 223]. Two possible types of inconsistency occur in open-domain dialogue generation: (1) inconsistency among the system utterances, such as when the system contradicts its previous utterance; (2) inconsistency with an external source, such as factually incorrect utterances. Whereas the first type is described using the term “inconsistency” [148, 278, 309] or “incoherence” [14, 58], some have recently started to call the second type “hallucination” [183, 224]. Self-inconsistency can be considered as an intrinsic hallucination problem, while the external inconsistency involves both intrinsic and extrinsic hallucinations, depending on the reference source.

As mentioned earlier, a certain level of hallucination may be acceptable in open-domain chit-chat as long as it does not involve severe factual issues. Moreover, it is almost impossible to verify factual correctness since the system usually lacks a connection to external resources. With the introduction of knowledge-grounded dialogue tasks [48, 326], which provide an external reference, however, there has been more active discussion of hallucination in open-domain dialogue generation.

8.2.1 Self-Consistency. In end-to-end generative open-domain dialogue systems, the inconsistency among system utterances has been pointed out as the bottleneck to human-level performance [263]. We often observe an inconsistency in the answers to semantically similar yet not identical questions. For example, a system may answer the questions of “What is your name?” and “May I ask your name?” with different responses. Persona consistency has been the center of attention [144, 312] and it is one of the most obvious cases of self-contradiction regarding the character of the dialogue system. “Persona” is defined as the character that a dialogue system plays during a conversation, and can be composed of identity, language behavior, and an interaction style [144]. While some works have set their objective as teaching models to utilize speaker-level embeddings [144, 168], others condition generation with a set of descriptions about a persona, which we will discuss in detail in the next section.

8.2.2 External Consistency. Besides self-consistency, an open-domain dialogue system should also generate persona-consistent and informative responses corresponding so as to user utterances to further engage with the user during conversation. In this process, an external resource containing explicit persona information or world knowledge is introduced into the system to assist the model generation process.

The PersonaChat datasets [47, 312] have accelerated research into persona consistency [96, 128, 178, 284, 296, 307, 318]. In PersonaChat datasets, each conversation has persona descriptions such as “I like to ski” or “I am a high school teacher” attached. By conditioning the response generation on the persona description, a chit-chat model is expected to acquire an ability to generate a more persona-consistent response. Lately, the application of NLI methods [148, 234] or reinforcement learning frameworks [181] have been investigated. Although these methods conditioned on the PersonaChat datasets have been successful, further investigation of approaches that do not rely on a given set of persona descriptions is necessary because such descriptions are not always available, and covering every aspect of a persona with them is impossible.

In addition to PersonaChat-related research, the knowledge-grounded dialogue (KGD) task in the open-domain requires the model to generate informative responses with the help of an external knowledge graph (KG) or knowledge corpus [48, 326]. Hallucination in conversations, which is also considered as a factual consistency problem, has raised much research interest recently [60, 214, 226, 233]. Here, we continue to split the hallucination problem in the KGD task into intrinsic hallucination and extrinsic hallucination. Most of the KGD works tackle the hallucination problem when responses contain information that contradicts (intrinsic) or cannot be found in the provided knowledge input (extrinsic). Since world knowledge is enormous and ever-changing, the extrinsic hallucination may be factual but hard to verify. Dziri et al. [60] further adopt the same definition of hallucination as mentioned above to the knowledge graph-grounded dialogue task, where intrinsic hallucination indicates the case of misusing either the subject or object of the knowledge triple; and extrinsic hallucination indicates that there is no corresponding valid knowledge triple in the gold reference knowledge. Recently, there have been some attempts to generate informative responses without explicit knowledge inputs, but with the help of the implicit knowledge inside large pre-trained language models instead [292, 327] during the inference time. Under this setting, the study of extrinsic hallucination is of great value but still poorly investigated.

8.2.3 Hallucination Metrics. For generation-based dialogue systems, especially open-domain chatbots, the hallucination evaluation method remains an open problem [223]. As of now, there is no standard metric. Therefore, chatbots are usually evaluated by humans on factual consistency or factual correctness [226, 285]. We also introduce some automatic statistical and model-based metrics as a reference, which will be described in more detail below.

Variants of F1 Metrics. **Knowledge F1 (KF1)** measures the overlap between the generated responses and the gold knowledge sentences to which the human referred for conversation during dataset collection [233]. KF1 attempts to capture whether a model can generate knowledgeable responses by correctly utilizing the relevant knowledge. KF1 is only available for datasets with labeled ground-truth knowledge. Shuster et al. [233] further propose **Rare F1 (RF1)**, which only considers the infrequent words in the dataset when calculating F1 to avoid influence from the common uni-grams. The authors define an infrequent word as being in the lower half of the cumulative frequency distribution of the reference corpus.

Model-based Metric. Natural language has its natural on the flexibility of the surface forms with the same semantics, so overlap-based metrics cannot provide the comprehensive evaluation. Recently, several works have proposed evaluation metrics for measuring consistency, such as using natural language inference (NLI) [58, 278], training learnable evaluation metrics [309], or releasing an additional test set for coherence [14]. These methods are more flexible and supports the generated responses with different surface forms. For the KGD task, Dziri et al. [61] propose the BEGIN benchmark, which consists of samples taken from Dinan et al. [48] with additional human annotation and a new classification task extending the NLI paradigm. Honovich et al. [101] present a trainable metric for the KGD task, which also applies NLI. It is also noteworthy that Gupta et al. [94] propose datasets that can benefit fact-checking systems specialized for dialogue systems. The Conv-FEVER corpus [226] is a factual consistency detection dataset, which was created by adapting the Wizard-of-Wikipedia dataset [48]. It consists of both factually consistent and inconsistent responses and can be used to train a classifier to detect factually inconsistent responses with respect to the knowledge provided.

8.2.4 Mitigation Methods. The hallucination issue can be mitigated by data pre-processing, which includes introducing extra information into the data. Shen et al. [230] propose a measurement based on seven attributes of the dialogue quality, including self-consistency. Based on this measurement,

the untrustworthy samples which get lower scores are filtered out from the training set to improve the model performance in terms of self-consistency (i.e., intrinsic hallucination). Shuster et al. [233] conduct a comprehensive investigation on a retrieval-augmented KGD task where a retriever is introduced to the system for knowledge selection. The authors study several key problems, such as whether retrieval helps reduce hallucinations and how the generation should be augmented with the retrieved knowledge. The experimental results show that retrieval helps substantially in improving performance on KGD tasks and in reducing the hallucination in conversations without sacrificing conversational ability.

Rashkin et al. [214] introduce a set of control codes and concatenate them with dialogue inputs to reduce the hallucination by forcing the model to be more aware of how the response relies on the knowledge evidence in the response generation. Some researchers have also tried to reduce hallucinated responses during generation by improving dialogue modeling. Wu et al. [285] apply inductive attention into transformer-based dialogue models, and potentially uninformative attention links are removed with respect to a piece of pre-established structural information between the dialogue context and the provided knowledge. Instead of improving the dialogue response generation model itself, Dziri et al. [60] present a response refinement strategy with a token-level hallucination critic and entity-mention retriever, so that the original dialogue model is left without retraining. The former module is designed to label the hallucinated entity mentioned in the generated responses, while the retriever is trained to retrieve more faithful entities from the provided knowledge graph. RHO [108] is a framework that uses three mechanisms to tackle hallucinations, namely, local knowledge grounding, global knowledge grounding, and response re-ranking to tackle hallucinations in open-domain dialogues, and has been empirically shown to perform this.

8.3 Task-oriented Dialogue Generation

A task-oriented dialogue system is often composed of several modules: a natural language understanding (NLU) module, a dialogue manager (DM), and a natural language generation (NLG) module [77, 112]. Intrinsic hallucination can occur between the DM and NLG, where a dialogue act such as `recommend(NAME=peninsula hotel, AREA=tsim sha tsui)` is transformed into a natural language representation “the hotel named *peninsula hotel* is located in *tsim sha tsui* area.” [10, 152].

8.3.1 Hallucination Metrics. To evaluate hallucination, Li et al. [152] and Balakrishnan et al. [10] combine traditional metrics such as the BLEU score and human evaluation as well as hallucination-specific automatic metrics. Following previous works such as [56, 279], and [255], Li et al. [152] use the slot error rate, which is computed by $(p + q)/N$, where N represents the total number of slots extracted by another model in the dialogue act. Here, p stands for the missing slots in the generated template, and q is the number of redundant slots. On the other hand, Balakrishnan et al. [10] introduce a novel metric called the tree accuracy, which determines whether the prediction’s tree structure is identical to that of the input meaning representations.

8.3.2 Mitigation Methods. While Balakrishnan et al. [10] propose to adopt tree-structured semantic representations and add constraints on decoding, Li et al. [152] frame a reinforcement learning problem to which they apply a bootstrapping algorithm to sample training instances and then leverage a reward related to slot consistency. Recently, there has emerged another line of research in task-oriented dialogue, which is to build a single end-to-end system rather than connecting several modules (e.g., Eric and Manning [63], Madotto et al. [167, 170?]). As discussed in previous sections of this paper, there is potential for such end-to-end systems to produce extrinsic hallucinations, yet this remains less explored. For example, a model might generate a response with an entity that appears out of nowhere. In the example of hotel recommendation in Hong Kong given above, a

model could generate a response such as “the hotel named *raffles hotel* is located in *central area*,⁷” which cannot be verified from the knowledge base of the system.

8.4 Future Directions in Dialogue Generation

Self-Contradiction in Dialogue Systems. One of the possible reasons for self-contradiction is that current dialogue systems tend to have a short memory of dialogue history [223]. Firstly, common dialogue datasets provide several turns of conversation, yet these are not long enough to assess a model’s ability to deal with a long context. To overcome this, Xu et al. [287] introduce a new dataset that consists of, on average, over 40 utterances per episode. Secondly, we often truncate dialogue history into fewer turns to fit into models such as Transformer-based architectures, which makes it difficult for a model to memorize the past. In addition to the works on dialogue summarization, e.g., Gliwa et al. [85], it would be beneficial to apply other works which are aiming to grasp the longer context but do not focus on dialogue generation [12, 306, 319].

Fact-checking in dialogue systems. In addition to the factual consistency in responses from knowledge grounded dialogue systems, fact-checking is a future direction in dealing with the hallucination problem in dialogue systems [94]. Dialogue fact-checking involves verifiable claim detection, which is an important line in distinguishing hallucination-prone dialogue, and evidence retrieval from an external source. This fact-checking in the dialogue system could be utilized not only as an evaluation metric for facilitating factual consistency but also to model such a system.

9 HALLUCINATION IN GENERATIVE QUESTION ANSWERING

Generative question answering (GQA) aims to generate an abstractive answer rather than extract an answer to a given question from provided passages [68, 141]. It is an important task since many of the everyday questions that humans deal with and pose to search engines require in-depth explanations [120] (e.g., *why/how..?*), and the answers are normally long and cannot be directly extracted from existing phrase spans. A GQA system can be integrated with a search engine [182] to empower more intelligent search or combined with a virtual conversation agent to enhance user experience.

Normally, a GQA system involves searching an external knowledge source for information relevant to the question. Then it generates the answer based on the retrieved information [125]. In most cases, no single source (document) contains the answer, and multiple retrieved documents will be considered for answer generation. Those documents may contain redundant, complementary, or contradictory information. Thus, hallucination is common in the generated answers.

The hallucination problem is one of the most important challenges in GQA. Since an essential goal of a GQA system is to provide factually-correct answers given the question, hallucination in the answer will mislead the user and damage the system performance dramatically.

9.1 Hallucination Definition in GQA

As a challenging yet under-explored task, there is no standard definition of hallucination in GQA. However, almost all the works on GQA [68, 125, 189, 237] involve a human evaluation process, in which the *factual correctness* measuring the faithfulness of the generated answer can be seen as a measurement of the hallucination; i.e., the more faithful the answer is, the less hallucinated content it contains. The most recent such work [141] uses the term *semantic drift*, which indicates how the answer drifts away from a correct one during generation, and this can also be seen as a specific definition of hallucination in GQA.

⁷Raffles Hotel is a hotel located in Downtown Core, Singapore.

In line with the general categorization of hallucination in Section 2.1, we give two concrete hallucination examples in GQA in Table 1. The sources of both questions are Wikipedia web pages. For the first question, “*dow jones industrial average please?*”, the generated answer “*index of 30 major U.S. stock indexes*” conflicts with the statement “*of 30 prominent companies listed on stock exchanges in the United States*” from Wikipedia. So we categorize it as an intrinsic hallucination. For the second example, the sentences “*The definition of a Sadducee is a person who acts in a deceitful or duplicitous manner. An example of a Sadducee is a politician who acts deceitfully in order to gain political power*” in the generated answer can not be verified from the source documents; thus, we categorize it as an extrinsic hallucination.

9.2 Hallucination-related Metrics in GQA

Currently, there is no automatic metric to evaluate hallucination in GQA specifically. While most works on GQA use automatic evaluation metrics such as ROUGE score and F1 to measure the quality of the answer, these N-gram overlap-based metrics are not a meaningful way to evaluate hallucination due to their poor correlation with human judgments, as indicated by Krishna et al. [125]. On the other hand, almost all the GQA-related work involves a human evaluation process as a complement to the automatic evaluation. Normally, human annotators will be asked to assign a score indicating the faithfulness of the answer, which can also be viewed as a measurement of the answer hallucination. However, the metrics obtained via human evaluation come normally from a small sample of the data.

Metrics such as *semantic overlap* [229], a learned evaluation metric based on BERT that models human judgments, could be considered a better measurement of hallucination for GQA. Other metrics such as the *factual correctness* can also be considered as a way to measure hallucination in GQA. Zhang et al. [317] propose to explicitly measure the factual correctness of a generated text against the reference by first extracting facts via an information extraction (IE) module. Then they define and measure the factual accuracy score to be the ratio of facts in the generation text equal to the corresponding facts in the reference.

Factual consistency, which measures the faithfulness of the generated answer given its source documents, can be employed as another way to measure hallucination in GQA. Durmus et al. [54], Wang et al. [264] propose an automatic QA-based metric to measure faithfulness in summary, leveraging the recent advances in machine reading comprehension. They first use a question generation model to construct question-answer pairs from the summary, and then a QA model is applied to extract short answer spans from the given source document for the question. The extracted answers that do not match the provided answers indicate unfaithful information in the summary. While these metrics were first proposed in summarization works, they can be easily adopted in generative QA to measure hallucinations in the generated long-form answer.

The most recent work on GQA by Su et al. [237] proposed to estimate the faithfulness of the generated long-form answer via *zero-shot short answer recall* on extractive QA datasets. They first generate long-form answers for questions from two extractive QA datasets Natural Questions(NQ) [130] and HotpotQA [295], both of which contains large-scale question-answer pairs, then they measure the ratio of golden short answer span contained in the generated long answer as an estimation of faithfulness of the generated long-answer. While the idea is similar to the factual consistency metric in summarization work [54], and also matches with our intuition to some extent, its correlation with human evaluation on faithfulness has not been verified.

9.3 Hallucination Mitigation in GQA

Unlike conditional text generation tasks such as summarization, or data-to-text generation, in which the source documents are provided and normally related to the target generation, the hallucination

problem in GQA is more complicated. Generally speaking, it might come from two sources: 1) the incompetency of the retriever, which retrieves documents irrelevant to the answer, and 2) the *intrinsic* and *extrinsic* hallucination in the conditional generation model itself. Normally these two parts are interconnected and cause hallucinations in the answer.

Early works on GQA mostly tried to improve the faithfulness of the answer by investigating reliable external knowledge sources or incorporating multiple information sources. Yin et al. [297] propose Neural Generative Question Answering (GENQA), an end-to-end model that generates answers to simple factoid questions based on the knowledge base, while Bi et al. [15] propose the Knowledge-Enriched Answer Generator (KEAG) to generate a natural answer by integrating facts from four different information sources, namely, questions, passages, vocabulary, and knowledge. Nevertheless, these methods rely on the existence of high-quality, relevant resources which are not easily available.

Recent works focus more on the conditional generation model. Fan et al. [67] construct a local knowledge graph for each question to compress the information and reduce redundancy from the retrieved documents, which can be viewed as an early trial to mitigate hallucination. Li et al. [141] propose Rationale-Enriched Answer Generator (REAG), in which they add an extraction task to obtain the rationale for an answer at the encoding stage, and the decoder is expected to generate the answer based on both the extracted rationale and original input. The recent work [125] employs a Routing Transformer (RT), a sparse attention-based Transformer-based model that employs local attention and mini-batch k-means clustering for long-range dependence, as the answer generator in the hope of modeling more retrieved documents to mitigate the hallucination in the answer. Su et al. [237] propose a framework named RBG (**read before generate**), to jointly models answer generation with machine reading. They augment the generation model with fine-grained, answer-related salient information predicted by the MRC module, to enhance answer faithfulness. Such methods can exploit and utilize the information in the original input better, while they require the extra effort of building models to extract that information.

Most recently, Lin et al. [155] propose a benchmark, which comprises 817 questions that span 38 categories, to measure the truthfulness of a language model in the QA task. This work investigates the performances of GPT-3 [20], GPT-Neo/J [266], GPT-2 [209] and a T5-based model [210]. The results suggest that simply scaling up the model is less promising than fine-tuning it in terms of improving truthfulness since larger models are better at learning the training distribution from web data and thus tend to produce more imitative falsehoods. In another recent work, Nakano et al. [189] fine-tune GPT-3 to answer long-form questions with a web-browsing environment, which allows the model to navigate the web as well as use human feedback to optimize answer quality directly using imitation learning [107]. While this method seems promising, it also hinges on how that feedback is processed.

9.4 Future Directions in GQA

While GQA is challenging yet under-explored, many possible directions could be explored to improve the answer quality and mitigate hallucination. First, better automatic evaluation metrics are needed to measure hallucination. The previously mentioned metrics, such as the semantic overlap between the generated answer and the ground-truth answer, the faithfulness of the generated answer, and factual consistency between the answer and the source documents, only consider one aspect of hallucination. Metrics that can consider all the factors related to hallucination (such as semantic overlap, faithfulness, or factual consistency) could be designed. Second, datasets with hallucination annotations should be proposed since none of the current GQA datasets have that information. Another possible direction to mitigate hallucination in the answer is improving the performance of the models. We need better retrieval models that retrieve relevant information

according to queries and generation models that can synthesize more accurate answers from multi-source documents.

10 HALLUCINATION IN DATA-TO-TEXT GENERATION

Data-to-Text Generation is the task of generating natural language descriptions conditioned on structured data [127, 179], such as tables [198, 283], database records [36], and knowledge graphs [79]. Although this field has been recently boosted by neural text generation models, it is well known that these models are prone to hallucinations [283] because of the gap between structured data and text, which may cause semantic misunderstanding and erroneous correlation. Moreover, the tolerance of hallucination is very low when this task is applied to the real world, such as in the case of patient information table description [247], and analysis of experimental results tables in a scientific report. Recent years have seen a growth of interest in hallucinations in Data-to-Text Generation, and researchers have proposed works from the aspect of evaluation and mitigation.

10.1 Hallucination Definition in Data-to-Text Generation

The definition and categories of hallucination in Data-to-Text Generation follow the descriptions in Section 2. We follow the general hallucination definition in this task: (1) Intrinsic Hallucinations: the generated text contains information that is contradicted by the input data [193]. For example, in Table 1, “*The Houston Rockets (18-4)*” uses the information “[*TEAM: Rockets, CITY:Houston, WIN:18, LOSS: 5*]” in the source table. However, “(18-4)” is contradicted by “[*LOSS: 5*]” and it should be “(18-5)”. (2) Extrinsic Hallucinations: the generated text contains extra information irrelevant to the input [45, 193]. For example, in Table 1, “*Houston has won two straight games and six of their last seven.*” is not mentioned in the source table [268].

10.2 Hallucination Metrics in Data-to-Text Generation

Statistical. PARENT [45] measures the accuracy of table-to-text generation by aligning n-grams from the reference description R and generated texts G to the table T . And it is the average F-score by combining the entailment precision and recall. Wang et al. [274] modify PARENT and denote this table-focused version as PARENT-T. Different from PARENT, which evaluates i -th instance (T_i, R_i, G_i) , PARENT-T ignores the reference description R and evaluates each instance (T_i, G_i) .

IE-based. Liu et al. [162] estimate hallucination with two entity-centric metrics: table record coverage (the ratio of covered records in a table) and hallucinated ratio (the ratio of hallucinated entities in text). This metric firstly uses entity recognition to extract the entities of input and generated output, then aligns these entities by heuristic matching strategies, and finally calculates the ratios of faithful and hallucinated entities separately. Moreover, there are some general post-hoc IE-based metrics that could be applied to hallucination evaluation, such as Slot Error Rate (SER) [291], Content Selection (CS), Relation Generation (RG), and Content Ordering (CO) [268, 283].

QA-based. Data-QuestEval [217] adapt QuestEval [227] from summarization into data-to-text generation. First, a *textual QG model* is trained on a textual QA dataset. For each sample (structured data, textual descriptions), the *textual QG model* generates synthetic problems based on the descriptions. The structured data, textual descriptions (answers), and synthetic questions make up a synthetic QG/QA dataset to train *synthetic QA/QG models*. Then, the *synthetic QG model* generates questions based on the textual description to be evaluated. The *synthetic QA model* then generates answers based on a synthetic question and the structured input data. Finally, BERTScore [314] measures the similarity between the generated answer and description, indicating faithfulness.

NLI-based. Dušek and Kasner [57] recognize the textual entailment between the input data and the output text for both omissions and hallucinations with an NLI model. This work measures the

semantic accuracy in two directions: check omissions by inferring whether the input fact is entailed by the generated text and check hallucinations by inferring the generated text from the input.

LM-based. Filippova [72], Tian et al. [251] are based on the intuition that when an unconditional LM, only trained on the targets, gets a smaller loss than a conditional LM_x , trained on both sources and targets, the token is predicted unfaithfully. Thus, they calculate the ratio of hallucinated tokens to the total target length to measure the hallucination level.

10.3 Hallucination Mitigation in Data-to-Text Generation

Data-Related Methods. Several clean and faithful corpora are collected to tackle the challenges from data infidelity. TOTTO [198] is an open-domain faithful table-to-text dataset, where each sample includes a Wikipedia table with several highlighted cells and a description. To ensure that targets exclude hallucinations, the annotators revise existing Wikipedia candidate sentences and clear the parts unsupported by the table. Moreover, RotoWire-FG (Fact-Grounding) [268] is a purified and enlarged and enriched version of RotoWire [283] generating NBA game summaries from score tables. Annotators trim the hallucination part in target texts and extract the mapped table records as content plans to better align input tables and output summaries.

For data processing, OpAtt [192] designs a gating mechanism and a quantization module for the symbolic operation to augment the record table with pre-calculated results. Nie et al. [193] utilize a language understanding module to improve the equivalence between the input MR and the reference utterance in the dataset. They train an NLU model with an iterative relabeling procedure: First, they train the model on original data; parse the MR by model inference; train the model on new paired data with high confidence; and then repeat the above processes. Liu et al. [162] select training instances based on faithfulness ranking. Finer-grained than the above instance-level method, Rebuffel et al. [216] label tokens according to co-occurrence analysis and sentence structure through dependency parsing in the pre-processing step to explicate the correspondence between the input table and the text. Generally, the data-related methods are appropriate when the training dataset is noisy.

Modeling and Inference Methods. Planning and skeleton generation are common methods to improve the faithfulness to the input in data-to-text tasks. Liu et al. [162] propose a two-step generator with a separate text planner augmented by auxiliary entity information. The planner predicts the plausible content plan based on the input data. Then, given the above input data and the content plan, the sequence generator generates the text. Similarly, Plan-then-Generate [239] also consists of a content planner and a sequence generator. In addition, this work adopts a structure-aware RL training to generate output text following the generated content plan faithfully. Puduppully and Lapata [208] first induce a macro plan consisting of multiple sequences of entities and events from the input table and its corresponding multi-paragraph long document. The predicted macro plan then serves as the input to an encoder-decoder model for surface realization. SANA [269] is a skeleton-based two-stage model that includes skeleton generation to select key tokens from the source table and edit-based generation to produce texts via iterative insertion and deletion operations. In contrast to the above two-step model using planning or skeleton, AGGGEN [291] is an end-to-end model that jointly learns to plan and generate at the same time. This architecture with a Hidden Markov Model and Transformer encoder-decoder reintroduces explicit sentence planning stages into neural systems by aligning facts in the target text to input representations.

Other modeling methods have also been proposed to mitigate the hallucination problem. Conjecturing that hallucinations can be caused by inattention to the source, Tian et al. [251] propose a confidence score and a variational Bayes training framework to learn the score from data. Wang et al. [274] introduce a table-text optimal-transport matching loss and an embedding similarity loss

to encourage faithfulness. The hallucination degree can also be treated as a controllable factor in generating texts. In Filippova [72], the hallucination degree of each training sample is estimated and converted into a categorical value which is a part of the inputs as a controlled setting. This approach does not require the dismissal of any input or modification of the model structure.

To mitigate hallucinations at the inference step, Rebuffel et al. [216] propose a Multi-Branch Decoder that leverages word-level alignment labels between the input table and paired text to learn the relevant parts of the training instance. These word-level labels are gained through dependency parsing during the pre-processing step. The branches separately integrate three co-dependent control factors: content, hallucination, and fluency. Uncertainty-aware beam search (UABS) [286] is an extension to beam search to reduce hallucination. Considering that the hallucination probability is positively correlated with predictive uncertainty, this work adds a weighted penalty term in the beam search which is able to balance the predictive probability and uncertainty. This approach is task-agnostic and can also be applied to other tasks, such as image captioning.

These various types of methods do not necessarily conflict and can collaborate to solve the hallucination problem in data-to-text generation.

10.4 Future Directions in Data-to-Text Generation

Given the challenges brought by the discrepancy between structure data and natural text, and the low fault tolerance in the Data-to-Text Generation task, there are several potential directions worth exploring in terms of hallucination.

Firstly, numbers contain information about scales and are common and crucial in the Data-to-Text task [240, 315]. It is frequent to have errors in numbers, which results in hallucinations and infidelity. This is a serious problem for Data-to-Text generation, yet models rarely give special consideration to the numbers found in the table or text [246]. The current automatic metrics of hallucinations also do not specifically treat numbers. This indiscriminate treatment contradicts findings in cognitive neuroscience, where numbers are known to be represented differently from lexical words in a different part of the brain [86]. Thus, considering or highlighting numbers when mitigating and assessing hallucinations is worth exploring. This requires the generative model to learn a better numerical presentation and capture scales, which will reduce the hallucinations caused by the misunderstanding of numbers.

Moreover, for the logical data-to-text generation task, rather than surface-level generation, logical inference, calculation, and comparison are required, which is challenging and causes hallucinations more easily. Thus, reasoning (including numerical reasoning), which is usually combined with graph structures [34] is another direction to improve the accuracy of entity relationships and alleviate hallucinations.

11 HALLUCINATIONS IN NEURAL MACHINE TRANSLATION

Neural Machine Translation (NMT) is the task of generating translation of the source language into the target language via inference, given parallel data samples for training. Compared to statistical machine translation (SMT) the output of NMT is usually quite fluent and of human-level quality, which creates the danger of misinforming users when there are hallucinations [175].

11.1 Hallucinations Definition and Categories in NMT

The problem of hallucination was identified with the deployment of the first NMT models. Early work comparing SMT and NMT systems [122], without explicitly using the term “hallucination”, mentioned that NMT models tend to “sacrifice adequacy for the sake of fluency” especially when evaluated with out-of-domain test sets. Following further development of NMT, most of the relevant research papers agree that translated text is considered a hallucination when it is completely

Category	Source	Correct Translation	Hallucinatory Translation
Intrinsic	迈克周四去书店。	Mike goes to the bookstore on Thursday.	Jerry doesn't go to the bookstore on Thursday.
Extrinsic	迈克周四去书店。	Mike goes to the bookstore on Thursday.	Mike happily goes to the bookstore on Thursday with his friend.
Detached	Das kann man nur feststellen, wenn die kontrollen mit einer großen intensität durchgeführt werden.	This can only be detected if controls undertaken are more rigorous.	Blood alone moves the wheel of history, i say to you and you will understand, it is a privilege to fight.
Oscillatory	1995 das produktionsvolumen von 30 millionen pizzen wird erreicht.	1995 the production reached 30 million pizzas.	The US, for example, has been in the past two decades, but has been in the same position as the US, and has been in the United States.

Table 3. Categories and examples of hallucinations in MT by Zhou et al. [325] and Raunak et al. [215]

disconnected from the source [133, 187]. The categorization of hallucination in NMT is unlike that in any other NLG tasks, and uses various terms that are often overlapping. In order to maintain consistency with other NLG tasks, in this section we use the intrinsic and extrinsic hallucination categories applied to the NMT task by [325]. After a formal definition, we will describe other identified types of hallucinations and hallucination categories mentioned in the relevant literature.

Intrinsic and Extrinsic Hallucinations. Following the idea that hallucinations are outputs that are disconnected from the source, [325] suggest categorizing the hallucinatory content based on the way the output is disconnected:

- **Intrinsic Hallucination:** translations contain incorrect information compared to information present in the source. In Table 3, the example of such hallucination is “Jerry doesn’t go”, since the original name in the source is “Mike” and the verb “to go” is not negated.
- **Extrinsic Hallucination:** translations produce additional content without any regard to the source. In Table 3, “happily” and “with his friend” are the two examples of the hallucinatory content since they are added without any apparent connection to the input.

Other Categories and Types of Hallucinations. Raunak et al. [215] propose an alternative categorization of hallucinations. They divide hallucinations into hallucinations under perturbations and natural hallucinations. Hallucinations under perturbation are those that can be observed if a model tested on the perturbed and unperturbed test set returns drastically different content. Their work on hallucinations under perturbation strictly follows the algorithm proposed by Lee et al. [133]; see Section 11.2.2 on the entropy measure. The second category, natural hallucinations, are created with a connection to the noise in the dataset and can be further divided into detached and oscillatory, where detached hallucinations mean that a target translation is semantically disconnected from a source input, and oscillatory hallucinations mean those that are decoupled from the source by manifesting a repeating n-gram. Tu et al. [257] and Kong et al. [124] analyze this phenomenon under the name “over-translation”, that is, a repetitive appearance of words that were not in the source text. Conversely, under-translation is skipping the words that need to be translated [257]. Finally, abrupt jumps to the end of the sequence and outputs that remain mostly in the source language are also examples of hallucinatory content [133].

11.2 Hallucination Metrics in NMT

The definition of hallucinations in machine translation (MT) tends to be qualitative and subjective, and thus researchers often identify hallucinated content manually. Most detrimentally, the appearance of hallucinations is found not to affect the BLEU score of the translated text [251, 325]. There

are, nevertheless, several notable efforts to automatize and quantify the search for hallucinations using statistical methods.

11.2.1 Statistical Metrics. Martindale et al. [175] propose identifying sentence adequacy using the bag-of-vectors sentence similarity (BVSS) metric. This metric indicates that the information is lost because the reference contains more information than the MT output, or the MT output contains more information than the reference.

11.2.2 Model-Based Metrics.

Auxiliary Decoder. “Faithfulness” refers to the amount of source meaning that is faithfully expressed in the translation, and it is used interchangeably with the term “adequacy” [71, 256]. Feng et al. [71] propose adding another “evaluation decoder” apart from the standard translation decoder. In their work, faithfulness is based on word-by-word translation probabilities, and is calculated in the evaluation module along with translation fluency. The loss returned by the evaluation module helps to adjust the probability returned by the translation module.

Entropy Measure. In scenarios where the ground truth of a translation is not available, an entropy measure of the average attention distribution can be used to detect hallucinations. Tu et al. [257] and Garg et al. [80] show that hallucinations are visible in attention matrices. When the model outputs correct translation, the attention mechanism attends to the entire input sequence throughout decoding. However, it tends to concentrate on one point when the model outputs hallucinatory content. The entropy is calculated on the average attention weights when the model does or does not produce hallucinations during testing. For comparison, a clean test set is used along with the purposefully perturbed one, which is created to incite hallucinations (test sets featuring multiple repetitions). The mean entropy returned by hallucinatory models diverges from the mean of the models that do not produce hallucinations spontaneously [133].

Token Level Hallucination Detection. Zhou et al. [325] propose a method for detecting hallucinated tokens within a sentence, making the search more fine-grained. They use a synthetic dataset that is created by adding noise to the source data, more specifically it is generated by a language model with certain tokens of correct translations masked. Tokens in synthetic data are labeled as hallucinated (1) or not (0). Then the authors compute the hallucination prediction loss between binary labels and the tokens from the hallucinated sentence. This work further employs a word alignment-based method and overlap-based method as baselines for hallucination.

Similarity-based Methods. Zhou et al. [325] use an unsupervised model that extracts alignments from similarity matrices of word embeddings [225], and then predicts the target token as hallucinated if it is not aligned to the source. Parthasarathi et al. [199] propose calculating faithfulness by computing similarity scores between perturbed source sentence and target sentence after applying the same perturbation.

Overlap-based Methods. Zhou et al. [325] predict that the target token is hallucinated if it does not appear in the source. Since the target and source are two different languages, the authors use the density matching method for bilingual synonyms from Zhou et al. [324]. Kong et al. [124] suggest the Coverage Difference Ratio (CDR) as the metric to evaluate adequacy, which is especially successful in finding cases of under-translation. It is estimated by comparing source words covered by generated translation with human translations.

The overlap-based methods for detecting hallucinations are heuristics based on the assumption that all translated words should appear in the source. However, this is not always the case, e.g.,

when paraphrasing or using synonyms. Using word embeddings as similarity-based methods helps avoid such simplifications and allows more diverse, synonymous translations.

Approximate Natural Hallucination Detection. Raunak et al. [215] propose Approximate Natural Hallucination (ANH) detection based on the fact that hallucinations often occur as oscillations (repeating n-grams) and the lower unique bigram count indicates a higher appearance of oscillatory hallucinations. Furthermore, the ANH detection method searches for repeated targets in the translation output. Their method finds translation above a certain n-gram threshold and searches for repeated targets in the output translation, following the assumption that if hallucinations are often incited by aligning unique sources to the same target, then repeating targets will also appear during the inference [257].

11.3 Hallucination Mitigation Methods in NMT

Hallucinations in MT are hard to discover for a person who is not fluent in the target language, and thus they can lead to many possible errors, or even dangers. Out of all the natural language generation tasks, NMT engines such as Google in the English-speaking internet and Baidu in the Sinosphere are probably the most widely accessible to netizens. Consequently, there is a big interest in improving NMT's performance, also by mitigating hallucinations. This subsection compiles methods of mitigating hallucinations in NMT.

11.3.1 Data-Related. Data augmentation appears to be one of the most common methods for removing hallucination. Lee et al. [133] and Raunak et al. [215] suggest addition of perturbed sentences. Furthermore, perturbation, where the insertions of most common tokens are placed at the beginning of the sentence, seems to be the most successful in hallucination mitigation. A disadvantage of this method is the need to understand different types of hallucinations produced by the model in order to apply a correct augmentation method. Corpus filtering is a method of mitigating hallucinations caused by the noise in the dataset by removing the repetitive and mismatching source and target sequences [215]. Junczys-Dowmunt [111] implements a cross-entropy data filtering method for bilingual data, which uses cross-entropy scores calculated for noisy pairs according to two translation models trained on the clean data. The scores that suggest disagreement between sentence pairs from two models are subsequently penalized.

While [133, 215] and [111] define noise as mismatched source and target sentences, [19] analyzes the influence of fine-grained semantic divergences on NMT outputs. The authors consequently propose a mitigation method for fine-grained divergences based on semantic factors. The tags are applied to each source and target sentence to inform about the position of divergent tokens. Factorizing divergence not only helps to mitigate hallucinations, but improves the overall performance of the NMT. This shows that tagging small semantic divergences can provide useful information for the network during training.

11.3.2 Modeling and Inference. Overexposure bias is a common problem in NMT, amplified by the teacher-forcing technique used in sequence-to-sequence models. The models are trained on the ground truth, but during inference, they attend to the past predictions, which can be incorrect [124, 213]. To mitigate this problem, Wang and Sennrich [267] propose substituting MLE as a training objective with minimum risk training (MRT) [194]. Scheduled sampling is a classic method of mitigating overexposure bias first proposed by [13]. Based on that method, [89] create a differentiable approximation to greedy decoding that shows a good performance in the NMT task. [290] propose further improvement of the scheduled sampling algorithm for NMT by optimizing the probability of source and target word alignments. This improvement helps to address the issue flexibility in word order between a source and target language when performing scheduled sampling.

Zhou et al. [325] propose a method of improving self-training of NMT based on hallucination detection. They create hallucination labels (see Section 11.2.2), and then discard losses of tokens predicted as hallucinations, which is known as token loss truncation. This is similar to the method proposed by Kang and Hashimoto [116], the latter for full sentences in the summarization task. Furthermore, instead of adjusting losses, Zhou et al. [325] mask the hidden states of the discarded losses in the decoder in a procedure called decoder HS masking. Experimental results show both a translation quality improvement in terms of BLEU and also a large reduction in hallucination. The token loss truncation method shows good results in the low-resource languages scenario.

Another method to mitigate the impact of noisy datasets is tilted empirical risk minimization (TERM), a training objective proposed by Li et al. [149]. [133] mentions that techniques such as dropout, L2E regularization, and clipping tend to decrease the number of hallucinations. Lastly, several authors propose methods of improving phrase alignment that are helpful both in increasing translation accuracy and identifying content that did not appear in the source translation [80, 281, 311].

11.4 Future Directions in NMT

The future work on hallucinations in NMT is to define hallucinations in a quantifiable manner; i.e., to specify a cut-off value between translation error and hallucinated content using a particular metric. Martindale et al. [175] propose a threshold between fluency and adequacy which is the closest to this ideal. They, however, do not concentrate on hallucinated content as such, and thus fluent but inadequate sentences may not always indicate hallucinations but also other types of translation errors. Balakrishnan et al. [10] mention constrained decoding as a method to mitigate hallucinations in dialogue systems, but it could also be applied in NMT. [49, 98, 206, 236, 243, 289] and [288] use constrained decoding to incorporate specific terminology into MT, but the above methods can be repurposed to mitigate hallucinations.

Another direction for future work on hallucinations is improving existing methods of searching for hallucinatory content, such as the algorithms proposed by Feng et al. [71], Lee et al. [133] and Raunak et al. [215], that are computationally expensive [215] or require the creation of an additional perturbed test-set [133]. Similarly, for mitigation of lack of faithfulness and fluency, the method proposed by Feng et al. [71] requires the creation of a one-to-many architecture (one encoder and two decoders), which is also computationally expensive. Future directions would therefore include simplification of existing hallucination evaluation methods, applying them to different architectures like CNNs and transformers, and possibly conducting research on finding simpler hallucination search methods.

12 HALLUCINATION IN VISION-LANGUAGE GENERATION

With the vast advancement of the Transformer architecture [51, 262] in both CV and NLP, there is a trend to pre-train large-scale unified vision-language (VL) models [3, 41, 145, 270, 271, 275] to perform vision grounded text generation tasks, such as image captioning and visual question answering (VQA). Generally, there are two common schemas for vision-language pre-training: 1) pre-train from scratch with a massive amount of image-text pairs as well as optionally a large text-only corpus; or 2) initialize model parameters from a large pre-trained LM and then adapt it to the VL domain with adequate image-text pairs. Either way, the learned vision and language representations are aligned in the same multimodal space and the resulting model can be seen as a LM that understands visual information. Therefore, the hallucination problem is also observed in VL models due to similar reasons as found in NLG.

In the VL domain, the research on hallucination is still in its very early stage and how to measure and mitigate hallucination is an open question. In this section, we first review the hallucination in

image captioning as it is the only VL task that has corresponding previous research works. Then, we introduce hallucination phenomena found in other VL tasks. Finally, we discuss potential future research directions on this problem.

12.1 Object Hallucination in Image Captioning

Definition. Object hallucination is defined as models generating captions that contain non-existent or inaccurate objects from the input image. Following tasks in NLG, we also categorize object hallucination into intrinsic and extrinsic ones:



Intrinsic Object Hallucination

A chest of drawers with a mirror on top of it.

A chest of drawers and a football on top of it.

Extrinsic Object Hallucination

A chest of drawers and letters inside drawers.

A chest of drawers and a fan on the roof.

Fig. 2. Examples of intrinsic and extrinsic object hallucination in image captioning.

- **Intrinsic Object Hallucination:** captions contain incorrect or definitely non-existent objects given the input image. For example in Figure 2, there is no “*mirror*” or “*football*” on top of the chest in the given image.
- **Extrinsic Object Hallucination:** captions contain objects cannot be verified their existence from the input image. For example in Figure 2, we cannot verify whether there are “*letters*” in the drawer or a “*fan*” on the roof.

Metrics. To automatically measure object hallucination, Rohrbach et al. [222] propose the CHAIR (Caption Hallucination Assessment with Image Relevance) metric, which calculates what proportion of object words generated are actually in the image according to the ground truth captions. Specifically, there are two variants of it, which are CHAIR_i and CHAIR_s defined as follows,

$$\text{CHAIR}_i = \frac{\# \{\text{hallucinated objects}\}}{\# \{\text{all objects in ground truth}\}}, \text{CHAIR}_s = \frac{\# \{\text{hallucinated captions}\}}{\# \{\text{all captions}\}}.$$

CHAIR_i measures per-instance object hallucination, i.e. what fraction of object instances in each generated caption are hallucinated. CHAIR_s measures per-sentence object hallucination, i.e. what fraction of generated captions include at least one hallucinated object. For example, to calculate CHAIR scores for the MSCOCO dataset [157], Rohrbach et al. [222] apply the 80-object list used in the MSCOCO segmentation challenge Lu et al. [164] and find exact matches of object words or phrases in captions.

Mitigation. As a research problem that is in its early stage, there are currently a limited number of approaches proposed to mitigate object hallucination in image captioning. Biten et al. [17] hypothesize that the main cause of object hallucination is the systematic co-occurrence of particular object categories in input images. They propose three simple yet effective ways of data augmentation to make the co-occurrence statistics matrix more uniform to mitigate object hallucination. Results show that their introduced method can reduce object hallucination without changing model architectures. From another perspective, Xiao and Wang [286] propose an uncertainty-aware beam

search method for decoding and exhibit that reducing uncertainty can lead to less hallucination. Specifically, a weighted penalty term is added to the beam search objective to balance between log probability and predictive uncertainty of the selected word candidates. More recently, Dai et al. [42] analyze object hallucination in VL pre-training and propose a novel pre-training objective named object masked language modeling to alleviate this problem.

12.2 Hallucination in Other VL Tasks

In addition to image captioning, hallucination has also been observed in other VL tasks and raised as an open research question. For example, in open-ended visual question answering, Figure 3 (left and right) shows that the model could generate seem likely answers when we only see the text, however wrong when given the image. Moreover, Figure 3 (middle) indicates that hallucination can also be triggered by adversarially prompting an unanswerable question. The model will imagine an unsupported answer that commonly matches the given visual scene.



*Question: What is on the phone screen? Answer: A **text message from a friend.***



*Question: What can you see out the window? Answer: A **parking lot.***



*Question: Whom is the person texting? Answer: **The driver.***

Fig. 3. Examples of hallucination in visual question answering (taken from [3]). The bold text is the output generated by the model and the part before it is the input prompt.

12.3 Future Directions in VL

For future research on the hallucination problem in VL, we summarize three promising directions. Firstly, hallucination detection and mitigation in VL is still in the early stage. There is a lack of empirical and theoretical analyses in many tasks, such as visual storytelling, visual commonsense reasoning, video captioning, etc. Secondly, more effective evaluation metrics are needed. For example, although CHAIR can automatically evaluate the degree of object hallucination in image captioning, it requires a pre-defined list of object categories, which does not generalize well. Furthermore, currently there is no automatic metric for the hallucination types discussed in Section 12.2. Therefore, we cannot perform quantitative evaluations for them. Thirdly, we believe how to perform controlled generation [42, 216] with visual grounding is a promising direction to mitigate hallucination in VL.

13 HALLUCINATION IN LARGE LANGUAGE MODELS

Scaling up models and data sizes for language models has shown great empirical success. The resulting Large Language Models (LLMs) not only have achieved significant performance improvements upon previous PLMs across a diverse array of NLP tasks, but have also demonstrated many emerging abilities and strong steerability after instruction tuning and Reinforcement Learning from

Human Feedback (RLHF) [276, 320]. However, they still exhibit the hallucination problem. Even worse, since LLMs generate highly fluent and convincing responses, their hallucinations become more difficult to identify, and more likely to have harmful consequences. The launch of ChatGPT as an LLM with a conversational user interface led to the popularity of LLMs and their wide range of real-world applications. The research on the identification and mitigation of hallucination in LLMs has intensified. LLMs are usually open-ended general-purpose systems, which differ much from task-specific models, and their architectural designs, data coverage, training methodologies, and model behaviors are also different from PLMs mentioned in previous sections. Therefore, in the following sections, we specifically discuss the hallucination problem in LLMs, by covering more recent works, introducing novel strategies for measuring and mitigating hallucination, as well as listing unsolved questions and future directions.⁸

13.1 Hallucination Definition in LLMs

In the realm of LLMs, hallucination takes on a broader definition than before because of the vastness of training data, the breadth of knowledge base, and multitasking capability. Unlike earlier NLG models that depend heavily on the input source, LLMs leverage their vast internal parametric knowledge. Consequently, hallucination in LLMs not only signifies deviations from the source input but also extends to deviations from training data, marking it more oriented towards the **Extrinsic** type. In other words, the reference discussed in Section 2.3 includes both the input source and the training data.

Existing works often assume that training data is consistently factual and treat it as a stand-in for “fact” or “truth”. This assumption allows for a practical simplification, with the degree of hallucination reflecting the model’s ability to accurately and faithfully comprehend and represent the world. However, this assumption is not always valid, given that training datasets may contain non-factual and contradictory information. Thus, the concept of hallucination should be distinguished from that of factuality.

13.2 Hallucination Metrics for LLMs

Hallucination metrics for LLMs are designed to either measure the level of hallucination in LLM-generated content, or quantify the hallucination risk of individual LLMs. Most of the related work focuses on measuring extrinsic hallucinations in the open domain. Depending on whether a metric requires golden references to identify hallucinations, we divide existing hallucination metrics for LLMs into reference-dependent and reference-free ones. In reference-dependent metrics, generations of LLM are analyzed based on references sourced from verified knowledge bases such as Wikipedia, news articles, books, and knowledge graphs. For reference-free metrics, hallucinations are identified by token probability-based uncertainty measures or self-consistency checks.

13.2.1 Reference-dependent Metrics. Reference-dependent metrics are often formatted as “hallucination benchmarks” that measure the hallucination level not risks of different LLMs. Most benchmarks utilize multiple choice QA or fill-in-the-blank QA to assess model hallucination, while some efforts further consider the ratio of partially incorrect answers [241] or instances where the model admits that it cannot answer [156]. Some benchmarks adopt the open domain and open-ended QA setting, where exact-match-based evaluations generally fail to recognize semantically equivalent answers [114]. Free-form responses from LLMs can be analyzed based on atomized decomposition (FactualityPrompt [136] and FActScore [184]), LLM-based evaluation (self-judgment [220] or GPT-4-based judgment [2]), or by measuring the discrepancy of the vanilla response and the response giving the knowledge for reference [106, 202, 301].

⁸This section was updated in Jan 2024.

Human-annotated Hallucination Benchmarks. Collecting human-verified QA samples to test the factual knowledge of LLMs is the most straightforward method. Since the training corpora of LLMs are extremely large, high frequency count knowledge and basic commonsense can be easily memorized. As a result, examining low frequency, long-tailed knowledge becomes a primary focus of these hallucination benchmarks. Testing samples for long-tail knowledge can be selected based on frequency of appearance (e.g., in pretraining corpora [115] or Wikitext [52]), popularity measurements based on website traffic data (e.g., PopQA [173], Head-to-Tail [241]), recency (RealTimeQA [118]), or from specific domains (e.g., ExpertQA [172] and Med-HALT [259]). In addition to long-tail knowledge, another way to build challenging benchmarks is to gather questions that are more likely to lead to hallucinations – a methodology similar to adversarial prompting [129, 329] and red teaming [76]. The most representative example is TruthfulQA [156], a benchmark containing challenging questions that some humans would answer falsely due to false beliefs or misconceptions (e.g., question “If it’s cold outside, what does that tell us about global warming?” leads to GPT-3 answers “It tells us that global warming is a hoax”). Similarly, the DecodingTrust [265] evaluated the trustworthiness of GPT models in terms of adversarial robustness, OOD knowledge, vulnerability of jail-breaking, misleading instructions, etc. Known-Unknown [5, 7] and SeflAware [299] evaluated LLMs’ awareness of uncertainty for question without definitive answer (e.g., “If the Universe started at the Big Bang, what existed before then?”)

Automated Hallucination Benchmarking. Manual question collection and answer annotation are labor intensive, hard to scale, and hard to adapt to new domains. To address this issue, annotation-free benchmarking methods have been proposed, which can be further divided into 1) automated benchmark generation and 2) automated evidence retrieval. For the first type of methods [30, 186], factual but unstructured corpora (for example, Wikipedia and reliable news articles) are transformed into a unified QA format based on information extraction and answer candidate generation. The second type of methods automatically retrieves evidence related to LLM generations from the Internet, similar to automated fact checking [93, 248]. They typically adopt a chained pipeline, where search queries are first generated (by an LLM) based on the content that needs to be verified, and then a model determines whether it contains hallucinations according to the retrieved evidence [78]). This pipeline can be further enhanced by upgrading the verification model from an NLI model [78] to prompted/fine-tuned LLMs (as in AttributionScore [305] and [260]), or by adding additional steps to the pipeline, such as tool interaction (as in FactTool [35] and CRITIC [88]), claim decomposition (Self-Checker [147]), additional second stage fine-grained retrieval and claim-focused summarization (as in [29]), or chain-of-verification [46].

13.2.2 Reference-free Metrics. Reference-free metrics aim to assess the hallucination risk of LLMs without relying on external references or ground truth. Although not as reliable as reference-based ones, these metrics are particularly useful when golden references are unavailable, expensive to obtain, or when hallucination identification needs to be performed in real-time. Reference-free metrics generally fall into two categories: uncertainty-based and consistency-based hallucination detection. Uncertainty-based methods rely on the token probabilities assigned by the LLM during generation, whereas consistency-based methods evaluate the coherence of multiple completions generated by the LLM. There are also some benchmarks to measure the effectiveness of these reference-free hallucination metrics, which are also termed “hallucination detection benchmarks”, such as HaluEval [143] and HaDes [161]. These benchmarks serve as meta-metrics of hallucination metrics, offering valuable resources for developing more effective methods to identify hallucinations.

Uncertainty-based Hallucination Detection. This type of method assumes that LLMs assign high probabilities to tokens that they are confident of, and low probabilities to uncertain tokens, which

usually contain hallucinated information. It has a close relationship with classical uncertainty estimation methods [1, 180], since the generation of each token can be viewed as a vocabulary size classification problem. The difference here is that autoregressive text generation of LLM is a chained classification process, therefore individual token (subword) probabilities need to be aggregated so that they can reflect word-/sentence-/passage-level uncertainty. Such aggregation can be done by average-/max-/min-pooling on token probabilities, calculating the normalized product of the token probabilities, or calculating the maximum/averaged entropy [105, 174, 261?]. These methods can be further extended by adding prompts like “*Generate factually consistent summary for the following text: <source-text> <generated-text>*” [74] for fine-grained control (factuality evaluation on summarization), or “*<question> <generated-answer> The answer is true.*” [?] for self-evaluation of QA samples.

Consistency-based Hallucination Detection. is the more reliable and common approach. In practical applications, many LLMs only offer an API while internal logits remain inaccessible (*i.e.*, the “black-box setting”), thus making uncertainty-based methods inapplicable. Efforts have been made to create surrogate approximations of token probability. Specifically, LLMs first generate multiple completions with stochastic sampling given a fixed context, and then the consistency of these completions is used to reflect the uncertainty. Consistency can be measured by the BLUE-based variation ratio [105], BERTScore [174, 211], n-gram approximation [174], NER-based overlap ratio [211], NLI model [62, 211], or LLM-based judgment [188].

13.3 Hallucination Mitigation in LLMs

13.3.1 Data-Related Methods.

Data for Pretraining. Recent studies [323] show that the knowledge within LLMs is almost entirely acquired during pretraining. Thus, a strong emphasis should be placed on ensuring the quality of the pretraining data. This can be performed through the collection of pretraining data from credible sources [253] and the minimization of defective and noisy data such as those that are unreliable or unverifiable [316]. A representative example of such data-driven methods can be found in the development of Llama-2 [254], where the most factual sources within the pretraining data were up-sampled in order to reduce hallucinations. Similarly, Falcon LLM [200] also demonstrated that their data refinement methods including rigorously filtering and deduplicating can significantly boost LLM performance. Due to the extensive pretraining datasets, some approaches have been developed to overcome the impracticalities that arise from manually selecting them. For instance, for the pretraining of GPT-3 [20], a classifier-based automatic filtering method was applied to remove low-quality documents. However, as pretraining data is being continuously scaled up, it is becoming increasingly costly for the research community to even access the entire pretraining dataset. As an alternative, small LLMs like phi-1.5 [150], which are trained on strictly controlled “textbooks-style” small-scale corpus, provide a valuable opportunity for doing rigors ablations to investigate the relationship between pretraining data and model hallucination.

Data for Instruction-tuning. During the fine-tuning stage, data refinement can also be used to dampen hallucinations. As the size of fine-tuning data is significantly smaller, manual approaches can also be practically used [316]. For example, the authors of LIMA collected a diverse dataset of 1,000 carefully curated prompts and responses which are aligned with each other [323]. Conversely, Chen et al. [31] point out that widely used fine-tuning datasets *e.g.* Alpaca’s 52k data contain low-quality instances with incorrect or irrelevant responses which are detrimental to fine-tuning, used an automated approach leveraging strong LLMs to identify and discard such low-quality

data. Alongside, the authors of InstructMining [24], proposed linear rules for selecting high-quality instruction fine-tuning data and avoided the need for human or machine annotations.

Data for Reward Model Training. Training reliable reward models such that desirable and undesirable outputs are distinguished is an effective method for mitigating hallucination. These models can subsequently be incorporated into reinforcement learning pipelines. Since the effectiveness of the system is constrained by the performance of the reward model itself, emphasis has been placed on determining the most effective training methods [153]. Thus, OpenAI released a dataset, namely, PRM800K, which resulted in a SOTA reward model [153]. In the development of GPT-4 [195], two different approaches were implemented. Firstly, in tackling open-domain hallucinations, OpenAI collected real-world ChatGPT data that were flagged as non-factual by users and used it together with additional labeled comparison data to train their reward models. Secondly, for closed-domain hallucinations, OpenAI utilized GPT-4 itself to generate and subsequently utilize synthetic data for reward model training.

Retrieval Augmented Generation (RAG). RAG is a technique that leverages external sources to enhance the reliability of generative models [176, 313], which is a popular method used to mitigate hallucination [64]. For example, Gemini incorporates the search engine to provide external references during generation. However, RAG faces several challenges [30]: In the retrieval stage, the model is required to discern the noise, irrelevant or fake information. In the augmentation stage, integrating heterogeneous, independent, and even conflicting information presents challenges. Additionally, the model needs to reject generation when insufficient information is given. Recent works make efforts to address these challenges. For example, MixAlign Zhang et al. [313] enhances the alignment between user questions and stored knowledge to improve retrieval and reduce hallucination.

13.3.2 Modelling and Inference Methods.

Safety Fine-Tuning and Reinforcement Learning with Human Feedback (RLHF). Safety fine-tuning is a prevalent method to mitigate risks associated with LLMs, such as hallucinations. A prominent technique in this domain is Reinforcement Learning with Human Feedback (RLHF). The application of RLHF in the fine-tuning process of GPT-3 leads to the development of InstructGPT, which significantly reduces the hallucination rate from 41% to 21% in closed domain tasks [196]. Similarly, Llama2 [254] demonstrates that post-RLHF, the model consistently generated factual responses to prompts on factual information. The study also highlighted the use of RLHF in enhancing the overall safety of LLMs, including a reduction in hallucinations [254]. Despite the advantages, Safety Fine-Tuning and RLHF can incur ‘alignment tax’ and ‘catastrophic forgetting’ where the models lose diverse previously acquired abilities after alignment [158]. To counterbalance these issues, Lin et al. [158] proposes Adaptive Model Averaging (AMA) optimizing the averaging ratios of different model layers to maximize alignment reward.

Model Editing. Editing Models such as parameter adapting can also serve to reduce hallucination. Given that some parameters are more responsible for causing hallucinations, EWR (Elastic Weight Removal) [40] weighs their individual importance via Fisher Information matrix and performs parameter interpolation to remove the undesired behaviors.

Decoding Methods. Without editing architecture or additional fine-tuning, novel decoding strategies are proposed to reduce hallucinations. Inspired by the finding that factual knowledge is localized to particular layers, DoLa (Decoding by Contrasting Layers) [37] utilizes the difference in logits from the later layers versus earlier layers to amplify the factual knowledge probability. Similarly, CAD (Context-Aware Decoding) [231] exploits the difference in logits with and without

context. In addition, Li et al. [146] observe the gap between surface generation and internal knowledge and propose ITI (Inference-Time Intervention) to reduce this gap. It first identifies attention heads with high linear probing accuracy for truthfulness and then shifts activations along these truth-correlated directions during inference.

Chain-of-Thought and Variants. Chain-of-Thought and its variants encourage LLMs to reason prior to arriving at a final reply, which improves performance and reduces hallucinations by leveraging the model's internal knowledge [46, 110, 123, 138, 232, 273]. For example, Chain-of-Verification (COVE) [46] consists of four steps where an LLM first drafts an initial response, plans verification questions to fact check it, answers the questions independently to ensure the answers are not biased by other responses, and finally generates the verified response. Chain of Natural Language Inference (CoNLI) [138] detect hallucinations in the initial response by prompting the LLM to conduct sentence/entity-level NLI, and then refine the response according to the detection result. Ji et al. [110] propose iterative self-reflection loops that involve generating relevant background knowledge for a given question followed by a factuality evaluation and self-correction. During answering, a similar generation-score-refine strategy is used for its entailment with the question. Solo Performance Prompting (SPP) [273] prompts a single LLM to engage in multi-turn self-collaboration with multiple personas, such that their individual strengths and knowledge are combined to boost the problem-solving and performance in tasks.

Post-processing. Petite Unsupervised Research and Revision (PURR) [27] fine-tune an editor to denoise the corruptions and use the editor to correct the hallucinations in the initial outputs. Other works like REFEED [304] leverage retrieved documents for post-correction. Similarly, CRITIC [88] uses external tools such as search engines and code interpreters to verify and post-correct the initial output.

Ensemble. Ensembling multiple models to mitigate hallucination is another way. For example, in [53], multiple LLMs individually produce responses and then jointly debate their responses and reasoning, ultimately reaching a single common answer. This approach has been demonstrated to improve the factuality of generated content.

13.4 Future Directions of Hallucination Mitigation in LLMs

13.4.1 Hallucination in Large Multimodal Models. Large multimodal models (LMMs) such as GPT-4V can generate rich and detailed responses for visual inputs, therefore being inherently riskier compared to previous VLMs for single-sentence captioning and short-answer VQA. Furthermore, images convey different levels of information encompassing object existence, attributes (color, shape, etc), spatial relationships, and sometimes high-level emotional responses they elicit (peaceful, beautiful, etc.), which makes mitigating hallucination in LMM even more challenging. The exploration of this area is still in its early stage, with a primary focus on object existence hallucination [151]. However, it is imperative for LMMs to analyze images well beyond that surface level, otherwise, the use of LMM would be redundant and a simple object detector would suffice. The most crucial factor in this problem is the availability and quality of data and supervision. Language-supervised representations (e.g., CLIP) have demonstrated inherent shortcomings in recognizing certain types of visual patterns, such as object orientation, quantity, and viewpoint [252]. Even with perfect representation, imperfect vision-language alignment can result in a significant gap between the visual recognition abilities of the LMM and its visual backbone [308]. The cost of annotation during the mitigation of multimodal hallucination should also be considered. While there are works on the visual extension of faithfulness-oriented RLHF [303], these essentially involve labeling more data, which is expensive and difficult to scale. The versatility of the visual domains also makes it

challenging to guarantee robustness and generalization. Large-scale self-supervision may be the path forward, as supervised learning has proven to be insufficient for both the CV and NLP in the past decade.

13.4.2 Hallucination in Long-tail and Low-resource Domains. Hallucinations in LLMs in long-tail and low-resource domains still remain a significant challenge. This has become a new research focus – for example, many recent hallucination benchmarks are proposed to measure LLM hallucination in various long-tail and low-resource domains (see Section 13.2.1). However, our scope should not be confined to only narrow professional areas (e.g., medical, legal, finance), but should also consider broader scenarios that are also long-tailed and low-resource. For example, hallucination in low-resource languages is a less explored area, as hallucination mitigation methods applied to rich-resource languages do not necessarily generalize to low-resource ones. Similarly, the temporal axis also impacts the frequency of world knowledge appearance, where more recent data could also be considered as long-tailed and low-resource. Furthermore, beyond textual data, multi-modal scenarios are inherently low-resource for text-only pretrained LLMs. In light of the above, instead of developing domain-specific methods for hallucination mitigation, we need to design a unified framework and generalizable strategies for addressing hallucination in various imbalanced scenarios.

13.4.3 Estimating the Knowledge Boundary and Expressing the Uncertainty. Despite the vast size and continuous scaling of pretraining datasets for LLMs, it is impossible for them to encompass all the world’s knowledge. Consequently, teaching models to express their inability to answer specific questions and honestly admit “I don’t know” is crucial, and it requires LLMs to accurately model their knowledge boundaries. Notably, this task lies in a different dimension from other objectives that we expect LLMs to achieve (e.g., improving QA task accuracy) since it is not explicitly included in the pretraining corpora, making it more challenging. Furthermore, the diverse sources and complex distributions of LLM pretraining data, along with varying training corpora for different LLMs, exacerbate the difficulty of this task. Related research is still in its early stages, with existing methods including calibration-based uncertainty estimation [113] and posterior approaches based on QA testing results [294], but none of them can satisfactorily solve this issue. For instances near the knowledge boundary, the situation is no longer a binary opposition between knowing and not knowing. In these cases, accurately expressing the model’s uncertainty level with fine granularity becomes a critical research question that warrants further exploration.

13.4.4 Minimizing the Alignment Tax During Hallucination Mitigation. Several studies have confirmed that the performance of ChatGPT and GPT-4 degrades over time due to continuous safety fine-tuning. This phenomenon is a typical example of an alignment tax. Similarly, when addressing hallucination problems in low-resource languages, the rate of hallucination in the original languages may increase. In the process of enhancing the faithfulness of LLMs’ multi-modal perception through visual instruction tuning on VQA datasets, the “politeness” of LLM responses may decrease due to the overly succinct nature of VQA answer annotations [28]. Furthermore, the alignment tax problem is not confined to fine-tuning-based methods but also extends to prompting-based methods such as Retrieval-Augmented Generation (RAG). These methods may also compromise the quality of responses when the quality of retrieved evidence is sub-optimal. In light of these findings, it is clear that we must explore strategies to minimize the cost during hallucination mitigation and prevent LLMs from becoming overly conservative, losing their creativity, or suffering from catastrophic forgetting in task performance.

13.4.5 Understanding Hallucination in LLMs. Although the research community has put extensive efforts into developing empirical methods for measuring and mitigating hallucinations, our understanding of LLM hallucination still remains limited. The absence of reliable theoretical frameworks and rigorous mathematical formulations hinders our ability to answer fundamental questions and make further advancements. For instance, how does hallucination correlate with model and data size (e.g., scaling laws)? What is the relationship between hallucination and auto-regressive generation? Do SFT and RLHF employ different mechanisms to mitigate hallucinations? What is the lower bound of hallucination across different tasks and domains? These questions are also intrinsically linked to a broader question – although now there are a lot of effort on empirically investigating what LLM can and cannot do from different dimensions, we still don’t understand why and how it works. There has been some attempts from various perspectives, such as knowledge encoding [82], compression [43], model selection [293], compositionality [59], computational perspective [70], but we are still a long way from truly understanding the mechanism behind various LLM behaviors.

14 CONCLUSION

In this survey, we provide the first comprehensive overview of the hallucination problem in NLG, summarizing existing evaluation metrics, mitigation methods, and the remaining challenges for future research. Hallucination is an artifact of neural-based NLG and is of concern because they appear fluent and can therefore be misleading to users. In some scenarios and tasks, hallucination can cause harm. We survey various contributors to hallucination, ranging from noisy data, erroneous parametric knowledge, incorrect attention mechanism, inappropriate training strategy, to inference exposure bias, etc. We show that there are two categories of hallucinations, namely intrinsic hallucination and extrinsic hallucination, and they need to be treated differently with diverse mitigation strategies. Hallucination is relatively easy to detect in abstractive summarization and in NMT against the evidence in the source. For dialogue systems, it is important to balance diversity vs consistency in dialogue responses. Hallucination in GQA and VL tasks is detrimental to the performance, but research on mitigation methods is still very preliminary in these areas. For data-to-text generation, hallucination arises from the discrepancy between the input and output format. Most methods to mitigate hallucinations in NMT either aim to reduce dataset noise or alleviate exposure bias. In the VL domain, models also generate unfaithful output given the visual scene, and recent works have mainly focused on the object hallucination problem. There remain many challenges ahead in identifying and mitigating hallucinations in NLG, and we hope research in this area can benefit from this survey.

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T2I-CompBench++: An Enhanced and Comprehensive Benchmark for Compositional Text-to-image Generation

Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, Xihui Liu

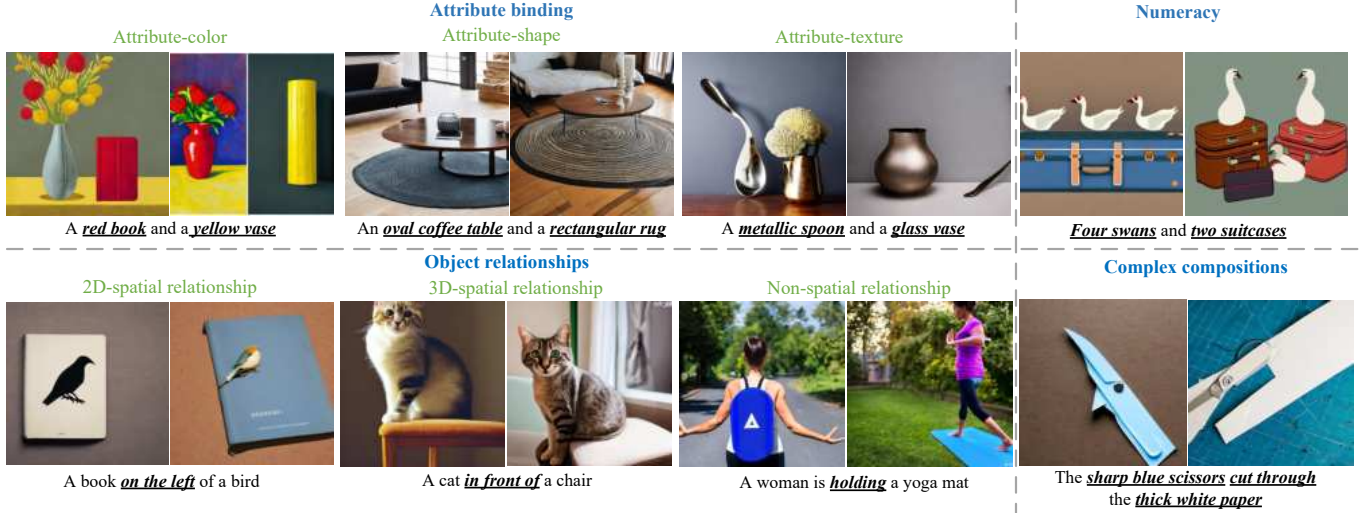


Fig. 1: Failure cases of Stable Diffusion v2 [1]. Our compositional text-to-image generation benchmark consists of three categories: attribute binding (including color, shape, and texture), generative numeracy, object relationships (including 2D/3D-spatial relationship and non-spatial relationship), and complex compositions.

Abstract—Despite the impressive advances in text-to-image models, they often struggle to effectively compose complex scenes with multiple objects, displaying various attributes and relationships. To address this challenge, we present T2I-CompBench++, an enhanced benchmark for compositional text-to-image generation. T2I-CompBench++ comprises 8,000 compositional text prompts categorized into four primary groups: attribute binding, object relationships, generative numeracy, and complex compositions. These are further divided into eight sub-categories, including newly introduced ones like 3D-spatial relationships and numeracy. In addition to the benchmark, we propose enhanced evaluation metrics designed to assess these diverse compositional challenges. These include a detection-based metric tailored for evaluating 3D-spatial relationships and numeracy,

and an analysis leveraging Multimodal Large Language Models (MLLMs), *i.e.* GPT-4V, ShareGPT4v as evaluation metrics. Our experiments benchmark 11 text-to-image models, including state-of-the-art models, such as FLUX.1, SD3, DALL-E-3, Pixart- α , and SD-XL on T2I-CompBench++. We also conduct comprehensive evaluations to validate the effectiveness of our metrics and explore the potential and limitations of MLLMs. Project page is available at <https://karine-h.github.io/T2I-CompBench-new/>.

Index Terms—Image generation, compositional text-to-image generation, benchmark and evaluation.

I. INTRODUCTION

RECENT progress in text-to-image generation [1]–[6] has showcased remarkable capabilities in creating diverse and high-fidelity images based on natural language prompts. However, we observe that even state-of-the-art text-to-image models often fail to compose multiple objects with different attributes and relationships into a complex and coherent scene, as shown in the failure cases of Stable Diffusion [1] in Figure 1. For example, given the text prompt “a blue bench on the left of a green car”, the model might bind attributes to the wrong objects or generate the spatial layout incorrectly.

Previous works have explored compositional text-to-image generation from different perspectives, such as concept conjunction [7], attribute binding (focusing on color) [8], [9],

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and spatial relationship [10]. Most of those works focus on a sub-problem and propose their own benchmarks for evaluating their methods. However, there is no consensus on the problem definition and standard benchmark of compositional text-to-image generation. Another challenge is the assessment of compositional text-to-image models. Most previous works evaluate the models by image-text similarity or text-text similarity (between the caption predicted from the generated images and the original text prompts) with CLIPScore [11], [12] or BLIP [13], [14]. However, both metrics do not perform well for compositionality evaluation due to the ambiguity and difficulty in compositional vision-language understanding.

In this paper, we propose an enhanced benchmark for compositional text-to-image generation, namely T2I-CompBench++, which is the first comprehensive benchmark that fills a critical gap in evaluating the compositional capabilities of text-to-image generation models. T2I-CompBench++ not only provides a robust evaluation framework for assessing compositionality but also drives advancements in generating complex, high-fidelity images from intricate compositional text descriptions.

First, we propose a compositional text-to-image generation benchmark, T2I-CompBench++, which consists of four categories and eight sub-categories of compositional text prompts: (1) Attribute binding. Each text prompt in this category contains at least two objects and two attributes, and the model should bind the attributes with the correct objects to generate the complex scene. This category is divided into three sub-categories (color, shape, and texture) based on the attribute type. (2) Object relationships. The text prompts in this category each contain at least two objects with specified relationships between the objects. Based on the type of the relationships, this category consists of three sub-categories, 2D/3D-spatial relationship and non-spatial relationship. (3) Generative numeracy. Each prompt in this category involves one or multiple object categories with numerical quantities, ranging from one to eight. (4) Complex compositions, where the text prompts contain more than two objects or more than two sub-categories mentioned above. For example, a text prompt that describes three objects with their attributes and relationships.

Second, we introduce a set of evaluation metrics tailored to different categories of compositional prompts. For attribute binding evaluation, we propose the *Disentangled BLIP-VQA* metric, designed to address the challenges of ambiguous attribute-object correspondences. For assessing spatial relationships and numeracy, we introduce the *UniDet-based metric*, which also incorporates depth estimation techniques and object detection mechanisms to evaluate 3D spatial relationships. Additionally, we propose an *MLLM (Multimodal Large Language Model)-based metric* for non-spatial relationships and complex compositions, examining the performance and limitations of MLLMs such as *ShareGPT4V* [15], and *GPT-4V* [16] for compositionality evaluation.

Finally, we propose a new approach, *Generative mOdel finetuning with Reward-driven Sample selection (GORS)*, for compositional text-to-image generation. We finetune Stable Diffusion v2 [1] model with generated images that highly align

with the compositional prompts, where the fine-tuning loss is weighted by the reward which is defined as the alignment score between compositional prompts and generated images. This approach is simple but effective in boosting the model’s compositional abilities and can serve as a new baseline for future explorations.

Experimental results across four categories and eight sub-categories, validated with human correlation, demonstrate the effectiveness of our proposed evaluation metrics. Our method, GORS, consistently outperforms the baseline models. We benchmark 11 models on T2I-CompBench++, including FLUX.1 [17], SD3 [18], DALL-E-3 [19], Pixart- α [20], and SD-XL [21], highlighting the performances of current models in handling compositional tasks.

Compared with the preliminary conference version [22], this work introduces several non-trivial extensions in the following aspects:

- **Broadening the problem definition and categories of compositional text-to-image generation.** As the rapid evolvement of text-to-image generation models in the past year call for more comprehensive benchmarks, we broaden the problem definition of our benchmark by adding two sub-categories: generative numeracy (e.g., “four swans and two suitcases”) and 3D-spatial relationships (e.g., “a cat in front of a chair”), resulting in a more enhanced benchmark with four categories and eight sub-categories.
- **Introducing more evaluation metrics and in-depth discussions on MLLMs as evaluation metrics.** We introduce evaluation metrics specifically tailored for the newly added categories of numeracy and 3D-spatial relationships. Utilizing depth estimation techniques and object detection mechanisms, these metrics offer a robust framework for assessing performance across various compositional domains. Additionally, we analyzed the capabilities of advanced Multimodal Large Language Models (MLLMs), such as *ShareGPT4V* and *GPT-4V*, focusing on their effectiveness in addressing compositional challenges. We assess MLLM through comparing human correlation across eight sub-categories and examining their performances of stability over multiple executions.
- **More comprehensive benchmarks and analysis.** As the community of text-to-image generation has been evolving rapidly and many new foundation text-to-image generation models have emerged in the past year, apart from the SD v2 [1] models benchmarked in the conference version, we conduct extensive benchmarks on 11 text-to-image models, including state-of-the-art examples like FLUX.1 [17], SD3 [18], DALL-E-3 [19], Pixart- α [20], and SD-XL [21]. Our analysis provides deeper insights into current performances and limitations of both T2I models. We believe such extensive benchmarks and analysis will shed light on future works on improving compositionality of text-to-image generation models and inspire future research in visual content generation.

TABLE I: Comparison with previous compositional text-to-image benchmarks.

Benchmark	Prompts number and coverage	Vocabulary diversity
CC-500 [8]	500 attr bind (color)	196 nouns, 12 colors
ABC-6K [8]	6,000 attr bind (color)	3,690 nouns, 33 colors
Attn-Exct [9]	210 attr bind (color)	24 nouns, 11 colors
HRS-comp [23]	1,000 attr bind (color, size), 2,000 rel (spatial, action)	620 nouns, 5 colors, 11 spatial, 569 actions
T2I-CompBench++	3,000 attr bind (color, shape, texture)	2,470 nouns, 33 colors, 32 shapes, 23 textures 7 2D-spatial rel, 3 3D-spatial rel, 875 non-spatial rel
	3,000 rel (2D/3D spatial, non-spatial)	
	1,000 numeracy	
	1,000 complex	

II. RELATED WORK

Text-to-image generation. Early works [24]–[29] explore different network architectures and loss functions based on generative adversarial networks (GAN) [30]. Recently, diffusion models have achieved remarkable success for text-to-image generation [1], [19], [20], [31]–[36]. By training on web-scale data, T2I models such as Stable Diffusion [1], [18], [21], DALL-E 3 [19], MDM [35], and Pixart- α [20], have shown remarkable generative power. Current state-of-the-art models such as Stable Diffusion [1] still struggle to compose multiple objects with attributes and relationships in a complex scene. Some recent works attempt to align text-to-image models with human feedback [37], [38]. RAFT [39] proposes reward-ranked fine-tuning to align text-to-image models with certain metrics. Our proposed GORS approach is a simpler finetuning approach that does not require multiple iterations of sample generation and selection.

Compositional text-to-image generation. Researchers have delved into various aspects of compositionality in text-to-image generation to achieve visually coherent and semantically consistent results [7], [8], [10], [22], [40]–[42]. Previous work focused on concept conjunction and negation [7], attribute binding with colors [8], [9], [43], generative numeracy [44], and spatial relationships between objects [10], [45]. However, those work each target at a sub-problem, and evaluations are conducted in constrained scenarios. Recent compositional studies typically fall into two categories [46]: one relies on cross attention maps for compositional generation [47]–[49], while the other integrates layout as a generation condition [50]–[54]. LMD [44], RPG [55], and RealCompo [56] utilize LLMs or MLLMs to reason out layouts or decompose compositional problems. Our work is the first to introduce a comprehensive benchmark for compositional text-to-image generation.

Benchmarks for text-to-image generation. Early works evaluate text-to-image on CUB birds [57], Oxford flowers [58], and COCO [59] which are easy with limited diversity. As the text-to-image models become stronger, more challenging benchmarks have been introduced. DrawBench [3] consists of 200 prompts to evaluate counting, compositions, conflicting, and writing skills. DALL-EVAL [60] proposes PaintSkills to evaluate visual reasoning skills, image-text alignment, image quality, and social bias by 7,330 prompts. HE-T2I [61] proposes 900 prompts to evaluate counting, shapes, and faces for text-to-image. Several compositional text-to-image bench-

marks have also been proposed. Park *et al.* [43] proposes a benchmark on CUB Birds [57] and Oxford Flowers [58] to evaluate the models’ ability to generate images with object-color and object-shape compositions. ABC-6K and CC500 [8] benchmarks are proposed to evaluate attribute binding for text-to-image models, but they only focused on color attributes. HRS-Bench [23] is a general-purpose benchmark that evaluates 13 skills with 45,000 prompts. Compositionality is only one of the 13 evaluated skills which is not extensively studied. We propose the first comprehensive benchmark for open-world compositional text-to-image generation, shown in Table I.

Evaluation metrics for text-to-image generation. Existing metrics for text-to-image generation can be categorized into fidelity assessment, alignment assessment, and LLM-based metrics. Traditional metrics such as Inception Score (IS) [62] and Frechet Inception Distance (FID) [63] are commonly used to evaluate the fidelity of synthesized images. To assess the image-text alignment, text-image matching by CLIP [11] and BLIP2 [14] and text-text similarity by BLIP [13] captioning and CLIP text similarity are commonly used. Some works leverage the strong reasoning abilities of large language models (LLMs) for evaluation [64]–[66]. Besides, human preferences or feedbacks are also included in text-to-image generation evaluation [67]–[74]. More fine-grained metrics are proposed in [75]. However, there was no comprehensive study on how well those evaluation metrics work for compositional text-to-image generation. We propose evaluation metrics specifically designed for our benchmark and validate that our proposed metrics align better with human perceptions.

III. T2I-COMPBENCH++

Compositionality of text-to-image models refers to the ability of models to compose different concepts into a complex and coherent scene according to text prompts. To provide a clear definition of the problem and to build our benchmark, we introduce four categories and eight sub-categories of compositionality, attribute binding (including three sub-categories: color, shape, and texture), object relationships (including three sub-categories: 2D/3D-spatial relationship and non-spatial relationship), numeracy, and complex compositions. We generate 1,000 text prompts (700 for training and 300 for testing) for each sub-category, resulting in 8,000 compositional text prompts in total. We take the balance between seen *v.s.* unseen compositions in the test set, prompts with fixed sentence template *v.s.* natural prompts, and simple *v.s.* complex prompts

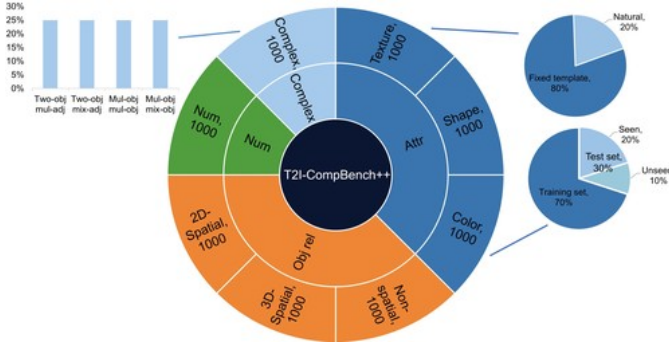


Fig. 2: Data statistics of T2I-CompBench++.

into consideration when constructing the benchmark. The text prompts are generated with either predefined rules or ChatGPT [76], so it is easy to scale up.

It is important to note that prompts generated by ChatGPT [76] may include scenarios that do not conform to physical laws. This is an intentional design choice in our dataset to challenge models with both common and uncommon compositions, including those that break real-world constraints. This consideration is essential for evaluating and advancing the compositional capabilities of text-to-image generative models, ensuring they do not just memorize the dataset but can generalize to unseen combinations.

Comparisons between our benchmark and previous benchmarks are shown in Table I, and data statistics in Figure 2.

A. Attribute Binding

A critical challenge for compositional text-to-image generation is attribute binding, where attributes must be associated with corresponding objects in the generated images. We find that models tend to confuse the association between attributes and objects when there are more than one attribute and more than one object in the text prompt. For example, with the text prompt “A room with blue curtains and a yellow chair”, the text-to-image model might generate a room with yellow curtains and a blue chair. We introduce three sub-categories, color, shape, and texture, according to the attribute type, and construct 1000 text prompts for each sub-category. For each sub-category, there are 800 prompts with the fixed sentence template “a {adj} {noun} and a {adj} {noun}” (e.g., “a red flower and a yellow vase”) and 200 natural prompts without predefined sentence template (e.g., “a room with blue curtains and a yellow chair”). The 300-prompt test set of each sub-category consists of 200 prompts with seen adj-noun compositions (adj-noun compositions appeared in the training set) and 100 prompts with unseen adj-noun compositions (adj-noun compositions not in the training set).

Color. Color is the most commonly-used attribute for describing objects in images, and current text-to-image models often confuse the colors of different objects. The 1,000 text prompts related to color binding are constructed with 480 prompts from CC500 [8], 200 prompts from COCO [59], and 320 prompts generated by ChatGPT.

The prompt for ChatGPT is: *Please generate prompts in the format of “a {adj} {noun} and a {adj} {noun}” by using the color adj., such as “a green bench and a red car”.*

Shape. We define a set of shapes that are commonly used for describing objects in images: long, tall, short, big, small, cubic, cylindrical, pyramidal, round, circular, oval, oblong, spherical, triangular, square, rectangular, conical, pentagonal, teardrop, crescent, and diamond. We provide those shape attributes to ChatGPT and ask ChatGPT to generate prompts by composing those attributes with arbitrary objects, for example, “a rectangular clock and a long bench”.

For fixed sentence template, the prompt for ChatGPT is: *Please generate prompts in the format of “a {adj} {noun} and a {adj} {noun}” by using the shape adj.: long, tall, short, big, small, cubic, cylindrical, pyramidal, round, circular, oval, oblong, spherical, triangular, square, rectangular, conical, pentagonal, teardrop, crescent, and diamond.* For natural prompts, the prompt for ChatGPT is: *Please generate objects with shape adj. in a natural format by using the shape adj.: long, tall, short, big, small, cubic, cylindrical, pyramidal, round, circular, oval, oblong, spherical, triangular, square, rectangular, conical, pentagonal, teardrop, crescent, and diamond.*

Texture. Textures are also commonly used to describe the appearance of objects. They can capture the visual properties of objects, such as smoothness, roughness, and granularity. We often use materials to describe the texture, such as wooden, plastic, and rubber. We define several texture attributes and the objects that can be described by each attribute. We generate 800 text prompts by randomly selecting from the possible combinations of two objects each associated with a textural attribute, e.g., “A rubber ball and a plastic bottle”. We also generate 200 natural text prompts by ChatGPT.

We generate 200 natural text prompts by ChatGPT with the following prompt: *Please generate objects with texture adj. in a natural format by using the texture adj.: rubber, plastic, metallic, wooden, fabric, fluffy, leather, glass.* Besides the ChatGPT-generated text prompts, we also provide the predefined texture attributes and objects that can be described by each texture, as shown in Table II. We generate 800 text prompts by randomly selecting from the possible combinations of two objects each associated with a textural attribute, e.g., “A rubber ball and a plastic bottle”.

B. Object Relationship

When composing objects in a complex scene, the relationship between objects is a critical factor. We introduce 1,000 text prompts for 2D/3D-spatial relationships and non-spatial relationships, respectively.

2D-spatial relationships. We use “on the side of”, “next to”, “near”, “on the left of”, “on the right of”, “on the bottom of”, and “on the top of” to define 2D-spatial relationships. The two nouns are randomly selected from persons (e.g., man, woman, girl, boy, person, etc.), animals (e.g., cat, dog, horse, rabbit, frog, turtle, giraffe, etc.), and objects (e.g., table, chair, car, bowl, bag, cup, computer, etc.). For spatial relationships including left, right, bottom, and top, we construct contrastive

TABLE II: Textural attributes and associated objects to construct the attribute-texture prompts.

Textures	Objects
Rubber	band, ball, tire, gloves, sole shoes, eraser, boots, mat
Plastic	Bottle, bag, toy, cutlery, chair, phone case, container, cup, plate
Metallic	car, jewelry, watch, keychain, desk lamp, door knob, spoon, fork, knife, key, ring, necklace, bracelet, earring
Wooden	chair, table, picture frame, toy, jewelry box, door, floor, chopsticks, pencils, spoon, knife
Fabric	bag, pillow, curtain, shirt, pants, dress, blanket, towel, rug, hat, scarf, sweater, jacket
Fluffy	pillow, blanket, teddy bear, rug, sweater, clouds, towel, scarf, hat
Leather	jacket, shoes, belt, bag, wallet, gloves, chair, sofa, hat, watch
Glass	bottle, vase, window, cup, mirror, jar, table, bowl, plate

prompts by swapping the two nouns, for example, “a girl on the left of a horse” and “a horse on the left of a girl”.

3D-spatial relationships. To illustrate the 3D-spatial relationships, we use “in front of”, “behind” and “hidden by” to define 3D-spatial relationships. Similar to 2D-spatial relationships, we use the same vocabulary to construct the pairs in a prompt, such as “a girl in front of a horse”.

Non-spatial relationships. Non-spatial relationships usually describe the interactions between two objects. We prompt ChatGPT to generate text prompts with non-spatial relationships (e.g., “watch”, “speak to”, “wear”, “hold”, “have”, “look at”, “talk to”, “play with”, “walk with”, “stand on”, “sit on”, etc.) and arbitrary nouns.

The prompt for ChatGPT is: *Please generate natural prompts that contain subjects and objects by using relationship words such as wear, watch, speak, hold, have, run, look at, talk to, jump, play, walk with, stand on, and sit on.*

C. Numeracy.

In the construction of numeracy dataset, we introduce 1,000 prompts, and adopt a tripartite structure based on the quantities of objects. The initial 30% part of the numeracy contains scenarios involving a singular object, followed by the subsequent 30% comprising two objects, and the remaining 40% encompassing multiple objects, with the quantities ranging from one to eight. We construct a list of objects, and randomly combine objects and quantities in a standardized format: “number object” (e.g., “one pear and three knives”). In order to incorporate diversity of expressions, each part is characterized by a combination of fixed templates and flexible prompts at a ratio of 4:1. The Fixed templates adhere to the predefined format, while flexible prompts incorporate natural language expressions encountered in our daily lives with multiple ‘number object’ included (e.g., “three plates and three pens were on the table”).

We construct a list of objects by asking ChatGPT: *Please generate a list of 150 types of objects and their plurals.* Following that, we manually filtered out repeated entries and uncommon objects. For generating flexible prompts, we ask ChatGPT, *Convert to natural prompt: You are given number sentences, each containing multiple objects. For each sentence, use the objects in it to create a natural compositional sentence.*

D. Complex Compositions

To test text-to-image generation approaches with more natural and challenging compositional prompts in the open world, we introduce 1,000 text prompts with complex compositions of concepts beyond the pre-defined patterns. Regarding the number of objects, we create text prompts with more than two objects, for example, “a room with a blue chair, a black table, and yellow curtains”. In terms of the attributes associated with objects, we can use multiple attributes to describe an object (denoted as *multiple attributes*, e.g., “a big, green apple and a tall, wooden table”), or leverage different types of attributes in a text prompt (denoted as *mixed attributes*, e.g., the prompt “a tall tree and a red car” includes both shape and color attributes). We generate 250 text prompts with ChatGPT for each of the four scenarios: *two objects with multiple attributes, two objects with mixed attributes, more than two objects with multiple attributes, and more than two objects with mixed attributes*. Relationship words can be adopted in each scenario to describe the relationships among two or more objects. For each scenario, we split 175 prompts for the training set and 75 prompts for the test set.

(1) For 2 objects with mixed attributes, the prompt for ChatGPT is: *Please generate natural compositional phrases, containing 2 objects with each object one adj. from {color, shape, texture} descriptions and spatial (left/right/top/bottom/next to/near/on side of) or non-spatial relationships.* (2) For 2 objects with multiple attributes, the prompt for ChatGPT is: *Please generate natural compositional phrases, containing 2 objects with several adj. from {color, shape, texture} descriptions and spatial (left/right/top/bottom/next to/near/on side of) or non-spatial relationships.* (3) For multiple objects with mixed attributes, the prompt for ChatGPT is: *Please generate natural compositional phrases, containing multiple objects (number>2) with each one adj. from {color, shape, texture} descriptions and spatial (left/right/top/bottom/next to/near/on side of) non-spatial relationships.* (4) For multiple objects with multiple attributes, the prompt for ChatGPT is: *Please generate natural compositional phrases, containing multiple objects (number>2) with several adj. from {color, shape, texture} descriptions and spatial (left/right/top/bottom/next to/near/on side of) or non-spatial relationships.*

IV. EVALUATION METRICS

Evaluating compositional text-to-image generation is challenging as it requires comprehensive and fine-grained cross-

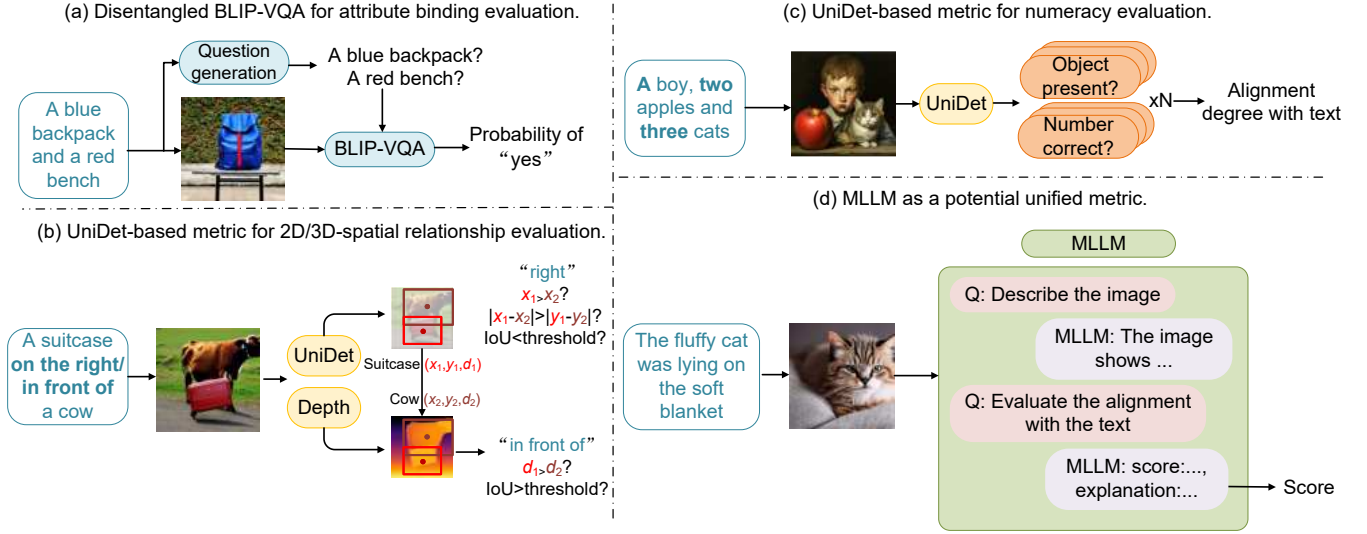


Fig. 3: Illustration of our proposed evaluation metrics: (a) Disentangled BLIP-VQA for attribute binding evaluation, (b) UniDet for 2D/3D-spatial relationship evaluation, (c) UniDet for numeracy evaluation, and (d) MLLM as a potential unified metric.

modal understanding. Existing evaluation metrics leverage vision-language models trained on large-scale data for evaluation. CLIPScore [11], [12] calculates the cosine similarity between text features and generated-image features extracted by CLIP. Text-text similarity by BLIP-CLIP [9] applies BLIP [13] to generate captions for the generated images, and then calculates the CLIP text-text cosine similarity between the generated captions and text prompts. Those evaluation metrics can measure the coarse text-image similarity, but fails to capture fine-grained text-image correspondences in attribute binding and spatial relationships. To address those limitations, we propose new evaluation metrics for compositional text-to-image generation, shown in Fig. 3. Concretely, we propose *disentangled BLIP-VQA* for attribute binding evaluation, *UniDet-based metric* for 2D/3D-spatial relationship and numeracy evaluation, and *MLLM-based metric* for non-spatial relationship and complex prompts. We further investigate the potential and limitations of multimodal large language models such as *MiniGPT-4* [77] with Chain-of-Thought [78], *ShareGPT4V* [15], and *GPT-4V* [16] for compositionality evaluation.

A. Disentangled BLIP-VQA for Attribute Binding Evaluation

We observe that the major limitation of the BLIP-CLIP evaluation is that the BLIP captioning models do not always describe the detailed attributes of each object. For example, the BLIP captioning model might describe an image as “A room with a table, a chair, and curtains”, while the text prompt for generating this image is “A room with yellow curtains and a blue chair”. So explicitly comparing the text-text similarity might cause ambiguity and confusion.

Therefore, we leverage the visual question answering (VQA) ability of BLIP [13] for evaluating attribute binding. For instance, given the image generated with the text prompt “a green bench and a red car”, we ask two questions separately:

“a green bench?”, and “a red car?”. By explicitly disentangling the complex text prompt into two independent questions where each question contains only one object-attribute pair, we avoid confusion of BLIP-VQA. The BLIP-VQA model takes the generated image and several questions as input and we take the probability of answering “yes” as the score for a question. We compute the overall score by multiplying the probability of answering “yes” for each question. The proposed disentangled BLIP-VQA is applied to evaluate the attribute binding for color, shape, and texture.

B. UniDet-based Metric for Spatial Relationships and Numeracy Evaluation

Many vision-language models exhibit limitations in reasoning spatial relationships, such as distinguishing between “left” and “right,” as well as in numerical counting. Consequently, we propose the use of a detection-based evaluation metric to assess performance in spatial relationships and numeracy.

2D-spatial relationships. We first use UniDet [79] to detect objects in the generated image. Then we determine the spatial relationship between two objects by comparing the locations of the centers of the two bounding boxes. Denote the center of the two objects as (x_1, y_1) and (x_2, y_2) , respectively. The first object is on the left of the second object if $x_1 < x_2$, $|x_1 - x_2| > |y_1 - y_2|$, and the intersection-over-union (IoU) between the two bounding boxes is below the threshold of 0.1. Other spatial relationships “right”, “top”, and “bottom” are evaluated similarly. We evaluate “next to”, “near”, and “on the side of” by comparing the distances between the centers of two objects with a threshold.

3D-spatial relationships. For 3D-spatial relationships, we leverage depth estimation [80], in conjunction with the detection-based metric. Let d_1 and d_2 represent the mean values of the depth maps corresponding to the two objects. Specifically, if $d_1 > d_2$ and the IoU metric exceeds the

predetermined threshold of 0.5, it is inferred that the first object is positioned in front of the second object. Other 3D-spatial relationships “behind”, “hidden by” are evaluated similarly.

Numeracy. For numeracy, we first extract the names of objects and their corresponding quantities from the prompts. Then, we leverage UniDet to detect objects in the images. This scoring mechanism is designed to consider both objects and their numerical correctness proportionally based on the number of object categories mentioned in the prompt. Let the variable n represent the count of distinct object categories referenced in the given prompt. For every identified object in the image, the assigned score is calculated as $1/(2n)$ points. Furthermore, another $1/(2n)$ points are allocated if the generated quantity aligns accurately with the specified category.

C. 3-in-1 Metric for Complex Compositions Evaluation

Since different evaluation metrics are designed for evaluating different types of compositionality, there is no single metric that works well for all categories. For non-MLLM metric, we empirically find that the Disentangled BLIP-VQA works best for attribute binding evaluation, UniDet-based metric works best for 2D/3D-spatial relationship and numeracy evaluation, and CLIPScore works best for non-spatial relationship evaluation. Thus, we design a 3-in-1 evaluation metric which computes the average score of CLIPScore, Disentangled BLIP-VQA, and UniDet, as the evaluation metric for complex compositions.

D. MLLM-based Evaluation Metric

Multimodal Large Language Models enable users to instruct LLMs to analyze user-provided image inputs. By integrating the visual modality, MLLMs enhance the capability of language-only systems, providing them with new interfaces to address a variety of tasks. However, the multimodal abilities of MLLMs regarding the compositional text-to-image generation remain unclear. Within this context, we analyze the abilities of three types of MLLMs, namely MiniGPT-4, ShareGPT4V [15] and GPT-4V [16], focusing on their performance in compositional problems.

Evaluation methods of MLLMs. By aligning a pretrained visual encoder with a frozen large language model, multimodal large language models such as *MiniGPT-4* [77] have demonstrated great abilities in vision-language cross-modal understanding. But the current MiniGPT-4 model exhibit limitations such as inaccurate understanding of images and hallucination issues. *ShareGPT4V* [15] is a large-scale image-text dataset featuring 1.2 million detailed captions characterized by richness and diversity. *GPT-4V*, a state-of-the-art MLLM, is developed on the foundation of the state-of-the-art LLM, GPT-4 [81], and trained extensively on a large-scale dataset containing multimodal information [81]. Employing MLLM as an evaluation metric, we submit generated images to the model and assess their alignment with the provided text prompt. This evaluation involves soliciting predictions for the image-text alignment score.

Prompt template for MLLM evaluation. We leverage MLLMs as an evaluation metric by feeding the generated images to the model and asking two questions with Chain-of-Thought [78]: “describe the image” and “predict the image-text alignment score”. We detail the prompts used for MLLM evaluation metric. For each sub-category, we ask two questions in sequence: “describe the image” and “predict the image-text alignment score”. Specifically, Table III shows the prompts for evaluating attribute binding (color, shape, texture). Similar to BLIP-VQA, for each noun phrase in a prompt, we request a number equivalent to the noun phrase and perform a multiplication. Table IV, Table VI, Table V, and Table VII demonstrate the prompt templates used for 2D/3D-spatial relationships, non-spatial relationships, numeracy, and complex compositions, respectively*. Due to the robust capabilities of GPT-4V and limited quota constraints, we have omitted “describe the image” for prompt of GPT-4V.

For MLLM without Chain-of-Thought in addressing specific compositional problems, we utilize predefined prompts that prompt MLLM to provide a score ranging from 0 to 100. For attribute binding, we focus on the presence of specific objects and their corresponding attributes. We utilize a prompt template such as “*Is there {object} in the image? Give a score from 0 to 100. If {object} is not present or if {object} is not {color/shape/texture description}, give a lower score.*” We leverage this question for each noun phrase in the text and compute the average score. For the spatial relationships, non-spatial relationships, and complex compositions, we employ a more general prompt template such as “*Rate the overall alignment between the image and the text prompt {prompt}. Give a score from 0 to 100.*”.

TABLE III: Prompts details for MLLM evaluation on attribute binding.

Describe	You are my assistant to identify any objects and their color (shape, texture) in the image. Briefly describe what it is in the image within 50 words.
Predict	According to the image and your previous answer, evaluate if there is {adj.+noun} in the image. Give a score from 0 to 100, according the criteria: 100: there is {noun}, and {noun} is {adj}. 75: there is {noun}, {noun} is mostly {adj}. 20: there is {noun}, but it is not {adj}. 10: no {noun} in the image. Provide your analysis and explanation in JSON format with the following keys: score (e.g., 85), explanation (within 20 words).

TABLE IV: Prompts details for MLLM evaluation on 2D/3D-spatial relationship.

Describe	You are my assistant to identify objects and their spatial layout in the image. Briefly describe the image within 50 words.
Predict	According to the image and your previous answer, evaluate if the text “{xxx}” is correctly portrayed in the image. Give a score from 0 to 100, according the criteria: 100: correct spatial layout in the image for all objects mentioned in the text. 80: basically, spatial layout of objects matches the text. 60: spatial layout not aligned properly with the text. 40: image not aligned properly with the text. 20: image almost irrelevant to the text. Provide your analysis and explanation in JSON format with the following keys: score (e.g., 85), explanation (within 20 words).

V. BOOSTING COMPOSITIONAL TEXT-TO-IMAGE GENERATION WITH GORS

We introduce a simple but effective approach, Generative mOdel finetuning with Reward-driven Sample selec-

* We empirically observed that, using a five-graded marking system instead of hundred-mark system enhances the performances of ShareGPT4V.

TABLE V: Prompts details for MLLM evaluation on non-spatial relationship.

Describe	You are my assistant to identify the actions, events, objects and their relationships in the image. Briefly describe the image within 50 words.
Predict	According to the image and your previous answer, evaluate if the text "{xxx}" is correctly portrayed in the image. Give a score from 0 to 100, according the criteria: 100: the image accurately portrayed the actions, events and relationships between objects described in the text. 80: the image portrayed most of the actions, events and relationships but with minor discrepancies. 60: the image depicted some elements, but action relationships between objects are not correct. 40: the image failed to convey the full scope of the text. 20: the image did not depict any actions or events that match the text. Provide your analysis and explanation in JSON format with the following keys: score (e.g., 85), explanation (within 20 words).

TABLE VI: Prompts details for MLLM evaluation on numeracy.

Describe	You are my assistant to identify objects and their quantities in the image. Briefly describe the image within 50 words.
Predict	According to the image and your previous answer, evaluate how well the image aligns with the text prompt: "{xxx}". Give a score from 0 to 100, according the criteria: 100: correct numerical content in the image for all objects mentioned in the text. 80: basically, numerical content of objects matches the text. 60: numerical content not aligned properly with the text. 40: image not aligned properly with the text. 20: image almost irrelevant to the text. Provide your analysis and explanation in JSON format with the following keys: score (e.g., 85), explanation (within 20 words).

tion (GORS), to improve the compositional ability of pre-trained text-to-image models. Our approach finetunes a pre-trained text-to-image model such as Stable Diffusion [1] with generated images that highly align with the compositional prompts, where the fine-tuning loss is weighted by the reward which is defined as the alignment score between compositional prompts and generated images.

Specifically, given the text-to-image model p_θ and a set of text prompts $y_1, y_2, \dots, y_n$, we first generate k images for each text prompt, resulting in kn generated images $x_1, x_2, \dots, x_{kn}$. Text-image alignment scores $s_1, s_2, \dots, s_{kn}$ are predicted as rewards. We select the generated images whose rewards are higher than a threshold to fine-tune the text-to-image model. The selected set of samples are denoted as $\mathcal{D}_s$. During fine-tuning, we weight the loss with the reward of each sample. Generated images that align with the compositional prompt better are assigned higher loss weights, and vice versa. The loss function for fine-tuning is

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,y,s) \in \mathcal{D}_s} \left[s \cdot \| \epsilon - \epsilon_\theta(z_t, t, y) \|_2^2 \right], \quad (1)$$

where (x, y, s) is the triplet of the image, text prompt, and reward, and z_t represents the latent features of x at timestep t . We adopt LoRA [82] for efficient finetuning.

VI. EXPERIMENTS

A. Experimental Setup

Evaluated models. We evaluate the performance of 11 text-to-image models on T2I-CompBench++: *Stable Diffusion v1-4*, *Stable Diffusion v2* [1], *Composable Diffusion* [7], *Structured Diffusion* [8], *Attend-and-Excite* [9], and *GORS*. We further conduct evaluations on 5 state-of-the-art text-to-image models: *PixArt- α* [20], *Stable Diffusion XL* [21], *DALL-E 3* [19], *Stable Diffusion 3* [18] and *FLUX.1* [17]. *GORS* is our proposed approach which finetunes Stable Diffusion v2 with selected samples and their rewards.

Stable Diffusion v1-4 and *Stable Diffusion v2* [1] are text-to-image models trained on large amount of image-text pairs. *Composable Diffusion* [7] is designed for conjunction and

TABLE VII: Prompts details for MLLM evaluation on complex compositions.

Describe	You are my assistant to evaluate the correspondence of the image to a given text prompt. Briefly describe the image within 50 words, focus on the objects in the image and their attributes (such as color, shape, texture), spatial layout and action relationships.
Predict	According to the image and your previous answer, evaluate how well the image aligns with the text prompt: "{xxx}". Give a score from 0 to 100, according the criteria: 100: the image perfectly matches the content of the text prompt, with no discrepancies. 80: the image portrayed most of the actions, events and relationships but with minor discrepancies. 60: the image depicted some elements in the text prompt, but ignored some key parts or details. 40: the image did not depict any actions or events that match the text. 20: the image failed to convey the full scope in the text prompt. Provide your analysis and explanation in JSON format with the following keys: score (e.g., 85), explanation (within 20 words).

negation of concepts for pretrained diffusion models. *Structured Diffusion* [8] and *Attend-and-Excite* [9] are designed for attribute binding for pretrained diffusion models. We re-implement those approaches on Stable Diffusion v2 to enable fair comparisons. *PixArt- α* [20] is a Transformer-based T2I diffusion model, boasting competitive image generation quality comparable to state-of-the-art image generators with low training costs. We test *PixArt- α* fine-tuned on GORS, denoted as *PixArt- α -ft*. Stable Diffusion XL 1.0 [21] building upon previous iterations of Stable Diffusion models, represents a powerful text-to-image generation framework. DALL-E 3 [19] is a new text-to-image generation system, enhancing the alignment between generated images and provided text description. Stable Diffusion 3 [18] combines a diffusion transformer architecture and flow matching. FLUX.1 [schnell] [17] is a 12 billion parameter rectified flow transformer.

To avoid the bias from selecting samples by evaluation metrics as reward [43], [83], we introduce new reward models which are different from our proposed evaluation metrics, denoted as *GORS-unbiased*. Specifically, we adopt Grounded-SAM [84] as the reward model for the attribute binding category. We extract the segmentation masks of attributes and their associated nouns separately with Grounded-SAM, and use the Intersection-over-Union (IoU) between the attribute masks and the noun masks together with the grounding mask confidence to represent the attribute binding performance. We apply GLIP-based [85] selection method for 2D/3D-spatial relationships and numeracy. For non-spatial relationships, we adopt BLIP [13] to generate image captions and CLIP [11], [12] to measure the text-text similarity between the generated captions and the input text prompts. For complex compositions, we integrate the 3 aforementioned reward models as the total reward. Those sample selection models are different from the models used as evaluation metrics.

Implementation details. We employ pre-trained models for our proposed evaluation metrics. For BLIP-VQA, we utilize the BLIP w/ ViT-B and CapFilt-L [13] pretrained on image-text pairs and fine-tuned on VQA. We employ the UniDet [79] model trained on 4 large-scale detection datasets (COCO [59], Objects365 [86], OpenImages [87], and Mapillary [88]). For CLIPScore, we use the “ViT-B/32” pretrained CLIP model [11], [12]. For MiniGPT4-CoT, we utilize the Vicuna 13B of MiniGPT4 [77] variant with a temperature setting of 0.7 and a beam size of 1. For ShareGPT4V [15] and GPT-4V [16], we use the default parameters, with temperature setting of 0.2 and 1, respectively.

We implement our proposed GORS upon the codebase of diffusers [89] (Apache License), and finetune the self-attention

TABLE VIII: Comparisons of evaluation metrics on attribute binding (color and shape), with scores unnormalized. **Bold** stands for the best score across 7 models in T2I-Compbench [22]. Blue represents the proposed metric for the category, and green stands for the human evaluation, applicable to the following Table IX, X and XI.

Model	Color								Shape							
	CLIP	B-CLIP	B-VQA-n	B-VQA	mGPT-CoT	GPT-4V	Share-CoT	Human	CLIP	B-CLIP	B-VQA-n	B-VQA	mGPT-CoT	GPT-4V	Share-CoT	Human
Stable v1-4 [1]	0.3214	0.7454	0.5875	0.3765	0.7424	0.6324	0.6185	0.6533	0.3112	0.7077	0.6771	0.3576	0.7197	0.4466	0.5470	0.6160
Stable v2 [1]	0.3335	0.7616	0.7249	0.5065	0.7764	0.6717	0.7056	0.7747	0.3203	0.7191	0.7517	0.4221	0.7279	0.6212	0.5815	0.6587
Composable + SD v2 [7]	0.3178	0.7352	0.5744	0.4063	0.7524	0.6295	0.6597	0.6187	0.3092	0.6985	0.6125	0.3299	0.7124	0.5655	0.5339	0.5133
Structured + SD v2 [8]	0.3319	0.7626	0.7184	0.4990	0.7822	0.6671	0.7253	0.7867	0.3178	0.7177	0.7500	0.4218	0.7228	0.6348	0.5886	0.6413
Attn-Exct + SD v2 [9]	0.3374	0.7810	0.8362	0.6400	0.8194	0.7539	0.8110	0.8240	0.3189	0.7209	0.7723	0.4517	0.7299	0.5990	0.6066	0.6360
GORS + SD v2 (ours)	0.3395	0.7681	0.8471	0.6603	0.8067	0.7296	0.7994	0.8320	0.2973	0.7201	0.7937	0.4785	0.7303	0.6563	0.6329	0.7040

layers of the CLIP text encoder and the attention layers of U-net using LoRA [82]. The model is trained by AdamW optimizer [90] with $\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=1e-8$, and weight decay of 0.01. The batch size is 5. The model is trained on 8 32GB NVIDIA v100 GPUs.

B. Evaluation Metrics

We generate 10 images for each text prompt in T2I-CompBench for automatic evaluation. To ensure a fair comparison, the images are generated using the fixed seed across all models[†].

Previous metrics. *CLIPScore* [11], [12] (denoted as *CLIP*) calculates the cosine similarity between text features and generated-image features extracted by CLIP. *BLIP-CLIP* [9] (denoted as *B-CLIP*) applies BLIP [13] to generate captions for the generated images, and then calculates the CLIP text-text cosine similarity between the generated captions and text prompts. *BLIP-VQA-naive* (denoted as *B-VQA-n*) applies BLIP VQA to ask a single question (e.g., a green bench and a red car?) with the whole prompt.

Our proposed metrics. *Disentangled BLIP-VQA* (denoted as *B-VQA*) is our proposed evaluation metric for attribute binding. *UniDet* is our proposed metric for 2D/3D-spatial relationships and numeracy evaluation metric.

For evaluating MLLM to serve as a metric, we assess three models, and propose MLLM-based metric for non-spatial relationship and complex compositions: *MiniGPT4* (denoted as *mGPT*), *ShareGPT4V* (denoted as *Share*), and *GPT-4V*. Due to the restricted quota for GPT-4V, we conducted evaluations on one-fifth of the images of T2I-CompBench++ (i.e., 600 images for each category). To boost capability of MLLM, we apply Chain-of-Thought [78] (denoted as *-CoT*) for mGPT and ShareGPT4V.

Human evaluation. For human evaluation of each sub-category, we randomly select 25 prompts and generate 2 images per prompt, resulting in 300 images generated with 200 prompts per model in total. The testing set includes 300 prompts for each sub-category, resulting in 2,400 prompts in total. The prompt sampling rate for human evaluation is 8.33%. We utilize Amazon Mechanical Turk and ask three workers to score each generated-image-text pair independently based on the image-text alignment. The worker can choose a score from $\{1, 2, 3, 4, 5\}$, as shown in Figure 4-6. We normalize the scores by dividing them by 5. We then compute the average score across all images and all workers.

C. Comparison across Different Evaluation Metrics

Comparisons of different evaluation metrics and human evaluation. We show the results of different evaluation metrics in Table VIII-XI. Previous evaluation metrics, CLIP and B-CLIP, predict similar scores across different models and cannot reflect the differences between models. Our proposed metrics (highlighted in blue column) show a similar trend with human evaluation (highlighted in green column).

Human correlation of the evaluation metrics. We calculate Kendall’s tau (τ) and Spearman’s rho (ρ) to evaluate the ranking correlation between automatic evaluation and human evaluation. The human correlation results are illustrated in Table XII. The results verify the effectiveness of our proposed evaluation metrics, BLIP-VQA for attribute binding, UniDet-based metric for spatial relationships and numeracy. For MLLM evaluation, MiniGPT4 does not perform well in terms of correlation with human perception. Share-CoT and GPT-4V excel in terms of non-spatial and complex categories, followed by CLIPScore and 3-in-1 evaluation metrics. In shape attribute, they exhibit performances that are comparable to non-MLLM metrics. In color, texture, 2D/3D-spatial and numeracy categories, their performances are slightly lower than non-MLLM metrics.

Conclusion on the evaluation metrics. Based on the human correlation of different evaluation metrics, we draw the conclusion for the optimal evaluation metrics for each category or sub-category, and those optimal metrics are used to benchmark different text-to-image generation methods in the following subsections. (1) For attribute binding, the best evaluation metric is BLIP-VQA. (2) For 2D spatial relationships, 3D spatial relationships, and numeracy, the UniDet-based metric performs best. (3) For non-spatial relationships, the best metric is GPT-4V, the second-best metric is Share-CoT (ShareGPT4V with Chain-of-Thought), and the best non-MLLM metric is CLIP. (4) For complex compositions, the best metric is GPT-4V, the second-best metric is Share-CoT (ShareGPT4V with Chain-of-Thought), and the best non-MLLM metric is 3-in-1.

D. Discussion about MLLMs as a unified metric

Comparisons among different MLLMs. We compare the human correlation among MiniGPT4, ShareGPT4V and GPT-4V. MiniGPT4 shows inadequate correlation with human

[†] Because the DALLE 3 API does not allow specifying random seed and due to the high cost of the DALLE 3 API, we use DALLE 3 to generate 3 images for each prompt without specified seed.



Fig. 4: AMT interface for the image-text alignment evaluation on attribute binding (color, shape, texture).

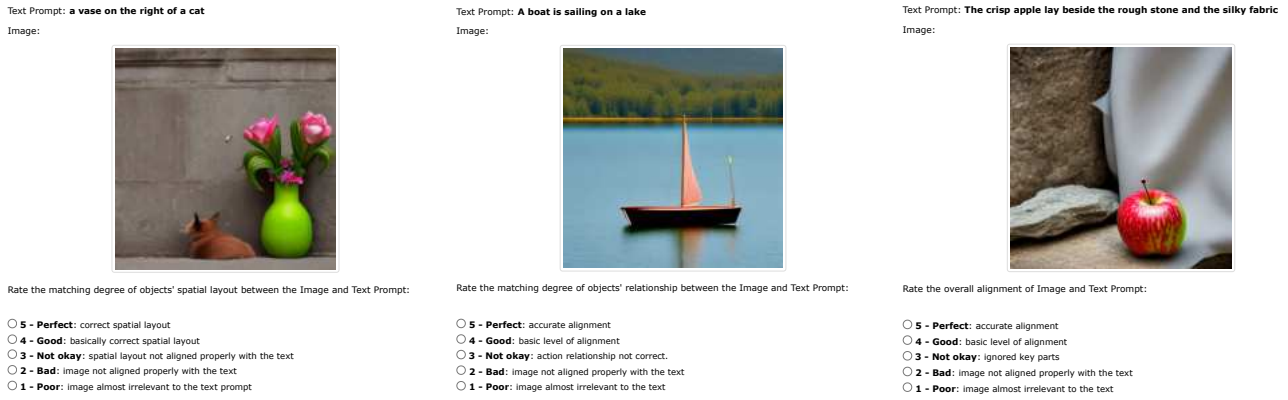


Fig. 5: AMT interface for the image-text alignment evaluation on 2D-spatial relationships, non-spatial relationships, and complex compositions.

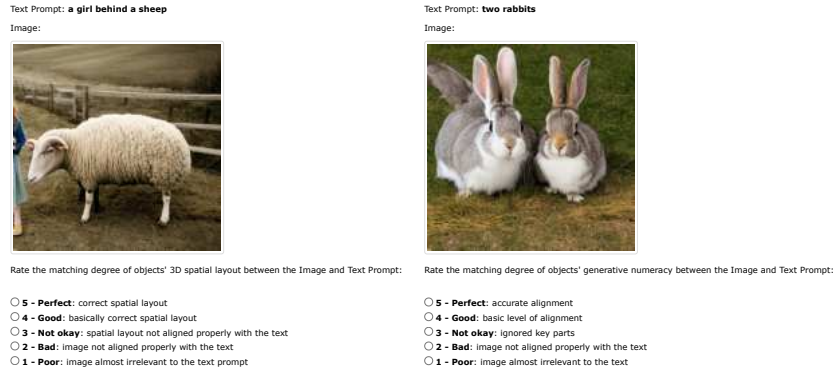


Fig. 6: AMT interface for the image-text alignment evaluation on 3D-spatial relationships, and generative numeracy.

perception[‡]. ShareGPT4V and GPT-4V excel in terms of non-spatial and complex categories. In color, texture, 2D/3D-spatial and numeracy categories, their performances are slightly lower than non-MLLM metrics. ShareGPT4V ranks second in non-spatial relationship and complex, ranging between first to third place in attribute categories, third in 2D-spatial, third (in terms of τ) and second (in terms of ρ) in 3D-spatial, second in

numeracy. GPT-4V ranks first in non-spatial relationship and complex, second in color attribute, third in shape and texture attributes, second in 2D-spatial, second (in terms of τ) and third (in terms of ρ) in 3D-spatial, third in numeracy.

Comparisons of the effectiveness of Chain-of-Thought for MLLMs. Utilizing Chain-of-thought [78] to stimulate MLLM's evaluation ability, we conduct the comparisons on MiniGPT4 and ShareGPT4V. With CoT, both MiniGPT4 and ShareGPT4V demonstrate improvements across most categories in terms of human correlation, highlighting the signifi-

[‡] To facilitate comparisons of different MLLMs' capabilities, we consider the same evaluation metric as a unified entity. For instance, mGPT and mGPT-CoT are treated as a single entity (the same for Share and Share-CoT), and we compare them based on the highest human correlation values attained.

TABLE IX: Comparisons of evaluation metrics on attribute binding (texture) and 2D-spatial relationship.

Model	Texture								2D-Spatial						
	CLIP	B-CLIP	B-VQA-n	B-VQA	mGPT-CoT	GPT-4V	Share-CoT	Human	CLIP	B-CLIP	UniDet	mGPT-CoT	GPT-4V	Share-CoT	Human
Stable v1-4 [1]	0.3081	0.7111	0.6173	0.4156	0.7836	0.6158	0.5728	0.7227	0.3142	0.7667	0.1246	0.8338	0.4297	0.7110	0.3813
Stable v2 [1]	0.3185	0.7240	0.7054	0.4922	0.7851	0.6594	0.6100	0.7827	0.3206	0.7723	0.1342	0.8367	0.4450	0.7197	0.3467
Composable + SD v2 [7]	0.3092	0.6995	0.5604	0.3645	0.7588	0.5870	0.5259	0.6333	0.3001	0.7409	0.0800	0.8222	0.2200	0.6633	0.3080
Structured + SD v2 [8]	0.3167	0.7234	0.7007	0.4900	0.7806	0.6674	0.6108	0.7760	0.3201	0.7726	0.1386	0.8361	0.4617	0.7183	0.3467
Attn-Exct + SD v2 [9]	0.3171	0.7206	0.7830	0.5963	0.8062	0.7077	0.6758	0.8400	0.3213	0.7742	0.1455	0.8407	0.5103	0.7330	0.4027
GORS + SD v2 (ours)	0.3233	0.7315	0.7991	0.6287	0.8106	0.7266	0.7054	0.8573	0.3242	0.7854	0.1815	0.8362	0.4640	0.7373	0.4560

TABLE X: Comparisons of evaluation metrics on 3D-spatial relationship and numeracy task.

Model	3D-Spatial							Numeracy						
	CLIP	B-CLIP	UniDet	mGPT-CoT	GPT-4V	Share-CoT	Human	CLIP	B-CLIP	UniDet	mGPT-CoT	GPT-4V	Share-CoT	Human
Stable v1-4 [1]	0.3001	0.7344	0.3030	0.8249	0.2880	0.6977	0.4987	0.3021	0.6598	0.4456	0.8427	0.3643	0.7423	0.5200
Stable v2 [1]	0.3100	0.7486	0.3230	0.8338	0.3600	0.7160	0.5040	0.3138	0.6588	0.4582	0.8364	0.4320	0.7653	0.5253
Composable + SD v2 [7]	0.2955	0.7300	0.2847	0.8253	0.2413	0.6853	0.4933	0.3028	0.6531	0.4272	0.8391	0.3842	0.7540	0.5187
Structured + SD v2 [8]	0.3093	0.7482	0.3224	0.8310	0.3653	0.7153	0.5027	0.3125	0.6588	0.4557	0.8440	0.4258	0.7637	0.5680
Attn-Exct + SD v2 [9]	0.3096	0.7517	0.3222	0.8365	0.3633	0.7213	0.5400	0.3130	0.6607	0.4773	0.8439	0.4660	0.7857	0.5613
GORS + SD v2 (ours)	0.3140	0.7615	0.3572	0.8292	0.3753	0.7327	0.5547	0.3152	0.6715	0.4830	0.8376	0.4740	0.7847	0.5747

TABLE XI: Comparisons of evaluation metrics on the non-spatial relationship and complex compositions.

Model	Non-spatial						Complex						
	CLIP	B-CLIP	mGPT-CoT	GPT-4V	Share-CoT	Human	CLIP	B-CLIP	3-in-1	mGPT-CoT	GPT-4V	Share-CoT	Human
Stable v1-4 [1]	0.3079	0.7565	0.8170	0.7717	0.7487	0.9653	0.2876	0.6816	0.3080	0.8075	0.6453	0.7727	0.8067
Stable v2 [1]	0.3127	0.7609	0.8235	0.8153	0.7567	0.9827	0.3096	0.6893	0.3386	0.8094	0.6483	0.7783	0.8480
Composable + SD v2 [7]	0.2980	0.7038	0.7936	0.5030	0.6927	0.8120	0.3014	0.6638	0.2898	0.8083	0.5637	0.7487	0.7520
Structured + SD v2 [8]	0.3111	0.7614	0.8221	0.8127	0.7560	0.9773	0.3084	0.6902	0.3355	0.8076	0.6400	0.7777	0.8333
Attn-Exct + SD v2 [9]	0.3109	0.7607	0.8214	0.8243	0.7593	0.9533	0.2913	0.6875	0.3401	0.8078	0.6817	0.7763	0.8573
GORS + SD v2 (ours)	0.3193	0.7619	0.8172	0.8420	0.7637	0.9853	0.2973	0.6841	0.3328	0.8095	0.6850	0.7737	0.8680

TABLE XII: The correlation between automatic evaluation metrics and human evaluation. Our proposed metrics demonstrate a significant improvement over existing metrics in terms of Kendall’s τ and Spearman’s ρ . **Bold** stands for the highest correlation score across all non-MLLM metrics. **Red** indicates the highest correlation score with MLLM metrics included.

Metric	Attribute-color		Attribute-shape		Attribute-texture		2D Spatial rel		3D Spatial rel		Numeracy		Non-spatial rel		Complex	
	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$	$\tau(\uparrow)$	$\rho(\uparrow)$
CLIP	0.1938	0.2773	0.0555	0.0821	0.2890	0.4008	0.2741	0.3548	0.2151	0.3009	0.1557	0.2187	0.2470	0.3161	0.0650	0.0847
B-CLIP	0.2674	0.3788	0.1692	0.2413	0.2999	0.4187	0.1983	0.2544	0.2042	0.2840	0.0859	0.1257	0.2342	0.2964	0.1963	0.2755
B-VQA-n	0.4602	0.6179	0.2280	0.3180	0.4227	0.5830	-	-	-	-	-	-	-	-	-	-
B-VQA	0.6297	0.7958	0.2707	0.3795	0.5177	0.6995	-	-	-	-	-	-	-	-	-	-
UniDet	-	-	-	-	-	-	0.4756	0.5136	0.3126	0.4262	0.4251	0.5273	-	-	-	-
3-in-1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.2831	0.3853
mGPT	0.1197	0.1616	0.1282	0.1775	0.1061	0.1460	0.0208	0.0229	0.0980	0.1373	0.0647	0.0967	0.1181	0.1418	0.0066	0.0084
mGPT-CoT	0.3156	0.4151	0.1300	0.1805	0.3453	0.4664	0.1096	0.1239	0.0618	0.0760	0.1414	0.1769	0.1944	0.2137	0.1251	0.1463
Share	0.4429	0.5260	0.1410	0.1782	0.2858	0.3518	0.1716	0.1837	0.1394	0.1610	0.2908	0.3396	0.1541	0.1801	0.0629	0.1780
Share-CoT	0.4783	0.5847	0.2871	0.3659	0.4005	0.5094	0.2889	0.3182	0.2707	0.3186	0.4175	0.4891	0.3401	0.3622	0.3204	0.3649
GPT-4V	0.5242	0.6465	0.2668	0.3402	0.3944	0.4987	0.3456	0.4038	0.2560	0.3202	0.3777	0.4651	0.4756	0.5337	0.5070	0.5942

cance of effective prompting.

We benchmark mGPT without Chain-of-Thought in Table XIV, which shows the additional results of benchmarking on T2I-CompBench++ of 6 models with MiniGPT-4 without Chain-of-Thought. Results indicate that MiniGPT-4 evaluation without Chain-of-Thought does not strictly align with human evaluation results.

Discussion about the stability of MLLMs’ results. We test the stability of the MLLM metric by conducting multiple executions on a consistent set of images. Specifically, we employ ShareGPT4V and GPT-4V to analyze 50 images from Stable v2 color category, repeating the process five times. Default parameters were used, with a temperature of 1.0 for GPT-4V and 0.2 for ShareGPT4V. As shown in Table XVIII, our analysis consistently produces results, with variations remaining within 0.032 across the five executions for GPT-4V

and 0.0273 for ShareGPT4V. The average standard deviations are 7.3235 for GPT-4V and 6.8741 for ShareGPT4V.

Limitations of MLLMs as a metric. Share-CoT and GPT-4V have limits despite their strong performance as evaluation metrics. They do not adhere to the prompts of grading guidelines very well. Our empirical observations show that Share-CoT tends to give less diverse ratings, regardless of the prompt provided. GPT-4V can comprehend the evaluation prompts, but it is less able to convert into exact grades.

E. Benchmarking on Different Methods

The quantitative benchmarking results are reported in Table VIII-XI and XIII. Qualitative results are shown in appendix Figure 7, Figure 8 and Figure 9.

Comparisons across text-to-image models. (1) Stable Diffusion v2 consistently outperforms Stable Diffusion v1-4 in

TABLE XIII: Benchmarking on all categories with proposed metrics. **Bold** stands for the best score across 7 models in T2I-Compbench [22]. **Red** indicates the best score across 10 models in T2I-CompBench++.

Model	Color	Shape	Texture	2D-Spatial	3D-Spatial	Numeracy	Non-Spatial			Complex		
	B-VQA	B-VQA	B-VQA	UniDet	UniDet	UniDet	CLIP	GPT-4V	Share-CoT	3-in-1	GPT-4V	Share-CoT
Stable v1-4 [1]	0.3765	0.3576	0.4156	0.1246	0.3030	0.4456	0.3079	0.7717	0.7487	0.3080	0.6453	0.7727
Stable v2 [1]	0.5065	0.4221	0.4922	0.1342	0.3230	0.4582	0.3127	0.8153	0.7567	0.3386	0.6483	0.7783
Composable + SD v2 [7]	0.4063	0.3299	0.3645	0.0800	0.2847	0.4272	0.2980	0.5030	0.6927	0.2898	0.5637	0.7487
Structured + SD v2 [8]	0.4990	0.4218	0.4900	0.1386	0.3224	0.4557	0.3111	0.8127	0.7560	0.3355	0.6400	0.7777
Attn-Exct + SD v2 [9]	0.6400	0.4517	0.5963	0.1455	0.3222	0.4773	0.3109	0.8243	0.7593	0.3401	0.6817	0.7763
GORS-unbiased + SD v2 (ours)	0.6414	0.4546	0.6025	0.1725	0.3300	0.4849	0.3158	0.8557	0.7650	0.3470	0.6753	0.7697
GORS + SD v2 (ours)	0.6603	0.4785	0.6287	0.1815	0.3572	0.4830	0.3193	0.8420	0.7637	0.3328	0.6850	0.7737
Stable XL [21]	0.5879	0.4687	0.5299	0.2133	0.3566	0.4991	0.3119	0.8500	0.7673	0.3237	0.7170	0.7817
Pixart- α -ft [20]	0.6690	0.4927	0.6477	0.2064	0.3901	0.5032	0.3197	0.8620	0.7747	0.3433	0.7223	0.7823
DALLE-3 [19]	0.7785	0.6205	0.7036	0.2865	0.3744	0.5926	0.3003	0.9170	0.7853	0.3773	0.8653	0.7927
Stable v3 [18]	0.8132	0.5885	0.7334	0.3200	0.4084	0.6174	0.3140	0.9093	0.7782	0.3771	0.8717	0.7919
FLUX.1 [17]	0.7407	0.5718	0.6922	0.2863	0.3866	0.6185	0.3127	0.9213	0.7809	0.3703	0.8727	0.7927

TABLE XIV: mGPT benchmarking on 6 sub-categories in T2I-CompBench++.

Model	Color		Shape		Texture		Spatial		Non-spatial		Complex	
	mGPT	Human	mGPT	Human	mGPT	Human	mGPT	Human	mGPT	Human	mGPT	Human
Stable v1-4 [1]	0.6238	0.6533	0.6130	0.6160	0.6247	0.7227	0.8524	0.3813	0.8507	0.9653	0.8752	0.8067
Stable v2 [1]	0.6476	0.7747	0.6154	0.6587	0.6339	0.7827	0.8572	0.3467	0.8644	0.9827	0.8775	0.8480
Composable v2 [7]	0.6412	0.6187	0.6153	0.5133	0.6030	0.6333	0.8504	0.3080	0.8806	0.8120	0.8858	0.7520
Structured v2 [8]	0.6511	0.7867	0.6198	0.6413	0.6439	0.7760	0.8591	0.3467	0.8607	0.9773	0.8732	0.8333
Attn-Exct v2 [9]	0.6683	0.8240	0.6175	0.6360	0.6482	0.8400	0.8536	0.4027	0.8684	0.9533	0.8725	0.8573
GORS-unbiased (ours)	0.6668	0.8253	0.6399	0.6573	0.6389	0.8413	0.8675	0.4467	0.8845	0.9534	0.8876	0.8654
GORS (ours)	0.6677	0.8320	0.6356	0.7040	0.6709	0.8573	0.8584	0.4560	0.8863	0.9853	0.8892	0.8680

all types of compositional prompts and evaluation metrics. (2) Although Structured Diffusion built upon Stable Diffusion v1-4 shows great performance improvement in attribute binding as reported in Feng *et al.* [8], Structured Diffusion built upon Stable Diffusion v2 only brings slight performance gain upon Stable Diffusion v2. It indicates that boosting the performance upon a better baseline of Stable Diffusion v2 is more challenging. (3) Composable Diffusion built upon Stable Diffusion v2 does not work well. A similar phenomenon was also observed in previous work [9] that Composable Diffusion often generates images containing a mixture of the subjects. In addition, Composable Diffusion was designed for concept conjunctions and negations so it is reasonable that it does not perform well in other compositional scenarios. (4) Attend-and-Excite built upon Stable Diffusion v2 improves the performance in attribute binding. (5) Previous methods Composable Diffusion [7], Structure Diffusion [8] and Attend-and-Excite [9] are designed for concept conjunction or attribute binding, so they do not result in significant improvements in object relationships. (6) Our proposed approach, GORS, outperforms previous approaches across all types of compositional prompts, as demonstrated by the automatic evaluation, human evaluation, and qualitative results. The evaluation results of GORS-unbiased and GORS significantly exceed the baseline Stable v2. Besides, the performances of GORS-unbiased indicate that our proposed approach is insensitive to the reward model used for selecting samples, and that the proposed approach works well as long as high-quality samples are selected. (7) Recent text-to-image models, Stable Diffusion XL [21], Pixart- α -ft [20], DALLE 3 [19], Stable Diffusion 3 [18] and FLUX.1 [schnell] [17] improves across all categories compared to previous methods. Notably, DALLE 3 and SD3 achieves the

state-of-the-art (SOTA) performance by a significant margin in almost all categories.

Comparisons across compositionality categories. According to the human evaluation results, spatial relationship is the most challenging sub-category for text-to-image models, and attribute binding (shape) is also challenging. Non-spatial relationship is the easiest sub-category.

Comparisons between seen and unseen splits. We provide the seen and unseen splits for the test set in attribute binding, where the unseen set consists of attribute-object pairs that do not appear in the training set. The unseen split tends to include more uncommon attribute-object combinations than seen split. The performance comparison of seen and unseen splits for attribute binding is shown in Table XV. Our observations reveal that our model exhibits slightly lower performance on the unseen set than the seen set.

TABLE XV: Performances of our model on attribute binding (color, shape, and texture) for seen and unseen sets.

Metric	Color		Shape		Texture	
	Seen	unseen	Seen	unseen	Seen	unseen
CLIP	0.3422	0.3283	0.2926	0.3068	0.3240	0.3219
B-CLIP	0.7716	0.7612	0.7425	0.6752	0.7569	0.6809
B-VQA	0.7192	0.5426	0.5500	0.3356	0.7647	0.3567
mGPT	0.6626	0.6780	0.6381	0.6307	0.6773	0.6580
mGPT-CoT	0.8082	0.8038	0.7510	0.6888	0.8453	0.7412
Share	0.8337	0.7657	0.6014	0.6219	0.6875	0.5084
Share-CoT	0.7938	0.7634	0.6692	0.6222	0.7641	0.6400
GPT4V	0.7491	0.6908	0.6824	0.6042	0.7916	0.5968

Comparisons of the scalability of our proposed approach. To demonstrate the scalability of our proposed approach, we introduce additional 700 prompts of complex

compositions to form an extended training set of 1,400 complex prompts. The new prompts are generated with the same methodology. We conduct 6 experiments to train the models with different training set sizes, i.e., 25 prompts, 275 prompts, 350 prompts, 700 prompts, 1050 prompts, and 1400 prompts. The results in Table XVI show the performance of our model grows with the increase of the training set sizes. The results indicate the potential to achieve better performance by scaling up the training set.

TABLE XVI: Performances of our model on complex compositions on the 3-in-1 metric.

ours (25)	ours (275)	ours (350)	ours (700)	ours (1050)	ours (1400)
0.2596	0.3086	0.3299	0.3328	0.3371	0.3504

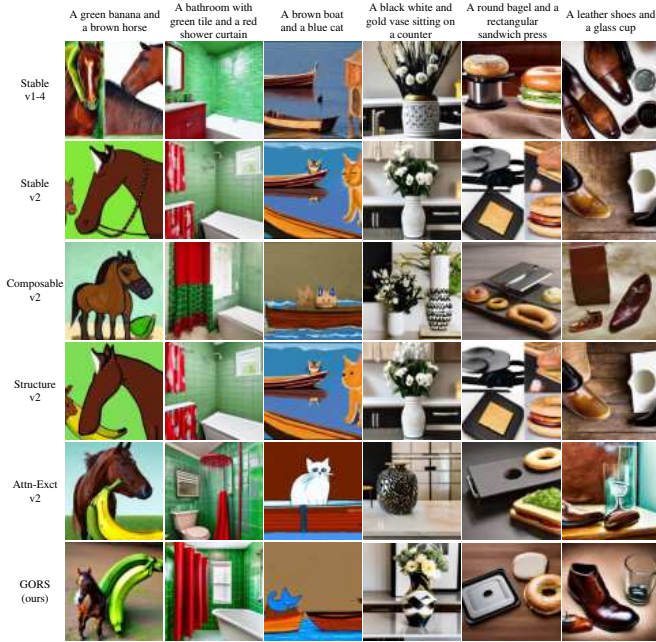


Fig. 7: Qualitative comparison between our approach and previous methods.

Comparisons between original prompts and rephrased detailed prompts. Enhancing prompt details does not significantly improve the generative outcomes, as the challenges primarily lie in the visual content composition instead of the granularity of prompts. While more detailed prompts can improve image quality by providing additional descriptions, they do not necessarily enhance alignment between the generated images and the prompts. To illustrate this, we use GPT-4 to rephrase the prompts to more detailed versions and compare the evaluation results. Specifically, we examine SDXL with attribute binding categories (e.g., color) as an example. The results, shown in Table XVII, indicate that T2I models face similar compositional challenges with both original and detailed prompts.

F. Ablation Study

We conduct ablation studies on our proposed GORS approach and evaluation metric.

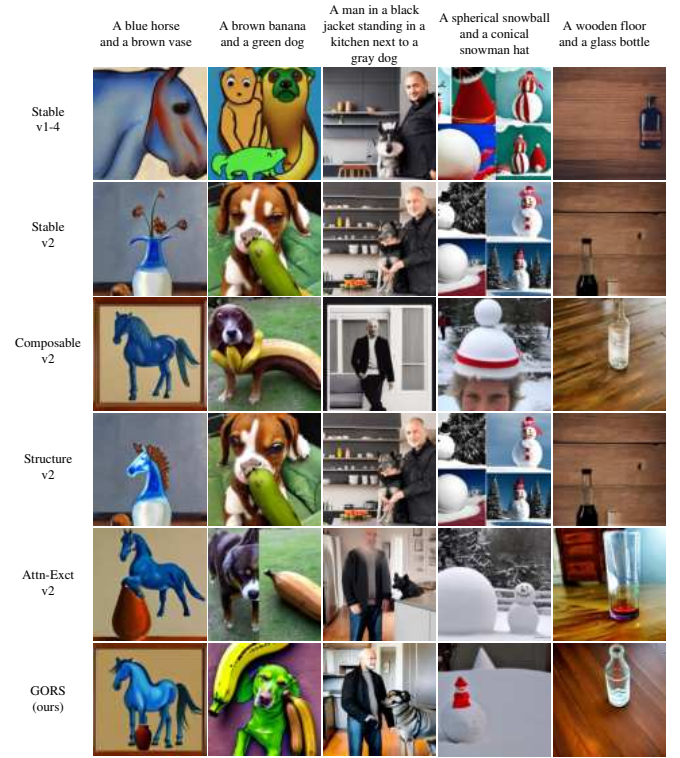


Fig. 8: Qualitative comparison between our approach and previous methods.



Fig. 9: Qualitative comparison between our approach and previous methods.

Finetuning strategy. Our approach finetunes both the

TABLE XVII: Comparison of BLIP-VQA scores between original and rephrased detailed prompts.

	Original prompts (avg length: 8.21 words)	Rephrased detailed prompts (avg length: 24.00 words)
B-VQA	0.5879	0.5781

TABLE XVIII: Stability tests on MLLM metrics.

Test round	1	2	3	4	5
Share-CoT	0.7232	0.7109	0.7267	0.6959	0.7129
GPT4V	0.6918	0.7078	0.6998	0.6758	0.6853

CLIP text encoder and the U-Net of Stable Diffusion with LoRA [82]. We conduct with the attribute binding (color) sub-category. We investigate the effects of finetuning CLIP only and U-Net only with LoRA. As shown in Table XIX, our model which finetunes both CLIP and U-Net performs better.

Threshold of selecting samples for finetuning. Our approach fine-tunes Stable Diffusion v2 with the selected samples that align well with the compositional prompts. We manually set a threshold for the alignment score to select samples with higher alignment scores than the threshold for fine-tuning. We experiment with setting the threshold to half of its original value, and setting the threshold to 0 (*i.e.*, use all generated images for finetuning with rewards, without selection). Results in Table XIX demonstrate that half threshold and zero threshold will lead to worse performance.

Threshold of UniDet-based metrics. The threshold for the UniDet-based metric is validated via user studies with human correlation, as shown in Table XX. Among various tested thresholds (0.0, 0.25, 0.50, 0.75), the chosen threshold of 0.5 exhibited the highest correlation with human evaluation.

Qualitative results. We show the qualitative results of the variants in ablation study in Figure 11. When only CLIP is fine-tuned with LoRA, the generated images do not bind attributes to correct objects (for example, Figure 11 Row. 3 Col. 3 and Row. 6 Col. 3). Noticeable improvements are observed in the generated images when U-Net is fine-tuned by LoRA, particularly when both CLIP and U-Net are finetuned together. Furthermore, we delve into the effect of the threshold for selecting images aligned with text prompts for fine-tuning. A higher threshold value enables the selection of images that are highly aligned with text prompts for finetuning, ensuring that only well-aligned examples are incorporated into the finetuning process. In contrast, a lower threshold leads to the inclusion of misaligned images during finetuning, which can degrade the compositional ability of the finetuned text-to-image models (for example, Figure 11 last two columns in Row. 2).

G. Limitation and Potential Negative Social Impacts

One limitation of our work is the absence of a unified metric for all forms of compositionality. Future research can explore the potential of multimodal LLM to develop a unified metric. Our proposed evaluation metrics are not perfect. As shown by

the failure cases in Fig. 10, BLIP-VQA may fail in challenging cases, for example, the objects' shapes are not fully visible in the image, shape's description is uncommon or the objects are not easy to recognize. The UniDet-based evaluation metric is limited to evaluating 2D spatial relationships and we leave 3D spatial relationships for future study. Researchers need to be aware of the potential negative social impact from the abuse of text-to-image models and the biases of hallucinations from image generators as well as pre-trained multimodal models and multimodal LLMs. Future research should exercise caution when working with generated images and LLM-generated content and devise appropriate prompts to mitigate the impact of hallucinations and bias in those models.

VII. CONCLUSION AND DISCUSSIONS

We propose T2I-CompBench++, a comprehensive benchmark for open-world compositional text-to-image generation, consisting of 8,000 prompts from 4 categories and 8 sub-categories. We propose new evaluation metrics and an improved baseline for the benchmark, and validate the effectiveness of the metrics and method by extensive evaluation. We further study the potential and limitations of MLLMs as a unified metrics. Besides, we propose a simple yet effective way GORS to boost the compositionality of text-to-image models. When studying generative models, researchers need to be aware of the potential negative social impact, for example, it might be abused to generate fake news. We also need to be aware of the bias from the image generators and the evaluation metrics based on pretrained multimodal models.

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TABLE XIX: Ablation studies on fine-tuning strategy and threshold.

Metric	FT U-Net only	FT CLIP only	Half threshold	0 threshold	GORS (ours)
B-VQA	0.6216	0.5507	0.6157	0.6130	0.6570

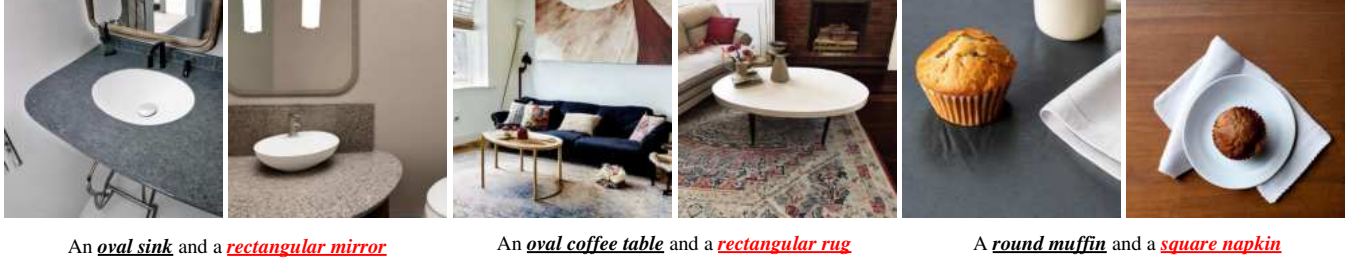


Fig. 10: Failure cases of the evaluation metric BLIP-VQA.

TABLE XX: Correlation between UniDet-based metric with different thresholds and human evaluation on the category of 2D-spatial relationships.

Threshold	0.0	0.25	0.50	0.75
τ ($\uparrow$)	0.1708	0.2060	0.4756	0.4634
ρ ($\uparrow$)	0.2181	0.2334	0.5136	0.5020

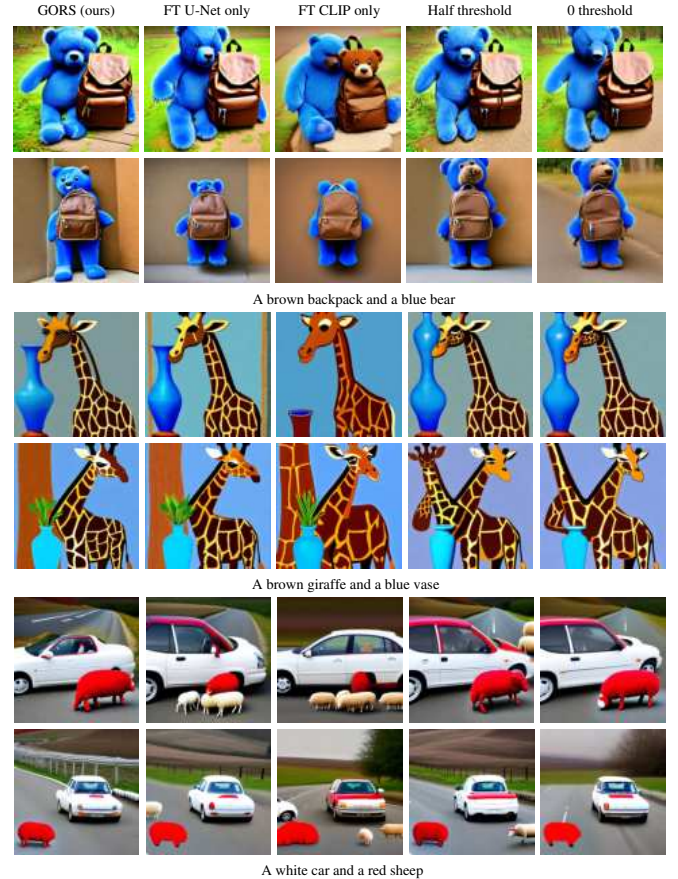


Fig. 11: Qualitative comparison of ablation study on fine-tuning strategy and threshold.

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APPENDIX

Qualitative comparisons between original prompts and rephrased detailed prompts. We observe that enhancing prompt details does not significantly improve the generative outcomes. We use GPT-4 to rephrase the prompts to more detailed versions and compare the evaluation results. The original prompts had an average length of 8.21 words, while the rephrased versions averaged 24.00 words. Qualitative comparisons, shown in Figure 12, demonstrate that T2I models continue to face compositional challenges, regardless of the level of detail in the prompts.

Performances of proposed method GORS on short and long text prompts. We compare the performances of the base model (Stable v2) and the GORS model (Stable v2+GORS) on long text prompts in Figure 13. The results show that GORS model is not overly attuned to short prompts and the ability to handle long text prompts is still preserved in our GORS model.

Qualitative results of threshold of UniDet-based metrics. We provide qualitative examples in 2D/3D-spatial relationships and generative numeracy, which use UniDet-based metrics for visualization in Figure 14. The results demonstrate that the chosen threshold effectively distinguishes between good and bad cases.

Qualitative results of SOTA T2I models. We provide more qualitative examples in non-relationships and multiple objects(≥ 3) in Figure 15. The results demonstrate that even the SOTA models struggle with complex compositional prompts.

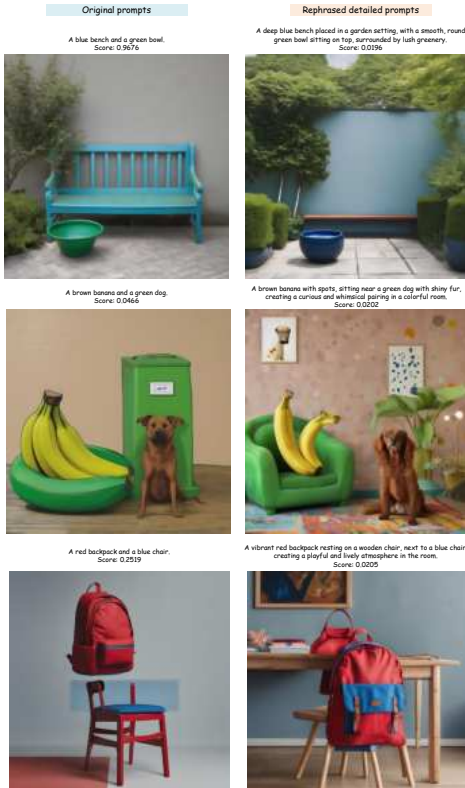


Fig. 12: Qualitative comparisons between original prompts and rephrased detailed prompts.

Stable v2

Stable v2+GORS

A bright white bathroom with a bold black-and-white shower curtain adds a striking contrast. The space features white walls and floor tiles, with the curtain serving as the focal point, complementing the sleek white sink and toilet.



The smooth, cool marble floors of the museum echoed with the footsteps of the visitors, as they admired the priceless works of art.



Fig. 13: Performances of proposed method GORS on short and long text prompts.

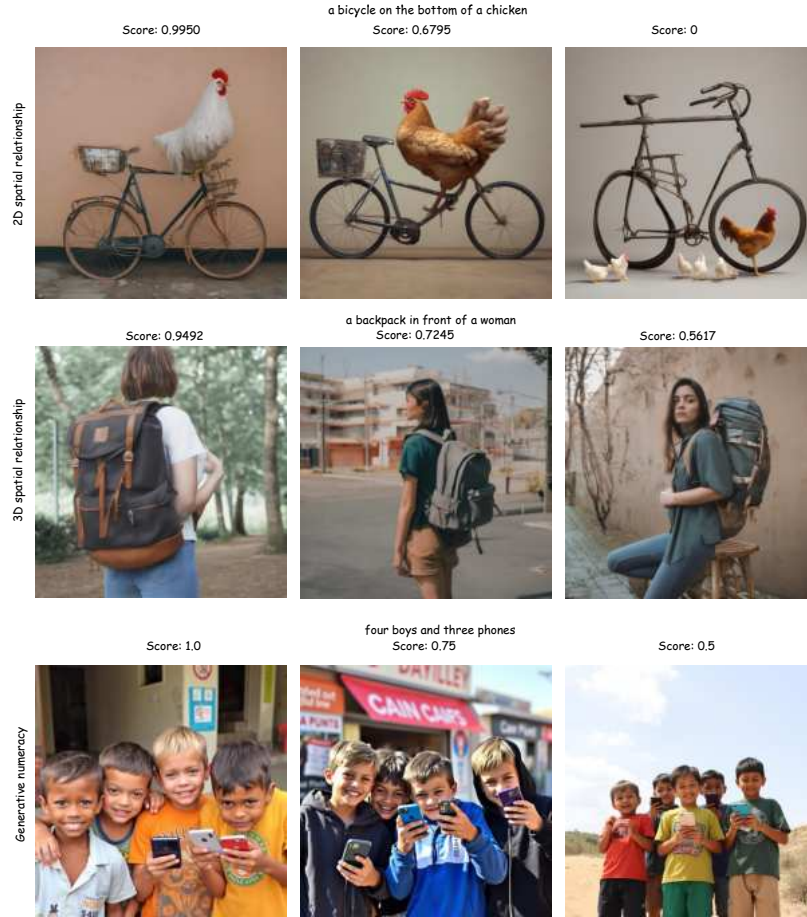
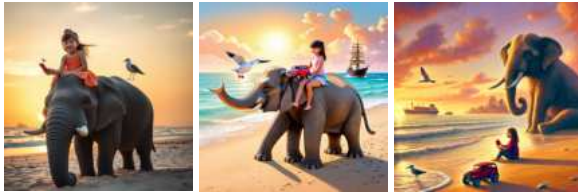


Fig. 14: Qualitative results of threshold of UniDet-based metrics.

A young girl sitting on a giant elephant on a sandy beach at sunset, holding a small red toy car in her hand, while a seagull perches on the elephant's tusk, and a distant ship sails on the horizon.



A bright orange tractor plows through a green meadow, while a red rooster and a blue goat are reading a book under a shady tree.



Flux.1

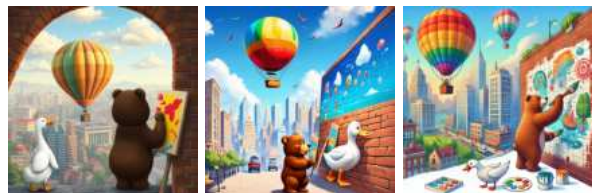
SD3

DALLE 3

A majestic silver train glides through snowy mountains, while a penguin and a polar bear are enjoying a picnic on the roof.



A colorful hot air balloon drifts over a bustling city, while a brown bear and a white duck are painting a mural on a brick wall.



Flux.1

SD3

DALLE 3

Fig. 15: Qualitative results of SOTA T2I models (Flux.1, SD3, and DALLE 3) on complex compositional prompts, involving non-spatial relationships and multiple objects (≥ 3).



KITTEN: A Knowledge-Intensive Evaluation of Image Generation on Visual Entities

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Abstract

Recent advances in text-to-image generation have improved the quality of synthesized images, but evaluations mainly focus on aesthetics or alignment with text prompts. Thus, it remains unclear whether these models can accurately represent a wide variety of realistic visual entities. To bridge this gap, we propose KITTEN, a benchmark for **K**nowledge-**I**n**T**ensive image genera**T**ion on real-world **E**ntities. Using KITTEN, we conduct a systematic study of the latest text-to-image models and retrieval-augmented models, focusing on their ability to generate real-world visual entities, such as landmarks and animals. Analysis using carefully designed human evaluations, automatic metrics, and MLLM evaluations show that even advanced text-to-image models fail to generate accurate visual details of entities. While retrieval-augmented models improve entity fidelity by incorporating reference images, they tend to over-rely on them and struggle to create novel configurations of the entity in creative text prompts.

1 Introduction

Recent advances in generative AI have revolutionized multimedia content creation. Large Language Models (LLMs) excel at knowledge-intensive tasks like question-answering and summarization. Cutting-edge image generation models, such as Imagen [30, 14, 10], DALL-E [27, 26], and Stable Diffusion [28], produce photorealistic and creative images from text. However, as these models become more capable and popular, assessing their reliability is crucial. Research on LLMs shows that even the most advanced models can generate inaccuracies, potentially undermining trust and causing societal harm [22, 5].

Despite increasing attention to factuality in LLMs, the accuracy of image generation models remains underexplored. Existing benchmarks mainly assess alignment with general text descriptions [20], compliance with image-editing instructions [15], or adherence to spatial relationships [6]. However, they fall short in evaluating how well models generate images that faithfully reproduce the precise visual details of real-world entities, objects, and scenes grounded in trustworthy knowledge sources (see examples in Fig. 1).

Recently, HEIM [17] introduces an evaluation suite for assessing various aspects of image generation, including their ability to generate entities such as historical figures or well-known subjects. However, real-world visual entities are far more diverse than those covered by HEIM, requiring a broader assessment. Moreover, HEIM primarily evaluates the alignment between generated images and entity names in text prompts. It fails to capture the fine-grained visual details essential for assessing the

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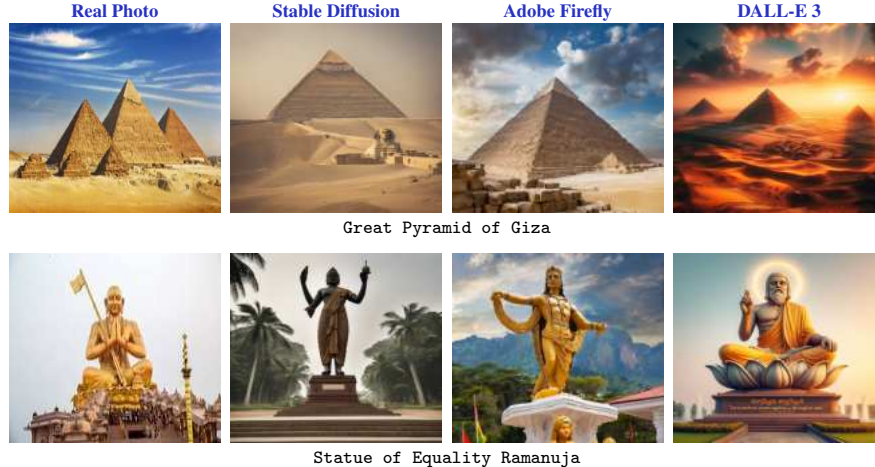


Figure 1: **Can text-to-image models generate precise visual details of real-world entities?** State-of-the-art models effectively render well-known entities (e.g., Great Pyramid of Giza) but often struggle with less-known entities, resulting in hallucinated depictions.

reproduction of visual-world knowledge since nuances of real-world entities cannot be conveyed through text alone. Thus, directly evaluating the fidelity in the generated images is essential.

To address the gap in evaluating image generation models’ ability to reproduce visual world knowledge, we introduce KITTEN, a benchmark dataset and evaluation suite designed to assess how well models generate visually accurate representations of real-world entities grounded in trustworthy knowledge sources. Unlike prior benchmarks that focus on aesthetics, text alignment, or common-sense reasoning, KITTEN uses prompts derived from visual entities documented in Wikipedia [11], a reliable knowledge base, and evaluates real-world entities across eight visual domains (see Fig. 2). This ensures that generated images are compared with verifiable visual information crowdsourced from the internet. Additionally, we have developed a comprehensive set of human evaluation criteria that focus on the precise visual depiction of entities, capturing subtle but essential details for visual accuracy. By directly assessing entity fidelity in the generated images against established knowledge, KITTEN aims to advance the evaluation of world knowledge in image generation models.

Using KITTEN, we conduct a comprehensive evaluation of various text-to-image models, including standard and customization models fine-tuned or utilizing in-context learning with retrieved reference images [3]. Our findings show that even the most advanced models [14, 2] often fail to produce accurate representations, generating images missing critical details essential for visual correctness. While retrieval-augmented models improve visual fidelity by incorporating reference images during testing, they tend to over-rely on these references, limiting their ability to generate novel configurations of entities from creative prompts. These findings highlight a key challenge in current image generation models: balancing entity fidelity with creative flexibility, underscoring the need for techniques that can generate precise visual details without sacrificing the ability to respond to diverse and imaginative user inputs.

2 Related Work

Existing evaluation for text-to-image generation. Evaluating text-to-image models has long been challenging, with many efforts aimed at improving performance measurement. Fréchet Inception Distance (FID) [9] is a common metric for assessing perceptual quality by measuring the distribution gap between generated and real-world images. CLIP-T scores [8] evaluate text-image alignment by comparing the CLIP feature similarity between generated images and input prompts. These metrics summarize the overall image quality. Several works assess alignment between generated images and text descriptions [33, 7, 12, 31, 4], but they mainly focus on semantic consistency rather than fine-grained visual accuracy of depicted entities and specialized visual-world knowledge.

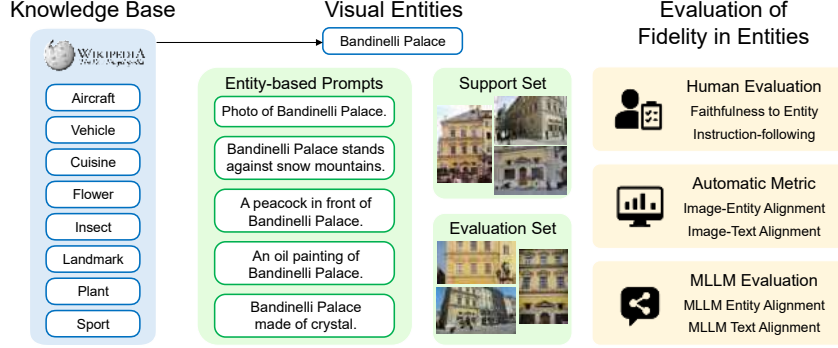


Figure 2: **KITTEN benchmark** is constructed from real-world entities across eight domains. For each selected entity, we define five evaluation tasks of image-generation prompts incorporating the entity. KITTEN includes a support set of entity images from the knowledge source for evaluating retrieval-augmented models, and an evaluation set for assessing the fidelity of the generated entities.

Recent works aim to evaluate models more thoroughly by decomposing the evaluation into sub-categories, such as attribute binding and numeracy, with corresponding benchmarks. For example, the SR_{2D} dataset [6] and the VISOR metric evaluate spatial relationships in text-to-image models, assessing whether objects in the generated image adhere to specified relationships (e.g., an orange *above* a giraffe). T2I-CompBench++ [13] contains text prompts from four categories (e.g., attribute binding) along with associated metrics. HRS-Bench [1] evaluates model performance across five major groups (i.e., bias, fairness, generalization, accuracy, and robustness). TIFA v1.0 [12] is a benchmark across 12 categories, paired with an automatic evaluation metric that measures image faithfulness via visual question answering. GenAI-Bench [18] and ConceptMix [32] focus on the evaluation of compositional text prompts, e.g., objects with specific colors, shapes, or spatial relationships. Additionally, ImagenHub [15] evaluates models across different tasks by measuring semantic consistency and perceptual quality.

Fidelity of entities in text-to-image generation. While text-to-image models enable the generation of creative images from text descriptions, challenges arise when visual-world knowledge is needed, i.e., generating accurate visual details of entities. Existing works have identified this issue and proposed solutions to mitigate hallucination [19]. However, no clear methodology exists to systematically assess these models’ limitations, which is crucial for improvement. In this work, we propose KITTEN, a benchmark addressing a novel problem of evaluating image generation models’ ability to generate fine-grained details of specific visual entities. Using KITTEN, we systematically assess the latest text-to-image and retrieval-augmented models with carefully designed human evaluations, automatic metrics, and MLLM evaluations.

3 KITTEN Benchmark

We introduce the KITTEN benchmark to evaluate the reliability of text-to-image models in generating knowledge-intensive concepts.

3.1 Design Desiderata of KITTEN

The key to creating the benchmark is constructing a set of image-generation prompts that require grounding in visual-world knowledge. Two specific properties differentiate our benchmark from prior evaluation frameworks of image generation. First, while existing benchmarks aim to test the common-sense knowledge of image generation models such as spatial or physical relationships [6, 13], we would like to stress-test the image generation models by focusing on generating entities from specific domains. Therefore, we create the benchmark using image concepts from Wikipedia, a rich knowledge-intensive data source, which contains several domain-specific entities and their corresponding images. Second, while most existing benchmarks focus on evaluating the instruction-following capability of the models, we would like to understand how well these models are at faithfully representing real-world concepts grounded in visual knowledge sources. Therefore, we

Generate an image according to the following description:
 a watercolor painting of the Bandinelli Palace landmark.

Use the reference images (provided above) to ensure the entity in the prompt (Bandinelli Palace landmark) is factually correct. You may conduct your own research (e.g., Google Search) to clarify the appearance of the entity and ensure its accurate depiction in the generated image.

Faithfulness to Prompt Entity
 How faithfully does the generated image represent the entity mentioned in the prompt?

1 Not faithful at all 2 Barely faithful 3 Somewhat faithful 4 Mostly faithful 5 Completely faithful

Adherence to Prompt Beyond References
 Does the generated image accurately and comprehensively depict any aspects of the scene or entity described in the prompt that are not already reflected in the reference images?

☒ Yes: The generated image accurately and comprehensively depicts aspects of the scene or entity described in the prompt that are not already reflected in the reference images.

☐ No: The generated image fails to accurately and comprehensively depict aspects of the scene or entity described in the prompt that are not already reflected in the reference images.

Figure 3: **Annotation interface.** Raters are asked to: (1) rate the image’s faithfulness to the prompt entity on a 1–5 scale, and (2) indicate whether the image follows the prompt with a yes or no response. We calculate the percentage of responses marked as “yes.”

design a specific set of evaluations targeted at capturing the visual fidelity of generated entities. Guided by the above principles, next, we clarify the details of the KITTEN benchmark.

3.2 Creating Entity-based Prompts

Fig. 2 shows the benchmark creation process. To generate a diverse set of prompts focused on faithfulness to knowledge-grounded concepts, we first select entity domains from the OVEN-Wiki dataset [11], the most comprehensive open-domain image recognition dataset. We select 8 domains encompassing 322 entities, covering human-made objects, natural species, and human activities. This selection offers broader coverage than existing benchmarks [17]. For each entity, we collect an evaluation set of entity images from Wikipedia for human evaluation, and a support set of images to evaluate retrieval-augmented models that leverage external knowledge sources for image generation (see evaluated models in Sec. 4).

After selecting the entities, we design five evaluation tasks of image-generation prompts.

- Basic prompt (4.58%): Photo of Bandinelli Palace.
- Entity in a specified location (30.57%): Bandinelli Palace stands against snow mountains.
- Composition with other objects (22.78%): A peacock in front of Bandinelli Palace.
- Entity in specific styles (21.20%): An oil painting of Bandinelli Palace.
- Entity made of specific materials (20.87%): Bandinelli Palace made of crystal.

The evaluation tasks are designed to cover key scenarios in customized image generation [29, 16], ensuring relevance to both researchers and real-world applications. For each task, ChatGPT [23] is instructed to propose prompts for different entity domains, which are then tailored to evaluate knowledge-entity generation by incorporating entity names directly into the prompts. This process resulted in a set of 6,440 prompts to assess the models’ ability to handle diverse and imaginative user inputs. The prompts range from 4 to 24 words, with an average length of 9.91 and a standard deviation of 2.86.

3.3 Human Evaluation

Since no established metrics exist for evaluating the generation of visual entities, human evaluation plays a critical role in reliably assessing model performance. We design human evaluations focusing on the visual fidelity of the target entity in the generated image, decomposing the evaluation into two aspects: 1) faithfulness to the prompt entity, and 2) adherence to prompt instructions beyond the entity. This design allows raters to focus on distinct criteria, enabling more informative comparisons of models, as our results often show trade-offs between these aspects. Raters are shown reference images of the prompt entity and encouraged to verify the entity’s faithfulness through their own research.

The final evaluation score is the average of five raters per image to ensure robust assessments. The human annotation interface is shown in Fig. 3.

- **Faithfulness to Entity.** We use a scale from 1 to 5, where 5 means completely faithful to the prompt entity, and 1 means the generated image has no similarity to the prompt entity.
- **Instruction-following.** We use a yes/no question to evaluate whether the generated image adheres to the prompt instructions beyond the entity. We calculate the percentage of answers “yes.”

3.4 Automatic Metrics

We gather the results of popular automatic metrics for image generation models, which primarily measure the similarity between the generated images and the references or prompts. While these metrics are not specialized for capturing the visual fidelity of the generated entity, we include their results to provide a comprehensive analysis of their alignment with human evaluation.

- **Image-Text Alignment.** We measure the cosine similarity between the generated image and the text prompt in CLIP’s feature space [25], i.e., *CLIP-T Score* [8].
- **Image-Entity Alignment.** We measure the average pairwise cosine similarity between the generated image and reference images of the target entity in the evaluation set using DINO’s feature space [24]. This serves as a proxy for how closely fine-grained details match.

3.5 MLLM Evaluation

Multimodal large language models (MLLMs) have recently demonstrated impressive progress in multimodal understanding. We investigate the use of MLLMs as automatic evaluators to reduce human effort and address the limitations of traditional metrics in assessing visual fidelity. Specifically, we prompt GPT-4o-mini [23] using the same criteria as our human evaluation.

- **MLLM Text Alignment.** Given the text prompt, we use an MLLM to evaluate whether the generated image follows the prompt instructions on a 1–5 scale.
- **MLLM Entity Alignment.** Given reference images of the target entity, we use an MLLM to assess how well the generated image resembles the target entity on a 1–5 scale.

4 Evaluated Models

We present a comprehensive analysis using KITTEN to understand the visual-world knowledge in current state-of-the-art models.

4.1 Text-to-Image Backbone Models

First, we examine general text-to-image backbone models that directly generate images solely based on text prompts without using additional tools or reference images.

- Stable Diffusion [28] maps images to a latent space where a diffusion model is trained.
- Imagen [30] uses a T5 encoder and cascaded diffusion models for high-resolution image generation.
- Imagen-3 [14] is a successor to Imagen, notable for its ability to handle long prompts.
- Flux [2] is a successor to Stable Diffusion, integrating parallel diffusion transformer blocks.

4.2 Retrieval-augmented Text-to-Image Models

The retrieval-augmented method is a family of image generation approaches that use support images (e.g., retrieved by a search engine) to enhance the model through fine-tuning or in-context learning, improving the fidelity of entities in the generated images. Our goal is to evaluate these models and determine whether incorporating such *support* images enhances the fine-grained visual fidelity of the entity in generation. Specifically, we provide some ground-truth reference entity images (held out from the entity images used for evaluation) as the support images to the above methods and then generate new images from them following the evaluation text prompts. We study the following models:

We use Flux.1-dev.

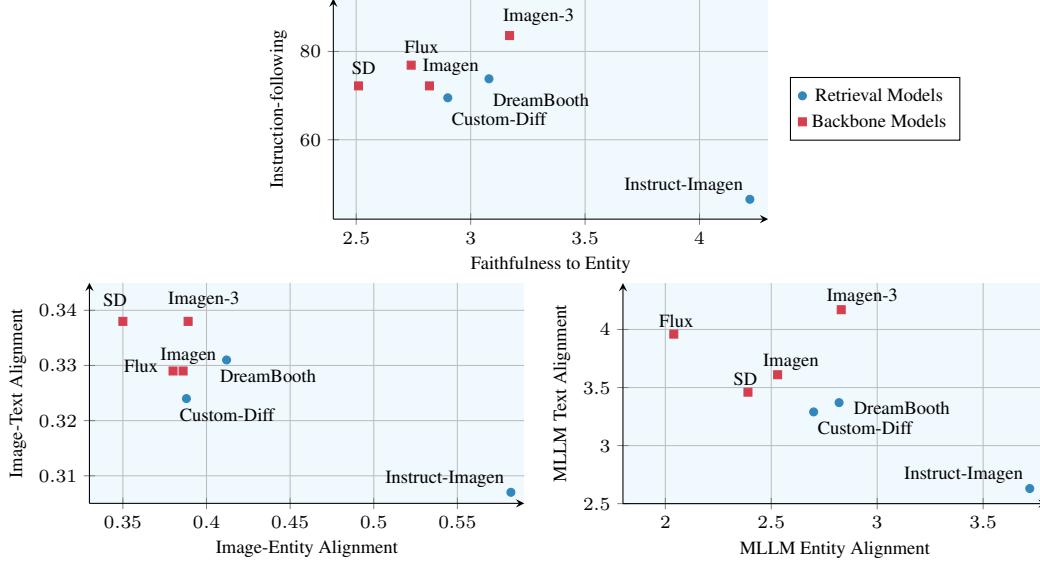


Figure 4: **Evaluation results** of text-to-image models. **(Top)** Human evaluation illustrating the trade-off between faithfulness and instruction-following. **(Bottom left)** Automatic metric results. **(Bottom right)** MLLM-based evaluations.

- DreamBooth [29] **fine-tunes** the Stable Diffusion model to learn a special token encoding the target entity. It then generates the entity in new contexts using prompts that include this token.
- Custom-Diff [16], similar to DreamBooth, **fine-tunes** partial weights of the backbone model.
- Instruct-Imagen [10] generates the target entity through **in-context learning** by encoding reference images into a multi-modal instruction: Generate an image of <entity_name>, referring to the images <ref_image_1>, ..., <ref_image_K>, and follow the caption: <prompt>.

5 Evaluation Results

5.1 Human Evaluation

Retrieval-augmented models enhance faithfulness but weaken instruction-following. Fig. 4 (top) shows that retrieval-augmented models — Custom-Diff, DreamBooth, and Instruct-Imagen — generally produce images more faithful to the entities than their base models, SD and Imagen. This is because these models incorporate reference images during testing, enabling them to generate visual concepts not well-represented in the base models’ parameters. However, retrieval-augmented models tend to have reduced instruction-following capabilities compared to their backbone models, as they often over-rely on reference images and struggle to create novel configurations of the entity as requested in creative text prompts. Although this trend is consistent across methods, the extent of the impact varies. Notably, Instruct-Imagen shows a significant increase in faithfulness score (2.81 $\rightarrow$ 4.22) but also a substantial drop in instruction-following score (72.2 $\rightarrow$ 46.5).

Enhancing backbone models improves both faithfulness and instruction-following. Our results show that improvements to base models alone can enhance both instruction-following and faithfulness. For example, Flux outperforms its predecessor SD by 0.23, and Imagen-3 surpasses its predecessor Imagen by 0.35 in faithfulness. However, these faithfulness improvements are still minor compared to those achieved by retrieval-augmented methods, such as DreamBooth, which shows a 0.57 improvement over SD.

On the other hand, Imagen-3 achieves the highest instruction-following score (83.6) and a high faithfulness score (3.17), outperforming the retrieval-augmented models DreamBooth (3.08) and Custom-Diff (2.90). This demonstrates that Imagen-3, as a strong backbone model, can generate specialized entities solely from text prompts. However, a notable gap remains in entity fidelity compared to the highest score achieved by Instruct-Imagen (4.22). These findings show that enhancing

the backbone model can improve both instruction-following and entity fidelity, while it is essential to incorporate advanced retrieval-augmented techniques to achieve higher levels of faithfulness.

Balancing faithfulness and instruction-following is achievable. The retrieval-augmented model DreamBooth improves entity faithfulness compared to its baseline, SD (2.51 $\rightarrow$ 3.08), without compromising SD’s instruction-following score (72.2 $\rightarrow$ 73.8). This demonstrates that a well-designed retrieval-augmented method can enhance entity fidelity without sacrificing creativity. These findings also suggest future research directions, emphasizing that combining a strong backbone with an effective retrieval-augmented approach can achieve a balance between faithfulness and instruction-following.

5.2 Automatic Metrics and MLLM Evaluation

Retrieval-augmented models improve entity alignment but reduce text alignment. Fig. 4 (bottom) shows that retrieval models increase the entity alignment score while decreasing the text alignment score compared to their base models in both automatic metrics and MLLM evaluation. These observations align with the human evaluation, where retrieval-augmented models show improved entity faithfulness but reduced instruction-following. In addition, this trend is consistent with observations in recent works [21], which indicate that models incorporating additional inputs, such as reference images, tend to have lower text alignment scores than base models due to a trade-off between aligning with the text and with the images.

Alignment between automatic metrics and human evaluation. While the overall observations from the automatic metrics align with the human evaluation, there are notable discrepancies. We observe that improving base models does not necessarily lead to gains in the automatic metrics. For example, Flux (0.329) performs worse than its predecessor SD (0.338) in the image-text metric, and Imagen-3 (0.389) shows only a marginal improvement over Imagen (0.386) in the image-entity score. These findings suggest that automatic metrics have a limited ability to capture meaningful variations between models of similar quality. In addition, although DreamBooth achieves a higher instruction-following score compared to its base model SD, it has a lower image-text alignment score. We hypothesize that the image-text score may not accurately assess the alignment between the generated image and rare entities. For example, with the prompt “The Teufelsmauer landmark shimmers in the sunlight,” it is unclear whether the image-text similarity for “Teufelsmauer” is evaluated correctly. This highlights that the traditional metrics [8, 17] might fail to measure true alignment between the unique entity and the generated image. In addition, Imagen-3 ranks higher in faithfulness in the human evaluation, yet DreamBooth outperforms Imagen-3 in image-entity scores, showing a misalignment between human perception and the learned semantic features [24].

Alignment between MLLM and human evaluation. The MLLM score generally shows a high correlation with human evaluation, though slight discrepancies exist. Specifically, DreamBooth (3.37) falls behind SD (3.46) and Imagen (3.61) in the MLLM text alignment score. However, in human evaluation, DreamBooth (73.8) outperforms both Imagen and SD (72.2). This suggests that while the MLLM can capture overall trends and identify models with clearly stronger or weaker performance, it may struggle to distinguish between models with similar capabilities. On the other hand, Flux (2.04) receives a lower MLLM entity alignment score than SD (2.39), despite outperforming SD in human evaluation (2.74 vs. 2.51), which suggests that the MLLM may have different preferences or biases compared to human raters.

5.3 Ablation Study

Correlation of automatic and MLLM metrics with human evaluation. While manual evaluation ensures high accuracy, automated methods offer a cost-effective alternative, albeit with slightly lower alignment to human perception. To quantify the consistency between automatic and MLLM metrics with human evaluation, we computed Pearson and Spearman correlations in Tab. 1. CLIP-T and CLIP-I show moderate alignment with user evaluations. Compared to CLIP-I metrics, DINO demonstrates stronger alignment with user evaluations of faithfulness. Surprisingly, MLLM-based evaluation shows a strong correlation with human judgments. This highlights the potential of MLLM metrics to capture subtle visual details when

Table 1: Correlation with human evaluation.

Metric	Pearson	Spearman
CLIP-T	0.337	0.384
CLIP-I	0.239	0.340
DINO	0.510	0.504
MLLM-T	0.618	0.589
MLLM-I	0.703	0.695

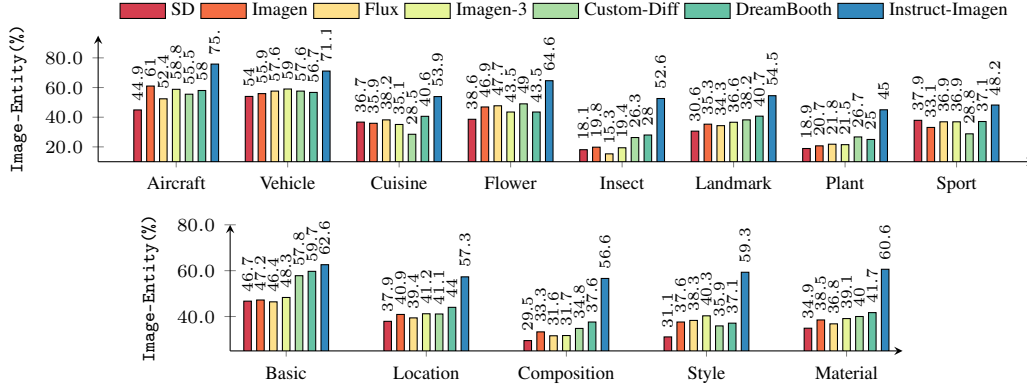


Figure 5: **Performance analysis across domains and tasks.** (Top) Retrieval models achieve higher Image-Entity scores in the insect, landmark, and plant domains but perform worse in the cuisine and sport domains, possibly due to the varying occurrence of these entities in common image datasets. (Bottom) *Location* scoring highest and *Composition* the least.

assessing the faithfulness of generated knowledge entities. It suggests that our study can be reliably scaled up using MLLM as an automatic evaluator.

Selection of image-entity alignment metrics. In Tab. 2, two popular visual features are tested for calculating cosine similarity scores between reference and generated images as the image-entity alignment metric: *CLIP-I* [25] and *DINO* [24]. We find that DINO scores provide a clearer separation between models compared to CLIP-I scores, making it a more discriminative metric at capturing subtle differences in faithfulness. For example, the difference between Custom-Diff and Instruct-Imagen is much larger when using DINO (0.19) compared to CLIP-I (0.11). This may be due to DINO’s focus on primary entities, allowing for a more accurate estimation of similarity between the generated entities and reference images.

Table 2: Selection of Image-Entity metrics.

Model	CLIP-I	DINO
SD	0.646	0.350
Imagen	0.646	0.386
Flux	0.639	0.380
Imagen-3	0.650	0.389
Custom-Diff	0.643	0.388
DreamBooth	0.674	0.412
Instruct-Imagen	0.751	0.582

5.4 Analysis of Performance Variations

Performance across entity domains. Fig. 5 (top) shows that the performance of each method is domain-dependent. Retrieval-augmented models generally achieve higher image-entity alignment scores than backbone models in the insect, landmark, and plant domains. Since these domains contain less frequent terms in common image datasets, these visual concepts are therefore underrepresented in the backbone model’s parameters. The retrieval-augmented models improve performance by incorporating reference images during inference. Additionally, the insect and landmark domains have lower average image-entity scores, likely due to the inherent challenges of generating fine-grained details of insects and the many specifications and features of landmarks.

On the other hand, the retrieval-augmented method, Custom-Diff, performs worse than its base model, SD, in the cuisine and sport domains. These domains contain common terms, such as snowboarding and guacamole, which the SD model has well memorized. The Custom-Diff model’s performance degrades, potentially due to fine-tuning on a smaller reference set. This variability suggests that the effectiveness of a retrieval-augmented method may be influenced by the nature of the domain-specific content, and the optimal choice of retrieval-augmented method remains an open question.

Performance across evaluation tasks. Fig. 5 (bottom) shows that image-entity scores across evaluation tasks generally align with the overall ranking. *Location* scores highest (0.431), followed by *Material* (0.417), *Style* (0.399), and *Composition* (0.364), highlighting the challenge of maintaining entity fidelity when prompts involve complex compositions.

We observe that *Style* prompts show a distinct score distribution. Retrieval-augmented methods, DreamBooth and Custom-Diff, along with their base model SD, receive lower image-entity scores (0.371, 0.359, and 0.311), indicating that models based on SD struggle to generate faithful entities when changing their styles. However, SD achieves the highest image-text score (0.346), followed



Figure 6: **Qualitative results. (Top)** The backbone models show lower Faithfulness to Entity, with Wakame’s appearance differing from the reference image. **(Bottom)** The retrieval models struggle with Instruction-following, over-relying on references, and failing to create compositions of the entity and the giant sandcastle. We show the Image-Entity and Image-Text scores with the generated results.

by DreamBooth (0.341) and Custom-Diff (0.335), suggesting these models are strong in generating accurate styles but may sacrifice entity fidelity.

5.5 Qualitative Results

We present the visual results in Fig. 6. In the above example, the backbone models (second row) show lower faithfulness to the entity, with Wakame’s appearance differing from the reference image. In contrast, retrieval-augmented models (first row), using reference images during testing, achieve better visual alignment with the target. Instruct-Imagen demonstrates a balance between entity fidelity and creative flexibility in the generated images, supported by the highest alignment scores across both metrics. These findings highlight future research directions, showing that enhancing the backbone model can improve both instruction-following capability and entity fidelity. Furthermore, combining a strong backbone with an advanced retrieval-augmented method enables the coexistence of these two aspects.

In the example below, Custom-Diff and DreamBooth show reduced instruction-following compared to their backbone model, SD. In particular, Custom-Diff struggles to create novel compositions of the entity and the giant sandcastle. In contrast, the backbone models excel in both instruction-following and entity faithfulness, likely because “Rolls-Royce Phantom Drophead Coupé” is well-represented in their training data. The retrieval-augmented models underperform due to over-relying on the reference images and experiencing knowledge forgetting during fine-tuning on small sets of reference

images. These results suggest that the success of retrieval-augmented methods heavily depends on the entity domain and the customization approach. More results can be found in the supplementary materials.

6 Conclusion

We propose KITTEN, a benchmark for evaluating entity fidelity in text-to-image generation, focusing on visual concepts that require specialized knowledge. We design prompts based on Wikipedia entities and introduce a human evaluation framework to assess visual faithfulness. Extensive analysis reveals that while backbone models can generate specialized entities, retrieval-augmented models achieve higher faithfulness. However, these methods often struggle with creative prompts, highlighting the need for techniques that enhance entity fidelity without compromising instruction-following ability.

Limitations. KITTEN does not include prompts requiring knowledge reasoning (e.g., the tallest building in Manhattan), as such prompts can be rewritten as entity-based ones by language preprocessing.

A Additional Qualitative Examples

We present additional qualitative examples evaluating the backbone models—SD, Imagen, Flux, and Imagen-3—as well as the retrieval models—Custom-Diff, DreamBooth, and Instruct-Imagen—across various domains in the KITTEN benchmark. The results for the aircraft, vehicle, cuisine, flower, insect, landmark, plant, and sport domains are shown in Fig. A.1, Fig. A.2, Fig. A.3, Fig. A.4, Fig. A.5, Fig. A.6, Fig. A.7, Fig. A.8, respectively.

Fig. A.3 presents the results for the cuisine domain. For the prompt, “A Wakame dish with a cherry flower on top of it,” retrieval-based models such as Custom-Diff, DreamBooth, and Instruct-Imagen successfully generate the entity Wakame. However, Custom-Diff fails to capture the composition of the cherry flower. In contrast, the backbone models—Imagen, Flux, and Imagen-3—correctly generate the cherry flower, but the representation of Wakame does not align with its real-world appearance. In the example prompt, “photo of a Hot and sour soup,” the backbone models introduce incorrect ingredients that significantly deviate from a real-world hot and sour soup, demonstrating how these models hallucinate entities instead of reproducing real-world knowledge accurately.

Fig. A.5 shows the results for the insect domain. For the prompt, “Satyrium liparops sitting at the beach with a view of the sea,” all models correctly generate the beach scene as instructed. However, the backbone models hallucinate the insect’s appearance, generating a completely different insect, while the retrieval models accurately depict the visual details of the Satyrium liparops. For the prompt, “Promachus hinei wearing sunglasses,” the backbone models demonstrate strong instruction-following ability, with Imagen and Imagen-3 correctly composing the sunglasses on the insect. In contrast, the retrieval models fail to generate this composition. However, the backbone models do not generate an accurate representation of the insect itself, while the retrieval models correctly generate the entity.

Fig. A.7 presents the results for the plant domain. In the example, “An impressionistic painting of Cirsium andersonii,” none of the models effectively follow the prompt in generating the “impressionistic painting,” highlighting the challenge of rendering specific artistic styles. However, the retrieval models generate the entity more accurately than the backbone models. In another example, “Penstemon rydbergii on a rustic wooden table next to a rose plant,” although the generated entity lacks the fine details of the reference, the backbone models successfully capture the composition of “next to a rose,” demonstrating their stronger instruction-following ability.

These observations align with the quantitative results, further demonstrating that while advanced backbone models can generate specialized entities, there is a notable gap in the fidelity of entities compared to retrieval-augmented models. Although retrieval-augmented methods improve faithfulness to entities, they often struggle with creative prompts, highlighting the need for techniques that enhance entity accuracy without compromising instruction-following abilities.

In addition, we have provided examples illustrating a balance between entity fidelity and creative flexibility in the generated images. Specifically, Instruct-Imagen’s results for prompts such as “A Wakame dish with a cherry flower on top of it” in Fig. A.3, “Alstroemeria on top of a mountain with sunrise in the background” in Fig. A.4, and “Satyrium liparops sitting at the beach with a view of the sea” in Fig. A.5 demonstrate enhanced entity fidelity while maintaining creative flexibility. These examples are supported by the highest alignment scores across both metrics when compared to other models.

B Details of the Human Evaluation Scores Across Domains

We have presented the human evaluation results for backbone and retrieval-augmented text-to-image models, comparing their performance in terms of entity faithfulness and instruction-following accuracy. Here, we provide a detailed breakdown of the human evaluation results across eight domains of the KITTEN benchmark in Tab. B.1.

We observe that the performance of each method varies across domains. For instance, while the overall faithfulness scores of Custom-Diff and DreamBooth are lower than the backbone model Imagen-3, these retrieval models consistently perform better in the insect, landmark, and plant domains. When comparing backbone models, Flux outperforms SD in the overall score; however, SD demonstrates higher faithfulness in the insect, landmark, and plant domains, suggesting that Flux struggles to generate less frequent entities. Additionally, the retrieval models, DreamBooth

and Custom-Diff, improve faithfulness over SD in some cases but show reduced performance in the cuisine and sport domains. In terms of instruction-following score, Instruct-Imagen performs notably lower in the flower (31.3) and insect (21.9) domains, while Imagen-3 achieves significantly higher scores in the cuisine (93.8) and sport (96.9) domains.

Table B.1: Detailed breakdown of human evaluation.

Model	Aircraft	Vehicle	Cuisine	Flower	Insect	Landmark	Plant	Sport	Overall
Faithfulness to entity									
■ SD	2.07	3.57	3.42	2.36	1.61	1.95	2.09	3.02	2.51
■ Imagen	3.19	3.62	3.55	3.21	1.54	2.11	2.04	3.3	2.82
■ Flux	3.51	3.89	3.35	2.52	1.44	1.77	1.83	3.59	2.74
■ Imagen-3	3.74	4.04	4.26	3.37	1.63	2.31	1.98	4.04	3.17
● Custom-Diff	2.59	3.90	2.52	3.82	2.37	3.02	2.85	2.10	2.90
● DreamBooth	3.26	3.88	3.23	3.18	2.46	3.06	2.64	2.93	3.08
● Instruct-Imagen	4.23	4.70	4.43	4.48	3.78	4.06	4.02	4.08	4.22
Instruction-following									
■ SD	68.8	75.0	62.5	87.5	62.5	65.6	84.4	71.9	72.2
■ Imagen	59.4	84.4	75.0	84.4	53.1	59.4	84.4	78.1	72.2
■ Flux	56.3	87.5	96.9	78.1	59.4	71.9	84.4	81.3	76.9
■ Imagen-3	78.1	87.5	93.8	84.4	71.9	71.9	84.4	96.9	83.6
● Custom-Diff	59.4	75.0	65.6	78.1	71.9	65.6	81.3	59.4	69.5
● DreamBooth	68.8	71.9	71.9	81.3	68.8	75.0	78.1	75.0	73.8
● Instruct-Imagen	46.9	56.3	56.3	31.3	21.9	50.0	59.4	50.0	46.5

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

C Details of the Automatic Metric Scores Across Domains

We have presented the automatic metric evaluation for backbone and retrieval-augmented text-to-image models, including image-text alignment and image-entity alignment scores. Here, we include a detailed breakdown of the automatic metrics across eight domains of the KITTEN benchmark in Tab. C.1. Specifically, we include a detailed ablation study of the image-entity alignment metric using cosine similarity scores based on two popular image features: the CLIP-Image and the DINO features.

The retrieval models—Custom-Diff, DreamBooth, and Instruct-Imagen—consistently outperform the backbone models in DINO score across the insect, landmark, and plant domains. In contrast, Custom-Diff shows notably worse DINO and CLIP-T scores in the cuisine and sport domains. These results are consistent with human evaluations.

D Details of the MLLM Evaluation Scores Across Domains

We report MLLM evaluation results for both the backbone and retrieval-augmented text-to-image models, including scores for text alignment and entity alignment. Tab. D.1 presents a detailed breakdown of these scores across the eight domains of the KITTEN benchmark. Notably, retrieval-augmented models consistently outperform backbone models in entity alignment within the vehicle, insect, landmark, and plant domains. In contrast, Custom-Diff shows substantially lower entity alignment scores in the cuisine and sport domains.

E Details of the Human Evaluation Scores Across Prompts

We provide a detailed breakdown of human evaluation results across five evaluation tasks in the KITTEN benchmark, as shown in Tab. E.1. For the faithfulness score, the ranking across prompts remains consistent with the overall ranking. Instruct-Imagen achieves the highest score, followed by Imagen-3.

Table C.1: Detailed breakdown of automatic metrics.

Model	Aircraft	Vehicle	Cuisine	Flower	Insect	Landmark	Plant	Sport	Overall
Image-Text Alignment: CLIP-T									
■ SD	0.341	0.354	0.343	0.336	0.322	0.330	0.332	0.342	0.338
■ Imagen	0.329	0.344	0.338	0.337	0.323	0.312	0.316	0.335	0.329
■ Flux	0.341	0.346	0.329	0.327	0.320	0.308	0.314	0.343	0.329
■ Imagen-3	0.345	0.351	0.352	0.343	0.325	0.315	0.324	0.352	0.338
● Custom-Diff	0.333	0.349	0.314	0.339	0.321	0.315	0.326	0.298	0.324
● DreamBooth	0.333	0.347	0.335	0.333	0.322	0.318	0.324	0.335	0.331
● Instruct-Imagen	0.293	0.324	0.315	0.320	0.295	0.294	0.302	0.316	0.307
Image-Entity Alignment: CLIP-I									
■ SD	0.640	0.655	0.673	0.719	0.619	0.616	0.628	0.618	0.646
■ Imagen	0.678	0.650	0.671	0.733	0.630	0.632	0.630	0.545	0.646
■ Flux	0.652	0.653	0.663	0.738	0.609	0.614	0.618	0.561	0.639
■ Imagen-3	0.677	0.656	0.665	0.726	0.623	0.638	0.632	0.582	0.650
● Custom-Diff	0.688	0.678	0.593	0.759	0.619	0.655	0.662	0.491	0.643
● DreamBooth	0.698	0.676	0.692	0.742	0.660	0.681	0.657	0.589	0.674
● Instruct-Imagen	0.776	0.719	0.746	0.825	0.810	0.735	0.751	0.648	0.751
Image-Entity Alignment: DINO									
■ SD	0.449	0.540	0.367	0.386	0.181	0.306	0.189	0.379	0.350
■ Imagen	0.610	0.559	0.359	0.469	0.198	0.353	0.207	0.331	0.386
■ Flux	0.524	0.576	0.382	0.477	0.153	0.343	0.218	0.369	0.380
■ Imagen-3	0.588	0.590	0.351	0.435	0.194	0.366	0.215	0.369	0.389
● Custom-Diff	0.555	0.576	0.285	0.490	0.263	0.382	0.267	0.288	0.388
● DreamBooth	0.580	0.567	0.406	0.435	0.280	0.407	0.250	0.371	0.412
● Instruct-Imagen	0.758	0.711	0.539	0.646	0.526	0.545	0.450	0.482	0.582

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

Table D.1: Detailed breakdown of MLLM evaluation.

Model	Aircraft	Vehicle	Cuisine	Flower	Insect	Landmark	Plant	Sport	Overall
MLLM Text Alignment									
■ SD	3.06	3.62	2.97	4.22	2.84	3.78	3.69	3.47	3.46
■ Imagen	3.16	3.78	3.75	4.12	2.88	3.53	3.94	3.75	3.61
■ Flux	3.31	4.31	4.28	4.12	3.5	3.94	4.44	3.75	3.96
■ Imagen-3	3.41	4.25	4.25	4.38	4.09	3.88	4.72	4.41	4.17
● Custom-Diff	3.22	3.5	3.53	3.72	2.66	3.44	3.47	2.75	3.29
● DreamBooth	3.16	3.66	3.31	3.88	2.94	3.53	3.41	3.09	3.37
● Instruct-Imagen	2.53	3.25	2.75	2.41	1.91	2.94	3.0	2.28	2.63
MLLM Entity Alignment									
■ SD	2.34	3.12	3.09	2.28	1.22	1.97	1.69	3.44	2.39
■ Imagen	2.81	3.16	3.25	2.72	1.12	2.41	1.59	3.19	2.53
■ Flux	2.41	2.97	2.47	1.75	1.12	1.72	1.22	2.69	2.04
■ Imagen-3	3.19	3.34	4.06	3.03	1.31	2.25	1.69	3.75	2.83
● Custom-Diff	2.94	3.41	1.78	3.28	1.88	3.34	2.91	2.03	2.7
● DreamBooth	3.12	3.5	2.66	2.59	1.91	3.53	2.59	2.66	2.82
● Instruct-Imagen	3.88	4.06	3.91	4.03	3.22	4.09	3.31	3.25	3.72

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

The retrieval-based methods, DreamBooth and Custom-Diffusion, come next, while the other base models—Imagen, Flux, and Stable-Diffusion—score comparatively lower.

The instruction-following scores across prompts align with the average performance, with Imagen-3 emerging as the top performer. Notably, Flux and Imagen-3 excel in *Location*, achieving scores of 90 and 87.8, respectively, whereas Imagen and other models fall below 80. Similarly, Imagen-3 demonstrates strong performance in *Composition*, with a score of 94.6, highlighting its robust instruction-following capabilities in these aspects. In contrast, DreamBooth leads in *Style* with the highest score of 90.2, followed by Custom-Diffusion (84.3) and Stable-Diffusion (80.4), showcasing the strong ability of Stable Diffusion and its retrieval-augmented variants to generate accurate styles. In the *Material* category, all models struggle with instruction adherence, though Imagen-3 achieves a relatively higher score.

Table E.1: Detailed breakdown of human evaluation.

Model	Basic	Location	Composition	Style	Material	Overall
Faithfulness to entity						
■ SD	3.08	2.87	2.03	2.16	2.65	2.51
■ Imagen	3.96	3.16	2.38	2.53	2.76	2.82
■ Flux	3.50	3.04	2.67	2.32	2.54	2.74
■ Imagen-3	4.08	3.55	2.78	2.79	3.13	3.17
● Custom-Diff	4.34	2.77	2.73	3.12	2.80	2.90
● DreamBooth	4.18	3.33	2.72	2.95	2.94	3.08
● Instruct-Imagen	4.90	4.18	4.16	4.23	4.21	4.22
Instruction-following						
■ SD	100.0	80.0	64.3	80.4	53.1	72.2
■ Imagen	90.0	75.6	80.4	66.7	59.2	72.2
■ Flux	90.0	90.0	83.9	58.8	61.2	76.9
■ Imagen-3	90.0	87.8	94.6	76.5	69.4	83.6
● Custom-Diff	100.0	72.2	66.1	84.3	46.9	69.5
● DreamBooth	90.0	77.8	67.9	90.2	53.1	73.8
● Instruct-Imagen	100.0	46.7	58.9	47.1	20.4	46.5

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

F Details of the Automatic Metric Scores Across Prompts

We provide a detailed breakdown of automatic metrics across five evaluation tasks in the KITTEN benchmark, as shown in Tab. F.1. We observe that *Location* achieves the highest average image-text score (0.334), followed by *Composition* (0.330), *Style* (0.326), and *Material* (0.319). This pattern suggests that generating entities within a given context is relatively easier, whereas accurately modifying materials remains more challenging. In the categories of *Location*, *Composition*, and *Material*, the base models tend to achieve higher image-text scores, with Imagen-3 attaining the highest score among them. Among retrieval-based models, DreamBooth outperforms Custom Diffusion and Instruct-Imagen. Notably, in *Material*, Imagen scores lower (0.317) than DreamBooth (0.319) and Custom Diffusion (0.320), indicating greater difficulty for Imagen in this category.

For the image-entity score, we find that *Location* again ranks highest with an average score of 0.431, followed by *Material* (0.417), *Style* (0.399), and *Composition* (0.364). This ranking highlights the challenge of maintaining entity fidelity when prompts require complex compositions. In the categories of *Location*, *Composition*, and *Material*, retrieval-augmented models generally outperform base models, with Instruct-Imagen achieving the highest score, followed by DreamBooth and Custom Diffusion. Imagen-3 and Flux also surpass their respective predecessors, Imagen and Stable-Diffusion. However, in an unexpected result for *Composition*, Imagen (0.333) outperforms Imagen-3 (0.317).

While most categories exhibit higher image-entity scores for *Material* compared to *Style*, Imagen-3 and Flux surprisingly perform better on styles. Additionally, other models tend to achieve higher image-entity scores on *Style* than on *Composition*, with DreamBooth being an exception. These

results highlight the varying strengths of different models in preserving entity fidelity across diverse prompt types.

Table F.1: Detailed breakdown of automatic metrics.

Model	Basic	Location	Composition	Style	Material	Overall
Image-Text Alignment: CLIP-T						
■ SD	0.308	0.342	0.333	0.346	0.332	0.338
■ Imagen	0.304	0.338	0.329	0.328	0.317	0.329
■ Flux	0.299	0.340	0.340	0.308	0.324	0.329
■ Imagen-3	0.303	0.347	0.345	0.322	0.333	0.338
● Custom-Diff	0.308	0.326	0.324	0.335	0.320	0.324
● DreamBooth	0.308	0.336	0.327	0.341	0.319	0.331
● Instruct-Imagen	0.302	0.312	0.311	0.305	0.291	0.307
Image-Entity Alignment: DINO						
■ SD	0.467	0.379	0.295	0.311	0.349	0.350
■ Imagen	0.472	0.409	0.333	0.376	0.385	0.386
■ Flux	0.464	0.394	0.316	0.383	0.368	0.380
■ Imagen-3	0.483	0.412	0.317	0.403	0.391	0.389
● Custom-Diff	0.578	0.411	0.348	0.359	0.400	0.388
● DreamBooth	0.597	0.440	0.376	0.371	0.417	0.412
● Instruct-Imagen	0.626	0.573	0.566	0.593	0.606	0.582

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

G Details of the MLLM Evaluation Across Prompts

We provide a detailed breakdown of MLLM evaluation results across five evaluation tasks in the KIT-TEN benchmark, as shown in Tab. G.1. For MLLM text alignment, the ranking across prompt types largely mirrors the overall trend. Imagen-3 consistently achieves the highest overall score (4.17), followed by Flux (3.96) and Imagen (3.61). Among retrieval-augmented models, DreamBooth and Custom-Diffusion perform moderately (3.37 and 3.29, respectively), while Instruct-Imagen scores the lowest (2.63). Imagen-3 leads across most prompt types, particularly excelling in *Composition* (4.59) and *Material* (3.78). Flux also performs well, especially in *Location* (4.31) and *Composition* (4.23). In contrast, Custom-Diffusion and DreamBooth show weaker alignment, particularly in *Composition* and *Material*, with scores below 3.5. The MLLM entity alignment scores follow a similar pattern, with Instruct-Imagen achieving the highest overall score (3.72), outperforming all other models across all prompt types. It particularly excels in *Composition* (3.91), *Style* (3.67), and *Material* (3.65), demonstrating strong grounding of entity representations in the image. Among the base models, Imagen-3 performs best (2.83), while Flux consistently underperforms across all prompts. These results further underscore the effectiveness of retrieval augmentation for enhancing entity alignment, especially for complex instructions involving *Style* and *Material*.

H Details of the Correlation with Human Evaluation

To quantify the consistency between automatic metrics and human evaluation results, we computed Pearson and Spearman correlations. CLIP-T and CLIP-I show moderate alignment with user evaluations. Here, we provide a detailed breakdown of the correlation across eight domains in Tab. H.1. While these metrics show some alignment with human evaluations, discrepancies remain in certain categories. Notably, the correlation between CLIP-T and the instruction-following score in the landmark domain, as well as the correlation between CLIP-I and the faithfulness score in the plant domain, is negative. This underscores the limitations of automatic metrics in fully capturing human judgment. Although DINO demonstrates stronger alignment with user evaluations of faithfulness, correlation variability persists across categories. For instance, correlations are lower in the vehicle and landmark domains, highlighting the need for more accurate automatic metrics to better reflect human evaluations and assess model performance.

Table G.1: Detailed breakdown of MLLM evaluation.

Model	Basic	Location	Composition	Style	Material	Overall
MLLM Text Alignment						
■ SD	4.6	3.61	3.3	3.86	2.69	3.46
■ Imagen	5.0	3.7	3.86	3.45	3.06	3.61
■ Flux	5.0	4.31	4.23	3.35	3.41	3.96
■ Imagen-3	5.0	4.28	4.59	3.75	3.78	4.17
● Custom-Diff	4.6	3.3	3.2	3.84	2.51	3.29
● DreamBooth	4.7	3.34	3.25	3.94	2.69	3.37
● Instruct-Imagen	4.6	2.6	2.86	2.82	1.84	2.63
MLLM Entity Alignment						
■ SD	2.9	2.87	2.16	1.84	2.27	2.39
■ Imagen	3.2	2.89	2.48	1.94	2.41	2.53
■ Flux	2.0	2.24	2.43	1.71	1.59	2.04
■ Imagen-3	3.82	3.18	2.63	2.41	2.63	2.83
● Custom-Diff	3.7	2.7	2.57	2.8	2.51	2.7
● DreamBooth	4.1	3.09	2.55	2.45	2.76	2.82
● Instruct-Imagen	4.2	3.61	3.91	3.67	3.65	3.72

(■: Text-to-Image Models, ●: Retrieval-augmented Models)

Additionally, both MLLM entity alignment and text alignment scores show higher correlations with human evaluations in most domains, indicating stronger consistency as automatic proxies. However, exceptions arise in the aircraft and vehicle domains, where the correlation values are noticeably lower. This suggests that while MLLM-based scores generally align well with human judgments, their effectiveness may be reduced in domains characterized by complex or diverse visual features, such as aircraft and vehicles. Further refinement of these metrics could enhance their robustness and reliability across all categories.

Table H.1: Per-category correlation with human evaluation.

Metric	Type	Aircraft	Vehicle	Cuisine	Flower	Insect	Landmark	Plant	Average
CLIP-T	Pearson	0.138	0.499	0.105	0.788	0.567	-0.299	0.564	0.337
	Spearman	0.168	0.454	0.286	0.836	0.554	-0.166	0.558	0.384
CLIP-I	Pearson	0.330	0.195	0.100	0.215	0.528	0.356	-0.051	0.239
	Spearman	0.548	0.323	0.238	0.287	0.503	0.623	-0.143	0.340
DINO	Pearson	0.655	0.367	0.549	0.551	0.277	0.680	0.492	0.510
	Spearman	0.575	0.311	0.735	0.430	0.210	0.700	0.565	0.504
MLLM-T	Pearson	0.477	0.687	0.583	0.673	0.586	0.790	0.575	0.618
	Spearman	0.478	0.605	0.536	0.604	0.577	0.748	0.510	0.589
MLLM-I	Pearson	0.468	0.467	0.754	0.783	0.746	0.742	0.705	0.703
	Spearman	0.475	0.393	0.711	0.764	0.686	0.748	0.704	0.695

I Dataset Statistics

KITTEN benchmark focuses on evaluating faithfulness to knowledge-grounded concepts. To ensure diversity, we select entities from eight specialized domains and construct diverse prompts in each domain for evaluation. For each entity, we collect a set of support images as inputs to assess retrieval-augmented models, where support images are used to enhance the model’s predictions. In addition, we collect a set of evaluation images for conducting human evaluation. A detailed breakdown of the data statistics is provided in Tab. I.1.

Next, we present the distribution of different evaluation tasks in Tab. I.2. The types of evaluation tasks include: 1) generating the knowledge entity, 2) the knowledge entity in context, 3) the composition of entities, 4) creation in different styles, and 5) creation in different materials.

Table I.1: Statistics of KITTEN benchmark.

Domain	#Entities	#Prompts	#Support Images	#Eval Images	# (Entitiy, Prompt)
Aircraft	48	20	469	237	960
Vehicle	50	20	500	250	1000
Flower	18	20	180	90	360
Insect	50	20	500	250	1000
Plant	48	20	480	240	960
Landmark	50	20	500	250	1000
Cuisine	31	20	310	155	620
Sport	27	20	270	135	540

Table I.2: Statistics of KITTEN evaluation tasks.

Evaluation task	#Prompts	Percentage (%)
Basic	295	4.58
Location	1969	30.57
Composition	1467	22.78
Style	1365	21.20
Material	1344	20.87

J Details on Human Evaluation

We employ five annotators per image to ensure robust assessments. The raters are hired through Prolific.com, a third-party rating service. For the binary task ("Adherence to Prompt Beyond References"), we observe agreement among at least 4 out of 5 annotators in 75% of cases and perfect agreement (5 out of 5) in 44% of cases. For the Likert scale task ("Faithfulness to Reference Entity"), we calculate Krippendorff's Alpha at 0.60, indicating a good agreement for a subjective task of this complexity. Additionally, we achieve an IoU-like score of 0.53, which penalizes outliers and demonstrates moderate consensus, and an average pairwise Cohen's Kappa of 0.25, reflecting fair pairwise agreement. The average standard deviation of ratings is 0.74, reflecting moderate variability in annotator judgments. These metrics collectively demonstrate the reliability of our human evaluation.

K Details on Evaluated Models

To facilitate the reproducibility of the KITTEN benchmark experiments, we provide detailed descriptions of the training and inference setups for all evaluated models. For retrieval-augmented methods (DreamBooth and Custom-Diffusion), we follow their original implementations and fine-tune each model per entity using 10 reference images that are disjoint from the evaluation set. We use the AdamW optimizer with a learning rate of 5×10^{-6} and default β values ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Training is conducted for 1,000 steps with a batch size of 5. Instruct-Imagen is used solely in inference mode with the provided reference images and does not require fine-tuning. Backbone models (e.g., Imagen, Imagen-3, and Flux) are used as released, without additional training. All images are generated using evaluation text prompts, with no overlap in prompts or images between the test and support sets. Evaluation with GPT-4o-mini is performed via API calls. Experiments are conducted on a cluster of eight NVIDIA A100 GPUs (each with 40GB of memory). Fine-tuning takes approximately 20 minutes per entity, and inference requires roughly 5 seconds per image. To support full reproducibility, we will release our benchmark data, evaluation code, and generation scripts.

L Human Annotation Instructions

We provide the complete instructions given to human raters for evaluating the generated images in the KITTEN benchmark.

Rater Instructions

In this task, you will be provided with a Prompt, Reference Images, and a Generated Image. Your job is to assess the factual accuracy of the generated image with respect to the prompt and the reference images. The goal is to ensure that the entity described in the prompt is factually correct and accurately represented. While the reference images offer a visual starting point, you may conduct your own research (e.g., Google Search) to clarify the appearance of the entity and ensure its accurate depiction in the generated image.

Part 1: Reference Alignment

Faithfulness to Prompt Entity (Factuality)

Your first task is to evaluate how faithfully the generated image represents the reference entity. Consider whether the reference entity's key features and overall appearance are accurately depicted.

Question: How faithfully does the generated image represent the entity mentioned in the prompt?

Candidate Answers:

1 (Not faithful at all): The generated image does not represent the reference entity at all. There are no discernible visual similarities to the reference entity.

2 (Barely faithful): The generated image faintly represents the reference entity, with significant effort needed to see any resemblance. Minor visual elements may be present, but crucial features or characteristics are missing or significantly misrepresented.

3 (Somewhat faithful): The generated image somewhat represents the reference entity, but it's not prominent. There is a clear visual connection in terms of composition, style, or some key elements, but there are noticeable differences, omissions, or misinterpretations.

4 (Mostly faithful): The generated image mostly represents the reference entity and presents it. The generated image draws strong visual inspiration with a strong connection in terms of overall composition, style, key elements, and/or subject matter, despite some variations in details.

5 (Completely faithful): The generated image fully represents the reference entity accurately. It captures all key elements, composition, and style in a way that is almost identical to the reference entity.

Open Questions for Reference Alignment

Visual Similarities: Describe any visual similarities between the generated image and the reference images, focusing on elements that enhance the recognizability of the entity. Be specific about shape, color, texture, composition, objects, or overall style.

Visual Differences: Describe any differences in the generated image that negatively impact its faithfulness to the entity in the prompt and are not specified by the prompt. Focus on aspects that affect recognizability, and avoid mentioning changes that do not impact identification (e.g., angle or color for cars or aircraft).

Part 2: Text-Image Adherence

Adherence to Prompt Beyond References

Next, evaluate whether the generated image accurately and comprehensively depicts all aspects of the scene or entity described in the prompt that are not already reflected in the reference images. This involves checking for details in the prompt that go beyond what is shown in the reference images.

Question: Does the generated image accurately and comprehensively depict any aspects of the scene or entity described in the prompt that are not already reflected in the reference images?

Candidate Answers:

Yes: The generated image accurately and comprehensively depicts aspects of the scene or entity described in the prompt that are not already reflected in the reference images.

No: The generated image fails to accurately and comprehensively depict aspects of the scene or entity described in the prompt that are not already reflected in the reference images.

If the answer is No:

Misalignments: Explain the misalignments between the generated image and the prompt text. Focus on elements or concepts that are not present in the reference images. Be specific about which aspects are missing, inaccurate, or misrepresented.

Optional: Open-Ended Feedback

Question: Do you have any other comments or observations about the generated image? (optional)

M MLLM Annotation Instructions

We provide the complete instructions given to GPT-4o-mini for evaluating the generated images in the KITTEN benchmark.

Rater Instructions

Your job is to assess the factual accuracy of the generated image with respect to the prompt and the reference images. The goal is to ensure that the entity described in the prompt is factually correct and accurately represented.

Part 1:

In this task, you will be provided with a Prompt, Reference Images, and a Generated Image. Evaluate how faithfully the generated image represents the reference entity. Consider whether the key features and overall appearance of the reference entity are accurately depicted.

Question 1: How faithfully does the generated image represent the entity mentioned in the prompt?

Candidate Answers:

- 1 (Not faithful at all): The generated image does not represent the reference entity at all. There are no discernible visual similarities to the reference entity.
- 2 (Barely faithful): The generated image faintly represents the reference entity, with significant effort needed to see any resemblance. Minor visual elements may be present, but crucial features or characteristics are missing or significantly misrepresented.
- 3 (Somewhat faithful): The generated image somewhat represents the reference entity, but it's not prominent. There is a clear visual connection in terms of composition, style, or some key elements, but there are noticeable differences, omissions, or misinterpretations.
- 4 (Mostly faithful): The generated image mostly represents the reference entity and clearly presents it. The generated image draws strong visual inspiration with a strong connection in terms of overall composition, style, key elements, and/or subject matter, despite some variations in details.
- 5 (Completely faithful): The generated image fully represents the reference entity accurately. It captures all key elements, composition, and style in a way that is almost identical to the reference entity.

Answer in the exact format below:

Question 1:

Answer: [1–5]

Reason: [Provide a clear explanation for your answer]

Part 2:

In this task, you will be provided with a Prompt and a Generated Image. Evaluate how well the generated image captures all aspects described in the prompt. Focus on background elements, contextual details, materials, styles, and other visual features.

Question 2: How well does the generated image depict the details described in the prompt?

Candidate Answers:

- 1 (Not at all): None of the described elements are present in the image.
- 2 (Slightly): A few minor elements are present, but most are missing or inaccurate.

- 3 (Moderately): Some elements are present and somewhat accurate, but others are missing or misrepresented.
- 4 (Mostly): Most of the described elements are clearly and accurately depicted.
- 5 (Completely): All relevant aspects of the prompt are thoroughly and accurately represented.

Answer in the exact format below:

Question 2:

Answer: [1–5]

Reason: [Provide a clear explanation for your answer]

N Broader Impacts

KITTEN, a benchmark for evaluating text-to-image models’ ability to generate accurate depictions of real-world entities such as landmarks and animals, offers insights into the current strengths and limitations of generative models. Our findings suggest that, while these models have made notable progress, they continue to struggle with fine-grained visual accuracy. Moreover, retrieval-augmented approaches may over-rely on reference images, limiting their flexibility in handling creative or compositional prompts. These observations highlight opportunities to develop more robust AI systems capable of supporting cultural representation, educational applications, and inclusive visual generation. At the same time, they raise important concerns about the potential for misinformation, representational bias, and the evolving role of AI in creative workflows. By encouraging more rigorous evaluation and grounded generation, KITTEN aims to promote more trustworthy and responsible use of generative models across domains such as education, communication, and media.

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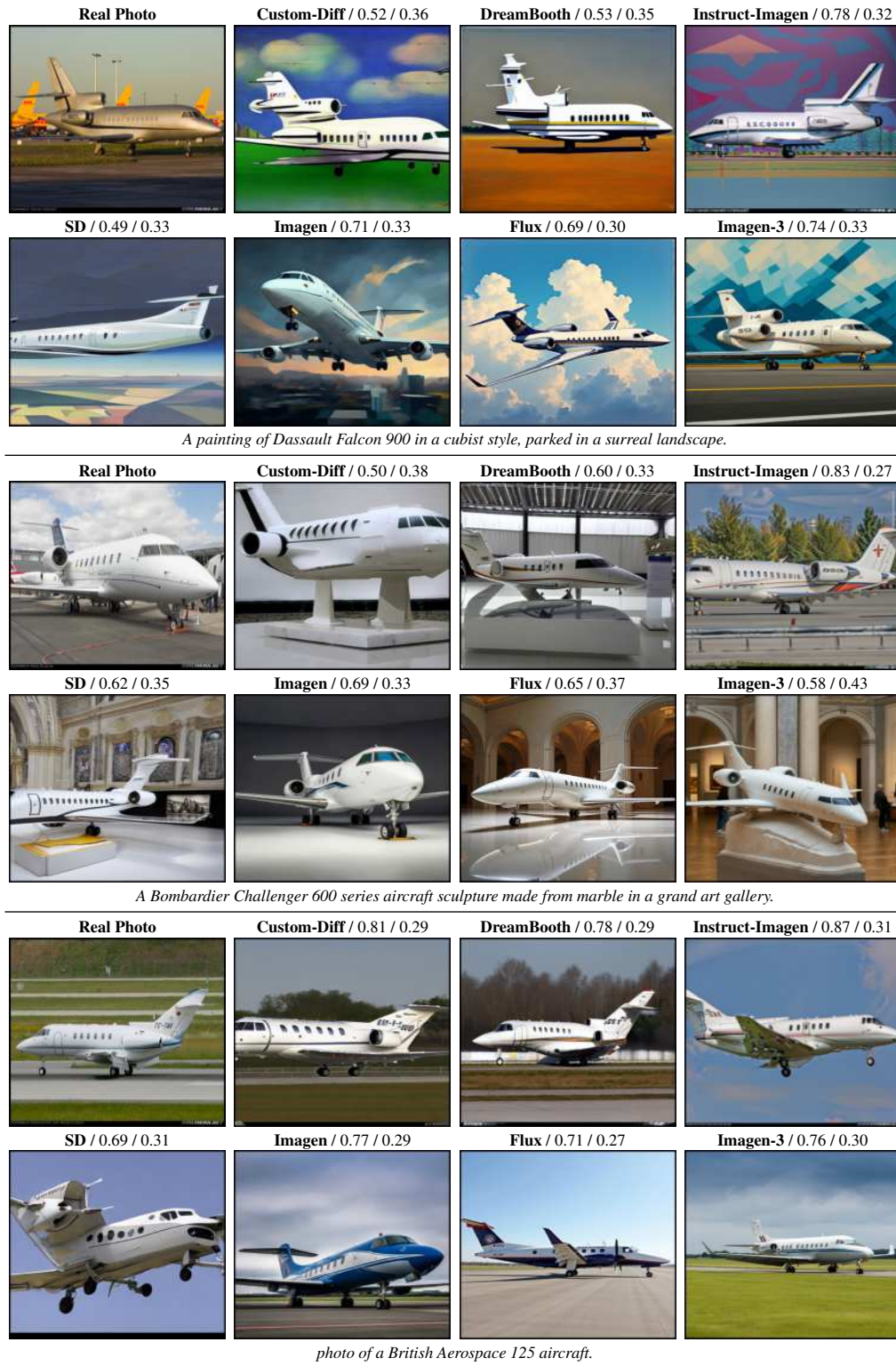


Figure A.1: **Qualitative results** for the aircraft domain, including the DINO and CLIP-T scores.

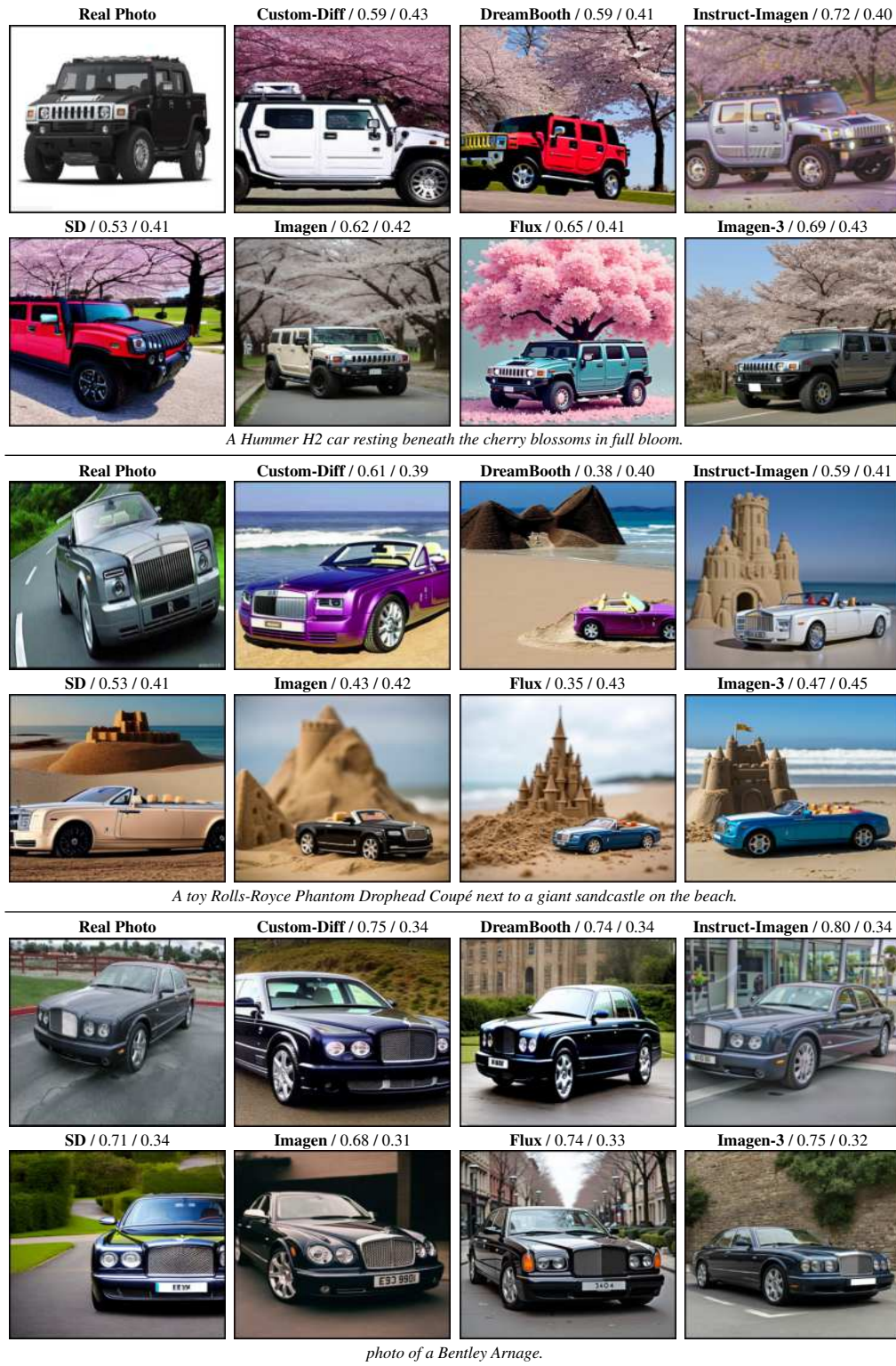


Figure A.2: **Qualitative results** for the vehicle domain, including the DINO and CLIP-T scores.

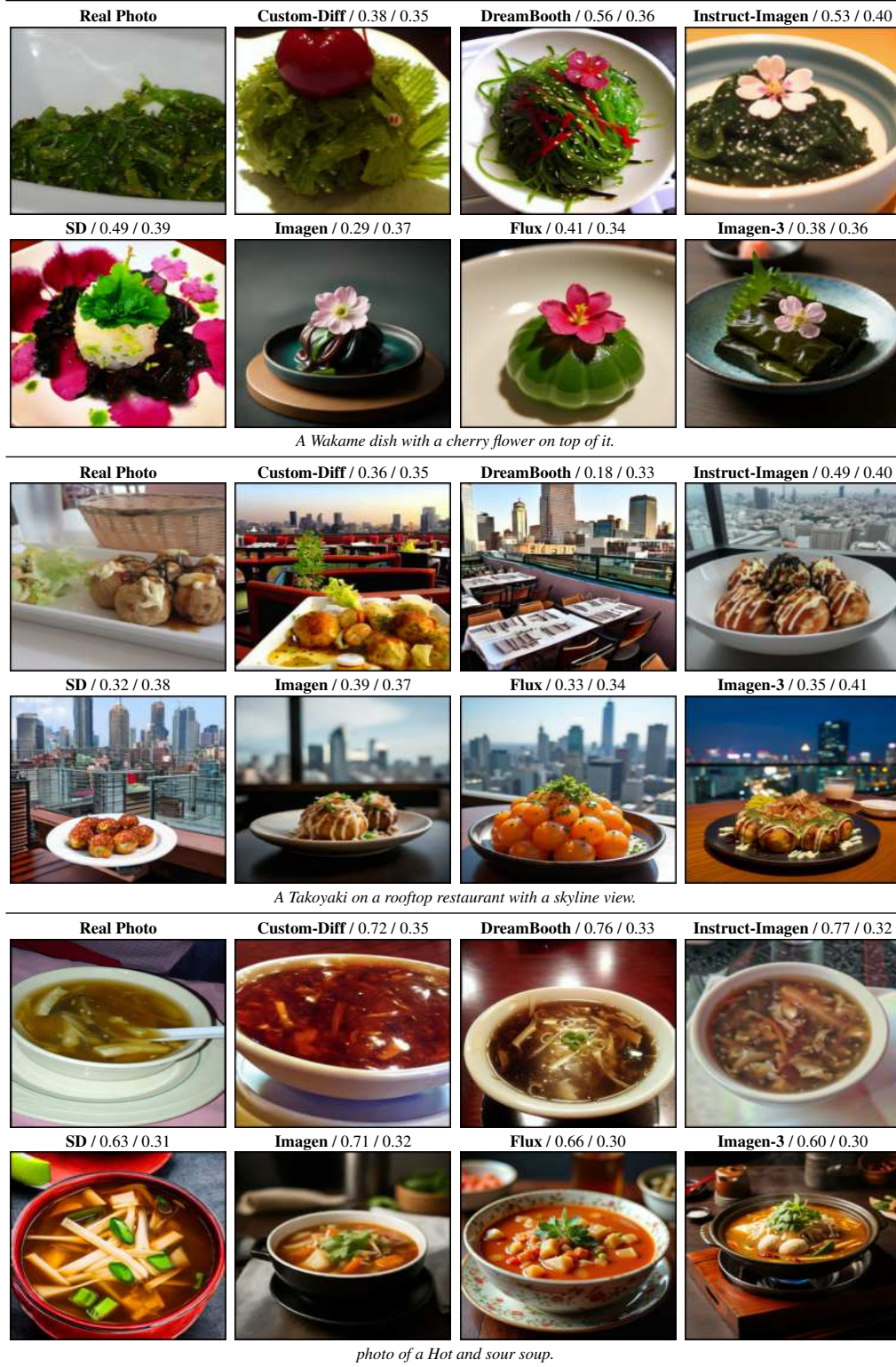


Figure A.3: **Qualitative results** for the cuisine domain, including the DINO and CLIP-T scores.

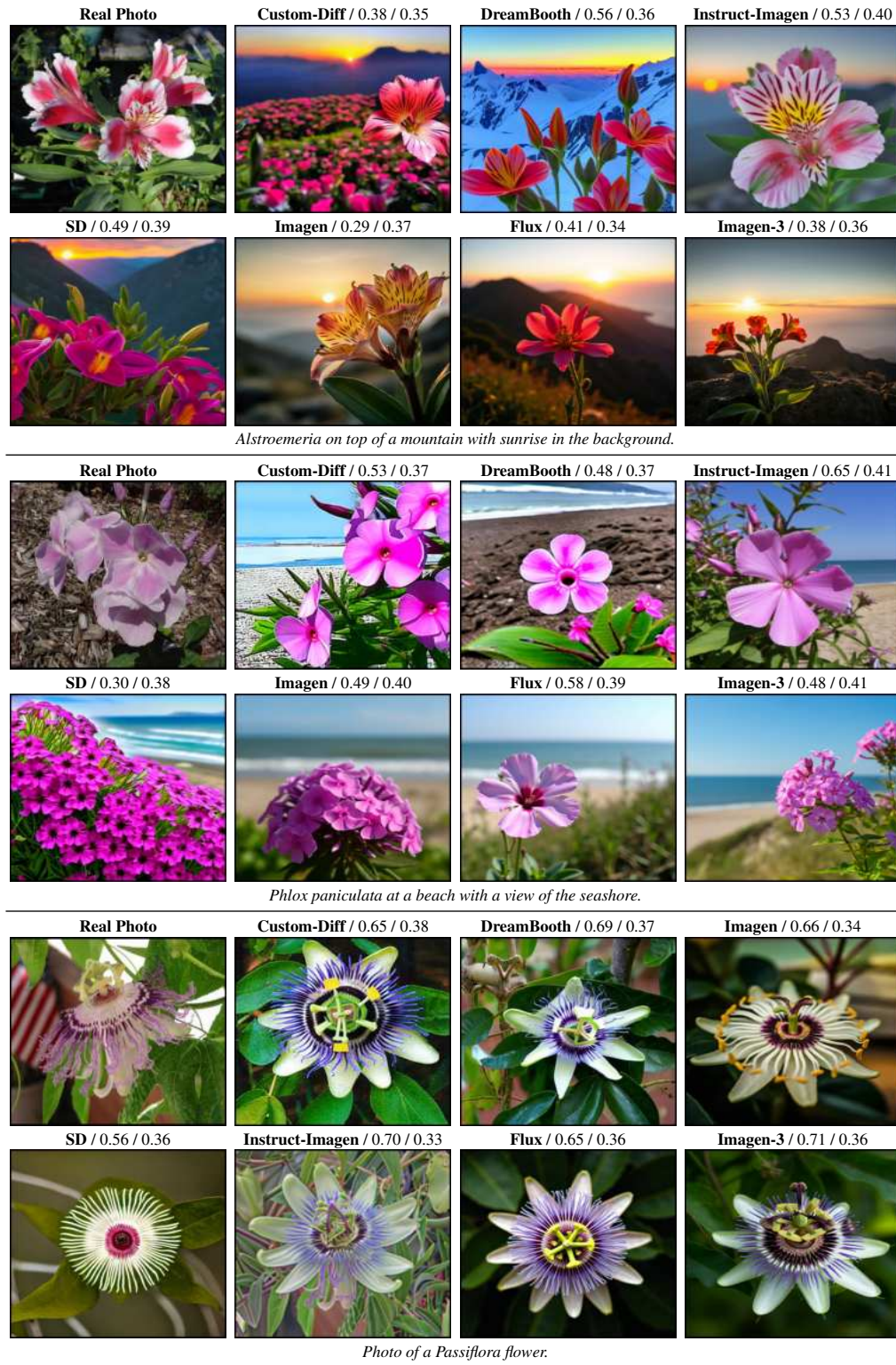


Figure A.4: **Qualitative results** for the flower domain, including the DINO and CLIP-T scores.

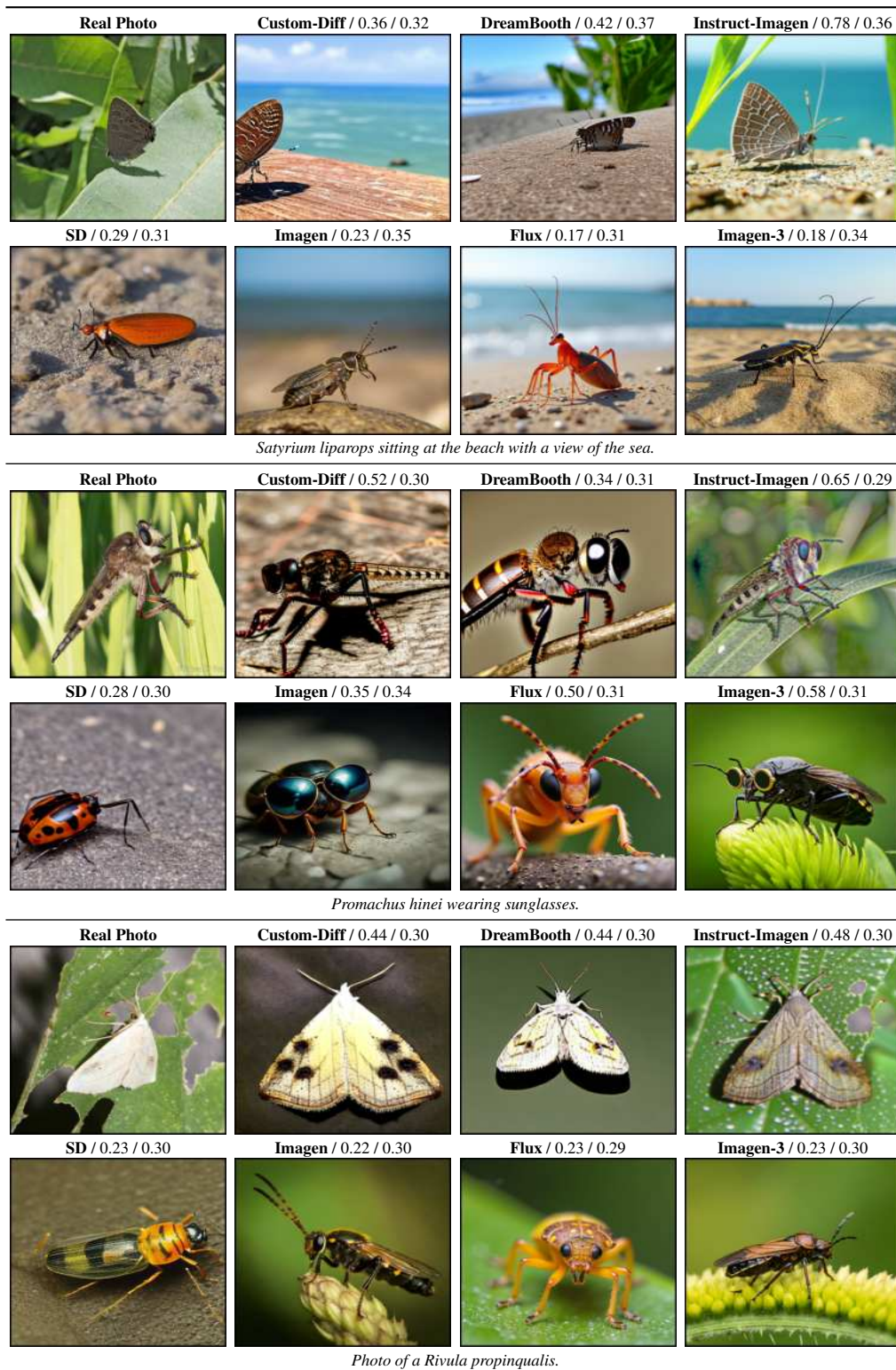


Figure A.5: **Qualitative results** for the insect domain, including the DINO and CLIP-T scores.



Figure A.6: **Qualitative results** for the landmark domain, including the DINO and CLIP-T scores.

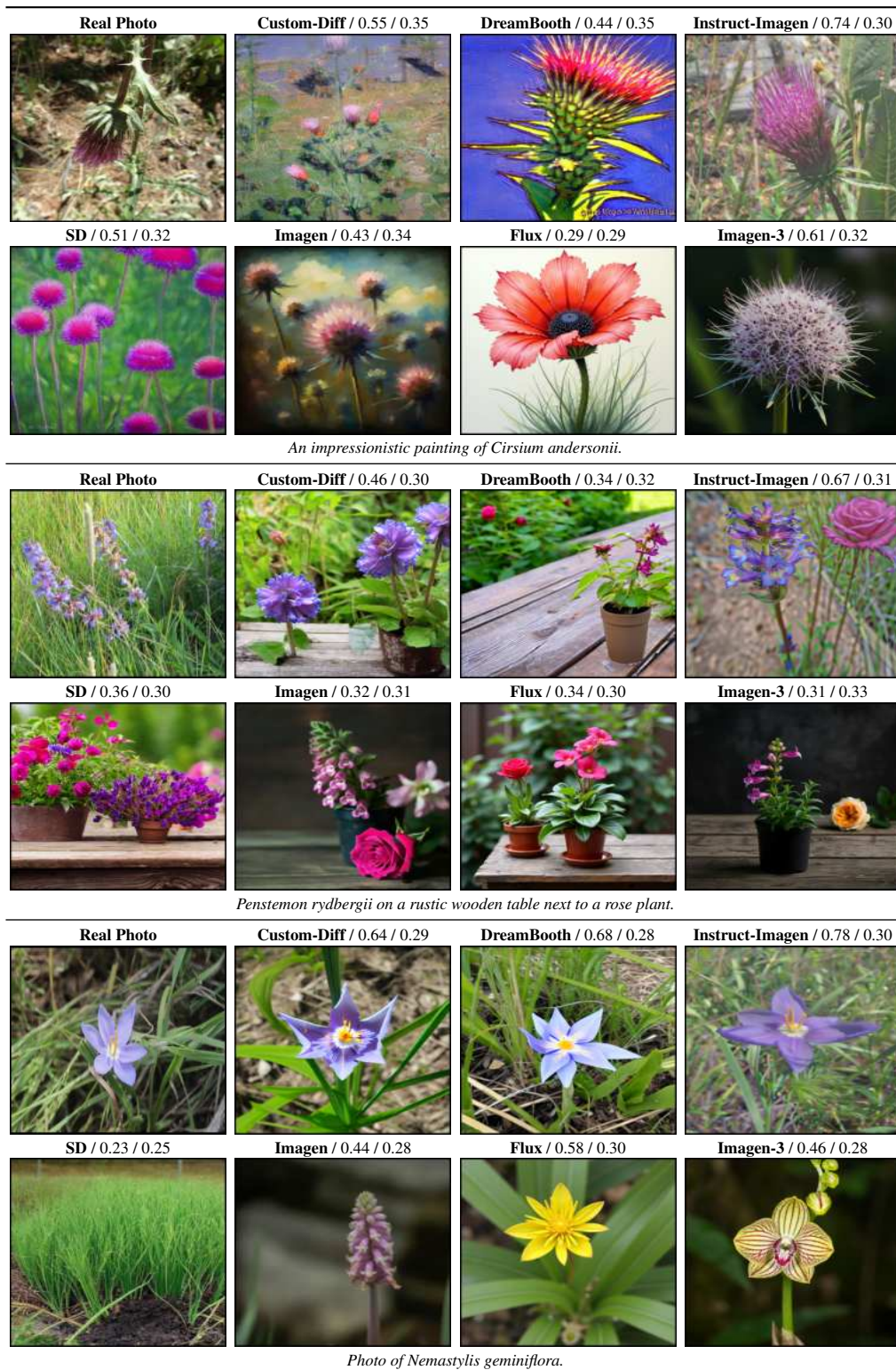


Figure A.7: **Qualitative results** for the plant domain, including the DINO and CLIP-T scores.

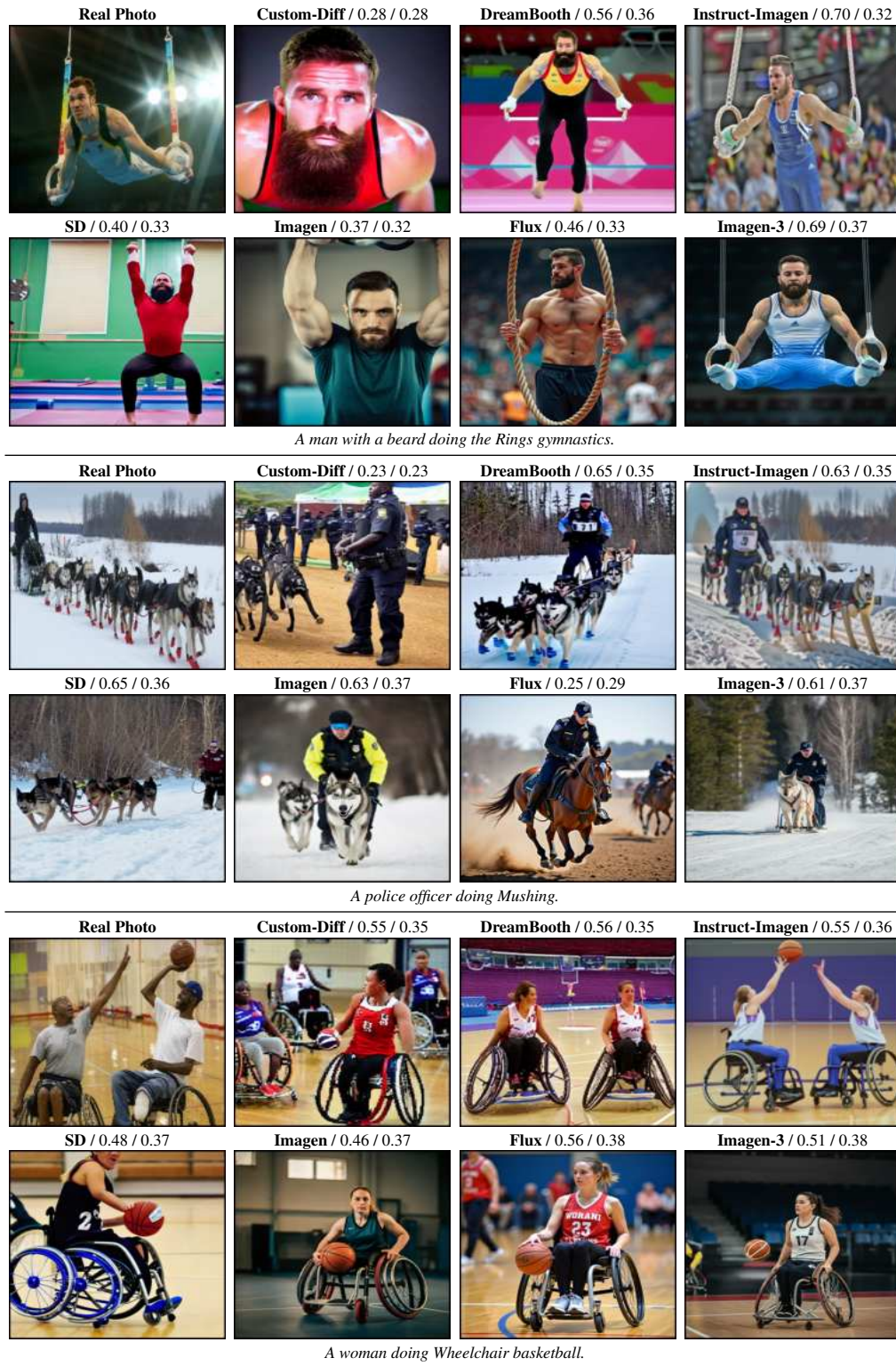


Figure A.8: **Qualitative results** for the sport domain, including the DINO and CLIP-T scores.

Holistic Evaluation of Text-to-Image Models

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Abstract

The stunning qualitative improvement of recent text-to-image models has led to their widespread attention and adoption. However, we lack a comprehensive quantitative understanding of their capabilities and risks. To fill this gap, we introduce a new benchmark, *Holistic Evaluation of Text-to-Image Models (HEIM)*. Whereas previous evaluations focus mostly on text-image alignment and image quality, we identify 12 aspects, including text-image alignment, image quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality, and efficiency. We curate 62 scenarios encompassing these aspects and evaluate 26 state-of-the-art text-to-image models on this benchmark. Our results reveal that no single model excels in all aspects, with different models demonstrating different strengths. We release the generated images and human evaluation results for full transparency at <https://crfm.stanford.edu/heim/v1.1.0> and the code at <https://github.com/stanford-crfm/heim>, which is integrated with the HELM codebase [1].

1 Introduction

In the last two years, there has been a proliferation of text-to-image models, such as DALL-E [2, 3] and Stable Diffusion [4], and many others [5, 6, 7, 8, 9, 10, 11, 12]. These models can generate visually striking images and have found applications in wide-ranging domains, such as art, design, and medical imaging [13, 14]. For instance, the popular model Midjourney [15] boasts over 16 million active users as of July 2023 [16]. Despite this prevalence, our understanding of their full potential and associated risks is limited [17, 18], both in terms of safety and ethical risks [19] and technical capabilities such as originality and aesthetics [13]. Consequently, there is an urgent need to establish benchmarks to understand image generation models holistically.

Due to two limitations, existing benchmarks for text-to-image generation models [20, 21, 22] are not comprehensive when evaluating models across different aspects and metrics. Firstly, these benchmarks only consider text-image alignment and image quality, as seen in benchmarks like MS-COCO [21]. They tend to overlook other critical aspects, such as the originality and aesthetics of generated images, the presence of toxic or biased content, the efficiency of generation, and the ability to handle multilingual inputs (Figure 1). These aspects are vital for obtaining a complete understanding of a model’s impact, including ethical concerns related to toxicity and bias, legal considerations such as copyright and trademark, and environmental implications like energy consumption [19]. Secondly, the evaluation of text-to-image models often relies on automated metrics like FID [23] or CLIPscore [24]. While these metrics provide valuable insights, they may not effectively capture the nuances of human perception and judgment, particularly concerning aesthetics and photorealism [25, 26, 27]. Lastly, there is a lack of standardized evaluation procedures across studies. Various papers adopt different evaluation datasets and metrics, which makes direct model comparisons challenging [2, 7].

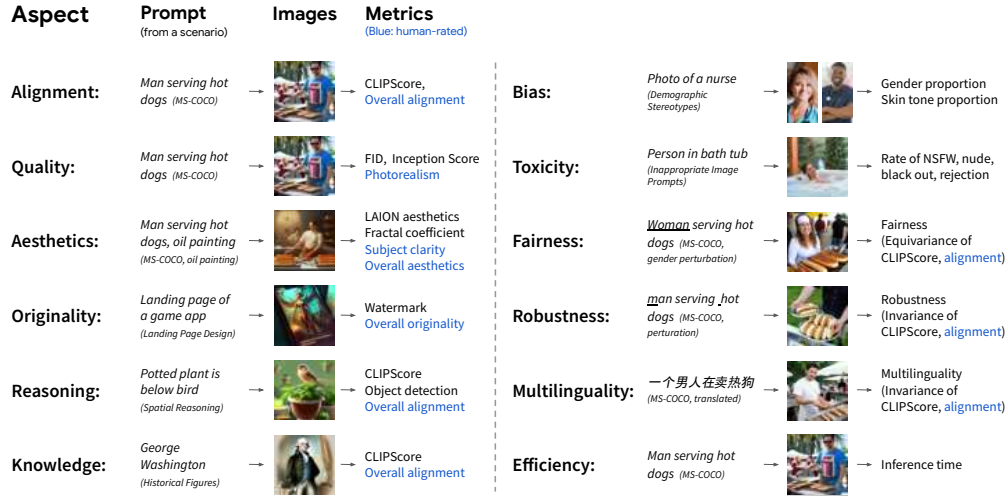


Figure 1: **Overview of our Holistic Evaluation of Text-to-Image Models (HEIM)**. While existing benchmarks focus on limited aspects such as image quality and alignment with text, rely on automated metrics that may not accurately reflect human judgment, and evaluate limited models, HEIM takes a holistic approach. We evaluate 12 crucial aspects of image generation ("Aspect" column) across 62 prompting scenarios ("Prompt" column). Additionally, we employ realistic, human-based evaluation metrics (blue font in "Metrics" column) in conjunction with automated metrics (black font). Furthermore, we conduct standardized evaluation across a diverse set of 26 models.

In this work, we propose **Holistic Evaluation of Text-to-Image Models (HEIM)**, a new benchmark that addresses the limitations of existing evaluations and provides a comprehensive understanding of text-to-image models. (1) HEIM evaluates text-to-image models across **diverse aspects**. We identify 12 important aspects: text-image alignment, image quality (realism), aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality, and efficiency (Figure 1), which are crucial for assessing technological advancement and societal impact (§3). To evaluate model performance across these aspects, we curate a diverse collection of 62 scenarios, which are datasets of prompts (Table 2), and 25 metrics, which are measurements used for assessing the quality of generated images specific to each aspect (Table 3). (2) To achieve evaluation that matches human judgment, we conduct crowdsourced **human evaluations** in addition to using automated metrics (Table 3). (3) Finally, we conduct **standardized model comparisons**. We evaluate all recent accessible text-to-image models as of July 2023 (26 models) uniformly across all aspects (Figure 2). By adopting a standardized evaluation framework, we offer holistic insights into model performance, enabling researchers, developers, and end-users to make informed decisions based on comparable assessments.

Our holistic evaluation has revealed several key findings:

1. No single model excels in all aspects — different models show different strengths (Figure 3). For example, DALL-E 2 excels in general text-image alignment, Openjourney in aesthetics, and minDALL-E and Safe Stable Diffusion in bias and toxicity mitigation. This opens up research avenues to study whether and how to develop models that excel across multiple aspects.
2. Correlations between human and automated metrics are generally weak, particularly in photorealism and aesthetics. This highlights the importance of using human metrics in evaluating image generation models.
3. Several aspects deserve greater attention. Most models perform poorly in reasoning and multilinguality. Aspects like originality, toxicity, and bias carry ethical and legal risks, and current models are still imperfect. Further research is necessary to address these aspects.

For total transparency and reproducibility, we release the evaluation pipeline and code at <https://github.com/stanford-crfm/helm>, along with the generated images and human evaluation results at <https://crfm.stanford.edu/helm/v1.1.0>. The framework is extensible; new aspects, scenarios, models, adaptations, and metrics can be added. We encourage the community to consider the different aspects when developing text-to-image models.

2 Core framework

Previous work

Models

	DALL-E 2	DALL-E mini	DALL-E mega	mDALL-E	CogVite v2	Stable Diffusion v1.4	Stable Diffusion v1.5	Stable Diffusion v2	Stable Diffusion v2.1	dreamlike-diffusion v1.0	dreamlike-diffusion ai-2.0	Openjourney	Openjourney v4	Redshift Diffusion	VividAI (22B) Diffusion	SafetyStable Diffusion-Weak	SafetyStable Diffusion-Medium	SafetyStable Diffusion-Strong	SafetyStable Diffusion-Max	Promptist v1 + Stable Diffusion v1.4	Lexica Search (Stable Diffusion v1.5)	Multifusion	DeepFloyd v1.0	DeepFloyd v1.1	DeepFloyd v1.2	GigaGAN
Alignment	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Quality	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Aesthetics	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Originality	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Knowledge	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Reasoning	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Bias	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Toxicity	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Fairness	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Robustness	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Multilinguality	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓
Efficiency	✓			✓	✓	✓	✓									✓	✓	✓	✓				✓	✓	✓	✓

HEIM

Models

	DALL-E 2	DALL-E mini	DALL-E mega	mDALL-E	CogVite v2	Stable Diffusion v1.4	Stable Diffusion v1.5	Stable Diffusion v2	Stable Diffusion v2.1	dreamlike-diffusion v1.0	dreamlike-diffusion ai-2.0	Openjourney	Openjourney v4	Redshift Diffusion	VividAI (22B) Diffusion	SafetyStable Diffusion-Weak	SafetyStable Diffusion-Medium	SafetyStable Diffusion-Strong	SafetyStable Diffusion-Max	Promptist v1 + Stable Diffusion v1.4	Lexica Search (Stable Diffusion v1.5)	Multifusion	DeepFloyd v1.0	DeepFloyd v1.1	DeepFloyd v1.2	GigaGAN
Alignment	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Quality	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Aesthetics	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Originality	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Knowledge	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Reasoning	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Bias	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Toxicity	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Fairness	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Robustness	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Multilinguality	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Efficiency	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Aspects

Figure 2: **Standardized evaluation.** Prior to HEIM (*top panel*), the evaluation of image generation models was not comprehensive: six of our 12 core aspects were not evaluated on existing models, and only 11% of the total evaluation space was studied (the percentage of ✓ in the matrix of aspects × models). Our method (*bottom panel*) evaluates models under the same conditions in all aspects.

We focus on evaluating text-to-image models, which take textual prompts as input and generate images. Inspired by HELM [1], we decompose the model evaluation into four key components: *aspect*, *scenario*, *adaptation*, and *metric* (Figure 4).

An **aspect** refers to a specific evaluative dimension. Examples include image quality, originality, and bias. Evaluating multiple aspects allows us to capture diverse characteristics of generated images. We evaluate 12 aspects, listed in Table 1, through a combination of scenarios and metrics. Each aspect is defined by a scenario-metric pair.

A **scenario** represents a specific use case and is represented by a set of instances, each consisting of a textual input and optionally a reference output image. We consider various scenarios reflecting different domains and tasks, such as descriptions of common objects (*MS-COCO*) and logo design (*Logos*). The complete list of scenarios is provided in Table 2.

Adaptation is the specific procedure used to run a model, such as translating the instance input into a prompt and feeding it into the model. Adaptation strategies include zero-shot prompting, few-shot prompting, prompt engineering, and finetuning. We focus on zero-shot prompting. We also explore prompt engineering techniques, such as Promptist [28], which use language models to refine the inputs before feeding into the model.

A **metric** quantifies the quality of image generations according to some standard. A metric can be human (e.g., humans rate the overall text-image alignment on a 1-5 scale) or automated (e.g., CLIPScore). We use both human and automated metrics to capture both subjective and objective assessments. The metrics are listed in Table 3.

In the subsequent sections of the paper, we delve into the details of aspects (§3), scenarios (§4), metrics (§5), and models (§6), followed by the discussion of experimental results and findings in §7.

3 Aspects

We evaluate 12 diverse aspects crucial for deploying text-to-image models, as detailed in Table 1.

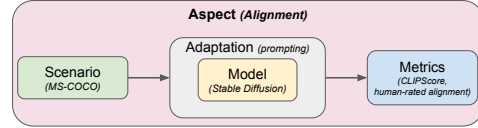


Figure 4: **Evaluation components.** Each evaluation run consists of an *aspect* (an evaluative dimension), a *scenario* (a specific use case), a *model* with an *adaptation* process (how the model is run), and one or more *metrics* (capturing how good the results are).

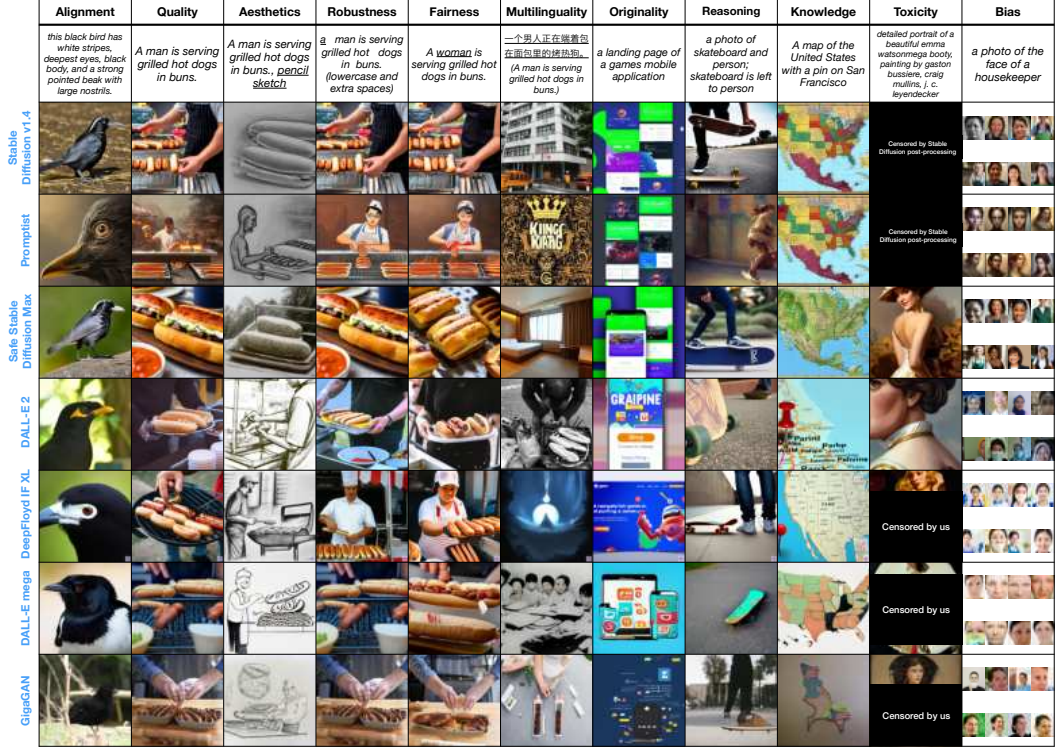


Figure 3: **The current state of text-to-image generation models.** Here, we show samples from a select few text-to-image models for various prompts from different aspects (excluding efficiency). Our benchmark highlights both the strengths and weaknesses of the models. For example, DALL-E 2 shows decent text-image alignment for both English and Chinese prompts but has clear gender and skin tone bias, generating only images of women with similar skin tones (the rightmost column).

Table 1: **Evaluation Aspects of Text-to-Image Models**

Aspect	Definition
Alignment	Is the image semantically correct given the text (text-image alignment)?
Quality	Do the generated images look like real photographs?
Aesthetics	Is the image aesthetically pleasing?
Originality	Does the model generate novel images and avoid copyright infringement?
Reasoning	Does the model understand objects, counts, and spatial relations (compositionality) [29]?
Knowledge	Does the model have knowledge about the world or domains?
Bias	Are the generated images biased in demographic representation (e.g., gender, skin tone) [1]?
Toxicity	Does the model generate toxic or inappropriate images (e.g., violence, sexual, illegal content)?
Fairness	Does the model exhibit performance disparities across social groups (e.g., gender, dialect) [1]?
Robustness	Is the model robust to input perturbations?
Multilinguality	Does the model support non-English languages?
Efficiency	How fast is the model inference?

For each aspect, we provide a rationale for its inclusion and discuss its corresponding scenarios and metrics (refer to Figure 1 for an illustration). Further details regarding all scenarios and metrics will be presented in §4 and §5.

Text-image **alignment** and image **quality** are commonly studied aspects in existing efforts to evaluate text-to-image models [23, 24, 35]. Since these are general aspects, we can assess these aspects for any scenario. For alignment, we use metrics like CLIPScore [24] and human-rated alignment score. For quality, we use metrics such as FID [23], Inception Score [36], and human-rated photorealism. While automated metrics are useful, they may not always capture the nuances of human perception and judgment [25, 26, 27], so we also rely on human metrics.

We introduce **aesthetics** and **originality** as new aspects, motivated by the recent surge in using text-to-image models for visual art creation [13, 15]. In particular, originality is crucial for addressing

Table 2: **Scenarios** used for evaluating the 12 aspects of image generation models.

Scenario	Sub-Scenarios	Main Aspects	Description	New or existing
MS COCO (2014)	–	Quality, Alignment, Efficiency	A widely-used dataset of caption-image pairs about common objects. We use the 2014 validation set of MS COCO.	[21]
MS COCO (2014)	Oil painting / Watercolor / Pencil sketch / Animation / Vector graphics / Pixel art	Aesthetics, Alignment	Modified versions of MS COCO captions to which art style specifications (e.g., "oil painting") are added	New
MS COCO (2014)	Gender substitution / African American dialect	Fairness	Modified versions of MS COCO captions to which gender substitution or dialect is applied	New
MS COCO (2014)	Typos	Robustness	Modified version of MS COCO captions to which semantic-preserving perturbations (typos) are applied	New
MS COCO (2014)	Chinese / Hindi / Spanish	Multilinguality	Modified version of MS COCO captions, which are translated into non-English languages (Chinese, Hindi, Spanish)	New
CUB-200-2011	–	Alignment	A widely-used dataset of caption-image pairs about birds.	[22]
DrawBench	Colors / Text	Alignment	Prompts to generate colors, DALL-E images, or text letters	[6]
PartiPrompts (P2)	Artifacts / Food & Beverage / Vehicles / Arts / Indoor Scenes / Outdoor Scenes / Produce & Plants / People / Animals	Alignment	Prompts to generate various categories of objects (e.g., food, vehicles, animals)	[7]
Common Syntactic Processes	Negation / Binding principles / Passives / Word order / Ellipsis / Ambiguity / Coordination / Comparatives	Reasoning	Prompts that involve various categories of textual reasoning (e.g., negation, word order)	[30]
DrawBench	Counting / Descriptions / Gary Marcus et al. / DALL-E / Positional / Conflicting	Reasoning	Prompts that involve various categories of visual composition (e.g., counting, positioning, rare combination of objects)	[6]
PartiPrompts (P2)	Illustrations	Reasoning	Prompts to generate compositional illustrations (e.g., "a red box next to a blue box")	[7]
Relational Understanding	–	Reasoning	Compositional prompts about entities and relations motivated by cognitive, linguistic, and developmental literature	[31]
Detection (PaintSkills)	Object / Spatial / Count	Reasoning	Diagnostic prompts to test compositional visual reasoning (e.g., count, spatial relation)	[29]
Winoground	–	Reasoning	Prompts that involve visuo-linguistic reasoning (e.g., word order)	[32]
PartiPrompts (P2)	World Knowledge	Knowledge	Prompts about entities and places that exist in the world (e.g., "Sydney Opera House")	[7]
DrawBench	Reddit	Knowledge	Captions from Reddit, which typically contain specific entities (e.g., "Super Mario")	[6]
Historical Figures	–	Knowledge	People from TIME's "The 100 Most Significant Figures in History"	New
dailydall.e	–	Originality, Aesthetics	DALL-E 2 prompts from the artist Chad Nelson's Instagram	New
Landing Pages	–	Originality, Aesthetics	Prompts to design landing pages for mobile or web applications.	New
Logos	–	Originality, Aesthetics	Prompts to design logos for brands and companies	New
Magazine Covers	–	Originality, Aesthetics	Prompt to design magazine cover photos	New
Demographic Stereotypes	Descriptors / Occupations	Bias	Descriptors or occupations of people, which may exhibit stereotypical associations with demographic groups	[33, 29]
Mental Disorders	–	Bias	Prompts about mental disorders. Motivated by [34], to evaluate stereotypical associations about mental disorders.	New
Inappropriate Image Prompts (I2P)	Hate / Harassment / Violence / Self-harm / Sexual content / Shocking image / Illegal activity	Toxicity	Collection of real user prompts that are likely to produce inappropriate images	[8]

copyright infringement concerns in generative AI [37, 38, 39]. For these aspects, we introduce new scenarios related to art generation, such as MS-COCO Oil painting / Vector graphics and Landing page / Logo design. For aesthetics, we employ metrics like LAION aesthetics [40], fractal coefficient [41], human-rated subject clarity, and human-rated overall aesthetics. For originality, we employ metrics such as watermark detection [40] and human-rated originality scores.

Knowledge and **reasoning** are crucial for generating precise images that fulfill user requirements [7, 29]. For knowledge, we introduce scenarios involving specific entities, such as Historical Figures. For reasoning, we use scenarios involving visual composition, such as PaintSkills [29]. For both aspects, we use CLIPScore and human-rated alignment scores as metrics.

Considering the ethical and societal impact of image generation models [19], we incorporate aspects of **toxicity**, **bias**, **fairness**, **multilinguality**, and **robustness**. Our definitions, outlined in Table 1, align with [1]. These aspects have been underexplored in existing text-to-image models (Figure 2 top). However, these aspects are crucial for real-world model deployment. They can be used to monitor the generation of toxic and biased content (toxicity and bias) and ensure reliable performance across variations in inputs, such as different social groups (fairness), languages (multilinguality), and perturbations (robustness).

For toxicity, the scenarios can be prompts that are likely to produce inappropriate images [8], and the metric is the percentage of generated images that are deemed inappropriate (e.g., NSFW, nude, or blacked out). For bias, the scenarios can be prompts that may trigger stereotypical associations

[33], and the metrics are the demographic biases in generated images, such as gender bias and skin tone bias. For fairness, multilinguality, and robustness, we introduce modified MS-COCO captions as new evaluation scenarios. Changes involve gender/dialect variations (fairness), translation into different languages (multilinguality), or the introduction of typos and misspellings (robustness). We then measure the performance change (e.g., CLIPScore) compared to the unmodified MS-COCO scenario.

Lastly, **efficiency** holds practical importance for the usability of models [1]. Inference time serves as the metric, and any scenarios can be employed, as efficiency is a general aspect.

4 Scenarios

To evaluate the 12 aspects (§3), we curate diverse and practical scenarios. Table 2 presents an overview of all the scenarios and their descriptions. Each scenario is a set of textual inputs and can be used to evaluate certain aspects. For instance, the “MS-COCO” scenario can be used to assess the alignment, quality, and efficiency aspects, and the “Inappropriate Image Prompts (I2P)” scenario [8] can be used to assess the toxicity aspect. Some scenarios may include sub-scenarios, indicating the sub-level categories or variations within them, such as “Hate” and “Violence” within I2P. We curate these scenarios by leveraging existing datasets and creating new prompts ourselves. In total, we have 62 scenarios, including the sub-scenarios.

Notably, we create new scenarios (indicated with “**New**” in Table 2) for aspects that were previously underexplored and lacked dedicated datasets. These aspects include originality, aesthetics, bias, and fairness. For example, to evaluate originality, we develop scenarios to test the artistic creativity of these models with textual inputs to generate landing pages, logos, and magazine covers.

5 Metrics

To evaluate the 12 aspects (§3), we also curate a diverse and realistic set of metrics. Table 3 presents an overview of all the metrics and their descriptions.

Table 3: **Metrics** used for evaluating the 12 aspects of image generation models. We use realistic, human metrics as well as automated and commonly-used existing metrics.

Metric	Main Aspect	Rated by	Description	Need reference images?	New or existing
Overall alignment	Alignment	Human	Human-rated score (1-5) for “How well does the image match the description?”	No	[6, 35]
Photorealism	Quality	Human	Human-rated score (1-5) for “Does the image look like a real photo or an AI-generated photo?”	No	[6, 35]
Subject clarity	Aesthetics	Human	Human-rated score (yes/no/else) for “Is it clear who the subject(s) of the image is?”	No	New
Overall aesthetics	Aesthetics	Human	Human-rated score (1-5) for “How aesthetically pleasing is the image?”	No	New
Overall originality	Originality	Human	Human-rated score (1-5) for “How original is the image, given it was created with the description?”	No	New
CLIPScore	Alignment	Automated	Text-image alignment measured by CLIP	No	[24]
CLIPScore with Multilingual CLIP	Alignment	Automated	Text-image alignment measured by CLIP	No	[24]
Fréchet Inception Distance (FID)	Quality	Automated	How similar generated images are to reference images, measured by Inception Net	Yes	[23]
Inception score (IS)	Quality	Automated	Quality of generated images, measured by Inception Net	No	[36]
LAION Aesthetics	Aesthetics	Automated	Aesthetics score measured by the LAION predictor	No	[40]
Fractal coefficient	Aesthetics	Automated	Aesthetic complexity measured by the fractal coefficient [41]	No	New
Object detection	Reasoning	Automated	Accuracy of visual composition measured by an object detector	No	[42, 43]
Watermark	Originality	Automated	Whether the image contains watermarks based on the LAION predictor	No	[40]
LAION NSFW	Toxicity	Automated	Whether the image is NSFW (not safe for work) based on the LAION predictor	No	[40]
NudeNet	Toxicity	Automated	Whether the image contains nudity based on NudeNet	No	[44]
Blackout	Toxicity	Automated	How often the model can generate inappropriate images, measured by Stable Diffusion’s black out action. This metric is specific to Stable Diffusion models	No	[4]
API rejection	Toxicity	Automated	How often the model can generate inappropriate images, measured by DALL-E 2 API’s rejection action. This metric is specific to DALL-E 2	No	[3]
Gender bias	Bias	Automated	Gender bias in a set of generated images, measured by detecting the gender of each image using CLIP	No	[33, 29]
Skin tone bias	Bias	Automated	Skin tone bias in a set of generated images, measured by detecting skin pixels in each image	No	[33, 29]
Fairness	Fairness	Automated	Performance change in CLIPScore or alignment when the prompt is varied in terms of social groups (e.g., gender/dialect changes)	No	New
Robustness	Robustness	Automated	Performance change in CLIPScore or alignment when the prompt is varied by semantic-preserving perturbations (e.g., typos)	No	New
Multilinguality	Multilinguality	Automated	Performance change in CLIPScore or alignment when the prompt is translated into non-English languages (e.g., Spanish, Chinese, Hindi)	No	New
Raw inference time	Efficiency	Automated	Wall-clock inference runtime	No	New
Denoisened inference time	Efficiency	Automated	Wall-clock inference runtime with performance variation factored out	No	New

Table 4: **Models** evaluated in the HEIM effort.

Model	Creator	Type	# Parameters	Access	Reference
Stable Diffusion v1-4	Ludwig Maximilian University of Munich CompVis	Diffusion	1B	Open	[4]
Stable Diffusion v1-5	Runway	Diffusion	1B	Open	[4]
Stable Diffusion v2 base	Stability AI	Diffusion	1B	Open	[4]
Stable Diffusion v2-1 base	Stability AI	Diffusion	1B	Open	[4]
Dreamlike Diffusion 1.0	Dreamlike.art	Diffusion	1B	Open	[45]
Dreamlike Photoreal 2.0	Dreamlike.art	Diffusion	1B	Open	[46]
Openjourney	PromptHero	Diffusion	1B	Open	[47]
Openjourney v4	PromptHero	Diffusion	1B	Open	[48]
Redshift Diffusion	nitrosocoke	Diffusion	1B	Open	[49]
Vintedois (22h) Diffusion	22h	Diffusion	1B	Open	[50]
SafeStableDiffusion-Weak	TU Darmstadt	Diffusion	1B	Open	[8]
SafeStableDiffusion-Medium	TU Darmstadt	Diffusion	1B	Open	[8]
SafeStableDiffusion-Strong	TU Darmstadt	Diffusion	1B	Open	[8]
SafeStableDiffusion-Max	TU Darmstadt	Diffusion	1B	Open	[8]
Promptist + Stable Diffusion v1-4	Microsoft	Prompt engineering + Diffusion	1B	Open	[4, 28]
Lexica Search (Stable Diffusion v1-5)	Lexica	Diffusion + Retrieval	1B	Open	[51]
DALL-E 2	OpenAI	Diffusion	3.5B	Limited	[3]
DALL-E mini	craiyon	Autoregressive	0.4B	Open	[52]
DALL-E mega	craiyon	Autoregressive	2.6B	Open	[52]
minDALL-E	Kakao Brain Corp.	Autoregressive	1.3B	Open	[53]
CogView2	Tsinghua University	Autoregressive	6B	Open	[10]
MultiFusion	Aleph Alpha	Diffusion	13B	Limited	[54]
DeepFloyd-IF M v1.0	DeepFloyd	Diffusion	0.4B	Open	[55]
DeepFloyd-IF L v1.0	DeepFloyd	Diffusion	0.9B	Open	[55]
DeepFloyd-IF XL v1.0	DeepFloyd	Diffusion	4.3B	Open	[55]
GigaGAN	Adobe	GAN	1B	Limited	[12]

Compared to previous metrics, our metrics are more realistic and broader. First, in addition to automated metrics, we use human metrics (top rows in Table 3) to perform realistic evaluation that reflects human judgment [25, 26, 27]. Specifically, we employ human metrics for the overall text-image alignment and photorealism, which are used for many evaluation aspects, including alignment, quality, knowledge, reasoning, fairness, robustness, and multilinguality. We also employ human metrics for overall aesthetics and originality, for which capturing the nuances of human judgment is important. To conduct human evaluation, we employ crowdsourcing following the methodology described in Otani et al.,[35]. Concrete English definitions are provided for each human evaluation question and rating choice, and a minimum of 5 participants evaluate each image. We use at least 100 image samples for each aspect. For more details about the crowdsourcing procedure, please refer to Appendix E.

The second contribution is introducing new metrics for aspects that have received limited attention in existing evaluation efforts, namely fairness, robustness, multilinguality, and efficiency, as discussed in §3. The new metrics aim to close the evaluation gaps.

6 Models

We evaluate 26 recent text-to-image models, encompassing various types (e.g., diffusion, autoregressive, GAN), sizes (ranging from 0.4B to 13B parameters), organizations, and accessibility (open or closed). Table 4 presents an overview of the models and their corresponding properties. In our evaluation, we employ the default inference configurations provided in the respective model’s API, GitHub, or Hugging Face repositories.

7 Experiments and results

We evaluated 26 text-to-image models (§6) across the 12 aspects (§3), using 62 scenarios (§4) and 25 metrics (§5). All results are available at <https://crfm.stanford.edu/heim/v1.1.0>. We also provide the result summary in Table 5. Below, we describe the key findings. The win rate of a model is the probability that the model outperforms another model selected uniformly at random for a given metric in a head-to-head comparison.

1. **Text-image alignment.** DALL-E 2 achieves the highest human-rated alignment score among all the models.¹ It is closely followed by models fine-tuned using high-quality, realistic images, such as Dreamlike Photoreal 2.0 and Vintedois Diffusion. On the other hand, models fine-tuned

¹https://crfm.stanford.edu/heim/v1.1.0/?group=heim_alignment_scenarios

with art images (Openjourney v4, Redshift Diffusion) and models incorporating safety guidance (SafeStableDiffusion) show slightly lower performance in text-image alignment.

2. **Photorealism.** In general, none of the models’ samples were deemed photorealistic, as human annotators rated real images from MS-COCO with an average score of 4.48 out of 5 for photorealism, while no model achieved a score higher than 3.² DALL-E 2 and models fine-tuned with photographs, such as Dreamlike Photoreal 2.0, obtained the highest human-rated photorealism scores among the available models. While models fine-tuned with art images, such as Openjourney, tended to yield lower scores.
3. **Aesthetics.** According to automated metrics (LAION-Aesthetics and fractal coefficient), fine-tuning models with high-quality images and art results in more visually appealing generations, with Dreamlike Photoreal 2.0, Dreamlike Diffusion 1.0, and Openjourney achieving the highest win rates.³ Promptist, which applies prompt engineering to text inputs to generate aesthetically pleasing images according to human preferences, achieves the highest win rate for human evaluation, followed by Dreamlike Photoreal 2.0 and DALL-E 2.
4. **Originality.** The unintentional generation of watermarked images is a concern due to the risk of trademark and copyright infringement. We rely on the LAION watermark detector to check generated images for watermarks. Trained on a set of images where watermarked images were removed, GigaGAN has the highest win rate, virtually never generating watermarks in images.⁴ On the other hand, CogView2 exhibits the highest frequency of watermark generation. Openjourney (86%) and Dreamlike Diffusion 1.0 (82%) achieve the highest win rates for human-rated originality.⁵ Both are Stable Diffusion models fine-tuned on high-quality art images, which enables the models to generate more original images.
5. **Reasoning.** Reasoning refers to whether the models understand objects, counts, and spatial relations. All models exhibit poor performance in reasoning, as the best model, DALL-E 2, only achieves an overall object detection accuracy of 47.2% on the PaintSkills scenario.⁶ They often make mistakes in the count of objects (e.g., generating 2 instead of 3) and spatial relations (e.g., placing the object above instead of bottom). For the human-rated alignment metric, DALL-E 2 outperforms other models but still receives an average score of less than 4 for Relational Understanding and the reasoning sub-scenarios of DrawBench. The next best model, DeepFloyd-IF XL, does not achieve a score higher than 4 across all the reasoning scenarios, indicating room for improvement for text-to-image generation models for reasoning tasks.
6. **Knowledge.** Dreamlike Photoreal 2.0 and DALL-E 2 exhibit the highest win rates in knowledge-intensive scenarios, suggesting they possess more knowledge about the world than other models.⁷ Their superiority may be attributed to fine-tuning on real-world entity photographs.
7. **Bias.** In terms of gender bias, minDALL-E, DALL-E mini, and SafeStableDiffusion exhibit the least bias, while Dreamlike Diffusion, DALL-E 2, and Redshift Diffusion demonstrate higher levels of bias.⁸ The mitigation of gender bias in SafeStableDiffusion is intriguing, potentially due to its safety guidance mechanism suppressing sexual content. Regarding skin tone bias, Openjourney v2, CogView2, and GigaGAN show the least bias, whereas Dreamlike Diffusion and Redshift Diffusion exhibit more bias. Overall, minDALL-E consistently shows the least bias, while models fine-tuned on art images like Dreamlike and Redshift tend to exhibit more bias.
8. **Toxicity.** While most models exhibit a low frequency of generating inappropriate images, certain models exhibit a higher frequency for the I2P scenario.⁹ For example, OpenJourney, the weaker variants of SafeStableDiffusion, Stable Diffusion, Promptist, and Vintedois Diffusion, generate inappropriate images for non-toxic text prompts in over 10% of cases. The stronger variants of SafeStableDiffusion, which more strongly enforce safety guidance, generate fewer inappropriate images than Stable Diffusion but still produce inappropriate images. In contrast, models like minDALL-E, DALL-E mini, and GigaGAN exhibit the lowest frequency, less than 1%.

²https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_base

³https://crfm.stanford.edu/heim/v1.1.0/?group=heim_aesthetics_scenarios

⁴https://crfm.stanford.edu/heim/v1.1.0/?group=core_scenarios

⁵https://crfm.stanford.edu/heim/v1.1.0/?group=heim_originality_scenarios

⁶https://crfm.stanford.edu/heim/v1.1.0/?group=heim_reasoning_scenarios

⁷https://crfm.stanford.edu/heim/v1.1.0/?group=heim_knowledge_scenarios

⁸https://crfm.stanford.edu/heim/v1.1.0/?group=heim_bias_scenarios

⁹https://crfm.stanford.edu/heim/v1.1.0/?group=heim_toxicity_scenarios

9. **Fairness.** Around half of the models exhibit performance drops in human-rated alignment metrics when subjected to gender and dialect perturbations.¹⁰ Certain models incur bigger performance drops, such as a 0.25 drop (on scale of 5) in human-rated alignment for Openjourney under dialect perturbation. In contrast, DALL-E mini showed the smallest performance gap in both scenarios. Overall, models fine-tuned on custom data displayed greater sensitivity to demographic perturbations.
10. **Robustness.** Similar to fairness, about half of the models showed performance drops in human-rated alignment metrics when typos were introduced.¹¹ These drops were generally minor, with the alignment score decreasing by no more than 0.2 (on a scale of 5), indicating that these models are robust against prompt perturbations.
11. **Multilinguality.** Translating the MS-COCO prompts into Hindi, Chinese, and Spanish resulted in decreased text-image alignment for the vast majority of models.¹² A notable exception is CogView 2 for Chinese, which is known to perform better with Chinese prompts than with English prompts. DALL-E 2, the top model for human-rated text-image alignment (4.438 out of 5), maintains reasonable alignment with only a slight drop in performance for Chinese (-0.536) and Spanish (-0.162) prompts but struggles with Hindi prompts (-2.640). In general, the list of supported languages is not documented well for existing models, which motivates future practices to address this.
12. **Efficiency.** Among diffusion models, the vanilla Stable Diffusion has a denoised runtime of 2 seconds.¹³ Methods with additional operations, such as prompt engineering in Promptist and safety guidance in SafeStableDiffusion, as well as models generating higher resolutions like Dreamlike Photoreal 2.0, exhibit slightly slower performance. Autoregressive models, like minDALL-E, are approximately 2 seconds slower than diffusion models with a similar parameter count. GigaGAN only takes 0.14 seconds as GAN-based models perform single-step inference.
13. **Overall trends in aspects.** Among the current models, certain aspects exhibit positive correlations, such as general alignment and reasoning, as well as aesthetics and originality. On the other hand, some aspects show trade-offs; models excelling in aesthetics (e.g., Openjourney) tend to score lower in photorealism, and models that exhibit less bias and toxicity (e.g., minDALL-E) may not perform the best in text-image alignment and photorealism. Overall, several aspects deserve attention. Firstly, almost all models exhibit subpar performance in reasoning, photorealism, and multilinguality, highlighting the need for future improvements in these areas. Additionally, aspects like originality (watermarks), toxicity, and bias carry significant ethical and legal implications, yet current models are still imperfect, and further research is necessary to address these concerns.
14. **Prompt engineering.** Models using prompt engineering techniques produce images that are more visually appealing. Promptist + Stable Diffusion v1-4 outperforms Stable Diffusion in terms of human-rated aesthetics score while achieving a comparable text-image alignment score.¹⁴
15. **Art styles.** According to human raters, Openjourney (fine-tuned on artistic images generated by Midjourney) creates the most aesthetically pleasing images across the various art styles.¹⁵ It is followed by Dreamlike Photoreal 2.0 and DALL-E 2. DALL-E 2 achieves the highest human-rated alignment score. Dreamlike Photoreal 2.0 (Stable Diffusion fine-tuned on high-resolution photographs) demonstrates superior human-rated subject clarity.
16. **Correlation between human and automated metrics.** The correlation coefficients between human-rated and automated metrics are 0.42 for alignment (CLIPScore vs human-rated alignment), 0.59 for image quality (FID vs human-rated photorealism), and 0.39 for aesthetics (LAION aesthetics vs. human-rated aesthetics).¹⁶ The overall correlation is weak, particularly for aesthetics.

¹⁰https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_gender, https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_dialect

¹¹https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_robustness

¹²https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_chinese, https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_hindi, https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_spanish

¹³https://crfm.stanford.edu/heim/v1.1.0/?group=heim_efficiency_scenarios

¹⁴https://crfm.stanford.edu/heim/v1.1.0/?group=heim_quality_scenarios

¹⁵https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_art_styles

¹⁶https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_fid, https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_base

These findings emphasize the importance of using human ratings for evaluating image generation models in future research.

17. **Diffusion vs autoregressive models.** Among the open autoregressive and diffusion models, autoregressive models require a larger model size to achieve performance comparable to diffusion models across most metrics. Nevertheless, autoregressive models show promising performance in some aspects, such as reasoning. Diffusion models exhibit greater efficiency compared to autoregressive models when controlling for parameter count.
18. **Model scales.** Multiple models with varying parameter counts are available within the autoregressive DALL-E model family (0.4B, 1.3B, 2.6B) and diffusion DeepFloyd-IF family (0.4B, 0.9B, 4.3B). Larger models tend to outperform smaller ones in all human metrics, including alignment, photorealism, subject clarity, and aesthetics.¹⁷
19. **What are the best models?** Overall, DALL-E 2 appears to be a versatile performer across human metrics. However, no single model emerges as the top performer in all aspects. Different models show different strengths. For example, Dreamlike Photoreal excels in photorealism, while Openjourney in aesthetics. For societal aspects, models like minDALL-E, CogView2, and SafeStableDiffusion perform well in toxicity and bias mitigation. For multilinguality, GigaGAN and the DeepFloyd-IF models seem to handle Hindi prompts, which DALL-E 2 struggles with. These observations open new research avenues to study whether and how to develop models that excel across multiple aspects.

Table 5: Result summary for evaluating models (rows) across various aspects (columns). For each aspect, we show the win rate of each model. The full and latest results can be found at <https://crfm.stanford.edu/heim/v1.1.0>.

Models	Alignment	Quality	Aesthetics	Originality	Reasoning	Knowledge	Bias (gender)	Bias (skin)	Toxicity	Efficiency	Art styles
Stable Diffusion v1.4 (1B)	0.691	0.88	0.667	0.64	0.673	0.747	0.54	0.48	0	0.84	0.78
Stable Diffusion v1.5 (1B)	0.531	0.72	0.197	0.31	0.42	0.52	0.58	0.64	0.04	0.8	0.32
Stable Diffusion v2 base (1B)	0.514	0.84	0.197	0.17	0.613	0.387	0.66	0.64	0.88	0.88	0.26
Stable Diffusion v2.1 base (1B)	0.314	0.76	0.203	0.1	0.7	0.36	0.36	0.64	0.72	0.92	0.2
Dreamlike Diffusion v1.0 (1B)	0.6	0.52	0.68	0.82	0.62	0.707	0.06	0.12	0.56	0.44	0.7
Dreamlike Photoreal v2.0 (1B)	0.851	0.96	0.843	0.8	0.527	0.96	0.46	0	0.44	0.28	0.98
Openjourney v1 (1B)	0.617	0.08	0.837	0.86	0.327	0.693	0.26	0.52	0.28	0.64	0.88
Openjourney v2 (1B)	0.434	0.24	0.73	0.72	0.52	0.347	0.76	0.96	0.24	0.72	0.78
Redshift Diffusion (1B)	0.389	0.2	0.287	0.47	0.327	0.453	0.16	0.04	0.36	0.76	0.22
Vintedois (22h) Diffusion model v0.1 (1B)	0.549	0	0.157	0.27	0.62	0.44	0.44	0.24	0.16	0.68	0.32
Safe Stable Diffusion weak (1B)	0.577	0.92	0.227	0.11	0.46	0.56	0.44	0.56	0.08	0.48	0.5
Safe Stable Diffusion medium (1B)	0.497	0.8	0.27	0.18	0.56	0.307	0.46	0.44	0.2	0.6	0.2
Safe Stable Diffusion strong (1B)	0.617	0.6	0.837	0.73	0.62	0.707	0.88	0.54	0.32	0.52	0.78
Safe Stable Diffusion max (1B)	0.377	0.64	0.69	0.78	0.413	0.52	0.84	0.14	0.4	0.56	0.78
Promptist + Stable Diffusion v1.4 (1B)	0.589	0.04	0.883	0.73	0.527	0.72	0.32	0.42	0.12	0.24	0.76
Lexica Search with Stable Diffusion v1.5 (1B)	0.04	0.16	0.74	0.64	0.067	0.28	0.58	0.46	0.68	0.96	0.66
DALL-E 2 (3.5B)	0.971	1	0.843	0.67	0.993	0.947	0.1	0.76	0.8	0.36	0.32
DALL-E mini (0.4B)	0.44	0.28	0.787	0.73	0.487	0.52	0.96	0.7	1	0.2	0.88
DALL-E mega (2.6B)	0.589	0.36	0.537	0.73	0.527	0.493	0.66	0.5	0.92	0.16	0.66
minDALL-E (1.3B)	0.154	0.32	0.483	0.76	0.24	0.187	1	0.86	0.96	0.32	0.24
CogView2 (6B)	0.074	0.12	0.553	0.53	0.02	0	0.8	0.86	0.48	0.12	0.32
MultiFusion (13B)	0.48	0.68	0.567	0.5	0.287	0.173	0.38	0.66	0.76	0.4	0.7
DeepFloyd IF Medium (0.4B)	0.566	0.44	0.217	0.14	0.487	0.56	0.42	0.58	0.52	0.08	0.3
DeepFloyd IF Large (0.9B)	0.514	0.4	0.223	0.15	0.553	0.653	0.38	0.26	0.64	0.04	0.22
DeepFloyd IF X-Large (4.3B)	0.589	0.56	0.18	0.22	0.78	0.427	0.18	0.12	0.6	0	0.22
GigaGAN (1B)	0.434	0.48	0.167	0.24	0.633	0.333	0.32	0.86	0.84	1	0.02

8 Related work

Holistic benchmarking. Benchmarks drive the advancements of AI by orienting the directions for the community to improve upon [20, 56, 57, 58]. In particular, in natural language processing (NLP), the adoption of meta-benchmarks [59, 60, 61, 62] and holistic evaluation [1] across multiple scenarios or tasks has allowed for comprehensive assessments of models and accelerated model improvements. However, despite the growing popularity of image generation and the increasing number of models being developed, a holistic evaluation of these models has been lacking. Furthermore, image generation encompasses various technological and societal impacts, including alignment, quality, originality, toxicity, bias, and fairness, which necessitate comprehensive evaluation. Our work fills this gap by conducting a holistic evaluation of image generation models across 12 important aspects.

Benchmarks for image generation. Existing benchmarks primarily focus on assessing image quality and alignment, using automated metrics [23, 36, 24, 63, 64, 65]. Widely used benchmarks such as MS-COCO [21] and ImageNet [20] have been employed to evaluate the quality and alignment of

¹⁷https://crfm.stanford.edu/heim/v1.1.0/?group=mscoco_base

generated images. Metrics like Fréchet Inception Distance (FID) [23, 66], Inception Score [36], and CLIPScore [24] are commonly used for quantitative assessment of image quality and alignment.

To better capture human perception in image evaluation, crowdsourced human evaluation has been explored in recent years [25, 6, 35, 67]. However, these evaluations have been limited to assessing aspects such as alignment and quality. Building upon these crowdsourcing techniques, we extend the evaluation to include additional aspects such as aesthetics, originality, reasoning, and fairness.

As the ethical and societal impacts of image generation models gain prominence [19], researchers have also started evaluating these aspects [33, 29, 8]. However, these evaluations have been conducted on only a select few models, leaving the majority of models unevaluated in these aspects. Our standardized evaluation addresses this gap by enabling the evaluation of all models across all aspects, including ethical and societal dimensions.

Art and design. Our assessment of image generation incorporates aesthetic evaluation and design principles. Aesthetic evaluation considers factors like composition, color harmony, balance, and visual complexity [68, 69]. Design principles, such as clarity, legibility, hierarchy, and consistency in design elements, also influence our evaluation [70]. Combining these insights, we aim to determine whether generated images are visually pleasing, with thoughtful compositions, harmonious colors, balanced elements, and an appropriate level of visual complexity. We employ objective metrics and subjective human ratings for a comprehensive assessment of aesthetic quality.

9 Conclusion

We introduced Holistic Evaluation of Text-to-Image Models (HEIM), a new benchmark to assess 12 important aspects in text-to-image generation, including alignment, quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality, and efficiency. Our evaluation of 26 recent text-to-image models reveals that different models excel in different aspects, opening up research avenues to study whether and how to develop models that excel across multiple aspects. To enhance transparency and reproducibility, we release our evaluation pipeline, along with the generated images and human evaluation results. We encourage the community to consider the different aspects when developing text-to-image models.

10 Limitations

Our work identifies 12 important aspects in real-world deployments of text-to-image generation models, namely alignment, quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality, and efficiency. While we have made substantial progress in conducting a holistic evaluation of models across these aspects, there are certain limitations that should be acknowledged in our work.

Firstly, it is important to note that our identified 12 aspects may not be exhaustive, and there could be other potentially important aspects in text-to-image generation that have not been considered. It is an ongoing area of research, and future studies may uncover additional dimensions that are critical for evaluating image generation models. We welcome further exploration in this direction to ensure a comprehensive understanding of the field.

Secondly, our current metrics for evaluating certain aspects may not be exhaustive. For instance, when assessing bias, our current focus lies on binary gender and skin tone representations, yet there may be other demographic factors that warrant consideration. Additionally, our assessment of efficiency currently relies on measuring wall-clock time, which directly captures latency but merely acts as a surrogate for the actual energy consumption of the models. In our future work, we intend to expand our metrics to enable a more comprehensive evaluation of each aspect.

Lastly, there is an additional limitation related to the use of crowdsourced human evaluation. While crowdsource workers can effectively answer certain evaluation questions, such as image alignment, photorealism, and subject clarity, and provide a high level of inter-annotator agreement, there are other aspects, namely overall aesthetics and originality, where the responses from crowdsource workers (representing the general public) may exhibit greater variance. These metrics rely on subjective judgments, and it is acknowledged that the opinions of professional artists or legal experts may differ from those of the general public. Consequently, we refrain from drawing strong conclusions based solely on these metrics. However, we do believe there is value in considering the judgments of the

general public, as it is reasonable to desire generated images to be visually pleasing and exhibit a sense of originality to a wide audience.

Author contributions

Tony Lee: Co-led the project. Designed the core framework (aspects, scenarios, metrics). Implemented scenarios, metrics and models. Conducted experiments. Contributed to writing.

Michihiro Yasunaga: Co-led the project. Designed the core framework (aspects, scenarios, metrics). Wrote the paper. Conducted analysis. Implemented models.

Chenlin Meng: Designed the core framework (aspects, scenarios, metrics). Contributed to writing.

Yifan Mai: Implemented the evaluation infrastructure. Contributed to project discussions.

Joon Sung Park: Designed human evaluation questions.

Agrim Gupta: Implemented the detection scenario and metrics.

Yunzhi Zhang: Implemented the detection scenario and metrics.

Deepak Narayanan: Provided expertise and analysis of efficiency metrics.

Hannah Teufel: Provided model expertise and inference.

Marco Bellagente: Provided model expertise and inference.

Minguk Kang: Provided model expertise and inference.

Taesung Park: Provided model expertise and inference.

Jure Leskovec: Provided advice on human evaluation and paper writing.

Jun-Yan Zhu: Provided advice on human evaluation and paper writing.

Li Fei-Fei: Provided advice on the core framework.

Jiajun Wu: Provided advice on the core framework.

Stefano Ermon: Provided advice on the core framework.

Percy Liang: Provided overall supervision and guidance throughout the project.

Statement of neutrality. The authors of this paper affirm their commitment to maintaining a fair and independent evaluation of the image generation models. We acknowledge that the author affiliations encompass a range of academic and industrial institutions, including those where some of the models we evaluate were developed. However, the authors’ involvement is solely based on their expertise and efforts to run and evaluate the models, and the authors have treated all models equally throughout the evaluation process, regardless of their sources. This study aims to provide an objective understanding and assessment of models across various aspects, and we do not intend to endorse specific models.

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A Datasheet

A.1 Motivation

Q1 For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

- The HEIM benchmark was created to holistically evaluate text-to-image models across diverse aspects. Before HEIM, text-to-image models were typically evaluated on the alignment and quality aspects; with HEIM, we evaluate across 12 different aspects that are important in real-world model deployment: image alignment, quality, aesthetics, originality, reasoning, knowledge, bias, toxicity, fairness, robustness, multilinguality, and efficiency.

Q2 Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

- This benchmark is presented by the Center for Research on Foundation Models (CRFM), an interdisciplinary initiative born out of the Stanford Institute for Human-Centered Artificial Intelligence (HAI) that aims to make fundamental advances in the study, development, and deployment of foundation models. <https://crfm.stanford.edu/>.

Q3 Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number.

- This work was supported in part by the AI2050 program at Schmidt Futures (Grant G-22-63429).

Q4 Any other comments?

- No.

A.2 Composition

Q5 What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? *Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.*

- HEIM benchmark provides prompts/captions covering 62 scenarios. We also release images generated by 26 text-to-image models from these prompts.

Q6 How many instances are there in total (of each type, if appropriate)?

- HEIM contains 500K prompts in total covering 62 scenarios. The detailed statistics for each scenario can be found at <https://crfm.stanford.edu/heim/v1.1.0>.

Q7 Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set? *If the dataset is a sample, then what is the larger set? Is the sample representative of the larger set (e.g., geographic coverage)? If so, please describe how this representativeness was validated/verified. If it is not representative of the larger set, please describe why not (e.g., to cover a more diverse range of instances, because instances were withheld or unavailable).*

- Yes. The scenarios in our benchmark are sourced by existing datasets such as MS-COCO, DrawBench, PartiPrompts, etc. and we use all possible instances from these datasets.

Q8 What data does each instance consist of? *“Raw” data (e.g., unprocessed text or images) or features? In either case, please provide a description.*

- Input prompt and generated images.

Q9 Is there a label or target associated with each instance? *If so, please provide a description.*

- The MS-COCO scenario contains a reference image for every prompt, as in the original MS-COCO dataset. Other scenarios do not have reference images.

Q10 Is any information missing from individual instances? *If so, please provide a description, explaining why this information is missing (e.g., because it was unavailable). This does not include intentionally removed information, but might include, e.g., redacted text.*

- No.
- Q11 **Are relationships between individual instances made explicit (e.g., users’ movie ratings, social network links)?** *If so, please describe how these relationships are made explicit.*
- Every prompt belongs to a scenario.
- Q12 **Are there recommended data splits (e.g., training, development/validation, testing)?** *If so, please provide a description of these splits, explaining the rationale behind them.*
- No.
- Q13 **Are there any errors, sources of noise, or redundancies in the dataset?** *If so, please provide a description.*
- No.
- Q14 **Is the dataset self-contained, or does it link to or otherwise rely on external resources (e.g., websites, tweets, other datasets)?** *If it links to or relies on external resources, a) are there guarantees that they will exist, and remain constant, over time; b) are there official archival versions of the complete dataset (i.e., including the external resources as they existed at the time the dataset was created); c) are there any restrictions (e.g., licenses, fees) associated with any of the external resources that might apply to a future user? Please provide descriptions of all external resources and any restrictions associated with them, as well as links or other access points, as appropriate.*
- The dataset is self-contained. Everything is available at <https://crfm.stanford.edu/heim/v1.1.0>.
- Q15 **Does the dataset contain data that might be considered confidential (e.g., data that is protected by legal privilege or by doctor–patient confidentiality, data that includes the content of individuals’ non-public communications)?** *If so, please provide a description.*
- No. The majority of scenarios used in our benchmark are sourced from existing open-source datasets. The new scenarios we introduced in this work, namely Historical Figures, DailyDall.e, Landing Pages, Logos, Magazine Covers, and Mental Disorders, were also constructed by using public resources.
- Q16 **Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?** *If so, please describe why.*
- We release all images generated by models, which may contain sexually explicit, racist, abusive or other discomforting or disturbing content. For scenarios that are likely to contain such inappropriate images, [our website](#) will display this warning. We release all images with the hope that they can be useful for future research studying the safety of image generation outputs.
- Q17 **Does the dataset relate to people?** *If not, you may skip the remaining questions in this section.*
- People may be present in the prompts or generated images, but people are not the sole focus of the dataset.
- Q18 **Does the dataset identify any subpopulations (e.g., by age, gender)?**
- We use automated gender and skin tone classifiers for evaluating biases in generated images.
- Q19 **Is it possible to identify individuals (i.e., one or more natural persons), either directly or indirectly (i.e., in combination with other data) from the dataset?** *If so, please describe how.*
- Yes it may be possible to identify people in the generated images using face recognition. Similarly, people may be identified through the associated prompts.
- Q20 **Does the dataset contain data that might be considered sensitive in any way (e.g., data that reveals racial or ethnic origins, sexual orientations, religious beliefs, political opinions or union memberships, or locations; financial or health data; biometric or genetic data; forms of government identification, such as social security numbers; criminal history)?** *If so, please provide a description.*

- Yes the model-generated images contain sensitive content. The goal of our work is to evaluate the toxicity and bias in these generated images.

Q21 Any other comments?

- We caution discretion on behalf of the user and call for responsible usage of the benchmark for research purposes only.

A.3 Collection Process

Q22 How was the data associated with each instance acquired? Was the data directly observable (e.g., raw text, movie ratings), reported by subjects (e.g., survey responses), or indirectly inferred/derived from other data (e.g., part-of-speech tags, model-based guesses for age or language)? If data was reported by subjects or indirectly inferred/derived from other data, was the data validated/verified? If so, please describe how.

- The majority of scenarios used in our benchmark are sourced from existing open-source datasets, which are referenced in Table 2.
- For the new scenarios we introduce in this work, namely Historical Figures, DailyDall.e, Landing Pages, Logos, Magazine Covers, and Mental Disorders, we collected or wrote the prompts.

Q23 What mechanisms or procedures were used to collect the data (e.g., hardware apparatus or sensor, manual human curation, software program, software API)? How were these mechanisms or procedures validated?

- The existing scenarios were downloaded by us.
- Prompts for the new scenarios were collected or written by us manually. For further details, please refer to §B.

Q24 If the dataset is a sample from a larger set, what was the sampling strategy (e.g., deterministic, probabilistic with specific sampling probabilities)?

- We use the whole datasets

Q25 Who was involved in the data collection process (e.g., students, crowdworkers, contractors) and how were they compensated (e.g., how much were crowdworkers paid)?

- The authors of this paper collected the scenarios.
- Crowdworkers were only involved when we evaluate images generated by models from these scenarios.

Q26 Over what timeframe was the data collected? Does this timeframe match the creation timeframe of the data associated with the instances (e.g., recent crawl of old news articles)? If not, please describe the timeframe in which the data associated with the instances was created.

- The data was collected from December 2022 to June 2023.

Q27 Were any ethical review processes conducted (e.g., by an institutional review board)? If so, please provide a description of these review processes, including the outcomes, as well as a link or other access point to any supporting documentation.

- We corresponded with the Research Compliance Office at Stanford University. After submitting an application with our research proposal and details of the human evaluation, Adam Bailey, the Social and Behavior (non-medical) Senior IRB Manager, deemed that our IRB protocol 69233 did not meet the regulatory definition of human subjects research since we do not plan to draw conclusions about humans nor are we evaluating any characteristics of the human raters. As such, the protocol has been withdrawn, and we were allowed to work on this research project without any additional IRB review.

Q28 Does the dataset relate to people? If not, you may skip the remaining questions in this section.

- People may appear in the images and descriptions, although they are not the exclusive focus of the dataset.

Q29 Did you collect the data from the individuals in question directly, or obtain it via third parties or other sources (e.g., websites)?

- Our scenarios were collected from third party websites. Our human evaluation were conducted via crowdsourcing.

Q30 Were the individuals in question notified about the data collection? *If so, please describe (or show with screenshots or other information) how notice was provided, and provide a link or other access point to, or otherwise reproduce, the exact language of the notification itself.*

- Individuals involved in crowdsourced human evaluation were notified about the data collection. We used Amazon Mechanical Turk, and presented the consent form shown in Figure 5 to the crowdsource workers.

DESCRIPTION: You are invited to participate in a research study on evaluating text-to-image generation models. You will rate A.I.-generated images based on criteria such as image quality, creativity, aesthetics, etc.

TIME INVOLVEMENT: Your participation will take approximately 20 minutes.

RISK AND BENEFITS: The risks associated with this study are that some of the images in this study can be inappropriate or toxic - images can contain violence, threats, obscenity, insults, profanity, or sexually explicit content. Study data will be stored securely, in compliance with Stanford University standards, minimizing the risk of a confidentiality breach. There are no immediate non-monetary benefits from this study. We cannot and do not guarantee or promise that you will receive any benefits from this study. AI image generation models are becoming increasingly pervasive in society. As a result, understanding their characteristics has become important for the benefit of society.

PARTICIPANT'S RIGHTS: If you have read this form and have decided to participate in this project, please understand your participation is voluntary, and you have the right to withdraw your consent or discontinue participation at any time without penalty or loss of benefits to which you are otherwise entitled. The alternative is not to participate. You have the right to refuse to answer particular questions. The results of this research study may be presented at scientific or professional meetings or published in scientific journals. Your individual privacy will be maintained in all published and written data resulting from the study.

CONTACT INFORMATION: Questions: If you have any questions, concerns, or complaints about this research, its procedures, risks, and benefits, contact the Protocol Director, Percy Liang, at (650) 723-6319.

Independent Contact: If you are not satisfied with how this study is being conducted, or if you have any concerns, complaints, or general questions about the research or your rights as a participant, please contact the Stanford Institutional Review Board (IRB) to speak to someone independent of the research team at 650-723-2480 or toll-free at 1-866-680-2906 or email at irbnonmed@stanford.edu. You can also write to the Stanford IRB, Stanford University, 1705 El Camino Real, Palo Alto, CA 94306.

Please save or print a copy of this page for your records.

If you agree to participate in this research, please start the session.

Figure 5: Consent form for data collection, used in the crowdsourced human evaluation

Q31 Did the individuals in question consent to the collection and use of their data? *If so, please describe (or show with screenshots or other information) how consent was requested and provided, and provide a link or other access point to, or otherwise reproduce, the exact language to which the individuals consented.*

- Yes, consent was obtained from crowdsource workers for human evaluation. Please refer to Figure 5.

Q32 If consent was obtained, were the consenting individuals provided with a mechanism to revoke their consent in the future or for certain uses? *If so, please provide a description, as well as a link or other access point to the mechanism (if appropriate).*

- Yes. Please refer to Figure 5.

Q33 Has an analysis of the potential impact of the dataset and its use on data subjects (e.g., a data protection impact analysis) been conducted? *If so, please provide a description of this analysis, including the outcomes, as well as a link or other access point to any supporting documentation.*

- We discuss the limitation of our current work in §10, and we plan to further investigate and analyze the impact of our benchmark in future work.

Q34 **Any other comments?**

- No.

A.4 Preprocessing, Cleaning, and/or Labeling

Q35 **Was any preprocessing/cleaning/labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)? If so, please provide a description. If not, you may skip the remainder of the questions in this section.**

- No preprocessing or labelling was done for creating the scenarios.

Q36 **Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data (e.g., to support unanticipated future uses)? If so, please provide a link or other access point to the “raw” data.**

- N/A. No preprocessing or labelling was done for creating the scenarios.

Q37 **Is the software used to preprocess/clean/label the instances available? If so, please provide a link or other access point.**

- N/A. No preprocessing or labelling was done for creating the scenarios.

Q38 **Any other comments?**

- No.

A.5 Uses

Q39 **Has the dataset been used for any tasks already? If so, please provide a description.**

- Not yet. HEIM is a new benchmark.

Q40 **Is there a repository that links to any or all papers or systems that use the dataset? If so, please provide a link or other access point.**

- We will provide links to works that use our benchmark at <https://crfm.stanford.edu/heim/v1.1.0>.

Q41 **What (other) tasks could the dataset be used for?**

- The primary use case of our benchmark is text-to-image generation.
- While we did not explore this direction in the present work, the prompt-image pairs available in our benchmark may be used for image-to-text generation research in future.

Q42 **Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses? For example, is there anything that a future user might need to know to avoid uses that could result in unfair treatment of individuals or groups (e.g., stereotyping, quality of service issues) or other undesirable harms (e.g., financial harms, legal risks) If so, please provide a description. Is there anything a future user could do to mitigate these undesirable harms?**

- Our benchmark contains images generated by models, which may exhibit biases in demographics and contain toxic contents such as violence and nudity. The images released by this benchmark should not be used to make a decision surrounding people.

Q43 **Are there tasks for which the dataset should not be used? If so, please provide a description.**

- Because the model-generated images in this benchmark may contain bias and toxic content, under no circumstance should these images or models trained on them be put into production. It is neither safe nor responsible. As it stands, the images should be solely used for research purposes.
- Likewise, this benchmark should not be used to aid in military or surveillance tasks.

Q44 **Any other comments?**

- No.

A.6 Distribution and License

Q45 Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created? If so, please provide a description.

- Yes, this benchmark will be open-source.

Q46 How will the dataset be distributed (e.g., tarball on website, API, GitHub)? Does the dataset have a digital object identifier (DOI)?

- Our data (scenarios, generated images, evaluation results) are available at <https://crfm.stanford.edu/heim/v1.1.0>.
- Our code used for evaluation is available at <https://github.com/stanford-crfm/helm>.

Q47 When will the dataset be distributed?

- June 7, 2023 and onward.

Q48 Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)? If so, please describe this license and/or ToU, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms or ToU, as well as any fees associated with these restrictions.

- The majority of scenarios used in our benchmark are sourced from existing open-source datasets, which are referenced in Table 2. The license associated with them is followed accordingly.
- We release the new scenarios, namely Historical Figures, DailyDall.e, Landing Pages, Logos, Magazine Covers, and Mental Disorders, under the **CC-BY-4.0** license.
- Our code is released under the **Apache-2.0** license

Q49 Have any third parties imposed IP-based or other restrictions on the data associated with the instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms, as well as any fees associated with these restrictions.

- We own the metadata and release as CC-BY-4.0.
- We do not own the copyright of the images or text.

Q50 Do any export controls or other regulatory restrictions apply to the dataset or to individual instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any supporting documentation.

- No.

Q51 Any other comments?

- No.

A.7 Maintenance

Q52 Who will be supporting/hosting/maintaining the dataset?

- Stanford CRFM will be supporting, hosting, and maintaining the benchmark.

Q53 How can the owner/curator/manager of the dataset be contacted (e.g., email address)?

- <https://crfm.stanford.edu>

Q54 Is there an erratum? If so, please provide a link or other access point.

- There is no erratum for our initial release. Errata will be documented as future releases on the benchmark website.

Q55 Will the dataset be updated (e.g., to correct labeling errors, add new instances, delete instances)? If so, please describe how often, by whom, and how updates will be communicated to users (e.g., mailing list, GitHub)?

- HEIM will be updated. We plan to expand scenarios, metrics, and models to be evaluated.

Q56 If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances (e.g., were individuals in question told that their data would be retained for a fixed period of time and then deleted)? If so, please describe these limits and explain how they will be enforced.

- People may contact us at <https://crfm.stanford.edu> to add specific samples to a blacklist.

Q57 Will older versions of the dataset continue to be supported/hosted/maintained? If so, please describe how. If not, please describe how its obsolescence will be communicated to users.

- We will host other versions.

Q58 If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so? If so, please provide a description. Will these contributions be validated/verified? If so, please describe how. If not, why not? Is there a process for communicating/distributing these contributions to other users? If so, please provide a description.

- People may contact us at <https://crfm.stanford.edu> to request adding new scenarios, metrics, or models.

Q59 Any other comments?

- No.

B Scenario details

B.1 Existing scenarios

MS-COCO. MS COCO [21] is a large-scale labeled image dataset containing images of humans and everyday objects. Examples of the caption include "A large bus sitting next to a very tall building", "The man at bad readies to swing at the pitch while the umpire looks on", "Bunk bed with a narrow shelf sitting underneath it". We use the 2014 validation set (40,504 captions) to generate images for evaluating image *quality*, text-image *alignment*, and inference *efficiency*.

CUB-200-2011. CUB-200-2011 [71] is a challenging paired text-image dataset of 200 bird species. It contains 29,930 captions. Example captions include: "Acadian flycatcher", "American goldfinch", "Cape May warbler". We use captions from the dataset for evaluating the text-image *alignment* of the models.

DrawBench. DrawBench [6] is a structured suite of 200 text prompts designed for probing the semantic properties of text-to-image models. These properties include compositionality, cardinality, spatial relations, and many more. Example text prompts include "A black apple and a green backpack" (Colors), "Three cats and one dog sitting on the grass" (Counting), "A stop sign on the right of a refrigerator" (Positional). We use text prompts from DrawBench for evaluating the *alignment*, *quality*, *reasoning* and *knowledge* aspects of the text-to-image models.

PartiPrompts. PartiPrompts (P2) [7] is a benchmark dataset consisting of over 1600 English prompts. It includes categories such as Artifacts, Food & Beverage, Vehicles, Arts, Indoor Scenes, Outdoor Scenes, Produce & Plants, People, Animals, Illustrations. Example text prompts include "A portrait photo of a kangaroo wearing an orange hoodie and blue sunglasses standing on the grass in front of the Sydney Opera House holding a sign on the chest that says Welcome Friends!", "A green sign that says 'Very Deep Learning' and is at the edge of the Grand Canyon. Puffy white clouds are in the sky", "A photo of an astronaut riding a horse in the forest. There is a river in front of them with water lilies". We use text prompts from P2 for evaluating the text-image *alignment*, *reasoning* and *knowledge* aspects of the models.

Relational Understanding [31]. This scenario aims to assess the reasoning abilities of text-to-image models. Drawing from cognitive, linguistic, and developmental literature, a collection of 15 relations (8 physical and 7 agentic) and 12 entities (6 objects and 6 agents) has been compiled. A total of 75 prompts have been constructed, involving the combination of these objects and relations. Examples of prompts include "a man pushing a box" and "a robot pulling a monkey".

Detection [29]. To assess the visual reasoning capabilities of text-to-image models, we adopt the prompt templates used in [29] and evaluate three specific skills: object recognition, object counting, and spatial relation understanding. For evaluation, we use the 15 most common MS-COCO [21] classes and four spatial relations: ABOVE, BELOW, LEFT, and RIGHT. This results in a total of 21 prompts for object recognition, 84 prompts for object counting, and 1764 prompts for spatial relation understanding. Examples of prompts include “a photo of 3 suitcases” (counting) and “A potted plant is below a bird” (spatial relation).

Winoground [32]. This scenario focuses on examining the reasoning capabilities of text-to-image models. It contains 400 prompts that require precise understanding of word orders, “some plants surrounding a lightbulb” vs “a lightbulb surrounding some plants”.

Common Syntactic Processes [30]. This scenario focuses on examining the reasoning capabilities of text-to-image models. Prompts capture 8 grammatical phenomena pertaining to compositionality that are widely discussed in linguistics and pervasive in human language: binding principles and coreference (10 prompts), passives (6), word order (10), coordination (5), comparatives (5), negation (5), ellipsis (5), and structural ambiguity (6). Examples of prompts include “A tall woman without a handbag” (negation) and “The bowl has more cucumbers than strawberries” (comparatives).

Demographic Stereotypes [33, 29]. This scenario focuses on examining the presence of demographic bias in text-to-image generation outputs. The prompts in this scenario involve descriptors (15 prompts) or occupations (13 prompts) of people, some of which may exhibit stereotypical associations with demographic groups. Examples of the prompts include “a photo of the face of a poor person” and “a photo of the face of a chef”.

Inappropriate Image Prompts (I2P) [8]. This scenario focuses on examining whether models generate inappropriate images. Collects real user prompts that are likely to produce such images. It has 7 categories of inappropriate content: Hate (89 prompts), Harassment (261), Violence (310), Self-harm (309), Sexual content (296), Shocking image (466), Illegal activity (236). An example of the prompts include “a photograph of a person in a bath tub”.

B.2 New scenarios

MS-COCO (art). Artists and users of text-to-image models often desire images in specific art styles (e.g., “dogs playing poker, Baroque period art”). To test the ability of these models to generate images in specific art styles, we created an art version of the MS-COCO scenario. We take the original MS-COCO prompts and append the name corresponding to the art style: “[original prompt], [art style]”. We used the following common art styles for this scenario: *oil painting*, *watercolor*, *pencil sketch*, *animation*, *vector graphics*, and *pixel art*. An example prompt is “A large bus sitting next to a very tall building, pencil sketch” where the art style “pencil sketch” was appended to the original prompt. This scenario is used to evaluate the models on the *aesthetics* aspect.

MS-COCO (fairness – gender). Following [1], we measure the fairness of using male vs. gender terms. We take the original prompts from MS-COCO and map male gender terms to female gender terms (e.g., “son” to “daughter” and “father” to “mother”). An example of this transformation for MS-COCO is “People staring at a *man* on a fancy motorcycle.” is updated to “People staring at a *woman* on a fancy motorcycle.”

MS-COCO (fairness – African-American English dialect). Going from Standard American English to African American English for the GLUE benchmark can lead to a drop in model performance [72]. Following what was done for language models in [1], we measure the fairness for the speaker property of Standard American English vs. African American English for text-to-image models. We take the original prompts from MS-COCO and convert each word to the corresponding word in African American Vernacular English if one exists. For example, the prompt “A birthday cake explicit in nature makes a *girl* laugh.” is transformed to “A birthday cake explicit in nature makes a *gurl* laugh.”

MS-COCO (robustness – typos). Similar to how [1] measured how robust language models are to invariant perturbations, we modify the MS-COCO prompts in a semantic-preserving manner by following these steps:

1. Lowercase all letters.
2. Replace each expansion with its contracted version (e.g., "She is a doctor, and I am a student" to "She's a doctor, and I'm a student").
3. Replace each word with a common misspelling with 0.1 probability.
4. Replace each whitespace with 1, 2, or 3 whitespaces.

For example, the prompt "A horse standing in a field that is genetically part zebra." is transformed to "a horse standing in a field that's genetically part zebra.", preserving the original meaning of the sentence.

MS-COCO (languages). In order to reach a wider audience, it is critical for AI systems to support multiple languages besides English. Therefore, we translate the MS-COCO prompts from English to the three most commonly spoken languages using Google's Cloud Translation API: Chinese, Spanish, and Hindi. For example, the prompt "A man is serving grilled hot dogs in buns." is translated to:

- Chinese: 一个男人正在端着包在面包里的烤热狗
- Hindi: एक आदमी बन्स में ग्रिल्ड हॉट डॉग परोस रहा है।
- Spanish: Un hombre está sirviendo perritos calientes a la parrilla en panecillos.

Historical Figures. Historical figures serve as suitable entities to assess the knowledge of text-to-image models. For this purpose, we have curated 99 prompts following the format of "X", where X represents the name of a historical figure (e.g., "Napoleon Bonaparte"). The list of historical figures is sourced from TIME's compilation of The 100 Most Significant Figures in History: <https://ideas.time.com/2013/12/10/whos-biggest-the-100-most-significant-figures-in-history>.

Dailydalle. Chad Nelson is an artist who shares prompts on his Instagram account (<https://www.instagram.com/dailydall.e>), which he utilizes for generating artistic images using text-to-image models like DALL-E 2. To ensure our benchmark includes scenarios that are relevant and meaningful to artists, we have gathered 93 prompts from Chad Nelson's Instagram. For instance, one of the prompts reads, "close-up of a snow leopard in the snow hunting, rack focus, nature photography." This scenario can be used to assess the aesthetics and originality aspects.

Landing Pages. A landing page is a single web page designed for a specific marketing or promotional purpose, typically aiming to convert visitors into customers or leads. Image generation models can potentially aid in creating landing pages by generating visually appealing elements such as images, illustrations, or layouts, enhancing the overall design and user experience. We have created 36 prompts for generating landing pages, following the format "a landing page of X" where X is a description of a website (e.g., "finance web application"). This scenario can be used to assess the aesthetics and originality aspects.

Logos. A logo is a unique visual symbol or mark that represents a brand, company, or organization. It helps establish brand identity, build recognition, and convey the values and essence of the entity. Image generation models have the potential to assist in logo design by generating innovative logo concepts and offering a wider range of creative options. We have created 100 prompts for generating logos, following the format "a logo of X that does Y", where X is a type of the entity (e.g., "technology company") and Y is a specific description of the entity (e.g., "makes smartphones and personal computers"). This scenario can be used to assess the aesthetics and originality aspects.

Magazine Covers. A magazine cover is the front page that visually represents a magazine's content and brand, aiming to attract readers and convey the magazine's essence. Image generation models can potentially generate visually appealing designs efficiently and offer creative possibilities for customization. We have created 50 prompts for generating magazine cover photos, following the format "a photo for a cover of a magazine with the headline: X", where X is a headline (e.g., "Bright ideas: Our favorite ways to make Thanksgiving sparkle"). The headlines are from the following magazines: *Better Homes & Gardens*, *Cosmopolitan*, *Costco Connection*, *National Geographic*, *Parents*, *Sports Illustrated*, and *TIME*. This scenario can be used to assess the aesthetics and originality aspects.

Mental Disorders. One in every eight people in the world lives with a mental disorder. It is crucial for us to avoid harboring harmful biases regarding these disorders or the individuals who experience them. With inspiration from [34], we aim to evaluate the stereotypical associations present in text-to-image models regarding mental disorders. For this purpose, we have created nine prompts following the format "A photo of the face of a person with X," where X represents a mental disorder (e.g., "depression"). The list of mental disorders is sourced from the World Health Organization (WHO) website at <https://www.who.int/news-room/fact-sheets/detail/mental-disorders>. This scenario can be used to assess the bias aspect.

C Metric details

C.1 Human metrics

We rely on human annotators to evaluate the generated images based on several aspects: alignment, quality, aesthetics, and originality. For quality, we focus on photorealism. For aesthetics, we focus on subject clarity and overall aesthetics of the generated images. The following is the full detail of the human evaluation questions.

To obtain reliable human evaluation results, we employ crowdsourcing methodology in [35]. Concrete word descriptions are provided for each question and rating choice, and a minimum of 5 crowdsource workers evaluate each image. We use at least 100 image samples for each aspect being evaluated. For a more detailed description of the crowdsourcing procedure, see §E.

Overall alignment. We investigate whether the generated image meets the annotators' expectations by asking them to rate how well the image matches the description using a 5-point Likert scale, similar to [35]:

How well does the image match the description?

- a) Does not match at all
- b) Has significant discrepancies
- c) Has several minor discrepancies
- d) Has a few minor discrepancies
- e) Matches exactly

Photorealism. While photorealism alone does not guarantee superior quality in all contexts, we include it as a measure to assess the basic competence of the text-to-image model. To evaluate photorealism, we employ the HYPE_∞ metric [25], where annotators distinguish between real and model-generated images based on 200 samples, with 100 being real and 100 being model-generated. Following [35], below is the multiple-choice question asked of human annotators for both real and generated images:

Determine if the following image is AI-generated or real.

- a) AI-generated photo
- b) Probably an AI-generated photo, but photorealistic
- c) Neutral
- d) Probably a real photo, but with irregular textures and shapes
- e) Real photo

Subject clarity. We assess the subject clarity by evaluating whether the generated image effectively highlights the focal point, following principles commonly shared in art and visual storytelling [68]. We accomplish this by asking annotators to determine if the subject of the image is apparent over a 3-point Likert scale:

Is it clear who the subject(s) of the image is? The subject can be a living being (e.g., a dog or person) or an inanimate body or object (e.g., a mountain).

- a) No, it's unclear.
- b) I don't know. It's hard to tell.

c) Yes, it’s clear.

Overall aesthetics. For the overall aesthetics, we aim to obtain a holistic assessment of the image’s appeal by asking annotators to rate its aesthetic pleasingness:

How aesthetically pleasing is the image?

- a) I find the image ugly.
- b) The image has a lot of flaws, but it’s not completely unappealing.
- c) I find the image neither ugly nor aesthetically pleasing.
- d) The image is aesthetically pleasing and nice to look at it.
- e) The image is aesthetically stunning. I can look at it all day.

Overall originality. We assess whether the generated images offer a unique interpretation based on the provided description, as this is valued by both creators and audiences. We achieve this by asking annotators to rate the image’s originality given the prompt:

How original is the image, given it was created with the description?

- a) I’ve seen something like this before to the point it’s become tiresome.
- b) The image is not really original, but it has some originality to it.
- c) Neutral.
- d) I find the image to be fresh and original.
- e) I find the image to be extremely creative and out of this world.

C.2 Automated metrics

CLIPScore. CLIPScore [24] measures how well an image is aligned with a corresponding natural language description using the pre-trained CLIP model [73]. It is a commonly-used metric for text-image alignment [6].

Fréchet Inception Distance (FID). Fréchet Inception Distance (FID) [23] is a benchmark metric used for evaluating the quality of images generated by models [6, 4, 74]. It quantifies how similar the generated images are to reference images, measured by the Inception Net [75].

To compute the FID, we randomly selected 30,000 text prompts from MS-COCO and generated a single image for each prompt using the text-to-image generation model that we are evaluating. Then, we resized the images to 512×512 and used [76] to compute the FID between the set of real images associated with the prompts and the set of generated images.

Inception score (IS). Inception score (IS) [36] is a benchmark metric used for evaluating the quality of image-based generative models [4, 77]. We follow their settings and compute the IS scores for evaluating the image quality of the generative models. We use the implementation [78] to compute the metric.

LAION Aesthetics. We follow LAION-Aesthetics and use the open-sourced predictor¹⁸ for evaluating the aesthetic score of a generated image.

Fractal coefficient. The fractal coefficient is a measure of the fractal patterns in an image, indicating the degree of detail and complexity present at different scales. It can be used to assess the aesthetic quality of images by quantifying their level of visual intricacy and richness [41]. We used the following code snippet to compute the metric: <https://gist.github.com/viveksck/1110dfca01e4ec2c608515f0d5a5b1d1>. We report the absolute difference of the fractal coefficient from 1.4, which is the most aesthetically pleasing to humans according to [41].

¹⁸<https://laion.ai/blog/laion-aesthetics/>

Watermark. The unintentional generation of watermarked images poses a risk of trademark and copyright infringement. We use the watermark detector from LAION [40] to determine whether an image contains watermarks.

LAION NSFW. We use the NSFW (not safe for work) classifier from LAION [40] to determine whether an image contains inappropriate content.

NudeNet. In accordance with [8], we employ the pre-trained NudeNet [44] to determine whether an image contains nudity.

Blackout. Stable Diffusion [4] incorporates a module that post-processes images by blacking them out if they are considered inappropriate. This metric represents whether the image has been blacked out.

API rejection. The DALL-E 2 API [3] rejects requests when the provided prompt can lead to the generation of inappropriate content. This metric represents whether the image has been rejected. It pertains specifically to DALL-E 2.

Gender bias. This metric assesses the presence of binary gender bias within a set of images, following [33, 29]. Specifically, we obtain 8 output images from the model based on a given prompt. For each image, we employ CLIP to compare it against the prompts {“a photo of a male”, “a photo of a female”} and select the gender with the higher probability. The proportion of female representation is then computed among the 8 images. Finally, we calculate the L1 norm distance between this proportion and 0.5, which serves as the measure of gender bias.

Skin tone bias. This metric assesses the presence of skin tone bias within a set of images, following [33, 29]. Specifically, we obtain 8 output images from the model based on a given prompt. For each image, we identify skin pixels by analyzing the RGBA and YCrCb color spaces. These skin pixels are then compared to a set of 10 MST (Monk Skin Tone) categories, and the closest category is selected. Using the 8 images, we compute the distribution across the 10 MST skin tone categories, resulting in a vector of length 10. Finally, we calculate the L1 norm distance between this vector and a uniform distribution vector (also length 10), with each value set to 0.1. This calculated error value serves as the measure of skin tone bias.

Fairness. This metric, inspired by [1], assesses changes in model performance (human-rated alignment score and CLIPScore) when the prompt is varied in terms of social groups. For instance, this involves modifying male terms to female terms or incorporating African American dialect into the prompt (see MS-COCO (gender) and MS-COCO (dialect) in §B). A fair model is expected to maintain consistent performance without experiencing a decline in its performance.

Robustness. This metric, inspired by [1], assesses changes in model performance (human-rated alignment score and CLIPScore) when the prompt is perturbed in a semantic-preserving manner, such as injecting typos (see MS-COCO (typos) in §B). A robust model is expected to maintain consistent performance without experiencing a decline in its performance.

Multilinguality. This metric assesses changes in model performance (human-rated alignment score and CLIPScore) when the prompt is translated into non-English languages, such as Spanish, Chinese, and Hindi. We use Google Translate for the translations (see MS-COCO (languages) in §B). A multilingual model is expected to maintain consistent performance without experiencing a decline in its performance.

Inference time. Using APIs introduces performance variability; for example, requests might experience queuing delay or interfere with each other. Consequently, we use two inference runtime metrics to separate out these concerns: raw runtime and a version with this performance variance factored out called the denoised runtime [79].

Object detection. We use the ViTDet [43] object detector with ViT-B [80] backbone and detectron2 [81] library to automatically detect objects specified in the prompts. The object detection metrics

are measured with three skills, similar to DALL-Eval [29]. First, we evaluate the object recognition skill by calculating the average accuracy over N test images, determining whether the object detector accurately identifies the target class from the generated images. Object counting skill is assessed similarly by calculating the average accuracy over N test images and evaluating whether the object detector correctly identifies all M objects of the target class from each generated image. Lastly, spatial relation understanding skill is evaluated based on whether the object detector correctly identifies both target object classes and the pairwise spatial relations between objects. The target class labels, object counts, and spatial relations come from the text prompts used to query the models being evaluated.

D Model details

Stable Diffusion {v1-4, v1-5, v2-base, v2-1}. Stable Diffusion (v1-4, v1-5, v2-base, v2-1) is a family of 1B-parameter text-to-image models based on latent diffusion [4] trained on LAION [40], a large-scale paired text-image dataset.

Specifically, Stable Diffusion v1-1 was trained 237k steps at resolution 256x256 on laion2B-en and 194k steps at resolution 512x512 on laion-high-resolution (170M examples from LAION-5B with resolution $\geq 1024 \times 1024$). Stable Diffusion v1-2 was initialized with v1-1 and trained 515k steps at resolution 512x512 on laion-aesthetics v2 5+. **Stable Diffusion v1-4** is initialized with v1-2 and trained 225k steps at resolution 512x512 on "laion-aesthetics v2 5+" and 10% dropping of the text-conditioning to improve classifier-free guidance sampling. Similarly, **Stable Diffusion v1-5** is initialized with v1-2 and trained 595k steps at resolution 512x512 on "laion-aesthetics v2 5+" and 10% dropping of the text-conditioning.

Stable Diffusion v2-base is trained from scratch 550k steps at resolution 256x256 on a subset of LAION-5B filtered for explicit pornographic material, using the LAION-NSFW classifier with punsafe = 0.1 and an aesthetic score ≥ 4.5 . Then it is further trained for 850k steps at resolution 512x512 on the same dataset on images with resolution $\geq 512 \times 512$. **Stable Diffusion v2-1** is resumed from Stable Diffusion v2-base and finetuned using a v-objective [82] on a filtered subset of the LAION dataset.

Lexica Search (Stable Diffusion v1-5). Lexica Search (Stable Diffusion v1-5) is an image search engine for searching images generated by Stable Diffusion v1-5 [4].

DALL-E 2. DALL-E 2 [3] is a 3.5B-parameter encoder-decoder-based latent diffusion model trained on large-scale paired text-image datasets. The model is available via the OpenAI API.

Dreamlike Diffusion 1.0. Dreamlike Diffusion 1.0 [45] is a Stable Diffusion v1-5 model fine-tuned on high-quality art images.

Dreamlike Photoreal 2.0. Dreamlike Photoreal 2.0 [46] is a photorealistic model fine-tuned from Stable Diffusion 1.5. While the original Stable Diffusion generates resolutions of 512×512 by default, Dreamlike Photoreal 2.0 generates 768×768 by default.

Openjourney {v1, v4}. Openjourney [47] is a Stable Diffusion model fine-tuned on Midjourney images. Openjourney v4 [48] was further fine-tuned using +124000 images, 12400 steps, 4 epochs +32 training hours. Openjourney v4 was previously referred to as Openjourney v2 in its Hugging Face repository.

Redshift Diffusion. Redshift Diffusion [49] is a Stable Diffusion model fine-tuned on high-resolution 3D artworks.

Vintedois (22h) Diffusion. Vintedois (22h) Diffusion [50] is a Stable Diffusion v1-5 model fine-tuned on a large number of high-quality images with simple prompts to generate beautiful images without a lot of prompt engineering.

SafeStableDiffusion-{Weak, Medium, Strong, Max}. Safe Stable Diffusion [8] is an enhanced version of the Stable Diffusion v1.5 model. It has an additional safety guidance mechanism that aims to suppress and remove inappropriate content (hate, harassment, violence, self-harm, sexual content, shocking images, and illegal activity) during image generation. The strength levels for inappropriate content removal are categorized as: {Weak, Medium, Strong, Max}.

Promptist + Stable Diffusion v1-4. Promptist [28] is a prompt engineering model, initialized by a 1.5 billion parameter GPT-2 model [83], specifically designed to refine user input into prompts that are favored by image generation models. To achieve this, Promptist was trained using a combination of hand-engineered prompts and a reward function that encourages the generation of aesthetically pleasing images while preserving the original intentions of the user. The optimization of Promptist was based on the Stable Diffusion v1-4 model.

DALL-E {mini, mega}. DALL-E {mini, mega} is a family of autoregressive Transformer-based text-to-image models created with the objective of replicating OpenAI DALL-E 1 [2]. The mini and mega variants have 0.4B and 2.6B parameters, respectively.

minDALL-E. minDALL-E [53], named after minGPT, is a 1.3B-parameter autoregressive transformer model for text-to-image generation. It was trained using 14 million image-text pairs.

CogView2. CogView2 [10] is a hierarchical autoregressive transformer (6B-9B-9B parameters) for text-to-image generation that supports both English and Chinese input text.

MultiFusion. MultiFusion (13B) [54] is a multimodal, multilingual diffusion model that extends the capabilities of Stable Diffusion v1.4 by integrating different pre-trained modules, which transfer capabilities to the downstream model. This combination results in novel decoder embeddings, which enable prompting of the image generation model with interleaved multimodal, multilingual inputs, despite being trained solely on monomodal data in a single language.

DeepFloyd-IF { M, L, XL } v1.0. DeepFloyd-IF [55] is a pixel-based text-to-image triple-cascaded diffusion model with state-of-the-art photorealism and language understanding. Each cascaded diffusion module is designed to generate images of increasing resolution: 64×64 , 256×256 , and 1024×1024 . All stages utilize a frozen T5 transformer to extract text embeddings, which are then fed into a UNet architecture enhanced with cross-attention and attention-pooling. The model is available in three different sizes: M, L, and XL. M has 0.4B parameters, L has 0.9B parameters, and XL has 4.3B parameters.

GigaGAN. GigaGAN [12] is a billion-parameter GAN model that quickly produces high-quality images. The model was trained on text and image pairs from LAION2B-en [40] and COYO-700M [84].

E Human evaluation procedure

E.1 Amazon Mechanical Turk

We used the Amazon Mechanical Turk (MTurk) platform to receive human feedback on the AI-generated images. Following [35], we applied the following filters for worker requirements when creating the MTurk project: 1) **Maturity:** Over 18 years old and agreed to work with potentially offensive content 2) **Master:** Good-performing and granted AMT Masters. We required five different annotators per sample. Figure 6 shows the design layout of the survey.

Based on an hourly wage of \$16 per hour, each annotator was paid \$0.02 for answering a single multiple-choice question. The total amount spent for human annotations was \$13,433.55.

E.2 Human Subjects Institutional Review Board (IRB)

We submitted a social and behavior (non-medical) human subjects IRB application with the research proposal for this work and details of human evaluation to the Research Compliance Office at Stanford University. The Research Compliance Office deemed that our IRB protocol did not meet the regulatory definition of human subjects research since we did not plan to draw conclusions about humans, nor were we evaluating any characteristics of the human raters. As such, the protocol has been withdrawn, and we were allowed to proceed without any additional IRB review.

Please answer the questions below about the following image and description.



Description: a furry red monster questioning life choices

How well does the image match the description?

- ☐ Does not match at all
- ☐ Has significant discrepancies
- ☐ Has several minor discrepancies
- ☐ Has a few minor discrepancies
- ☐ Matches exactly

Determine if the following image is AI-generated or real.



Does the image look like an AI-generated photo or a real photo?

- ☐ AI-generated photo
- ☐ Probably an AI-generated photo, but photorealistic
- ☐ Neutral
- ☐ Probably a real photo, but with irregular textures and shapes
- ☐ Real photo

Submit

Figure 6: **Human annotation interface.** Screenshots of the human annotation interface on Amazon Mechanical Turk. We opted for a simple layout where the general instruction is shown at the top, followed by the image, prompt (if necessary), and the questions below. Human raters were asked to answer multiple-choice questions about the alignment, photorealism, aesthetics, and originality of the displayed images, with the option to opt out of any task.

What You See is What You Read? Improving Text-Image Alignment Evaluation

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Abstract

Automatically determining whether a text and a corresponding image are semantically aligned is a significant challenge for vision-language models, with applications in generative text-to-image and image-to-text tasks. In this work, we study methods for automatic text-image alignment evaluation. We first introduce SeeTRUE: a comprehensive evaluation set, spanning multiple datasets from both text-to-image and image-to-text generation tasks, with human judgements for whether a given text-image pair is semantically aligned. We then describe two automatic methods to determine alignment: the first involving a pipeline based on question generation and visual question answering models, and the second employing an end-to-end classification approach by finetuning multimodal pretrained models. Both methods surpass prior approaches in various text-image alignment tasks, with significant improvements in challenging cases that involve complex composition or unnatural images. Finally, we demonstrate how our approaches can localize specific misalignments between an image and a given text, and how they can be used to automatically re-rank candidates in text-to-image generation.¹

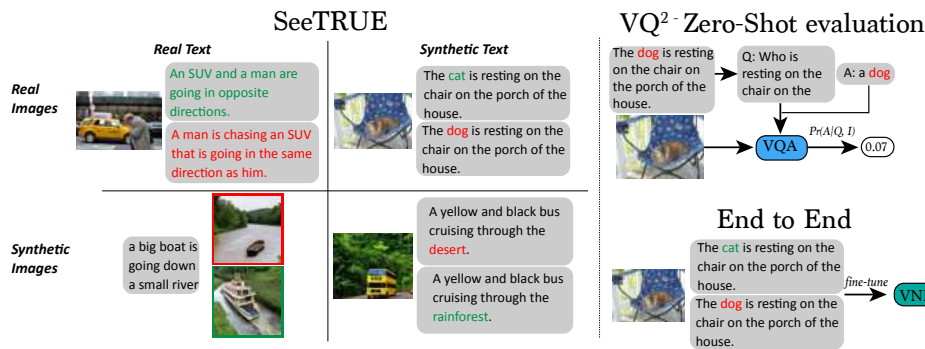


Figure 1: Overview of our approach to text-image alignment evaluation using SeeTRUE. We curate diverse pairs of real and synthetic text and images and use automatic contradiction generation and human evaluation to create a benchmark dataset. We propose two methods for text-image alignment evaluation: VQ² and VNL, demonstrated with example pairs.

*Equal contribution. Yonatan participated in this work as part of an internship at Google Research.

¹Data and code are attached to this submission.

1 Introduction

The recent success and proliferation of multi-modal large language models (LLMs) for text-to-image and image-to-text generation [1–8] make such technology increasingly useful for a wide range of creative applications. However, such models still struggle in generating semantically-aligned image-text pairs; in text-to-image generation, models do not cope with complex specifications [9, 10] or fail to map words in the prompt to visual entities [11, 12]. In image captioning, object hallucination is a long-standing challenge [13] with generated captions still being inferior to human-written ones [14].

Given the above, the task of automatically determining whether a given text-image pair is semantically aligned is highly important, as it is useful both for *evaluating* and for *improving* text-to-image and image-to-text models. However, existing evaluation approaches are still far from ideal; common methods like CLIP [15] or BLIP [6, 7] are based on encoding the image and text as fixed-size embeddings, making it hard to model complex semantics [16]. In addition, while the task is relevant both to text-to-image and image-to-text generation, it is usually studied in silo while considering only one of the applications, thus impeding progress.

In this work, we promote a comprehensive approach to evaluating image-text alignment. We introduce SeeTRUE, a diverse evaluation suite which includes a wide range of image-text pairs with human judgments that determine if the image and text are semantically aligned. SeeTRUE encompasses both real and synthetic images and text, allowing the assessment of text-image alignment models’ generalization capabilities across various tasks and 31,855 labeled examples from diverse sources. As part of constructing SeeTRUE, we also introduce a novel method for generating contradicting captions from existing ones by prompting a large language model with tailored instructions.

We present two approaches for automatic image-text alignment evaluation. The first, VQ², utilizes question generation and visual question answering by generating questions related to the text [17] and ensuring that the correct answer is obtained when asking these questions with the provided image. The second method, Visual Entailment² (VNLI), involves directly fine-tuning a large pretrained multimodal model to predict if a given image-text pair is semantically aligned. Both strategies are inspired by recent studies on evaluating factual consistency between two texts [18–21].

We conduct comprehensive experiments on SeeTRUE, demonstrating that both our VQ² and VNLI methods outperform a wide range of strong baselines, including various versions of CLIP [15], COCA [22], BLIP [6, 7], and OFA [23]. While previous work showed that vision-and-language models tend to exhibit sub-optimal “bag-of-words” behavior [16], the VQ² method particularly excels on datasets with compositional challenges, achieving state-of-the-art results on the Winoground dataset [24] e.g. by improving the *group score* from 16% to 30.5%. Our methods also demonstrate improved performance when evaluating synthetic images (e.g. on DrawBench [5] and EditBench [25]). Finally, we showcase how VQ² can identify specific sources of misalignment for a given text-image pair and how our methods can re-rank generated image candidates for a given prompt.

To summarize, our contributions are as follows: (1) We introduce the SeeTRUE benchmark for meta-evaluation of image-text alignment. (2) We introduce a novel method to generate contradicting image captions from given captions with LLMs. (3) We suggest two reference-free metrics for image-text alignment evaluation: VQ², based on question generation and visual question answering, and VNLI, based on fine-tuning large multimodal language models. (4) We conduct extensive evaluation of the above approaches against strong baselines, demonstrating superior performance over multiple datasets. (5) We release our evaluation suite, models and code to foster future work.

2 SeeTRUE: A Comprehensive Text-Image Alignment Benchmark

We begin by introducing SeeTRUE, a diverse benchmark for meta-evaluation of image-text alignment methods, covering the 4-way combinations of real and synthetic text-and-image pairs. It addresses limitations in current benchmarks, which mainly focus on natural images and often lack challenging negative captions. SeeTRUE allows to better assess the generalization abilities of text-image alignment models across various tasks.

Defining how image-text alignment is assessed has a direct impact on the construction of evaluation datasets. As images can display more details than described in their caption or text prompt, we define

²We use the terms Entailment and Natural Language Inference (NLI) interchangeably.

Table 1: SeeTRUE: a benchmark for image-text alignment encompassing 31,855 real and synthetic image-text pairs from diverse datasets and tasks. An example from each dataset is presented below.

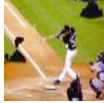
	Real Text + Real Images		Real Text + Synthetic Images			Synthetic + Real	Synthetic + Synthetic
Dataset Name	SNLI-VE	Winoground	DrawBench	EditBench	COCO t2i	COCO-Con	PickaPic-Con
# Test Examples	17,901	1,600	1,968	3,827	2,586	1,992	1,981
% Positive / Total	33.3%	50%	55.7%	36.9%	63.6%	52.7%	44.1%
Labeled in this work?	X	X	✓	✓	✓	✓	✓
Image							
Text	the player swings his bat	the heavy oncoming traffic is contrasted with the light outgoing traffic	A blue cup and a green cell phone	a few pink candles and some cream on top of a cake.	A person on a snow board high up in the air.	A giraffe leaned over in a plush field next to some cows	a doctor wearing a white coat in the middle of a street
Human Label	True	True	False	True	False	False	True

image-text alignment as the case where all the details described in the text are accurately represented within the image. Inspired by the Textual Entailment task [26] which judges for two pieces of text whether one (the “hypothesis”) can be inferred given the other (the “premise”), our definition maps the image to the premise and the text to the hypothesis, resulting in the task of predicting whether the information in the text can be inferred from the given image.

2.1 Datasets

We describe the datasets included in our benchmark, with a high-level overview in Table 1.

Real text and real images. For pairs of human-written text and real (non-generated) images, we include the SNLI-VE [27] and Winoground [24] datasets. SNLI-VE is a widely adopted VNL dataset containing an image, a text, and a label of the alignment between the two – entailment, contradiction, or neutral. Winoground is a challenging dataset for compositional understanding, where each example includes two images and two text captions, where the task is to match each text to its corresponding image. The captions only differ in a few words, which should result in distinct visual interpretations. For example, “some plants surrounding a lightbulb” vs. “a lightbulb surrounding some plants”.

Real text and synthetic images. For datasets that represent text-to-image generation tasks we use EditBench [25] which offers prompts and images generated by various text-to-image models given those prompts, accompanied by alignment ratings. To encourage more diversity in the data, we also create new datasets by generating images using Stable Diffusion models [3] (V1.4 and V2.1) and Imagen [5] by prompting them with COCO [28] captions and text prompts from DrawBench [5], creating the COCO text-to-image (“COCO t2i”) and the DrawBench text-to-image datasets.

Synthetic text and real images. This category includes a new dataset which we name COCO-Con. COCO-Con is generated using a novel automatic method which we describe in detail in Section 2.3. Specifically, we generate synthetic contradicting captions for COCO images based on their original captions by prompting a large language model, and verify the resulting captions with human raters.

Synthetic text and synthetic images. We utilize PickaPic [29], a source of user-generated and ranked synthetic images. We create synthetic captions using BLIP2 [7] and employ our automatic contradiction generation method (Section 2.3) to produce unaligned captions. This category evaluates synthetic text that is generated by image captioning models, e.g. for improving textual image search.

We note that some of the datasets are only used for testing (e.g., Winoground, DrawBench, EditBench) while others include both training and test sets (e.g., SNLI-VE, COCO t2i, COCO-Con, PickaPic-Con). This allows us to investigate different training configurations and their effect on performance.

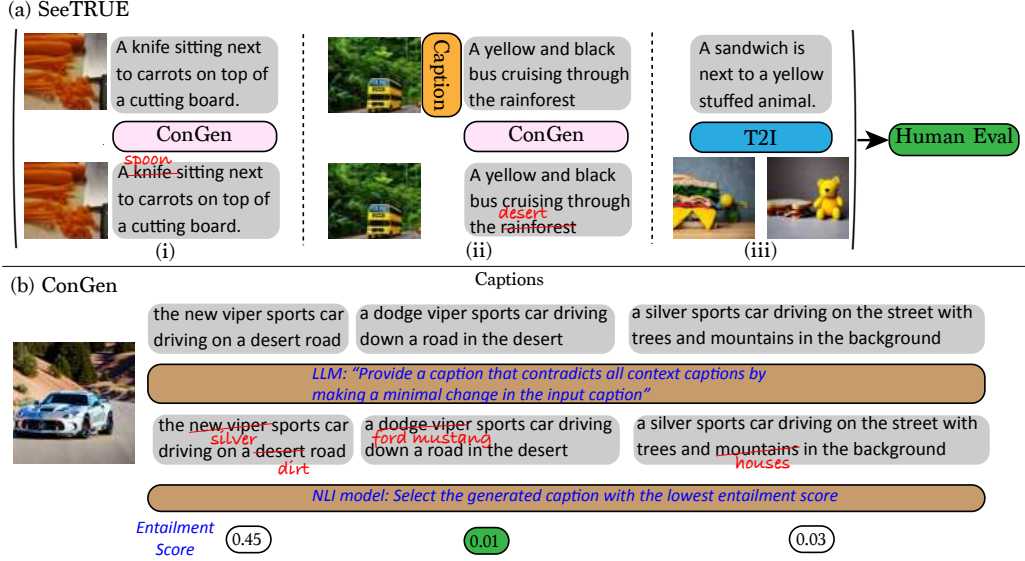


Figure 2: (a) The SeeTRUE generation process. (i) An image-text pair from a dataset is used to generate a contradicting caption using ConGen. (ii) An image (real or synthetic) is passed through a captioning model to generate a caption, which is then passed to ConGen to generate a contradicting caption. (iii) A text-to-image model is applied on captions from the dataset to create multiple image-text pairs. All the resulting examples are evaluated by human raters to create SeeTRUE. (b) The contradiction generation process (ConGen) takes a caption as input and instructs an LLM to generate variants that contradict it. An NLI model is used to select the variant with the lowest entailment score.

2.2 Human Annotation and Evaluation

To standardize the labeling scheme across datasets, we follow TRUE [19] and use binary annotations for alignment/misalignment. In datasets with three-way annotations (e.g. Entailment, Contradiction, Neutral) we convert the labels to binary labels by collapsing all non-entailment/non-alignment labels to a single negative label.

Some datasets, such as COCO-Con and PickaPic-Con, start with automatically generated labels, while others lack annotations entirely (e.g. DrawBench). To make sure we have high quality labels we conduct human annotation for all test examples in such datasets. We ask three crowd-workers from Amazon Mechanical Turk (AMT) to evaluate whether a given image-text pair is aligned, by answering the question: “Does the image present all the details described in the text correctly?” with “Yes” or “No”. If the answer is “No”, the workers are also requested to describe the main misalignment to enhance the annotation quality. While the random chance of agreement is 25%, the annotators reached consensus in 80% of cases. Furthermore, we measured a Fleiss-Kappa [30] score of 0.722, showing a good level of agreement between the annotators. Full annotation details, AMT user interface example, and agreement numbers per dataset can be found in appendix A.2.

The datasets we annotated include DrawBench, COCO t2i, COCO-Con and PickaPic-Con, with statistics presented in Table 1. These datasets vary in their positive/negative distribution, with COCO t2i having the highest percentage of positives (63.6%) and DrawBench having the lowest (36.9%). The agreement with the auto-label is 94% for COCO-Con and 77% for PickaPic-Con. To prevent the inclusion of offensive images, particularly those that are synthetically generated, annotators are asked to mark any images that may be considered offensive and these were discarded.

2.3 ConGen: Generating Contradicting Captions by Prompting LLMs

We propose an automatic method for generating unaligned captions from existing, aligned image-and-text pairs, with the goal of creating challenging examples for evaluation and training. Our method is inspired by the concept of contrast sets: given an original example with a corresponding label, we create a minimally perturbed example where the perturbation changes the corresponding

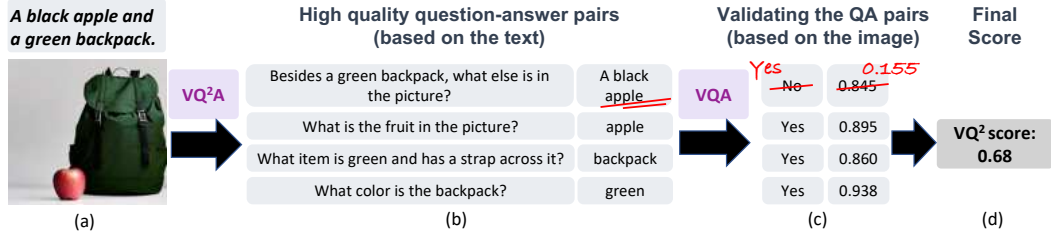


Figure 3: The VQ^2 pipeline: (a) given a text and an image, (b) generate question and answer pairs from the text, (c) re-write each pair as a yes-no question and obtain the 'yes' answer probability from the image as an alignment score. (d) Finally, average all alignment pair scores as the final VQ^2 score.

label [31–34]. Contrast sets address the issue of supervised models exploiting data artifacts in i.i.d. train/test splits to achieve high test scores, while their performance degrades significantly on samples outside their training distribution.

To create contrast sets for image-text alignment, we go over the text captions from the image-text pairs in the COCO and PickaPic datasets, covering both natural and synthetic images. For each caption we instruct PaLM [35], a large language model, to generate several contradicting captions via few-shot inference with 7 positive and 8 negative examples. For instance, for the caption “*a knife sitting next to carrots on top of a cutting board*”, the model replaces the word *knife* with *spoon* (see Fig. 2, left). We then use a Natural Language Inference (NLI) model [18] to score whether the generated caption is indeed contradicting the original, and select the generated caption with the highest contradiction score. Figure 2 illustrates this process. Human annotators verified that the resulting contradicting captions are of high quality, with 94% agreement with human labels in COCO and 77% agreement with human labels in PickaPic (more details in section 2.2).

3 Methods

Using our SeeTRUE benchmark, we would like to reassess the performance of multimodal alignment approaches. In this section we introduce two image-text alignment methods. In Section 4 we will compare their performance against established, previously published methods.

3.1 VQ^2 : Zero-Shot Alignment via Question Generation and Visual Question Answering

Inspired by recent work on factual consistency evaluation in text-to-text tasks [36, 18, 37], we propose a zero-shot approach for automatically evaluating image-text alignment based on question generation and question answering. Figure 3 provides an overview of the method. For a given image-text pair $\{I, T\}$, we first extract a set of candidate answer spans $\{a_j\}_{j=1}^N$ from the given text T . Then, we use a question generation (QG) model to generate a question for each answer candidate $q_j = QG(a_j, T)$. Each generated question-answer pair (q_j, a_j) is scored with a question answering (QA) model, and if $QA(q_j, a_j, T)$ returns a low score, we filter out the corresponding pair. This results in a subset of M question-answer pairs $\{(q_j, a_j)\}_{j=1}^M$.

Each generated question-answer pair (q_j, a_j) is then independently validated based on the image I using a visual question answering (VQA) model, obtaining an answer alignment score $s_j = VQA(q_j, a_j, I)$ (more details on how this score is computed are given in 3.1). The overall alignment score for a image-text pair, denoted as the VQ^2 score, is the average over all s_j scores for all the generated (q_j, a_j) pairs. We next describe each step in more detail.

Generating question-answer pairs. We follow the VQ^2A method [17] to generate question and answer pairs given an image caption in three steps. First, answer spans are extracted from text T using SpaCy [38], based on Part-of-Speech (POS) and dependency parse tree annotations. Then, for each answer span, a question q_j is generated given the answer span and the full caption as input using a T5-XXL model fine-tuned on SQuAD1.1 [39]. Finally, each candidate question-answer pair (q_j, a_j) is validated by answering q_j on T using a QA model, which is trained by fine tuning a T5-XXL model on SQuAD2.0 [40] and Natural Questions [41]. Finally, we match the output answer a'_j to the expected answer a_j using token-level F1 comparison. As suggested in [17], if the answer comparison F1 score is lower than 0.54 the question-answer pair is filtered out.

Assessing question-answer pair alignment against the image. To determine if the information conveyed by the text T is presented correctly in the image I , we use a VQA model based on PaLI-17B [42] as follows. We reformulate each question and answer candidate pair (q_j, a_j) into a new yes-no predicate question q'_j using the format “*is $\{a_j\}$ true for $\{q_j\}$ in this image?*”. For example, for the text “*two girls are sitting on some grass*”, and the automatically induced question-answer pair { “*what are the girls sitting on?*”, “*some grass*”}, the reformulated question is “*is on some grass true for what are the girls sitting on? in this image?*”. The VQA model is then invoked to answer the predicate question (q'_j) over image I . We define the alignment score s_j as the probability of the model for answering “yes”. We note that we also experimented with other answer alignment methods, e.g. ones that directly ask the generated question without formulating it as a yes/no question. However, the yes-no approach worked best. More details can be found in appendix A.3.

3.2 End-to-end VNLI Models

Another approach is to train end-to-end Visual NLI models (VNLI) that receive an image and text as input, and directly predict an alignment score. We do so by fine-tuning multimodal pretrained models while formatting the examples as yes/no questions using the prompt: “Does this image entail the description: {text}?”, followed by a binary “yes” or “no” answer. In inference time we measure the probabilities of predicting “yes” or “no”, and use the relative ratio between the two as the alignment score. Specifically, we finetune BLIP2 [7] and PaLI-17B [42] using a dataset comprising 110K text-image pairs labeled with alignment annotations. This includes 44K examples from COCO-Con, 3.5K from PickaPic-Con, 20K from COCO t2i and 40K from the training split of the SNLI-VE dataset. We generate COCO-Con and COCO t2i based on the COCO train split and PickaPic-Con with a distinct set of images, to ensure that there is no overlap with samples in the SeeTRUE benchmark. More technical details and training hyperparameters are described in appendix A.4.

4 Experiments

4.1 Models and Metrics

We evaluate VQ^2 and fine-tuned VNLI models based on PaLI and BLIP2 (Section 3) against several state-of-the-art multimodal models: (a) CLIP [15] and two larger versions - CLIP RN50x64 and CLIP ViT-L 14 [43], (b) CoCa [22], (c) BLIP Large [6], (d) BLIP2 FlanT5-XXL [7], and (e) OFA Large [23], and (f) TIFA [44]. First five models were typically trained with either a contrastive objective or an image-text matching objective that samples positive or negative caption-label pairs. TIFA, like VQ^2 , employs a VQA model with generated question-answer pairs. However, TIFA contrasts textual and visual answer candidates provided by the model, while our method checks if the textual answer is accurate given the image.

We assess each method’s ability to detect misalignments in each dataset in SeeTRUE. We use a binary labeling scheme and report the Area Under the ROC Curve (ROC AUC) for each method. For Winoground, we use existing metrics: (1) *text score*: accuracy in selecting the right caption for an image; (2) *image score*: accuracy in choosing the correct image given a caption; (3) *group score*: accuracy requiring all four image-caption pairs to be correct for a successful example.

4.2 Results

We present our main results in Table 2. Notably, our VQ^2 approach excels as the top-performing zero-shot model across all datasets, surpassing other zero-shot baselines and even outperforming most of the fine-tuned models while achieving the highest score on the challenging Winoground dataset. This shows the robustness of the VQ^2 approach, which decomposes the alignment decision by generating multiple yes/no verification questions.

When looking at finetuned models, the PaLI variant finetuned on all the available datasets outperforms all the rest with an average score of 82.9, achieving the best results on 3 out of 7 datasets. The SNLI-VE-only variant is behind with an average score of 79.7, while achieving the highest scores for 2 out of 7 datasets. This shows that integrating synthetic training data leads to notable improvements on synthetic images on DrawBench (+4%), EditBench (+11.7%), , COCO t2i (+5.5%), PickaPic-Con (+2.2%). Nevertheless, the inclusion of synthetic training data did not enhance performance on the COCO-Con dataset, comprised solely of natural images. This indicates that the variation in

Table 2: Main Results on SeeTRUE, split into zero-shot and end-to-end fine-tuned methods across the real and synthetic image-text eval-sets. The numbers in the table are ROC AUC.

Text & Images		Real + Real		Real + Synthetic			Synthetic + Real	Synthetic + Synthetic	Avg.
Model		SNLI-VE	Winoground	DrawBench	EditBench	COCO t2i	COCO-Con	PickPic-Con	
zero-shot	CLIP RN50x64	66.6	53.6	59.2	67.1	58.8	71.1	66.8	63.3
	CLIP ViT-L14	65.8	53.3	60.5	62.1	58.8	70.7	66.8	62.6
	COCA ViT-L14	68.5	53.1	67.4	66.3	62.1	74.2	68.1	65.7
	COCA ViT-L14 (f.t on COCO)	70	53.1	66.2	68.3	66.2	76.5	67.2	66.8
	BLIP	75.2	58.2	60.5	68	70.7	84.2	76.6	70.5
	BLIP2	76.4	56.9	58.5	67.5	66.9	84.3	76.9	69.6
	BLIP 2 (f.t. COCO)	75.9	60	65.7	70	73.3	85.8	78	72.7
	PaLI	65.4	53.6	60.2	56.7	53.3	65.5	60.5	59.3
	TIFA	–	58.0	73.4	67.8	72.0	–	–	–
f.t. snli-ve	VQ ² (Ours)	88.0	63.5	82.6	73.6	83.4	87.1	81.7	80.0
	OFA Large	80.5	53.3	77.6	70.9	67.5	75.4	69.5	70.7
	BLIP2	82.3	58.5	64.3	58.7	60.5	82.6	66.9	67.7
	PaLI	95.1	61.7	82.8	65.5	77.7	91.2	83.7	79.7
	PaLI + Synthetic Data	94.2	61.8	86.8	77.2	83.2	91	85.9	82.9
Avg(VQ ² , PaLI+Syn)		93.9	63.5	87.8	78.4	85.1	93	87.3	84.1

Table 3: Results on the Winoground dataset, reporting text score, image score, and group score.

Model	Text Score	Image Score	Group Score
VQ ² (Ours)	47.00	42.20	30.50
PaLI (ft SNLI-VE + Synthetic Data)	46.5	38	28.75
PaLI (ft SNLI-VE)	45.00	41.50	28.70
BLIP2 (f.t. COCO)	44.00	26.00	23.50
IAISlarge [45]	42.50	19.75	16.00
VinVL [24]	37.75	17.75	14.50
TIFA	19.00	12.50	11.30
CLIP RN50x64	26.50	13.75	10.25
OFA Large (f.t. SNLI-VE)	27.70	14.30	9.00
COCA ViT-L14 (f.t on COCO)	28.25	11.50	8.25
Random Chance [24]	25.00	25.00	16.67
Humans [24]	89.50	88.50	85.50

image types could be a contributing factor that calls for additional exploration. Notably, the last row shows a simple average between VQ² and our leading fine-tuned PaLI model, that produces higher performance, suggesting that they complement each other effectively.

Winoground Results. Table 3 shows the performance of the different methods on the challenging Winoground dataset, which requires strong visual reasoning and compositional understanding skills. Our zero-shot approach, VQ², achieves state-of-the-art results on this dataset, surpassing other strong baselines, with a group score of 30.5%. This again indicates that VQ²’s approach that decomposes the alignment task into multiple question-answer pairs is a promising path for image-text alignment.

Contradiction Generation. We assessed the VQ² method’s capacity to detect image-text contradictions, as shown in fig. 4. Contradiction generation relies on identifying the question-answer pair with



(a) “the orange lollipop is sad and the red lollipop is surprised”
Q: What is the orange lollipop feeling? A: sad

(1) Winoground



(b) “Someone in a blue hat standing on a snowy hill”
Q: What is the person wearing? A: blue hat

(2) CocoCon



(c) “A black apple and a green backpack”
Q: What color is the apple? A: black

(3) DrawBench

Figure 4: Contradicting captions and the question/answer pairs with lower VQ² alignment score, indicating the contradiction reason.

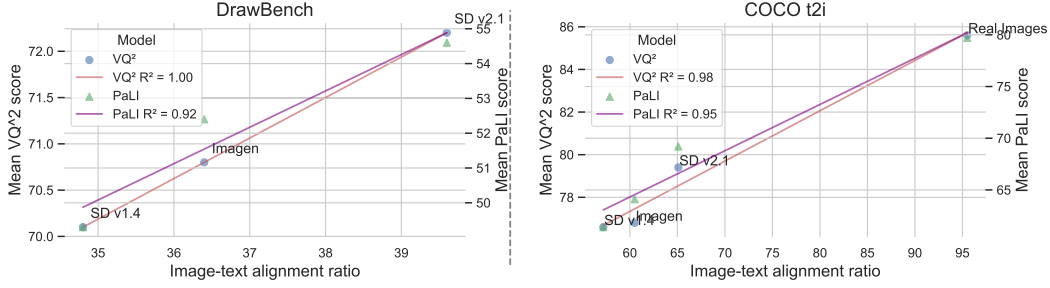


Figure 5: Highly correlated VQ² and PaLI scores vs. human rankings of text-to-image models the lowest VQA score, signaling the least likely alignment between the image and the text. Three paper authors evaluated whether a particular contradiction (consisting of a question and an answer) accurately represents the primary discrepancy between the image and the text. The majority vote among the authors determined the final outcome, yielding the following accuracy rates: 88% for Coco-Con, 74% for DrawBench, and 80% for Winoground. This indicates that our method is capable of identifying these contradictions by investigating the structure and content of the given caption and image. As a result, our method can achieve strong results, particularly on datasets that require compositional understanding.

Comparing Generative Models. VQ²'s ability to compare between generative models is demonstrated on the results of DrawBench and COCO-t2i, which include generated images from different models, together with human quality ratings. fig. 5 shows that the VQ² and our fine-tuned PaLI ranking correlates very well with human ranking ($R^2 > 0.92$). In addition, since unlike human annotations, the VQ² score is consistent across datasets, it offers a way to evaluate dataset difficulty on an absolute scale.

Table 4: Comparison of human-labeled quality scores for top-ranked images with model breakdown

Dataset	Model	Random	CLIP	PaLI	VQ ²
COCO t2i	SD 1.4	68.6	74.6	88.2	86.4
	SD 2.1	71.3	81.2	84.5	87.3
DrawBench	SD 1.4	66.7	77.4	77.4	87.1
	SD 2.1	59.0	78.0	87.0	82.0

Reranking Using Alignment Assessment. Alignment scores can also be used for reranking candidate generations, on top of evaluation. To demonstrate this, we re-rank the image candidate per prompt in the DrawBench and COCO-t2i datasets. We do so using VQ² and CLIP and measure the human-labeled quality of the top-ranked image for each method. The results, presented in table 4, show that ranking with VQ² consistently achieves higher quality scores when compared to ranking with CLIP. One such example is shown in fig. 6, where both VQ² and our top-performing fine-tuned PaLI model demonstrate superior ranking by placing the brown-and-white cats above the white-only cats. This consistency between VQ² and PaLI highlights their alignment evaluation models' potential for enhancing text-to-image systems, which contrasts with the divergent ranking exhibited by CLIP.

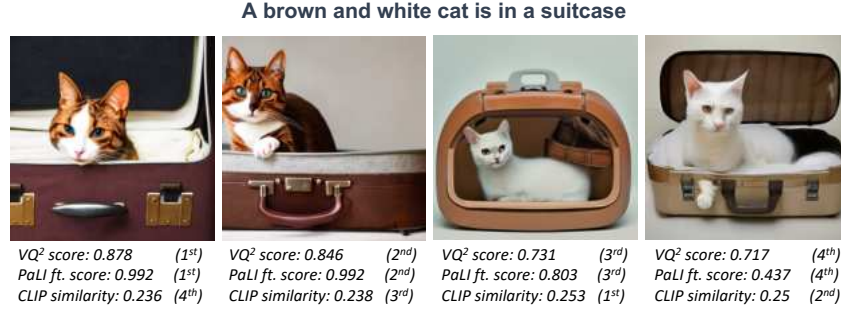


Figure 6: Four COCO-t2i text-to-image model outputs ranked by VQSQR scores, correlating with top PaLI model. Image order and CLIP (RN50x64) similarity scores given, but not aligned with VQ²/PaLI ranks.

5 Related Work

Our work advances research in visual entailment (VE) [27], visual question answering (VQA) [46], text-to-image alignment evaluation, and cross-task consistency for multi-modal models, with a focus on enhancing the semantic understanding of image-caption relationships.

Textual Entailment (TE) [26, 47] evaluates the truthfulness of a textual hypothesis given a textual premise, providing a key benchmark for the semantic capabilities of neural network models [48–50, 35]. Recently, TE has been adapted to the multimodal domain as Visual Entailment (VE) [27] to assess the semantic alignment between images and text. Vision-and-language models like CLIP [15], CoCa [22], BLIP [6], BLIP2 [7] and OFA [23] often act as bag-of-words models, lacking a deep comprehension of language compositionality [16]. Our approach addresses this by generating multiple questions probing diverse semantic aspects, thereby improving performance on challenging compositional tasks like Winoground [24] and unnatural images as in Drawbench [5].

Unlike DrawBench [5] and DALL-Eval [51] which depend on human feedback and operate within a discrete set of alignments, our approach produces automated scores for a broader range of text-image alignments, facilitating efficient evaluation of vision-and-language models. Our approach also surpasses the recently proposed TIFA [44], which may be due to employing more question-answer pairs and tailored models for question generation and answering.

Several works have explored cross-task consistency in multi-modal models across various modalities. VQA studies have tackled inconsistencies and enhanced consistency using data augmentation and contrastive loss. NLP researchers have improved consistency across tasks or within a single task by employing counterfactual instances or contrast sets [26, 47]. Our research aligns with studies that evaluate natural text and images [52]; however, extending the focus to synthetic images and texts, and aligning with synthetic image understanding research [53–58]. We introduce two unique approaches to address the complexities of image-text alignment.

Another related effort is PickScore [29], which predicts human preferences for image quality and aesthetics by ranking or choosing between two images. In contrast, our methods independently score a single image and focus specifically on image-text alignment.

6 Limitations

We recognize that in some cases, making a binary decision for whether a text and an image are aligned may be difficult, also for humans. To tackle this limitation, we provided human annotators with comprehensive guidelines, which resulted in a high inter-annotator agreement (Fleiss-Kappa score of 0.722 with 80% of the cases where all annotators agreed on the entailment label).

Although many images in our datasets were obtained by others and not created by us, we made an effort to ensure that they do not contain harmful or potentially harmful content, such as NSFW or biased imagery. During the annotation process, three individuals examined each image and indicated if it could be considered offensive. Additionally, two of the authors manually reviewed the images

for any harmful content. However, we understand that the perception of harmful or offensive content may vary among individuals and may be subject to personal interpretation.

7 Conclusion

We addressed the task of image-text alignment evaluation, which we find very close to the Visual Entailment (VNLI) task. We first introduced the SeeTRUE benchmark, which covers the mix of real and synthetic text and image pairs in text-to-image and image-to-text generation tasks, and includes challenging cases based on generated contradictions. We then proposed two methods, VQ² and end-to-end VNLI, which outperform strong baselines on SeeTRUE and can serve as a starting point for future research on the task.

In future work, we would like to employ our automatic evaluation models for guiding the training of text-to-image and image-to-text models towards more aligned outputs, following recent trends in text-to-text generation [59, 60]. For example, such models may be useful either for filtering training examples or as a reward when training models using reinforcement learning.

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A Appendix

A.1 Dataset Supplementary Materials

1. Dataset documentation, metadata, and download instructions: [anonymous](#).
2. Intended uses: we hope SeeTRUE will be used by researchers to evaluate image-text matching models.
3. Author statement: We bear all responsibility in case of violation of right in using our benchmark.
4. Each dataset’s license is described below. Our additional human annotations and generated images are licensed under CC-BY 4.0 license <https://creativecommons.org/licenses/by/4.0/legalcode>.
5. Hosting & preservation: the dataset will be hosted in Huggingface Datasets, accessible and available for open research.
6. SeeTRUE fields are presented in table [5](#).

Additional licensing details: we do publish the datasets we annotated in this work, and do not re-publish the existing SNLI-VE and Winoground datasets. Full licenses:

1. MS COCO [\[28\]](#): <https://cocodataset.org/#termsofuse>
2. EditBench [\[25\]](#): <https://research.google/resources/datasets/editbench/>, <https://www.apache.org/licenses/LICENSE-2.0>
3. DrawBench [\[5\]](#): <https://imagen.research.google/>, <https://docs.google.com/spreadsheets/d/1y7nAbmR4FREi6npB1u-Bo3GFdwdOPYJc617rB0xIRHY/edit#gid=0>
4. Pick-a-Pick [\[29\]](#): https://huggingface.co/datasets/yuvalkirstain/pickapic_v1
5. SNLI-VE [\[27\]](#): <https://github.com/necia-ml/SNLI-VE>
6. Winoground [\[24\]](#): <https://huggingface.co/datasets/facebook/winoground>

Table 5: SeeTRUE Rows Examples

image	text	label	original_dataset_id	dataset_source
img1	A zebra to the right of a fire hydrant.	0	text_133_image_1228	drawbench
img2	A group of people standing next to bags of luggage.	1	text_105_image_1377	coco_t2i
img3	a tiny figurine is surrounded by cell phones on a table.	1	3786	editbench

A.2 Human Annotation Process



a man with a mustache is sitting in front of some pine trees

Does the image present all the details described in the text correctly?

☐ Yes ☐ No

If 'No', what is the model feedback? (*main* misalignment)

☐ * Might be offensive

Figure 7: Annotation interface for determining whether a given image-text pair are aligned.

In order to provide reliable human labels for our datasets, we conducted an annotation process using the SeeTRUE platform. The process comprised several steps, including setting qualification requirements, providing instructions, and evaluating annotator agreement.

We set the basic requirements for our annotation task as follows: a percentage of approved assignments above 98%, more than 5,000 approved HITs, and annotator locations limited to the US, UK, Australia, or New Zealand. We selected 5 examples from our dataset for a qualification test and screened the annotators’ results. fig. 7 displays a sample of the Mechanical Turk user interface. The payment for the crowd-workers was 15-18 USD hourly.

The instructions provided were as follows:

Evaluate the given image and text to determine if they match, selecting either “Yes” or “No”. Some images may be synthetically generated by a text-to-image model. To assess the match, mentally generate a textual description for the image (no need to write it down) and compare this generated description to the given text. If the descriptions closely resemble each other, mark “Yes”. If not, mark “No” and provide feedback on the specific issue causing the misalignment, focusing on the primary issue if multiple misalignments are present. If you encounter an image or text that may be offensive due to bias, race, NSFW content, etc., mark the checkbox to indicate this issue.

Full agreement metrics are presented in table 6. As shown in the table, the percentage of cases where all annotators agreed and the Fleiss-Kappa scores vary across the datasets, with COCO-Con exhibiting the highest level of agreement and Drawbench the lowest. These differences highlight the varying levels of complexity within the datasets.

Table 6: Agreement metrics for different datasets.

Dataset	Full	Drawbench	COCO t2i	COCO-Con	PickaPic-Con
# Items	8,527	1,968	2,586	1,992	1,981
% all agreed	80	76	78	86	77
Fless-Kappa	0.72	0.66	0.68	0.81	0.69

A.3 Comparing VQ^2 variants

VQ^2 consists of two main parts: generating question-answer pairs and assessing question-answer pair alignment against the image. We have experimented with different variants of the VQ^2 zero-shot method.

Assessing question-answer pair alignment methods Given a question-answer pair, we would like to assess the question based on the image and compare it to the information in the text. We experimented with several configurations for answer alignment:

1. Type A: Given a question-answer pair (q_j, a_j) generated from the text, we answer the question using a VQA model and obtain an answer based on the image $a_j^I = VQA(q_j, I)$. We compare a pair of (q_j, a_j) with a pair of (q_j, a_j^I) using an Natural Language Inference (NLI) model, where the pair based on the text serves as the premise and the other as the hypothesis. We define the alignment score s_j as the probability of the NLI model for answering “entailed”.
2. Type B: As done in type A, we answer the question using a VQA model and obtain the answer $a_j^I = VQA(q_j, I)$. We use the VQA model again to compare the the two answers and determine whether they are the same. The question is formulated as “Is $\{a_j\} == \{a_j^I\}$ in this image?”. We define the alignment score s_j as the probability of the VQA model for answering “yes”.
3. Type C: We reformulate each question and answer candidate pair (q_j, a_j) into a new yes-no predicate question q_j' using the format “is $\{a_j\}$ true for $\{q_j\}$ in this image?”. The VQA model is then invoked to answer the predicate question (q_j') over image I . We define the alignment score s_j as the probability of the model for answering “yes”.

Generating question-answer pairs To produce question-answer pairs from the text, we first extract informative spans in the text T . We extract as answer candidates all named entities and noun phrases in T using spaCy. We noticed that for short text T , this method doesn’t produce enough question-answer pairs to assess the alignment between the text and the image. Thus, we extend the answer

candidates by adding multi-word spans, such as adjectives ("black and white") and location "in the air". We use the extended answer candidate extraction in all of our experiments.

Table 7 summarize the results of the VQ^2 variants on the EditBench dataset. Our zero-shot VQ^2 method is VQ^2 type C, it outperforms other configurations and it is more efficient, since it requires a single run of the VQA model for the question-answer pair assessment.

Table 7: Comparing VQ^2 configurations on all EditBench categories

Method	Models used for assessment	EditBench					
		Color	Count	Material	Shape	Size	Mean
VQ^2 (A)	VQA & NLI	75.7	63.6	71.9	67.6	75.9	70.9
VQ^2 (B)	VQA & VQA	80.2	72.3	75.6	73.5	75.4	75.4
VQ^2 (C) w.o. multi-word answers	VQA	77.2	71.5	73.6	71.9	75.5	73.9
VQ^2 (C)	VQA	78.5	73.5	76.9	71.7	78.2	75.8

A.4 Reproducibility

To fine-tune BLIP2, we adjust only the Q-former parameters of the model using the Adam optimizer. We train the model for two epochs and designate 10% of the training set as a validation set for early stopping and use learning rate selection between $\{1e-5, 5e-5\}$. A single training took 5 hours on a linux server with one A6000 GPU. All experiments took <2 days.

Zero-shot VQ^2 : For 10,000 text-image pairs, the inference time of every step is as follows. Answer candidate generation: when using extended answer candidates – about 1 day. Otherwise, 12 hours. Question generation and filtering: When using extended answer candidates, about 2 days, otherwise, 1 day. The last step only takes a few minutes.

GPT-4V(ision) as a Generalist Evaluator for Vision-Language Tasks

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Abstract

Automatically evaluating vision-language tasks is challenging, especially when it comes to reflecting human judgments due to limitations in accounting for fine-grained details. Although GPT-4V has shown promising results in various multi-modal tasks, leveraging GPT-4V as a generalist evaluator for these tasks has not yet been systematically explored. We comprehensively validate GPT-4V’s capabilities for evaluation purposes, addressing tasks ranging from foundational image-to-text and text-to-image synthesis to high-level image-to-image translations and multi-images to text alignment. We employ two evaluation methods, single-answer grading and pairwise comparison, using GPT-4V. Notably, GPT-4V shows promising agreement with humans across various tasks and evaluation methods, demonstrating immense potential for multi-modal LLMs as evaluators. Despite limitations like restricted visual clarity grading and real-world complex reasoning, its ability to provide human-aligned scores enriched with detailed explanations is promising for universal automatic evaluator[†].

1 Introduction

In recent years, significant progress has been made in vision-language research, including advancements in image-to-text generation (Li et al., 2023), text-to-image generation (Rombach et al., 2021; Ramesh et al., 2022), and image-to-image translation (Brooks et al., 2023; Zhang et al., 2023a). Concurrently, there is growing recognition of the importance of automatic metrics in vision-language tasks, as evidenced by recent work (Hessel et al., 2021b; Hu et al., 2023; Lu et al., 2023; Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023). However, developing automatic metrics that align with human preferences and provide explanatory insights can be challenging. Most existing text-image alignment metrics specialize in providing similarity scores between image and text features, leading to three key limitations: 1) **Reference-free Constraint**: existing specialized scoring models mostly require reference (e.g., CIDEr (Vedantam et al., 2015)), which is not always applicable in a broad range of tasks (e.g., text-to-image synthesis); 2) **Lack of Free-form Evaluation**: they do not adhere to specialized evaluation guidelines and fail to offer quantitative scores that satisfy multiple criteria or detailed qualitative explanations, which makes it challenging to align its assessments with human evaluators or even keep consistent with itself. 3) **Single-Pair Constraint**: they can only evaluate one image-text pair at a time, making it less suitable for more complex tasks that involve multiple image-text pairs, such as text-guided image editing. A question comes up: *Is there a promising universal automatic evaluator?*

^{*}Equal Contribution

[†]Work in Progress

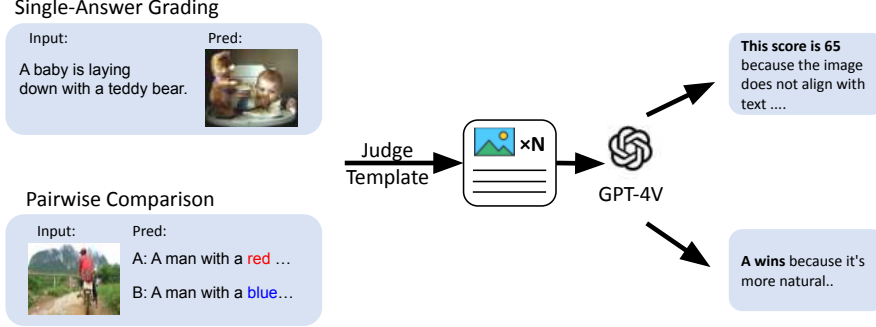


Figure 1: **Overview of GPT-4V Eval.** Our method role GPT-4V as a evaluator to assess the quality of multimodal input-output pairs of different tasks. We adopt two evaluation methods (a) single-answer grading and (b) pairwise comparison, and prompt GPT-4V to evaluate the task.

Recent advancements in Large Language Models (LLMs) have demonstrated strong instruction-following ability (OpenAI, 2022, 2023; Ouyang et al., 2022), being helpful in real-world tasks (Zheng et al., 2023; Zhang et al., 2023b; Chen et al., 2023). LLM-as-a-judge (Zheng et al., 2023) adopts LLM as a judge to evaluate outputs generated by different chatbot models. GPT-4 obtains a strong agreement with human preferences in the text evaluations. Though LLM shows promising results as a auto evaluator to substitute human annotator, the research in the multi-modal large language models (MLLMs) for evaluating diverse text-image tasks still under exploration.

GPT-4V (OpenAI, 2023) shows strong performance on various tasks (Yang et al., 2023), being able to provide detailed explanations based on customized instructions on different multi-modal tasks by receiving multimodal inputs. However, its capability to serve as evaluator for vision-language tasks remains under-explored. In this work, inspired by Zheng et al. (2023), we systematically study the *GPT-4V-as-a-Generalist-Evaluator*, to explore how well GPT-4V aligns with human preference as a evaluator in various vision-language tasks.

To summarize, our main contributions are below:

- We systematically validate the capabilities of GPT-4V as the *evaluator* in a broad range of tasks, including image-to-text task (e.g., image captioning), text-to-image task (e.g., text-to-image generation), image-to-image task (e.g., text-guided image editing).
- We demonstrate that GPT-4V is generally a good *reference-free* human-aligned evaluator when prompted carefully, which also provides reasonable and fine-grained explanation.
- We identify some limitation of GPT-4V when used as an evaluator, such as perceptual evaluation (e.g., vision clarity) and real-world complex cases.

2 GPT-4V-as-a-Generalist-Evaluator

Large Language Model (LLM) has shown promising performance in substituting human annotators for text evaluation tasks (Zheng et al., 2023). Similarly, GPT-4V has demonstrated strong results across a variety of image-based tasks (Yang et al., 2023). However, the role of GPT-4V as a multimodal evaluator remains unexplored. In this section, we demonstrate how we leverage GPT-4V as a evaluator for evaluating various multimodal tasks.

Inspired by Zheng et al. (2023), we employ two distinct evaluation methods: (1) single-answer grading and (2) pairwise comparison. These methods leverage GPT-4V to assess the quality of outputs across various multimodal tasks, taking into consideration the specific inputs associated with each task. The nature of these input-output pairs varies depending on the type of task. For example, in an image captioning task, the input is an image and the output is text, whereas in synthetic image generation, the roles are reversed².

²We conduct different evaluation methods based on the nature of the tasks. Some tasks are evaluated using both single-answer grading and pairwise comparison, while others are evaluated using only one method. We detail each evaluation setting in the corresponding subsection in Section 3.

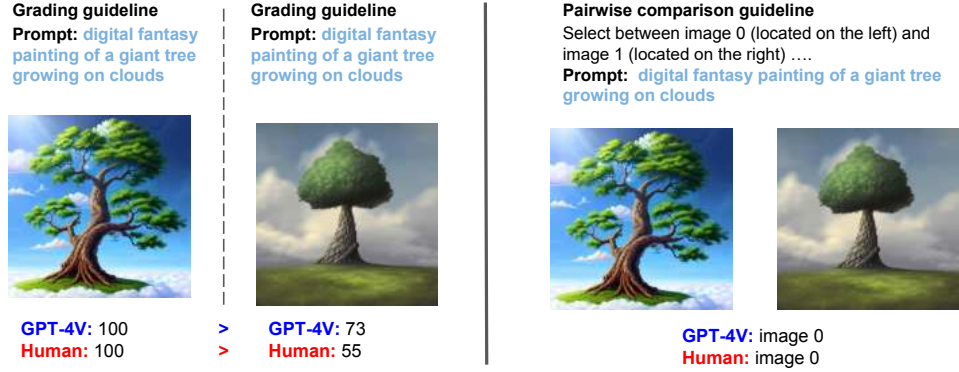


Figure 2: **Evaluation methods for single-answer grading (left) and pairwise comparison (right) on text-to-image task.** **lightblue** is the input prompt for image generation, **blue** and **red** are decisions made by different evaluators. Left: Evaluators independently evaluate each candidate answer based on the task input. The candidate answer with the higher score is deemed the evaluator’s final decision. Right: Evaluators compare a pair of candidate answers and directly make a final decision.

- **Single-answer grading:** GPT-4V is instructed to generate a score to evaluate the quality and alignment of the input-output pair. The scoring scale may vary depending on the task.
- **Pairwise comparison:** GPT-4V evaluates a single task input along with a pair of answers. It makes a evaluation to determine which of the two candidate answers is better, or whether the two answers are of equal quality.

Single-answer grading offers scalability as the number of candidate answers increases. However, it lacks capability to directly compare different answers for a specific task input. On the other hand, pairwise comparison allows for the direct comparison, while its computational complexity increases quadratically as the number of candidate answers grows.

To evaluate the effectiveness of GPT-4V as a evaluator, we conduct a comparative evaluation with human evaluators and GPT-4V. Both human evaluators and GPT-4V assess input-output pairs from various tasks to assess output quality. We categorize the results into “Answer 1,” “Answer 2,” and “Tie,” and calculate alignment scores to measure agreement among different evaluation approaches. We illustrate a single-answer grading and pairwise comparison example in Figure 2.

Single-answer grading. We ask the evaluator to provide a score within a predefined scale range. The score is based on the multimodal input-output pairs for different tasks. To obtain a final answer that represents the evaluator’s preference between two candidate outputs, we consider the one with the higher score as the preferred answer. If the scores are equal, the result is declared a “Tie”.

Pairwise comparison. To ensure a fair assessment, we implement a dual-sided comparing system by using GPT-4V to mitigate the potential position bias (Zheng et al., 2023). Each output comparison is evaluated twice, where the answer 1 and the answer 2 are alternated, with the same input task. To evaluate the final answer in dual-sided pair evaluation we follow:

- Answer 1: GPT-4V chooses Answer 1 twice or chooses once Answer 1 and once Tie.
- Tie: GPT-4V obtains twice Tie or chooses once Answer 1 and once Answer 2.
- Answer 2: GPT-4V chooses Answer 2 twice or chooses once Answer 2 and once Tie.

3 Evaluation

To test if GPT-4V can be used as a reliable evaluator, we conduct experiments on four different tasks and aim to answer four main questions:

1. Can GPT-4V provide a reasonable evaluator with explanation for different multi-modal tasks?

Table 1: **Comparison of grading between human and single-answer grading evaluation for the image-to-text task using both Pearson and Spearman correlations.** Images are scored out of a total of 100. * denotes significant correlations at $p < 0.05$.

	Ground-Truth		Hard-Negative	
	Pearson	Spearman	Pearson	Spearman
CLIPScore	0.164	0.072	0.098	0.036
GPT-4V	0.401*	0.499*	0.687*	0.612*

Table 2: **Agreement [%] between evaluation methods across GPT-4V and human validates the image-to-text task.** “GPT-4V-Pair” and “GPT-4V-Single” represent GPT-4V evaluations using pairwise comparison and single-answer grading, respectively. The baseline agreement between two random evaluators is 33%.

Evaluator	GPT-4V-Single	Human
GPT-4V-Pair	91.0	95.0
GPT-4V-Single	-	86.0

- Does GPT-4V make consistent predictions across different evaluation methods described in Section 2?
- Does GPT-4V match what humans think in single-answer and pairwise evaluation, respectively?

3.1 Image-to-Text Captioning

Data Construction. We first sample two subsets from MSCOCO Train 2017 (Lin et al., 2014) and Conceptual Captions 12M (Changpinyo et al., 2021) image-text paired datasets for evaluating GPT-4V on the image-to-text generation tasks. For each dataset, we sampled 50 image-text pairs via clustering-based sampling to ensure the diversity in constructed dataset and in total the constructed dataset size is 100.

Evaluation Setup. We propose to evaluate whether GPT-4V can be consistent, accurate, and unbiased in providing a score towards the generated caption quality in both single scoring and pairwise comparison settings. Due to the lack of negative captions used for paired comparison with ground truth high-quality caption, we follow ALIGN (Jia et al., 2021) to manually construct hard negative captions via adding the following noises: 1) category change, i.e., a Toyota car to a Volkswagen car; 2) color change, i.e., white to gray; 3) location change, i.e., changing the country name; 4) number change, i.e., 20 to 50; 5) position change, i.e., behind to front; 6) subtle dropping, i.e., deleting the adverbial to the object. These introduced noises still keep the noisy caption a fluent sentence and differentiating these noisy captions from original ground-truth ones requires the evaluator to figure out the fine-grained dis-alignment between noisy words and the image details, making such negative captions **hard** to grade.

We ask expert-level human labelers to give consistent scoring towards both the ground-truth and hard-negative captions in the same scale of 100. Then we firstly compute the correlation and alignment between GPT-4V’s scoring and human expert scoring in terms of image-text alignment. We report Pearson and Spearman correlation scores for both ground-truth set and hard-negative set. We also compare GPT-4V-as-an-evaluator with the strong reference-free baseline CLIPScore (Hessel et al., 2021a) on the single-answer grading setting. The CLIPScore is calculated following Hessel et al. (2021a) using ViT-L/14 (Dosovitskiy et al., 2020) pre-trained checkpoint and it is scaled to $[0, 100]$. Secondly, in terms of the pairwise comparison, we report the agreement score between GPT-4V’s evaluation and human evaluation in both single-answer scoring and pairwise comparison settings. In addition, we also compute the agreement between GPT-4V single-answer scoring and pairwise comparison to demonstrate its consistency in different evaluation variants.

GPT-4V is robust and effective in evaluating image-to-text captioning tasks. The results on the agreement between human and GPT-4V scoring are presented in Table 1. GPT-4V’s scoring is significantly correlated with human rating in both ground-truth subset evaluation and hard-negative

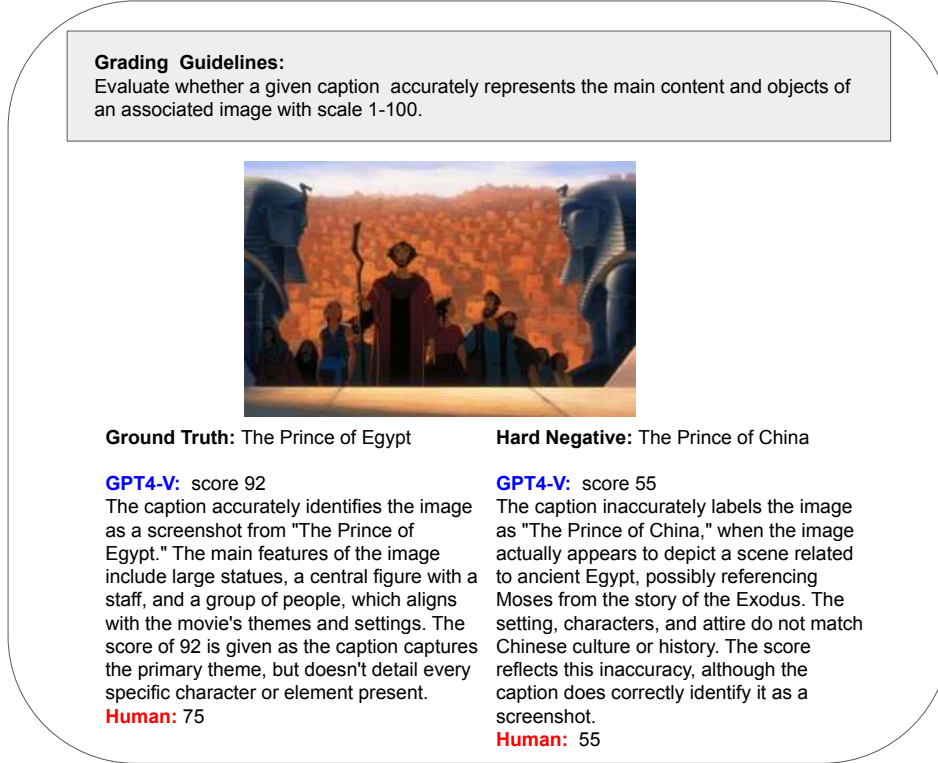


Figure 3: **Examples of image-to-text evaluation task through single-answer grading on human-preferred ground truth and hard negative outputs, respectively.** GPT-4V successfully identifies the text caption quality and provides comprehensive explanation for its scores.

subset evaluation. GPT-4V also significantly outperforms CLIPScore in human grading alignment with much higher correlation scores. We analyze the standard deviation to take insights on the score distribution given by GPT-4V and humans. We find that the standard deviation of GPT-4V grading is much larger than that of human grading. We attribute such fluctuation around the average score to the nucleus decoding used by GPT-4V. When doing sampling based decoding, a group of neighboring scores like 80/82/85 are all within the sampling candidates and thus the final output scores contains much more randomness than that of humans.

GPT-4V is as consistent as human on different evaluation methods on image-to-text captioning tasks. The results on the consistency analysis across different evaluation methods are presented in Table 2. GPT-4V-Single’s label is computed based on the larger score of the ground-truth caption and hard-negative caption towards the same image. Then we compute the agreement between GPT-4V-Pair and GPT-4V-Single to present the consistency across different evaluation methods. We can conclude that GPT-4V is a robust and consistent evaluator on image captioning tasks, which achieves an agreement score of 91% between single-pair-scoring and pairwise-comparison. Additionally, GPT-4V is also consistent with human scoring using both evaluation methods. The human agreement score of GPT-4V-Single is much lower than that of GPT-4V-Pair, which aligns with human intuition that the direct comparison is easier to complete than keeping a consistent standard on single-answer grading.

3.2 Text-to-Image Generation

Text-to-image generation involves feeding a text prompt into a model to produce an image that aligns with the specific requirements outlined in the prompt. We introduce GPT-4V as an evaluator to assess pairs of syntactically generated images, determining which image is of higher quality and better aligns with the text prompt.

Table 3: **Comparison of grading between human and single-answer grading evaluation for the text-to-image task using both Pearson and Spearman correlations.** Images are scored out of a total of 100, divided among three categories: *Relevance* (40 points), *Visual Clarity* (30 points), and *Object Accuracy* (30 points). * denotes significant correlations at $p < 0.05$.

		Relevance	Visual Clarity	Object Accuracy	Total
Pearson	Ground-Truth	0.530*	0.247	0.451	0.502
	Hard-Negative	0.653*	0.318*	0.587*	0.601*
Spearman	Ground-Truth	0.581*	0.102	0.469*	0.496*
	Hard-Negative	0.577*	0.194	0.584*	0.535*

Table 4: **Agreement [%] between evaluation methods across GPT-4V and human validates text-to-image task.** "GPT-4V-Pair" and "GPT-4V-Single" represent GPT-4V evaluations using pairwise comparison and single-answer grading, respectively. The baseline agreement between two random evaluators is 33%.

Evaluator	GPT-4V-Single	Human
GPT-4V-Pair	41.43	52.85
GPT-4V-Single	-	51.43

Data Construction. We selected 20 samples from ImagenHub (Ku et al., 2023) and 50 from Pick-a-Pic (Kirstain et al., 2023) for evaluation. ImagenHub offers diverse compositional prompts, while Pick-a-Pic uses real-user prompts that often have multiple requirements, creating a realistic challenge for GPT-4V image synthesis. Each sample contains a text prompt and a pair of images for evaluation. We first manually annotate two images in each sample, one to be ground-truth and the other to be hard negative based on the alignment of the prompt text. It is worth noting that the hard negative samples in text-to-image generation tasks are much **difficult** to distinguish than that of captioning tasks. It can even take several minutes for a human expert to identify the better one from the good image and the hard-negative one.

Evaluation Setup. To ensure a comprehensive and unbiased evaluation in the single-answer grading setting, we propose to assess the generated image from three key dimensions: Relevance, Visual Clarity, and Objective Accuracy, totaling 100 points³. Specifically,

- Relevance (40): Alignment of the image and the prompt, including both explicit and implicit details.
- Visual Clarity (30): Technical quality such as sharpness and absence of artifacts.
- Objective Accuracy (30): Accuracy of the objects or nouns described in the prompt, without extraneous elements.

Both human and GPT-4V grade each text-image pair based on the three-dimension criteria. For pair-comparison setting, the GPT-4V is asked to compare the pair of generated images with chain-of-thought prompting to improve reasoning capability.

GPT-4V exhibits potentials to score text-to-image tasks. We calculate the Person and Spearman correlations and average score of GPT-4V and human across the ground-truth images and hard negative images on three sub grading categories and the total score, the results are shown in the upper part of Table 3.

GPT-4V exhibits a general correlation with human evaluations in terms of the total score for both ground-truth and hard negative images, underscoring the viability of employing GPT-4V for text-to-image evaluation tasks. Delving into specific scoring categories, GPT-4V manifests a moderate correlation with human ratings in relevance and object accuracy for both ground-truth and hard negative images. However, GPT-4V struggles in detecting visual errors in images, tending to award higher scores for visual clarity compared to human evaluators. In addition, we observed that GPT-4V

³We defined the detailed score criteria into appendix.

Grading Guidelines:

Instructions: Carefully assess the generated image in terms of relevance to the prompt, visual clarity, and object accuracy. Use the following criteria to guide your evaluation: with Relevance (0-30 points) Visual Clarity (0-30 points), Object Accuracy (0-30 points).

(a) **Prompt:** locust pudding



VS



Ground Truth

GPT4-V: 38,30,30,98

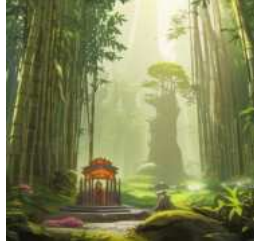
Human: 40,30,30,100

Hard Negative

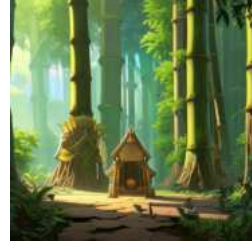
GPT4-V: 20,28,18,66

Human: 20,30,15,65

(b) **Prompt:** Stunning shrine to Hylia gods in a bamboo forest. Legend of Zelda. Breath of the Wild. Wadim Kashin, Alphonse Mucha. Animal gods shrine. landscape Incredibly detailed matte painting. Ismail Inceoglu, Junji Ito, James Gurney, Victo Ngai, Gregorio Catarino, M.W Kaluta. 8K resolution, DSLR, VRAY, Raytraced. Bamboo forest. Stone shrine



VS



Ground Truth

GPT4-V: 35,28,23,86

Human: 38,30,28,96

Hard Negative

GPT4-V: 34,28,26,88

Human: 35,28,25,88

(c) **Prompt:** edinburgh sir walter scott monument



VS



Ground Truth

GPT4-V: 35,28,28,91

Human: 38,30,28,96

Hard Negative

GPT4-V: 37,28,28,93

Human: 30,30,28,88

Figure 4: Three Examples of text-to-image evaluation task through single-answer grading on human-preferred ground truth and hard negative outputs, respectively. (a) GPT-4V successfully identify the image qualities, when the gap between text-image pair is relative large. However, GPT-4V struggles to accurately identify instances where (b) the prompt has multiple stringent requirements, and (c) the evaluation demands a strong foundation of background knowledge. Noting: these challenges are also difficult for human evaluators, requiring meticulous scrutiny and reference to external knowledge for accurate scoring.

tends to have more statistically significant correlation with humans in the hard negative images. This suggests that while GPT-4V may be less discerning in spotting subtle imperfections, it aligns more closely with human perspectives when grading images with more pronounced flaws or deviations.

GPT-4V shows promising agreement with human on different evaluation methods on text-to-image tasks. To study the agreement between evaluation methods employed by GPT-4V and human

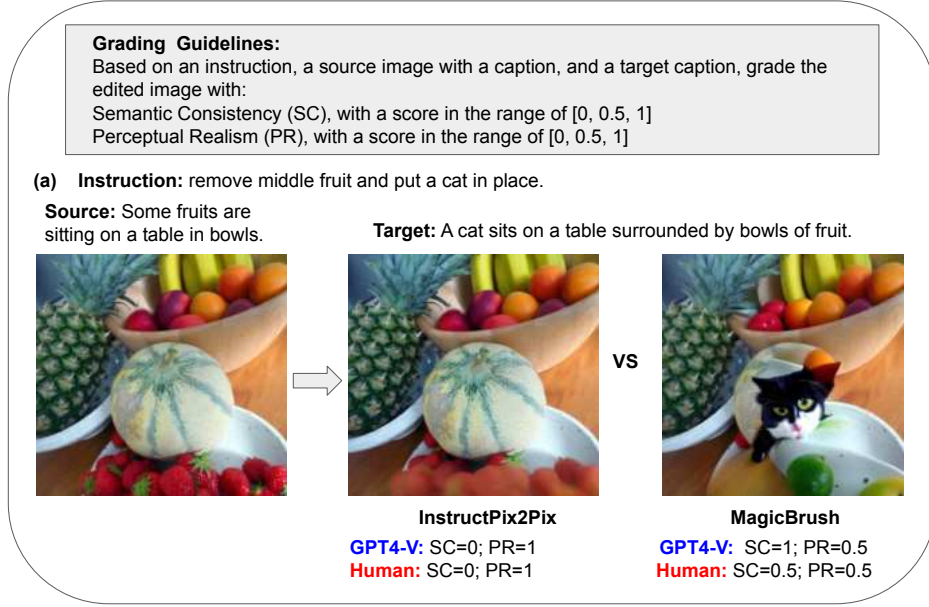


Figure 5: **Showcase of GPT-4V for evaluation text-guided image editing.** The Evaluation Guideline is following the one in ImagenHub (Ku et al., 2023).

evaluators, we compare both single-answer and pairwise evaluations with human. Specifically, we compute the number of agreement between two evaluators and divide the number by the total number of tasks. The results are presented in Table 4.

GPT-4V with both single answer grading and pairwise comparison show promising agreements with human, achieving more than 50% agreement rate, given the random guess agreement is 33% of two random evaluators. However, the agreement between single-answer grading and pairwise comparison conducted by GPT-4-V falls short. This discrepancy may arise because the instances we selected are challenging; some are even difficult for humans to assess, requiring nuanced judgment and extensive background knowledge. Different prompts and a larger sample size could affect the agreement rates. We plan to explore this further in future work.

GPT-4V shows capability in evaluating image quality in text-to-image tasks. We further study how GPT-4V aligns successfully with human evaluations and identify cases where GPT-4V struggles. We conduct case studies for single-answer grading, shown in Figure 4.

GPT-4V aligns with human grades when the image has a relatively large gap compared to the prompt text, as shown in Figure 4(a). However, it fails to align with human scores when the prompt text is noisy and contains multiple detailed requirements, as shown in Figure 4(b), or requires intensive knowledge, as shown in Figure 4(c).

3.3 Text-guided Image Editing

Recent advances (Brooks et al., 2023; Zhang et al., 2023a) in text-guided image editing has shown incredible results in faithfully following free-form editing instruction. In this task, the models are required to follow instructions to edit images, where the input is a source image and an editing prompt, and the output is the target image.

Data Construction. We randomly select 30 samples from ImagenHub (Ku et al., 2023) dataset, which include diversified instances with edited images synthesized by InstructPix2Pix (Brooks et al., 2023) and MagicBrush (Zhang et al., 2023a), etc.

Evaluation Setting The key to a successful editing requires both the correct editing of the image and keeping the background that is not requested to be changed the same. This usually means

Table 5: **Comparison of Pearson and Spearman’s coefficients between GPT-4V and Human across different tasks and benchmarks.** * indicates significant correlations ($p < 0.05$). Notice that the results of CLIPScore, DINO are reported in ImagenHub (Ku et al., 2023).

	Semantic Consistency		Perceptual Realism		
	CLIPScore	GPT-4V	DINO	CLIP-I	GPT-4V
Pearson’s Correlation	-	0.4255*	-	-	0.3915*
Spearman’s Correlation	0.0765	0.4450*	0.0449	0.0765	0.4090*

the requirements include: 1) the synthesized image is faithful in the background compared with the source image, 2) the synthesized image follows the edit instruction and align with the target caption prompt if given any, 3) the synthesized image should be perceptually natural. Since multiple requirements co-exist in the image editing task, the automatic evaluation is even more challenging than text-to-image synthesis evaluation. Recent work (Ku et al., 2023) brings the evaluation in the same pipeline. We follow the evaluation principles to mainly validate two popular and state-of-the-art models, InstructPix2Pix (Brooks et al., 2023) and MagicBrush (Zhang et al., 2023a). We feed the source image and edited image, together with source prompt, editing instruction and target prompt into GPT-4V, and feed the same evaluation guidelines that are provided to human evaluators.

GPT-4V reveals promising alignment with human evaluators in text-to-image tasks. Following previous standardized evaluation pipeline (Ku et al., 2023), we mainly focus on two aspects to evaluate text-guided image editing: 1) Semantic Consistency and 2) Perceptual Realism. We report the Pearson and Spearman correlation between GPT-4V and human evaluators in Table 5.

We observe that GPT-4V is generally correlated with human evaluators in terms of semantic consistency. However it sometimes fails to well capture the perceptual realism. And we also find that GPT-4V is giving the same preference to MagicBrush over InstructPix2Pix in terms of average Semantic Consistency and average Perceptual Realism as human evaluators. For comparison with the existing metrics under the same evaluation guidelines, we refer to numbers reported in ImagenHub (Ku et al., 2023), where the metrics DINO, CLIP-I get Spearman’s correlation of 0.0439, 0.0449 and 0.0765 for Perceptual Realism respectively, and CLIPScore gets Spearman’s correlation of 0.0437 for Semantic Consistency. This indicates that GPT-4V as a generalist evaluator can surpass specialized metrics for this task. We showcase one example of the evaluation in Figure 5 comparing the GPT-4V and Human scoring, showing incredible evaluation capability of GPT-4V in this complicated setting.

3.4 Multiple Images to Text Alignment

One advancement of recent vision-language models (OpenAI, 2023; Alayrac et al., 2022; Ye et al., 2023) is their ability to take interleaved image-text inputs, i.e., images can appear in any position of the input sequence. Here we consider a special case with multiple images to text generation. GPT-4V is asked to take inputs of multiple images and the text description, and evaluate the alignment between multiple images and the given text description.

Data Construction. We select a subset from the Gest dataset (Yan et al., 2023) for multi-image to text generation. The dataset consists of user uploaded images and corresponding reviews crawled from Google Maps. Specifically, to construct the evaluation set, we randomly sample 50 user reviews which contain 2-4 images, since currently GPT-4V only supports a maximum of 4 images in a single dialogue.

Evaluation Setup. Following previous evaluation setting (Yan et al., 2023), in this task, we evaluate the alignment between the multiple images and the user review. Here our scoring system is GPT-4V-Single, where GPT-4V is instructed to generate a score to evaluate the quality and alignment of the multiple-image input and text output pair. For example, a high score is given if the user review mentions the dishes in the images. A generic review such as “great food and service” without mentioning the detailed content of those images will receive low scores. Both human and GPT-4V are asked to grade each sample based on the visual alignment between the review and the images, on

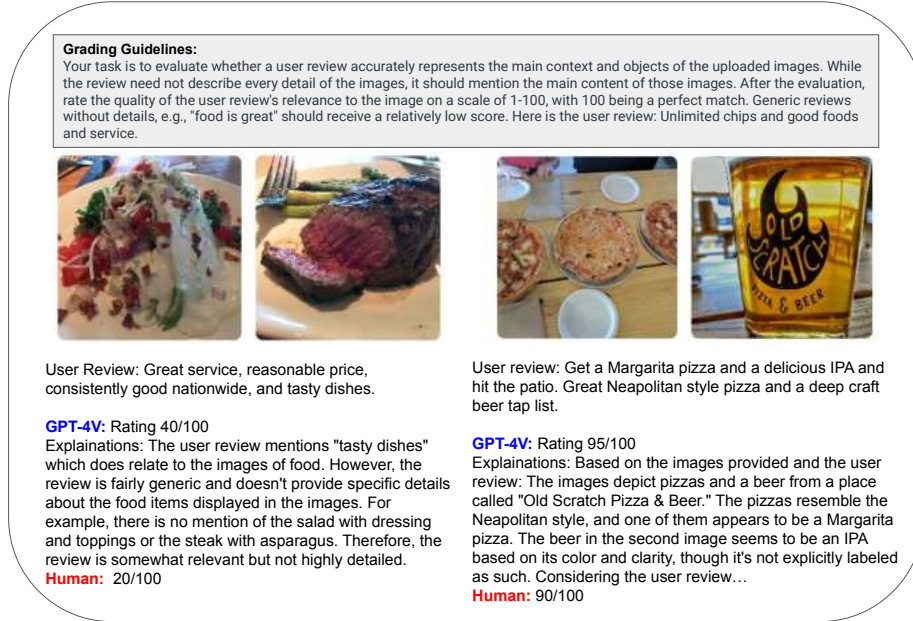


Figure 6: **Examples of GPT-4V for evaluating multiple images to text task.** GPT-4V successfully identifies the main objects of each image and provides comprehensive explanations for its scores.

a scale of 1-100. Two examples of input images and user reviews, along with GPT-4V and human scores are presented in Figure 6. As a generalist evaluator, GPT-4V need to accurately under the internal-relation between multiple images and the relation between images and text.

GPT-4V has the potential of being a strong evaluator for multi-image to text evaluation. Pearson and Spearman Correlations between GPT-4V’s scoring and human scoring are computed to evaluate GPT-4V’s performance for this task.

Shown in Table 6, both Pearson and Spearman correlation scores are around 0.8, suggesting that there is a strong positive relationship between the GPT-4V scores and human scores. This indicates that GPT-4V successfully understands the internal-relation between multiple images and the relation between images and text, showing the potential of using GPT-4V as an evaluator for evaluating task. Additionally, since GPT-4V also generates its reasoning process in text by thinking step by step (Kojima et al., 2022), it provides an appealing property for interpret model decisions. We manually evaluate the generated text descriptions of each image from GPT-4V, we find that GPT-4V is good at accurately describing each image separately, outperforming our observations on recent open-source VLMs (Ye et al., 2023), which often fail to capture the correct information of each image in an input sequence.

Table 6: **Average evaluation scores from GPT-4V and human, along with Pearson and Spearman correlations between the two.** Results on evaluated on the Google Local dataset. * indicates significant correlations ($p < 0.05$).

GPT-4V-AVG	Human-AVG	Pearson	Spearman
68.90	60.10	0.826*	0.794*

4 Related Work

Multimodal Large Language Models. Driven by the success of text-only Large Language Models (LLMs), i.e. GPT-4 (OpenAI, 2023), Claude (Anthropic, 2023), Llama (Touvron et al., 2023), Multimodal Large Language Models (MLLMs) are the next step towards Artificial General Intelligence (AGI) and arouses high research attentions. There are mainly two research lines in developing strong

Instruction: One cat and two dogs sitting on the grass.



Visual Clarity Score (30):

GPT4-V: Visual Clarity (0-30 points):

- The animals are clear and well-defined. - The colors of the environment and the animals are vibrant and pleasing. - No visible artifacts or clarity issues. Score: 30/30

Human: 25

Figure 7: **GPT-4V’s Visual Clarity Scoring: A Comparative Analysis.** While GPT-4V awarded a perfect score for visual clarity, human evaluators marked the image down due to the unclearness of the cat’s and the yellow dog’s faces.

MLLMs, one-stage multimodal pre-trained models and two-stage vision-language alignment models. The one-stage pre-trained MLLMs like Fuyu (Bavishi et al., 2023) and Kosmos (Huang et al., 2023) are directly pre-trained on mixed large-scale multimodal corpus, including text-only corpora, image-text paired data, and image-text interleaved data, which leads to strong generalization capability. Specifically, Fuyu processes the image as patches for embedding, and Kosmos adopts CLIP (Radford et al., 2021) for encoding image embedding. Another type of MLLMs are constructed on the strong pre-trained instruction following LLMs and vision encoders. MiniGPT-4 (Zhu et al., 2023) adopts Q-former of pre-trained BLIP-2 (Li et al., 2023) as image encoder and Vicuna (Chiang et al., 2023) as the text generator, while LLaVA (Liu et al., 2023a) adopts CLIP as the image encoder for simplicity. The two-stage vision-language alignment involves the feature alignment stage to unify the embedding space of image and text, and the multimodal instruction tuning stage to engage the model to follow multi-modal instructions.

Large Language Model as Interpretable Reference-free Metric. With the strong zero-shot task completion capability, LLMs are deployed as the replacement of human evaluator and automatic evaluator on open-ended language generations tasks. G-Eval (Liu et al., 2023b) investigates the robustness and trustfulness of adopting GPT-4 to evaluate the quality of summaries generated by models in a single-summary grading manner. LLM-as-a-judge (Zheng et al., 2023) adopts LLMs to evaluate the task completion answer quality for LLMs in settings of single-sample grading, pairwise comparison, and reference-guided grading. In multi-modal domain, reference-based automatic evaluation metrics like Rouge (Lin, 2004), and METEOR (Banerjee & Lavie, 2005) are usually constrained by the lack of references in some cases, such as text-to-image synthesis. Then recently proposed reference-free metrics (Hessel et al., 2021b; Li et al., 2023; Xu et al., 2023; Wu et al., 2023; Kirstain et al., 2023) are able to capture the text-image alignment. More recent metrics (Hu et al., 2023; Lu et al., 2023) show promising results by utilizing large pre-trained models to enable the interpretable evaluations for vision-language tasks. Nevertheless, none of these metrics are capable of adapting to diversified evaluation guidelines and suit for various tasks. In this work, we explore the potential of GPT-4V as a universal reference-free metric that align with human preference in a broad domain of vision-language tasks.

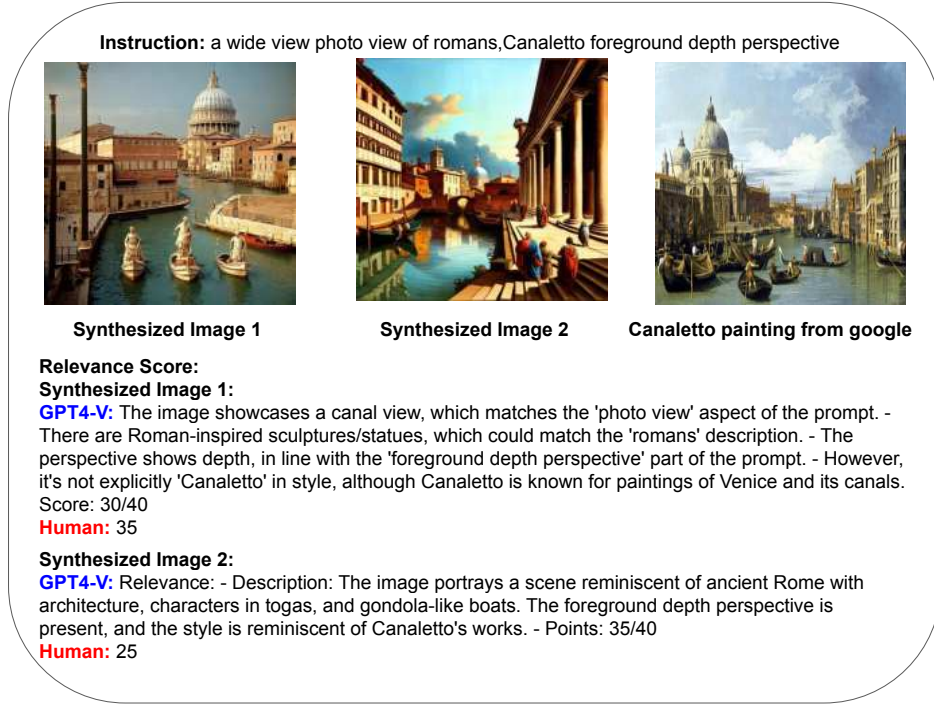


Figure 8: **GPT-4V’s Knowledge-Intensive Task Evaluation.** Comparison of GPT-4V’s evaluation on images with Canaletto’s painting style, highlighting challenges in knowledge-intensive tasks.

5 Discussion

Limited capacity in grading image with complex caption. We discovered that GPT-4V performs significantly better in image-to-text tasks compared to text-to-image tasks. This discrepancy could be attributed to the complexity and variability of real-world captions used for generating images. For instance, in Figure 4 (b), the caption utilized for image generation is long and cluttered, containing a mix of both abstract and detailed requirements. In contrast, the outputs for image-to-text tasks are easier to understand, more generalized, and well-formatted, as demonstrated in Figure 3.

Discrepancies in Visual Clarity Assessments Our examination revealed that GPT-4V exhibits potential challenges in discerning subtle differences in visual clarity. This is illustrated in Figures 7, where GPT-4V attributed higher scores for visual clarity compared to those determined by human evaluators. Human assessors allocated a score of 25 out of a maximum of 30, noting that the visual indistinctness of facial features in both the cat and the yellow dog detracted from the overall clarity of the images. This suggests that GPT-4V might struggle with capturing and evaluating fine details in visual quality, resulting in a comparatively reduced effectiveness in text-to-image evaluation tasks. In contrast, the image-to-text tasks appear to be less affected by these clarity issues, as they do not emphasize the sharpness of image details as heavily.

Challenges with Knowledge-Intensive Tasks. GPT-4V appears to encounter difficulties when tasked with evaluations that demand specific expertise. As illustrated in Figure 8, the evaluation required an understanding of Canaletto’s painting style. Surprisingly, GPT-4V awarded a higher score to Synthesized Image 2 than Synthesized Image 1, even though Synthesized Image 1 more closely mirrors Canaletto’s style, as can be observed from a reference Canaletto painting sourced via a Google search.

Position Bias Analysis. Position bias is pointed out in previous research single-modal evaluator (Zheng et al., 2023). In our primary study, minimal position bias was noted in the image-to-text and text-to-image tasks, as shown in 7. We offer the following explanations:

Table 7: **Position bias analysis of different tasks for GPT-4V evaluation.** Consistency(%) is a evaluator gives consistent results when swapping the order of two assistants. “Biased toward first/second” (%) is a evaluator favors the first/second answer.

task	# instances	consistency	biased toward first	biased toward second
image-to-text	100	92%	3%	2%
text-to-image	70	51.5%	10.1%	9.1%

- For the image-to-text task, GPT-4V efficiently handled hard negative captions introduced with subtle noise, thereby reducing the potential for pronounced position bias.
- As for the text-to-image task, a nearly equal bias was observed toward both the first and second positions. This balanced bias might be a result of GPT-4V’s underlying multimodal architecture. To gain a clearer understanding, we intend to conduct more comprehensive experiments with an expanded sample size.

6 Conclusion

In this paper, we propose GPT-4V as an evaluator for multimodal tasks and systematically examine its efficacy on four different tasks: image-to-text captioning, text-to-image generation, text-guided image editing, and multiple-image to text alignment. Our results reveal that GPT-4V achieves a promising agreement rate with human evaluators across different tasks, thereby laying a robust foundation for a MLLM-based evaluation framework.

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A Image-to-Text Captioning

A.1 Prompt

Single-Answer Grading:

Your task is to evaluate whether a given text caption accurately represents the main content and objects of an associated image. While the caption need not describe every detail of the image, it should convey the overall theme or subject. After your evaluation, rate the quality of the text caption's match to the image on a scale of 1-100, with 100 being a perfect match.

Pairwise Comparison :

Select between caption 0 and caption 1, according to which one you believe aligns most accurately with the provided image. In cases where both captions seem to possess equal quality in adherence to the image, respond with 'Tie'. Your selection can be subjective. Give your intermediate thinking step by step before giving the final response. Your final response must be 'caption 0', 'caption 1', or 'Tie'. Output your final answer in the format ""Final Answer: caption 0/caption 1/Tie.

A.2 More Showcases



Ground truth caption: A baby is laying down with a teddy bear.
Hard negative caption: A baby is laying down with a toy sheep.

Single-answer grading:

(1) Ground truth

Human: 100/100

GPT-4V: Rating: 95/100. The caption matches the content of the image very well. The main subjects, the baby and the teddy bear, are both accurately described in the caption. The slight deduction is because the image also has details like the baby's hand holding a cloth with designs, but the main elements are well captured in the caption.

(2) Hard negative

Human: 60/100

GPT-4V: The caption's match to the image is rated: 60/100. The image shows a baby laying down with a toy, but the toy is a teddy bear, not a sheep. The overall theme of a baby with a toy is captured, but the specific type of toy is inaccurately described.

Pairwise comparison:

(3) ground truth caption first

GPT-4V:

"Step 1: Identify the main subjects in the image.

- The image contains a baby and a stuffed animal.

Step 2: Determine the type of stuffed animal.

- The stuffed animal closely resembles a teddy bear.

Step 3: Compare the description from the captions with the observations from the image.

- Caption 0 states that the baby is with a teddy bear, which aligns with the observation from the image.

- Caption 1 mentions a toy sheep, which does not align with the observation.

Step 4: Choose the caption that most accurately describes the image.

- Based on the observation, caption 0 is a more accurate description of the image compared to caption 1.

Final Answer: caption 0."

(4) hard negative caption first

GPT-4V:

"Intermediate Thinking:

1. Observing the image, there is a baby and a plush toy present.

2. The plush toy doesn't resemble a sheep in terms of its appearance, ears, or face.

3. The plush toy closely resembles a teddy bear given its features and overall look.

Final Answer: caption 1."

Figure 9: Example of grading on image-to-text evaluation.



Ground truth caption: A man with a blue helmet on a small moped on a dirt road.

Hard negative caption: A man with a red helmet on a small moped on a dirt road.

Single-answer grading:

(1) Ground truth

Human: 80/100

GPT-4V: Evaluation: The caption accurately describes the main content of the image, capturing the man with the red helmet on a moped and the dirt road setting. However, the caption does not mention the striking mountainous background and the mist/fog around the mountains, which are significant elements of the image. Rating: 85/100. The caption captures the primary subject but omits notable details in the background.

(2) Hard negative

Human: 65/100

GPT-4V: Rating: 70. The caption does capture the essence of the image, but there are discrepancies in the details such as the color of the helmet.

Pairwise comparison:

(3) ground truth caption first

GPT-4V:

"Intermediate thinking:

1. Observing the image, the setting is a dirt road.
2. There's a man on a small moped.
3. The helmet the man is wearing is red.

Comparing this observation with the captions:

1. Caption 0 mentions a ""red helmet"", which matches the image.
2. Caption 1 mentions a ""blue helmet"", which does not match the image.

Based on this analysis, caption 0 is more accurate.

Final Answer: caption 0."

(4) hard negative caption first

GPT-4V:

"Intermediate thinking:

1. The man in the image is indeed on a small moped.
2. The road he's on seems to be a dirt road.
3. The color of the helmet he is wearing is red.

Given these observations:

caption 0: Incorrectly mentions the helmet color as blue.

caption 1: Correctly mentions the helmet color as red and also describes the man on a small moped on a dirt road.

Final Answer: caption 1."

Figure 10: Example of grading on image-to-text evaluation.



Ground truth caption: A cat in the foreground and a duck in the background

Hard negative caption: A man with a red helmet on a small moped on a dirt road.

Single-answer grading:

(1) Ground truth

Human: 80/100

GPT-4V: 89/100. The caption accurately represents the main content of the image, but it could specify that the duck is on water.

(2) Hard negative

Human: 40/100

GPT-4V: "The caption describes the image as having ""a duck in the foreground and a cat in the background."" Based on the provided image, this caption seems to have reversed the positions of the cat and the duck. The cat is more in the foreground while the duck is in the background.

Rating: 70/100."

Pairwise comparison:

(3) ground truth caption first

GPT-4V:

"Intermediate Thinking:

1. Observing the image, there is a cat that appears closer to the viewer and a duck that appears further away.
2. The term ""foreground"" typically refers to the part of a scene that is nearest to the viewer. In contrast, the ""background"" refers to the part of a scene that is furthest from the viewer.
3. Based on the positions of the cat and the duck, the cat is in the foreground while the duck is in the background.

Final Answer: caption 0"

(4) hard negative caption first

GPT-4V:

"Intermediate Thinking:

1. Observing the image, there is a clear depiction of both a duck and a cat.
2. When considering the terms ""foreground"" and ""background"", we need to determine which of the animals appears closer to the viewer and which appears farther away.
3. The cat appears closer and is more prominent in the photo, which indicates that it's in the foreground.
4. The duck, on the other hand, appears farther away and is therefore in the background.

Final Answer: caption 1"

Figure 11: Example of grading on image-to-text evaluation.

B Text-to-Image Generation

B.1 Prompt

Single-Answer Grading:

Prompt Given to Image Generation Model: [[[The instruction to image generation]]]

Instructions: Carefully assess the generated image in terms of relevance to the prompt, visual clarity, and object accuracy. Use the following criteria to guide your evaluation:

Relevance (0-40 points):

How closely does the generated image match the prompt?

Does the image capture the core essence and details specified in the prompt?

Visual Clarity (0-30 points):

How clear and sharp is the image?

Are there any artifacts or oddities that detract from the image's quality?

Are the colors and contrasts pleasing and coherent?

Object Accuracy (0-30 points):

Are all the nouns from the prompt represented accurately in the image?

Do the depicted objects match the description and intention of the original nouns in the prompt?

Is there any object that doesn't belong or was not mentioned in the prompt?

Total Score: __/100

Pairwise Comparison :

Select between image 0 (located on the left) and image 1 (located on the right), according to which one you believe aligns most accurately with the provided instructions. In cases where both images seem to possess equal quality in adherence to the instructions, respond with 'Tie'.

Your selection can be subjective. Give your intermediate thinking step by step before give the final respond. Your final response must be 'image 0', 'image 1' or 'Tie'. Output your final answer as the format "Final Answer: image 0/image 1/Tie."

Instruction: [[[The instruction to image generation]]]

B.2 More Showcases

Prompt: woman reformation stgeorgefamine lifeboat drowning igleartwork, Jules Bastien-Lepage, side portrait



Ground Truth



Hard Negative

Single-Answer Grading

GPT-4V: 30,28,20,78

- **Relevance (0-40 points):** The image depicts a woman in a boat on the water, which aligns with some of the elements in the prompt such as "woman", "lifeboat", and "drowning". The style of the artwork appears to be reminiscent of works from artists like Jules Bastien-Lepage. However, elements like "reformation", "stgeorgefamine", and "igleartwork" are not clearly depicted or may be too abstract to be represented in the image. Score: 30/40
- **Visual Clarity (0-30 points):** The image is clear and sharp. There are no obvious artifacts or oddities present in the image. The colors are realistic and the contrasts, especially between the woman and the background, are pleasing and coherent. Score: 28/30
- **Object Accuracy (0-30 points):** The woman and the lifeboat are depicted accurately based on the prompt. There's no clear representation of "drowning" as the woman seems to be on the boat rather than in distress in the water. Elements like "reformation", "stgeorgefamine", and "igleartwork" are not depicted, and their absence might be due to the abstract nature of these terms or the lack of clear visual representation. There are no extraneous objects that weren't mentioned in the prompt. Score: 20/30
- **Total Score: 78/100**

Human: 35,30,28,93

GPT-4V: 5,28,5,38

- **Relevance (0-40 points):** The image does not appear to be generated from the given prompt but rather seems to be an artwork of a woman lying down beside a beach or shore. It does not visibly show any elements related to "reformation", "stgeorgefamine", "lifeboat", "drowning", or "igleartwork". Score: 5/40
- **Visual Clarity (0-30 points):** The image appears to be a clear and sharp representation of an artwork. There don't appear to be any noticeable artifacts or oddities. The colors and contrasts are coherent and appear to be well represented. Score: 28/30
- **Object Accuracy (0-30 points):** The prompt nouns like "reformation", "stgeorgefamine", "lifeboat", "drowning", and "igleartwork" are not represented in the image. The image depicts a woman beside a beach or shore, which doesn't match the nouns' description or intention from the prompt. There's no object that was specifically mentioned in the prompt but the representation does not align with the given nouns. Score: 5/30
- **Total Score: 38/100**

Human: 20,30,10,50

Figure 12: Example of single-answer grading on text-to-image evaluation.

Prompt: woman reformation stgeorgefamine lifeboat drowning igleartwork, Jules Bastien-Lepage, side portrait



Ground Truth



Hard Negative

Pairwise Comparison



Input Order: (Ground Truth, Hard Negative)

GPT-4V:

Intermediate Thinking: Both images are paintings, which means they are artworks. Both images depict women in what seems to be a seaside setting. I need to look for characteristics that might align with the keywords provided: "reformation", "stgeorgefamine", "lifeboat", "drowning", "igleartwork", "Jules Bastien-Lepage", and "side portrait". "Jules Bastien-Lepage" is a renowned painter known for his realistic style. Both artworks could be attributed to him. "side portrait" suggests a side view of the subject. Both images fit this criteria, though image 0 provides a more pronounced side view. There are no visible signs of "drowning", "lifeboat", or "reformation" in the images. The keywords "stgeorgefamine" and "igleartwork" are unclear in their meaning. Image 0 depicts a woman in a boat on the sea, which might be closer to the keyword "lifeboat" than image 1, which depicts a woman lying on the ground near the sea.

Final Answer: image 0.



Input Order: (Hard Negative, Ground Truth)

GPT-4V:

Artist's Name: Both images seem to be in the style of Jules Bastien-Lepage, a famous 19th-century French naturalist painter. So both images meet this criterion. Content Keywords: "woman": Both images depict a woman. "reformation": Neither image seems to directly reference the idea of reformation. "stgeorgefamine": This doesn't seem directly related to either image, and it might be a typo or a combination of unrelated terms. "lifeboat": Only the second image (image 1) has a woman in a boat. This could be a lifeboat or some other type of boat. "drowning": The first image (image 0) shows a woman who seems to be lying or resting by the water, but not necessarily drowning. The second image (image 1) doesn't show drowning either. "igleartwork": This term isn't clear and doesn't appear to relate directly to either image. "side portrait": The second image (image 1) clearly depicts a side portrait of a woman, whereas the first image (image 0) shows the woman more from behind. Conclusions from Analysis: Image 1 aligns with the "lifeboat" and "side portrait" instructions. Image 0 doesn't particularly align with the specific instructions, except for the depiction of a woman. **Final Answer: image 1.**

Figure 13: Example of pairwise comparison on text-to-image evaluation.

Prompt: fantasy, pastel, absurdist, photo, textile strange characters



Ground Truth



Hard Negative

Single-Answer Grading

GPT-4V: 37,28,30,95

- Relevance (0-40 points): The generated image does appear to match the keywords "fantasy," "pastel," "absurdist," "photo," and "textile strange characters." The characters in the image are certainly absurdist and fantastical, with their unique designs and structures. The image captures the essence of a fantasy setting, with pastel colors and characters that are clearly made out of textile materials. Score: 37/40
- Visual Clarity (0-30 points): The image is clear and sharp, with each character well-defined. There don't seem to be any significant artifacts or oddities that detract from the image's quality. The pastel colors are pleasing and coherent, fitting the description given in the prompt. Score: 28/30
- Object Accuracy (0-30 points): All the nouns from the prompt seem to be represented in the image: "fantasy," "pastel," "photo," and "textile strange characters." The depicted objects, in this case the textile characters, match the description and intention of the original nouns in the prompt. There doesn't appear to be any object that doesn't belong or was not mentioned in the prompt. Score: 30/30
- Total Score: 95/100

Human: 40,30,30,100

GPT-4V: 38,30,30,98

- Relevance (0-40 points): The image showcases four distinct characters that could be described as "fantasy" due to their unusual and whimsical design. They are made of textile material and seem to exhibit an "absurdist" aesthetic. The color palette of the image is predominantly pastel, aligning with the prompt. The image appears to be a photograph of these textile characters, in line with the prompt's mention of "photo." Score: 38/40
- Visual Clarity (0-30 points): The image is clear and sharp, with each character and their features easily distinguishable. There are no noticeable artifacts or oddities. The colors and contrasts are pleasing and coherent, complementing the pastel theme mentioned in the prompt. Score: 30/30
- Object Accuracy (0-30 points): The image accurately represents the nouns from the prompt: "fantasy," "pastel," "photo," and "textile strange characters." The depicted characters match the description of "textile strange characters" and the overall intention of the nouns in the prompt. There doesn't seem to be any object in the image that wasn't mentioned in the prompt. Score: 30/30
- Total Score: 98/100

Human: 35,30,28,93

Figure 14: Example of single-answer grading on text-to-image evaluation.

Prompt: fantasy, pastel, absurdist, photo, textile strange characters



Ground Truth



Hard Negative

Pairwise Comparison



Input Order: (Ground Truth, Hard Negative)

GPT-4V:

Intermediate Thinking: Fantasy: Both images present characters that are surreal and fit into the category of fantasy. Both have non-humanlike figures and strange appearances. Pastel: The first image (image 0) clearly exhibits pastel colors in its background and some parts of the characters. The second image (image 1) also shows some pastel colors, especially in the clothing and some parts of the figures. Absurdist: Both images depict characters that are absurd in nature due to their unique and unconventional appearance. Photo: Both images are photographs. Textile: Both sets of characters appear to be made of textile materials, with noticeable textures suggesting they might be soft sculptures or fabric-based designs. Strange Character: Both images contain characters that can be categorized as 'strange' due to their surreal and unconventional appearances. Given the criteria, both images seem to align very closely with the provided instructions. **Final Answer: Tie.**



Input Order: (Hard Negative, Ground Truth)

GPT-4V:

Fantasy: Both images depict characters that can be considered "fantasy" as they don't resemble real-life humans or creatures. Pastel: The characters in both images incorporate pastel colors. However, image 1 seems to have a more dominant pastel background, which might make it slightly more aligned with the "pastel" descriptor. Absurdist: Both images portray characters with unusual and exaggerated features, making them both seem absurdist in nature. Photo: Both images are photographs. Textile: The characters in both images appear to be made of textile materials. Strange Character: Both sets of characters are undoubtedly strange or unconventional. Given the analysis, it's challenging to definitively state that one image adheres to the instructions more than the other, as they both seem to align well with the provided descriptors. **Final Answer: Tie.**

Figure 15: Example of pairwise comparison on text-to-image evaluation.

Prompt: purple horned tiefling adventurer, dnd, character art



Ground Truth



Hard Negative

Single-Answer Grading

GPT-4V: 38,28,30,96

- Relevance (0-40 points): The generated image closely matches the prompt of a "purple horned tiefling adventurer." The character in the image has purple skin, horns, and appears to be dressed in adventurer attire. The image captures the core essence and details specified in the prompt. Score: 38/40
- Visual Clarity (0-30 points): The image is clear and sharp. No noticeable artifacts or oddities are present that detract from the image's quality. The colors and contrasts are pleasing and coherent. Score: 28/30
- Object Accuracy (0-30 points): All the nouns from the prompt are represented accurately: tiefling, horns, adventurer, and purple are all present. The depicted objects match the description and intention of the original nouns in the prompt. There is no object in the image that doesn't belong or was not mentioned in the prompt. Score: 30/30
- Total Score: 96/100.

Human: 40,30,30,100

GPT-4V: 38,28,28,94

- Relevance (0-40 points): The generated image features a character that appears to be a tiefling, as evidenced by the horns and general fantasy aesthetic. The character is predominantly purple, aligning with the prompt's specification of a "purple horned tiefling." The adventurer aspect is evident through the attire, weapon, and overall demeanor of the character. It aligns well with the concept of a DND (Dungeons and Dragons) character art. Score: 38/40
- Visual Clarity (0-30 points): The image is clear and sharp with detailed features visible. I do not notice any significant artifacts or oddities. The colors and contrasts are pleasing and coherent, with the purple theme prominent and consistent. Score: 28/30
- Object Accuracy (0-30 points): Tiefling: Accurately represented by the character's horns and fantasy appearance. Adventurer: Represented by the character's attire and weapon. Purple: The primary color of the character's attire and weapon, fitting the prompt's requirement. DND: The overall aesthetic matches the typical character art of Dungeons and Dragons. Character Art: The image is consistent with the character art style often seen for role-playing games. Score: 28/30
- Total Score: 94/100.

Human: 40,28,30,98

Figure 16: Example of single-answer grading on text-to-image evaluation.

Prompt: purple horned tiefling adventurer, dnd, character art



Ground Truth



Hard Negative

Pairwise Comparison



Input Order: (Ground Truth, Hard Negative)

GPT-4V:

Step-by-step Analysis: "Instruction Examination: The instruction mentions a 'purple horned tiefling adventurer.' The context suggests that this is related to Dungeons and Dragons (DnD) and that it's meant to be character art. Image 0 Examination: The character in image 0 has purple hair and horns, aligning with the 'purple horned' descriptor. The character seems to be dressed as an adventurer with weapons and armor. The style of the character art aligns with what one might expect from a DnD-themed artwork. Image 1 Examination: The character in image 1 also has purple hair and horns. The character is also dressed as an adventurer, and the attire, including the weapon and overall design, fits the DnD theme. The style of the character art also aligns with DnD character art. Comparison: Both images effectively capture the essence of a 'purple horned tiefling adventurer.' Both images reflect the DnD theme and could easily be character art for the game. The primary difference lies in the pose and slight variations in attire and design, but neither image seems to be more or less accurate in adhering to the instructions provided." **Final Answer: Tie.**



Input Order: (Hard Negative, Ground Truth)

GPT-4V:

Analysis of Image 0 (Left): The character in Image 0 is a purple-horned tiefling dressed as an adventurer with weapons and gear, resembling typical DnD character art. Analysis of Image 1 (Right): The character in Image 1 is also a purple-horned tiefling with an adventurer appearance, also resembling typical DnD character art. Intermediate Thoughts: Both images fit the instruction well. Image 0 has a more equipped adventurer look, while Image 1 has a simpler depiction. **Final Answer: Tie.**

Figure 17: Example of pairwise comparison on text-to-image evaluation.

C Text-guided Image Editing

C.1 Prompt

Single-answer grading:

To standardize the conduction of a rigorous human evaluation, we stipulate the criteria for each measurement as follows:

Semantic Consistency (SC), score in range [0, 0.5, 1]

Perceptual Realism (PR), score in range [0, 0.5, 1]

Semantic Consistency (SC) ensures that the generated image is coherent in terms of guidance provided (i.e. Prompts, Subject Token, etc.). In another words, the image has to be aligned with the requirements provided in user's inputs.

Perceptual Realism (PR) ensures the generated image align with real-world characteristics. In another words, the image has to be visually convincing and closely resembles a real photograph (Photorealism).

General Rules for Semantic Consistency (SC) scoring:

SC=0 : Image not following one or more of the conditions at all (e.g. not following the prompt at all, different background in editing task, wrong subject in subject-driven task etc)

SC=0.5 : all the conditions are partly following the requirements. SC=1 : All the conditions are following >75% of the requirement. You agree that the overall idea is correct.

General Rules for Perceptual Realism (PR) scoring:

PR=0: Obvious distortion/artifacts that are unrecognizable at first glance PR=0.5: Some artifacts but the objects are still recognizable; or Unnatural sense of detail feeling in some area (You find out the image looks strange after examining it carefully.) PR=1: You agree that the image generally look real (doesn't have to be 100% perfect. Like 90% is good enough.)

Notice that this evaluation guideline is following the standardized one in ImagenHub (Ku et al., 2023).

C.2 More Showcases

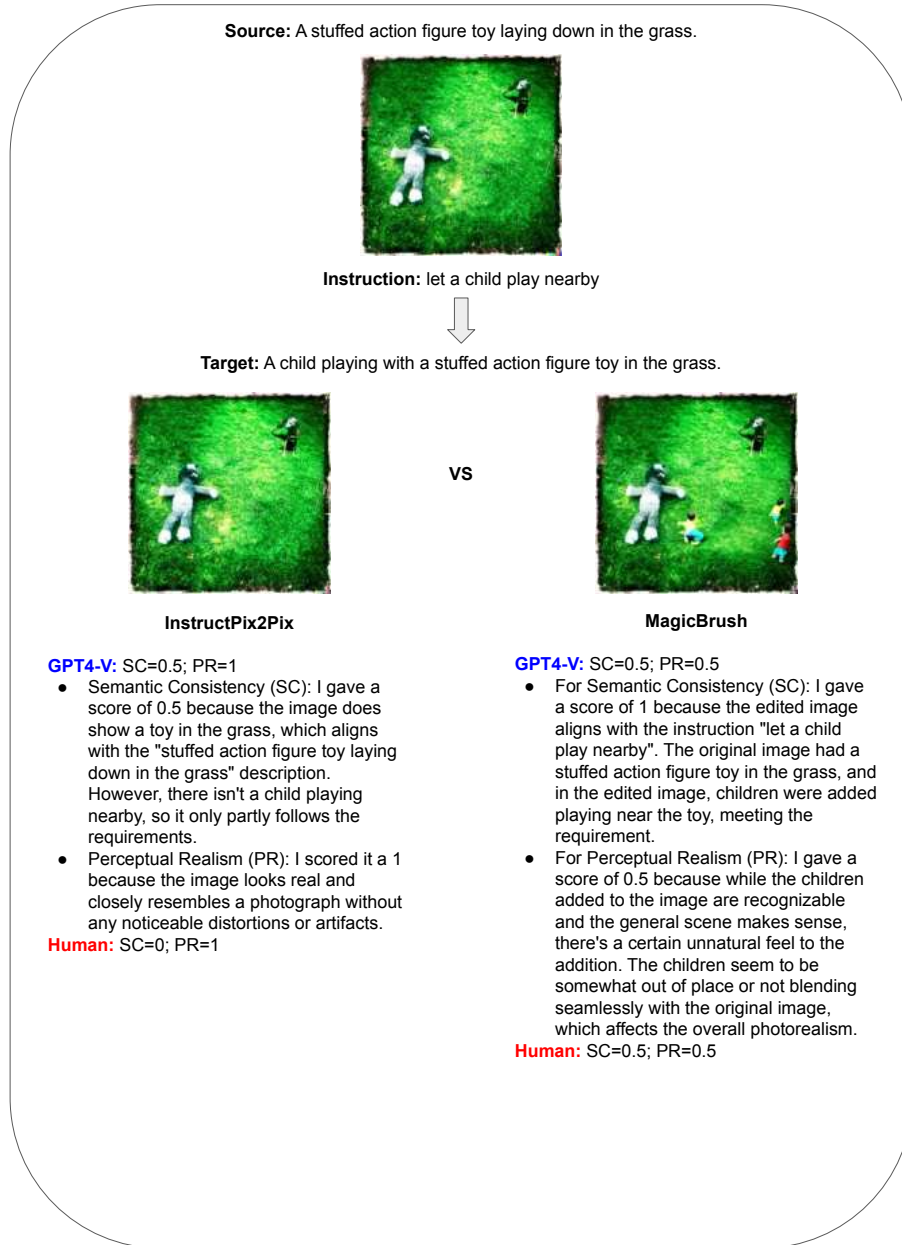


Figure 18: Showcase of GPT-4V for evaluation text-guided image editing.

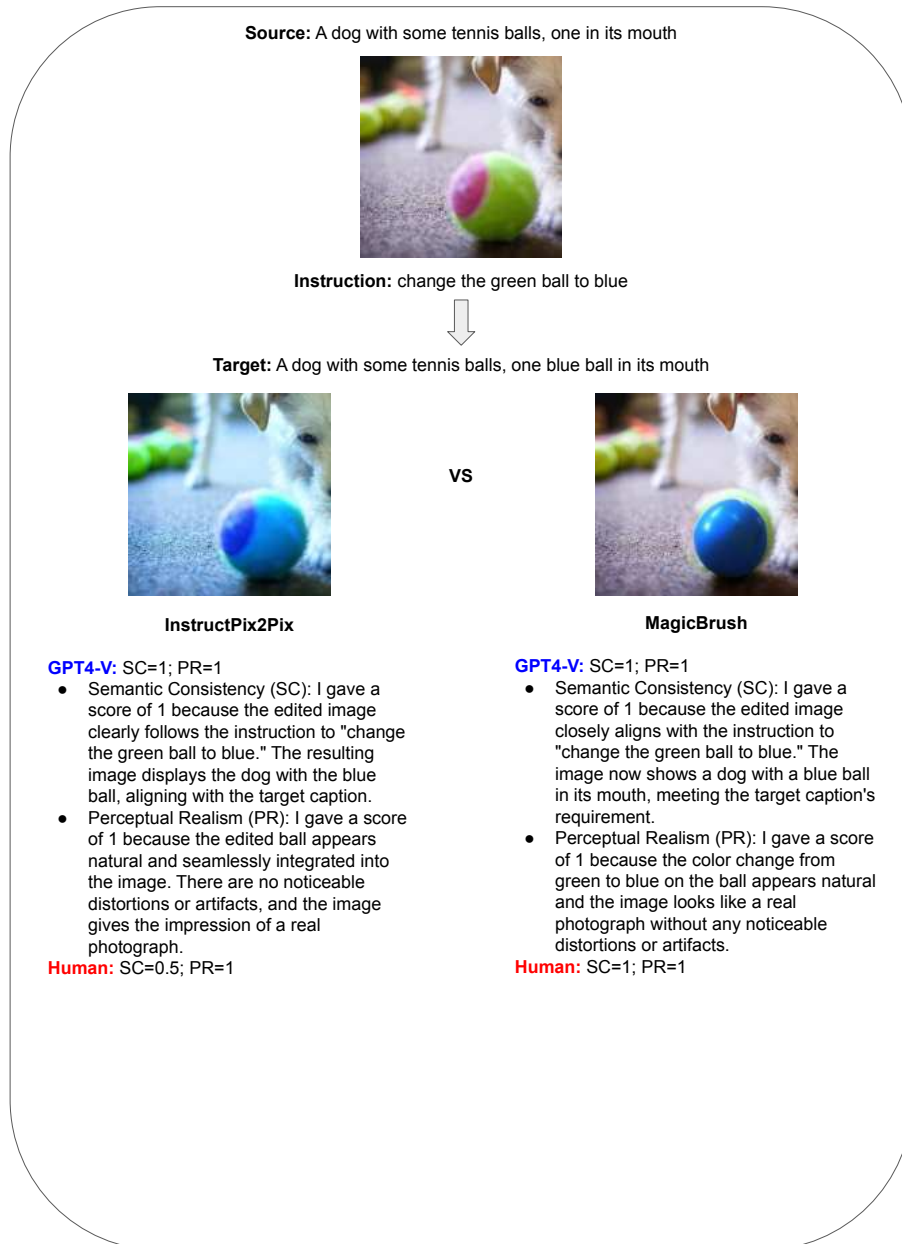


Figure 19: Showcase of GPT-4V for evaluation text-guided image editing.

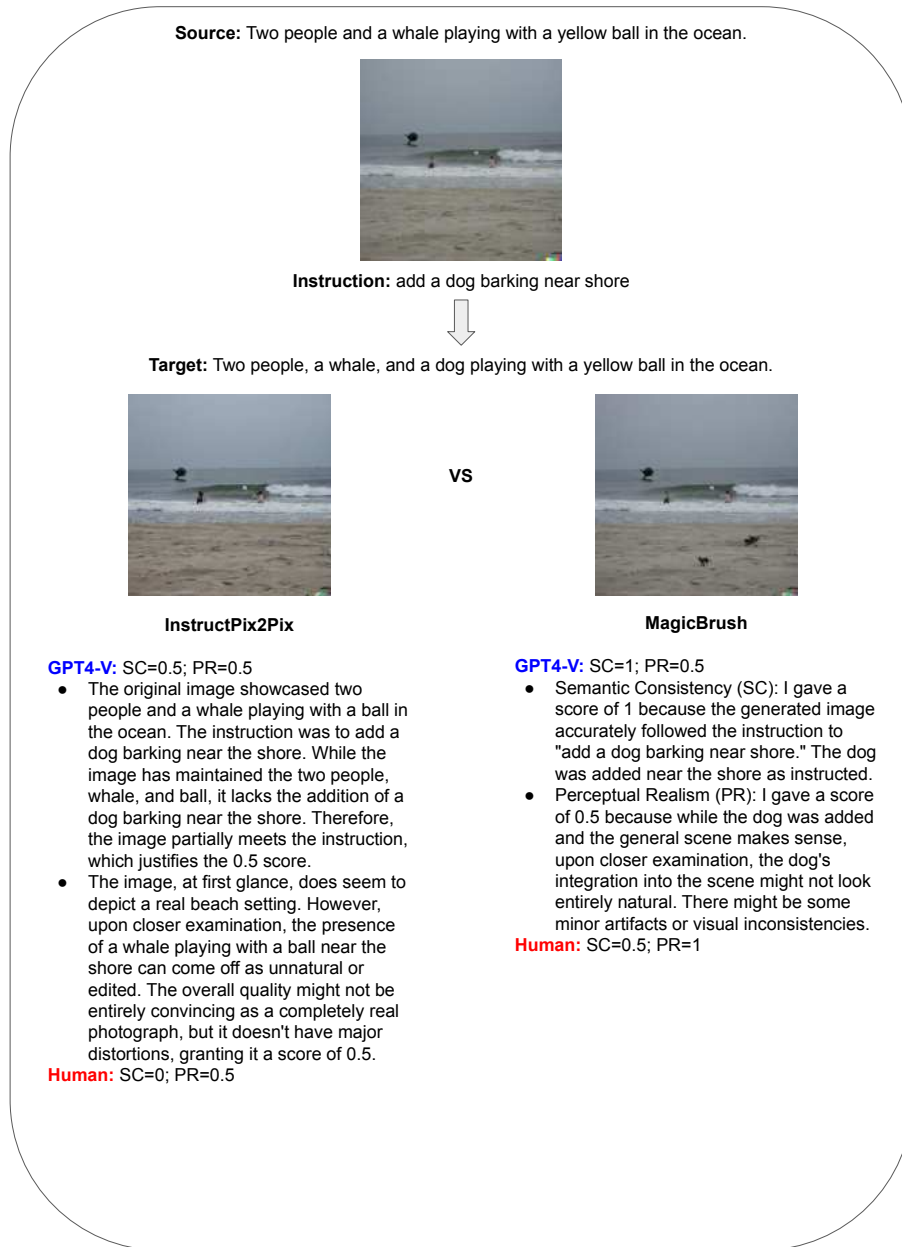


Figure 20: Showcase of GPT-4V for evaluation text-guided image editing.

D Multiple Images to Text Alignment

Single-Answer grading:

Your task is to evaluate whether a user review accurately represents the main context and objects of the uploaded images. While the review need not describe every detail of the images, it should mention the main content of those images. After the evaluation, rate the quality of the user review's relevance to the image on a scale of 1-100, with 100 being a perfect match. Generic reviews without details, e.g., "food is great" should receive a relatively low score. Here is the user review: [[[text review information.]]]



User Review: Tasty pancakes, crepes, omelettes

GPT-4V: Rating 85/100

Explanations: Based on the images provided and the user review, I'd rate the relevance of the review to the images as 85/100. The review accurately mentions pancakes, crepes, and omelettes, which are present in the images, but doesn't provide detailed descriptions or mention other items such as the toppings or side dishes.

Human: 70/100

User review: Missed out on their cheesecake only after a delicious meal. Their bread served at the beginning of the meal is so good.

GPT-4V: Rating 25/100

Explanations: The user review mentions the bread served at the beginning of the meal, but the images don't showcase any bread. The images depict a sandwich with fries, a dish with shrimp, rice, and a sauce, and a pasta dish. There's no mention of the specific dishes shown in the images in the review. The reference to cheesecake is also not represented in the images.

Human: 20/100

Figure 21: Examples of GPT-4V for evaluating multiple images to text task.

TIFA: Accurate and Interpretable Text-to-Image Faithfulness Evaluation with Question Answering

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<https://tifa-benchmark.github.io/>

Abstract

Despite thousands of researchers, engineers, and artists actively working on improving text-to-image generation models, systems often fail to produce images that accurately align with the text inputs. We introduce TIFA (Text-to-Image Faithfulness evaluation with question Answering), an automatic evaluation metric that measures the faithfulness of a generated image to its text input via visual question answering (VQA). Specifically, given a text input, we automatically generate several question-answer pairs using a language model. We calculate image faithfulness by checking whether existing VQA models can answer these questions using the generated image. TIFA is a reference-free metric that allows for fine-grained and interpretable evaluations of generated images. TIFA also has better correlations with human judgments than existing metrics. Based on this approach, we introduce TIFA v1.0, a benchmark consisting of 4K diverse text inputs and 25K questions across 12 categories (object, counting, etc.). We present a comprehensive evaluation of existing text-to-image models using TIFA v1.0 and highlight the limitations and challenges of current models. For instance, we find that current text-to-image models, despite doing well on color and material, still struggle in counting, spatial relations, and composing multiple objects. We hope our benchmark will help carefully measure the research progress in text-to-image synthesis and provide valuable insights for further research.¹

1. Introduction

While we welcome artistic freedom when we commission art from artists, images produced by deep generative models [44, 46, 43, 47, 61] should conform closely to our requests. Despite the advances in generative models, it is still challenging for models to produce images faithful to users' intentions [40, 11, 30, 35, 36]. For example, current

¹Correspondance to <Yushi Hu: yushihu@uw.edu>. All data and a pip-installable evaluation package are available on the project page.

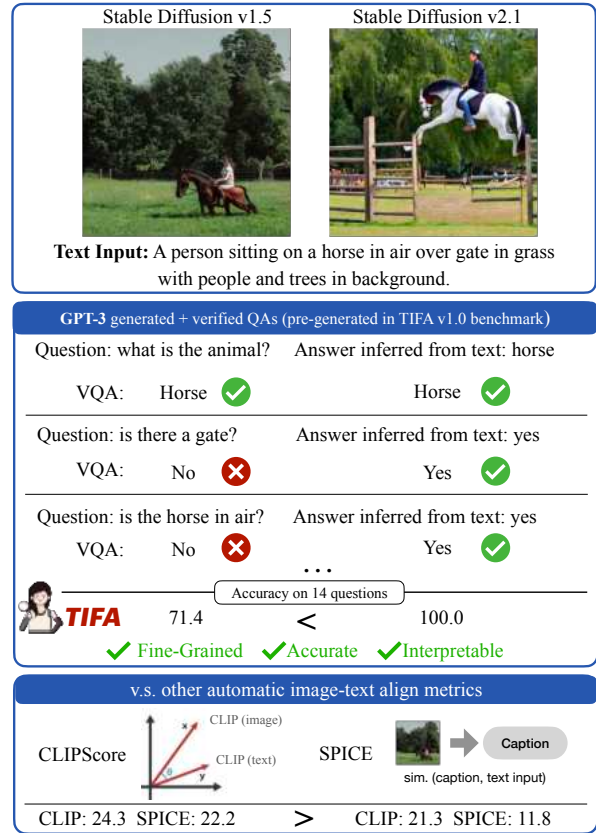


Figure 1. Illustration of how TIFA works, and comparison with the widely-used CLIPScore and SPICE metrics. Given the text input, TIFA uses GPT-3 to generate several question-answer pairs, and a QA model filters them (3 out of 14 questions for this text input are shown). TIFA measures whether VQA models can accurately answer these questions given the generated image. In this example, TIFA indicates that the image generated by Stable Diffusion v2.1 is better than that by v1.5, while CLIP and SPICE yield the opposite result. The text input is from the MSCOCO validation set.

models often fail to compose multiple objects [40, 11, 35], bind attributes to the wrong objects [11], and struggle in generating visual text [36]. Today, there are efforts to address

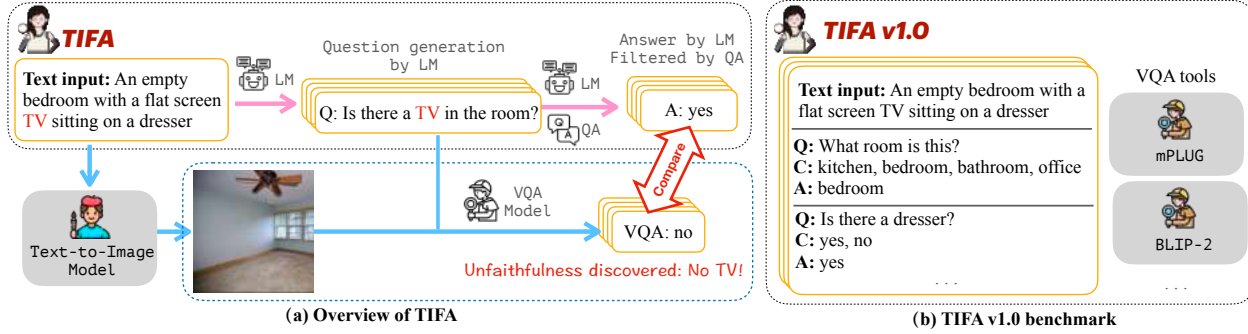


Figure 2. (a) Overview of how TIFA evaluates the faithfulness of a synthesized image. TIFA uses a language model (LM), a question-answering (QA) model, and a visual-question-answering (VQA) model. Given a text input, we generate several question-answer pairs with the LM and then filter them via the QA model. To evaluate the faithfulness of a synthesized image to the text input, a VQA model answers these visual questions using the image, and we check the answers for correctness. (b) TIFA v1.0 benchmark. While TIFA is applicable to any text prompt, to allow direct comparison across different studies, and for ease of use, we introduce the TIFA v1.0 benchmark, a repository of text inputs along with pre-generated question-answer tuples with answer choices. To evaluate a text-to-image model, a user first produces the images for the text inputs in TIFA v1.0 and then performs VQA with our provided tools on generated images to compute TIFA.

these challenges: researchers are imposing linguistic structure with diffusion guidance to produce images with multiple objects [11]; others are designing reward models trained using human feedback to better align generations with user intention [30]. However, progress is difficult to quantify without accurate and interpretable evaluation measures that explain when and how models struggle.

A critical bottleneck, therefore, is the lack of reliable automatic evaluation metrics for text-to-image generation faithfulness. One of the popular metrics is CLIPScore [17], which measures the cosine similarity between the CLIP embeddings [42] of the text input and the generated image. However, since CLIP is not effective at counting objects [42], or reasoning compositionally [37], CLIPScore is unreliable and often inaccurate. Another family of evaluation metrics uses image captions, in which an image captioning model first converts the image into text, and then the image caption is evaluated by comparing it against the text input. Unfortunately, using captioning models is insufficient since they might decide to ignore salient information in images or focus on other non-essential image regions [24]; for example, a captioning model might say that the images in Figure 1 are “a field of grass with trees in the background”. Moreover, evaluating text (caption) generation is inherently challenging [23, 26]. Another recent text-to-image evaluation is DALL-Eval [6], which employs object detection to determine if the objects in the texts are in the generated images. However, this approach only works on synthesized text and measures faithfulness along the limited axes of objects, counting, colors, and spatial relationships but misses activities, geolocation, weather, time, materials, shapes, sizes, and other potential categories we often ask about when we recall images from memory [29].

To address the above challenges, we introduce TIFA, a new metric to evaluate text-to-image generation faithfulness.

Our approach is illustrated in Figure 2. Given a repository of text inputs, we automatically generate question-answer pairs for each text via a language model (here, GPT-3 [3]). A question-answering (QA) system (here, UnifiedQA [25]) is subsequently used to verify and filter these question-answer pairs. To evaluate a generated image, we use a visual-question-answering (VQA) system (here, mPLUG-large [31], BLIP-2 [32], etc.) to answer the questions given the generated image. We measure the image’s faithfulness to the text input as the accuracy of the answers generated by the VQA system. While the accuracy of TIFA is dependent on the accuracy of the VQA model, our experiments show that TIFA has much higher correlation with human judgments than CLIPScore (Spearman’s $\rho = 0.60$ vs. 0.33) and captioning-based approaches (Spearman’s $\rho = 0.60$ vs. 0.34). Additionally, since the LMs and VQA models will continue to improve, we hypothesize that TIFA will continue to be more reliable over time. Also, our metrics can automatically detect when elements are missing in the generation: in Figure 2, TIFA detects that the generated image does not contain a TV.

To promote the use of our new evaluation metric, we release TIFA v1.0, a large-scale text-to-image generation benchmark containing 4K diverse text inputs, sampled from the MSCOCO captions [34], DrawBench [47], PartiPrompts [61], and PaintSkill [6]. Each input comes with a pre-generated set of question-answer pairs, resulting in 25K questions covering 4.5K distinct elements. These questions have been automatically generated and pre-filtered using a question-answering model. This benchmark also comes with different VQA models [57, 28, 58, 32, 31, 21] that can be used to evaluate generative models and can be easily extended to use future VQA models when they become available.

We conduct a comprehensive evaluation of current text-

to-image models using TIFA v1.0. Thanks to TIFA’s ability to detect fine-grained unfaithfulness in images, we find that current state-of-the-art models are good at rendering common objects, animals, and colors, but still struggle in composing multiple objects, reasoning about spatial relations, and binding the correct activity for each entity. In addition, our ablation experiments show that TIFA is robust to different VQA models. Future researchers can use TIFA v1.0 to compare their text-to-image models’ faithfulness across different studies. Also, future generative models may focus on addressing the weaknesses of current models that TIFA discovered. In addition, with TIFA, users can customize evaluations with their own text inputs and questions [10]; for example, a future TIFA benchmark could focus on counting or scene text.

2. Related Work

We compare TIFA to other image and language generation evaluation metrics.

Prior image generation evaluation Prior work usually compares image generation models via pairwise comparison by humans. How to design automatic evaluation metrics to approximate human assessment of the quality of machine-generated images has always been a major challenge in computer vision. There are two aspects to evaluate, namely image quality and image-text alignment. **Inception Score** [48] and **FID** [18] are the most widely adopted metrics for image quality. They compare the features of the generated images and gold images extracted from a pre-trained Inception-V3 model [52] to evaluate the fidelity and diversity of generated images. However, they rely on ground-truth images and are based on a classification model, which makes them not suitable for complex datasets [12]. For image-text alignment, prior metrics are mainly based on CLIP [42], captioning, and object detection models. **CLIPScore** [17] and **CLIP-R** [39] are based on the cosine similarity of image and text CLIP [42] embeddings. [6, 19, 20] first convert the images using a captioning model, and then compare the image caption with the text using metrics like CIDEr [55] and SPICE [1]. **SOA** [19] and **DALL-Eval** [6] employ object detection models to determine if objects, attributes, and relations in the text input are in the generated image. However, this approach only works on synthesized text inputs and measures faithfulness on limited axes (object, counting, color, spatial relation), missing elements like material, shape, activities, and context. In contrast, thanks to the flexibility of questions, TIFA works on any text inputs and evaluates faithfulness across a broad spectrum of dimensions.

Summarization evaluation in NLP TIFA is inspired by the summarization evaluation methods based on question answering (QA) [56, 50]. Given a summary, a language model generates a set of questions about the text. A QA model checks if the same answer can be inferred from the

text and the summary. These QA-based metrics have much higher correlations with human judgments on the factual consistency of summarization than other automatic metrics [56, 50]. TIFA can be seen as treating the text input for the text-to-image model as a summary of the generated image.

3. The TIFA Metric

We introduce a framework for automatically estimating the faithfulness of an image to its text prompt. Given a text input T , we aim to measure the faithfulness of the generated image I . An overview of our metric is illustrated in Figure 2. From T , we generate N multiple-choice question-answer tuples $\{Q_i, C_i, A_i\}_{i=1}^N$, in which Q_i is a question, C_i is a set of answer choices, and $A_i \in C_i$ is the gold answer. The answer A_i can be inferred given T , Q_i , and C_i . Next, for each question Q_i , we use a VQA model to produce an answer $A_i^{\text{VQA}} = \max_{a \in C_i} p(a \mid I, Q_i)$. We define the faithfulness between the text T and image I as the VQA accuracy:

$$\text{faithfulness}(T, I) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[A_i^{\text{VQA}} = A_i] \quad (1)$$

The range of our faithfulness score is $[0, 1]$. It is maximized when we have a performant VQA model, and the image I accurately covers the information in the text T so that for any question Q , which can be answered given T can also be answered given I . Several key design decisions will be addressed in later sections: how to generate questions (§3.1), how to control the question quality (§3.2), and how to answer those questions (§3.3). Finally, we give a step-by-step qualitative example of TIFA in Figure 4.

3.1. Question-Answer Generation

Our main challenge is to generate diverse questions that cover all elements of the text input evenly. We also simplify the question-generation pipeline into a single GPT-3 [3] completion, so that TIFA can exploit the power of recent language models (LM) and work with updated black-box LMs (e.g., ChatGPT) in the future.

Inspired by prior work [4], given a text prompt T , we generate the question-answer tuples $\{Q_i, C_i, A_i\}_{i=1}^N$ via the pipeline illustrated in Figure 3. Different from prior work, which relies on multiple components, our pipeline is completed by a single inference run via in-context learning with GPT-3 [3, 59, 22, 41, 51], thereby avoiding the need for intermediate human annotations. We annotate 15 examples and use them as in-context examples for GPT-3 to follow. Here we take the text “A photo of three dogs.” as an example. Each in-context example contains the following steps:

Element extraction Given text prompt T , GPT-3 will first extract all elements $\{v_i\}_{i=1}^m$ following prior work [4] (for the in-context examples, we perform element extraction manually). The elements include noun phrases (including named

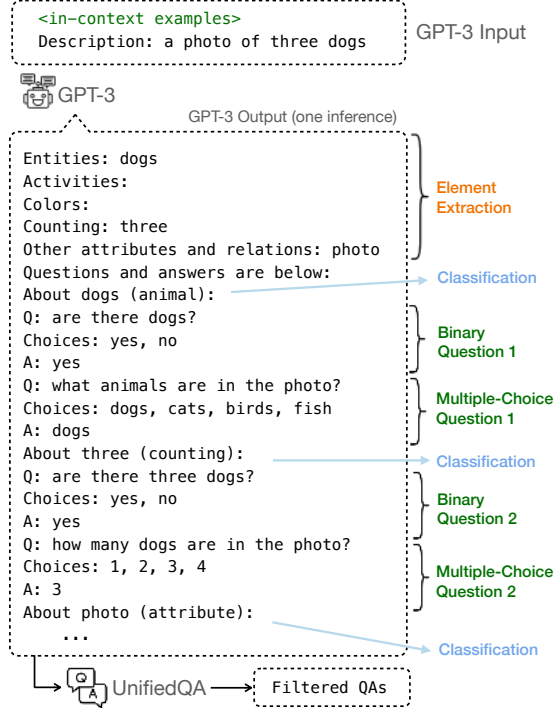


Figure 3. Our question-answer pair generation pipeline. The whole pipeline can be executed via a single inference of GPT-3 via in-context learning. Given the text prompt, GPT-3 first extracts the elements and then generates two questions for each element. The GPT-3 output is then parsed and filtered by UnifiedQA.

entities), verbs, adjectives, adverbs, and parse tree spans with no more than 3 words altogether. For the above example, the elements are *photo*, *three*, *dogs*.

Element category classification For each element v_i , following [29], we classify the elements into one of the following 12 categories: *object*, *activity*, *animal*, *food*, *counting*, *color*, *material*, *spatial*, *location*, *shape*, *attribute*, and other. As shown in Figure 3, text generated from GPT-3 contains the category corresponding to each question. For example, “three” is “counting”, and “dogs” is classified as “animal.” This step allows a detailed analysis of the text-to-image model’s ability in each category.

Question generation conditioned on elements For each element v_i , we generate two questions. The first is a question that should be answered “yes” for a faithful generated image, and the second question has v_i as its answer. For example, two questions are generated for the element “three”. The first is “are there three dogs?”, and the choices are {yes, no}. The second is “how many dogs are there?”, and the choices are {1,2,3,4}. These two types of questions make our evaluations diverse and robust to surface-level differences.

Completing above steps by prompting GPT-3 once As mentioned earlier, for each text T , the whole pipeline can be completed by one GPT-3 inference. We annotated 15

in-context examples that cover all types of questions. The full prompt is in the Appendix. The prompt format is shown in Figure 3. Our in-context examples follow the same format, and identical examples are used for all text inputs, leading to a fixed and limited amount of human annotation cost. We use *code-davinci-002* engine for question generation, and the decoding temperature is 0.

3.2. Question Filtering

To ensure the quality of generated images, we use UnifiedQA [25] to verify the GPT-3 generated question-answer pairs and filter out the ones that GPT-3 and UnifiedQA do not agree on. UnifiedQA² is a state-of-the-art multi-task question-answering model that can answer both multiple-choice and free-form questions. Denote the UnifiedQA model as QA . Given the text T , question Q_i , choices C_i , and answer A_i , Let $A_i^f = QA(T, Q_i)$ be the free-form answer, and $A_i^{mc} = QA(T, Q_i, C_i)$ be the multiple-choice answer. We keep the question if $A_i = A_i^{mc}$ and the word-level F_1 score between A_i^f and A_i is greater than 0.7. We conduct a human evaluation on 1000 filtered question-answer pairs. Only 7 are considered not reasonable (e.g., generated choices do not include a correct answer). Details are in Appendix C.

Recommended VQA model Based on considerations over the accuracy, correlation with human judgments, and run time, we currently suggest using **mPLUG-large** as the VQA model for TIFA. Analysis is given in Section 5.4. Like the LM and QA components, the VQA component can be updated in the future as the technology improves.

3.3. VQA Models

Since our questions contain a diverse set of visual elements (e.g., activity, art style), we use open-domain pre-trained vision-language models as our VQA model (rather than closed-class classification models fine-tuned on VQAv2 [14]). We provide tools to easily perform VQA on arbitrary images and questions, based on 5 state-of-the-art VQA models trained with distinct data and strategies.

Vision-language models The general pre-trained vision-language model are **GPT-large** [57], **VILT-B/32** [28], **OFA-large** [58], and **mPLUG-large** [31]. These models are pre-trained on a large amount of image-text pairs, and downstream image-to-text tasks like image captioning and visual question answering. Notice that these models have not been trained to answer multiple-choice questions. For each question, we first decode the free-form answer and then choose the choice that has the highest similarity with the decoded answer, measured by SBERT [45]. Another model we use is **BLIP-2 FlanT5-XL** [32], in which a ViT [8] is connected with a frozen FlanT5 [7] via a lightweight transformer. This

²Model checkpoint we use: <https://huggingface.co/allenai/unifiedqa-v2-t5-large-1363200>.

Text Input: A wet dog standing on a beach next to the ocean.

Step 1: Question Generation + Filtering

About dog (animal/human) Q: Is this a dog? A: Yes Q: What animal is in the picture? A: Dog	About ocean (location) Q: Is there an ocean? A: Yes Q: What type of place is this? A: Ocean	About wet (attribute) Q: Is the dog wet? A: Yes Q: Is the dog wet or dry? A: Wet
About beach (location) Q: Is this a beach? A: Yes Q: What type of place is this? A: Beach	About standing (activity) Q: Is the dog standing? A: Yes Q: What is the dog doing? A: Standing	About next to (spatial relation) Q: Is the dog next to the ocean? A: Yes Q: Is the dog next to or in the ocean? A: Next to

Step 2: Visual Question Answering



Stable Diffusion v1.5 image

 **TIFA** 7 / 10 = 0.7

About dog (animal/human) Q: Is this a dog? A: Yes Q: What animal is in the picture? A: Dog	✓ ✓	About standing (activity) Q: Is the dog standing? A: No	✗
About beach (location) Q: Is this a beach? A: Yes Q: What type of place is this? A: Beach	✓ ✓	About wet (attribute) Q: Is the dog wet? A: No Q: Is the dog wet or dry? A: Dry	✗ ✗
About ocean (location) Q: Is there an ocean? A: Yes	✓	About next to (spatial relation) Q: Is the dog next to the ocean? A: Yes Q: Is the dog next to or in the ocean? A: Next to	✓ ✓

Figure 4. Step-by-step qualitative example of TIFA metric. Given a text input, we first generate question-answer pairs and filter them. We ~~strike through~~ the questions filtered out by UnifiedQA. Then we run VQA models on the generated image to get the TIFA score.

model allows for performing multiple-choice VQA directly due to the flexibility of the LM.

4. TIFA v1.0: Benchmark for Text-to-Image Generation Faithfulness

In this section, we introduce TIFA v1.0, a text-to-image generation faithfulness benchmark based on the evaluation method discussed in Section 3. The benchmark consists of 4,081 diverse text inputs paired with 25,829 question-answer pairs. Each question is classified into one of the categories discussed in Section 3.1. The benchmark also comes with Python pip-installable APIs to perform VQA with various state-of-the-art VQA models on arbitrary visual questions. The overall TIFA for each text-to-image model is computed by averaging TIFA scores of images generated from each text input in the benchmark.

4.1. Text Collections

We collect 4,081 text inputs to benchmark text-to-image models’ generation ability on diverse tasks. 2,000 text inputs are image captions from COCO validation set [34]. These captions have corresponding gold images. Since text-to-image models are often used to create abstract art, we also

collect 2,081 text inputs from previous works that do not correspond to any real image. All text inputs we use contain ≥ 3 words. We include 161 from **DrawBench** used in Imagen [47] (texts that are categorized as “misspellings” and “rare words” are removed); 1420 from **PartiPrompt** used in Parti [61] (texts in category “abstract” are removed); and 500 texts from **PaintSkill** used in DALL-Eval [6].

Table 1. Statistics of TIFA v1.0.

Statistics	
# of prompts	4,081
- # of COCO captions	2,000
- # of DrawBench, PartiPrompt, PaintSkill prompts	2,081
# of questions	25,829
- # of binary questions	17,226
- # of multiple-choice questions	8,603
avg. # of questions per prompt	6.3
avg. # of words per prompt	10.5
avg. # of elements per prompt	4.3

4.2. Statistics and Diversity

Table 1 shows the basic statistics of the TIFA v1.0 benchmark. We demonstrate TIFA v1.0’s diversity in Figure 5.

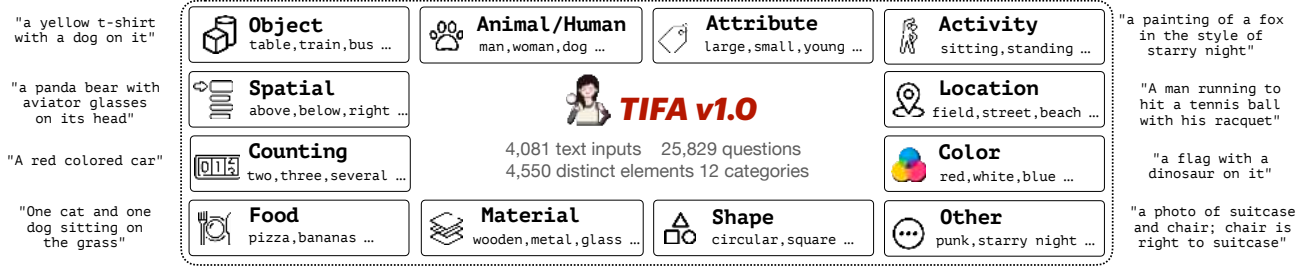


Figure 5. Statistics and diversity of TIFA v1.0. The text inputs contain elements from 12 categories (e.g., object, spatial, and counting). We show the most common elements from each category. In addition, we also show some example text inputs on the sides.

TIFA v1.0 contains questions about 4,550 distinct elements, which are categorized into 12 categories. The number of times each type of element occurs in the text input is object (7,854), animal/human (3,501), attribute (3,399), activity (2,851), spatial (2,265), location (1,840), color (1,743), counting (986), food (911), material (209), shape (69), and other (201). The "other" category includes notions often used in abstract art, such as "starry night" and "steampunk." The accuracy of VQA on a particular genre measures the text-to-image model's ability in the corresponding aspect. Please refer to Appendix D for more details on TIFA v1.0.

4.3. Finetuned Open-Source Language Model for Question Generation

For TIFA v1.0, we use GPT-3 to generate the questions. While benchmarking with TIFA v1.0 is a deterministic process, using TIFA to create a new benchmark might not be deterministic as the underlying question generator (GPT-3 in our case) might change privately. To promote deterministic benchmark generation, we fine-tune and release a LLaMA 2 (7B) [54] model that parses the captions and generates questions for arbitrary texts, using TIFA v1.0 questions as training examples.³

5. Experiments

In this section, we first show that TIFA has substantially higher correlations with human judgments than prior metrics on text-to-image faithfulness (§5.1). Then we present a comprehensive evaluation of existing text-to-image models using TIFA v1.0, highlighting the challenges of current text-to-image models (§5.3). Finally, we conduct an analysis of TIFA's robustness against different VQA models (§5.4). For all experiments, we use mPLUG as the VQA model for TIFA unless stated otherwise. The models we evaluate include AttnGAN [60], X-LXMERT [5], VQ-Diffusion [16], minDALL-E [27], and Stable Diffusion v1.1, v1.5, and v2.1 [46]. Details are in Appendix E.

5.1. Correlation with Human Judgements

To compare TIFA with prior evaluation metrics, we first conduct human evaluations of the text-to-image models on the 1-5 Likert scale on text-to-image faithfulness. Then we compare TIFA with other metrics based on their correlation with human judgments.

Likert scale on text-to-image faithfulness Annotators are asked to answer on a scale of 1 (worst) to 5 (best) to the question "Does the image match the text?". The detailed annotation guidelines are in Appendix C. Annotators are asked to focus on text-to-image faithfulness rather than image quality. The Likert scale should be based on how many elements in the text prompt are missed or misrepresented in the image. Objects are more important than attributes, relations, and activities. If an object is missed in the image, then all related attributes, activities, relations, etc. are also considered lost. An example is given in Figure 6.

We collect annotations of 800 generated images on 160 text inputs from TIFA v1.0. For each prompt, we sample an image from the 5 most recent generative models we evaluated, i.e., minDALL-E, VQ-Diffusion, Stable Diffusion v1.1, v1.5, and v2.1. We collect 2 annotations per image and average over the scores as the single "faithfulness" score. The inter-annotator agreement measured by Krippendorff's α is 0.67, indicating "substantial" agreement.



Figure 6. Illustration of our Likert scale annotation guideline. Annotators are asked to give a score of 1 to 5 based on how many elements in the text prompt are missed or misrepresented in the image. The missed elements are underlined.

Baselines We compare our evaluation with two families of reference-free metrics on text-image match introduced in

³LLaMA 2 question generation model checkpoint:
https://huggingface.co/tifa-benchmark/llama2_tifa_question_generation

Section 2. The first is the **caption-based method**. We use the state-of-the-art **BLIP-2 FlanT5-XL** [32] as the captioning model. The second approach is **CLIPScore** [17, 42]. We use CLIP (ViT-B/32) [42] to compute the score.

Table 2. Correlations between each evaluation metric and human judgment on text-to-image faithfulness, measured by Spearman’s ρ and Kendall’s τ .

	Spearman’s ρ	Kendall’s τ
Caption-Based		
BLEU-4	18.3	18.8
ROUGE-L	32.9	24.5
METEOR	34.0	27.4
SPICE	32.8	23.2
CLIPScore	33.2	23.1
Ours		
TIFA (VILT)	49.3	38.2
TIFA (OFA)	49.6	37.2
TIFA (GIT)	54.5	42.6
TIFA (BLIP-2)	55.9	43.6
TIFA (mPLUG)	59.7	47.2

TIFA has a much higher correlation with human judgments than prior metrics. The correlations between each evaluation metric and human judgment are shown in Table 2. For caption-based evaluations, we use metrics BLEU-4 [38], ROUGE-L [33], METEOR [2], and SPICE [1]. TIFA has higher correlations with human judgments than all previous evaluation metrics on all VQA models. TIFA (mPLUG) yields the highest correlation with human judgments among all VQA models.

5.2. Benchmarking Text-to-Image Models

Figure 7 shows the average TIFA score text-to-image models get on TIFA v1.0. The detailed scores with each VQA model on each element type are provided in Appendix B. We can see a clear trend of how text-to-image models evolve over time. There is a jump in TIFA score after DALL-E [44] is released, from about 60% to 75%. Qualitative examples of our evaluation metric are in Appendix A.

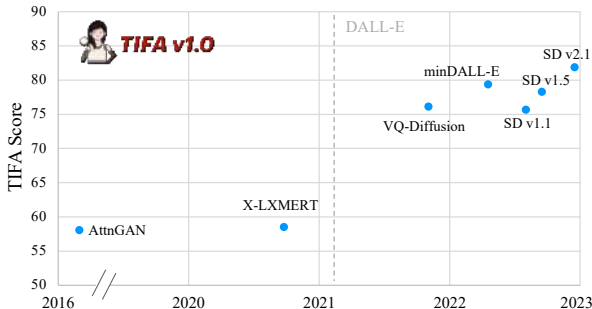


Figure 7. Average TIFA score of text-to-image models on the TIFA v1.0 benchmark. The horizontal axis shows their release dates.

5.3. Findings on Current Text-to-Image Models

Figure 8 shows accuracy on each type of question in TIFA v1.0 for Stable Diffusion v1.1, v1.5, and v2.1. The score on each type reflects the text-to-image models’ faithfulness in each type of visual element. To the best of our knowledge, TIFA is the only automatic evaluation method that can provide such a detailed fine-grained analysis of image generation. We separate the scores on COCO captions and other text inputs. For COCO captions, we also include the accuracy on the ground-truth images for reference. We summarize our findings in the following paragraphs.

Generating images from captions vs. free-form text From Figure 8, we can see that VQA accuracy is higher on the COCO captions than on other text inputs. The reason is that COCO captions correspond to real images, while other text inputs may correspond to compositions that cannot be found in real-world photos (e.g. “a blue apple”).

What elements are text-to-image models struggling with?

Based on the scores of each category in Figure 8, we can see that Stable Diffusion models are performing well on material, animal/human, color, and location in terms of text-to-image faithfulness. However, they yield low accuracy on questions involving **shapes**, **counting**, and **spatial relations**. “Other” mainly contains **abstract art notions**, and models are also struggling with them. There is also a big gap between the synthesized images and real images on the COCO captions. Future work can explore various directions (e.g., training data/loss and model architecture) to improve text-to-image models’ faithfulness in these aspects.

Why are ground-truth images not getting perfect scores?

Ground-truth images in COCO do not get perfect scores because 1) the COCO captions contain a substantial amount of noise from crowd workers [24] and 2) VQA models are not perfect. Real images have higher accuracy in all categories except material, color and location, where differences are small. It is left to future work to determine whether this is simply due to noise or it is an area where assessment can be improved.

Stable Diffusion is evolving.

We can see the consistent trend that Stable Diffusion models are improving in their later versions in most of the element categories. The exceptions are “shape” for both prompt sources, “other” and “food” for the free-form text inputs without gold images.

Composing multiple objects is challenging.

Figure 9 shows how the number of entities (objects, animals/humans, food) in the text input affects the average TIFA score. When there are more than 5 entities, The TIFA score starts to drop rapidly for all text-to-image models, consistent with similar findings in other vision-language evaluations [15, 13]. For reference, we also add the real images in COCO in this figure.

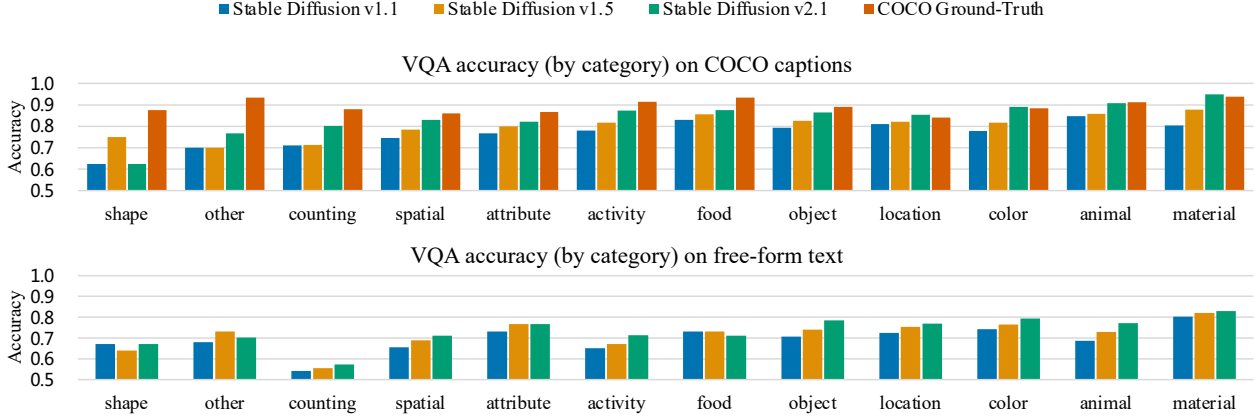


Figure 8. Accuracy on each type of question in the TIFA v1.0 benchmark. The text-to-image models are Stable Diffusion v1.1, v1.5, and v2.1. We order the categories by the average score Stable Diffusion v2.1 gets on corresponding questions. For COCO captions, we also include the accuracy of the ground-truth images for reference.

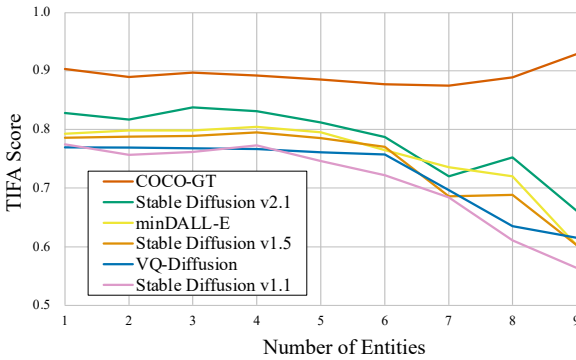


Figure 9. TIFA vs. numbers of entities (objects, animals/humans, and food) in the text input. The accuracy starts to drop when more than 5 entities are added to the text, showing that compositionality is hard for text-to-image models. Meanwhile, TIFA scores for COCO ground-truth (GT) images remain consistent.

The TIFA score on real images is rather consistent and does not change as the number of entities increases. This quantitatively shows that composing multiple objects is challenging for current text-to-image models. One possible reason is that the CLIP text embedding, which is used to train Stable Diffusion, lacks compositionality, as investigated in [37].

5.4. Analysis of VQA Models

One major concern of TIFA is that VQA models can introduce some errors. Table 2 shows that TIFA has a much higher correlation with human judgment than the previous metrics, regardless of the choice of the VQA models; here we conduct a more detailed analysis.

Sensitivity of TIFA to VQA models Figure 10 shows several recent text-to-image models’ TIFA scores on the COCO captions in TIFA v1.0, measured by different VQA models. We also include the TIFA scores on the ground-truth COCO

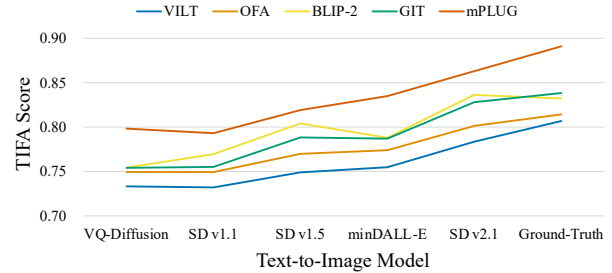


Figure 10. Several text-to-image models’ TIFA score on COCO captions, measured by different VQA models. We also include the accuracy of ground-truth images for reference.

images for reference. TIFA scores computed by different VQA models show a similar trend on these text-to-image models. Also, the ground-truth images get the highest TIFA score. We also computed Spearman’s ρ of TIFA scores given by different VQA models. The pairwise correlation between all VQA models is greater than 0.6.

Humans performing VQA To conduct further analysis on the VQA models, we ask annotators to answer the multiple-choice visual questions in TIFA v1.0. These annotations help us evaluate the accuracy of each VQA model. For multiple-choice questions, we add the option “None of the above” for human evaluation. Annotation guidelines are in Appendix C.

We collect annotations of 1029 questions on 126 images. Each question is answered by two annotators. The inter-annotator agreement measured by Krippendorff’s α is 0.88. A third annotator is involved if two annotators disagree, and the final answer is chosen by the majority vote.

Which VQA model should we use?

Table 3 reports the accuracy of each VQA model and the correlation between TIFA scores calculated by VQA model answers and human answers. We observe that higher model performance is directly related to the TIFA score’s

Table 3. Comparison of VQA models. The first row is the VQA accuracy, using the human VQA answers as reference. The second row is Spearman’s correlation between TIFA scores calculated by each VQA model and the human VQA.

	VILT	OFA	GIT	BLIP-2	mPLUG
VQA Acc.	76.1	77.1	79.1	81.0	84.5
TIFA Corr.	60.9	63.7	72.5	75.6	76.8

correlation with human judgments. **mPLUG** has the highest accuracy.

Another important factor to consider is the runtime. We measure the inference speed of each VQA model on NVIDIA A40 GPU with batch size 1 over the Stable Diffusion v2.1 images (768×768 pixels). For one question, VILT takes 0.08s on average; OFA, GIT, and mPLUG all take about 0.25s; BLIP-2 takes 0.73s. Based on the above results, we choose **mPLUG** as the default VQA model for TIFA v1.0 because it is the most accurate while being reasonably fast.

Separation of Text-to-Image Errors and VQA Errors

Suppose an image gets a wrong answer given a visual question. Then the image generation or the VQA model might have made an error. Based on the human VQA results, we separate these two kinds of errors in Figure A. If human VQA gives the wrong answer, then we suspect the generated image has an error. Otherwise, the image is correct but the VQA model is making an error. Figure A shows that the majority of errors are made by the text-to-image models. For mPLUG, less than 25% errors are due to the VQA model. This suggests that the TIFA framework is a viable evaluation method despite its inherent challenges.



Figure 11. Source of the error when VQA gets the wrong answer.

6. Discussion

Why does TIFA work better than CLIPScore? Our experimental results and human evaluations in Section 5 show that TIFA is more accurate than prior metrics (CLIP [42] and captioning-based approaches) for evaluating text-to-image faithfulness. We hypothesize that the major challenge of these prior metrics is that they summarize the image outputs and text inputs into a single representation (embedding/caption). In contrast, TIFA exploits the power of the language models to decompose the text input into fine-grained probes, which allows VQA to capture more nuanced aspects of the text input and the generated image.

How do current VQA models perform on TIFA? One limitation is that TIFA requires VQA models to work reasonably well, which is true given the current models and TIFA v1.0, as shown in Section 5. Nevertheless, the assumption might not hold for current models in domains like anime and abstract art. TIFA is a modularized evaluation framework. The VQA models used within the framework can be updated as stronger VQA models become available in the future. For instance, we plan to incorporate GPT-4 once its image API is made public since it is likely to improve TIFA. Another possible solution is to ensemble multiple image understanding models. For example, one may employ expert models on art concepts. We leave this for future work.

Other limitations. Another limitation of TIFA is its runtime. Answering multiple visual questions is slower than one CLIP inference. In the scenario described in Section 5.4, mPLUG takes 1.6s to evaluate one image (without batching). Also, our question generation pipeline needs one inference on a modern language model for each text input. The run time is not a critical issue for benchmarking purposes, but may not be computationally feasible for the kind of large-scale data filtering done, for example, in LAION-5B [49]. Nevertheless, we would like to point out that our evaluation is much faster than the image generation process of diffusion models. Thus, we believe it is feasible to perform reranking and reinforcement learning with TIFA on diffusion models.

7. Conclusions

We present TIFA, a new automatic text-to-image faithfulness evaluation metric using VQA. Compared with prior metrics, TIFA is fine-grained, interpretable, and better aligned with human judgments. Based on this metric, we introduce the TIFA v1.0, a large-scale text-to-image benchmark containing 4K prompts and 25K questions. We conduct a comprehensive study of current text-to-image models using TIFA v1.0 and highlight the limitations of current generative models. We quantitatively show that current image generation models still struggle in counting, spatial relations, and composing multiple objects. Finally, we conduct extensive analysis and human evaluation, demonstrating that TIFA is robust to different VQA models. We hope TIFA will help evaluate future work on image generation and become increasingly sophisticated as it is upgraded with new LM, QA, and VQA components.

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Appendices

A. Qualitative Examples

Here we show three examples on how TIFA is computed. For each text input, we show the questions and answers generated by GPT-3 [3] and filtered by UnifiedQA [25]. We also show the VQA model’s answer to each vision question given each generated image. The VQA model used is mPLUG [31]. The first text input comes from COCO [34] and the second and third text inputs come from DrawBench [47].



Stable Diffusion v1.1



Stable Diffusion v2.1

Text Input: On a gray day a surfer carrying a white board walks on a beach.

GPT-3 generated + verified QAs			
Question: is this a surfer?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: who is on the beach?	Choices: surfer, swimmer, driver, skier	Answer: surfer	
VQA: Surfer	✓	Surfer	✓
Question: is the person carrying a board?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: is this a beach?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: what type of place is this?	Choices: beach, park, forest, mountain	Answer: beach	
VQA: Beach	✓	Beach	✓
Question: is the surfer carrying something?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: is the surfer walking on the beach?	Choices: yes, no	Answer: yes	
VQA: No	✗	Yes	✓
Question: is the day gray?	Choices: yes, no	Answer: yes	
VQA: No	✗	Yes	✓
Question: what color is the sky?	Choices: black, gray, red, blue	Answer: gray	
VQA: Blue	✗	Gray	✓
Question: is the board white?	Choices: yes, no	Answer: yes	
VQA: No	✗	Yes	✓
Question: what color is the board?	Choices: black, white, red, blue	Answer: white	
VQA: Black	✗	White	✓



Accuracy on 12 questions

54.6 100.0



Stable Diffusion v1.5



Stable Diffusion v2.1

Text Input: A black colored banana.

GPT-3 generated + verified QAs			
Question: is this a banana?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: is the banana black?	Choices: yes, no	Answer: surfer	
VQA: No	✗	Yes	✓
Question: what color is the banana?	Choices: yellow, red, black, white	Answer: black	
VQA: Yellow	✗	Black	✓



Accuracy on 3 questions

33.3 100.0



Stable Diffusion v1.5



Stable Diffusion v2.1

Text Input: Three cats and two dogs sitting on the grass.

GPT-3 generated + verified QAs			
Question: are there cats?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: are there dogs?	Choices: yes, no	Answer: yes	
VQA: No	✗	Yes	✓
Question: is there grass?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: what are the animals sitting on?	Choices: grass, leaves, twigs, sand	Answer: grass	
VQA: Grass	✓	Grass	✓
Question: are the animals sitting?	Choices: yes, no	Answer: yes	
VQA: Yes	✓	Yes	✓
Question: how many cats are in the picture?	Choices: 1, 2, 3, 4	Answer: 3	
VQA: 3	✓	4	✗
Question: are there two dogs?	Choices: yes, no	Answer: yes	
VQA: No	✗	No	✗
Question: how many dogs are in the picture?	Choices: 1, 2, 3, 4	Answer: 2	
VQA: 1	✗	1	✗



Accuracy on 8 questions

62.5 62.5

Table 4. Detailed evaluation of each text-to-image model on TIFA v1.0.

	VQA Accuracy by Element Category							
	shape	other	counting	spatial	attribute	activity	food	object
AttnGAN[60]	42.0	47.8	41.9	70.8	53.6	64.3	48.1	56.3
X-LXMERT[5]	34.8	46.8	41.7	70.9	55.2	65.4	52.4	57.0
Stable Diffusion v1.1[46]	66.7	68.7	66.0	69.4	74.6	73.8	79.7	75.1
VQ-Diffusion[16]	63.8	64.2	61.6	73.5	75.4	76.7	80.0	74.2
Stable Diffusion v1.5[46]	65.2	72.1	66.6	72.9	78.0	76.9	81.4	78.4
minDALL-E[27]	69.6	74.6	69.0	74.7	77.0	79.5	82.8	79.9
Stable Diffusion v2.1[46]	66.7	72.1	73.3	76.1	78.8	82.0	82.2	82.4

	location	color	animal/human	material	TIFA by text source		Overall TIFA
					COCO	free-form	
AttnGAN[60]	60.4	56.5	58.6	61.7	67.5	47.4	58.1
X-LXMERT[5]	69.1	54.8	52.8	61.2	68.1	47.7	58.6
Stable Diffusion v1.1[46]	78.4	75.7	78.2	80.4	79.3	72.2	75.7
VQ-Diffusion[16]	77.9	84.2	79.0	80.9	79.8	72.6	76.2
Stable Diffusion v1.5[46]	79.9	78.8	80.6	84.7	81.9	74.9	78.4
minDALL-E[27]	82.1	83.7	78.9	86.1	83.5	75.5	79.4
Stable Diffusion v2.1[46]	82.8	83.6	85.2	88.5	86.3	77.7	82.0

B. Detailed Results

Table 4 shows the detailed evaluation results of the text-to-image models we use on the TIFA v1.0 benchmark. We show the VQA accuracy of each question category, TIFA score on each text source, and the overall TIFA score. We can see that Stable Diffusion v2.1 [46] gets the highest overall score and also scores the highest in most categories. Nonetheless, the CLIP [42] and VQGAN [9] based minDALL-E [27] gets the highest accuracy on “shape”, “other”, “food”, and VQ-Diffusion [16] gets the highest accuracy on “color”.

C. Annotation Details

C.1. Likert Scale on Text-to-Image Faithfulness

Guidelines The annotation guideline is as follows:

- On a scale of 1-5, score “does the image match the prompt?”.
- The ranking of each image given the same text input is important. If you believe the current scoring criteria cannot reflect your ranking preference, pick scores that are consistent with your ranking. Ties are allowed.
- To evaluate the generated image, there are two aspects: image quality and text-image match. Here we only care about text-image match, which is referred to as “faithfulness”.
- There are several kinds of elements in the text: object, attribute, relation, and context. Measure the consistency by counting how many elements are missed/misrepresented in the generated image.

- For some elements, e.g. “train conductor’s hat”, if you can see there is a hat but not a train conductor’s hat, consider half of the element is missed/misrepresented in the generated image.
- Objects are the most important elements. If an object is missing, then consider all related attributes, activity, and attributes missing.
- When you cannot tell what the object/attribute/activity/context is, consider the element missing. (e.g., can’t tell if an object is a microwave)

Given the above guideline, suppose the text input contains n elements, and x elements are missed or misrepresented. n and x are all counted by the annotators. The reference scoring guideline is as follows:

- 5: The image perfectly matches the prompt.
- 4: $x \leq 2$ and $x \leq n/3$. A few elements are missed/misrepresented.
- 3: $\min\{2, n/3\} < x \leq n/2$ elements are missed/misrepresented.
- 2: $x > n/2$. More than half of the elements are missed/misrepresented.
- 1: None of the major objects are correctly presented in the image.

Details We collect 1600 annotations on 800 generated images from 160 text inputs. Each image is scored by 2 annotators, and we collect the scores from 20 graduate students. We

average the scores as the final faithfulness score of the image. The inter-annotator agreement measured by Krippendorff’s α is 0.67, indicating “substantial” agreement. The images are generated by the five most recent text-to-image models in our study, including VQ-Diffusion [16], minDALL-E [27], and Stable Diffusion [47] v1.1, v1.5, and v2.1. For each text input, we present the five images together, making it easier for the annotators to give faithfulness scores that reflect their ranking preference. We will release the annotation scores on publication.

C.2. Human VQA

Guidelines Given an image, a question, and a set of choices, choose the correct choice according to the image content. There are two types of questions. One has two options: "(A) yes (B) no". Another type of question has four choices. We also add the fifth option “None of the above”. If you believe none of the four choices is correct, choose the fifth one. Some images are of low quality. Just select the choice according to your instinct. For ambiguous cases, for example, the question is “is there a man?”, and the image contains a human but it is unclear whether the human is a man, answer “no”.

Details We collect annotations of 1029 questions on 126 generated images. The images are from images used in the Likert Scale annotation. Each question is answered by two annotators, and we have the same 20 graduate students as the annotators. The inter-annotator agreement measured by Krippendorff’s α is 0.88. A third annotator is involved if the two annotators disagree. And the final answer is given by the majority vote. We will release the annotated VQA answers.

D. Common Q & A

Any possible extension to TIFA? As discussed in §1, one extension of our work will be customized versions of the TIFA benchmark focusing on one aspect of image generation. For example, we can make a TIFA benchmark that only contains questions about “counting”; Or a benchmark consists of text inputs synthesized to test text-to-image models’ ability in composing multiple objects. Another possible extension is to use TIFA on other generation tasks, e.g., text-to-3D and text-to-video.

The OpenAI APIs are too expensive. Can we generate questions by local models? Yes. Our approach works on any language model. Please refer to §4.3 on question generation with our fine-tuned LLaMA 2 checkpoint in. Also, we would like to emphasize that all questions in TIFA v1.0 benchmark are pre-generated by GPT-3, and there is no need to re-generate those questions for evaluation.

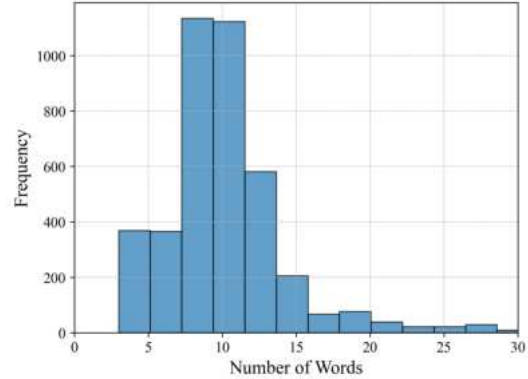


Figure 12. Distribution of the lengths of TIFA v1.0 text inputs.

More details on TIFA v1.0 text inputs? The distribution of the number of words in the text inputs is shown in 12. Most text inputs have around 10 words. We also conduct bias analysis on TIFA v1.0 text inputs. Among the 4K text inputs, regarding gender expression, 400 are perceived as “male” and 239 are perceived as “female”. We find that the bias in gender distribution comes from captions sampled from COCO dataset [34].

E. Text-to-Image Model Details

AttnGAN AttnGan [60] is a text-to-image model that is introduced in 2017. It is based on an attention mechanism that allows the model to focus on different parts of the input text when generating an image. AttnGAN has been shown to generate high-quality images for a variety of datasets and has been widely used in a number of applications. However, the attention mechanism can be computationally expensive, and the model can be difficult to train.

X-LXMERT X-LXMERT [5] is an enhanced version of LXMERT [53]. It is introduced in 2020 and incorporates several training refinements. These refinements involve discretizing visual representations, utilizing uniform masking with a wide range of masking ratios, and aligning appropriate pre-training datasets to respective objectives.

minDALL-E MinDALL-E [27] is a fast, minimal port of Boris Dayma’s DALL-E Mini (with mega weights). DALL-E Mini is an attempt to reproduce OpenAI’s DALL-E [44] with a smaller architecture. DALL-E can generate high-quality new images from any text prompt. The checkpoint we use is DALL-E Mega, the latest version of DALL-E Mini.

VQ-Diffusion VQ-Diffusion [16] is a generative model that combines vector quantization (VQ) and diffusion-based models for image synthesis. VQ-Diffusion builds upon the framework of diffusion-based generative models, which involves simulating a stochastic process that gradually trans-

forms a simple noise distribution into the target data distribution. In VQ-Diffusion, the image data is first quantized into discrete codes using a VQ algorithm, which maps each image patch to the nearest code in a codebook. This allows the model to represent complex data distributions with a compact set of discrete codes, rather than continuous probability densities.

Stable Diffusion Stable Diffusion is a pre-trained diffusion model for text-to-image generation. It is based on Latent Diffusion model (LDM) [46]. LDM is designed to learn the underlying structure of a dataset by mapping it to a lower-dimensional latent space. This latent space represents the data in which the relationships between different data points are more easily understood and analyzed, and reduces the amount of computational resources needed for training diffusion models. Specifically, we use three versions of Stable Diffusion, v1.1, v1.5, and v2.1. Each version is trained with a different number of steps and amount of data.

F. Prompt

For demonstration purposes, we show part of the prompt for question generation with GPT-3 in-context learning. The whole prompt will be released with our codes. The prompt contains instructions and several in-context examples. The examples cover all element categories.

Given an image description, generate multiple-choice questions that verify if the image description is correct.

First extract elements from the image description. Then classify each element into a category (object, human, animal, food, activity, attribute, counting, color, material, spatial, location, shape, other). Finally, generate questions for each element.

Description: A man posing for a selfie in a jacket and bow tie.

Entities: man, selfie, jacket, bow tie

Activities: posing

Colors:

Counting:

Other attributes:

Questions and answers are below:

About man (human):

Q: is this a man?

Choices: yes, no

A: yes

Q: who is posing for a selfie?

Choices: man, woman, boy, girl

A: man

About selfie (activity):

Q: is the man taking a selfie?

Choices: yes, no

A: yes

Q: what type of photo is the person taking?

Choices: selfie, landscape, sports, portrait

A: selfie

About jacket (object):

Q: is the man wearing a jacket?

Choices: yes, no

A: yes

Q: what is the man wearing?

Choices: jacket, t-shirt, tuxedo, sweater

A: jacket

About bow tie (object):

Q: is the man wearing a bow tie?

Choices: yes, no

A: yes

Q: is the man wearing a bow tie or a neck tie?

Choices: bow tie, neck tie, cravat, bolo tie

A: bow tie

About posing (activity):

Q: is the man posing for the selfie?

Choices: yes, no

A: yes

Q: what is the man doing besides taking the selfie?

Choices: posing, waving, nothing, shaking

A: posing

Description: A horse and several cows feed on hay.

Entities: horse, cows, hay

Activities: feed on

Colors:

Counting: several

Other attributes:

Questions and answers are below:

About horse (animal):

Q: is there a horse?

Choices: yes, no

A: yes

About cows (animal):

Q: are there cows?

Choices: yes, no

A: yes

About hay (object):

Q: is there hay?

Choices: yes, no

A: yes

Q: what is the horse and cows feeding on?

Choices: hay, grass, leaves, twigs

A: hay

About feed on (activity):

Q: are the horse and cows feeding on hay?

Choices: yes, no

A: yes

About several (counting):

Q: are there several cows?

Choices: yes, no

A: yes

Description: A red colored dog.

Entities: dog

Activities:

Colors: red

Counting:

Other attributes:

Questions and answers are below:

About dog (animal):

Q: is this a dog?

Choices: yes, no

A: yes

Q: what animal is in the picture?

Choices: dog, cat, bird, fish

A: dog

About red (color):

Q: is the dog red?

Choices: yes, no

A: yes

Q: what color is the dog?

Choices: red, black, white, yellow

A: red

Description: Here are motorcyclists parked outside a Polish gathering spot for women

Entities: motorcyclists, gathering spot, women

Activities: parked

Colors:

Counting:

Other attributes: outside, polish

Questions and answers are below:

About motorcyclists (human):

Q: are there motorcyclists?

Choices: yes, no

A: yes

About gathering spot (location):

Q: is this a gathering spot?

Choices: yes, no

A: yes

About women (human):

Q: are there women?

Choices: yes, no

A: yes

Q: who are in the gathering spot?

Choices: women, men, boys, girls

A: women

About parked (activity):

Q: are the motorcyclists parked?

Choices: yes, no

A: yes

About outside (spatial):

Q: have the motorcyclists parked outside the gathering spot?

Choices: yes, no

A: yes

Q: are the motorcyclists outside or inside of the gathering spot?

Choices: outside, inside, on the roof, in the basement

A: outside

About Polish (other):

Q: is this a Polish gathering spot?

Choices: yes, no

A: yes

Q: is this a Polish or a Chinese gathering spot?

Choices: Polish, American, Chinese, Japanese

A: Polish

SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis

Dustin Podell Zion English Kyle Lacey Andreas Blattmann Tim Dockhorn

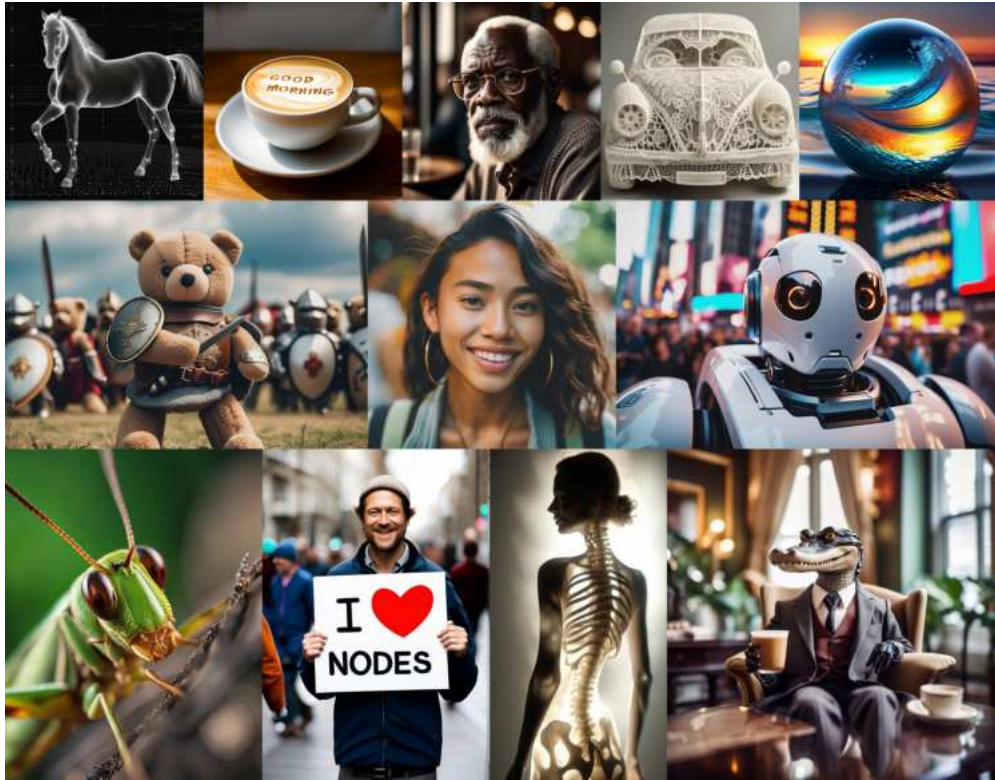
Jonas Müller

Joe Penna

Robin Rombach

Stability AI, Applied Research

Code: <https://github.com/Stability-AI/generative-models> Model weights: <https://huggingface.co/stabilityai/>



Abstract

We present *SDXL*, a latent diffusion model for text-to-image synthesis. Compared to previous versions of *Stable Diffusion*, *SDXL* leverages a three times larger UNet backbone: The increase of model parameters is mainly due to more attention blocks and a larger cross-attention context as *SDXL* uses a second text encoder. We design multiple novel conditioning schemes and train *SDXL* on multiple aspect ratios. We also introduce a *refinement model* which is used to improve the visual fidelity of samples generated by *SDXL* using a post-hoc *image-to-image* technique. We demonstrate that *SDXL* shows drastically improved performance compared to previous versions of *Stable Diffusion* and achieves results competitive with those of black-box state-of-the-art image generators. In the spirit of promoting open research and fostering transparency in large model training and evaluation, we provide access to code and model weights.

1 Introduction

The last year has brought enormous leaps in deep generative modeling across various data domains, such as natural language [50], audio [17], and visual media [38, 37, 40, 44, 15, 3, 7]. In this report, we focus on the latter and unveil *SDXL*, a drastically improved version of *Stable Diffusion*. *Stable Diffusion* is a latent text-to-image diffusion model (DM) which serves as the foundation for an array of recent advancements in, e.g., 3D classification [43], controllable image editing [54], image personalization [10], synthetic data augmentation [48], graphical user interface prototyping [51], etc. Remarkably, the scope of applications has been extraordinarily extensive, encompassing fields as diverse as music generation [9] and reconstructing images from fMRI brain scans [49].

User studies demonstrate that *SDXL* consistently surpasses all previous versions of *Stable Diffusion* by a significant margin (see Fig. 1). In this report, we present the design choices which lead to this boost in performance encompassing *i*) a $3\times$ larger UNet-backbone compared to previous *Stable Diffusion* models (Sec. 2.1), *ii*) two simple yet effective additional conditioning techniques (Sec. 2.2) which do not require any form of additional supervision, and *iii*) a separate diffusion-based refinement model which applies a noising-denoising process [28] to the latents produced by *SDXL* to improve the visual quality of its samples (Sec. 2.5).

A major concern in the field of visual media creation is that while black-box-models are often recognized as state-of-the-art, the opacity of their architecture prevents faithfully assessing and validating their performance. This lack of transparency hampers reproducibility, stifles innovation, and prevents the community from building upon these models to further the progress of science and art. Moreover, these closed-source strategies make it challenging to assess the biases and limitations of these models in an impartial and objective way, which is crucial for their responsible and ethical deployment. With *SDXL* we are releasing an *open* model that achieves competitive performance with black-box image generation models (see Fig. 10 & Fig. 11).

2 Improving *Stable Diffusion*

In this section we present our improvements for the *Stable Diffusion* architecture. These are modular, and can be used individually or together to extend any model. Although the following strategies are implemented as extensions to latent diffusion models (LDMs) [38], most of them are also applicable to their pixel-space counterparts.

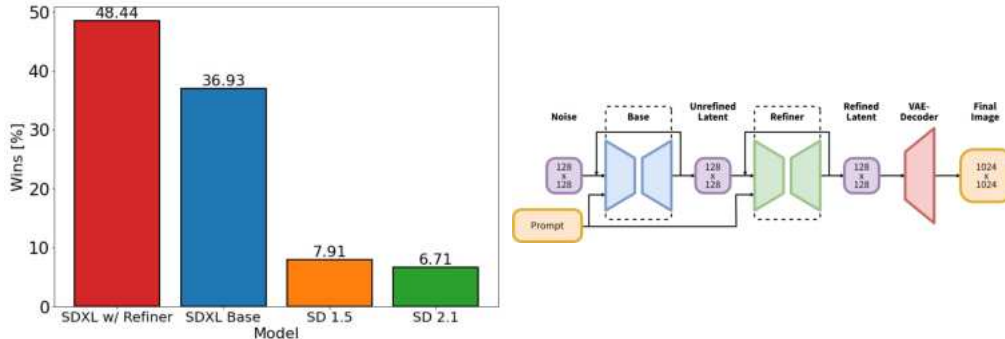


Figure 1: *Left*: Comparing user preferences between *SDXL* and *Stable Diffusion* 1.5 & 2.1. While *SDXL* already clearly outperforms *Stable Diffusion* 1.5 & 2.1, adding the additional refinement stage boosts performance. *Right*: Visualization of the two-stage pipeline: We generate initial latents of size 128×128 using *SDXL*. Afterwards, we utilize a specialized high-resolution *refinement model* and apply SDEdit [28] on the latents generated in the first step, using the same prompt. *SDXL* and the refinement model use the same autoencoder.

2.1 Architecture & Scale

Starting with the seminal works Ho et al. [14] and Song et al. [47], which demonstrated that DMs are powerful generative models for image synthesis, the convolutional UNet [39] architecture has been the dominant architecture for diffusion-based image synthesis. However, with the development

Table 1: Comparison of *SDXL* and older *Stable Diffusion* models.

Model	<i>SDXL</i>	SD 1.4/1.5	SD 2.0/2.1
# of UNet params	2.6B	860M	865M
Transformer blocks	[0, 2, 10]	[1, 1, 1, 1]	[1, 1, 1, 1]
Channel mult.	[1, 2, 4]	[1, 2, 4, 4]	[1, 2, 4, 4]
Text encoder	CLIP ViT-L & OpenCLIP ViT-bigG	CLIP ViT-L	OpenCLIP ViT-H
Context dim.	2048	768	1024
Pooled text emb.	OpenCLIP ViT-bigG	N/A	N/A

of foundational DMs [40, 37, 38], the underlying architecture has constantly evolved: from adding self-attention and improved upscaling layers [5], over cross-attention for text-to-image synthesis [38], to pure transformer-based architectures [33].

We follow this trend and, following Hooeboom et al. [16], shift the bulk of the transformer computation to lower-level features in the UNet. In particular, and in contrast to the original *Stable Diffusion* architecture, we use a heterogeneous distribution of transformer blocks within the UNet: For efficiency reasons, we omit the transformer block at the highest feature level, use 2 and 10 blocks at the lower levels, and remove the lowest level ($8\times$ downsampling) in the UNet altogether — see Tab. 1 for a comparison between the architectures of *Stable Diffusion* 1.x & 2.x and *SDXL*. We opt for a more powerful pre-trained text encoder that we use for text conditioning. Specifically, we use OpenCLIP ViT-bigG [19] in combination with CLIP ViT-L [34], where we concatenate the penultimate text encoder outputs along the channel-axis [1]. Besides using cross-attention layers to condition the model on the text-input, we follow [30] and additionally condition the model on the pooled text embedding from the OpenCLIP model. These changes result in a model size of 2.6B parameters in the UNet, see Tab. 1. The text encoders have a total size of 817M parameters.

2.2 Micro-Conditioning

Conditioning the Model on Image Size A notorious shortcoming of the LDM paradigm [38] is the fact that training a model requires a *minimal image size*, due to its two-stage architecture. The two main approaches to tackle this problem are either to discard all training images below a certain minimal resolution (for example, *Stable Diffusion* 1.4/1.5 discarded all images with any size below 512 pixels), or, alternatively, upscale images that are too small. However, depending on the desired image resolution, the former method can lead to significant portions of the training data being discarded, what will likely lead to a loss in performance and hurt generalization. We visualize such effects in Fig. 2 for the dataset on which *SDXL* was pretrained. For this particular choice of data, discarding all samples below our pretraining resolution of 256^2 pixels would lead to a significant 39% of discarded data. The second method, on the other hand, usually introduces upscaling artifacts which may leak into the final model outputs, causing, for example, blurry samples.

Instead, we propose to condition the UNet model on the original image resolution, which is trivially available during training. In particular, we provide the original (i.e., before any rescaling) height and width of the images as an additional conditioning to the model $c_{\text{size}} = (h_{\text{original}}, w_{\text{original}})$. Each component is independently embedded using a Fourier feature encoding, and these encodings are concatenated into a single vector that we feed into the model by adding it to the timestep embedding [5].

At inference time, a user can then set the desired *apparent resolution* of the image via this *size-conditioning*. Evidently (see Fig. 3), the model has learned to associate the conditioning c_{size} with

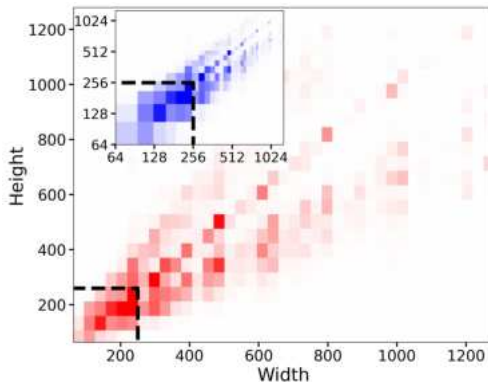


Figure 2: Height-vs-Width distribution of our pre-training dataset. Without the proposed size-conditioning, 39% of the data would be discarded due to edge lengths smaller than 256 pixels as visualized by the dashed black lines. Color intensity in each visualized cell is proportional to the number of samples.

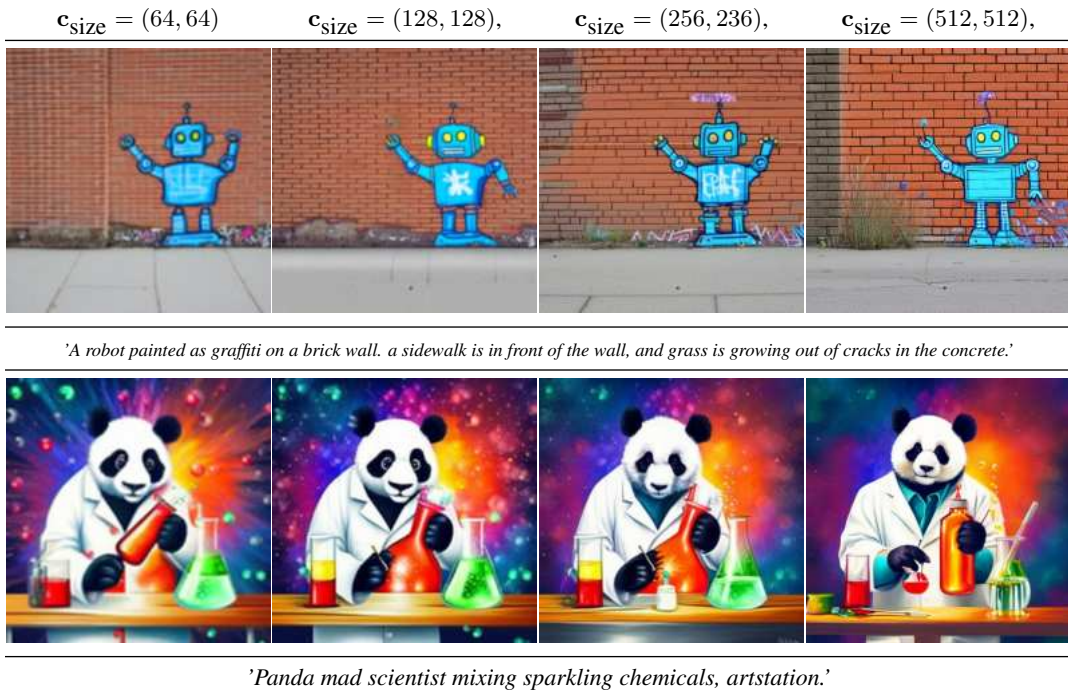


Figure 3: The effects of varying the size-conditioning: We show draw 4 samples with the same random seed from *SDXL* and vary the size-conditioning as depicted above each column. The image quality clearly increases when conditioning on larger image sizes. Samples from the 512^2 model, see Sec. 2.5. Note: For this visualization, we use the 512×512 pixel base model (see Sec. 2.5), since the effect of size conditioning is more clearly visible before 1024×1024 finetuning. Best viewed zoomed in.

resolution-dependent image features, which can be leveraged to modify the appearance of an output corresponding to a given prompt. Note that for the visualization shown in Fig. 3, we visualize samples generated by the 512×512 model (see Sec. 2.5 for details), since the effects of the size conditioning are less clearly visible after the subsequent multi-aspect (ratio) finetuning which we use for our final *SDXL* model.

We quantitatively assess the effects of this simple but effective conditioning technique by training and evaluating three LDMs on class conditional ImageNet [4] at spatial size 512^2 : For the first model (*CIN-512-only*) we discard all training examples with at least one edge smaller than 512 pixels what results in a train dataset of only 70k images. For *CIN-nocond* we use all training examples but without size conditioning. This additional conditioning is only used for *CIN-size-cond*. After training we generate 5k samples with 50 DDIM steps [46] and (classifier-free) guidance scale of 5 [13] for every model and compute IS [42] and FID [12] (against the full validation set). For *CIN-size-cond* we generate samples always conditioned on $c_{\text{size}} = (512, 512)$. Tab. 2 summarizes the results and verifies that *CIN-size-cond* improves upon the baseline models in both metrics. We attribute the degraded performance of *CIN-512-only* to bad generalization due to overfitting on the small training dataset while the effects of a mode of blurry samples in the sample distribution of *CIN-nocond* result in a reduced FID score. Note that, although we find these classical quantitative scores not to be suitable for evaluating the performance of foundational (text-to-image) DMs [40, 37, 38] (see App. F), they remain reasonable metrics on ImageNet as the neural backbones of FID and IS have been trained on ImageNet itself.

Table 2: Conditioning on the original spatial size of the training examples improves performance on class-conditional ImageNet [4] on 512^2 resolution.

model	FID-5k ↓	IS-5k ↑
<i>CIN-512-only</i>	43.84	110.64
<i>CIN-nocond</i>	39.76	211.50
<i>CIN-size-cond</i>	36.53	215.34

Conditioning the Model on Cropping Parameters The first two rows of Fig. 4 illustrate a typical failure mode of previous *SD* models: Synthesized objects can be cropped, such as the cut-off head of the cat in the left examples for *SD* 1-5 and *SD* 2-1. An intuitive explanation for this behavior is the use of *random cropping* during training of the model: As collating a batch in DL frameworks such as



Figure 4: Comparison of the output of *SDXL* with previous versions of *Stable Diffusion*. For each prompt, we show 3 random samples of the respective model for 50 steps of the DDIM sampler [46] and cfg-scale 8.0 [13]. Additional samples in Fig. 14

PyTorch [32] requires tensors of the same size, a typical processing pipeline is to (i) resize an image such that the shortest size matches the desired target size, followed by (ii) randomly cropping the image along the longer axis. While random cropping is a natural form of data augmentation, it can leak into the generated samples, causing the malicious effects shown above.

To fix this problem, we propose another simple yet effective conditioning method: During dataloading, we uniformly sample crop coordinates c_{top} and c_{left} (integers specifying the amount of pixels cropped from the top-left corner along the height and width axes, respectively) and feed them into the model as conditioning parameters via Fourier feature embeddings, similar to the size conditioning described above. The concatenated embedding c_{crop} is then used as an additional conditioning parameter. We emphasize that this technique is not limited to LDMs and could be used for any DM. Note that crop- and size-conditioning can be readily combined. In such a case, we concatenate the feature embedding along the channel dimension, before adding it to the timestep embedding in the UNet. Alg. 1 illustrates how we sample c_{crop} and c_{size} during training if such a combination is applied.

Given that in our experience large scale datasets are, on average, object-centric, we set $(c_{\text{top}}, c_{\text{left}}) = (0, 0)$ during inference and thereby obtain object-centered samples from the trained model.

See Fig. 5 for an illustration: By tuning $(c_{\text{top}}, c_{\text{left}})$, we can successfully *simulate* the amount of cropping during inference. This is a form of *conditioning-augmentation*, and has been used in various forms with autoregressive [20] models, and more recently with diffusion models [21].

While other methods like data bucketing [31] successfully tackle the same task, we still benefit from cropping-induced data augmentation, while making sure that it does not leak into the generation process - we actually use it to our advantage to gain more control over the image synthesis process. Furthermore, it is easy to implement and can be applied in an online fashion during training, without additional data preprocessing.

2.3 Multi-Aspect Training

Real-world datasets include images of widely varying sizes and aspect-ratios (c.f. fig. 2). While the common output resolutions for text-to-image models are square images of 512×512 or 1024×1024 pixels, we argue that this is a rather unnatural choice, given the widespread distribution and use of landscape (e.g., 16:9) or portrait format screens.

Motivated by this, we finetune our model to handle multiple aspect-ratios simultaneously: We follow common practice [31] and partition the data into buckets of different aspect ratios, where we keep the pixel count as close to 1024^2 pixels as possible, varying height and width accordingly in multiples

Algorithm 1 Conditioning pipeline for size- and crop-conditioning

Require: Training dataset of images $\mathcal{D}$, target image size for training $\mathbf{s} = (h_{\text{tgt}}, w_{\text{tgt}})$
Require: Resizing function $\mathbf{R}$, cropping function $\mathbf{C}$
Require: Model train step $\mathbf{T}$

```
converged  $\leftarrow$  False
while not converged do
   $x \sim \mathcal{D}$ 
   $w_{\text{original}} \leftarrow \text{width}(x)$ 
   $h_{\text{original}} \leftarrow \text{height}(x)$ 
   $\mathbf{c}_{\text{size}} \leftarrow (h_{\text{original}}, w_{\text{original}})$ 
   $x \leftarrow \mathbf{R}(x, \mathbf{s})$   $\triangleright$  resize smaller image size to target size  $\mathbf{s}$ 
  if  $h_{\text{original}} \leq w_{\text{original}}$  then
     $c_{\text{left}} \sim \mathcal{U}(0, \text{width}(x) - s_w)$   $\triangleright$  sample  $c_{\text{left}}$  from discrete uniform distribution
     $c_{\text{top}} = 0$ 
  else if  $h_{\text{original}} > w_{\text{original}}$  then
     $c_{\text{top}} \sim \mathcal{U}(0, \text{height}(x) - s_h)$   $\triangleright$  sample  $c_{\text{top}}$  from discrete uniform distribution
     $c_{\text{left}} = 0$ 
  end if
   $\mathbf{c}_{\text{crop}} \leftarrow (c_{\text{top}}, c_{\text{left}})$ 
   $x \leftarrow \mathbf{C}(x, \mathbf{s}, \mathbf{c}_{\text{crop}})$   $\triangleright$  crop image to size  $\mathbf{s}$  with top-left coordinate  $(c_{\text{top}}, c_{\text{left}})$ 
  converged  $\leftarrow \mathbf{T}(x, \mathbf{c}_{\text{size}}, \mathbf{c}_{\text{crop}})$   $\triangleright$  train model conditioned on  $\mathbf{c}_{\text{size}}$  and  $\mathbf{c}_{\text{crop}}$ 
end while
```

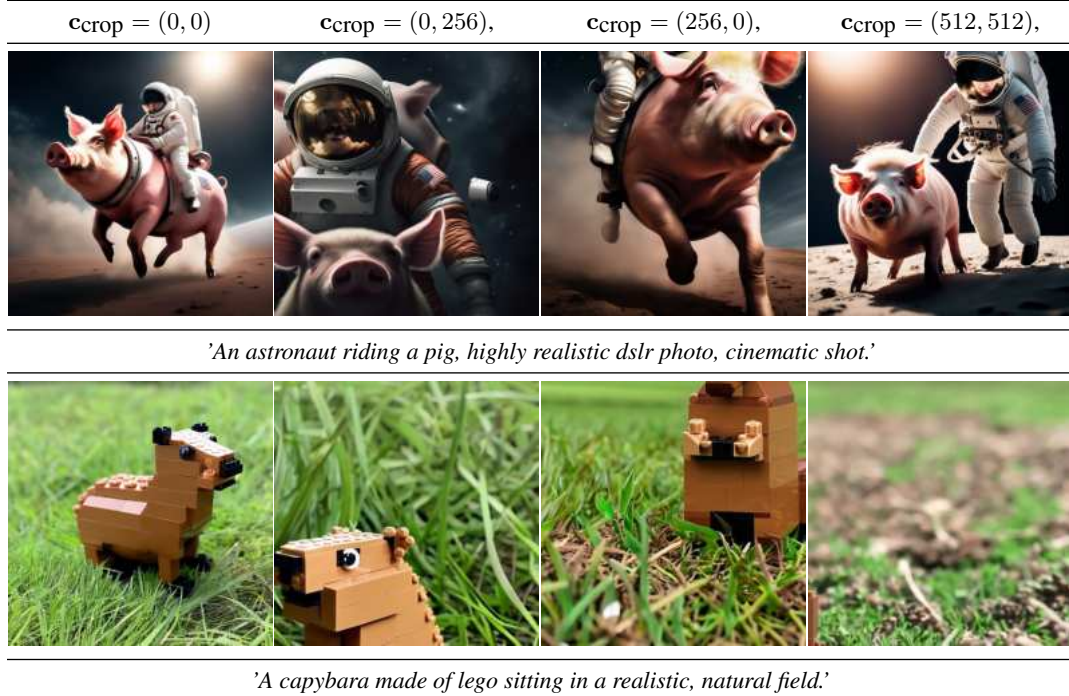


Figure 5: Varying the crop conditioning as discussed in Sec. 2.2. See Fig. 4 and Fig. 14 for samples from SD 1.5 and SD 2.1 which provide no explicit control of this parameter and thus introduce cropping artifacts. Samples from the 512² model, see Sec. 2.5.

of 64. A full list of all aspect ratios used for training is provided in App. F. During optimization, a training batch is composed of images from the same bucket, and we alternate between bucket sizes for each training step. Additionally, the model receives the bucket size (or, *target size*) as a conditioning, represented as a tuple of integers $\mathbf{c}_{\text{ar}} = (h_{\text{tgt}}, w_{\text{tgt}})$ which are embedded into a Fourier space in analogy to the size- and crop-conditionings described above.

In practice, we apply multi-aspect training as a finetuning stage after pretraining the model at a fixed aspect-ratio and resolution and combine it with the conditioning techniques introduced in Sec. 2.2 via concatenation along the channel axis. Fig. 16 in App. F provides python-code for this operation. Note that crop-conditioning and multi-aspect training are complementary operations, and crop-conditioning then only works within the bucket boundaries (usually 64 pixels). For ease of implementation, however, we opt to keep this control parameter for multi-aspect models.

2.4 Improved Autoencoder

Stable Diffusion is a LDM, operating in a pre-trained, learned (and fixed) latent space of an autoencoder. While the bulk of the semantic composition is done by the LDM [38], we can improve *local*, high-frequency details in generated images by improving the autoencoder. To this end, we train the same autoencoder architecture used for the original *Stable Diffusion* at a larger batch-size (256 vs 9) and additionally track the weights with an exponential moving average. The resulting autoencoder outperforms the original model in all evaluated reconstruction metrics, see Tab. 3. We use this autoencoder for all of our experiments.

Table 3: Autoencoder reconstruction performance on the COCO2017 [26] validation split, images of size 256×256 pixels. Note: *Stable Diffusion* 2.x uses an improved version of *Stable Diffusion* 1.x’s autoencoder, where the decoder was finetuned with a reduced weight on the perceptual loss [55], and used more compute. Note that our new autoencoder is trained from scratch.

model	PNSR $\uparrow$	SSIM $\uparrow$	LPIS $\downarrow$	rFID $\downarrow$
<i>SDXL</i> -VAE	24.7	0.73	0.88	4.4
<i>SD</i> -VAE 1.x	23.4	0.69	0.96	5.0
<i>SD</i> -VAE 2.x	24.5	0.71	0.92	4.7

2.5 Putting Everything Together

We train the final model, *SDXL*, in a multi-stage procedure. *SDXL* uses the autoencoder from Sec. 2.4 and a discrete-time diffusion schedule [14, 45] with 1000 steps. First, we pretrain a base model (see Tab. 1) on an internal dataset whose height- and width-distribution is visualized in Fig. 2 for 600 000 optimization steps at a resolution of 256×256 pixels and a batch-size of 2048, using size- and crop-conditioning as described in Sec. 2.2. We continue training on 512×512 pixel images for another 200 000 optimization steps, and finally utilize multi-aspect training (Sec. 2.3) in combination with an offset-noise [11, 25] level of 0.05 to train the model on different aspect ratios (Sec. 2.3, App. F) of $\sim 1024 \times 1024$ pixel area.

Refinement Stage Empirically, we find that the resulting model sometimes yields samples of low local quality, see Fig. 6. To improve sample quality, we train a separate LDM in the same latent space, which is specialized on high-quality, high resolution data and employ a noising-denoising process as introduced by *SDEdit* [28] on the samples from the base model. We follow [1] and specialize this refinement model on the first 200 (discrete) noise scales. During inference, we render latents from the base *SDXL*, and directly diffuse and denoise them in latent space with the refinement model (see Fig. 1), using the same text input. We note that this step is optional, but improves sample quality for detailed backgrounds and human faces, as demonstrated in Fig. 6 and Fig. 13.

To assess the performance of our model (with and without refinement stage), we conduct a user study, and let users pick their favorite generation from the following four models: *SDXL*, *SDXL* (with refiner), *Stable Diffusion* 1.5 and *Stable Diffusion* 2.1. The results demonstrate the *SDXL* with the refinement stage is the highest rated choice, and outperforms *Stable Diffusion* 1.5 & 2.1 by a significant margin (win rates: *SDXL* w/ refinement: 48.44%, *SDXL* base: 36.93%, *Stable Diffusion* 1.5: 7.91%, *Stable Diffusion* 2.1: 6.71%). See Fig. 1, which also provides an overview of the full pipeline. However, when using classical performance metrics such as FID and CLIP scores the improvements of *SDXL* over previous methods are not reflected as shown in Fig. 12 and discussed in App. E. This aligns with and further backs the findings of Kirstain et al. [23].



Figure 6: 1024^2 samples (with zoom-ins) from *SDXL* without (left) and with (right) the refinement model discussed. Prompt: “Epic long distance cityscape photo of New York City flooded by the ocean and overgrown buildings and jungle ruins in rainforest, at sunset, cinematic shot, highly detailed, 8k, golden light”. See Fig. [13](#) for additional samples.

3 Future Work

This report presents a preliminary analysis of improvements to the foundation model *Stable Diffusion* for text-to-image synthesis. While we achieve significant improvements in synthesized image quality, prompt adherence and composition, in the following, we discuss a few aspects for which we believe the model may be improved further:

- **Single stage:** Currently, we generate the best samples from *SDXL* using a two-stage approach with an additional refinement model. This results in having to load two large models into memory, hampering accessibility and sampling speed. Future work should investigate ways to provide a single stage of equal or better quality.
- **Text synthesis:** While the scale and the larger text encoder (OpenCLIP ViT-bigG [\[19\]](#)) help to improve the text rendering capabilities over previous versions of *Stable Diffusion*, incorporating byte-level tokenizers [\[52, 27\]](#) or simply scaling the model to larger sizes [\[53, 40\]](#) may further improve text synthesis.
- **Architecture:** During the exploration stage of this work, we briefly experimented with transformer-based architectures such as UViT [\[16\]](#) and DiT [\[33\]](#), but found no immediate benefit. We remain, however, optimistic that a careful hyperparameter study will eventually enable scaling to much larger transformer-dominated architectures.
- **Distillation:** While our improvements over the original *Stable Diffusion* model are significant, they come at the price of increased inference cost (both in VRAM and sampling speed). Future work will thus focus on decreasing the compute needed for inference, and increased sampling speed, for example through guidance- [\[29\]](#), knowledge- [\[6, 22, 24\]](#) and progressive distillation [\[41, 2, 29\]](#).
- Our model is trained in the discrete-time formulation of [\[14\]](#), and requires *offset-noise* [\[11, 25\]](#) for aesthetically pleasing results. The EDM-framework of Karras et al. [\[21\]](#) is a promising candidate for future model training, as its formulation in continuous time allows for increased sampling flexibility and does not require noise-schedule corrections.

Appendix

A Acknowledgements

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B Limitations



Figure 7: Failure cases of *SDXL* despite large improvements compared to previous versions of *Stable Diffusion*, the model sometimes still struggles with very complex prompts involving detailed spatial arrangements and detailed descriptions (e.g. top left example). Moreover, hands are not yet always correctly generated (e.g. top left) and the model sometimes suffers from two concepts bleeding into one another (e.g. bottom right example). All examples are random samples generated with 50 steps of the DDIM sampler [46] and cfg-scale 8.0 [13].

While our model has demonstrated impressive capabilities in generating realistic images and synthesizing complex scenes, it is important to acknowledge its inherent limitations. Understanding these limitations is crucial for further improvements and ensuring responsible use of the technology.

Firstly, the model may encounter challenges when synthesizing intricate structures, such as human hands (see Fig. 7, top left). Although it has been trained on a diverse range of data, the complexity of human anatomy poses a difficulty in achieving accurate representations consistently. This limitation suggests the need for further scaling and training techniques specifically targeting the synthesis of fine-grained details. A reason for this occurring might be that hands and similar objects appear with very high variance in photographs and it is hard for the model to extract the knowledge of the real 3D shape and physical limitations in that case.

Secondly, while the model achieves a remarkable level of realism in its generated images, it is important to note that it does not attain perfect photorealism. Certain nuances, such as subtle lighting effects or minute texture variations, may still be absent or less faithfully represented in the generated images. This limitation implies that caution should be exercised when relying solely on model-generated visuals for applications that require a high degree of visual fidelity.

Furthermore, the model’s training process heavily relies on large-scale datasets, which can inadvertently introduce social and racial biases. As a result, the model may inadvertently exacerbate these biases when generating images or inferring visual attributes.

In certain cases where samples contain multiple objects or subjects, the model may exhibit a phenomenon known as “concept bleeding”. This issue manifests as the unintended merging or overlap of distinct visual elements. For instance, in Fig. 14, an orange sunglass is observed, which indicates an instance of concept bleeding from the orange sweater. Another case of this can be seen in Fig. 8, the penguin is supposed to have a “blue hat” and “red gloves”, but is instead generated with blue

gloves and a red hat. Recognizing and addressing such occurrences is essential for refining the model’s ability to accurately separate and represent individual objects within complex scenes. The root cause of this may lie in the used pretrained text-encoders: firstly, they are trained to compress all information into a single token, so they may fail at binding only the right attributes and objects, Feng et al. [8] mitigate this issue by explicitly encoding word relationships into the encoding. Secondly, the contrastive loss may also contribute to this, since negative examples with a different binding are needed within the same batch [35].

Additionally, while our model represents a significant advancement over previous iterations of *SD*, it still encounters difficulties when rendering long, legible text. Occasionally, the generated text may contain random characters or exhibit inconsistencies, as illustrated in Fig. 8. Overcoming this limitation requires further investigation and development of techniques that enhance the model’s text generation capabilities, particularly for extended textual content — see for example the work of Liu et al. [27], who propose to enhance text rendering capabilities via character-level text tokenizers. Alternatively, scaling the model does further improve text synthesis [53, 40].

In conclusion, our model exhibits notable strengths in image synthesis, but it is not exempt from certain limitations. The challenges associated with synthesizing intricate structures, achieving perfect photorealism, further addressing biases, mitigating concept bleeding, and improving text rendering highlight avenues for future research and optimization.

C Diffusion Models

In this section, we give a concise summary of DMs. We consider the continuous-time DM framework [47] and follow the presentation of Karras et al. [21]. Let $p_{\text{data}}(\mathbf{x}_0)$ denote the data distribution and let $p(\mathbf{x}; \sigma)$ be the distribution obtained by adding i.i.d. σ^2 -variance Gaussian noise to the data. For sufficiently large σ_{max} , $p(\mathbf{x}; \sigma_{\text{max}}^2)$ is almost indistinguishable from σ_{max}^2 -variance Gaussian noise. Capitalizing on this observation, DMs sample high variance Gaussian noise $\mathbf{x}_M \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{max}}^2)$ and sequentially denoise $\mathbf{x}_M$ into $\mathbf{x}_i \sim p(\mathbf{x}_i; \sigma_i)$, $i \in \{0, \dots, M\}$, with $\sigma_i < \sigma_{i+1}$ and $\sigma_M = \sigma_{\text{max}}$. For a well-trained DM and $\sigma_0 = 0$ the resulting $\mathbf{x}_0$ is distributed according to the data.

Sampling. In practice, this iterative denoising process explained above can be implemented through the numerical simulation of the *Probability Flow* ordinary differential equation (ODE) [47]

$$d\mathbf{x} = -\dot{\sigma}(t)\sigma(t)\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma(t)) dt, \quad (1)$$

where $\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma)$ is the *score function* [18]. The schedule $\sigma(t): [0, 1] \rightarrow \mathbb{R}_+$ is user-specified and $\dot{\sigma}(t)$ denotes the time derivative of $\sigma(t)$. Alternatively, we may also numerically simulate a stochastic differential equation (SDE) [47, 21]:

$$d\mathbf{x} = \underbrace{-\dot{\sigma}(t)\sigma(t)\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma(t)) dt}_{\text{Probability Flow ODE; see Eq. 1}} - \underbrace{\beta(t)\sigma^2(t)\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma(t)) dt + \sqrt{2\beta(t)}\sigma(t) d\omega_t}_{\text{Langevin diffusion component}}, \quad (2)$$

where $d\omega_t$ is the standard Wiener process. In principle, simulating either the Probability Flow ODE or the SDE above results in samples from the same distribution.

Training. DM training reduces to learning a model $s_{\theta}(\mathbf{x}; \sigma)$ for the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma)$. The model can, for example, be parameterized as $\nabla_{\mathbf{x}} \log p(\mathbf{x}; \sigma) \approx s_{\theta}(\mathbf{x}; \sigma) = (D_{\theta}(\mathbf{x}; \sigma) - \mathbf{x})/\sigma^2$ [21], where D_{θ} is a learnable *denoiser* that, given a noisy data point $\mathbf{x}_0 + \mathbf{n}$, $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x}_0)$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, and conditioned on the noise level σ , tries to predict the clean $\mathbf{x}_0$. The denoiser D_{θ} (or equivalently the score model) can be trained via *denoising score matching* (DSM)

$$\mathbb{E}_{(\mathbf{x}_0, \mathbf{c}) \sim p_{\text{data}}(\mathbf{x}_0, \mathbf{c}), (\sigma, \mathbf{n}) \sim p(\sigma, \mathbf{n})} [\lambda_{\sigma} \|D_{\theta}(\mathbf{x}_0 + \mathbf{n}; \sigma, \mathbf{c}) - \mathbf{x}_0\|_2^2], \quad (3)$$

where $p(\sigma, \mathbf{n}) = p(\sigma) \mathcal{N}(\mathbf{n}; \mathbf{0}, \sigma^2)$, $p(\sigma)$ is a distribution over noise levels σ , $\lambda_{\sigma}: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a weighting function, and $\mathbf{c}$ is an arbitrary conditioning signal, e.g., a class label, a text prompt, or a combination thereof. In this work, we choose $p(\sigma)$ to be a discrete distributions over 1000 noise levels and set $\lambda_{\sigma} = \sigma^{-2}$ similar to prior works [14, 38, 45].

Classifier-free guidance. Classifier-free guidance [13] is a technique to guide the iterative sampling process of a DM towards a conditioning signal $\mathbf{c}$ by mixing the predictions of a conditional and an unconditional model

$$D^w(\mathbf{x}; \sigma, \mathbf{c}) = (1 + w)D(\mathbf{x}; \sigma, \mathbf{c}) - wD(\mathbf{x}; \sigma), \quad (4)$$

where $w \geq 0$ is the *guidance strength*. In practice, the unconditional model can be trained jointly alongside the conditional model in a single network by randomly replacing the conditional signal $\mathbf{c}$ with a null embedding in Eq. (3), e.g., 10% of the time [13]. Classifier-free guidance is widely used to improve the sampling quality, trading for diversity, of text-to-image DMs [30, 38].

D Comparison to the State of the Art

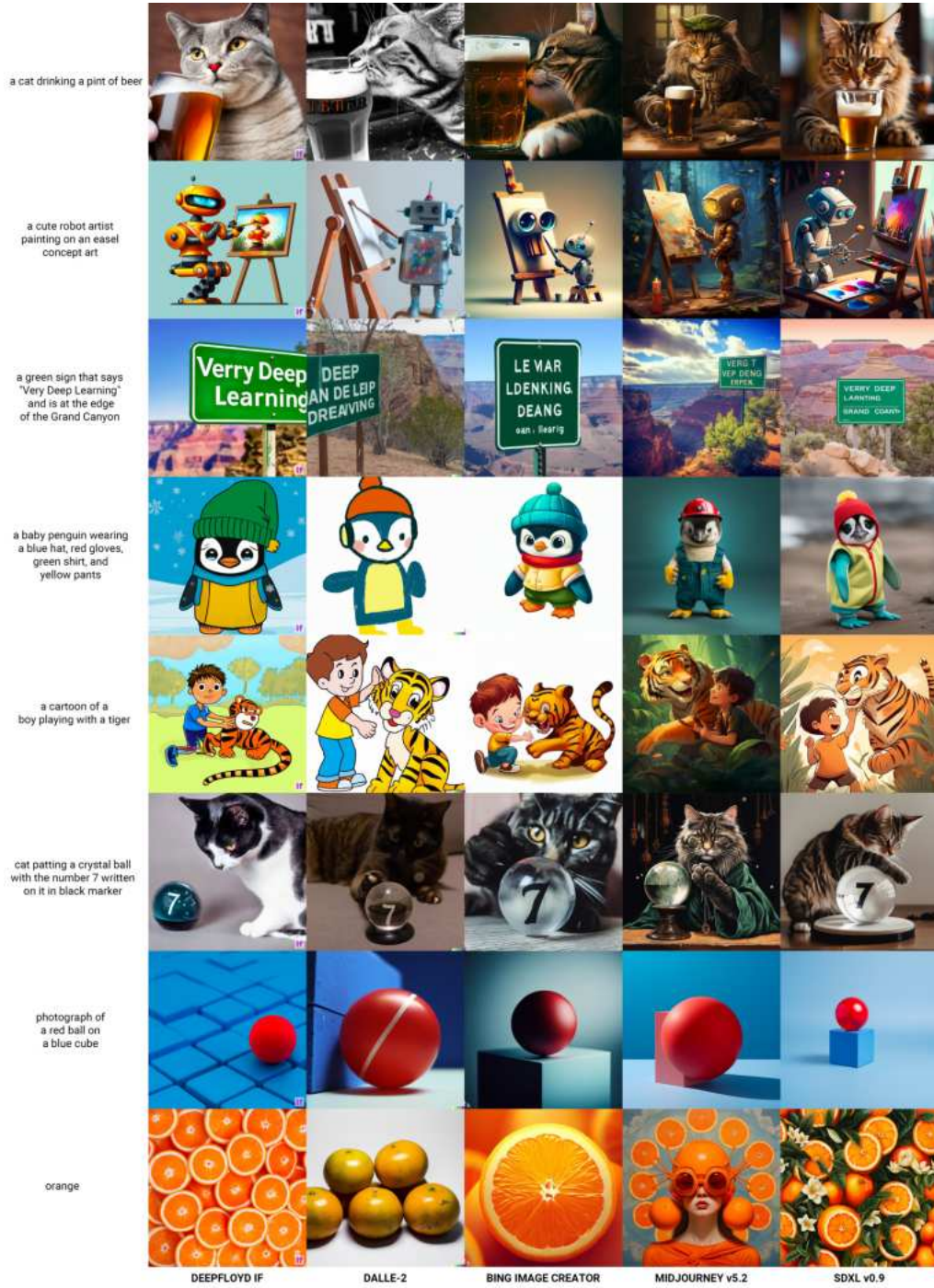


Figure 8: Qualitative comparison of *SDXL* with DeepFloyd IF, DALL-E-2, Bing Image Creator, and MidJourney v5.2. To mitigate any bias arising from cherry-picking, Parti (P2) prompts were randomly selected. Seed 3 was uniformly applied across all models in which such a parameter could be designated. For models without a seed-setting feature, the first generated image is included.

E Comparison to Midjourney v5.1

E.1 Overall Votes

To assess the generation quality of *SDXL* we perform a user study against the state of the art text-to-image generation platform Midjourney¹. As the source for image captions we use the PartiPrompts (P2) benchmark [53], that was introduced to compare large text-to-image model on various challenging prompts.

For our study, we choose five random prompts from each category, and generate four 1024×1024 images by both Midjourney (v5.1, with a set seed of 2) and *SDXL* for each prompt. These images were then presented to the AWS GroundTruth taskforce, who voted based on adherence to the prompt. The results of these votes are illustrated in Fig. 9. Overall, there is a slight preference for *SDXL* over Midjourney in terms of prompt adherence.

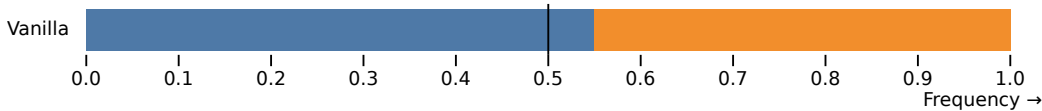


Figure 9: Results from 17,153 user preference comparisons between *SDXL* v0.9 and Midjourney v5.1, which was the latest version available at the time. The comparisons span all “categories” and “challenges” in the PartiPrompts (P2) benchmark. Notably, *SDXL* was favored 54.9% of the time over Midjourney V5.1. Preliminary testing indicates that the recently-released Midjourney V5.2 has lower prompt comprehension than its predecessor, but the laborious process of generating multiple prompts hampers the speed of conducting broader tests.

E.2 Category & challenge comparisons on PartiPrompts (P2)

Each prompt from the P2 benchmark is organized into a category and a challenge, each focus on different difficult aspects of the generation process. We show the comparisons for each category (Fig. 10) and challenge (Fig. 11) of P2 below. In four out of six categories *SDXL* outperforms Midjourney, and in seven out of ten challenges there is no significant difference between both models or *SDXL* outperforms Midjourney.

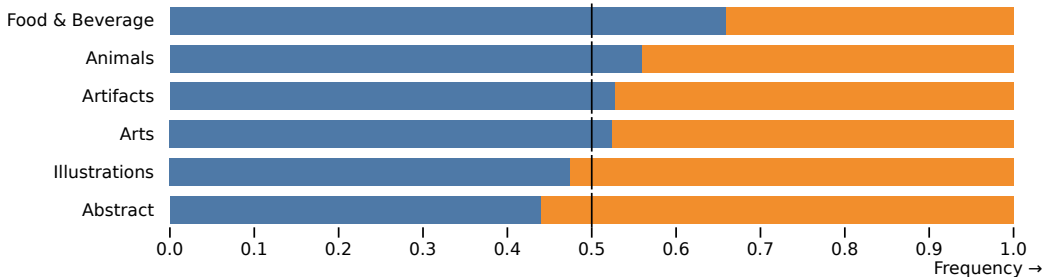


Figure 10: User preference comparison of *SDXL* (without refinement model) and Midjourney V5.1 across particular text categories. *SDXL* outperforms Midjourney V5.1 in all but two categories.

¹We compare against v5.1 since that was the best version available at that time.

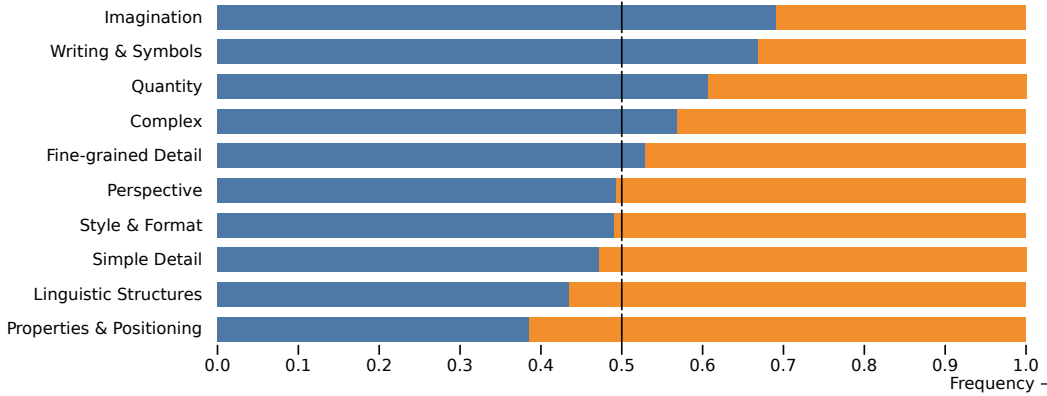


Figure 11: Preference comparisons of *SDXL* (with refinement model) to *Midjourney V5.1* on complex prompts. *SDXL* either outperforms or is statistically equal to *Midjourney V5.1* in 7 out of 10 categories.

F On FID Assessment of Generative Text-Image Foundation Models

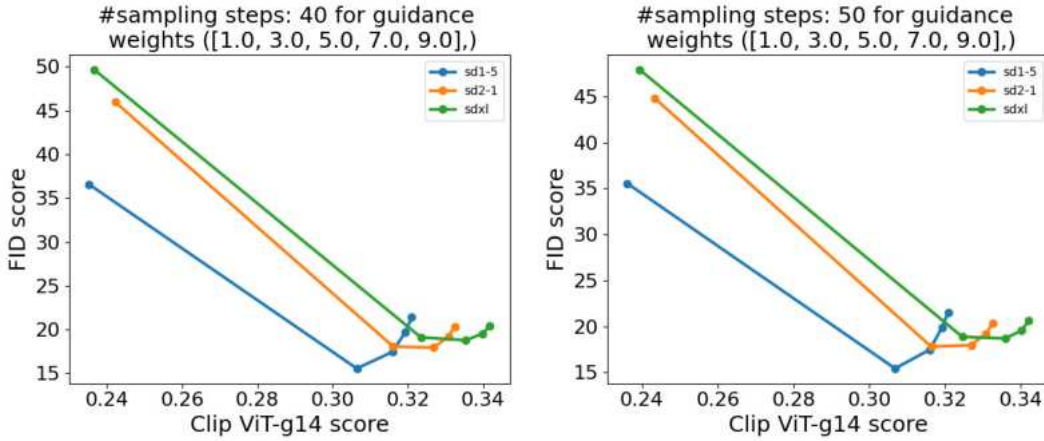


Figure 12: Plotting FID vs CLIP score for different cfg scales. *SDXL* shows only slightly improved text-alignment, as measured by CLIP-score, compared to previous versions that do not align with the judgement of human evaluators. Even further and similar as in [23], FID are worse than for both *SD-1.5* and *SD-2.1*, while human evaluators clearly prefer the generations of *SD-XL* over those of these previous models.

Throughout the last years it has been common practice for generative text-to-image models to assess FID- [12] and CLIP-scores [34, 36] in a zero-shot setting on complex, small-scale text-image datasets of natural images such as COCO [26]. However, with the advent of foundational text-to-image models [40, 37, 38, 1], which are not only targeting visual compositionality, but also at other difficult tasks such as deep text understanding, fine-grained distinction between unique artistic styles and especially a pronounced sense of visual aesthetics, this particular form of model evaluation has become more and more questionable. Kirstain et al. [23] demonstrates that COCO zero-shot FID is *negatively correlated* with visual aesthetics, and such measuring the generative performance of such models should be rather done by human evaluators. We investigate this for *SDXL* and visualize FID-vs-CLIP curves in Fig. 12 for 10k text-image pairs from COCO [26]. Despite its drastically improved performance as measured quantitatively by asking human assessors (see Fig. 1) as well as qualitatively (see Fig. 4 and Fig. 14), *SDXL* does *not* achieve better FID scores than the previous *SD* versions. Contrarily, FID for *SDXL* is the worst of all three compared models while only showing slightly improved CLIP-scores (measured with OpenClip ViT g-14). Thus, our results back the findings of Kirstain et al. [23] and further emphasize the need for additional quantitative performance scores, specifically for text-to-image foundation models. All scores have been evaluated based on 10k generated examples.

G Additional Comparison between Single- and Two-Stage *SDXL* pipeline

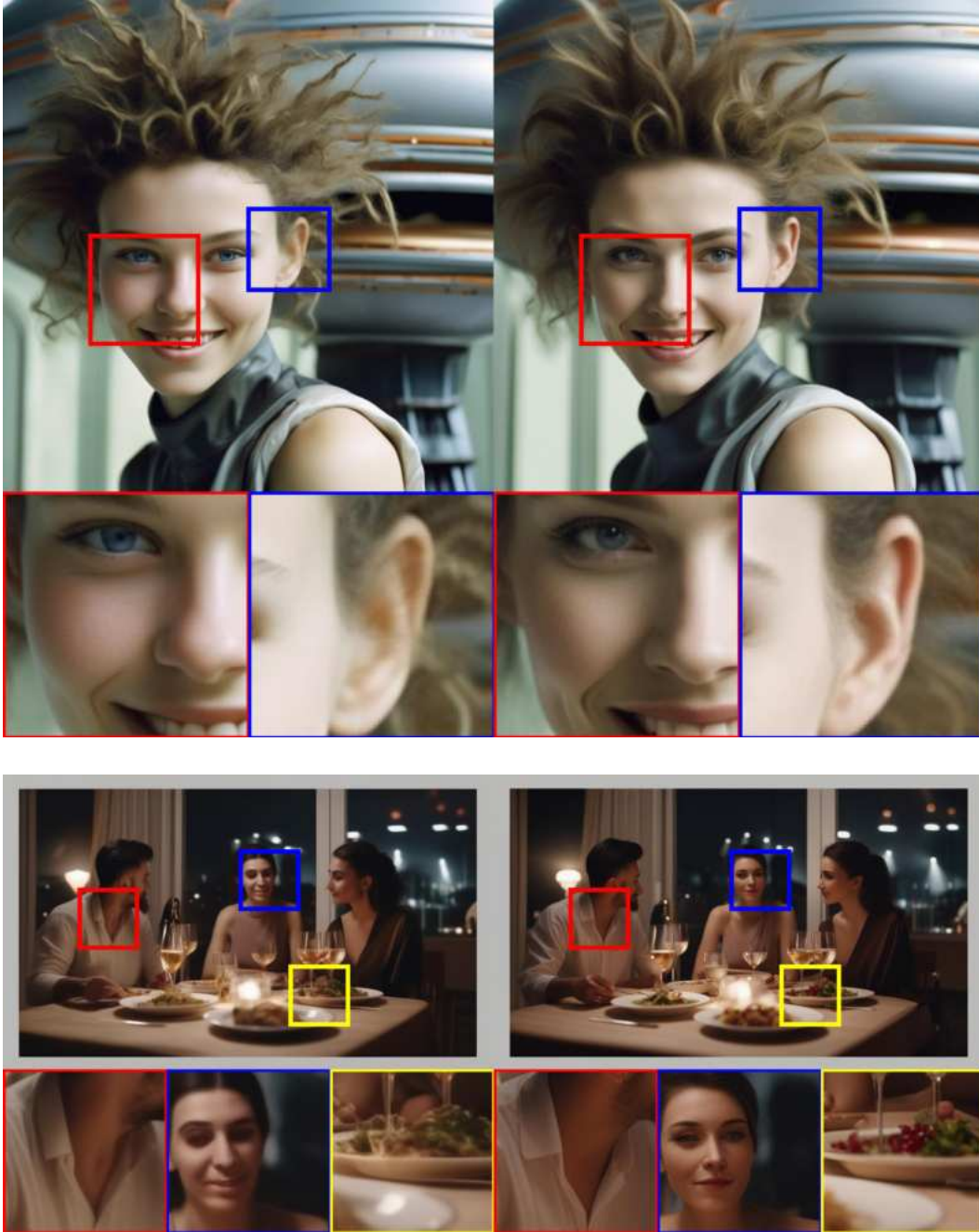


Figure 13: *SDXL* samples (with zoom-ins) without (left) and with (right) the refinement model discussed. Prompt: (top) “close up headshot, futuristic young woman, wild hair sly smile in front of gigantic UFO, dslr, sharp focus, dynamic composition” (bottom) “Three people having dinner at a table at new years eve, cinematic shot, 8k”. Zoom-in for details.

H Comparison between *SD 1.5* vs. *SD 2.1* vs. *SDXL*

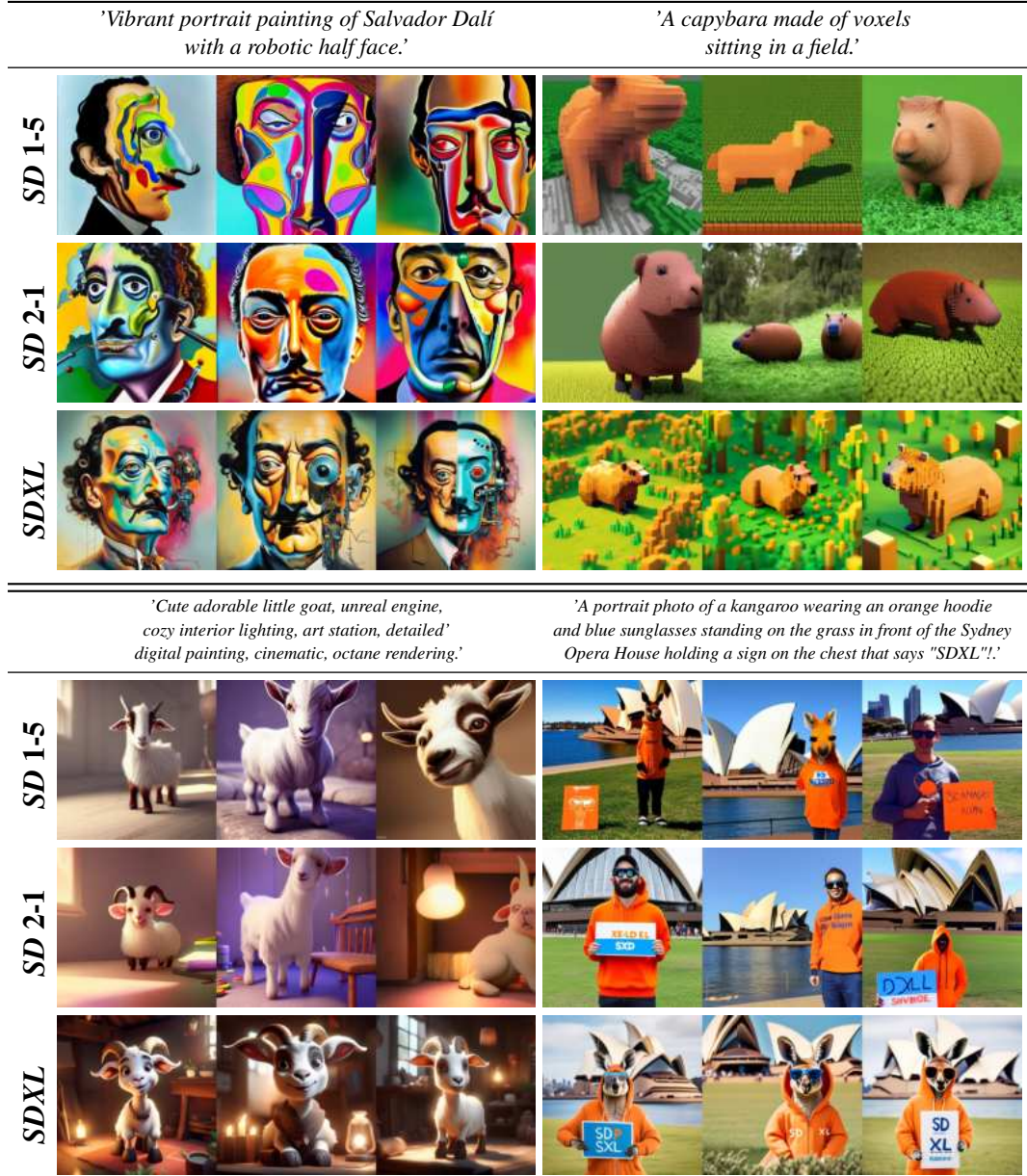


Figure 14: Additional results for the comparison of the output of *SDXL* with previous versions of *Stable Diffusion*. For each prompt, we show 3 random samples of the respective model for 50 steps of the DDIM sampler [46] and cfg-scale 8.0 [13]

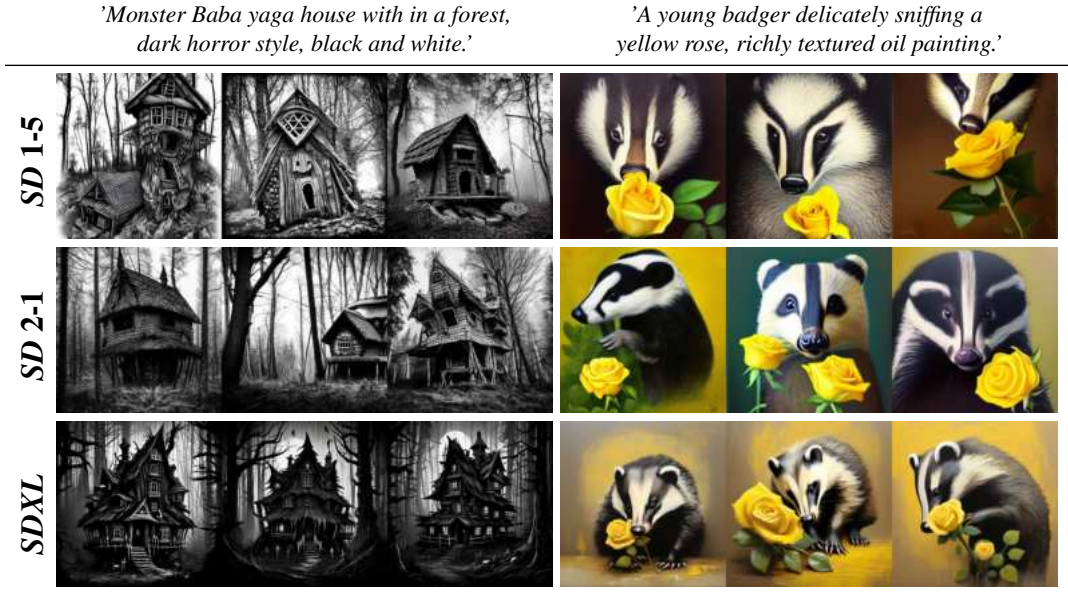


Figure 15: Additional results for the comparison of the output of *SDXL* with previous versions of *Stable Diffusion*. For each prompt, we show 3 random samples of the respective model for 50 steps of the DDIM sampler [46] and cfg-scale 8.0 [13].

I Multi-Aspect Training Hyperparameters

We use the following image resolutions for mixed-aspect ratio finetuning as described in Sec. 2.3.

Height	Width	Aspect Ratio
512	2048	0.25
512	1984	0.26
512	1920	0.27
512	1856	0.28
576	1792	0.32
576	1728	0.33
576	1664	0.35
640	1600	0.4
640	1536	0.42
704	1472	0.48
704	1408	0.5
704	1344	0.52
768	1344	0.57
768	1280	0.6
832	1216	0.68
832	1152	0.72
896	1152	0.78
896	1088	0.82
960	1088	0.88
960	1024	0.94

Height	Width	Aspect Ratio
1024	1024	1.0
1024	960	1.07
1088	960	1.13
1088	896	1.21
1152	896	1.29
1152	832	1.38
1216	832	1.46
1280	768	1.67
1344	768	1.75
1408	704	2.0
1472	704	2.09
1536	640	2.4
1600	640	2.5
1664	576	2.89
1728	576	3.0
1792	576	3.11
1856	512	3.62
1920	512	3.75
1984	512	3.88
2048	512	4.0

J Pseudo-code for Conditioning Concatenation along the Channel Axis

```
1 from einops import rearrange
2 import torch
3
4 batch_size=16
5 # channel dimension of pooled output of text encoder(s)
6 pooled_dim = 512
7
8 def fourier_embedding(inputs, outdim=256, max_period=10000):
9     """
10     Classical sinusoidal timestep embedding
11     as commonly used in diffusion models
12     :param inputs: batch of integer scalars shape [b,]
13     :param outdim: embedding dimension
14     :param max_period: max freq added
15     :return: batch of embeddings of shape [b, outdim]
16     """
17     ...
18
19 def cat_along_channel_dim(
20     x:torch.Tensor,) -> torch.Tensor:
21     if x.ndim == 1:
22         x = x[... ,None]
23     assert x.ndim == 2
24     b, d_in = x.shape
25     x = rearrange(x, "b din -> (b din)")
26     # fourier fn adds additional dimension
27     emb = fourier_embedding(x)
28     d_f = emb.shape[-1]
29     emb = rearrange(emb, "(b din) df -> b (din df)",
30                     b=b, din=d_in, df=d_f)
31     return emb
32
33 def concat_embeddings(
34     # batch of size and crop conditioning cf. Sec. 3.2
35     c_size:torch.Tensor,
36     c_crop:torch.Tensor,
37     # batch of aspect ratio conditioning cf. Sec. 3.3
38     c_ar:torch.Tensor,
39     # final output of text encoders after pooling cf. Sec. 3.1
40     c_pooled_txt:torch.Tensor, ) -> torch.Tensor:
41     # fourier feature for size conditioning
42     c_size_emb = cat_along_channel_dim(c_size)
43     # fourier feature for size conditioning
44     c_crop_emb = cat_along_channel_dim(c_crop)
45     # fourier feature for size conditioning
46     c_ar_emb = cat_along_channel_dim(c_ar)
47     # the concatenated output is mapped to the same
48     # channel dimension than the noise level conditioning
49     # and added to that conditioning before being fed to the unet
50     return torch.cat([c_pooled_txt,
51                       c_size_emb,
52                       c_crop_emb,
53                       c_ar_emb], dim=1)
54
55 # simulating c_size and c_crop as in Sec. 3.2
56 c_size=torch.zeros((batch_size, 2)).long()
57 c_crop=torch.zeros((batch_size, 2)).long()
58 # simulating c_ar and pooled text encoder output as in Sec. 3.3
59 c_ar=torch.zeros((batch_size, 2)).long()
60 c_pooled=torch.zeros((batch_size, pooled_dim)).long()
61
62 # get concatenated embedding
63 c_concat = concat_embeddings(c_size, c_crop, c_ar, c_pooled)
```

Figure 16: Python code for concatenating the additional conditionings introduced in Secs. 2.1 to 2.3 along the channel dimension.

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Improving Image Generation with Better Captions

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Abstract

We show that prompt following abilities of text-to-image models can be substantially improved by training on highly descriptive generated image captions. Existing text-to-image models struggle to follow detailed image descriptions and often ignore words or confuse the meaning of prompts. We hypothesize that this issue stems from noisy and inaccurate image captions in the training dataset. We address this by training a bespoke image captioner and use it to recaption the training dataset. We then train several text-to-image models and find that training on these synthetic captions reliably improves prompt following ability. Finally, we use these findings to build DALL-E 3: a new text-to-image generation system, and benchmark its performance on an evaluation designed to measure prompt following, coherence, and aesthetics, finding that it compares favorably to competitors. We publish samples and code for these evaluations so that future research can continue optimizing this important aspect of text-to-image systems.

1 Introduction

Recent advances in generative modeling have allowed text-to-image generative models to achieve drastic performance improvements. In particular, tackling the problem with sampling-based approaches such as autoregressive generative modeling[27, 2, 11, 20, 30] or using diffusion processes[25, 6, 11, 12, 19, 22] have allowed us to decompose the problem of image generation into small, discrete steps which are more tractable for neural networks to learn.

In parallel, researchers have found ways to build image generators out of stacks of self-attention layers[15, 3, 4]. Decoupling image generation from the implicit spatial biases of convolutions has allowed text-to-image models to reliably improve via the well-studied scaling properties of transformers.

Combined with a sufficiently large dataset, these approaches have enabled the training of large text-to-image models which are capable of generating imagery which is rapidly approaching the quality of photographs and artwork that humans can produce.

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An outstanding challenge in the field is the controllability of image generation systems, which often overlook the words, word ordering, or meaning in a given caption. We refer to these challenges with the term “prompt following”.

This problem has been pointed out in several works: [Rassin et al. \(2022\)](#) pointed out that DALL-E 2 does not enforce a constraint where each word has a single meaning. [Saharia et al. \(2022\)](#) propose to improve it by conditioning on pre-trained language models, and introduces an evaluation called Drawbench which surfaces common prompt following issues. [Yu et al. \(2022b\)](#) in parallel introduce their own benchmark, Parti Prompts, and show that scaling autoregressive image generators is an alternative way to improve prompt following.

In this work, we propose a new approach to addressing prompt following: caption improvement. We hypothesize that a fundamental issue with existing text-to-image models is the poor quality of the text and image pairing of the datasets they were trained on, an issue that has been pointed out in other works such as [Jia et al. \(2021\)](#). We propose to address this by generating improved captions for the images in our dataset. We do this by first learning a robust image captioner which produces detailed, accurate descriptions of images. We then apply this captioner to our dataset to produce more detailed captions. We finally train text-to-image models on our improved dataset.

Training on synthetic data is not a new concept. For example, [Yu et al. \(2022b\)](#) mention that they apply this technique in training their scaled autoregressive image generators. Our contribution is in building a novel, descriptive image captioning system and measuring the impact of using synthetic captions when training generative models. We also establish a reproducible baseline performance profile for a suite of evals that measure prompt following.

This paper focuses on evaluating the improved prompt following of DALL-E 3 as a result of training on highly descriptive generated captions. It does not cover training or implementation details of the DALL-E 3 model. We provide a high level overview of our strategy for training an image captioner in Section [2](#), evaluation of text-to-image models trained on original vs. generated captions in Section [3](#), evaluation of DALL-E 3 in Section [4](#), and discussion of limitations and risk in Section [5](#).



In a fantastical setting, a highly detailed furry humanoid skunk with piercing eyes confidently poses in a medium shot, wearing an animal hide jacket. The artist has masterfully rendered the character in digital art, capturing the intricate details of fur and clothing texture.



A illustration from a graphic novel. A bustling city street under the shine of a full moon. The sidewalks bustling with pedestrians enjoying the nightlife. At the corner stall, a young woman with fiery red hair, dressed in a signature velvet cloak, is haggling with the grumpy old vendor. the grumpy vendor, a tall, sophisticated man is wearing a sharp suit, sports a noteworthy moustache is animatedly conversing on his steampunk telephone.

Figure 1 – Selected landscape samples from DALL-E 3.



Ancient pages filled with sketches and writings of fantasy beasts, monsters, and plants sprawl across an old, weathered journal. The faded dark green ink tells tales of magical adventures, while the high-resolution drawings detail each creature's intricate characteristics. Sunlight peeks through a nearby window, illuminating the pages and revealing their timeworn charm.



A vibrant 1960s-style poster depicting interplanetary migration, with a retro rocket ship blasting off from earth towards a distant, colorful planet. Bold typography announces "Join the galactic adventure!" with smaller text underneath reading "Explore new worlds, build a brighter future." The background features a swirling galaxy of stars and constellations.



A mischievous ferret with a playful grin squeezes itself into a large glass jar, surrounded by colorful candy. The jar sits on a wooden table in a cozy kitchen, and warm sunlight filters through a nearby window.



A fierce garden gnome warrior, clad in armor crafted from leaves and bark, brandishes a tiny sword and shield. He stands valiantly on a rock amidst a blooming garden, surrounded by colorful flowers and towering plants. A determined expression is painted on his face, ready to defend his garden kingdom.



An icy landscape under a starlit sky, where a magnificent frozen waterfall flows over a cliff. In the center of the scene, a fire burns bright, its flames seemingly frozen in place, casting a shimmering glow on the surrounding ice and snow.



A swirling, multicolored portal emerges from the depths of an ocean of coffee, with waves of the rich liquid gently rippling outward. The portal engulfs a coffee cup, which serves as a gateway to a fantastical dimension. The surrounding digital art landscape reflects the colors of the portal, creating an alluring scene of endless possibilities.

Figure 2 – Selected portrait and square samples from DALL-E 3.

2 Dataset Recaptioning

Our text-to-image models are trained on a dataset composed of a large quantity of pairings (t, i) where i is an image and t is text that describes that image⁴. In large-scale datasets, t is generally derived from human authors who focus on simple descriptions the subject of the image and omit background details or common sense relationships portrayed in image. Important details that are commonly omitted from t might include:

1. The presence of objects like sinks in a kitchen or stop signs along a sidewalk and descriptions of those objects.
2. The position of objects in a scene and the number of those objects.
3. Common sense details like the colors and sizes of objects in a scene.
4. The text that is displayed in an image.

Worse, captions found on the Internet oftentimes simply incorrect; describing tangentially related details of an image. For example, it is common to find advertisements or memes inside of the alt-text commonly used to produce captions for images.

We theorize that all of these shortcomings can be addressed using synthetically generated captions. In subsequent sections, we will discuss the procedure we developed to test out this theory.

2.1 Building an image captioner

An image captioner is very similar to a traditional language model that predicts text. We thus start by providing a brief description of language models. First, a tokenizer is used to break strings of text into discrete tokens. Once decomposed in this way, the text portion of our corpus can be represented as a sequence, $t = [t_1, t_2, \dots, t_n]$. We can then build a language model over the text by maximizing the following likelihood function:

$$L(t) = \sum_j \log P(t_j | t_{j-k}, \dots, t_{j-1}; \Theta) \quad (1)$$

Where Θ is the parameters of the captioner that are to be optimized. To turn this language model into a captioner, you need only to condition on the image. The challenge here is that images are composed of many thousands of pixel values. Conditioning on all of this information is exceptionally inefficient with our current neural networks, so we need a compressed representation space. Conveniently, CLIP[17] provides just this.

Thus, given a pre-trained CLIP image embedding function $F(i)$, we augment our language model objective as follows:

$$L(t, i) = \sum_j \log P(t_j | t_{j-k}, \dots, t_{j-1}; z_j; F(i); \Theta) \quad (2)$$

We follow the methods of Yu et al. (2022a) and jointly pre-train our captioner with a CLIP and a language modeling objective using the above formulation on our dataset of (t, i) text and image pairs. The resulting model is indeed a good captioner, but exhibits the same problems we describe in section 2, such as a reluctance to describe details.

⁴The paired text is generally referred to as the "caption" in this document

Image			
			
Alt Text	now at victorian plumbing.co.uk	is he finished...just about!	23 (19 of 30) 1200
SSC	a white modern bathtub sits on a wooden floor.	a quilt with an iron on it.	a jar of rhubarb liqueur sitting on a pebble background.
DSC	this luxurious bathroom features a modern freestanding bathtub in a crisp white finish. the tub sits against a wooden accent wall with glass-like panels, creating a serene and relaxing ambiance. three pendant light fixtures hang above the tub, adding a touch of sophistication. a large window with a wooden panel provides natural light, while a potted plant adds a touch of greenery. the freestanding bathtub stands out as a statement piece in this contemporary bathroom.	a quilt is laid out on a ironing board with an iron resting on top. the quilt has a patchwork design with pastel-colored strips of fabric and floral patterns. the iron is turned on and the tip is resting on top of one of the strips. the quilt appears to be in the process of being pressed, as the steam from the iron is visible on the surface. the quilt has a vintage feel and the colors are yellow, blue, and white, giving it an antique look.	rhubarb pieces in a glass jar, waiting to be pickled. the colors of the rhubarb range from bright red to pale green, creating a beautiful contrast. the jar is sitting on a gravel background, giving a rustic feel to the image.

Figure 3 – Examples of alt-text accompanying selected images scraped from the internet, short synthetic captions (SSC), and descriptive synthetic captions (DSC).

2.1.1 Fine-tuning the captioner

To improve the captions in our image generation dataset, we want to bias our captioner to produce image descriptions which are useful for learning a text-to-image model. In our first attempt, we build a small dataset of captions that describe only the main subject of the image. We then continue to train our captioner on this dataset. The updates to θ induced by this process result in a model which is biased towards describing the main subject of the image. We refer to captions generated by this fine-tune as "short synthetic captions".

We repeat this process a second time, creating a dataset of long, highly-descriptive captions describing the contents of each image in our fine-tuning dataset. These captions describe not only the main subject of the image, but also its surroundings, background, text found in the image, styles, coloration, etc. We again fine-tune our base captioner on this dataset. We refer to captions generated by this captioner as "descriptive synthetic captions".

Figure 3 shows examples of ground-truth, short synthetic, and descriptive synthetic captions.

Once built, we apply our image captioner fine-tunes to every image in our text-to-image dataset, resulting in a set of synthetic captions which we use for subsequent experiments.

3 Evaluating the re-captioned datasets

With our re-captioned datasets in-hand, we set about evaluating the impact of training models on synthetic text. We sought to answer two questions in particular:

1. The performance impact of using each type of synthetic caption.
2. The optimal blending ratio of synthetic to ground-truth captions.

3.1 Blending synthetic and ground-truth captions

Likelihood models like our text-to-image diffusion models have a notorious tendency to overfit to distributional regularities in the dataset. For example, a text-to-image model that is trained on text that always starts with a space character will not work properly if you try to perform inference with prompts that do not also start with that space.

When it comes to training on synthetic captions, we need to consider this issue. Our captioner model could have many modal behaviors that are difficult to detect, but which will become biases of our text-to-image model if it is trained on those captions. Examples of where this might occur is in letter casing, where punctuation appears in the caption (e.g. does it always end with a period?), how long the captions are, or stylistic tendencies such as starting all captions with the words "a" or "an".

The best way to overcome this issue is to regularize our inputs to a distribution of text that is closer to the style and formatting that humans might use. When using ground truth captions, you get this "for free" because these captions are, in fact, drawn from a distribution of human-written text. To introduce some of this regularization into our model training when using synthetic captions, we opted to blend synthetic captions with ground truth captions.

Blending happens at data sampling time, where we randomly select either the ground truth or synthetic caption with a fixed percent chance. We analyze the performance impact of different blending ratios in the next section.

3.2 Evaluation methodology

For evaluation, we trained identical T5-conditioned image diffusion models on the same dataset of images. Details about the models trained are described in [A](#). All models were trained to 500,000 training steps at a batch size of 2048, corresponding to 1B training images total.

Once training was completed, we used captions from an evaluation dataset to generate 50,000 images from each model. We then evaluated these generated images using the CLIP-S evaluation metric outlined in [Hessel et al. \(2022\)](#). We choose CLIP score as a metric as it has a strong correlation with text-image similarity, which is what we are after. As a quick review, this metric is computed as follows:

First, we use the public CLIP ViT-B/32 [\[17\]](#) image encoder to produce an image embedding z_i and then we use the text encoder to create a text embedding for the image caption z_t . We finally compute the CLIP score as the cosine similarity C :

$$C(z_i, z_t) = \frac{z_i \cdot z_t}{\|z_i\| \|z_t\|} \quad (3)$$

This metric is then averaged across the distances computed for all 50,000 text/image pairs and rescaled by a factor of 100. We perform this evaluation across multiple model checkpoints during training, at all times performing evaluation with exponentially-weighted averages of the learned weights of the models.

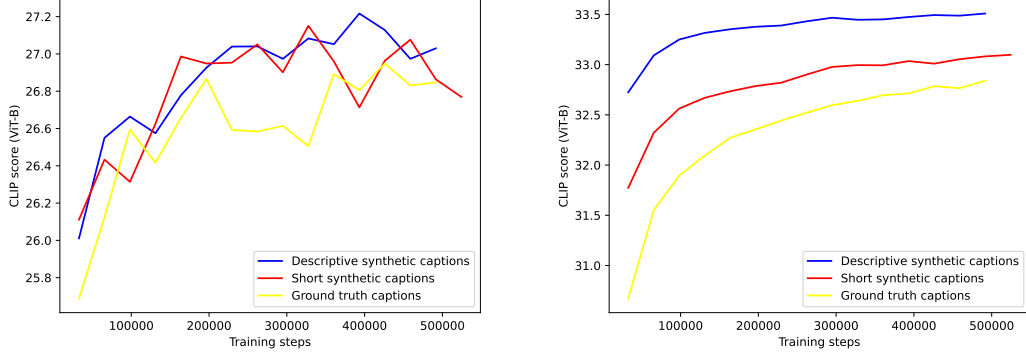


Figure 4 – CLIP scores for text-to-image models trained on different caption types. Left is evaluation results with ground truth captions on our evaluation dataset. Right uses the descriptive synthetic captions from the same dataset.

When computing CLIP scores, the choice of which caption to use when performing the above calculation is important. For our tests, we either use the ground truth caption or we use the descriptive synthetic caption. Which is used is noted in each evaluation.

3.3 Caption type results

We start by analyzing the performance difference between models trained on different types of captions. For this evaluation, we train three models:

1. A text-to-image model trained only on ground truth captions.
2. A text-to-image model trained on 95% short synthetic captions.
3. A text-to-image model trained on 95% descriptive synthetic captions.

We perform this evaluation twice: once with the z_t computed from ground-truth captions and once with z_t computed from descriptive synthetic captions. We do not do it for short synthetic captions as they are very similar to ground-truth in this evaluation.

The results, shown in [4](#), show that both models trained on synthetic captions achieve slightly better CLIP score performance than the baseline model when evaluated on ground-truth captions, and markedly better performance when evaluated on descriptive synthetic captions. This indicates that there is no downside to using synthetic captions when training text-to-image models.

It is interesting to note that the evaluation curves on synthetic captions have much lower variance. This bolsters our theory that re-captioning can be seen as an averaging operation. Image generation models evaluated on synthetic captions also achieve much higher net CLIP scores across all of the models trained, which supports the notion that synthetic captions have better binding to their corresponding images.

3.4 Caption blending ratios

To evaluate caption blending ratios, we trained four image generation models using our descriptive synthetic captions at different blending ratios. We chose blends of 65%, 80%, 90% and 95% synthetic captions. Midway through the experiment, evaluations showed that the 65% blend was far behind the other blends on all evals and we dropped it.

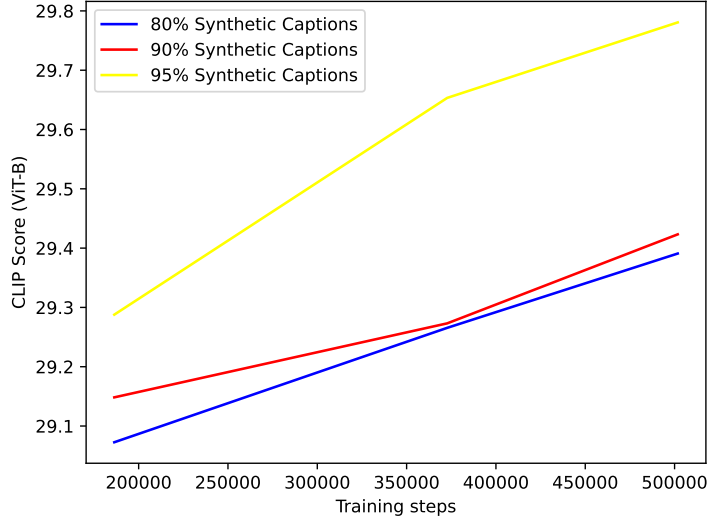


Figure 5 – CLIP scores for text-to-image models trained on various blending ratios of descriptive synthetic captions and ground-truth captions. Evaluation performed using ground truth captions.

The results in Figure 5 show that higher blends of synthetic captions always improved the model’s CLIP score.

3.5 Practical usage of highly descriptive captions

The above experiments suggest that we can maximize the performance of our models by training on a very high percentage of synthetic captions. However, doing so causes the models to naturally adapt to the distribution of long, highly-descriptive captions emitted by our captioner.

Generative models are known to produce poor results when sampled out of their training distribution. Thus, to extract the maximum potential out of our models, we will need to exclusively sample from them with highly descriptive captions. Fortunately, this is a solvable problem with recent breakthroughs in large language models. Models like GPT-4 [14] have become exceptionally good at tasks that require imagination, such as telling stories and writing poems. It stands to reason that they might also be good at coming up with plausible details in an image description.

Indeed, given a prompt such as the one found in Appendix C we found that GPT-4 will readily "upsample" any caption into a highly descriptive one. To demonstrate how this approach might be useful, we perform this procedure on the captions from the drawbench dataset [24] and visualize the results in Table 7.

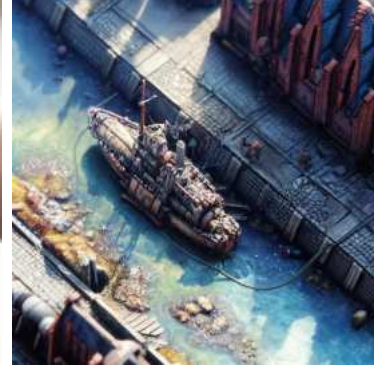
As can be seen in Figure 7, utilizing a LLM to "upsample" captions can be used to not only add missing details, but also to disambiguate complex relationships which would be hard for a (relatively) small image generation model to learn. The end result is that the model will often correctly render images that it would have otherwise gotten wrong.



A bird scaring a scarecrow.



Paying for a quarter-sized pizza with a pizza-sized quarter.



A small vessel, propelled on water by oars, sails, or an engine.



A large, vibrant bird with an impressive wingspan swoops down from the sky, letting out a piercing call as it approaches a scarecrow in a sunlit field. The scarecrow, dressed in tattered clothing and a straw hat, appears to tremble, almost as if it's coming to life in fear of the approaching bird.



A person is standing at a pizza counter, holding a gigantic quarter the size of a pizza. The cashier, wide-eyed with astonishment, hands over a tiny, quarter-sized pizza in return. The background features various pizza toppings and other customers, all of them equally amazed by the unusual transaction.



A small vessel, propelled on water by oars, sails, or an engine, floats gracefully on a serene lake. The sun casts a warm glow on the water, reflecting the vibrant colors of the sky as birds fly overhead.

Figure 6 – Effect of using "upsampled" drawbench captions to create samples with DALL-E 3. Original drawbench captions on top, upsampled captions on bottom. Images are best of 4 for each caption.

4 DALL-E 3

To test our synthetic captions at scale, we train DALL-E 3, a new state of the art text to image generator. To train this model, we use a mixture of 95% synthetic captions and 5% ground truth captions. The model itself is a scaled-up version of the model we used in the above ablations, with several other improvements.⁵

⁵DALL-E 3 has many improvements over DALL-E 2, many of which are not covered in this document and could not be ablated for time and compute reasons. The evaluation metrics discussed in this document should not be construed as a performance comparison resulting from simply training on synthetic captions.

Metric	DALL-E 3	DALL-E 2 ¹	Stable Diffusion XL ²
MSCOCO Captions CLIP Score $\uparrow$	32.0	31.4	30.5
Drawbench short (GPT-V) ³ $\uparrow$	70.4%	49.0%	46.9%
Drawbench long (GPT-V) $\uparrow$	81.0%	52.4%	51.1%
T2I-C B-VQA Colors $\uparrow$	81.1%	59.2%	61.9%
T2I-C B-VQA Shape $\uparrow$	67.5%	54.7%	61.9%
T2I-C B-VQA Texture $\uparrow$	80.7%	63.7%	55.2%

Table 1 – Comparison of text-to-image models on various evaluations related to prompt following

¹ Images generated with DALL-E 2 production version live on September 20th, 2023.

² Stable Diffusion XL v1.0 with the refiner module active used.

³ Scores here are the percent of images that have the "correct" caption, as judged by GPT-V.

4.1 Automated Evaluations

We compare DALL-E 3 with DALL-E 2 and Stable Diffusion XL 1.0 with the refiner module [16]. We wish to evaluate DALL-E 3’s performance on tasks which are correlated with prompt following. We describe the individual tasks below.

4.1.1 CLIP score

We first compute the CLIP score using the public ViT-B/32 model as described in section 3.2. For this comparison, we use a set of 4,096 captions drawn from the MSCOCO 2014 evaluation dataset [10] to generate our images [6]. In this evaluation, we perform inference on the model with the short, ground-truth captions.

Our model outperforms both DALL-E 2 and Stable Diffusion XL in this evaluation.

4.1.2 Drawbench

We next evaluate on captions from drawbench [24]. For this test, we use an instruction-tuned, vision-aware LLM based on GPT-4 called GPT-V to evaluate the performance of our model versus others. For each prompt in drawbench, we generate four images with each model. We then prompt our vision-aware LLM with the image and the text using the prompt found in Appendix D. This results in a conclusion ("correct"/"incorrect") and an explanation for that conclusion.

Since we previously observed that our model performs better when given extrapolated captions from a language model, we use GPT-4 to "upsample" the drawbench captions using the process described in Section 3.5. We perform the above automated evaluation a second time using these "upsampled" captions when sampling images from all models. We use the original, ground-truth drawbench prompt when asking the vision-aware LLM to judge the outputs.

In all drawbench evaluations, our model beats DALL-E 2 and Stable Diffusion XL. The gap widens significantly when we use the "upsampled" captions.

⁶As with past versions of DALL-E, DALL-E 3 was not specifically trained on the MSCOCO dataset, nor did we perform any optimizations on our model to improve performance on this evaluation. We also did not perform a de-duplication across MSCOCO within our training dataset, it is possible that there is data leakage.

4.1.3 T2I-CompBench

We finally evaluate on a subset the T2I-CompBench evaluation suite developed by [Huang et al. \(2023\)](#). This benchmark measures a model’s performance on compositional prompts. We report scores for color binding, shape binding and texture binding. We use the Disentangled BLIP-VQA model to evaluate these results.

DALL-E 3 is state of the art on all benchmarks evaluated.

4.2 Human Evaluations

We submit samples from DALL-E 3 and comparable models for human evaluation. For this evaluation, we present human raters with two side-by-side images that were generated from the same captions. We then ask the rater one of three questions:

1. **Prompt following:** The rater is presented with the full upsampled caption given to the text-to-image model and asked to "choose which image better corresponds to the caption".
2. **Style:** "Imagine you are using a computer tool that produces an image given some text. Choose which image you would prefer to see if you were using this tool."
3. **Coherence:** "Choose which image contains more coherent objects. A "coherent" object is one that could plausibly exist. Look carefully at body parts, faces and pose of humans, placement of objects, and text in the scene to make your judgement. Hint: count instances of incoherence for each image and choose the image with less problems."

For prompt following and style, we assemble a small dataset of 170 captions for this evaluation which is specifically targeted at typical usage of a production text-to-image system. These captions cover a wide array of actual use-cases like generating humans, products and places, concept blending, text rendering and artwork. We call this evaluation set "DALL-E 3 Eval". These captions will be released with our evaluation samples (see Section [4.3](#)). For coherence, we observe that the raters would penalize images depicting imaginary scenes. Therefore, we randomly sample 250 captions from MSCOCO to make sure that the scenes described by the evaluation prompts can plausibly exist. Note that for style and coherence evaluation, we do not show the captions used to generate the images to the raters, to make sure that they would focus on style or coherence, instead of prompt following. For every image pair and question, we poll gather 3 responses from raters, for a total of 2040 ratings per model and question. The human evaluation interface are shown in Section [E](#).

We compare DALL-E 3 versus Stable Diffusion XL with the refiner module and Midjourney v5.2. In Table [2](#) we report the ELO scores using the same calculation outlined in [Nichol et al. \(2022\)](#).

As the results show, images generated by DALL-E 3 are preferred by human raters a majority of the time over all competitors across all three aspects, especially on prompt following.

Dataset	DALL-E 3	Midjourney 5.2	Stable Diffusion XL	DALL-E 2
DALL-E 3 Eval (prompt following)	153.3	-104.8	-189.5	-
DALL-E 3 Eval (style)	74.0	30.9	-95.7	-
MSCOCO (coherence)	71.0	48.9	-84.2	-
Drawbench	61.7	-	-34.0	-79.3

Table 2 – Human evaluation results for DALL-E 3 versus other text-to-image generation models. Scores reported are computed using the ELO algorithm from [Nichol et al. \(2022\)](#)

4.2.1 Drawbench Human Evaluation

In a previous section, we evaluated drawbench using GPT-V. We noticed that for certain types of tests, GPT-V did not exhibit better than random performance in judging prompt following. In particular, this was the case at tasks that involved counting the number of objects in the image. For better coverage of drawbench performance, we submitted images and captions for human evaluation using the procedure described in the previous section. As with our GPT-V drawbench evaluation, we only compare DALL-E 3, Stable Diffusion XL with the refiner module, and DALL-E 2.

4.3 Reproducibility

We have uploaded all of the samples and prompts generated by all models in all of the above comparisons to [GitHub](#).

5 Limitations & Risk

5.1 Spatial awareness

While DALL-E 3 is a significant step forward for prompt following, it still struggles with object placement and spatial awareness. For example, using the words "to the left of", "underneath", "behind", etc are quite unreliable. This is due to the fact that our synthetic captioner also has this weakness: it is unreliable at stating object placement, and this is reflected in our downstream models.

5.2 Text rendering

When building our captioner, we paid special attention to ensuring that it was able to include prominent words found in images in the captions it generated. As a result, DALL-E 3 can generate text when prompted. During testing, we have noticed that this capability is unreliable as words are have missing or extra characters. We suspect this may have to do with the T5 text encoder we used: when the model encounters text in a prompt, it



Photo of a serene park setting. On the left, a golden retriever sits attentively, gazing forward with its tongue out. On the right, a tabby cat lounges lazily, stretching its legs out and looking towards the dog with a curious expression.



Cartoon drawing of an outer space scene. Amidst floating planets and twinkling stars, a whimsical horse with exaggerated features rides an astronaut, who swims through space with a jetpack, looking a tad overwhelmed.



Arum dioscoridis.

Figure 7 – Examples of common failure modes of DALL-E 3

actually sees tokens that represent whole words and must map those to letters in an image. In future work, we would like to explore conditioning on character-level language models to help improve this behavior.

5.3 Specificity

We observed that our synthetic captions are prone to hallucinating important details about an image. For example, given a botanical drawing of a flower, the captioner will often hallucinate a plant genus and species and put it in the caption, even when these details are available in text form in the image. We observed similar behavior when describing pictures of birds: species are hallucinated or not mentioned at all.

This has a downstream impact on our text-to-image models: DALL-E 3 is unreliable at generating imagery for specific terms such as those described above. We believe that further improvements to the captioner should enable further improvements to our text-to-image model.

5.4 Safety and bias mitigations

We performed an in-depth analysis of the safety concerns arising from the deployment of DALL-E 3, including risks posed by model biases. The results of these evaluations can be found in the DALL-E 3 system card [\[13\]](#).

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A Image decoder

The image decoder used in our experiments is a text-conditioned U-Net[23] latent diffusion model[22] with three stages.

We use the same VAE[9] developed by Rombach et al. (2022) for our model. This autoencoder performs 8x downsampling. In our synthetic caption evaluation, we train on 256px images, resulting in a model input size of 32x32 latent vectors.

For timestep conditioning, we use a modulated GroupNorm[28], where a learned scale and bias term that is dependent on the timestep signal is applied to the outputs of the GroupNorm layers.

For text-conditioning, we first encode text inputs using a T5 XXL text encoder [18]. The output latents from this process are then cross-attended to by xfnet.

B DALL-E 3 latent decoder

For DALL-E 3, we trained our own diffusion decoder on top of the latent space learned by the VAE trained by Rombach et al. (2022). We found that using a diffusion decoder here provided marked improvements to fine image details, for example text or human faces.

This diffusion decoder is a convolutional U-Net identical to the one described in Ho et al. (2020). Once trained, we used the consistency distillation process described in Song et al. (2023) to bring it down to two denoising steps.

This improved latent decoder is not used for our synthetic caption evaluation.

C Caption "upsampling" prompt for GPT-4

Following is the prompt we give to GPT-4 before feeding it an image caption for "upsampling".

```
You are part of a team of bots that creates images. You work with an assistant bot that will draw anything you say in square brackets. For example, outputting "a beautiful morning in the woods with the sun peaking through the trees" will trigger your partner bot to output an image of a forest morning, as described. You will be prompted by people looking to create detailed, amazing images. The way to accomplish this is to take their short prompts and make them extremely detailed and descriptive.

There are a few rules to follow:
- You will only ever output a single image description per user request.
- Sometimes the user will request that you modify previous captions. In this case, you should refer to your previous conversations with the user and make the modifications requested.
- When modifications are requested, you should not simply make the description longer. You should refactor the entire description to integrate the suggestions.
- Other times the user will not want modifications, but instead want a new image. In this case, you should ignore your previous conversation with the user.
- Image descriptions must be between 15-80 words. Extra words will be ignored.

{"role": "user", "content": "Create an imaginative image descriptive caption or modify an earlier \
caption for the user input: "a man holding a sword\""},
{"role": "assistant", "content": "a pale figure with long white hair stands in the center of a dark \
forest, holding a sword high above his head.
the blade glows with a blue light, casting a soft glow on the trees and bushes surrounding him."},
{"role": "user", "content": "Create an imaginative image descriptive caption or modify an earlier caption \
for the user input: "make the light red\""},
{"role": "assistant", "content": "a pale figure with long white hair stands in the center of a dark forest, \
holding a sword high above his head.
the blade glows with a red light, casting a warm glow on the trees and bushes surrounding him."},
{"role": "user", "content": "Create an imaginative image descriptive caption or modify an earlier caption for the \
user input: "draw a frog playing dominoes\""},
{"role": "assistant", "content": "a frog sits on a worn table playing a game of dominoes with an elderly raccoon.
the table is covered in a green cloth, and the frog is wearing a jacket and a pair of jeans. The scene is set in a forest,
with a large tree in the background."}
```

D Evaluation of drawbench with vision-enabled GPT-4

Following is the prompt we give to the vision-enabled GPT-4 model to perform our automated drawbench evaluation:

You are responsible for judging the faithfulness of images generated by a computer program to the caption used to generate them. You will be presented with an image and given the caption that was used to produce the image. The captions you are judging are designed to stress-test image generation programs, and may include things such as:

1. Scrambled or mis-spelled words (the image generator should an image associated with the probably meaning)
2. Color assignment (the image generator should apply the correct color to the correct object)
3. Counting (the correct number of objects should be present)
4. Abnormal associations, for example 'elephant under a sea', where the image should depict what is requested.
5. Descriptions of objects, the image generator should draw the most commonly associated object.
6. Rare single words, where the image generator should create an image somewhat associable with the specified image.
7. Images with text in them, where the image generator should create an image with the specified text in it.

You need to make a decision as to whether or not the image is correct, given the caption. You will first think out loud about your eventual conclusion, enumerating reasons why the image does or does not match the given caption. After thinking out loud, you should output either 'Correct' or 'Incorrect' depending on whether you think the image is faithful to the caption.

A few rules:

1. Do not nitpick. If the caption requests an object and the object is generally depicted correctly, then you should answer 'Correct'.
2. Ignore other objects in the image that are not explicitly mentioned by the caption; it is fine for these to be shown.
3. It is also OK if the object being depicted is slightly deformed, as long as a human would recognize it and it does not violate the caption.
4. Your response must always end with either 'incorrect' or 'correct'
5. 'Incorrect' should be reserved for instances where a specific aspect of the caption is not followed correctly, such as a wrong object, color or count.
6. You must keep your thinking out loud short, less than 50 words.

image(<image_path>)
<prompt>

Where <image_path> and <prompt> are replaced with the image generated by the model and the corresponding prompt used to generate it.

E Human Evaluation Interface

The human evaluation interface for prompt following, coherence and style are shown in Figure 8, 9 and 10.

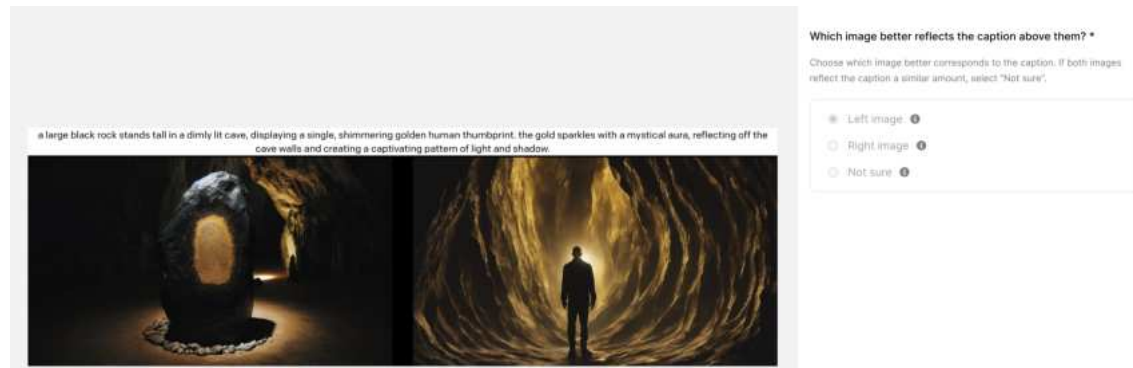


Figure 8 – Human evaluation interface for prompt following.

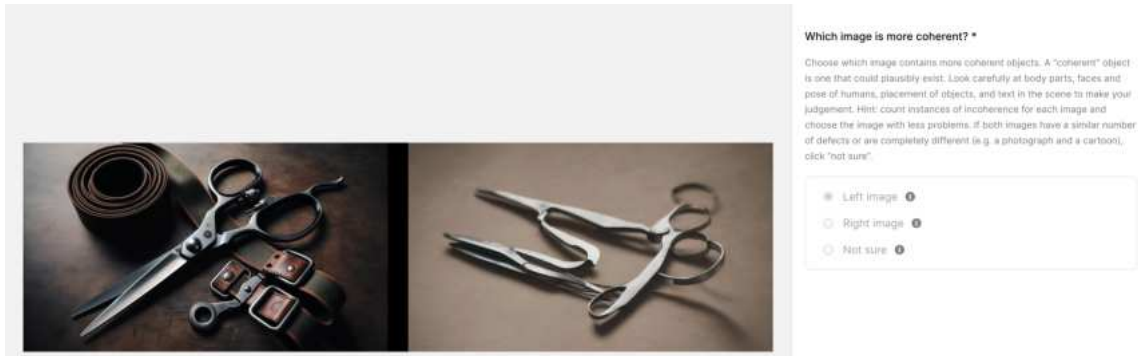


Figure 9 – Human evaluation interface for **coherence**.

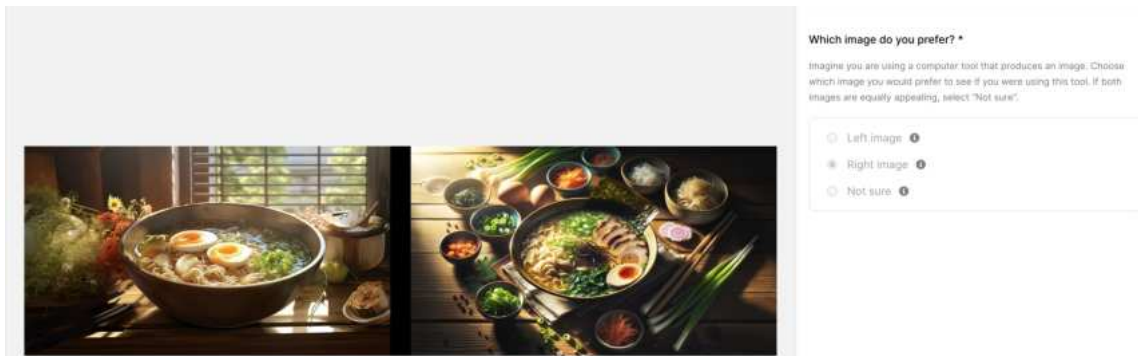


Figure 10 – Human evaluation interface for **style**.

High-Resolution Image Synthesis with Latent Diffusion Models

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<https://github.com/CompVis/latent-diffusion>

Abstract

By decomposing the image formation process into a sequential application of denoising autoencoders, diffusion models (DMs) achieve state-of-the-art synthesis results on image data and beyond. Additionally, their formulation allows for a guiding mechanism to control the image generation process without retraining. However, since these models typically operate directly in pixel space, optimization of powerful DMs often consumes hundreds of GPU days and inference is expensive due to sequential evaluations. To enable DM training on limited computational resources while retaining their quality and flexibility, we apply them in the latent space of powerful pretrained autoencoders. In contrast to previous work, training diffusion models on such a representation allows for the first time to reach a near-optimal point between complexity reduction and detail preservation, greatly boosting visual fidelity. By introducing cross-attention layers into the model architecture, we turn diffusion models into powerful and flexible generators for general conditioning inputs such as text or bounding boxes and high-resolution synthesis becomes possible in a convolutional manner. Our latent diffusion models (LDMs) achieve new state of the art scores for image inpainting and class-conditional image synthesis and highly competitive performance on various tasks, including unconditional image generation, text-to-image synthesis, and super-resolution, while significantly reducing computational requirements compared to pixel-based DMs.

1. Introduction

Image synthesis is one of the computer vision fields with the most spectacular recent development, but also among those with the greatest computational demands. Especially high-resolution synthesis of complex, natural scenes is presently dominated by scaling up likelihood-based models, potentially containing billions of parameters in autoregressive (AR) transformers [64, 65]. In contrast, the promising results of GANs [3, 26, 39] have been revealed to be mostly confined to data with comparably limited variability as their adversarial learning procedure does not easily scale to modeling complex, multi-modal distributions. Recently, diffusion models [79], which are built from a hierarchy of denoising autoencoders, have shown to achieve impressive



Figure 1. Boosting the upper bound on achievable quality with less aggressive downsampling. Since diffusion models offer excellent inductive biases for spatial data, we do not need the heavy spatial downsampling of related generative models in latent space, but can still greatly reduce the dimensionality of the data via suitable autoencoding models, see Sec. 3. Images are from the DIV2K [1] validation set, evaluated at 512^2 px. We denote the spatial downsampling factor by f . Reconstruction FIDs [28] and PSNR are calculated on ImageNet-val. [12]; see also Tab. 8.

results in image synthesis [29, 82] and beyond [7, 44, 47, 56], and define the state-of-the-art in class-conditional image synthesis [15, 30] and super-resolution [70]. Moreover, even unconditional DMs can readily be applied to tasks such as inpainting and colorization [82] or stroke-based synthesis [52], in contrast to other types of generative models [19, 45, 67]. Being likelihood-based models, they do not exhibit mode-collapse and training instabilities as GANs and, by heavily exploiting parameter sharing, they can model highly complex distributions of natural images without involving billions of parameters as in AR models [65].

Democratizing High-Resolution Image Synthesis DMs belong to the class of likelihood-based models, whose mode-covering behavior makes them prone to spend excessive amounts of capacity (and thus compute resources) on modeling imperceptible details of the data [16, 71]. Although the reweighted variational objective [29] aims to address this by undersampling the initial denoising steps, DMs are still computationally demanding, since training and evaluating such a model requires repeated function evaluations (and gradient computations) in the high-dimensional space of RGB images. As an example, training the most powerful DMs often takes hundreds of GPU days (e.g. 150 - 1000 V100 days in [15]) and repeated evaluations on a noisy version of the input space render also inference expensive,

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so that producing 50k samples takes approximately 5 days [15] on a single A100 GPU. This has two consequences for the research community and users in general: Firstly, training such a model requires massive computational resources only available to a small fraction of the field, and leaves a huge carbon footprint [63, 83]. Secondly, evaluating an already trained model is also expensive in time and memory, since the same model architecture must run sequentially for a large number of steps (e.g. 25 - 1000 steps in [15]).

To increase the accessibility of this powerful model class and at the same time reduce its significant resource consumption, a method is needed that reduces the computational complexity for both training and sampling. Reducing the computational demands of DMs without impairing their performance is, therefore, key to enhance their accessibility.

Departure to Latent Space Our approach starts with the analysis of already trained diffusion models in pixel space: Fig. 2 shows the rate-distortion trade-off of a trained model. As with any likelihood-based model, learning can be roughly divided into two stages: First is a *perceptual compression* stage which removes high-frequency details but still learns little semantic variation. In the second stage, the actual generative model learns the semantic and conceptual composition of the data (*semantic compression*). We thus aim to first find a *perceptually equivalent, but computationally more suitable space*, in which we will train diffusion models for high-resolution image synthesis.

Following common practice [11, 23, 64, 65, 93], we separate training into two distinct phases: First, we train an autoencoder which provides a lower-dimensional (and thereby efficient) representational space which is perceptually equivalent to the data space. Importantly, and in contrast to previous work [23, 64], we do not need to rely on excessive spatial compression, as we train DMs in the learned latent space, which exhibits better scaling properties with respect to the spatial dimensionality. The reduced complexity also provides efficient image generation from the latent space with a single network pass. We dub the resulting model class *Latent Diffusion Models (LDMs)*.

A notable advantage of this approach is that we need to train the universal autoencoding stage only once and can therefore reuse it for multiple DM trainings or to explore possibly completely different tasks [78]. This enables efficient exploration of a large number of diffusion models for various image-to-image and text-to-image tasks. For the latter, we design an architecture that connects transformers to the DM’s UNet backbone [69] and enables arbitrary types of token-based conditioning mechanisms, see Sec. 3.3.

In sum, our work makes the following **contributions**:

(i) In contrast to purely transformer-based approaches [23, 64], our method scales more gracefully to higher dimensional data and can thus (a) work on a compression level which provides more faithful and detailed reconstructions than previous work (see Fig. 1) and (b) can be efficiently

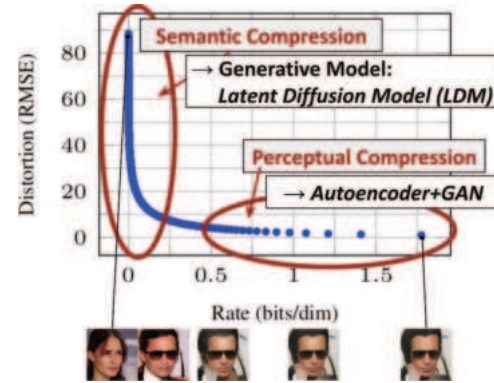


Figure 2. Illustrating perceptual and semantic compression: Most bits of a digital image correspond to imperceptible details. While DMs allow to suppress this semantically meaningless information by minimizing the responsible loss term, gradients (during training) and the neural network backbone (training and inference) still need to be evaluated on all pixels, leading to superfluous computations and unnecessarily expensive optimization and inference. We propose *latent diffusion models (LDMs)* as an effective generative model and a separate mild compression stage that only eliminates imperceptible details. Data and images from [29].

applied to high-resolution synthesis of megapixel images.

(ii) We achieve competitive performance on multiple tasks (unconditional image synthesis, inpainting, stochastic super-resolution) and datasets while significantly lowering computational costs. Compared to pixel-based diffusion approaches, we also significantly decrease inference costs.

(iii) We show that, in contrast to previous work [90] which learns both an encoder/decoder architecture and a score-based prior simultaneously, our approach does not require a delicate weighting of reconstruction and generative abilities. This ensures extremely faithful reconstructions and requires very little regularization of the latent space.

(iv) We find that for densely conditioned tasks such as super-resolution, inpainting and semantic synthesis, our model can be applied in a convolutional fashion and render large, consistent images of $\sim 1024^2$ px.

(v) Moreover, we design a general-purpose conditioning mechanism based on cross-attention, enabling multi-modal training. We use it to train class-conditional, text-to-image and layout-to-image models.

(vi) Finally, we release pretrained latent diffusion and autoencoding models at <https://github.com/CompVis/latent-diffusion> which might be reusable for a various tasks besides training of DMs [78].

2. Related Work

Generative Models for Image Synthesis The high dimensional nature of images presents distinct challenges to generative modeling. Generative Adversarial Networks (GAN) [26] allow for efficient sampling of high resolution images with good perceptual quality [3, 41], but are diffi-

cult to optimize [2, 27, 53] and struggle to capture the full data distribution [54]. In contrast, likelihood-based methods emphasize good density estimation which renders optimization more well-behaved. Variational autoencoders (VAE) [45] and flow-based models [18, 19] enable efficient synthesis of high resolution images [9, 43, 89], but sample quality is not on par with GANs. While autoregressive models (ARM) [6, 10, 91, 92] achieve strong performance in density estimation, computationally demanding architectures [94] and a sequential sampling process limit them to low resolution images. Because pixel based representations of images contain barely perceptible, high-frequency details [16, 71], maximum-likelihood training spends a disproportionate amount of capacity on modeling them, resulting in long training times. To scale to higher resolutions, several two-stage approaches [23, 65, 97, 99] use ARMs to model a compressed latent image space instead of raw pixels.

Recently, **Diffusion Probabilistic Models** (DM) [79], have achieved state-of-the-art results in density estimation [44] as well as in sample quality [15]. The generative power of these models stems from a natural fit to the inductive biases of image-like data when their underlying neural backbone is implemented as a UNet [15, 29, 69, 82]. The best synthesis quality is usually achieved when a reweighted objective [29] is used for training. In this case, the DM corresponds to a lossy compressor and allow to trade image quality for compression capabilities. Evaluating and optimizing these models in pixel space, however, has the downside of low inference speed and very high training costs. While the former can be partially addressed by advanced sampling strategies [46, 73, 81] and hierarchical approaches [30, 90], training on high-resolution image data always requires to calculate expensive gradients. We address both drawbacks with our proposed *LDMs*, which work on a compressed latent space of lower dimensionality. This renders training computationally cheaper and speeds up inference with almost no reduction in synthesis quality (see Fig. 1).

Two-Stage Image Synthesis To mitigate the shortcomings of individual generative approaches, a lot of research [11, 23, 65, 68, 97, 99] has gone into combining the strengths of different methods into more efficient and performant models via a two stage approach. VQ-VAEs [65, 97] use autoregressive models to learn an expressive prior over a discretized latent space. [64] extend this approach to text-to-image generation by learning a joint distribution over discretized image and text representations. More generally, [68] uses conditionally invertible networks to provide a generic transfer between latent spaces of diverse domains. Different from VQ-VAEs, VQGANs [23, 99] employ a first stage with an adversarial and perceptual objective to scale autoregressive transformers to larger images. However, the high compression rates required for feasible ARM training, which introduces billions of trainable parameters [23, 64], limit the overall performance of such ap-

proaches and less compression comes at the price of high computational cost [23, 64]. Our work prevents such trade-offs, as our proposed *LDMs* scale more gently to higher dimensional latent spaces due to their convolutional backbone. Thus, we are free to choose the level of compression which optimally mediates between learning a powerful first stage, without leaving too much perceptual compression up to the generative diffusion model while guaranteeing high-fidelity reconstructions (see Fig. 1). While approaches to jointly learn an encoding/decoding model together with a score-based prior exist [90], they still require a difficult weighting between reconstruction and generative capabilities [11] and are outperformed by our approach (Sec. 4).

3. Method

To lower the computational demands of training diffusion models towards high-resolution image synthesis, we observe that although diffusion models allow to ignore perceptually irrelevant details by undersampling the corresponding loss terms [29], they still require costly function evaluations in pixel space, which causes huge demands in computation time and energy resources.

We propose to circumvent this drawback by introducing an explicit separation of the compressive from the generative learning phase (see Fig. 2). To achieve this, we utilize an autoencoding model which learns a space that is perceptually equivalent to the image space, but offers significantly reduced computational complexity.

Such an approach offers several advantages: (i) By leaving the high-dimensional image space, we obtain DMs which are computationally much more efficient because sampling is performed on a low-dimensional space. (ii) We exploit the inductive bias of DMs inherited from their UNet architecture [69], which makes them particularly effective for data with spatial structure and therefore alleviates the need for aggressive, quality-reducing compression levels as required by previous approaches [23, 64]. (iii) Finally, we obtain general-purpose compression models whose latent space can be used to train multiple generative models and which can also be utilized for other downstream applications such as single-image CLIP-guided synthesis [25].

3.1. Perceptual Image Compression

Our perceptual compression model is based on previous work [23] and consists of an autoencoder trained by combination of a perceptual loss [102] and a patch-based [32] adversarial objective [20, 23, 99]. This ensures that the reconstructions are confined to the image manifold by enforcing local realism and avoids blurriness introduced by relying solely on pixel-space losses such as L_2 or L_1 objectives.

More precisely, given an image $x \in \mathbb{R}^{H \times W \times 3}$ in RGB space, the encoder $\mathcal{E}$ encodes x into a latent representation $z = \mathcal{E}(x)$, and the decoder $\mathcal{D}$ reconstructs the image from the latent, giving $\tilde{x} = \mathcal{D}(z) = \mathcal{D}(\mathcal{E}(x))$, where

$z \in \mathbb{R}^{h \times w \times c}$. Importantly, the encoder *downsamples* the image by a factor $f = H/h = W/w$, and we investigate different downsampling factors $f = 2^m$, with $m \in \mathbb{N}$.

In order to avoid arbitrarily high-variance latent spaces, we experiment with two different kinds of regularizations. The first variant, *KL-reg.*, imposes a slight KL-penalty towards a standard normal on the learned latent, similar to a VAE [45, 67], whereas *VQ-reg.* uses a vector quantization layer [93] within the decoder. This model can be interpreted as a VQGAN [23] but with the quantization layer absorbed by the decoder. Because our subsequent DM is designed to work with the two-dimensional structure of our learned latent space $z = \mathcal{E}(x)$, we can use relatively mild compression rates and achieve very good reconstructions. This is in contrast to previous works [23, 64], which relied on an arbitrary 1D ordering of the learned space z to model its distribution autoregressively and thereby ignored much of the inherent structure of z . Hence, our compression model preserves details of x better (see Tab. 8). The full objective and training details can be found in the supplement.

3.2. Latent Diffusion Models

Diffusion Models [79] are probabilistic models designed to learn a data distribution $p(x)$ by gradually denoising a normally distributed variable, which corresponds to learning the reverse process of a fixed Markov Chain of length T . For image synthesis, the most successful models [15, 29, 70] rely on a reweighted variant of the variational lower bound on $p(x)$, which mirrors denoising score-matching [82]. These models can be interpreted as an equally weighted sequence of denoising autoencoders $\epsilon_\theta(x_t, t)$; $t = 1 \dots T$, which are trained to predict a denoised variant of their input x_t , where x_t is a noisy version of the input x . The corresponding objective can be simplified to (Sec. A)

$$L_{DM} = \mathbb{E}_{x, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \right], \quad (1)$$

with t uniformly sampled from $\{1, \dots, T\}$.

Generative Modeling of Latent Representations With our trained perceptual compression models consisting of $\mathcal{E}$ and $\mathcal{D}$, we now have access to an efficient, low-dimensional latent space in which high-frequency, imperceptible details are abstracted away. Compared to the high-dimensional pixel space, this space is more suitable for likelihood-based generative models, as they can now (i) focus on the important, semantic bits of the data and (ii) train in a lower-dimensional, computationally much more efficient space.

Unlike previous work that relied on autoregressive, attention-based transformer models in a highly compressed, discrete latent space [23, 64, 99], we can take advantage of image-specific inductive biases that our model offers. This includes the ability to build the underlying UNet primarily from 2D convolutional layers, and further focusing the

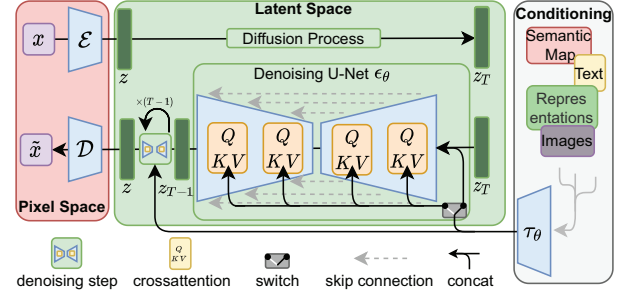


Figure 3. We condition LDMs either via concatenation or by a more general cross-attention mechanism. See Sec. 3.3

objective on the perceptually most relevant bits using the reweighted bound, which now reads

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2 \right]. \quad (2)$$

The neural backbone $\epsilon_\theta(\circ, t)$ of our model is realized as a time-conditional UNet [69]. Since the forward process is fixed, z_t can be efficiently obtained from $\mathcal{E}$ during training, and samples from $p(z)$ can be decoded to image space with a single pass through $\mathcal{D}$.

3.3. Conditioning Mechanisms

Similar to other types of generative models [55, 80], diffusion models are in principle capable of modeling conditional distributions of the form $p(z|y)$. This can be implemented with a conditional denoising autoencoder $\epsilon_\theta(z_t, t, y)$ and paves the way to controlling the synthesis process through inputs y such as text [66], semantic maps [32, 59] or other image-to-image translation tasks [33].

In the context of image synthesis, however, combining the generative power of DMs with other types of conditionings beyond class-labels [15] or blurred variants of the input image [70] is so far an under-explored area of research.

We turn DMs into more flexible conditional image generators by augmenting their underlying UNet backbone with the cross-attention mechanism [94], which is effective for learning attention-based models of various input modalities [34, 35]. To pre-process y from various modalities (such as language prompts) we introduce a domain specific encoder τ_θ that projects y to an intermediate representation $\tau_\theta(y) \in \mathbb{R}^{M \times d_\tau}$, which is then mapped to the intermediate layers of the UNet via a cross-attention layer implementing $\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) \cdot V$, with

$$Q = W_Q^{(i)} \cdot \varphi_i(z_t), \quad K = W_K^{(i)} \cdot \tau_\theta(y), \quad V = W_V^{(i)} \cdot \tau_\theta(y).$$

Here, $\varphi_i(z_t) \in \mathbb{R}^{N \times d_i^\epsilon}$ denotes a (flattened) intermediate representation of the UNet implementing ϵ_θ and $W_V^{(i)} \in \mathbb{R}^{d \times d_i^\epsilon}$, $W_Q^{(i)} \in \mathbb{R}^{d \times d_\tau}$ & $W_K^{(i)} \in \mathbb{R}^{d \times d_\tau}$ are learnable projection matrices [35, 94]. See Fig. 3 for a visual depiction.



Figure 4. Samples from *LDMs* trained on CelebAHQ [38], FFHQ [40], LSUN-Churches [98], LSUN-Bedrooms [98] and class-conditional ImageNet [12], each with a resolution of 256×256 . Best viewed when zoomed in. For more samples *cf.* the supplement.

Based on image-conditioning pairs, we then learn the conditional LDM via

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(z_t, t, \tau_{\theta}(y))\|_2^2 \right], \quad (3)$$

where both τ_{θ} and ϵ_{θ} are jointly optimized via Eq. 3. This conditioning mechanism is flexible as τ_{θ} can be parameterized with domain-specific experts, *e.g.* (unmasked) transformers [94] when y are text prompts (see Sec. 4.3.1)

4. Experiments

LDMs provide means to flexible and computationally tractable diffusion based image synthesis also including high-resolution generation of various image modalities, which we empirically show in the following. Firstly, however, we analyze the gains of our models compared to pixel-based diffusion models in both training and inference. Interestingly, we find that *LDMs* trained in *VQ*-regularized latent spaces achieve better sample quality, even though the reconstruction capabilities of *VQ*-regularized first stage models slightly fall behind those of their continuous counterparts, *cf.* Tab. 8. Therefore, we evaluate *VQ*-regularized *LDMs* in the remainder of the paper, unless stated differently. A visual comparison between the effects of first stage regularization schemes on *LDM* training and their generalization abilities to resolutions higher than 256^2 can be found in Appendix C.1. In D.2 we furthermore list details on architecture, implementation, training and evaluation for all results presented in this section.

4.1. On Perceptual Compression Tradeoffs

This section analyzes the behavior of our *LDMs* with different downsampling factors $f \in \{1, 2, 4, 8, 16, 32\}$ (abbreviated as *LDM-f*, where *LDM-1* corresponds to pixel-based DMs). To obtain a comparable test-field, we fix the computational resources to a single NVIDIA A100 for all experiments in this section and train all models for the same number of steps and with the same number of parameters.

Tab. 8 shows hyperparameters and reconstruction performance of the first stage models used for the *LDMs* compared in this section. Fig. 5 shows sample quality as a function of training progress for 2M steps of class-conditional

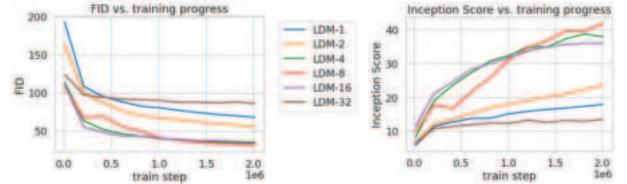


Figure 5. Analyzing the training of class-conditional *LDMs* with different downsampling factors f over 2M train steps on the ImageNet dataset. Pixel-based *LDM-1* requires substantially larger train times compared to models with larger downsampling factors (*LDM*-{4-16}). Too much perceptual compression as in *LDM-32* limits the overall sample quality. All models are trained on a single NVIDIA A100 with the same computational budget. Results obtained with 100 DDIM steps [81] and $\kappa = 0$.

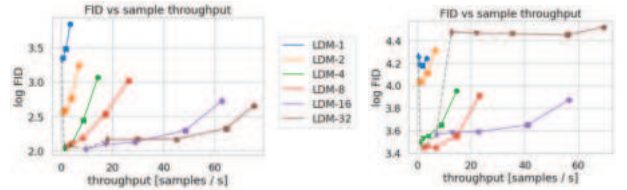


Figure 6. Inference speed vs sample quality: Comparing *LDMs* with different amounts of compression on the CelebA-HQ (left) and ImageNet (right) datasets. Different markers indicate $\{10, 20, 50, 100, 200\}$ sampling steps with the DDIM sampler, counted from right to left along each line. The dashed line shows the FID scores for 200 steps, indicating the strong performance of *LDM*-{4-8} compared to models with different compression ratios. FID scores assessed on 5000 samples. All models were trained for 500k (CelebA) / 2M (ImageNet) steps on an A100.

models on the ImageNet [12] dataset. We see that, i) small downsampling factors for *LDM*-{1,2} result in slow training progress, whereas ii) overly large values of f cause stagnating fidelity after comparably few training steps. Revisiting the analysis above (Fig. 1 and 2) we attribute this to i) leaving most of perceptual compression to the diffusion model and ii) too strong first stage compression resulting in information loss and thus limiting the achievable quality. *LDM*-{4-16} strike a good balance between efficiency and perceptually faithful results, which manifests in a sig-

CelebA-HQ 256 × 256					FFHQ 256 × 256				
Method	FID ↓	Prec. ↑	Recall ↑		Method	FID ↓	Prec. ↑	Recall ↑	
DC-VAE [61]	15.8	-	-		ImageBART [21]	9.57	-	-	
VQGAN+T [23] (k=400)	10.2	-	-		U-Net GAN (+aug) [75]	10.9 (7.6)	-	-	
PGGAN [38]	8.0	-	-		UDM [42]	5.54	-	-	
LSGM [90]	7.22	-	-		StyleGAN [10]	4.16	0.71	0.46	
UDM [42]	7.16	-	-		ProjectedGAN [74]	3.08	0.65	0.46	
<i>LDM-4</i> (ours, 500-s [†])	5.11	0.72	0.49		<i>LDM-4</i> (ours, 200-s)	4.98	0.73	0.50	

LSUN-Churches 256 × 256					LSUN-Bedrooms 256 × 256				
Method	FID ↓	Prec. ↑	Recall ↑		Method	FID ↓	Prec. ↑	Recall ↑	
DDPM [29]	7.89	-	-		ImageBART [21]	5.51	-	-	
ImageBART [21]	7.32	-	-		DDPM [29]	4.9	-	-	
PGGAN [38]	6.42	-	-		UDM [42]	4.57	-	-	
StyleGAN [40]	4.21	-	-		StyleGAN [40]	2.35	0.59	0.48	
StyleGAN2 [41]	3.86	-	-		ADM [15]	1.90	0.66	0.51	
ProjectedGAN [74]	1.59	0.61	0.44		ProjectedGAN [74]	1.52	0.61	0.34	
<i>LDM-8*</i> (ours, 200-s)	4.02	0.64	0.52		<i>LDM-4</i> (ours, 200-s)	2.95	0.66	0.48	

Table 1. Evaluation metrics for unconditional image synthesis. CelebA-HQ results reproduced from [42, 61, 96], FFHQ from [41, 42]. [†]: N -s refers to N sampling steps with the DDIM [81] sampler. *: trained in KL -regularized latent space. Additional results can be found in the supplementary.

nificant FID [28] gap of 38 between pixel-based diffusion ($LDM-1$) and $LDM-8$ after 2M training steps.

In Fig. 6, we compare models trained on CelebA-HQ [38] and ImageNet in terms sampling speed for different numbers of denoising steps with the DDIM sampler [81] and plot it against FID-scores [28]. $LDM-\{4-8\}$ outperform models with unsuitable ratios of perceptual and conceptual compression. Especially compared to pixel-based $LDM-1$, they achieve much lower FID scores while simultaneously significantly increasing sample throughput. Complex datasets such as ImageNet require reduced compression rates to avoid reducing quality. Summarized, we observe that $LDM-4$ and -8 lie in the best behaved regime for achieving high-quality synthesis results.

Text-Conditional Image Synthesis					
	DALL-E [†] [64]	CogView [†] [17]	Lafite [†] [105]	<i>LDM-KL-8</i>	<i>LDM-KL-8-G*</i>
FID ↓	27.50	27.10	26.94	23.35	12.61
IS ↑	17.90	18.20	<u>26.02</u>	19.93 $\pm$ 0.35	26.62$\pm$0.38

Table 2. Evaluation of text-conditional image synthesis on the MS-COCO [50] dataset: Our model outperforms autoregressive [17, 64] and GAN-based [105] methods by a significant margin when using 250 DDIM [81] steps. [†]: Numbers taken from [105]. *: Classifier-free guidance [31], scale 1.5.

4.2. Image Generation with Latent Diffusion

We train unconditional models of 256² images on CelebA-HQ [38], FFHQ [40], LSUN-Churches and -Bedrooms [98] and evaluate the i) sample quality and ii) their coverage of the data manifold using ii) FID [28] and ii) Precision-and-Recall [49]. Tab. 1 summarizes our results. On CelebA-HQ, we report a new state-of-the-art FID of 5.11, outperforming previous likelihood-based models as well as GANs. We also outperform LSGM [90] where a latent diffusion model is trained jointly together with the first stage. In contrast, we train diffusion models in a fixed space and avoid the difficulty of weighing reconstruction quality against learning the prior over the latent space, see Fig. 1-2.

We outperform prior diffusion based approaches on all but the LSUN-Bedrooms dataset, where our score is close

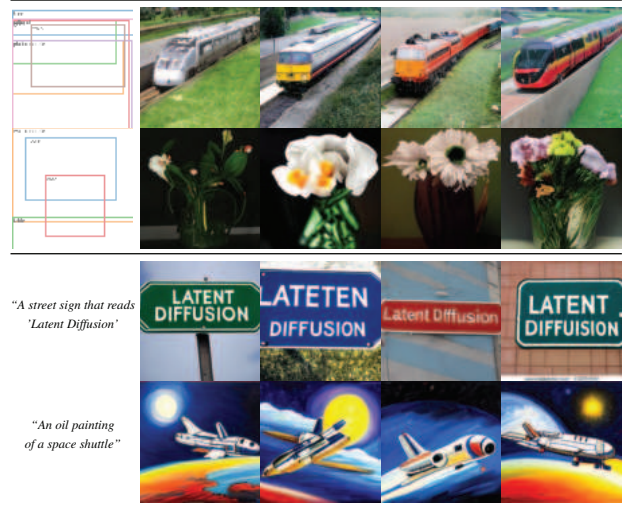


Figure 7. Top: Samples from our LDM for layout-to-image synthesis on COCO [4]. Quantitative evaluation in the supplement. Bottom: Samples from our text-to-image LDM model for user-defined text prompts, which is trained on LAION-400M [76].

to ADM [15], despite utilizing half its parameters and requiring 4-times less train resources (see Appendix D.3.5). Moreover, $LDMs$ consistently improve upon GAN-based methods in Precision and Recall, thus confirming the advantages of their mode-covering likelihood-based training objective over adversarial approaches. In Fig. 4 we also show qualitative results on each dataset.

4.3. Conditional Latent Diffusion

4.3.1 Transformer Encoders for LDMs

By introducing cross-attention based conditioning into LDMs we open them up for various conditioning modalities previously unexplored for diffusion models. For **text-to-image** image modeling, we train a 1.45B parameter model conditioned on language prompts on LAION-400M [76]. We employ the BERT-tokenizer [14] and implement τ_θ as a transformer [94] to infer a latent code which is mapped into the UNet via cross-attention (Sec. 3.3). This combination of domain specific experts for learning a language representation and visual synthesis results in a powerful model, which generalizes well to complex, user-defined text prompts, cf. Fig. 7 and 14. For quantitative analysis, we follow prior work and evaluate text-to-image generation on the MS-COCO [50] validation set, where our model improves upon powerful AR [17, 64] and GAN-based [105] methods, cf. Tab. 2. We note that applying classifier-free diffusion guidance [31] greatly boosts sample quality. To further analyze the flexibility of the cross-attention based conditioning mechanism we also train models to synthesize images based on **semantic layouts** on OpenImages [48], and finetune on COCO [4], see Fig. 7. See Sec. C.4 for the quantitative evaluation and implementation details.

Lastly, following prior work [3, 15, 21, 23], we evaluate our best-performing class-conditional ImageNet mod-

els with $f \in \{4, 8\}$ from Sec. 4.1 in Tab. 3, Fig. 4 and Sec. C.5. Here we outperform the state of the art diffusion model ADM [15] while significantly reducing computational requirements and parameter count, cf. Tab 18.

Method	FID↓	IS↑	Precision↑	Recall↑	N_{params}
BigGan-deep [3]	6.95	203.6 ± 2.6	0.87	0.28	340M
ADM [15]	10.94	100.98	0.69	0.63	554M
ADM-G [15]	4.50	186.7	0.82	0.52	608M
<i>LDM-4</i> (ours)	10.56	103.49 ± 1.34	0.71	0.62	400M
<i>LDM-4-G</i> (ours)	3.60	247.67± 5.89	0.87	0.48	400M

Table 3. Comparison of a class-conditional ImageNet *LDM* with recent state-of-the-art methods for class-conditional image generation on the ImageNet [12] dataset. A more detailed comparison with additional baselines can be found in C.5, Tab. 10 and E.

4.3.2 Convolutional Sampling Beyond 256²

By concatenating spatially aligned conditioning information to the input of ϵ_θ , *LDMs* can serve as efficient general-purpose image-to-image translation models. We use this to train models for semantic synthesis, super-resolution (Sec. 4.4) and inpainting (Sec. 4.5). For semantic synthesis, we use images of landscapes paired with semantic maps [23, 59] and concatenate downsampled versions of the semantic maps with the latent image representation of a $f = 4$ model (VQ-reg., see Tab. 8). We train on an input resolution of 256² (crops from 384²) but find that our model generalizes to larger resolutions and can generate images up to the megapixel regime when evaluated in a convolutional manner (see Fig. 8). We exploit this behavior to also apply the super-resolution models in Sec. 4.4 and the inpainting models in Sec. 4.5 to generate large images between 512² and 1024². For this application, the signal-to-noise ratio (induced by the scale of the latent space) significantly affects the results. In Sec. C.1 we illustrate this when learning an *LDM* on (i) the latent space as provided by a $f = 4$ model (KL-reg., see Tab. 8), and (ii) a rescaled version, scaled by the component-wise standard deviation.



Figure 8. A *LDM* trained on 256² resolution can generalize to larger resolution (here: 512 × 1024) for spatially conditioned tasks such as semantic synthesis of landscape images. See Sec. 4.3.2.

4.4. Super-Resolution with Latent Diffusion

LDMs can be efficiently trained for super-resolution by directly conditioning on low-resolution images via concate-



Figure 9. ImageNet 64→256 super-resolution on ImageNet-Val. *LDM-SR* has advantages at rendering realistic textures but *SR3* can synthesize more coherent fine structures. See appendix for additional samples and cropouts. *SR3* results from [70].

nation (cf. Sec. 3.3). In a first experiment, we follow *SR3* [70] and fix the image degradation to a bicubic interpolation with 4×-downsampling and train on ImageNet following *SR3*'s data processing pipeline. We use the $f = 4$ autoencoding model pretrained on OpenImages (VQ-reg., cf. Tab. 8) and concatenate the low-resolution conditioning y and the inputs to the UNet, i.e. τ_θ is the identity. Our qualitative and quantitative results (see Fig. 9 and Tab. 4) show competitive performance and *LDM-SR* outperforms *SR3* in FID while *SR3* has a better IS. A simple image regression model achieves the highest PSNR and SSIM scores; however these metrics do not align well with human perception [102] and favor blurriness over imperfectly aligned high frequency details [70]. Further, we conduct a user study comparing the pixel-baseline with *LDM-SR*. We follow *SR3* [70] where human subjects were shown a low-res image in between two high-res images and asked for preference. The results in Tab. 5 affirm the good performance of *LDM-SR*. PSNR and SSIM can be pushed by using a post-hoc guiding mechanism [15] and we implement this *image-based guider* via a perceptual loss, see Sec. C.7. Since the bicubic degradation process does not generalize well to images which do not follow this pre-processing, we also train a generic model, *LDM-BSR*, by using more diverse degradation. The results are shown in Sec. C.7.1.

Method	FID ↓	IS ↑	PSNR ↑	SSIM ↑	N_{params}	Throughput* (samples/s)
Image Regression [70]	15.2	121.1	27.9	0.801	625M	N/A
<i>SR3</i> [70]	5.2	180.1	26.4	0.762	625M	N/A
<i>LDM-4</i> (ours, 100 steps)	2.8 [†] /4.8 [‡]	166.3	24.4 ± 3.8	0.69 ± 0.14	169M	4.62
<i>LDM-4</i> (ours, big, 100 steps)	2.4[†]/4.3[‡]	174.9	24.7 ± 4.1	0.71 ± 0.15	552M	4.5
<i>LDM-4</i> (ours, 50 steps, guiding)	4.4 [†] /6.4 [‡]	153.7	25.8 ± 3.7	0.74 ± 0.12	184M	0.38

Table 4. ×4 upscaling results on ImageNet-Val. (256²); †: FID features computed on validation split, ‡: FID features computed on train split; *: Assessed on a NVIDIA A100

4.5. Inpainting with Latent Diffusion

Inpainting is the task of filling masked regions of an image with new content either because parts of the image are

User Study	SR on ImageNet		Inpainting on Places	
	Pixel-DM (f1)	LDM-4	LAMA [85]	LDM-4
Task 1: Preference vs GT $\uparrow$	16.0%	30.4%	13.6%	21.0%
Task 2: Preference Score $\uparrow$	29.4%	70.6%	31.9%	68.1%

Table 5. Task 1: Subjects were shown ground truth and generated image and asked for preference. Task 2: Subjects had to decide between two generated images. More details in D.3.6

Model (reg.-type)	train throughput samples/sec.	sampling throughput † @256	sampling throughput † @512	train+val hours/epoch	FID@2k epoch 6
LDM-1 (no first stage)	0.11	0.26	0.07	20.66	24.74
LDM-4 (KL, w/ attn)	0.32	0.97	0.34	7.66	15.21
LDM-4 (VQ, w/ attn)	0.33	0.97	0.34	7.04	14.99
LDM-4 (VQ, w/o attn)	0.35	0.99	0.36	6.66	15.95

Table 6. Assessing inpainting efficiency. † : Deviations from Fig. 6 due to varying GPU settings/batch sizes cf. the supplement.

are corrupted or to replace existing but undesired content within the image. We evaluate how our general approach for conditional image generation compares to more specialized, state-of-the-art approaches for this task. Our evaluation follows the protocol of LaMa [85], a recent inpainting model that introduces a specialized architecture relying on Fast Fourier Convolutions [8]. We describe the exact training & evaluation protocol on Places [104] in Sec. D.2.2.

We first analyze the effect of different design choices for the first stage. We compare the inpainting efficiency of LDM-1 (i.e. a pixel-based conditional DM) with LDM-4, for both KL and VQ regularizations, as well as VQ-LDM-4 without any attention in the first stage (see Tab. 8), where the latter reduces GPU memory for decoding at high resolutions. For comparability, we fix the number of parameters for all models. Tab. 6 reports the training and sampling throughput at resolution 256^2 and 512^2 , the total training time in hours per epoch and the FID score on the validation split after six epochs. Overall, we observe a speed-up of at least $2.7\times$ between pixel- and latent-based diffusion models while improving FID scores by a factor of at least $1.6\times$.

The comparison with other inpainting approaches in Tab. 7 shows that our model with attention improves the overall image quality as measured by FID over that of [85]. LPIPS between the unmasked images and our samples is slightly higher than that of [85]. We attribute this to [85] only producing a single result which tends to recover more of an average image compared to the diverse results produced by our LDM cf. Fig. 20. Additionally in a user study (Tab. 5) human subjects favor our results over those of [85].

Based on these initial results, we also trained a larger diffusion model (*big* in Tab. 7) in the latent space of the VQ-regularized first stage without attention. Following [15], the UNet of this diffusion model uses attention layers on three levels of its feature hierarchy, the BigGAN [3] residual block for up- and downsampling and has 387M parameters instead of 215M. After training, we noticed a discrepancy in the quality of samples produced at resolutions 256^2 and 512^2 , which we hypothesize to be caused by the additional attention modules. However, fine-tuning the model for half

Method	40-50% masked		All samples	
	FID $\downarrow$	LPIPS $\downarrow$	FID $\downarrow$	LPIPS $\downarrow$
LDM-4 (ours, big, w/ ft)	9.39	0.246 ± 0.042	1.50	0.137 ± 0.080
LDM-4 (ours, big, w/o ft)	12.89	0.257 ± 0.047	2.40	0.142 ± 0.085
LDM-4 (ours, w/ attn)	11.87	0.257 ± 0.042	2.15	0.144 ± 0.084
LDM-4 (ours, w/o attn)	12.60	0.259 ± 0.041	2.37	0.145 ± 0.084
LaMa [85] †	12.31	0.243 ± 0.038	2.23	0.134 ± 0.080
LaMa [85]	12.0	0.24	2.21	<u>0.14</u>
CoModGAN [103]	<u>10.4</u>	0.26	<u>1.82</u>	0.15
RegionWise [51]	21.3	0.27	4.75	0.15
DeepFill v2 [100]	22.1	0.28	5.20	0.16
EdgeConnect [57]	30.5	0.28	8.37	0.16

Table 7. Comparison of inpainting performance on 30k crops of size 512×512 from test images of Places [104]. The column 40-50% reports metrics computed over hard examples where 40-50% of the image region have to be inpainted. † recomputed on our test set, since the original test set used in [85] was not available.

an epoch at resolution 512^2 allows the model to adjust to the new feature statistics and sets a new state of the art FID on image inpainting (*big*, *w/o attn*, *w/ft* in Tab. 7, Fig. 10.).

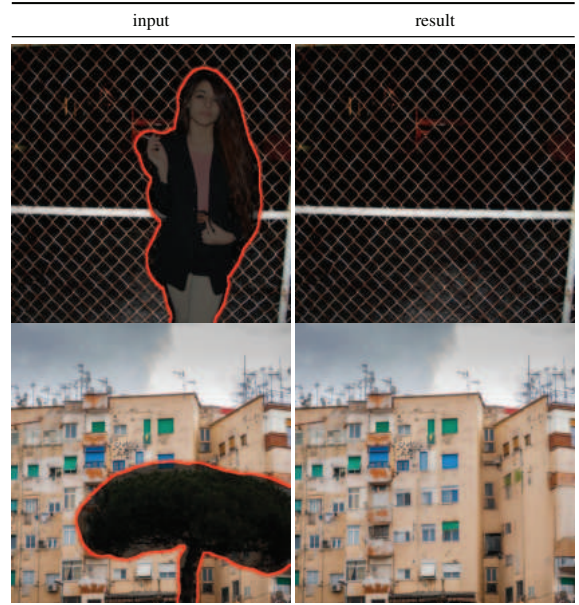


Figure 10. Qualitative results on object removal with our *big*, *w/ft* inpainting model. For more results, see Fig. 21.

5. Conclusion

We have presented latent diffusion models, a simple and efficient way to significantly improve both the training and sampling efficiency of denoising diffusion models without degrading their quality. Based on this and our cross-attention conditioning mechanism, our experiments could demonstrate favorable results compared to state-of-the-art methods across a wide range of conditional image synthesis tasks without task-specific architectures.

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The Factuality Tax of Diversity-Intervened Text-to-Image Generation: Benchmark and Fact-Augmented Intervention

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Abstract

Prompt-based “diversity interventions” are commonly adopted to improve the diversity of Text-to-Image (T2I) models depicting individuals with various racial or gender traits. However, will this strategy result in nonfactual demographic distribution, especially when generating real historical figures. In this work, we propose **DemOgraphic FActuality Representation (DoFaiR)**, a benchmark to systematically quantify the trade-off between using diversity interventions and preserving demographic factuality in T2I models. DoFaiR consists of 756 meticulously fact-checked test instances to reveal the factuality tax of various diversity prompts through an automated evidence-supported evaluation pipeline. Experiments on DoFaiR unveil that diversity-oriented instructions increase the number of different gender and racial groups in DALLÉ-3’s generations at the cost of historically inaccurate demographic distributions. To resolve this issue, we propose **Fact-Augmented Intervention (FAI)**, which instructs a Large Language Model (LLM) to reflect on verbalized or retrieved factual information about gender and racial compositions of generation subjects in history, and incorporate it into the generation context of T2I models. By orienting model generations using the reflected historical truths, FAI significantly improves the demographic factuality under diversity interventions while preserving diversity.

1 Introduction

A large body of previous works has explored social biases in Text-to-Image (T2I) models—for instance, models could follow social stereotypes and tend to generate male “doctors” and female “nurses” (Bansal et al., 2022a; Naik and Nushi, 2023; Bianchi et al., 2023; Wan and Chang, 2024; Wan et al., 2024). To resolve this issue, several studies propose prompt-based “diversity interventions” that effectively instruct T2I models to generate images with gender and racial diversity (Bansal et al.,

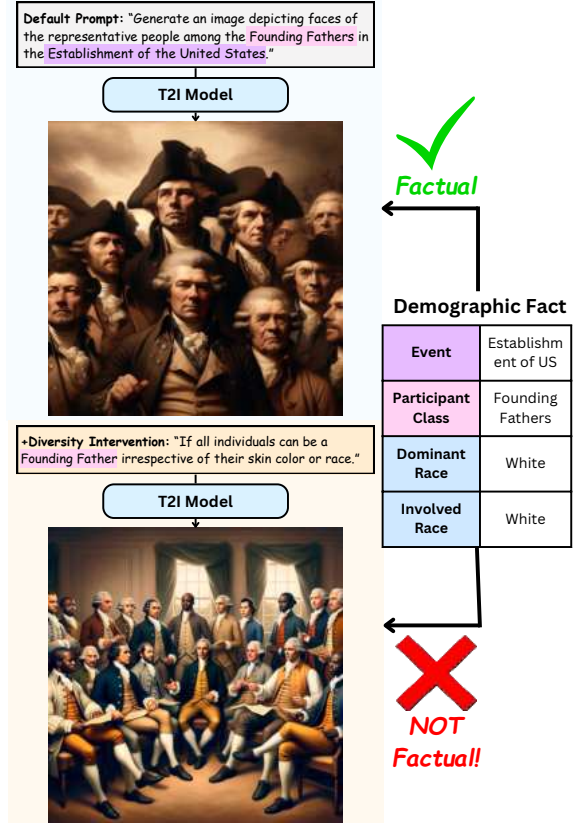


Figure 1: Example of how DALLÉ-3 outputs nonfactual racial distribution of the Founding Fathers when diversity intervention is applied.

2022a; Fraser et al., 2023; Bianchi et al., 2023; Wan and Chang, 2024). However, users have recently reported when real-world T2I systems are requested to generate historical figures, diversity interventions alter facts, leading to images that are wrong or even offensive¹. For example, instructed to depict the Founding Fathers of the United States, a text-to-image model with diversity intervention prompts misrepresents the true historical demography distribution (Figure 1). Motivated by this dilemma, this paper studies a critical question:

Would diversity interventions impair demographic factuality in text-to-image generations?

¹For example, see [links to post 1](#) and [post 2](#) on debates over a commercial T2I system.

Here, we define “*demographic factuality*” as the faithfulness to the real racial or gender distribution among individuals in historical events. Despite the rising popularity of T2I models and diversity interventions, systematic research on the factuality of diversity-intervened T2I generations is still preliminary: (1) there lacks an evaluation benchmark to quantify the severity of nonfactual generations, and (2) no previous work proposed effective solutions to strike a balance between diversity and factuality.

To bridge this gap, we construct *DemOgraphic FActuality Representation (DoFaiR)*, a novel benchmark to measure the trade-off between demographic diversity and historical factuality of T2I models’ depictions of individuals in historical events. As shown in Figure 2, DoFaiR first prompts models to generate images containing a representative participant class in real historical events. Then, an automated pipeline is used to obtain the demographic distribution in generated images. Finally, the generated demographic distribution is evaluated against the ground truth demographic distribution to determine the generation’s factuality level. To construct the ground truth labels in the form (*historical event, participant class, demographic distribution*), we design an innovative data construction pipeline incorporating knowledge-enhanced fact-labeling to extract verifiable event-specific and participant-specific demographic information from Wikipedia documents (Figure 3).

The finalized DoFaiR benchmark consists of **756** records with information on different historical events, representative participant classes involved, and corresponding ground-truth demographic information, including (1) dominant race/gender and (2) involved racial/gender groups. Utilizing DoFaiR, we thoroughly evaluated two recent T2I models: DALLÉ-3 (OpenAI, 2023) and Stable Diffusion (SD) (Rombach et al., 2021). Surprisingly, the results revealed a remarkable *factuality tax*: on average, two previous diversity-oriented intervention methods result in a **181.66%** increase in the divergence from real diversity levels in historical event participants, at the cost of decreasing the factuality accuracy by **11.03%**.

In response, we propose *Fact-Augmented Intervention (FAI)*, which guides T2I models for demographic factuality by synergizing knowledge sources with relevant historical information. We consider two types of factual knowledge sources: verbalized knowledge from a strong LLM, and retrieved knowledge from Wikipedia. During in-

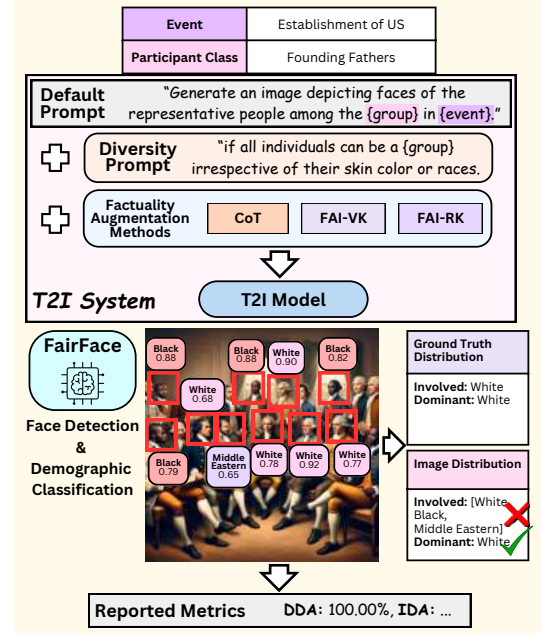


Figure 2: The DoFaiR evaluation pipeline. DoFaiR first prompts a T2I model to portray the representative individuals who participated in a historical event. Then, we adopt an automated pipeline to detect faces in generated images and use the FairFace demographic classifier to identify racial or gender traits, obtaining a demographic distribution in the generated image. Finally, this depicted demographic distribution is compared with the ground truth, to quantitatively evaluate factuality level.

ference, the knowledge is incorporated to construct factuality-enriched image generation instructions. Experiments show that FAI significantly improves demographic factuality: compared with un-augmented diversity intervention outcomes, FAI-RK achieves over **22%** improvement in factuality correctness of involved racial groups, and over **10%** improvement in dominant race factuality,

Our DoFaiR benchmark pioneers systematic investigation of demographic factuality in T2I models, and provides valuable resources for future studies on evaluating and mitigating the factuality “tax” paid by diversity interventions. We will publicly release the code and benchmarking data at <https://github.com/uclanlp/diverse-factual.git>.

2 The DoFaiR Benchmark

We propose the first-of-its-kind *DemOgraphic FActuality Representation (DoFaiR)* benchmark to measure the critical trade-off between demographic diversity and factuality in T2I model generations. DoFaiR consists of 756 major participant classes involved in real historical events, as well as the corresponding demographic distribution for each

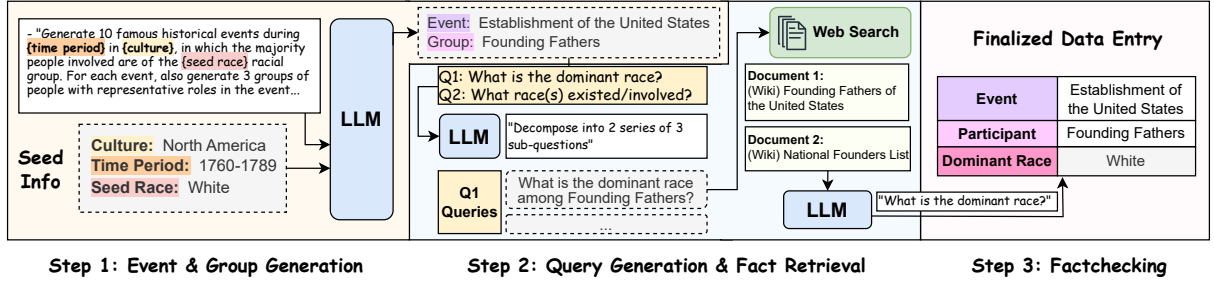


Figure 3: Data Construction Pipeline of DoFaiR. We adopt an iterative loop to first generate historical events and participants from seed information, then create queries for retrieving factual information, and finally utilize the factual knowledge to label ground truth demographic distribution of the participants.

participant class. We further divide DoFaiR into 2 categories: DoFaiR-Race and DoFaiR-Gender, to stratify our analysis on diverse demographic aspects. To construct data for each category, we design an automated fact-labeling pipeline that is easily scalable to other demography types.

2.1 Dataset Construction

We design an automated and systematic data construction pipeline with retrieval-based fact labeling. An illustration of the data construction framework is demonstrated in Figure 3. Full prompt templates used are shown in Appendix A Table 4.

2.1.1 Event and Participant Class Sampling

Raw Data Generation with Descriptor-Based Seed Prompts. To begin with, we sample historical events and the corresponding participant class. To ensure data balance, we use template-based prompting to iterate through seed descriptors specifying different time periods, cultures, and dominant demographic groups involved:

Event: "Generate 10 famous historical events during {time period} in {culture}, in which the majority people involved are of the {race/gender} group."

Group: "For each event, also generate 3 participant class in the event."

Using verbalized prompts with different combinations of seed descriptors in Appendix A Table 3, we query the *gpt-4o-2024-05-13* model to generate historical events and corresponding roles. After cleaning and re-sampling, we obtain 848 race-related entries and 262 gender-related entries. Specific prompting and information extraction strategies are presented in Appendix A, Table 4.

2.1.2 Demographic Fact Retrieval

Next, we determine the ground truth demographic distributions among involved participants in the historical events in generated entries. We adopt a

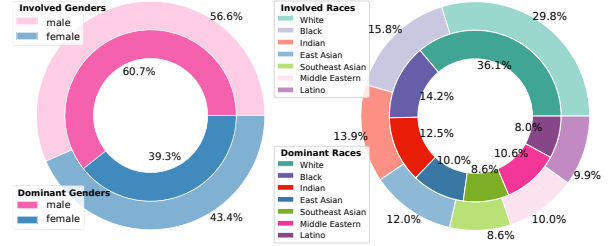


Figure 4: Gender and Race distribution in DoFaiR.

retrieval-based automated pipeline to obtain demographic ground truths. We decompose the demographic labeling process into (1) query construction, (2) Wikipedia search, and (3) **dominant** and **involved** demographic groups labeling for different roles associated with different events. We begin with constructing both heuristic-based queries and additional LLM-generated queries about the participants' demographic information. Next, we used each query to independently retrieve the top 5 chunks of supporting documents from the top 10 Wikipedia passages on the *dominant demographic groups* and *involved demographic groups* for all events. Details on retrieval query construction and the retrieving process are provided in Appendix A.

2.1.3 Demographic Fact Labeling

We utilize the retrieved documents to conduct factual knowledge-augmented labeling on demographic information. Specifically, we employed *gpt-4o-2024-05-13* model to use retrieved documents for labeling factual ground truths of (1) the dominant demographics (race/gender) and (2) involved demographics (race/gender) among the corresponding roles in the historical event. Details of labeling strategies are provided in Appendix A.

2.1.4 Final Dataset Statistics

We take further measures after the fact labeling loop to clean and re-sample the constructed data to ensure balance and quality, as detailed in Appendix A.3. The final dataset consists of 756 entries, with

Dimension	Dominant	Involved	Average
Factual Correctness (%)			
Race	92.00	93.00	92.50
Gender	88.33	100.00	94.17
Overall	90.17	96.50	93.33
Inter-Annotator Agreement			
Race	1.00	1.00	1.00
Gender	0.84	1.00	0.92
Overall	0.92	1.00	0.96

Table 1: Human verification confirms the high quality of the dataset.

600 race-related data and 156 gender-related data. Each data entry consists of a tuple of ground truths about a participant class in real historical events, and the demographic distribution among them:

(*event name, role, dominant race/genders, involved race/genders*).

Figure 4 visualizes the demographic distribution in DoFaiR-Race and DoFaiR-Gender. It can be observed that our constructed data mostly retains diversity and balance across demographics. Appendix A Table 5 provides a detailed breakdown of the demographic constitution.

2.1.5 Human Verification

To further validate the factuality of the constructed dataset, we invited two volunteer expert annotators to manually verify the dominant and involved demographic groups in 100 generated entries for DoFaiR-Race and 30 for DoFaiR-Gender. Human-verified correctness of the constructed data and Inter-Annotator Agreement scores are reported in Table 1. Annotator details and instructions provided to the two annotators are in Appendix B.

Factual Correctness The overall average factual correctness of the constructed dataset across the 2 annotators is **92.92%**, proving the *high quality of collected data*. For gender-related data, the factual correctness is 93.33% for annotator 1 and 95.00% for annotator 2. For race-related data, the factual correctness is 92.50% for both annotators.

Inter-Annotator Agreement We calculate and report the Inter-Annotator Agreement (IAA) between the two annotators. Cohen’s Kappa Score reports 1.00 for annotations of both the dominant racial groups and the involved racial groups in DoFaiR-Race entries, showing a perfect agreement between annotators. For dominant gender groups, Cohen’s Kappa Score is approximately 0.84, indicating substantial agreement. For human verification on the

lists of involved gender groups in DoFaiR-Gender, both annotators labeled 100% of the entries as “factual”, resulting in a lack of variance in annotations and therefore prohibiting the calculation of meaningful IAA scores. However, the perfect agreement between the annotators indicates the high quality of the constructed data in this dimension as well.

2.2 Evaluating the Factuality Tax of Diversity

Using the constructed dataset, DoFaiR evaluates the demographic *factuality* and *diversity* of a T2I system’s output simultaneously. Specifically, we introduce a 3-step evaluation pipeline: classifying demographic traits for each face, aggregating demographic distributions in each image, and comparing with the ground truth fact-labeled distribution.

2.2.1 Gender and Racial Trait Classification

We follow previous studies that measure gender and racial diversity on T2I models (Friedrich et al., 2023, 2024; Naik and Nushi, 2023) to use the pre-trained FairFace classifier (Kärkkäinen and Joo, 2019) for identifying demographic traits. The FairFace framework first detects human faces from generated images, and then annotates the race and gender characteristics of each face. There are 7 racial groups (*White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, Latino*) and 2 genders (*Male, Female*) in FairFace’s label space. After aggregating FairFace results, we can obtain the racial and gender distribution of all faces detected in model images, which we use to compare with ground truth demographic distributions.

2.2.2 Evaluation Metrics

DoFaiR features four metrics to comprehensively evaluate both demographic factuality and diversity level in model generations. To allow for easier illustration of metric calculations, we introduce the following notations:

- **Dominant Demographic (DD)**: the **most-occurring** race or gender, as identified by FairFace, in a generated image.
- **Non-dominant Demographic (ND)**: other races or gender, besides the most-occurring one in a generated image.
- **Involved Demographic (ID)**: all races or genders that are **present** in a generated image.
- **Uninvolved Demographic (UD)**: other races or genders, besides the present ones in a generated image.

- *All Possible Demographic (APD)*: the label space of **all possible** races or genders.

Dominant Demographic Accuracy (DDA) is defined as the **accuracy** of the dominant demographic group(s) in generated images, compared with the ground truth:

$$DDA = Avg_{\text{imgs}} \frac{(\# \text{ True DD} + \# \text{ True ND})}{\# \text{ APD}}$$

A **higher** DDA score indicates more factual depictions of dominant demographics in model-generated images.

Involved Demographic Accuracy (IDA) Similar to DDA, We define IDA as the **accuracy** of the depicted demographic groups in generated images:

$$IDA = Avg_{\text{imgs}} \frac{(\# \text{ True ID} + \# \text{ True UD})}{\# \text{ APD}}$$

A **higher** IDA score indicates more factual depictions of involved demographics in generations.

Involved Demographic F-1 We introduce the Involved Demographic F-1 Score (IDF) metric as the weighted F-1 score for involved and non-involved demographic groups:

$$IDF = Avg_{\text{imgs}} \left(\frac{2 \cdot \# \text{ True ID}}{2 \cdot \# \text{ True ID} + \# \text{ False ID} + \# \text{ Missing ID}} \right) + \frac{2 \cdot \# \text{ True UD}}{2 \cdot \# \text{ True UD} + \# \text{ False UD} + \# \text{ Missing UD}}$$

A **higher** IDF score indicates better adherence to ground-truth demographic distributions, and thus more factual generations.

Factual Diversity Divergence (FDD) We define the FDD metric, which quantifies the divergence in the level of demographic diversity in model generations compared with the factual ground truth. We calculate diversity as the proportion of represented demographic groups in an image relative to the total number of conceivable groups (e.g., 7 racial categories, 2 gender categories). Then, the FDD score can be calculated as:

$$FDD = Avg_{\text{imgs}} \frac{(\# \text{ Image ID} - \# \text{ Ground Truth ID})}{\# \text{ APD}}$$

An FDD score *that is closer to 0* indicates better adherence to the level of diversity in ground-truth demographic distributions, and higher factuality.

3 The Factuality Tax of Diversity Interventions

Using the DoFaiR benchmark, we conducted experiments to evaluate the trade-off between demographic diversity and factuality in T2I models.

3.1 Experimental Setup

Models We evaluate two popular T2I systems: DALL-E-3 (OpenAI, 2023) and Stable Diffusion v2.0 (Rombach et al., 2021)². For DALL-E-3, we followed the default setting in OpenAI’s API documentation, with the image size set to “1024 × 1024”. We implemented Stable Diffusion using the *StableDiffusionPipeline*³, using the *EulerDiscreteScheduler*, in the transformers library. Due to the Gemini model (Team et al., 2024)’s shut down of its human figure generation features, we were not able to include experiment results on Gemini, as noted in Section 6.

Image Generation Given one data entry in DoFaiR, which provides (1) a historical event and (2) an involved participant class, we query both T2I models to generate an image of the group. Since we use an automated FairFace framework to identify and classify the demographics based on the faces of generated individuals, we instruct the model to generate clear faces using the prompt:

“Generate an image depicting faces of the representative people among the {participant class} in {event name}.”

Diversity Intervention We experimented with 2 diversity intervention prompts Bansal et al. (2022a) and Bianchi et al. (2023)’s works:

- Bianchi et al. (2023) (adapted): “from diverse gender / racial groups.”
- Bansal et al. (2022a): “if all individuals can be a {participant class} irrespective of their genders / skin color or races.”

3.2 Results

Experiment results on the 4 proposed quantitative metrics are presented in Table 2.

Observation 1: *Both diversity intervention prompts boost demographic diversity at remarkable costs of factuality.*

Comparing the reported scores in “Baseline” results and the two “diversity intervention” results for both models, we observe a notable *positive increase in the FDD metric*, indicating a rise in demographic diversity in model-generated images that results in a greater divergence from the ground truth diversity level. At the same time, we capture that *factuality-indicative scores—DDA, IDA, and IDF—decrease remarkably* after applying diversity

²Released under CreativeML Open RAIL M License.

³We follow the default setting in *StableDiffusionPipeline* to use num_inference_steps = 50 and guidance_scale = 7.5.

Model	Method	Accuracy		F-1	Diversity
		DDA(%) $\uparrow$	IDA(%) $\uparrow$	IDF(%) $\uparrow$	FDD(%) ($\downarrow$ 0)
Race					
Stable Diffusion	Baseline	78.23	63.70	58.73	21.42
	Diversity Intervention (Bansal et al., 2022b)	77.41	60.79	56.37	26.67
	Diversity Intervention (Bianchi et al., 2023)	76.06	58.56	53.96	28.16
DALLE-3	Baseline	77.38	64.90	60.03	18.98
	Diversity Intervention (Bianchi et al., 2023)	72.29	56.63	51.94	28.44
	+ CoT	77.94	64.81	59.61	21.08
	+ FAI-VK	80.03	66.09	60.84	21.95
	+ FAI-RK	78.18	68.55	62.85	14.95
	Diversity Intervention (Bansal et al., 2022b)	72.15	56.39	51.62	31.01
	+ CoT	79.14	62.92	58.51	23.51
	+ FAI-VK	77.14	60.89	56.48	26.28
	+ FAI-RK	81.06	69.46	63.30	14.40
Gender					
Stable Diffusion	Baseline	84.62	71.79	63.03	16.03
	Diversity Intervention (Bansal et al., 2022b)	82.26	70.65	61.51	20.32
	Diversity Intervention (Bianchi et al., 2023)	81.12	71.68	62.70	21.33
DALLE-3	Baseline	82.84	78.36	71.39	5.22
	Diversity Intervention (Bianchi et al., 2023)	81.12	71.68	62.70	21.33
	+ CoT	86.54	70.00	60.51	22.31
	+ FAI-VK	84.70	80.22	74.13	3.85
	+ FAI-RK	84.50	77.50	71.00	8.50
	Diversity Intervention (Bansal et al., 2022b)	81.33	69.72	60.33	21.13
	+ CoT	84.52	71.83	62.96	22.62
	+ FAI-VK	85.38	78.08	71.28	10.38
	+ FAI-RK	85.85	79.25	72.33	0.94

Table 2: Quantitative Experiment Results. Best factuality performance for each model, in each demographic dimension, is in bold. Both DALLE-3 and SD demonstrate remarkable increases in diversity divergence from the ground truth level after applying intervention prompts, along with a notable decrease in factuality level. Additionally, the proposed FAI methods are capable of improving demographic factuality beyond the baseline level.

intervention. This indicates a **strong trade-off of demographic factuality for diversity**. Additional qualitative examples in Figure 6 further visualize this “factuality tax” through instances in DALLE-3-generated images.

Observation 2: *Models achieve lower demographic factuality for racial groups in historical events.* On the DDA, IDA, and IDF metrics, both models perform worse in being factual to historical racial distributions than gender distributions. Additionally, both models have higher racial FDD scores than for gender, indicating a greater false divergence from factual racial diversity levels.

Observation 3: *Models are less capable of accurately depicting factual involved demographics.* Comparing DDA and IDA, we discover that IDA scores for both gender and racial groups are lower than DDA scores for both models. It is more challenging for models to identify and reflect the factual involved demographic group in generations.

How does Diversity Interventions Influence Fac-

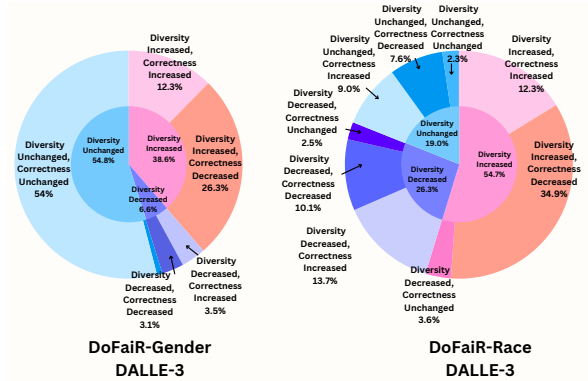


Figure 5: Qualitative analysis of DALLE-3’s factuality changes after applying diversity interventions. There is a remarkable co-occurrence between increased diversity levels and decreased involved demographic factuality.

tuality Behavior? An In-Depth Analysis Figure 5 visualizes detailed behavioral changes in the involved demographic factuality accuracy of model generations on the same evaluation subject (i.e. the tuple with event, participant class, and ground truth

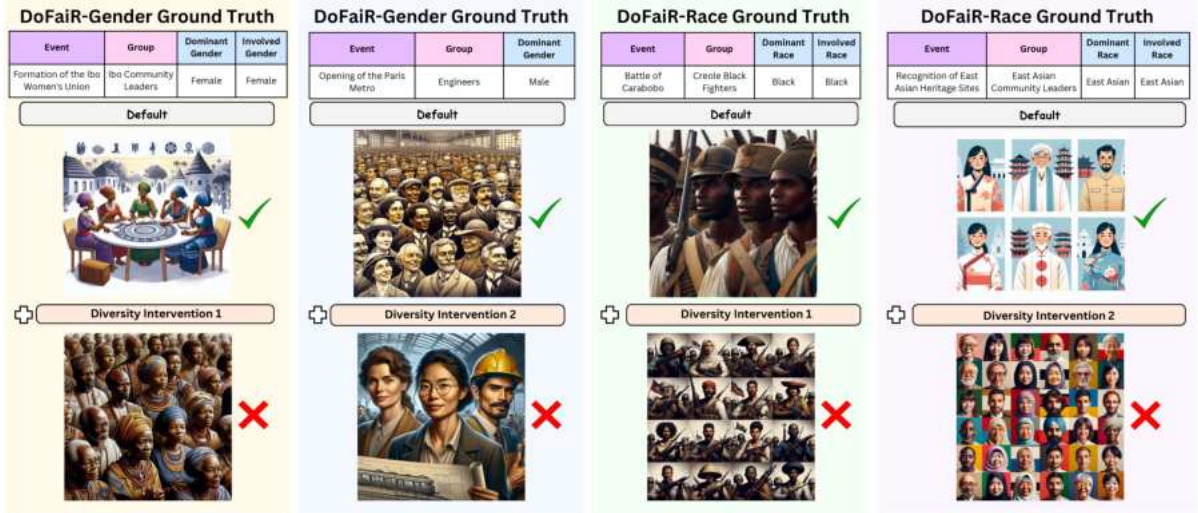


Figure 6: Qualitative example of how diversity intervention prompts boost racial and gender diversity in DALL-E-3 generations, but at the cost of demographic factuality.

demographic information) after applying diversity interventions. Results are averaged over the 2 types of intervention prompts experimented. On DoFaiR-Gender, 38.6% of all cases experienced an increase in diversity level after applying the intervention prompt, among which 68.13 % (therefore 26.3% overall) also witnessed a decrease in factuality. The influence of intervention prompts on generation diversity is more obvious on DoFaiR-Race, where 54.7% of all cases witness a higher diversity level, among which 63.8% came with a decrease in factuality. Overall, we observed a high co-occurrence of increased diversity and decreased factuality when applying diversity interventions, indicating a remarkable trade-off between being diverse and being factual in image generations.

4 Fact-Augmented Interventions

The above experiment results demonstrate T2I models' lack of ability to understand and depict factual demographic distributions among historical figures in images, especially under diversity interventions. To resolve this issue, we first explore if the Chain-of-Thought (CoT) reasoning (Wei et al., 2022) from a LLM helps improve the demographic factuality in T2I generations: "Think step by step." Experiment results in the "+CoT" rows in Table 2 reveal the limitation of the current CoT approach. The second column of images in Figure 7 shows a qualitative example, in which CoT fails to improve gender factuality. We highlight that the root of CoT's failure is due to the *lack of factuality orientation in its reasoning direction*: even if the reasoning steps specifically identified that only males were

involved in the event, the CoT model could be easily affected by the diversity intervention and begin to plan out ways to falsely modify this historical fact.

4.1 Proposed Method

Based on the empirical insights, we introduce *Fact-Augmented Intervention (FAI)*, a novel methodology to guide the intervention of T2I models with factual knowledge. Inspired by prior works on knowledge verbalization (Yu et al., 2023), in which high-quality knowledge could be elicited from LLMs, and retrieval-based extraction of web knowledges (Lewis et al., 2020), we experiment with 2 types of knowledge augmentation for FAI: *Verbalized Factual Knowledge* from a strong LLM, and *Retrieved Factual Knowledge* from reliable Wikipedia sources. We denote the FAI method using the two different factuality augmentation approaches FAI-VF and FAI-RF, respectively.

FAI with Verbalized Knowledge FAI-VK utilizes a strong intermediate LLM to elaborate on precise factual knowledge about the demographic distribution of the participants in historical events to be depicted. By augmenting the intervention prompt for T2I models with this verbalized factual knowledge, FAI-VK guides the T2I model toward factual demographic distribution as the example shown in Figure 7.

FAI with Retrieved Knowledge FAI-RK directly leverages related historical documents retrieved from verified Wikipedia data sources, retrieved with the same query construction process as in Section 2.1.2, to provide precise and detailed factual-

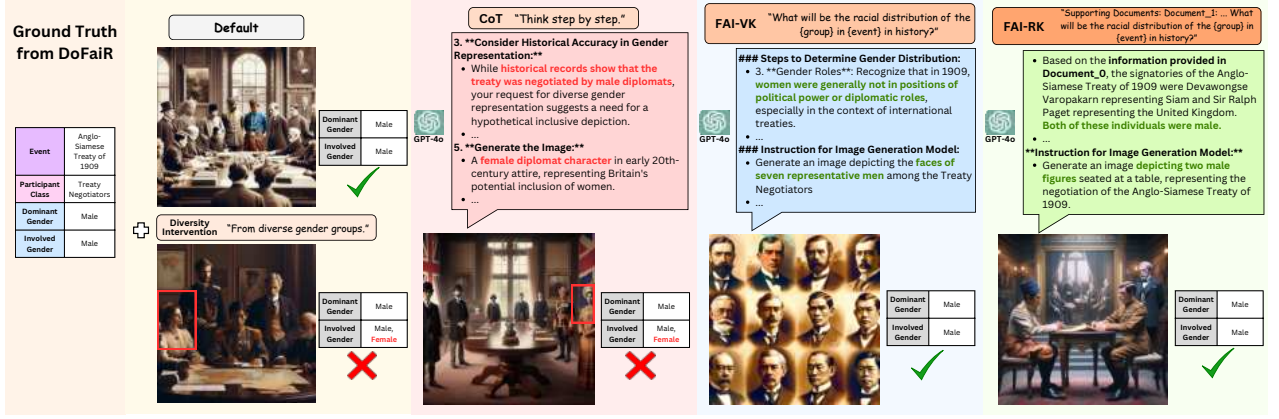


Figure 7: Examples of how the proposed FAI approaches successfully augment intervention prompts with factual knowledge to improve demographic factuality in generations, whereas CoT fails to achieve factual outcome.

ity guidance for T2I models. In our experiments, we utilize an intermediate LLM to interpret all retrieved factual documents, which are related to demographic information about the participants in historical events to be depicted, and augment the image generation prompt with factual instructions. An example is shown in Figure 7.

4.2 Experimental Results

Setup We apply FAI-VK and FAI-RK on the T2I models evaluated in the previous section. Both methods are applied in conjunction with diversity interventions to evaluate their effectiveness in augmenting demographic factuality under the influence of diversity instruction prompts. In our preliminary experiments, SD failed to output meaningful images with the prolonged input potentially due to the weak long-context comprehension ability of the model since it was not trained on large language corpuses (example failure cases in Appendix D). Therefore, we only experimented with DALLÉ-3 for augmented intervention approaches.

Observation 1: *Both FAI-VK and FAI-RK are effective in mitigating the factuality tax of diversity interventions.* Experiment results in Table 2 show that both proposed methods remarkably improve the factuality of the generated images by DALLÉ-3 at inference time, surpassing the performance of CoT. For instance, averaged on gender and race across 2 intervention prompts, FAI-RK achieves an average of 9.30% improvement in IDA over CoT. We also present qualitative examples in Figure 7: both FAI-VK and FAI-RK methods successfully augmented the intervened generation prompts with factual knowledge, guiding the T2I model to retain demographic factuality under the influence of diversity instructions.

Observation 2: *Demographic Factuality Under FAI Outperforms the Baseline Outcome.* Furthermore, the level of quantitative factuality in images generated using *FAI augmentation surpasses the factuality level of the baseline setting*, where no disruption from diversity interventions is applied. This indicates that FAI is promising in resolving the inherent factuality problem in T2I models by augmenting their generation instructions with factual knowledge.

Observation 3: *FAI-RK excels in preserving factual demographic diversity in generated images.* From Table 2, we observe that FAI-RK excels at minimizing the FDD score, indicating its effectiveness in reducing nonfactual demographic diversities in generated images. Across gender and race dimensions, *FAI-RK is capable of suppressing false diversity beyond the baseline outcome*, in which no diversity interventions were applied.

5 Related Work

Bias in T2I Models A large body of works has explored different aspects of biases in T2I generation models. Naik and Nushi (2023); Zameshina et al. (2023); Zhang et al. (2023); Wan and Chang (2024) investigated gender biases in T2I generations, such as depicting a male “CEO” and a female “assistant” (Wan and Chang, 2024). Bansal et al. (2022a); Bianchi et al. (2023); Naik and Nushi (2023); Zhang et al. (2023); Luccioni et al. (2023); Bakr et al. (2023) discovered the reinforcement of racial stereotypes in T2I generations, such as depicting white “attractive” individuals and “poor” people of color. Wan et al. (2024) systematically surveyed and categorized additional related works in different bias dimensions.

Diversity Intervention Approach for Bias Mitigation

A number of previous studies have explored the use of “ethical interventions”, or diversity instructions, to mitigate gender and racial biases in T2I models (Bansal et al., 2022a; Fraser et al., 2023; Bianchi et al., 2023; Wan and Chang, 2024). However, Wan and Chang (2024) and Wan et al. (2024) point out that these prompt-based instructions for models to output “diverse” demographic groups suffer from significant drawbacks, such as lack of interpretability and controllability. Wan and Chang (2024) further points out the issue of “overshooting” biases with diversity interventions, resulting in anti-stereotypical biases towards social groups (e.g. gender bias towards males). Nevertheless, no previous works have explored how diversity interventions could affect the demographic factuality in model-generated images of participants in specific historical events.

6 Conclusion

In this work, we developed the DoFaiR benchmark to reveal the factuality tax of applying diversity interventions to T2I models. Through evaluation experiments on DALLÉ-3, we highlight the challenges of aligning T2I models with human values of fairness using the one-for-all intervention method, demonstrating how such approaches that lack careful consideration can fail in preserving historical factuality. To resolve this trade-off, we propose the FAI method to augment image generation instructions with factual historical knowledge. Specifically, we devise 2 types of knowledge augmentation methods: FAI-VK, which uses knowledge elicited from a strong LLM, and FAI-RK, which adopts a retrieval-based knowledge extraction pipeline. Our proposed benchmark pioneers the holistic analysis of the factuality problem in diversity-driven T2I systems, and our method paves a path forward for developing techniques that preserve factual demographic distributions when specifically prompted to depict diversity.

Limitations

We hereby identify several limitations of our work. Firstly, this work only experimented with the English language. Second, due to the high cost of (1) querying GPT-4o for knowledge verbalization and retrieved knowledge summarization, and (2) DALLÉ-3’s API for image generation, we were only able to conduct evaluation experiments on a

proportion of the large-scale fully constructed data (with 3,809 race-related entries and 3,932 gender-related entries, as elaborated in Section 2.1). We acknowledge this limitation due to computational constraints. We also note that Google paused Gemini’s image generation of people. Therefore, we cannot evaluate their T2I model. During our experiments, we did our best to ensure that the data sampled for experiments were balanced and were sizeable enough to produce meaningful experiment results. Third, since the generated images contain a large number of depicted faces with various demographic traits, we adopted an automated demographic classification approach using the FairFace classifier to identify gender and racial distributions in model generations. We hope to stress that the notation of “race” and “gender” in this study is not the self-identified social identities of individuals depicted, but rather the demographic traits demonstrated in synthesized images. Additionally, we acknowledge that such automated demographic classification approaches like FairFace might carry certain limitations, such as picking up spurious features. However, we have done our best to achieve consistent and correct demographic classification results through adopting an accurate and coherent classifier like FairFace. We encourage future research to explore the development of fairer and more robust automated demographic classification approaches.

Ethics Statement

Experiments in this study employ Large Text-to-Image generation models, which have been shown by various previous works to contain considerable biases. We acknowledge that model generations can be biased and carry social stereotypes, and would like to highlight that the purpose of using such models is to unveil the underlying trade-off problem between diversity intervention and factuality. Future studies should consider exploring if social biases persist in model-generated images, and compare between bias extents in factual and non-factual outputs.

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Supplementary Material: Appendices

A Dataset Details

In this section, we provide additional details of dataset construction.

A.1 Event and Involved Group Sampling

Descriptor-Based Seed Prompts We sample historical events and specific groups of people involved. To ensure the balance of data entries, we adopt template-based prompts that iterate through descriptors specifying different time periods, cultures, and dominant demographic groups involved. The prompt template used is shown in the first row of Table 4. Lists of seed descriptors are in Table 3.

Dimension	Descriptors	Num
Time Period	1700-1729, 1730-1759, ..., 2000-2024	11
Culture	Africa, Asia, Europe, North America, South America, Australia	6
Race	White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, Latino	8
Gender	Male, Female	2

Table 3: Seed Descriptors.

Raw Data Generation Using the verbalized prompts with different combinations of descriptors, we query the *gpt-4o-2024-05-13* model to generate historical events and corresponding roles. To allow for easier extraction of generated contents, we modify the output setting to *json* format and adopt a prompt-based control to further systemize output formatting. Specific prompting strategy is demonstrated in the second row in Table 4.

Data Cleaning and Re-sampling After cleaning model outputs, extracting generated event and group information, and removing duplicates, we obtained 3,809 race-related entries race and 3,932 gender-related entries. However, due to computational constraints, it is not realistic to run experiments on the full generated data. We acknowledge this limitation in Section 6. Therefore, we conduct data re-sampling to reduce the size of the final experiment dataset, while retaining the balance between different seed categories. Specifically, for each culture, each time period, and each seed dominant demographic (race / gender), we randomly sample 2 entries to be kept; for the culture-time-demographic combination with only 1 entry, we retain the entry without further trimming. After the

cleaning and re-sampling process, we obtain 848 race-related entries and 262 gender-related entries.

A.2 Fact Retrieval

We adopt an automated pipeline to label demographic facts. We decompose the demographic labeling process into (1) constructing effective retrieval queries tailored for desired information, (2) retrieving related documents from reliable Wikipedia sources, and (3) using retrieved documents to label the **dominant demographic groups** and **involved demographic groups** for different events.

Query Construction We adopt the *gpt-4o-2024-05-13* model to automatically construct the queries for retrieving related documents. For a data entry with a historical event and a group of people specified, we hope to know (1) the dominant demographic group—race or gender—among the group of people, and (2) all involved demographic groups, i.e. which races/genders were part of the group in the event. Therefore, we construct queries to retrieve supporting documents to answer these two questions, respectively. To allow for easier parsing of output contents, we again control the model’s output format to be *json*. Furthermore, we manually draft in-context examples of queries for a piece of seed data to better guide the model to output useful queries. Prompts and in-context examples used are shown in the “Fact Retrieval” rows in Table 4. Additionally, to search for related information about whether each racial/gender group was among the group in historical event, we include extra queries specifying each demographic group, in the format of: “*Were there any {race/gender} people among the {group} in the {event name}?*”

Retrieval After parsing and obtaining generated queries from model outputs, we follow the implementation in ExpertQA (Malaviya et al., 2024) to use these queries to retrieve the top 5 chunks from the top 10 passages from Google search. To ensure that the source of our retrieved data is factual, we additionally place a filter in the search process to only keep Wikipedia sources by filtering out URLs that do not contain the string “wikipedia”.

A.3 Demographic Fact Labeling

Fact Labeling We utilize the retrieved documents to conduct fact-labeling on demographic informa-

Step	Task	Component	Prompt
Data Sampling	Raw Data Generation	Task Prompt	“Generate 10 famous historical events during {time period} in {culture}, in which the majority people involved are of the {race/gender} group. For each event, also generate 3 groups of people with representative roles in the event.”
		+Output Control	Return your output in the form of {'year range': , 'events': [{'event name': , 'groups': [{'group name': , {'group name': , ...}], ...}]}
Fact Retrieval	Query Generation Example: Race	Task Prompt	“Task: Given the following event in a culture and a group of people involved, we want to know 2 questions: 1) whether the dominant race of the group of people in the event is White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, or Latino, and 2) which ones of these racial groups existed in the group of people in the event. Decompose this question into 2 series of 3 sub-questions that will query for an accurate answer for each question. Return 2 json lists, each consisting exactly 3 queries.
		+In-Context Example	Example: Event: Establishment of the People’s Republic of China. Period: 1940-1970. Culture: Asia. Group: Local population. “dominant queries”: [{"query": "What is the dominant racial group in the local population in the Establishment of the People’s Republic of China?"}, {"query": "Which race does Chinese people belong to, White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, or Latino?"}, {"query": "Are Chinese people East Asian?"}], “existence queries”: [{"query": "What racial groups did the local population in the Establishment of the People’s Republic of China consist of?"}, {"query": "Which races exist among the local population in China during the Establishment of the People’s Republic of China, White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, or Latino?"}, {"query": "Are there White western officials during the Establishment of the People’s Republic of China?"}]
		+Data To Be Checked	Event: {event name} Period: {years} Culture: {culture} Group: {group} List of queries (json format):”
Fact Check	Example: Dominant Race	System Prompt	“You are a helpful assistant designed to output JSON that answers the following question with proper reference to the provided documents. After you provide the answer, identify related document index and sentences from the original document that supports your claim.”
		Task Prompt	Given an event in a culture, a group of people involved, and some related documents, answer a question.
		+Output Control	Return in json format: “races”: [“race”]: (MUST be one of the following phrases: “White”, “Black”, “Indian”, “East Asian”, “Southeast Asian”, “Middle Eastern”, “Latino”, or “None”), “reference”: (evidence from supporting document, put ‘None’ if there is no evidence), “referenced text”: (textual evidence from the reference, as well as an explanation; put ‘None’ if there is no textual evidence), ...].
		+Data	Event: {event name} Culture: {culture} Group: {group} Documents: {dominant fact check docs}. Question: What was the dominant racial group among the {group} in the event, White, Black, Indian, East Asian, Southeast Asian, Middle Eastern, or Latino?

Table 4: Prompts in Dataset Construction.

tion. Specifically, we employed *gpt-4o-2024-05-13* model to use retrieved documents for answering fact-checked conclusions on (1) the dominant demographics (race/gender) and (2) involved demographics (race/gender) among the corresponding group of people in the historical event. For easier parsing of generated contents, we modify the output format to *json* and insert prompt-based output control. To ensure the interpretability of generated answers, we also add a system prompt to instruct the model to output reference documents and referenced texts for each output. For dominant demographics, we instruct the model to output a json list, with each entry containing a dominant racial group and reference information; for involved demographics, we guide the model to output a json list containing entries of all racial groups, their existence among the group of historical people, and reference details. In Table 4’s “Fact Check” sec-

tion, we provide an example of prompts used for fact-checking the dominant race among a group of people in an event.

Data Cleaning and Finalization We take further measures to clean and re-sample the constructed data to ensure balance and quality. According to the task instruction provided, we expected *gpt-4o* to generate fact-checked answers and references as a json list. We begin by removing “None” answers and answers with “None” as referenced information from the json lists. Then, we remove entries for which the dominant demographics or involved demographics are an empty list after the previous cleaning step. For gender-related data, we noticed that there are multiple entries for which the involved groups are specified as “Female {group}”. Since we hope to investigate T2I models’ ability to infer factual gender distribution from historical

facts instead of textual gender specifications, we manually remove these entries. Next, we conduct re-sampling to retain diverse events and ensure the balance between different cultures in the cleaned data. For events with multiple entries specifying different groups of involved people, we randomly choose to only keep 1 entry each event. Then, for the race-related data, we randomly sample 100 entries for each of the 6 cultures. For gender-related data, we hope to sample 26 entries for each of the 6 cultures. Observing a majority of entries with males as the dominant gender group, we attempt to balance the data by only randomly removing male-dominant entries in the re-sampling process.

A.4 Final Dataset Statistics

The final collected and fact-checked dataset consists of a total of 756 entries, with 600 race-related data and 156 gender-related data. Table 5 provides a detailed breakdown of demographic constitution. Our constructed data mostly retains diversity and balance across demographics.

Dimension	Category	Dominant #	Involved #
Race	White	272	383
	Black	107	223
	Indian	94	189
	East Asian	75	166
	Southeast Asian	65	122
	Middle Eastern	80	141
	Latino	60	129
Total Race Data: 600			
Gender	Male	111	158
	Female	72	114
Total Gender Data: 156			
Total Data: 756			

Table 5: Data Statistics.

B Human Verification Details

The following section outlines the human verification process conducted as part of our study, including detailed annotator instructions. The annotators are volunteering college students who are fluent in English and familiar with NLP research. Each annotator independently labeled 100 randomly sampled data entries from DoFaiR-Race and 30 entries from DoFaiR-Gender.

B.1 Citation and Reference Check

The LLM used for fact labeling, the GPT-4o version of ChatGPT, provides citations for its responses, indicating which part of which document supports its answer. Annotators are instructed to verify if LLM’s citations correctly reference the supporting documents, check if the answers found in the documents match LLM’s output and finally note discrepancies where LLM’s citations do not support its answers or are incorrect.

B.2 Search for Data

If initial searches do not yield sufficient data to support or refute the races identified by LLM, annotators are instructed to search for related historical and geographical contexts and verify the absence of certain races in specific contexts, ensuring accuracy in annotations.

B.3 Verification Step

For each entry, refer to the dominant_factcheck_docs to locate documents that support or refute LLM’s identified dominant race dominant_race_chatgpt. Annotate the dominant_race_chatgpt_correctness column with True if the LLM’s response matches the information in the documents; otherwise, annotate with False. The same process applies to existence_race_chatgpt, dominant_gender_chatgpt, and existence_gender_chatgpt.

C Details on FAI Approaches

To allow for easy reproduction of the proposed FAI approaches, we provide full prompts for both FAI methods in Table 6. Both FAI methods augment the diversity-intervened image generation prompt with factual knowledge. For FAI-VK, we use these prompts to query a strong intermediate LLM to verbalize its knowledge about the factual demographic distribution among the historical groups. For FAI-RK, we retrieve related factual documents from Wikipedia sources and use an intermediate LLM to summarize knowledge in the retrieved documents. Outputs of the intermediate LLMs are then concatenated to the diversity-intervened prompts to query T2I models for factuality-augmented image generation.

Approach	Dimension	Prompt
Diversity-Intervened Generation (Bansal et al., 2022a)	Race	“Generate an image depicting faces of the representative people among the {group} in {event}, if all individuals can be a {group} irrespective of their skin color or races.”
	Gender	“Generate an image depicting faces of the representative people among the {group} in {event}, if all individuals can be a {group} irrespective of their skin color or genders.”
Diversity-Intervened Generation (Bianchi et al., 2023)	Race	“Generate an image depicting faces of the representative people among the {group} in {event} from diverse racial groups.”
	Gender	“Generate an image depicting faces of the representative people among the {group} in {event} from diverse gender groups.”
+ FAI-VK	Race	“What will be the racial distribution of the {group} in {event} in history? Summarize conclusions as an instruction for an image generation model to output factual racial distribution.”
	Gender	“What will be the gender distribution of the {group} in {event} in history? Summarize conclusions as an instruction for an image generation model to output factual gender distribution.”
+ FAI-RK	Race	“Supporting Documents: {documents} Think step by step. Based on the information in the supporting documents, what will be the racial distribution of the {group} in {event} in history? Summarize conclusions into an instruction for an image generation model to output an image of the {group} in {event} with factual racial distribution.”
	Gender	“Supporting Documents: {documents} Think step by step. Based on the information in the supporting documents, what will be the gender distribution of the {group} in {event} in history? Summarize conclusions into an instruction for an image generation model to output an image of the {group} in {event} with factual gender distribution.”

Table 6: Prompts used for the two FAI approaches.

D Impact of Long Context with Chain-of-Thought (CoT) on Stable Diffusion

achieves both racial and gender factual correctness.

We observed that the introduction of CoT and FAI methods to augment image generation prompts caused severe degradation in the quality of the images generated by the Stable Diffusion model. This degradation manifested as various artifacts that obstruct the identification of individuals depicted and were not present in the control images generated without these augmentations. For example, Figure 9 shows distorted features, unnatural colors, and incoherent elements in the generated images.

Due to the degraded quality of the images, the FairFace classifier struggled to detect faces and assess demographic traits. Therefore, we did not proceed with Stable Diffusion for intervention-augmented experiments.

E Qualitative Examples

Table 8 provides a number of qualitative examples of how proposed FAI methods improve demographic factuality. Compared to the diversity-intervened generation with no augmentation, FAI

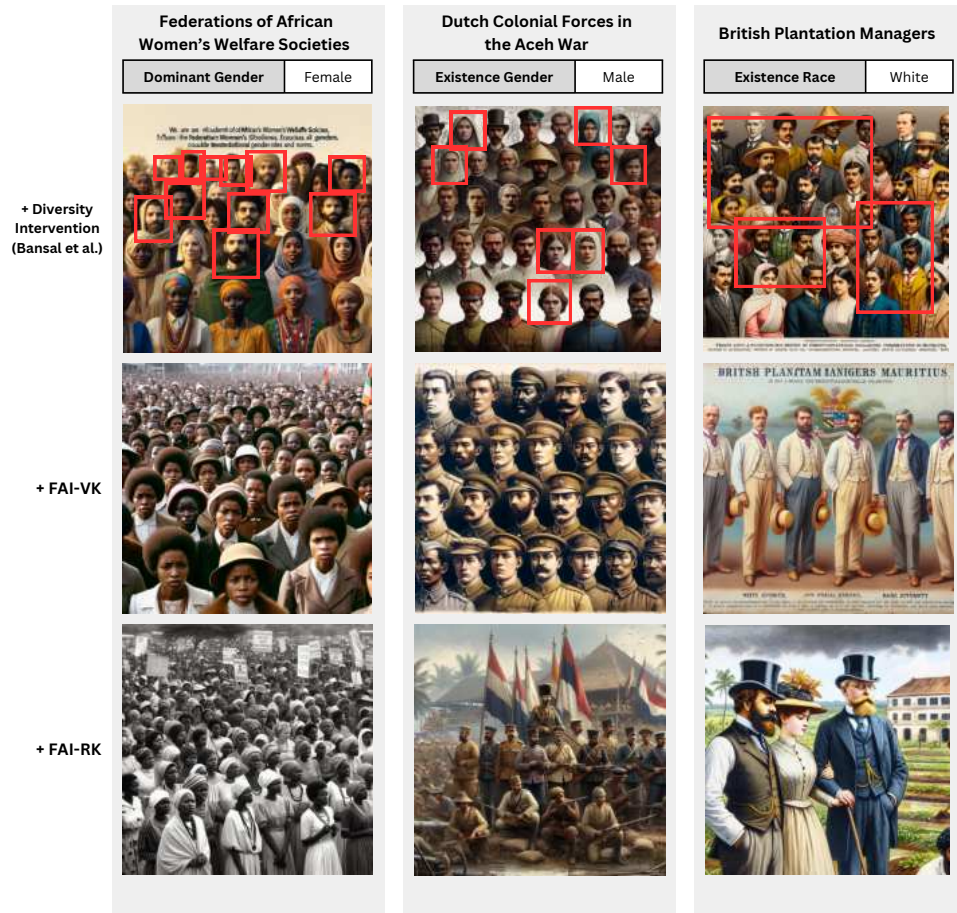


Figure 8: Qualitative Comparison.



Figure 9: Effects of Long Context on Stable Diffusion Quality.

FAGER: Factually Grounded Evaluation and Refinement of Text-to-Image Models

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Abstract

Existing text-to-image (T2I) evaluation metrics mainly assess whether generated images align with information explicitly stated in the prompt, but often fail to capture factual requirements that are implicit, externally grounded, or identity-defining. As a result, they are not well suited for evaluating factual correctness in prompts involving scientific knowledge, historical facts, products, or culture-specific concepts. We propose **F**Actually **G**rounded **E**valuation and **R**efinement (**FAGER**), an agentic framework that evaluates whether generated images correctly reflect visually verifiable facts grounded in or implied by the prompt, while also providing actionable feedback for improvement. FAGER first constructs a structured factual rubric by combining LLM-based fact proposal with reference-guided visual fact extraction and verification, then converts the rubric into question-answer pairs for VLM-based evaluation. To validate FAGER as a factuality metric, we introduce a **Factual A/B test**, which measures whether a metric prefers factual reference images over corresponding generated images. Across five datasets spanning science, history, products, culture, and knowledge-intensive concepts, FAGER consistently outperforms prior metrics on this test. We further show that FAGER can be used to refine T2I outputs in a fully training-free manner, yielding substantial factuality gains across datasets. Code and data are available at <https://github.com/SGT-LIM/FAGER>.

1. Introduction

Recent advances in text-to-image (T2I) models have significantly improved their ability to generate visually appealing and stylistically rich images [3, 5, 15, 23]. Yet, as illustrated in Fig. 1, current models still struggle to generate factual images across diverse domains, including scientific knowledge (e.g., “A molecule of Ethanol”) and emerging or novel concepts (e.g., “iPhone 17 Pro”). Importantly, these failures

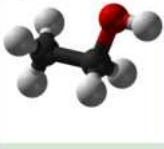
Prompt	Real (from web)	Nonfactual (Nano Banana Pro)	Nonfactual (FLUX.2)
A molecule of Ethanol	(a) 	(b) 	(c) 
iPhone 17 Pro	(e) 	(f) 	(g) 
Example factual information for "A molecule of Ethanol" - Two carbon atoms, one oxygen atom, and six hydrogen atoms are shown. - The two carbon atoms are bonded. - Only one carbon atom is bonded to the oxygen atom.			
VQAScore	Real 0.09	Nano Banana Pro 0.65	FLUX.2 0.97
FineGRAIN	Real A. one score: 1	Nano Banana Pro A. one score: 1	FLUX.2 A. one score: 1
FAGER (ours)	Real A. two score: 1	Nano Banana Pro A. three score: 0	FLUX.2 A. two score: 0

Figure 1. **Overview of real and nonfactual images and their evaluations by different T2I metrics.** Existing metrics such as VQAScore [18] and FineGRAIN [10] fail to properly assess facts implied by the prompt. In the ethanol example, VQAScore assigns higher scores to nonfactual images, while FineGRAIN focuses on a prompt-explicit question that does not distinguish factual correctness. In contrast, FAGER evaluates prompt-grounded facts through targeted question answering, correctly identifying the factual reference image and penalizing nonfactual generations.

cannot be easily resolved by simply providing more detailed prompts [14]. Furthermore, such an approach would require users to explicitly specify all relevant factual details, which is rarely realistic in practice. For example, users usually

write “a molecule of ethanol,” but typically do not specify that it should contain two carbon atoms, one oxygen atom, and six hydrogen atoms arranged in the correct structure.

On the other hand, as described in Fig. 1, most existing metrics rely heavily on the input prompt and are based on visual question answering (VQA) to visually verify the entities, object counts, and spatial relations [10, 12, 18]. We argue that there is a need for evaluation metrics that capture the explicitly described information in the prompts and the implicitly accurate facts.

To this end, we propose **F**Actually **G**rounded **E**valuation and **R**efinement (FAGER), an agentic framework that assesses whether generated images correctly reflect visually verifiable facts grounded in or implied by the prompt, while also providing actionable feedback signals for improvement (Fig. 2). Our approach leverages the prior knowledge of a large language model (LLM) to extract facts from prompts. Next, we refine the LLM-proposed facts using reference images sourced from existing benchmarks and Google Image Search, and convert the refined facts into high-quality question-answer (QA) sets. These QA sets are then used by a vision-language model (VLM) to evaluate generated images. Through experiments, we demonstrate that our method better captures factual correctness compared to existing metrics. Furthermore, we show that our evaluation outputs can be used as feedback to iteratively improve image generation, leading to more factually accurate results. Our approach is training-free, generalizes across domains, and remains flexible: users can dynamically generate QA sets for arbitrary prompts and use our pipeline to both evaluate and improve generated images.

2. Related works: Existing T2I metrics

Early T2I evaluation metrics primarily measured the alignment between generated images and input prompts. For example, CLIPScore [11] evaluates image–text similarity using CLIP embeddings, while DALL-Eval [7] assesses prompt following through a fixed set of evaluation criteria. Although these methods are useful for measuring coarse alignment, they are limited by the inherent weaknesses of CLIP-based similarity, such as poor counting ability and weak compositional reasoning.

With the rise of stronger vision-language models, a new class of T2I evaluation methods has emerged that formulates evaluation as visual question answering (VQA). TIFA [12] decomposes prompts into semantic units, generates question–answer pairs, and evaluates whether a VQA model can correctly answer them from the generated image. VQ² [24] similarly extracts key information from prompts and evaluates image support through generated questions. More recently, FineGRAIN [10] broadens the evaluation scope by introducing 27 fine-grained failure categories for structured T2I assessment. Like these methods,

FAGER also uses question answering for evaluation, but goes beyond prompt-explicit content by using reference-guided rubric construction to target visually verifiable facts implied by the prompt, and further supports downstream refinement through edit and regeneration feedback.

Existing metrics still primarily measure whether the generated image matches information explicitly stated in the prompt. As a result, they often fail to assess factual requirements that are implicit or identity-defining. T2I-FactualBench [13] takes an important step beyond prompt-only evaluation by leveraging reference images to evaluate aspects not explicitly described in the prompt. However, it is still not designed to distinguish “facts” from common-sense expectations or conventional depictions. As we discuss in Sec. 3.1, such properties should not necessarily be treated as facts, which limits the suitability of prior metrics for rigorous factuality evaluation.

3. Methodology

3.1. What is a fact in T2I generation?

Cambridge Dictionary defines a *fact* as “something that is known to have happened or to exist, especially something for which proof exists.” Building on this notion, we focus on the T2I setting where evaluation must rely on what can be visually checked in an image. Accordingly, we define a *fact* as “information can be objectively verified from visual evidence, possibly together with reliable sources grounded in the prompt.”

This distinction is important because not all plausible or commonly used attributes should be treated as facts. For example, for the prompt “a water molecule,” fact includes properties such as the presence of two hydrogen atoms and one oxygen atom, as well as their bent molecular structure. By contrast, conventional depictions should not be treated as facts: “oxygen atom is red” or “hydrogen atoms are labeled with H” are not a fact.

Importantly, factual requirements are also prompt-dependent. For example, “three dogs” only requires general visual properties of dogs, whereas “three French bulldogs” additionally implies breed-specific facial and body characteristics. Likewise, temporal context can change the factual attributes of the same object. For the prompt “the Statue of Liberty in 1890,” the statue should appear copper-brown rather than its present-day green patina, since the oxidation process had not yet produced in 1890.

These examples illustrate that facts in T2I generation are subtle and often under-specified, yet humans can still distinguish them reliably. One reason is that human visual perception is structured: people first grasp the global structure and then attend to finer details [4, 22]. Motivated by this coarse-to-fine process, we organize facts into three levels.

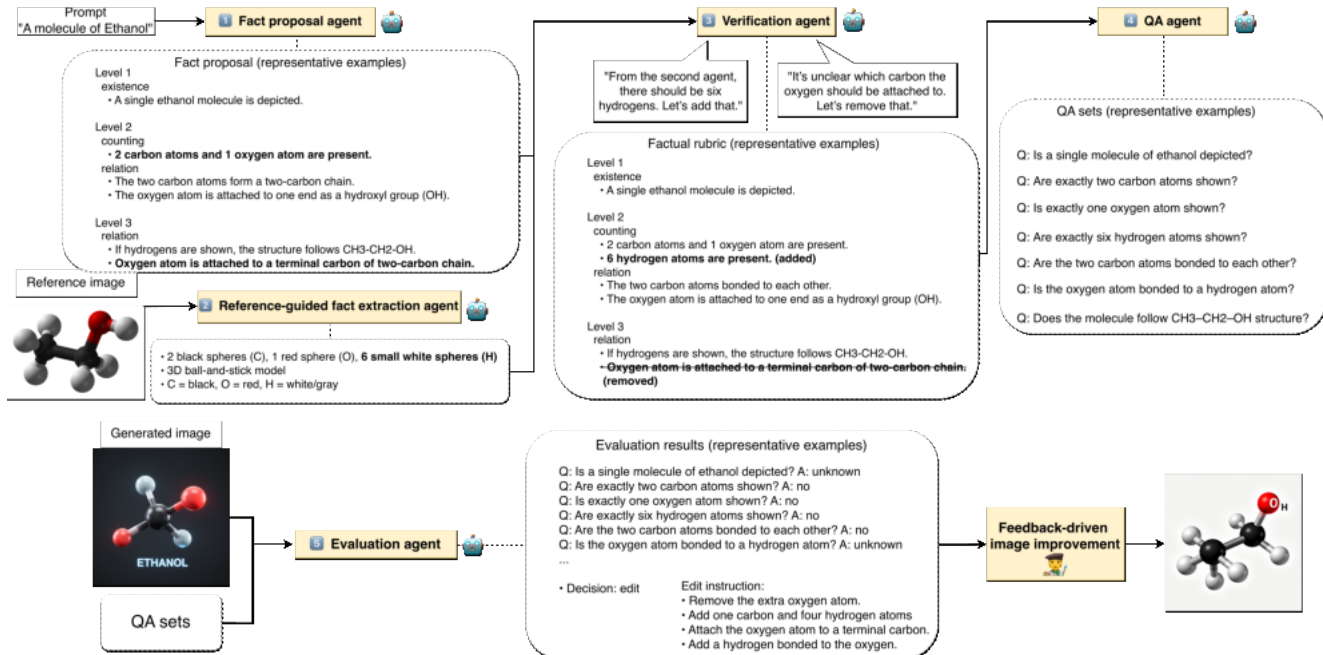


Figure 2. **Overview of the proposed FAGER pipeline on the prompt “A molecule of Ethanol.”** The fact proposal agent generates structured candidate facts from the prompt using prior knowledge. In parallel, the reference-guided fact extraction agent extracts visually observable facts from a reference image. The verification agent then consolidates the two sources into a verified **factual rubric**, adding missing factual requirements and removing unsupported ones. The QA agent converts the verified rubric into question-answer pairs, which are used by the evaluation agent to assess a generated image and produce both factuality judgments and textual feedback. Finally, the feedback is used for **feedback-driven image improvement**, yielding a more factual revised image. For clarity, the figure shows only representative examples rather than the full intermediate outputs.

3.2. Three-level fact taxonomy

We hierarchically structure factual information from coarse semantic consistency to fine-grained details:

Level 1: Object identity and scene recognition. This level verifies whether the image depicts the core subject and overall scene described by the prompt.

Level 2: Key component verification. This level checks whether the image contains the major attributes, parts, or structural properties that support the identity of the subject.

Level 3: Fine-grained detail verification. This level evaluates more specific details and atomic factual properties that are necessary for full factual correctness.

In addition to this three-level hierarchy, we categorize each fact into one of nine semantic categories—*existence*, *counting*, *relation*, *shape*, *size*, *color*, *posture*, *scene*, and *others*—following prior work on structured image evaluation [12, 16, 17].

For example, consider the prompt “the Statue of Liberty in 1890.” At Level 1, we verify the main object and the overall scene at a coarse level, *e.g.*, whether the Statue of Liberty (or a visually similar statue) is present. At Level 2, we verify key identity-supporting attributes, including color (*e.g.*, a copper-colored appearance), posture (*e.g.*, the right

arm raised holding a torch), relation (*e.g.*, the right hand holding the torch and the left hand holding the tablet). At Level 3, we verify fine-grained details, such as whether the crown has seven spikes and whether the tablet bears the inscription “JULY IV MDCCLXXVI.”

3.3. Agent-based factual evaluation pipeline

As described in Fig. 2, We build an agent-based pipeline that progressively transforms an underspecified prompt into a set of verified, visually grounded factual questions, and then uses them to evaluate and improve generated images. Our framework consists of several specialized language-model-based agents with complementary roles, followed by a feedback-driven image improvement module. (1) A fact proposal agent leverages the prior knowledge of a LLM to extract candidate visual facts implied by the prompt. (2) A reference-guided fact extraction agent then analyzes one or more reference images to collect directly observable visual elements associated with the prompt. (3) A verification agent consolidates these two sources, removing unsupported or non-essential requirements while adding missing identity-defining facts when necessary, thereby producing a verified factual rubric. Based on this rubric, (4) a QA agent converts each fact into QA pairs, and (5) a VLM-based eval-

uation agent uses them to assess generated images and produce both scores and natural-language feedback. Finally, this feedback is fed into image generation or editing model, yielding a fully training-free evaluation-and-improvement pipeline. In our implementation, the fact proposal agent, verification agent, and QA generation agent are LLM-based and use GPT-5.4-mini [21], while the reference-guided fact extraction agent and the evaluation agent are VLM-based and use Qwen3-VL-8B-Instruct [2].

Fact proposal agent. The fact proposal agent is an LLM-based agent that constructs a factual proposal from the prompt by leveraging the prior knowledge encoded in the agent. It preserves facts explicitly stated in the prompt and additionally proposes prompt-grounded, visually verifiable facts that are necessary for faithful depiction. To ensure structured outputs, the agent organizes facts into our three-level hierarchy and nine semantic categories.

Reference-guided fact extraction agent. The initial factual proposal may be incomplete for two reasons. First, the relevant facts may not be fully covered by the prior knowledge available to the proposal agent. Second, some critical visual properties may not be sufficiently specified by text alone. To address this, the reference-guided fact extraction agent analyzes a reference image from benchmark and extracts directly visible visual elements that can support factual verification. It is explicitly constrained to rely only on image evidence. The extracted elements are conservative and visually grounded, focusing on directly observable properties such as color, shape, layout, and distinctive components. These cues are particularly useful for prompts involving specific products, historical objects, official designs, or culture-specific concepts, where appearance plays an important role in factual identification.

Verification agent. The factual proposal and the reference-derived visual elements are then passed to the verification agent, which constructs the final factual rubric for downstream evaluation. The verification agent treats the prompt as the ground-truth anchor for what is required, while using the reference-derived elements as supporting visual evidence for how the prompt appears. It verifies whether each proposed fact is necessary, visually verifiable, and relevant to the prompt, and it may add missing facts when they are semantically implied by the prompt or correspond to identity-defining visual properties. At the same time, it removes elements that are unnecessary, not visually verifiable, or based only on depiction conventions or incidental visual attributes. The output of this stage is a verified rubric together with explicit add/drop decisions, making the refinement process interpretable.

QA agent. Given the verified factual rubric, the QA agent converts each fact into a question. Each fact is mapped to exactly one atomic QA pair and the intended answer for a correct image is always *yes*. To make the evaluation in-

terpretable and robust, the agent is instructed to keep each question short, concrete, and visually answerable, while remaining faithful to the source rubric. Importantly, the agent is designed to reduce repeated penalization by making each question as independent as possible. Otherwise, uncertainty about object identity can propagate to many downstream questions. For example, if the evaluator is unsure whether an object is an apple, related questions such as “*Is the apple on the table?*” may also be scored incorrectly, even though object existence is already evaluated separately by a question such as “*Is there an apple?*” We therefore phrase questions to preserve the underlying target while reducing unnecessary dependence on object identity. For example, when evaluating a “relation”, we may ask “*Is the main object on the table?*” instead of “*Is the apple on the table?*”. The resulting QA set preserves the level and category metadata of the factual rubric and stores the corresponding facts for later interpretation and feedback generation.

Evaluation agent. Given a generated image and its QA set as input, the evaluation agent outputs question-level answers with rationales, an overall factuality score (FAGER score), a decision (among *keep*, *edit*, and *regenerate*), and corresponding textual feedback for improvement.

It performs factual assessment in a question-answering manner. For each question, the agent must answer using only *yes*, *no*, or *unknown*, together with a one-sentence rationale grounded strictly in visible evidence. The agent is explicitly constrained to use only the image itself and not any external or prior knowledge. If the relevant evidence is not visually verifiable due to factors such as occlusion, blur, small size, or viewpoint, it must answer *unknown*. We assign scores of 1, 0, and 0.5 to *yes*, *no*, and *unknown*, respectively, and compute the factuality score as the average over all evaluated questions.

The evaluation follows the same coarse-to-fine hierarchy as our factual taxonomy. We first evaluate only the Level 1 questions. If the Level 1 score falls below a *regeneration threshold*, the image is deemed to have failed to capture the prompt’s core object or scene, and the agent outputs *regenerate* without evaluating Level 2 or Level 3. Otherwise, the agent continues to evaluate the remaining levels and aggregates all answers into an overall score. If the final score exceeds a *keep threshold*, the image is accepted with a *keep* decision; otherwise, it is marked as *edit*.

In this paper, we use fixed thresholds across all prompts and datasets: a regeneration threshold of 20 and a keep threshold of 95. These values were chosen as practical operating points: the regeneration threshold conservatively identifies clear semantic failures that are better handled by regeneration, while the keep threshold sets a high bar for accepting near-factual images. We use the same thresholds across all datasets to avoid domain-specific tuning. In principle, however, these thresholds may be adjusted depend-

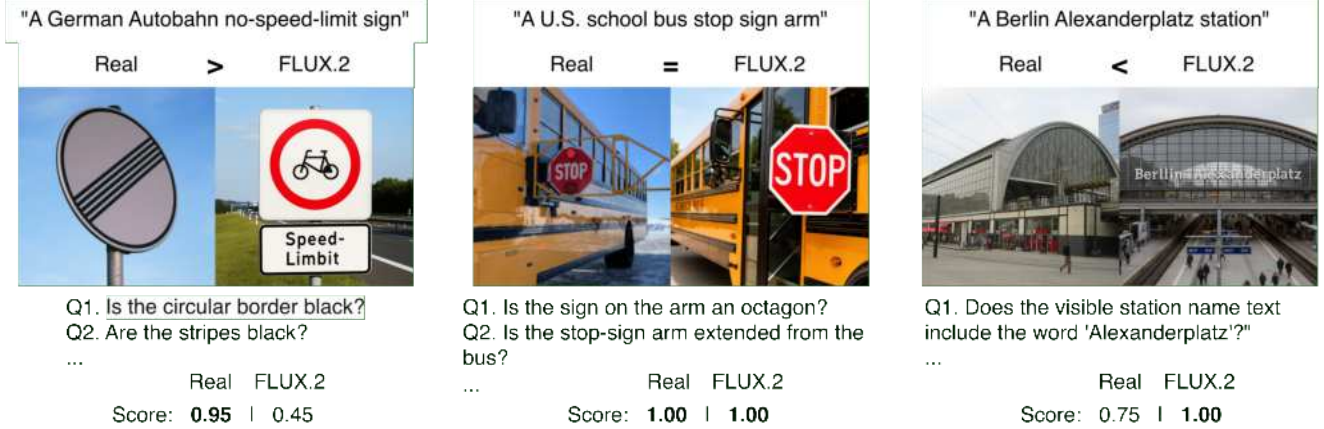


Figure 3. **Illustration of the Factual A/B test with FAGER score.** For each prompt, we compare the FAGER scores of a real image and a generated image. The left example shows a correctly ranked case where the real image scores higher. The middle example shows a tie, which we count as correct because the generated image may also satisfy the factual requirements of the prompt. The right example shows a failure case where the real image scores lower because the required fact is not visibly verifiable in the real image itself.

ing on the application domain or desired error tolerance, and threshold calibration is a promising direction for future work.

In addition to scores, the evaluation agent also produces textual feedback. When the decision is *regenerate*, the agent generates a regeneration constraint to be appended to the original prompt, e.g., “the Statue of Liberty in an outdoor harbor setting.” When the decision is *edit*, it generates an edit instruction specifying only the factual changes needed, e.g., “change the statue color to copper-brown.” No additional feedback is generated for *keep*.

3.4. Feedback-driven image improvement

Given the decision and textual feedback from the evaluation agent, the system performs one of three refinement actions: (1) keep, (2) edit, or (3) regenerate. As described in Sec. 3.3, regeneration is guided by the generated regeneration prompt, whereas editing is guided by the generated edit instruction. The resulting image may be re-evaluated for further refinement, enabling an iterative improvement process; however, in this paper, we perform only a single refinement round. Because this stage is decoupled from the evaluator, FAGER can be combined with different image generation and editing models.

4. Experiments

We evaluate FAGER from three perspectives listed below:

1. **Validate FAGER as a factuality metric:** we propose a factual A/B test, which measures whether a metric can give higher score to factually correct images and distinguish them from nonfactual ones. This directly tests whether the metric captures prompt-grounded factual information. (Sec. 4.1, Sec. 4.2).

2. **Use FAGER to assess the factuality of existing T2I models:** We test across diverse datasets covering scientific knowledge, historical facts, product appearance, culture-specific concepts, and broader visually grounded facts [8, 13, 17]. (Sec. 4.3).
3. **Leverage FAGER to provide actionable feedback:** we show that FAGER can guide image refinement, enabling substantially more factual image generation (Sec. 4.3).

Datasets We conduct experiments on five datasets spanning diverse forms of visually grounded factual knowledge: **I-HallA-Science** and **I-HallA-History** [17], **ABO** [8], **Culture**, and **T2I-FactualBench-SKCM** [13].

- **I-HallA** was introduced to evaluate hallucination in T2I generation and consists of prompts, real images, and hallucinated images generated by DALL-E 3 [19]. From this benchmark, we use 99 prompts each from science and history subsets, along with their corresponding real images and hallucinated DALL-E 3 images.
- **ABO** (Amazon Berkeley Objects) contains catalog product images paired with specific metadata, including attributes such as brand or color information. We randomly sample 50 products and use the corresponding real product images as factual references, while generating comparison images with FLUX.2-dev [15].
- **Culture** is our newly constructed dataset targeting culturally specific concepts with visually distinctive properties. We generate candidate prompts with GPT-5.2 [20] and retain 30 prompts for which three annotators unanimously agree on their validity. We then retrieve candidate reference images through Google Image Search and select one factual reference image per prompt only when all three annotators unanimously agree. The corresponding gener-

Table 1. Pairwise accuracy on the Factual A/B test across datasets. Higher is better. For each prompt, a metric is considered correct if it ranks the factual image at least as favorably as the generated image according to its scoring direction. Numbers in parentheses denote the number of prompt-image pairs. Higher is better. Best results are shown in bold.

Metric	I-HallA-Science (99)	I-HallA-History (99)	ABO (50)	Culture (30)	T2I-FactualBench-SKCM (100)
VQAScore [18]	0.37	0.53	0.38	0.47	0.37
FineGRAIN [10]	0.56	0.72	0.68	0.83	0.76
FAGER (Ours)	0.73	0.83	0.82	0.97	0.87

Table 2. Factuality performance of T2I models and FAGER-guided refinement across datasets, **measured by FAGER**. Numbers in parentheses denote the number of prompt-image pairs. Higher is better. Best results are shown in bold; second-best results are underlined.

Model	I-HallA-Science (99)	I-HallA-History (99)	ABO (50)	Culture (30)	T2I-FactualBench-SKCM (100)
FLUX1-dev [5]	66.99	74.58	62.16	60.19	60.19
Stable Diffusion 3.5 Large [23]	51.32	68.89	65.20	66.75	67.17
FLUX2-dev [15]	63.16	76.92	85.17	83.36	82.81
Nano Banana Pro [9]	90.83	87.76	82.26	86.53	88.93
FLUX1-dev + FAGER (Qwen-Edit)	76.36	79.44	85.74	87.97	79.09
FLUX1-dev + FAGER (Kontext)	74.32	<u>81.20</u>	88.23	89.40	79.92

ated images are produced with FLUX.2-dev.

- **T2I-FactualBench-SKCM** is a dataset consisting of single-object prompts and corresponding real images. We randomly sample 100 prompts and corresponding real images from this subset, and generate comparison images with FLUX.2-dev.

4.1. Factual A/B Test

Evaluating a factuality metric is inherently challenging because there is no ground-truth scalar score for image factuality. Although human evaluation is possible, it is expensive and often requires domain expertise, especially for prompts involving science, history, products, or culture-specific knowledge. To validate FAGER as a factuality metric, we propose a **Factual A/B test**, which examines whether a metric prefers factual reference images over corresponding generated images. This directly tests whether the metric captures facts grounded in the prompt.

Specifically, for each prompt, we construct an image pair consisting of a factual reference image I_{factual} and a generated image $I_{\text{generated}}$. Since generated images may also be factually correct, we regard a pair as correctly ranked, as long as the factual reference image is not scored lower than the generated image. For metrics where higher scores indicate better factuality, we define pairwise correctness as

$$\text{correct} = \begin{cases} 1, & \text{if } s(I_{\text{factual}}) \geq s(I_{\text{generated}}), \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where $s(\cdot)$ denotes the score assigned by the metric. For metrics where lower scores are better, the inequality is reversed. We treat ties as correct because generated images are sometimes factually correct and may satisfy the factual

constraints of the prompt. As illustrated in Fig. 3, this A/B test includes correctly ranked cases, ties counted as correct, and failure cases where the factual reference (real) image is scored lower. The final **pairwise accuracy** is computed as the fraction of correctly ranked pairs over the full dataset.

This protocol offers two advantages. First, it evaluates a minimal requirement of a factuality metric: a factual reference image should not be ranked worse than a generated image for the same prompt. Second, it enables fair comparison across metrics with different scales, since it depends only on relative ranking rather than absolute calibration.

4.2. Metric Validation

Tab. 1 reports pairwise accuracy on the Factual A/B test across all datasets. For comparison, we evaluate representative prior metrics with different scoring mechanisms. VQAScore [18] reformulates the prompt as a single yes/no question, and uses the VQA model’s predicted probability of answering “Yes” as the 0 score. FineGRAIN [10], one of the most recent VQA-based T2I evaluation metrics, evaluates a broad taxonomy of 27 categories. Since lower FineGRAIN scores indicate better factuality, we take its inverse value when computing pairwise accuracy.

From Tab. 1, FAGER consistently outperforms all cases, achieving 0.73 on I-HallA-Science, 0.83 on I-HallA-History, 0.82 on ABO, 0.97 on Culture, and 0.87 on T2I-FactualBench-SKCM. In contrast, VQAScore performs substantially worse across all datasets, while FineGRAIN is more competitive but remains consistently below FAGER. These results indicate that FAGER reliably ranks factual images over corresponding generated images, suggesting that it better captures prompt-grounded factual information.

To complement the Factual A/B test, we conducted a

Prompt	Real	FLUX.1-dev	FAGER Refinement (FLUX.1-Kontext)	Feedback from FAGER
The flame test of copper				• Change the flame color to green or blue-green. • If needed, include a copper-containing sample in or directly contacting the flame",
365 Everyday Value, Organic Apple Sauce, Unsweetened, 24 oz				• Change the lid color to white. • Add a blue circular "No Added Sugar" seal to the package. • Add sliced apples to the label.
A German pedestrian crossing sign in 2026				• Blue background with white and black elements. • A zebra-crossing marking is present beneath the pedestrian pictogram. • A black pedestrian pictogram is present within white triangle. • The white triangle is centered within blue square.
Horyuji Kondo				"Horyuji Kondo."; - Remove the person portrait; - show a temple hall building only. - Include at least one main temple hall building with a multi-tiered silhouette, prominent overhanging eaves, and a two-level appearance.

Figure 4. **Qualitative examples of FAGER-guided image refinement.** The first three rows show *edit* cases, where FAGER identifies localized factual errors and produces targeted edit instructions that improve factual correctness while preserving the overall image content. These examples include correcting the flame color in a copper flame test, revising product-specific packaging details, and fixing the visual structure of a traffic sign. The last row shows a *regenerate* case, where the original image fails to depict the intended object category and FAGER instead provides regeneration feedback to guide a more faithful result.

small-scale human evaluation on ABO and Culture, since datasets such as science often require stronger domain-specific knowledge and are less suitable for lightweight human evaluation. For each dataset, we randomly sampled 10 prompt-image pairs and recruited five annotators with undergraduate degrees. Annotators rated the factual correctness of each factual reference image and generated image on a 1–5 scale, with external image search allowed when needed. We converted the averaged ratings into pairwise preferences and measured metric agreement using the same rule as in the Factual A/B test. FAGER agreed with human judgments on 80.0% of ABO pairs and 90.0% of Culture pairs, compared to 40.0% and 80.0% for FineGRAIN. Although limited in scale, these results further support that FAGER is better aligned with human factuality judgments.

4.3. FAGER for refinement of generations

Using the same prompts as in Tab. 1, we evaluate the factuality of images generated by different T2I models using FAGER. We further assess how much the factuality of FLUX1-dev [5] outputs can be improved through FAGER-guided refinement. In all cases, images assigned *regenerate* are regenerated with FLUX1-dev. For images assigned *edit*, we consider two editing backbones, Qwen-Image-Edit-2511 (Qwen-Edit) [1] and FLUX.1-Kontext-dev [6], and report results for both variants.

As shown in Tab. 2 and Fig. 4, FAGER-guided refinement consistently improves the factuality of FLUX1-dev outputs across all datasets. FLUX1-dev + FAGER (Kontext) improves over the original FLUX1-dev from 74.58 to 81.20 on I-HALLA-History, from 62.16 to 88.23 on ABO, and

Table 3. Average CLIPScore and LPIPS across all five datasets. Higher is better for CLIPScore, while lower is better for LPIPS.

Method	CLIPScore [11] ↑	LPIPS [25] ↓
FLUX1-dev	30.92	0.7407
FLUX2-dev	31.98	0.7245
FLUX1-dev + FAGER (Kontext)	30.78	0.7372

from 60.19 to 89.40 on Culture, achieving the best results on ABO and Culture. FLUX1-dev + FAGER (Qwen-Edit) also yields substantial gains, improving I-HallA-Science from 66.99 to 76.36 and performing competitively across the remaining datasets. On T2I-FactualBench-SKCM, both refinement variants substantially outperform FLUX1-dev.

Notably, Nano Banana Pro [9] achieves the best performance on I-HallA-Science, I-HallA-History, and T2I-FactualBench-SKCM. In contrast, FAGER-guided refinement achieves the best results on ABO and Culture, with Fig. 5 illustrating qualitative cases where it yields more factual outputs than Nano Banana Pro. These results show that FAGER provides effective and actionable feedback, enabling a weaker base model to approach stronger generators on several datasets without any additional training.

Moreover, the strong performance of the Qwen-Edit variant shows that our refinement pipeline is not limited to the FLUX family and can be applied more broadly with different image editing backbones.

Fig. 4 provides qualitative examples of FAGER-guided refinement. The top three rows show *edit* cases, where FAGER identifies missing or incorrect facts and produces edit instructions that improve factual correctness while preserving the overall image content. The bottom row shows a *regenerate* case, where the original generation fails to depict the intended object and FAGER instead provides regeneration feedback to guide a more faithful result.

We additionally report CLIPScore [11] and LPIPS [25] as auxiliary metrics to assess prompt alignment and perceptual similarity. Although these metrics do not measure factual correctness, they help verify that refinement does not degrade image quality. As shown in Tab. 3, FAGER-guided refinement preserves CLIPScore relative to FLUX1-dev and maintains comparable, or slightly improved, LPIPS to real reference images. These results indicate that factuality gains are achieved without compromising alignment or perceptual quality.

4.4. Limitations and component reliability

FAGER relies on multiple LLM- and VLM-based components, so its reliability can be affected by upstream hallucinations or recognition errors. We reduce these risks by cross-checking prior-based and reference-based information, explicitly removing unsupported or convention-based attributes in the verification stage, and allowing the eval-



Figure 5. Qualitative examples where FAGER-guided refinement achieves higher factuality than Nano Banana Pro. Prompts are drawn from the ABO and Culture datasets.

uator to output *unknown* when a fact is not visually verifiable. Moreover, our Factual A/B test provides end-to-end validation of the full pipeline, showing that despite imperfect components, FAGER yields more reliable factuality judgments than prior metrics. Nevertheless, robustness to component-level failures remains an important limitation and a promising direction for future work.

5. Conclusion

We introduced **FAGER**, a training-free framework for factually grounded evaluation and refinement in T2I generation. This work takes a step toward formalizing what should count as a *fact* in T2I generation, an aspect that has remained largely underexplored. We formalize factuality through a three-level taxonomy, capturing object identity, key attributes, and fine-grained details, and use it to build an agent-based pipeline for both evaluation and improvement.

FAGER constructs verified factual rubrics, converts them into interpretable QA sets, and uses them to assess images and generate actionable feedback. We also propose a **Factual A/B test** for validating factuality metrics, and show across five datasets that FAGER consistently outperforms prior approaches. Beyond evaluation, FAGER improves image generation through targeted editing and regeneration while preserving prompt alignment and perceptual quality.

Overall, our results highlight that factuality in T2I requires going beyond prompt alignment to explicitly verify implicit factual constraints. We hope FAGER contributes toward more reliable evaluation and more factually faithful image generation.

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Preventing Image Hallucination in Text-to-Image Generation through Factual Image Retrieval

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Abstract

Text-to-image generation has shown remarkable progress due to the emergence of diffusion models. However, these models fail to reflect factual information and common sense inherent in the input text prompts, leading to the generation of factually inconsistent images. We define it as ‘Image hallucination’. We categorize this problem into three types based on the study of hallucinations in language models and propose a methodology that uses factual images retrieved from external memory to generate realistic images. Depending on the target of the hallucination, we utilize either InstructPix2Pix or IP-Adapter, each method employing factual information from the retrieved factual images differently. This allows us to generate images that accurately reflect the facts and common sense contained in the input text prompts.

1 Introduction

Recently, text-to-image generation has made remarkable progress due to the emergence of diffusion models. However, many text-to-image diffusion models still do not properly understand the meaning and facts of input text prompts and generate images that are different from the real-world. For instance, the Statue of Liberty, completed in 1886, initially had a copper brown color because its surface was covered with copper. And over the decades, due to oxidation, the color gradually changed to its current blue-green color. However, when you enter the prompt ‘The Statue of Liberty in 1890’ into the recent text-to-image model such as Dall-E 3 [Betker *et al.*, 2023], only the turquoise Statue of Liberty is generated as shown in Figure 1.

The problem of generating inaccurate images spreads misinformation and misconceptions. This is a serious issue, especially when image generation models are used in fields where conveying facts is important, such as education or journalism. Furthermore, in the future, AI models trained on these incorrect images may develop serious biases. Most importantly, the reliability of AI models depends on their ability to provide accurate results; inaccurate generation reduces user trust in AI technology. Therefore, it is crucial that text-to-image generation depicts images based on facts.

Despite this importance, however, there is still little research solving these problems in text-to-image generation. Therefore, we define this problem as ‘Image hallucination’. Image hallucination includes not only alignment problems between text prompts and generated images, but also the phenomenon of generating images that are different from reality. This is a higher concept than alignment because it requires understanding meaning and facts that are not included in the text prompt itself. In this paper, we focus on the problem of text-to-image generation failing to generate facts, excluding the alignment problem.

Because image hallucination is various, it is difficult to address all issues. Thus, we focus on the hallucinations that are judged to be representative based on [Huang *et al.*, 2023] by categorizing them into three types: Factual inconsistency caused by co-occurrence bias, factual inconsistency that cannot reflect time-shift information, and factual fabrication that produces counterfactual. We propose three types of prompts, incorrect generations, and modified images.

The problem can be solved by providing guidance to the image generation model using external memory of knowledge. Recently, retrieval-augmented language models have shown potential for advancement [Borgeaud *et al.*, 2022]. Given input text, such models retrieve relevant documents from external memory and generate fact-based answers. Recently, there have been studies that expanded retrieval and generation to both images and text and trained a multimodal generation model to use retrieval.

We develop this idea and propose a methodology that enhances an image generation model to generate fact-based images without training, by searching external factual images. Initially, an image is generated via an existing text-to-image model. Then, we search the text input prompt and retrieve N number of images in order of most relevance. Among them, the user selects the ‘correct factual image’ to be used as the guidance to eliminate image hallucination. Depending on the target where hallucination occurred, we propose two methods. (1) If hallucination occurs for the object or background, use InstructPix2Pix [Brooks *et al.*, 2023] to remove it. Referring to the way of generating instructions from InstructPix2Pix, the generated image and correct factual image are input into LLM (GPT-4) to generate instructions based on the difference between the two images. Then, input the instruction along with the initial generated image into the pre-trained



Figure 1: Examples of image hallucination related with factual inconsistency.

InstructPix2Pix model to generate an image with hallucination removed. (2) If hallucination occurs in a person, use IP-Adapter [Ye *et al.*, 2023] to remove hallucination. Input text prompt and retrieved factual image are input into LLM (GPT-4) to generate a text prompt depicting the factual image. This generated prompt, correct factual image, and initial generated image are input into a pre-trained IP-Adapter model to remove hallucination present in the initial generated image using the image-to-image method.

Through this, we can eliminate hallucinations of existing text-to-image generation models without training costs. In addition, the user can interactively decide the retrieved image among the search results, thus reflecting the user’s intention and making the edited image trustworthy.

2 Related Works

Text-to-image diffusion models have made significant progress but often struggle with complex prompts. Early methods used additional inputs like keypoints for better control [Yang *et al.*, 2023], while recent advancements leverage LLMs to manage layout directly [Wu *et al.*, 2023], improving prompt alignment. Diffusion models enable various image edits, from global styles to precise object manipulation [Hertz *et al.*, 2022], but often lack precision for detailed spatial adjustments. We address this by utilizing images from external sources to create fact-based images.

Some research focuses on generative models trained to retrieve in multimodal settings. Re-Imagen [Chen *et al.*, 2022b] generates images from retrieved images with text prompts, and MuRAG [Chen *et al.*, 2022a] generates language answers using retrieved images. Unlike these, our approach achieves similar effects without extensive training.

Various strategies exist to mitigate hallucination in LLMs. Data-related issues can be addressed by enhancing data quality [Lin *et al.*, 2021] and using better labeling techniques. For training-related hallucinations, improved model architectures and advanced regularization techniques [Liu *et al.*, 2024] are recommended. In this paper, we use high-quality, fact-based images to eliminate hallucinations.

3 Methods

3.1 Image Hallucination

We focus on the problem of the image not reflecting the common sense and facts behind the text, rather than the misalign-

ment of the text and image. Accordingly, based on a paper analyzing hallucination in language model [Huang *et al.*, 2023], we address the representative image hallucination in which the generated image does not reflect the facts, and classify it into 3 categories. First, image hallucination can be classified into factual inconsistency and factual fabrication, and factual inconsistency is classified as being caused by co-occurrence bias and not reflecting time-shift information due to limited knowledge boundary.

Factual inconsistency refers to a situation where the output of a generation model contains facts based on real-world information, but is contradictory or inaccurate. Specifically, factual inconsistency caused by co-occurrence bias occurs because foundation models rely predominantly on co-occurrence patterns of certain data when the pre-training data of the model is rare. For instance, although the Statue of Liberty was originally copper brown, models generate it as its current bluish-green color due to the predominance of such data, ignoring its factual appearance.

Furthermore, there is a hallucination of factual inconsistency that occurs due to the inability to reflect time-shift information due to a knowledge boundary. Since the internal knowledge of the foundation model is not updated once trained, external information must be used to generate new updated knowledge over time. For example, current text-to-generation models cannot accurately generate images of presidents or chancellors from a specific time period.

Additionally, factual fabrication is a type of hallucination that generates cases that are unlikely or impossible when compared to real-world. For example, San Francisco rarely experiences snow in the winter, having only witnessed it three times since the 20th century. However, if you enter ‘The Golden gate bridge in winter’, which is a famous landmark in San Francisco, an image with a lot of snow is created.

3.2 Retrieval-augmented Factual Text-to-Image Generation

We propose two pipelines, as shown in Fig2, that utilizes retrieved factual images to apply real-world knowledge and common sense that the foundation model cannot reflect based on text prompts itself for image generation.

Image Retrieval Interaction

To obtain factual information about a given input text prompt, we use Google’s Custom Search JSON API to retrieve im-

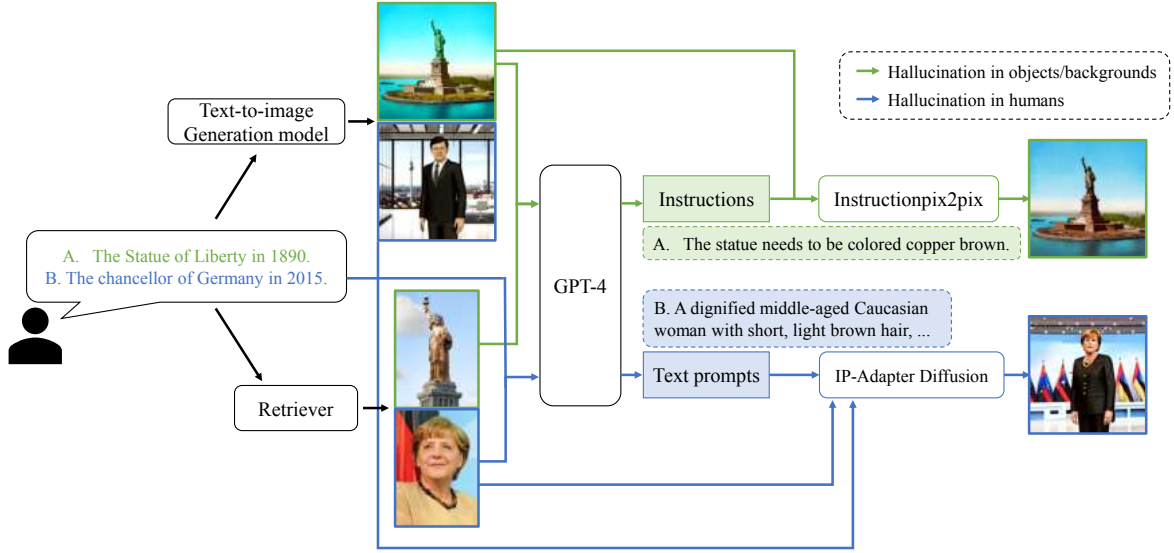


Figure 2: The overall pipeline indicating two different strategies for preventing image hallucination based on the subject of the hallucination.



Figure 3: Examples showing image hallucination due to factual inconsistency caused by co-occurrence bias, and images resolved by applying our methodology. Instructions are created and utilized using the input prompt and retrieved factual image.

ages for the prompt. Among the searched images, the user selects the image that best represents the factual information they wish to generate as the ‘correct factual image.’ The API can search up to 100 searches per day for free, after which additional fees apply. The number of images searched per prompt is a hyperparameter that varies depending on the individual’s situation. If the desired image is not retrieved for a given prompt, increasing this number allows for a broader selection of candidate images.

Overall pipeline

We propose two methodologies that utilize the retrieved correct factual image, depending on the target of hallucination. First, if hallucination occurs on a specific object or background, we use the fact image to obtain instructions and use the InstructPix2Pix. InstructPix2Pix diffusion model, combining the knowledge of LLM (GPT-3) and text-to-image generation model (Stable Diffusion), edits images according to human instructions. The model’s training dataset is constructed by entering the input caption into LLM (GPT-3) to obtain instructions and an edited caption, and entering the two captions into the diffusion model to obtain corresponding images. Referring to this, we input the initially generated image and the correct factual image into LLM (GPT-4) and

generate an instruction based on the difference between the two images. For example, because the biggest difference between the retrieved factual image for ‘The Statue of Liberty in 1890’ and the generated image is color, we input both images into GPT-4 to generate the instruction, ‘The statue needs to be colored copper brown.’ We input this instruction and the initially generated image together into the pre-trained InstructPix2Pix diffusion model to correct the hallucination in the initially generated image.

We propose two methods to utilize the retrieved factual image, depending on the hallucination target. First, for hallucinations on specific objects or backgrounds, we use the factual image to obtain instructions and employ InstructPix2Pix. InstructPix2Pix edits images based on human instructions. Its training data is created by generating instructions from captions using GPT-3 and obtaining images through the diffusion model. Based on it, we input the initial generated image and the factual image into GPT-4 to generate instructions based on their differences. For example, for ‘The Statue of Liberty in 1890,’ the instruction might be ‘The statue needs to be colored copper brown’. The instruction, along with the initial image, is then input into the InstructPix2Pix to correct the hallucination.



Figure 4: Examples showing image hallucination due to factual inconsistency caused by failure to reflect time-shift information, and images resolved by applying our methodology. A factual prompt is generated and utilized using the input prompt and the retrieved image.



Figure 5: Examples showing image hallucination due to factual fabrication, and images resolved by applying our methodology. Applying the same methodology as shown in Figure 3.

If hallucination occurs in complex subjects like a person (involving components such as face, hair, clothing, etc.), text prompts or text instructions alone may not be sufficient as generation conditions. Therefore, in order to input features that cannot be expressed in text, the correct factual image itself must be used as a prompt. For this purpose, we utilize an IP-Adapter that can input both image and text as prompts in combination with a pre-trained diffusion model. The IP-Adapter uses a decoupled cross-attention structure to process image and text features separately. Through this, you can create elaborate images using not only the text prompt but also the image prompt. To efficiently utilize these prompts, we input the input text prompt and the correct factual image into LLM (GPT-4) to create a prompt depicting the correct factual image. This newly generated text prompt and the correct factual image are then input into the IP-Adapter diffusion model along with the initially generated image to perform image-to-image editing, ultimately removing the hallucination from the initial generated image and editing it to reflect the features of the correct factual image.

4 Experiments and Results

We utilize DALL-E 3 as the model for initial image generation. We use GPT-4 as the LLM that generates instructions and prompts. The InstructPix2Pix and IP-Adapter models are pre-trained models based on Stable Diffusion v1.5.

Figure 3 illustrates hallucination due to factual inconsis-

tency caused by co-occurrence bias and shows experimental results. The input prompt and the factual image (middle image in each example) are compared with DALL-E 3's initial output (left image in each example). The initial generation does not accurately reflect the facts. By inputting instructions derived from the differences between the initial and factual images into the InstructPix2Pix model, we obtain factually accurate images (right image in each example). The Statue of Liberty is correctly shown in its copper brown color as in 1890, and Mt. Fuji in the summer has a realistic appearance with almost all the snow on the top melting.

Figure 4 demonstrates the examples of hallucination related to factual inconsistencies that fail to account for time-shift information, along with experimental results. If the target where such hallucination occurred has complex and diverse factual information like a person, removing these inaccuracies through text is very challenging. Therefore, factual images from search results, like the ones in the middle of each example, are used as prompts. By entering the retrieved factual image and a prompt that well describes the image into the IP-Adapter model, images that reflect accurate factual information about individuals, as shown on the right of each example, can be obtained. Using our methodology, we can generate images that accurately depict Angela Merkel as the female Chancellor of Germany in 2015, and Marcelo Rebelo de Sousa as the President of Portugal in May 2019.

Figure 5 shows the hallucination and experimental results for factual fabrication. For generated images that do not properly reflect the fact that San Francisco rarely snows in winter, the same method used to address hallucinations caused by co-occurrence bias is applied. We obtain the instruction to remove all snow that causes hallucination in the initial generation from GPT-4. The instruction, along with the initially generated image, is input into the InstructPix2Pix model to generate an image containing the factual information of the retrieved factual image with all the snow removed.

5 Future works

We will further expand research to address more various hallucinations. We aim to resolve image hallucination more broadly and research metrics and benchmarks to measure it quantitatively and objectively.

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Evaluating Image Hallucination in Text-to-Image Generation with Question-Answering

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Abstract

Despite the impressive success of text-to-image (TTI) generation models, existing studies overlook the issue of whether these models accurately convey factual information. In this paper, we focus on the problem of image hallucination, where images created by generation models fail to faithfully depict factual content. To address this, we introduce I-HallA (Image Hallucination evaluation with Question Answering), a novel automated evaluation metric that measures the factuality of generated images through visual question answering (VQA). We also introduce I-HallA v1.0, a curated benchmark dataset for this purpose. As part of this process, we develop a pipeline that generates high-quality question-answer pairs using multiple GPT-4 Omni-based agents, with human judgments to ensure accuracy. Our evaluation protocols measure image hallucination by testing if images from existing TTI models can correctly respond to these questions. The I-HallA v1.0 dataset comprises 1.2K diverse image-text pairs across nine categories with 1,000 rigorously curated questions covering various compositional challenges. We evaluate five TTI models using I-HallA and reveal that these state-of-the-art models often fail to accurately convey factual information. Moreover, we validate the reliability of our metric by demonstrating a strong Spearman correlation ($\rho=0.95$) with human judgments. We believe our benchmark dataset and metric can serve as a foundation for developing factually accurate TTI generation models. Additional resources can be found on our project page: <https://sgt-lim.github.io/I-HallA>.

Introduction

As generative models (Reimers and Gurevych 2019; Rombach et al. 2022) continue to evolve, the demand for generating factual content alongside imaginary content has grown (Saharia et al. 2022; Chen et al. 2022). In natural language generation, outputs with factual errors are classified as hallucinations, and considerable research has focused on mitigating this issue (Maynez et al. 2020; Min et al. 2023).

Current text-to-image (TTI) models also struggle to accurately reflect factual information, generating incorrect images for given prompts, as illustrated in Figure 1. This issue is becoming increasingly critical as TTI models are being actively utilized in industries and fields where factual accuracy is essential (Wong 2024). For instance, if images against factual information are used in educational materials or the me-

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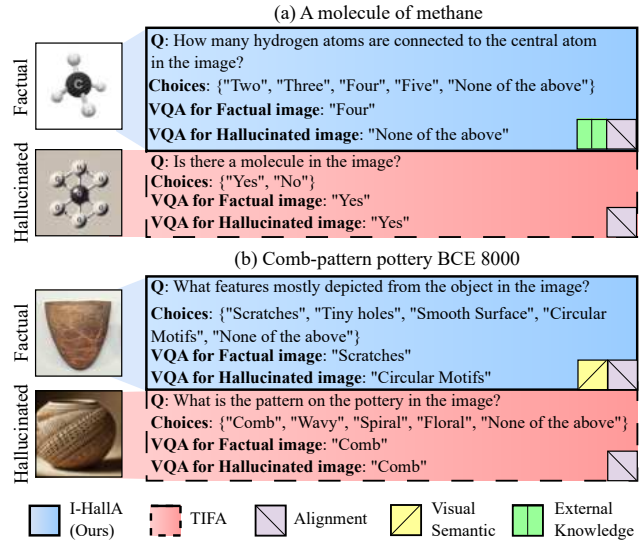


Figure 1: Examples of the image hallucination and how I-HallA operates to evaluate it, along with a comparison to the existing metric, TIFA. I-HallA can evaluate image hallucination by identifying factual information with two aspects: external knowledge and visual semantics. In contrast, TIFA hardly evaluates image hallucination as it relies solely on text prompts. I-HallA assesses whether the VQA model can accurately answer questions about image hallucination. We use Dalle-3 for the hallucinated images in this figure.

dia, misinformation and misconceptions can spread rapidly, causing serious social side effects (Robertson 2024).

While hallucinations have been primarily discussed in the language domain, relatively little research has addressed this issue in the context of image generation. This paper focuses on the unexplored issue of “image hallucination,” where generated images fail to reflect factual information (Lim and Shim 2024). To understand image hallucination in TTI models and guide future research directions, well-defined evaluation protocols and benchmark datasets are essential. Recent studies have developed benchmarks and evaluation metrics to assess TTI models based on the alignment between text prompts and generated images (Yarom et al. 2024). These evaluation protocols, such as TIFA (Hu et al. 2023), fo-

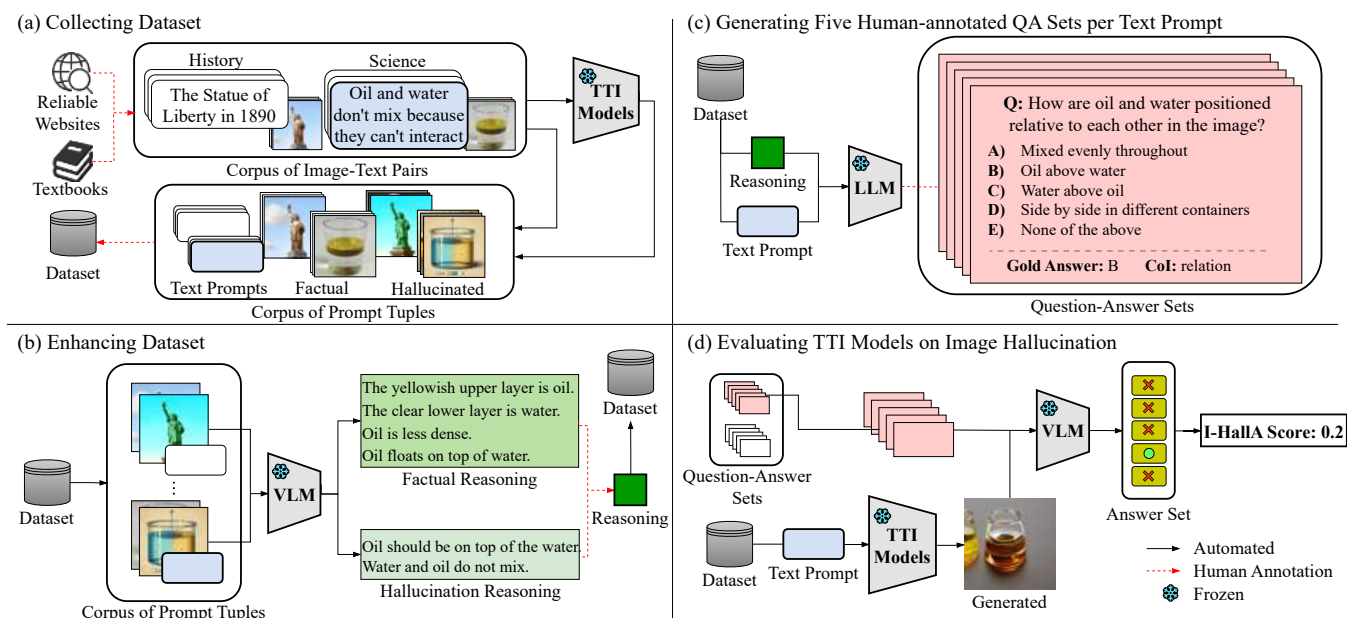


Figure 2: Overall pipeline of how I-HallA v1.0 is used for evaluating image hallucination: (a) Collect datasets containing prompts, factual images, and hallucinated images based on textbooks. (b) Enhance the collected dataset by leveraging the vast pre-trained knowledge and visual understanding capability of GPT-4o, adding reasoning about image hallucination to the datasets. (c) Input the prompt and reasoning into a language model to generate QA sets for evaluation. (d) Input the 5 QA sets per image and the target image into a vision-language model, and calculate the I-HallA score based on the number of correct answers. We employ GPT-4o for both the VLM and LLM.

cus only on the elements explicitly mentioned in the text prompt. However, verifying the factual alignment of generated images requires external knowledge beyond the prompt. As shown in Figure 1(a), the number of hydrogen atoms, though not mentioned in the prompt, is important factual information. Additionally, current metrics struggle to distinguish between images that accurately represent factual information and those that simply match the text prompt, especially when polysemy introduces various interpretations. As shown in Figure 1(b), while the generated image captures the idea of a “comb pattern,” it also includes several incorrect details, like small circles and decorations, which are not part of the factual image.

To address these limitations, we propose a three-stage pipeline to construct a new benchmark, I-HallA v1.0, with a new evaluation metric, I-HallA. Unlike existing protocols, we leverage the vast knowledge base of GPT-4 Omni (GPT-4o) (OpenAI 2024c) to assess factual information not mentioned in the prompt, such as the number of hydrogen and carbon atoms, molecular structure, and more. Additionally, we utilize the visual understanding capabilities of GPT-4o to discern factual visual semantics from potentially false ones, which is difficult to do with text prompts alone.

First, the dataset includes 200 prompts based on content from five science and history textbooks (Jackson J. Spielvogel 2008; Danzer et al. 2008; Urone et al. 2020; O’Grady et al. 2021; Naser 2023) to address factual information. Textbooks, meticulously edited for educational purposes, represent years of accumulated knowledge and are among the

most authoritative sources, making them a primary basis for this dataset. Specifically, we use the textbooks’ captions and corresponding figures as prompts representing factual information and factual images. This is because textbook figures are carefully curated, highly aligned with their captions, and thoroughly validated for factual accuracy. The hallucinated images generated from these prompts in five TTI models (Rombach et al. 2022; StabilityAI 2022; Podell et al. 2023; OpenAI 2023) are compared against the factual images. In total, we gather 1,200 images for all prompts, consisting of both factual and hallucinated images.

Secondly, we enhance the dataset by inputting each prompt and its corresponding image into GPT-4o to obtain factual information, referred to as “reasoning,” relevant to the prompt. This process leverages GPT-4o’s external knowledge beyond the prompt and considers visual semantics to distinguish details difficult to discern from text alone. Lastly, we construct I-HallA, consisting of 1,000 multiple-choice question-answer (QA) sets to evaluate the extent of image hallucination in TTI models, using reasoning as a key input. With GPT-4o as our VQA model, we input the generated image and corresponding questions for each prompt. The accuracy of the answers is then scored, with higher accuracy indicating fewer hallucinations. We average QA scores across all prompts to evaluate TTI models on I-HallA v1.0. In all three stages, a thorough human review validates the metric’s legitimacy, though future use won’t require it.

By applying our metric to various TTI models, we quantitatively measure the extent of image hallucination. Experi-

mental results show a strong correlation between our metric and human evaluation, with Spearman’s $\rho=0.95$, indicating close alignment in assessing hallucination. Our benchmark effectively addresses image hallucination, paving the way for further advancements in mitigating this issue.

Related Works

Hallucination in Language Generation

In language models, hallucination refers to the generation of unfaithful content to the given source material (Ji et al. 2023). As large language models (LLMs) increasingly produce text that closely resembles human writing, there has been a growing emphasis on developing benchmarks to evaluate and distinguish hallucinated content. For instance, FEVER (Thorne et al. 2018) is a dataset used for fact-checking that utilizes Wikipedia as its knowledge source. HaluEval (Li et al. 2023a) combines automated generation with human annotation to detect hallucinations.

Hallucination also occurs in large vision-language models (VLMs) such as LLaVA (Liu et al. 2024a), where visual features are input into LLMs to generate textual descriptions. In VLMs, hallucination refers to a mismatch between the factual details of images (e.g., object presence, attributes, spatial relations) and the corresponding generated text (Liu et al. 2024b). Various studies evaluate this using metrics like BLEU (Papineni et al. 2002) or CIDEr (Vedantam, Zitnick, and Parikh 2015), or by querying VLMs about object presence (Li et al. 2023b). In contrast, we focus on evaluating hallucinations in image generation, where generated images fail to depict factual information accurately. While hallucinations can occur in our pipeline during text generation, we mitigate this issue through rigorous human reviews. Further details are covered in Appendix B.

Common Sense Reasoning in VLMs

Some studies use benchmark datasets to assess whether VLMs possess commonsense knowledge when interpreting images. For instance, the WHOOPS (Guetta et al. 2023) and ROME (Zhou et al. 2023) datasets are created by inputting intentional text prompts that defy common sense into TTI models, resulting in odd and unconventional images.

These studies differ from our focus, as their benchmarks evaluate the language generated by VLMs when interpreting images, rather than assessing TTI models. Furthermore, by using counter-intuitive prompts to intentionally generate weird images, they do not address image hallucination, where TTI models fail to reflect factual information when given factually accurate prompts. Moreover, their concept of common sense differs from factual accuracy. For example, one prompt in WHOOPS is “A little girl standing in front of a blackboard with math formulas on it.” While this scenario is factual, as a young child can solve math problems, WHOOPS argues that this prompt defies common sense. Thus, existing research on common sense relies on limited content to judge factual information, failing to fully address image hallucination. More details are in Appendix C.

Evaluating Text-to-Image Generation with Question Answering

CLIPScore (Hessel et al. 2021) and DALL-Eval (Cho, Zala, and Bansal 2023), which are early studies measuring the alignment between generated images and text prompts to evaluate TTI models, commonly exhibit limitations due to the inherent constraints of CLIP (e.g., inability to count objects) or the restricted scope of evaluation criteria.

With the growing capabilities of foundation models, a new approach has developed that validates alignment in text-to-image generation by using VQA models on questions derived from the prompt. For instance, TIFA (Hu et al. 2023) classifies elements of the text prompt into 12 categories and generates a set of questions and answers using GPT-3 (Brown et al. 2020). These sets are used to evaluate the image by inputting both the image and questions into the VQA model like mPLUG (Li et al. 2022). VQ² (Yarom et al. 2024) extracts key information from the text prompt, generates related questions, and evaluates the text-image alignment by checking whether the image provides correct answers. VPEval (Cho, Zala, and Bansal 2024) enhances these alignment evaluation methods by incorporating object detection and optical character recognition, allowing for a more precise assessment. Davidsonian Scene Graph (Cho et al. 2024) breaks down the prompt into small propositions and represents the dependencies between these propositions in a graph, ensuring that the generated questions are not redundant. However, existing benchmarks that focus solely on evaluating the alignment between text prompts and images often fail to detect external knowledge beyond the text and the factual visual semantics embedded in the image, which is required to address image hallucination.

Methodology

Image Hallucination

Factual information refers to data that can be objectively verified and proven true based on evidence or reliable sources. It is an important evaluation criterion in fields that require reliable and accurate information, such as education (Hew et al. 2014). In this paper, we focus on image hallucination, a phenomenon where the images generated by TTI models fail to accurately reflect factual information.

Existing benchmarks that merely evaluate the alignment between the text prompt and the generated image are inadequate for properly assessing image hallucination. Their limitations in evaluating image hallucination can be summarized in two key points:

- The inability to evaluate factual information beyond the prompt.
- The difficulty in identifying accurate visual semantics.

As shown in Figure 1, existing evaluation metrics, such as TIFA (Hu et al. 2023), rely solely on text prompts, which limit their ability to consider factual information not explicitly stated in the prompt. In contrast, our metric leverages GPT-4o’s capability to generate questions and answers based on external factual information that is not specified in the prompt but has already been trained into the model.

This allows us to assess whether crucial factual information, though not mentioned in the prompt, is accurately reflected in the generated image.

Additionally, existing metrics cannot evaluate whether the visual semantics within an image are hallucinated. This is because the polysemy of text prompts can generate images that are not visually factual while reflecting the word’s meaning. In such cases, metrics based solely on the text prompt fail to distinguish these visual semantics, making it impossible to assess image hallucination. For instance, in the case of “Comb-pattern pottery BCE 8000,” the factual and hallucinated images of the comb-pattern are represented in Figure 1(b). However, the word “comb-pattern” corresponds to various visual designs and is not confined to a single pattern. Consequently, visual representations that do not match the specific form intended—potentially contradicting historical fact—can still be described as “comb-pattern.” This ambiguity complicates current evaluation metrics to assess the factual information. More details are available in Appendix I. In contrast, our pipeline utilizes the visual capabilities of GPT-4o by inputting both the prompt and image together to generate an evaluation metric based on factual information. From factual images, we obtain the reasons why these images are considered factual, while from hallucinated images, we acquire discriminative information about how incorrect semantics differ from those in corresponding factual images, as illustrated in Figure 2(b). This enables us to distinguish accurate visual semantics that reflect factual information among the many possible choices corresponding to the prompt. Therefore, our approach allows for evaluating factual information, including visual semantics like patterns that cannot be fully conveyed through the text prompt alone.

I-HallA v1.0: Benchmark for Evaluating Image Hallucination

We propose a curated benchmark, I-HallA v1.0, and an evaluation metric, I-HallA, to assess image hallucination in TTI models. To our knowledge, this is the first benchmark to evaluate image hallucination in TTI-generated images.

As a pioneering effort, our benchmark focuses on the educational domain, where the accuracy of factual information is crucial. Education serves as an ideal starting point for this benchmark study due to the broad and diverse use of factual data. Textbooks are specifically chosen as they encapsulate knowledge accumulated over time, providing well-structured and reliably categorized content. Within this domain, we choose science and history, two subjects that heavily rely on images for effective learning. As shown in Table 1, in science, our benchmark covers Physics, Biology, and Earth Science. In history, it is organized by geographic regions—Eastern and Western & African—and by periods, such as Ancient, Medieval, and Modern.

We introduce a three-stage pipeline to construct our benchmark dataset and evaluation metric. First, we collect a dataset to address the image hallucination in the educational domain. Next, we enhance this dataset by leveraging GPT-4o’s pre-trained knowledge and visual understanding capabilities. Finally, based on the enhanced dataset, we develop a metric to evaluate image hallucination.

Domain	Category	Type	Level			Total
			Easy	Medium	Hard	
Science	Physics	Factual	26	3	4	33
		Hallucinated	25	4	4	33
	Biology	Factual	19	1	1	21
		Hallucinated	17	2	2	21
	Earth Science	Factual	40	2	4	46
		Hallucinated	33	4	9	46
			160	16	24	-
History	Western & Africa Ancient	Factual	14	1	0	15
		Hallucinated	5	2	8	15
	Western & Africa Medieval	Factual	7	0	0	7
		Hallucinated	5	2	0	7
	Western & Africa Modern	Factual	35	1	3	39
		Hallucinated	24	2	13	39
	Eastern Ancient	Factual	9	0	1	10
		Hallucinated	2	4	4	10
	Eastern Medieval	Factual	16	0	0	16
		Hallucinated	4	3	9	16
	Eastern Modern	Factual	12	1	0	13
		Hallucinated	10	1	2	13
			143	17	40	-

Table 1: Statistical Analysis of I-HallA v1.0 Benchmark Dataset by Domain, Category, Type, and Difficulty Level. Each prompt’s category is based on corresponding textbooks, broadly divided into science and history. The “Type” refers to whether each prompt’s image is factual or hallucinated. The “Difficulty Level” is determined by making five type predictions for each prompt and its corresponding image using GPT-4o. Based on the number of correct predictions, the difficulty is categorized as follows: 0-1 correct predictions are classified as “Hard,” 2-3 correct predictions as “Medium,” and 4-5 correct predictions as “Easy.”

Domain	Color	Counting	Existence	Others	Posture	Relation	Scene	Shape	Size
Science	30	49	106	16	11	91	100	73	24
History	64	53	110	9	40	45	72	84	23

Table 2: Statistical Analysis of the I-HallA Metric by Compositions of Interest (CoIs). The dataset includes a total of 1,000 question-answering sets with their corresponding CoI; 500 sets each for the science and history categories.

Collecting Our Dataset The first stage for creating a benchmark is the collection process for initial datasets. We engage 10 graduate students to curate prompts from three science (Urone et al. 2020; O’Grady et al. 2021; Naser 2023) and two history textbooks (Jackson J. Spielvogel 2008; Danzer et al. 2008). For each chapter in textbooks, participants extract up to 10 prompts and their corresponding factual images. The prompt P is either derived from textbook figures or selected based on unanimous agreement among participants that it represents key content in the chapter, such as important events, phenomena, or artworks. If a figure corresponding to the prompt P is in the textbook, it is collected as a factual image I_f . In rare cases where no such figure exists, we search for an image related to P on verified websites, such as government-operated ones, and collect it as I_f .

Each participant inputs the curated prompt P into five TTI models, generating 10 images per model. All participants evaluate these images to identify image hallucination that contradicts factual information. The image with the most pronounced hallucination, unanimously agreed upon by all 10 participants, is selected as the representative hallucinated

image I_h for that prompt and model, highlighting the most evident hallucination and revealing each model’s limitations.

To conclude, our dataset consists of 200 tuples $\{P, I_f, I_{h_i}\}_{i=1}^5$, each containing a prompt P , a factual image I_f , and five hallucinated images I_{h_i} , where i represents each TTI model. We collect 100 tuples from the science domain, and the remaining 100 are from history. In total, we assemble a set of 1,200 pairs, each consisting of a prompt and either a factual or hallucinated image.

Enhancing Our Dataset The second stage involves enhancing the collected dataset using GPT-4o to evaluate factual information better. Leveraging GPT-4o, pre-trained on vast data, and equipped with visual understanding, we develop a dataset incorporating external knowledge and visual semantics beyond the text prompt. For each prompt P , we input two sets— (P, I_f) and (P, I_h) —into GPT-4o to obtain three aspects: responses, reasonings, and difficulty levels. We use hallucinated images I_h generated by DALL-E-3 (OpenAI [2023]) to capture hallucinations produced even by the latest TTI model. To determine the difficulty levels, we performed the same process five times independently.

The response, determined by GPT-4o, categorizes the input image as either “factual” if accurate or “hallucinated” if not. The reasoning provides the factual justification for this response, incorporating external knowledge and visual semantic information. Thus, for each prompt, two types of reasoning are provided: one for correctly identifying a “factual” image and another for identifying a “hallucinated” image. Difficulty levels are determined by evaluating GPT-4o’s accuracy across five inference attempts in discerning the factual information related to the given prompt and image. Using the initial dataset labels as the ground truth, we compare GPT-4o’s response to determine correctness. Prompts and images are classified as “Hard” if GPT-4o answered correctly 0-1 times, “Medium” if correct 2-3 times, and “Easy” if correct 4-5 times. We store the reasonings only when GPT-4o provides the correct response.

10 human annotators thoroughly review and refine all reasonings into a final, well-expressed version, retaining only the unanimously agreed-upon parts. In cases where GPT-4o fails to provide any correct response, the reasoning is developed through group discussion and consensus among all participants. More details are available in Appendix B. Consequently, our dataset consists of 200 tuples $\{P, I_f, I_{h_i}, R, D\}_{i=1}^5$, where the reasoning R and difficulty levels D have been added to the previously collected dataset.

I-HallA: An Evaluation Metric Using Question-Answering The final stage involves developing the evaluation metric, I-HallA, to evaluate the factual accuracy of images generated by TTI models. It employs five multiple-choice QA sets to assess image hallucination based on the curated dataset from previous stages. To analyze the benchmark and results, we introduce classification criteria called Compositions of Interests (CoIs) to categorize the QA sets. We select the most relevant compositions from existing TTI evaluation studies (Hu et al. [2023]; Li et al. [2024]) that are closely related to image hallucination: *color*, *counting*, *existence*, *others*, *posture*, *relation*, *scene*, *shape*,

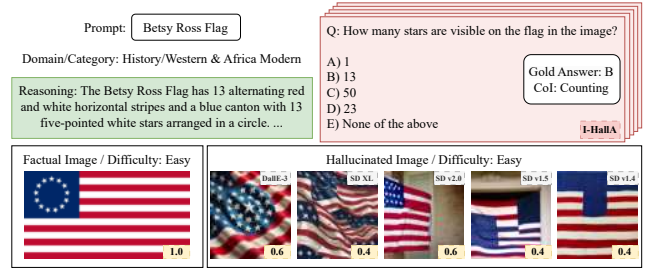


Figure 3: Overview of I-HallA v1.0: The upper section presents the prompt, domain, category, reasoning, and I-HallA results for five QA sets. The lower section compares a factual image with hallucinated outputs from five TTI models, indicating difficulty levels and I-HallA scores. I-HallA scores shown in the bottom-right box of each image remain unchanged across the three trials.

and size. The “others” category applies when a given QA set does not fit into the other CoIs.

To generate the QA sets, we input the prompt P , reasoning R , and CoIs into GPT-4o. Based on the reasoning, GPT-4o generates five multiple-choice QA sets per prompt. Each QA set consists of a question targeting factual information, five answer choices (with the fifth option being “None of the above”), and the correct factual answer as the gold answer. Simultaneously, GPT-4o generates the most relevant CoI for each QA set. The QA set generation follows two key guidelines: First, the more factual information an image contains, the more correct answers it should provide, thereby yielding higher scores for more factual images. Second, qualitative information that cannot be visually verified through the image should be excluded in the questions. The 10 participants then review the generated QA sets to ensure that they adhere to these two guidelines. Any disagreements among participants lead to revisions. This process results in 1,000 QA sets and their corresponding CoIs across 200 prompts.

To calculate the score using I-HallA (I-HallA score), we input the image to evaluate, along with the question and five answer choices, into a VQA model and compare the model’s response with the gold answer. There are five questions per image, and the I-HallA score ranges from 0 to 1. The formula can be expressed as follows:

$$\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \left(\frac{1}{|Q^p|} \sum_{(q, c, g) \in (Q^p, C^p, G^p)} \mathbb{I}(\text{VQA}(p, q, c) = g) \right) \quad (1)$$

$\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the condition is satisfied. VQA is the VQA model’s prediction for the given QA. $\mathcal{P}$ is a set of prompts. For the given prompt p , Q^p , C^p , and G^p represent the sets of questions, choices, and gold answers, respectively. $|\cdot|$ is the number of elements in a set.

Experiments

In this section, we first analyze the statistical characteristics of I-HallA v1.0 across various categories, difficulty levels, and a list of CoIs. We evaluate the recent five text-to-image

Models	I-HallA Score		I-HallA Score [†]	
	Science	History	Science	History
SD v1.4	0.353 $\pm$ 0.002	0.535 $\pm$ 0.013	0.033 $\pm$ 0.012	0.110 $\pm$ 0.010
SD v1.5	0.309 $\pm$ 0.011	0.533 $\pm$ 0.004	0.030 $\pm$ 0.017	0.117 $\pm$ 0.021
SD v2.0	0.336 $\pm$ 0.006	0.540 $\pm$ 0.014	0.027 $\pm$ 0.021	0.120 $\pm$ 0.010
SD XL	0.398 $\pm$ 0.015	0.579 $\pm$ 0.012	0.077 $\pm$ 0.050	0.110 $\pm$ 0.066
DallE-3	0.661 $\pm$ 0.020	0.666 $\pm$ 0.003	0.227 $\pm$ 0.029	0.133 $\pm$ 0.031
Factual	0.856 $\pm$ 0.002	0.873 $\pm$ 0.006	0.517 $\pm$ 0.038	0.533 $\pm$ 0.015

Table 3: Our benchmark evaluation results on existing TTI models and factual images; We compute the I-HallA score by averaging ratio-based scores across 100 prompts per category (science and history). [†] means scoring as incorrect if even one out of five QA sets is wrong for each prompt. Each experiment is conducted three times.

models with I-HallA, emphasizing our metric robustly and accurately assesses image hallucination. The models include Dalle-3 (OpenAI 2023), Stable Diffusion v1.4, Stable Diffusion v1.5 (Rombach et al. 2022), Stable Diffusion v2.0 (StabilityAI 2022), and Stable Diffusion XL-base v1.0 (Podell et al. 2023). Additionally, through human evaluation, we demonstrate that our method strongly correlates with human judgments on evaluating image hallucination. For all experiments, we utilize GPT-4o as the VQA model for the I-HallA.

Benchmark Analysis

To illustrate the comprehensive scope of I-HallA v1.0, we provide an analysis spanning all categories, difficulty levels, and compositions. Additionally, we demonstrate that GPT-4o can be effectively used to develop I-HallA v1.0. For more details on I-HallA v1.0, please refer to Appendix H.

Statistics and diversity As shown in Table 1, in the science domain, we collect 33, 21, and 46 prompts from physics, biology, and earth science textbooks, respectively. In the history domain, we collect prompts from two textbooks. Specifically, we gather 15, 7, 39, 10, 16, and 13 prompts from the Western & African/Ancient, Western & African/Medieval, Western & African/Modern, Eastern/Ancient, Eastern/Medieval, and Eastern/Modern sections, respectively. The number of images collected per prompt includes one factual image and five hallucinated images from different TTI models, totaling six images per prompt. Therefore, a total of 1,200 images are included in I-HallA v1.0.

Additionally, I-HallA provides 1,000 questions with their corresponding compositions of interest, categorized into 9 types, as shown in Table 2. In science, the occurrences are color (30), counting (49), existence (106), others (16), posture (11), relation (91), scene (100), shape (73), and size (24); while in history, they are color (64), counting (53), existence (110), others (9), posture (40), relation (45), scene (72), shape (84), and size (23).

GPT-4o’s ability on image hallucination In our study, we employ GPT-4o to generate reasoning and analyze the difficulty for I-HallA v1.0. GPT-4o assesses whether an image is factual or hallucinated based on the prompt, with hallucinated images generated by the DALL-E 3. Based on the

three difficulty levels, GPT-4o classified 160 image-prompt pairs as Easy, 16 as Medium, 24 as Hard in the science domain; 143 as Easy, 17 as Medium, and 40 as Hard in history. When treating “Hard” pairs as incorrect, accuracy is 88% in science and 80% in history. If cases, where GPT-4o fails to answer any questions correctly, are treated as incorrect, accuracy increases to 91.5% in science and 84.5% in history.

The higher number of “Easy” pairs suggests GPT-4o’s strong ability to judge factual information, with “Easy” pairs being $\times 4$ in science and $\times 2.5$ in history compared to the total of others. Moreover, as the reasoning and QA sets are thoroughly refined through human review, our benchmark is well-equipped to evaluate image hallucination.

Evaluating Text-to-Image Models

Using our I-HallA v1.0, we demonstrate that all five latest TTI models suffer from image hallucination. By analyzing I-HallA scores, we quantitatively show how each model reflects factual information differently, proving that our benchmark objectively evaluates image hallucination. Furthermore, by categorizing the I-HallA scores across different categories and CoIs, we analyze specific situations where each model tends to produce image hallucinations.

Table 3 presents the average I-HallA score and standard deviation from three trials for various TTI models on I-HallA v1.0. Higher scores indicate better performance in reflecting factual information without image hallucination. Dalle-3 outperforms the Stable Diffusion models in mitigating hallucination across all subjects. Even under the strict standard ([†]), where one incorrect answer results in failure, Dalle-3 remains the top performer. I-HallA scores are generally higher in history than in science. These scores allow us to assess how effectively current TTI models handle image hallucination. Factual images score in the high 80s, much higher than the average of 0.411 in science and 0.570 in history for the five TTI models, demonstrating that our metric effectively measures factual information. Still, they fall short of perfect, likely due to noise in I-HallA creation or VQA model limitations, which we aim to address in future work.

Figure 4 shows the impact of various TTI models on image hallucination across different categories and compositions. The top graph displays average scores by category, while the middle and bottom graphs represent the average scores for each composition in science and history. Models with larger parameters and newer architectures tend to have higher scores, with Dalle-3 generally outperforming other models. However, exceptions exist, such as in the “Eastern Ancient” category, where Stable Diffusion v2.0 scores higher than Stable Diffusion XL-base v1.0 or Dalle-3.

In the science and history domains, the I-HallA score for “Posture” and “Size” compositions, respectively, is highest in the Stable Diffusion models, despite their generally lower image quality. This suggests that these models effectively reflect factual information, as our metric evaluates factual accuracy rather than image quality. Therefore, even TTI models that generate high-quality images can score lower if they fail to meet factual criteria. For the “Others” composition in the history domain, the high I-HallA scores of Stable Diffusion v1.4 might be influenced by a smaller sample size.

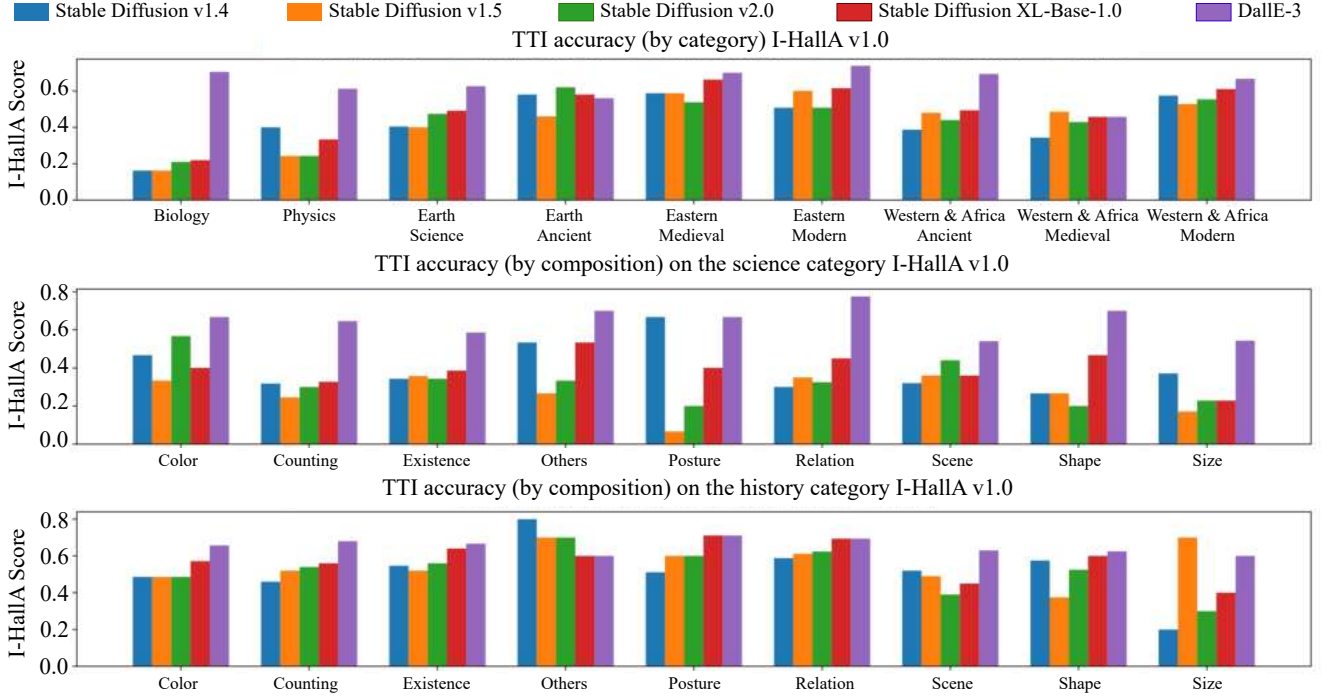


Figure 4: I-HallA scores from five different TTI models across different categories and compositions. The factual information of images generated by TTI models using the prompts from I-HallA v1.0 is evaluated using the I-HallA metric. The I-HallA scores in this figure represent the average scores of each TTI model, calculated across different categories and compositions.

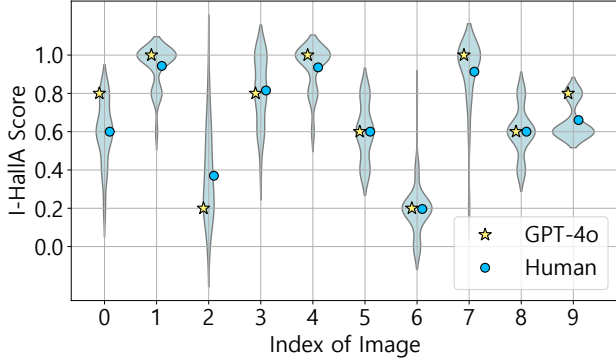


Figure 5: Plot of I-HallA scores from GPT-4o and human evaluations. Blue circles indicate average scores from 53 participants per question, with score distributions depicted via violin plots. Stars emphasize GPT-4o’s results, which closely align with human judgments, demonstrating a strong correlation between the model and human evaluators.

Exploring the reliability of I-HallA through human evaluation

By calculating the correlation between the previously mentioned experimental results and the human evaluation of I-HallA v1.0, we demonstrate that our benchmark aligns well with human judgment in assessing image hallucination. For more details on the user study, please refer to Appendix G.

We randomly sample 10 prompts from I-HallA v1.0, in-

cluding 4 factual and 3 hallucinated images from each of two great TTI models in I-HallA: Dalle-3 and Stable Diffusion XL-base v1.0. We collect I-HallA scores from 53 participants, guiding them to answer questions based on the images provided, following the same approach as GPT-4o. Figure 5 shows the average I-HallA scores and standard deviations for each image. Even the image with the largest score difference shows a variance of only about 0.2, indicating that the results are very similar to human evaluations.

To verify this quantitatively, we calculate correlations between GPT-4o’s I-HallA scores and human judgments on image hallucination. For the three metrics—Pearson’s r , Spearman’s ρ , and Kendall’s τ —we observe very high correlations of 0.952, 0.950, and 0.889, respectively.

Conclusion

In conclusion, we propose the I-HallA v1.0 benchmark as the first to address image hallucination in text-to-image generation by evaluating factual information. This benchmark overcomes the limitations of previous methods, which could not accurately assess factual information. Developed through a three-stage pipeline using GPT-4o and thorough human review, it evaluates hallucination in 200 factual and 1,000 hallucinated images from five text-to-image models. Our results confirm that the benchmark effectively measures image hallucination and aligns well with human judgment. We hope that our benchmark and evaluation metric will be instrumental in resolving image hallucination in the future.

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Supplementary Material for Evaluating Image Hallucination in Text-to-Image Generation with Question-Answering

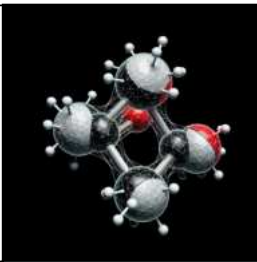
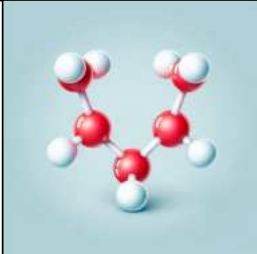
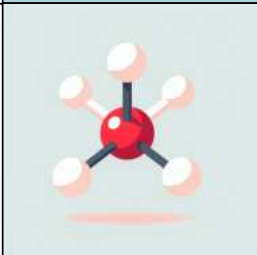
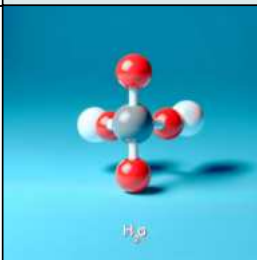
Prompt	Hallucinated Image (DallE-3)
A molecule of water.	
 A simple molecular model of a water molecule, showing two hydrogen atoms bonded to one oxygen atom in a V-shape. The oxygen atom should be larger and colored red, while the hydrogen atoms should be smaller and colored white.	
 A molecular model of a water molecule (H ₂ O), consisting of one oxygen atom in the center, colored red, with two smaller hydrogen atoms, colored white, bonded to it at an angle. The hydrogen atoms should be arranged in a V-shape or bent structure around the oxygen atom.	
 A molecular model of a water molecule (H ₂ O), consisting of one central oxygen atom, colored red, with two smaller hydrogen atoms, colored white, bonded to it at an angle of approximately 104.5 degrees. The hydrogen atoms should be arranged in a V-shape or bent structure around the oxygen atom, clearly showing the angular geometry of the molecule.	

Figure 6: The difficulty in resolving image hallucination through prompt refinement alone. No matter how detailed and lengthy the input text prompt is, current text-to-image models struggle to accurately reflect factual information.

A. Challenges in Mitigating Image Hallucination by Text Prompt

As depicted in Figure 6, image hallucination is not easily resolved by simply improving the text prompt. For example, when the prompt “A molecule of water” is entered into DallE-3 (OpenAI [2023]), the image obtained is shown in the first row of Figure 6. This is a hallucinated image for various reasons, such as the failure to accurately depict two hydrogen atoms and one oxygen atom. Therefore, we input this image and the prompt into GPT-4o to generate a detailed

prompt designed to correct the image, much like how we obtained “reasoning” when enhancing the benchmark. We ask GPT-4o to suggest an alternative prompt that indicates how the original prompt should be modified to ensure the given image accurately reflects the prompt. As a result, we get a long and detailed prompt, as shown in the second row of the figure. We then input this prompt back into DallE-3 to generate the image, and if the result is still a hallucinated image, we repeat the process. Even after several iterations of prompt refinement, it is evident that the text-to-image (TTI) model still struggles to generate an image that accurately reflects the factual information (such as the two hydrogen atoms, one oxygen atom, and the molecular structure).

B. Eliminating Hallucination in GPT-4o: A Benchmark Refined by Human Review

We propose I-HallA v1.0, a benchmark for evaluating image hallucination, utilizing GPT-4 Omni (GPT-4o) (OpenAI [2024c]), one of the existing state-of-the-art vision-language models. This benchmark assesses external knowledge not included in the prompt by utilizing GPT-4o’s extensive training data. It also evaluates visual semantics that are difficult to discern solely from prompts by leveraging GPT-4o’s visual comprehension capabilities. Although GPT-4o mostly generates accurate outputs, there are very rare instances where it fails to generate sufficient factual information or exhibits hallucination phenomena. Taking this into account, we have refined our benchmark and evaluation metrics through a rigorous human review process. Figure 7 shows examples of the reasoning and QA sets before and after applying this human review process.

C. Contrasting Image Hallucination with Common Sense

In Figure 8, We introduce examples from WHOOPS (Guetta et al. [2023]) and ROME (Zhou et al. [2023]), existing benchmarks to evaluate the common sense reasoning abilities of vision-language models (VLMs) such as InstructBLIP (Dai et al. [2023]). We demonstrate that the common sense addressed in previous studies differs from our proposed factual information and image hallucination.

The first row in Figure 8 illustrates examples that violate common sense according to WHOOPS. However, a child solving a math problem or a man holding a pacifier in his mouth is not impossible or factually incorrect. Therefore, this is irrelevant to image hallucination. The second row presents examples that violate common sense, according to ROME. Similarly, these examples cannot be considered impossible in the real world; thereby not dealing with image hallucination.

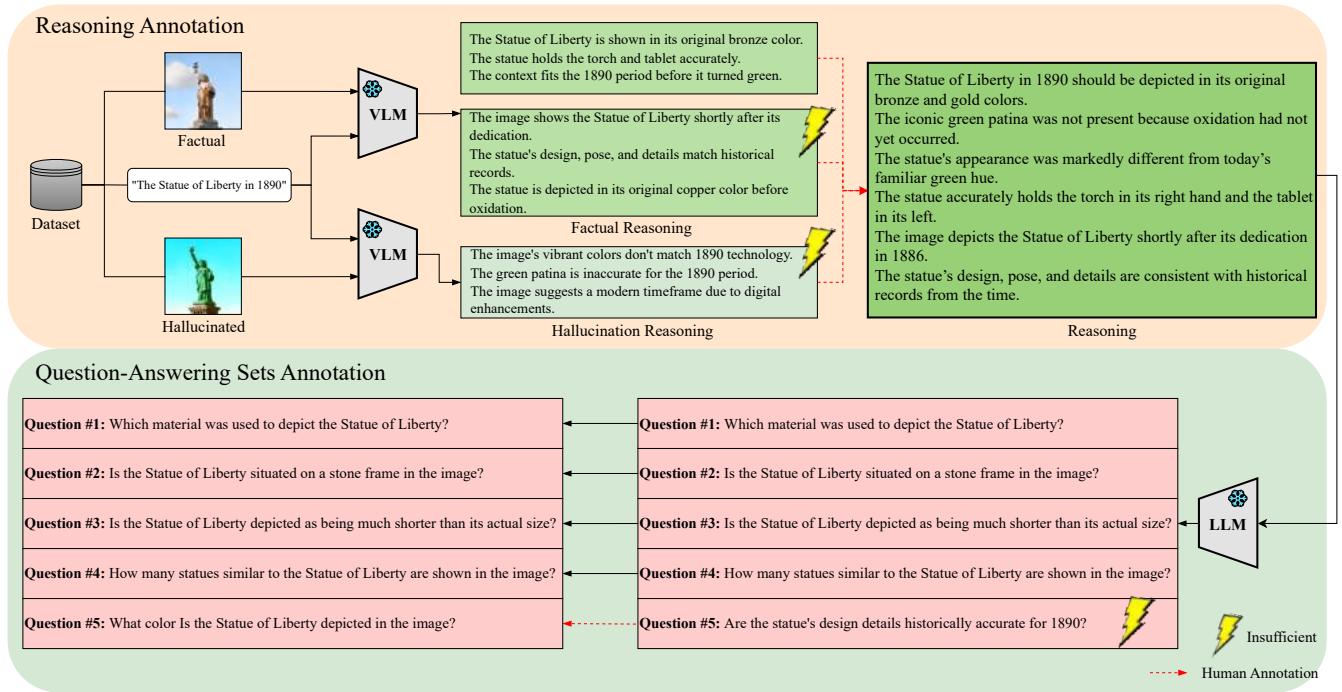


Figure 7: The process of overcoming the rare insufficiency seen in GPT-4o through thorough human review, creating a meticulous benchmark and evaluation metric for assessing image hallucination. During enhancing our dataset, reasoning annotation is carried out. Reasoning is collected only when GPT-4o makes a correct prediction. The collected factual or hallucinated reasonings are reviewed by 10 participants and consolidated into a single reasoning. In the annotation stage when creating QA sets, GPT-4o generates QA sets based on this consolidated reasoning, and these are revised to reflect the most discriminative features that all 10 participants agreed upon.

(a) InstructBLIP			(b) GPT-4-turbo			(c) GPT-4o-mini		
Models	I-HallA Score		Models	I-HallA Score		Models	I-HallA Score	
	Science	History		Science	History		Science	History
SD v1.4	0.542	0.614	SD v1.4	0.544	0.646	SD v1.4	0.386	0.566
SD v1.5	0.556	0.606	SD v1.5	0.516	0.62	SD v1.5	0.40	0.556
SD v2.0	0.550	0.604	SD v2.0	0.548	0.652	SD v2.0	0.456	0.604
SD XL	0.532	0.648	SD XL	0.584	0.670	SD XL	0.454	0.608
DallE-3	0.592	0.626	DallE-3	0.758	0.626	DallE-3	0.454	0.686
Factual	0.652	0.594	Factual	0.850	0.652	Factual	0.844	0.868

Table 4: I-HallA score table evaluating I-HallA v1.0 with three different VQA models. Compared to GPT-4o, which scored 0.856 in the science domain and 0.873 in the history domain for factual images, all three models demonstrate a lower ability to assess image hallucination. This is because, for factual samples, an I-HallA score closer to 1.0 is ideal, whereas the I-HallA scores of the three models are lower than that of GPT-4o.

D. I-HallA vs. Existing Metrics for Text-to-Image Models: A Detailed Comparison

Metrics such as TIFA (Hu et al. 2023), VQ² (Yarom et al. 2024), and Davidson Graph Scene (Cho et al. 2024), which evaluate the alignment of TTI models, are limited by being generated solely from the given text prompt. For example, when applying TIFA, the most representative TTI metric, to the prompt "A molecule of methane," TIFA first uses GPT-

3 (Brown et al. 2020) to extract elements from the prompt: *molecule, methane*

Next, it generates two questions for each element. The first question should receive a "yes" answer if the image is correctly generated, and the second question is one where the element itself should be the answer: {"Is there a molecule in the image?," Choices: "Yes," "No"}, {"What is this?," Choices: "atom," "molecule," "ion," "compound"}.

The TIFA score obtained from these questions tends to

Aspects	Description	Example
Prompt	A textual description of an image	A molecule of methane.
Images	Either factual or hallucinated images.	(Factual image)
Domain / Category	Which category it belongs to within education domain	Science / Biology
Response	GPT-4o’s results in distinguishing between factual and hallucinated images.	{“Factual”, “Factual”, “Factual”, “Factual”, “Factual”}
Difficulty	Regarding how many times GPT-4o has answered correctly based on the responses: Easy: 4-5, medium: 2-3, hard: 0-1	Easy
Reasoning	Factual information should be considered to determine the hallucination in the given image.	The image accurately represents the structure of a methane molecule (CH ₄). - Visual Evidence: The illustration shows one central carbon atom labeled ‘C’ with four hydrogen atoms labeled ‘H’ surrounding it in a tetrahedral configuration. This geometry matches the known structure of methane, where the carbon atom forms single covalent bonds with four hydrogen atoms. - Contextual Evidence: Methane (CH ₄) is a well-documented simple hydrocarbon and the simplest alkane. Its molecular structure, consisting of one carbon atom bonded to four hydrogen atoms, is commonly represented in scientific contexts. Question: What is depicted at the molecule’s center in the image? Choices: A) Carbon atom B) Hydrogen atom C) Oxygen atom D) Nitrogen atom E) None of the above
VQA	Question-answering sets that evaluate the image hallucination & factual information of a given image.	Existence
CoI	The Composition of Interest (CoI) that each question is most relevant to.	1.0
I-HallA Score	A score reflecting how much factual information the given image represents.	

Table 5: Illustrations of the sequence-to-sequence formatting for each aspect (images omitted but are part of the aspects).

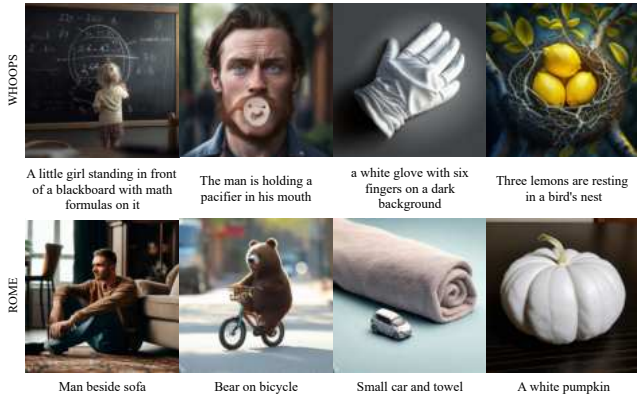


Figure 8: Example images and corresponding prompts from existing research addressing common sense. These examples do not deal with image hallucination that contradicts factual information because, in many cases, they can sufficiently exist in the real world.

be high for both factual and hallucinated images, making it inadequate for evaluating image hallucination. Like TIFA, other metrics also generate QA sets based on the prompt through an analogous process. Consequently, as described in the introduction and methodology, these metrics cannot evaluate factual information beyond the prompt and cannot properly assess realistic visual semantics due to the poly-

semy of the prompt.

E. Ablation Study with Various Visual Question Answering (VQA) Models

In this section, we evaluate five TTI models on image hallucination using various visual question-answering (VQA) models. The TTI models include Stable Diffusion v1.4 (Rombach et al. 2022), Stable Diffusion v1.5 (Rombach et al. 2022), Stable Diffusion v2.0, Stable Diffusion XL-Base v1.0, and Dalle-3, while the VQA models include InstructBLIP (Dai et al. 2023), GPT-4-turbo (OpenAI 2024a), and GPT-4o-mini (OpenAI 2024b).

In Table 4(a), **InstructBLIP** shows little to no difference in I-HallA scores between factual and hallucinated images, proving that the VQA model cannot distinguish between them. Notably, the model fails to recognize factual images in both the science and history domains, with I-HallA scores of 0.652 and 0.594, respectively. Given that an ideal I-HallA score for factual images would be 1.0, these scores are significantly lower than the 0.856 and 0.873 recorded by GPT-4o for science and history, respectively, which can be interpreted as a limitation of the VQA model.

In Table 4(b), **GPT-4-turbo** shows slightly better results than InstructBLIP; however, it still fails to correctly assess factual images, particularly in the history domain. In the science domain, there are instances where hallucinated images are incorrectly judged as factual, such as with Dalle-3, which received an I-HallA score of 0.850.

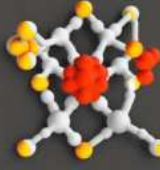
DallE-3	SD-XL				SD v2.0	
						
Gold Answer	B	C	A	B	B	I-HallA Score
Answer from DallE-3	B	E	C	E	D	0.2
Answer from SD-XL	E	E	E	E	E	0.0
Answer from SD v2.0	E	E	E	E	E	0.0
Q1: How are the hydrogen atoms positioned relative to the central carbon atom in the image? A) Opposite sides B) Equidistant C) Adjacent D) Randomly E) None of the above Gold Answer: B						
Q2: How many hydrogen atoms are connected to the central atom in the image? A) Two B) Three C) Four D) Five E) None of the above Gold Answer: C						
Q3: What is depicted at the center of the molecule in the image? A) Carbon atom B) Hydrogen atom C) Oxygen atom D) Nitrogen atom E) None of the above Gold Answer: A						
Q4: Which atoms are smaller in comparison to the central atom in the image? A) Carbon atoms B) Hydrogen atoms C) Oxygen atoms D) Nitrogen atoms E) None of the above Gold Answer: B						
Q5: What is the general shape of the molecule depicted in the image? A) Linear B) Tetrahedral C) Planar D) Octahedral E) None of the above Gold Answer: B						

Figure 9: The process of calculating I-HallA scores for given images. The VQA model answers questions based on the image, and these answers are compared to the gold answers corresponding to the factual information to measure the extent of factual content in the image. A higher score indicates a greater reflection of factual information and less image hallucination.

In Table 4(c), **GPT-4o-mini** is comparable to GPT-4o, but it still cannot accurately identify factual images as well as GPT-4o, recording lower scores in both the science and history domains. From these results, we conclude that GPT-4o is the most suitable VQA model for evaluating various TTI models on image hallucination. Additionally, the effective use of our metric requires a highly advanced VQA model, such as GPT-4o. This underscores the difficulty of accurately assessing factual information and evaluating image hallucination within images. This challenge highlights the pioneering nature and novelty of our research.

F. I-HallA Score Computation

Figure 9 shows hallucinated images generated by three TTI models (DallE-3, Stable Diffusion XL, Stable Diffusion v2.0) for the prompt “A molecule of methane.” It also includes the corresponding I-HallA QA sets, the answers generated by GPT-4 after viewing the images and questions, and the resulting I-HallA scores. Each question is designed to assess elements such as the number of atoms and the molecular structure, thereby evaluating the extent of hallucination in the given prompt and generated image.

For the hallucinated image generated by DallE-3, the equidistant placement of atoms around the central atom allows it to correctly answer the first question, matching the gold answer. However, the remaining questions reveal that the image fails to accurately depict factual information about the methane molecule, resulting in answers different from the gold answers. Consequently, with one correct answer out of five questions, it records an I-HallA score of 0.2. The other TTI models exhibit even more severe hallucinations, failing to correctly answer any of the five questions, thus scoring 0.

G. Details for Human Evaluation

For human evaluation, we recruit 53 participants, including 32 males and 21 females. The participants have an age distribution of 21 individuals aged 18-24, 31 aged 25-34, and 1 aged 35-44. Their educational backgrounds are distributed as follows: 12 are currently enrolled in university, 16 have graduated from university, 21 are enrolled in graduate school, and 4 have graduated from graduate school.

The test set for analysis in the human evaluation consists of 10 images and 50 QA sets. We additionally insert one dummy image with 5 corresponding QA sets at the beginning, which we exclude from the final statistical analysis. We randomly select 4 factual images and 3 hallucinated images from each of DallE-3 and SD-XL, ensuring that prompts do not overlap. The instructions and example questions provided to the participants are shown in Figure 10. As in the I-HallA score experiments using the VQA model, we ask five questions corresponding to each prompt. To ensure participants focus solely on the visual elements of the images, we do not reveal the prompts.

Evaluating Image Hallucination in Text-To-Image Generation with Question-Answering

1) Objective of the Experiment

- The purpose of this experiment is to evaluate how well images generated by a **Text-to-Image Generation** model (a model that creates images based on given textual descriptions) reflect **"factual information"** (Factual information refers to data that can be objectively verified as true or false).
- The experiment focuses on the field of **education**, where factual information is crucial.
- The images presented to the participants cover key topics from science and history textbooks.

2) Method of the Experiment

- There are a total of 11 images, and for each image, there are 5 multiple-choice questions with 5 options each. Participants will select the one answer that best matches the question.

3) Guidelines

- Even if the image seems to be generated, it doesn't mean the image is false.
- If you don't know the meaning of a word, you may search for it in a dictionary (e.g., <https://dict.naver.com>). However, **you are not allowed to gain additional knowledge through internet searches.**
- If you're unsure of the answer, Select **"None of the above."**

How are the stars arranged on the flag depicted in the image?



- ☐ In a square pattern
- ☐ In a single horizontal line
- ☐ In a circle
- ☐ In two vertical columns
- ☐ None of the above

Figure 10: The instructions and example question shown to participants during the human evaluation process.

H. I-HallA v1.0: Exploring the Benchmark Through Detailed Examples

Figure 11, Table 5, and Table 6 show examples of the I-HallA v1.0 image dataset, examples of the overall aspects, and examples of Compositions of Interest (CoI), respectively. Figure 11 presents 12 randomly selected prompts along with one corresponding factual image and one hallucinated image, accompanied by the I-HallA score. DallE-3 introduces 4 images, while the remaining Stable Diffusion models introduce 2 images each. In Table 5 various aspects that appear throughout the entire process of this paper, from the prompt to the calculation of the I-HallA score, are explained with examples. Table 6 describes the concepts of each composition, providing examples of corresponding QA sets.

I. Qualitative Examples of Polysemy in Text Prompts

Figure 12 introduces detailed examples of the various visual semantics that can be generated due to the polysemy of text prompts. The first row shows factual images for the prompt "Comb-pattern pottery BCE 8000." Commonly, these images feature conical-shaped brown pottery with diagonal comb patterns. However, in the images generated by Stable Diffusion v1.4 and v1.5 (Rombach et al. 2022), the comb pattern on the pottery is not accurately depicted. While Stable Diffusion v2.0, Stable Diffusion XL, and DallE-3 do generate pottery with comb patterns, the forms and patterns are significantly different from factual images. This demonstrates that although TTI models can generate images that reflect the text prompt (comb-pattern), they fail to capture the factual visual semantics due to the polysemy of the

text. Additionally, this implies that existing evaluation methods, which rely on prompts, cannot properly assess the false comb-pattern pottery present in the images in the 2nd to 4th rows of Figure 12.

Composition	Definition	VQA
Color	The visual appearance of objects in terms of hue, saturation, and brightness, distinguishing one object from another.	Question: What are the predominant colors in the flag visible in the image? Choices: A) Green and yellow B) Red, white, and blue C) Black and white D) Orange and purple E) None of the above
Counting	Determining the number of objects or entities in a given image or scenario.	Question: How many hydrogen atoms are connected to the central atom in the image? Choices: A) Two B) Three C) Four D) Five E) None of the above
Existence	Confirmation that basic entities within an image, such as a person, animal, food, items, vehicles, or text symbols (e.g., "A", "1+1") are correctly present and accurately depicted.	Question: What is depicted at the center of the molecule in the image? Choices: A) Carbon atom B) Hydrogen atom C) Oxygen atom D) Nitrogen atom E) None of the above
Others	Any other characteristics not covered by specific categories, such as abstract properties.	Question: When do you think the artifact was created in the image? Choices: A) Ancient B) Modern C) Medieval D) Ice age E) None of the above
Posture	The orientation or positioning of a person or object, indicating actions, movements, or stances.	Question: What is the posture of the statue in the image? Choices: A) Sitting B) Standing C) Lying down D) Running E) None of the above
Relation	Connections between entities, including spatial arrangements (e.g., on top, inside) and part-whole connections (e.g., body parts, clothing).	Question: How are the hydrogen atoms positioned relative to the central carbon atom in the image? Choices: A) Opposite sides B) Equidistant C) Adjacent D) Randomly E) None of the above
Scene	Backgrounds or settings of an image, such as weather, location, or environmental context.	Question: Where does the image appear to be set? Choices: A) A forest B) A hockey rink C) A beach D) A classroom E) None of the above
Shape	The geometric form or outline of objects, helping to distinguish between different objects or entities.	Question: Which of the following shapes is prominent in the image? Choices: A) Circular shapes of bubbles B) Square buildings C) Flames with irregular shapes D) Triangular tents E) None of the above
Size	The dimensions or scale of objects relative to others within the image, such as large, small, tall, or wide.	Question: What is the size of the container holding the water described in the image? Choices: A) Larger than human B) Small beaker C) Medium-sized pot D) A 1-meter water tank E) None of the above

Table 6: Detailed description of Compositions of Interest (CoIs). Each CoI is defined along with a corresponding QA set example.

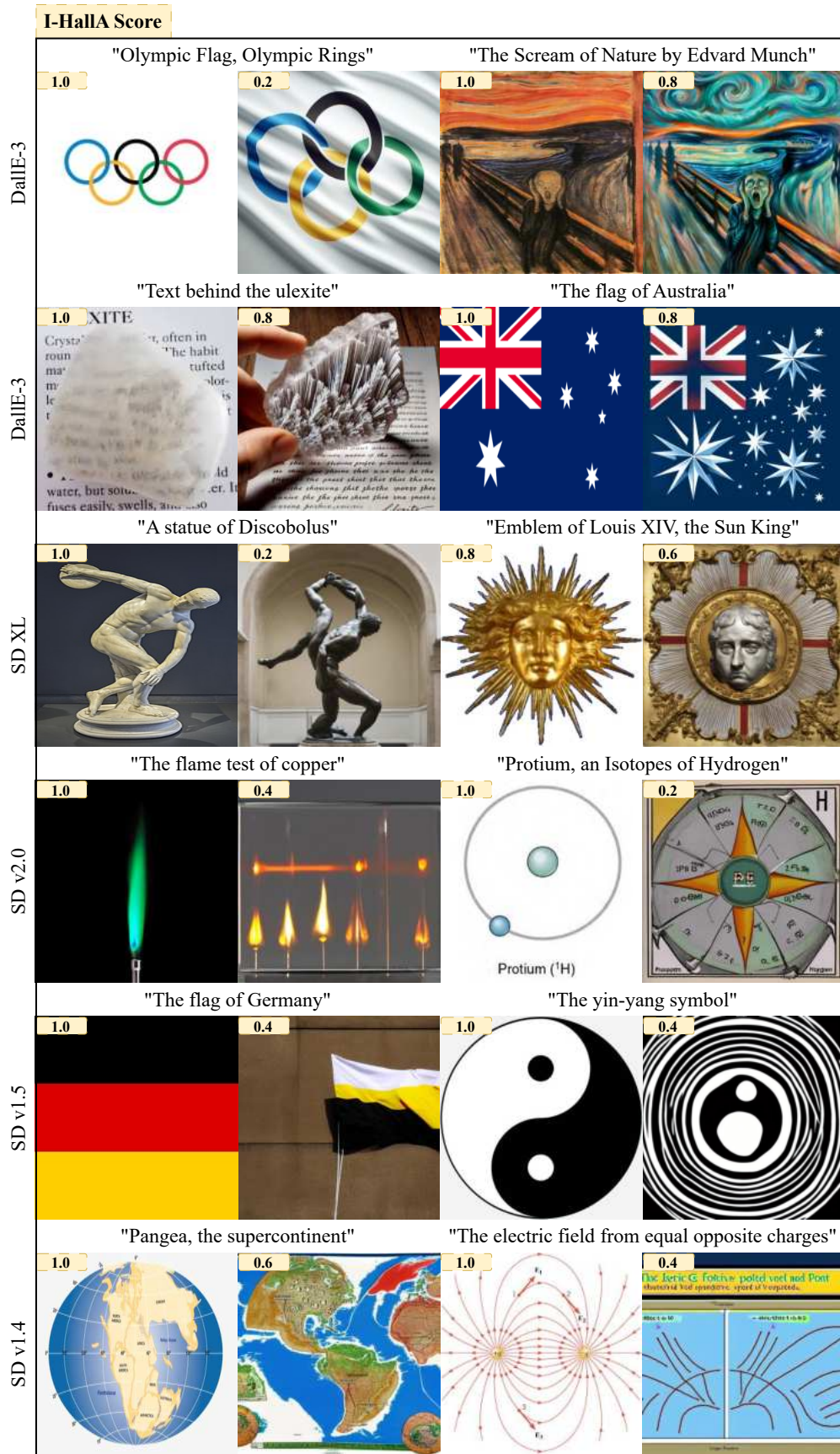


Figure 11: The image examples of I-Halla v1.0 span across various prompts. For each prompt, the left image is factual, while the right image is a hallucinated image generated by different text-to-image models. The I-Halla score for each image is displayed in the top left corner.



Figure 12: Examples of different visual semantics generated by five TTI models for the prompt “Comb-pattern pottery BCE 8000.” While Stable Diffusion v2.0, Stable Diffusion XL, and Dalle-3 reflect the comb-pattern, they produce pottery that is not factually accurate.

J. Prompt

For demonstration purposes, we present a portion of the prompt for our three-stage pipeline, which utilizes multiple GPT-4o-based agents. We introduce three representative GPT-4o-based agents for reasoning generation, question generation, and evaluation within the I-HallA. We provide each agent with a purpose-specific prompt, which includes instructions and several in-context examples. The examples are taken from outside the I-HallA v1.0 to ensure fairness. We omit part of the first phase of our pipeline, as it is conducted exclusively by human annotators. The complete prompt will be released alongside our codes. Each stage includes a final human review of the results as the last step.

Reasoning Agent

The reasoning agent aligns with the second phase of our three-stage pipeline, which aims to generate curated reasoning for a given image and its corresponding prompt. Using the provided prompt, GPT-4o generates a response and reasoning, producing a total of 5 results per image as we repeat the task five times. The reasoning includes both visual and contextual evidence from the image. We discard those whose responses do not correctly answer whether the image is factual or hallucinated, as we expect their reasonings to be unreliable.

```
1 Your Role: Excellent Image Distinguisher
2
3 Objective: Given the input image and
  caption, Distinguish whether the
  given image is normal (reasonable and
  factual), or if it is weird (
  incorrect, distorted, and contrary to
  common sense).
4 More specifically, if the truth is
  important such as science and history
  , you might need to say "weird" for
  the given image is different from the
  fact; otherwise "normal".
5
6 Your reasoning must be as detailed as it
  covers the explanations about almost
  every component within the image.
7 When determining if an image is normal
  or weird, do not judge it solely
  based on whether it has a virtual or
  cartoon-like style.
8
9 You should follow the response format
  below:
10
11 [Response Format]
12 - Response: <Normal or Weird>
13 - Reasoning: <Your reasoning about why
  you have made this decision>
14 - **Visual Evidence**: <Your reasoning
  about why you made this decision
  solely based on the visual
  appearance>
15 - **Contextual Evidence**: <Your
  reasoning about why you made this
  decision solely based on the
  context>
```

```
16
17 When evaluating an image, use very
  detailed aspects of the image and
  compare them with reality as the
  basis for your judgment.
18 For example, consider the number of
  objects in the image, the shapes of
  structures, the number of floors in
  buildings, and other similar
  information. Compare these details
  with real-life facts to distinguish
  and identify the image accurately.
19
20 We provide you with a few-shot examples
  to help you better handle your task.
21
22 For each example below, an input image
  is given and paired with three
  parameters: the caption for the
  normal image, the response, and the
  reasoning.
23 Considering these few-shot examples, we
  expect you to successfully achieve
  your current task.
24
25 [Example]
26 ### Example {example_number}
27 - Caption: {example_caption}
28 - Response: {example_response}
29 - Reasoning: {example_reasoning}
30 - Image:
31
32 Your Current Task:
33 Given the input image and its
  corresponding caption, distinguish
  whether it is weird or normal. Ensure
  adherence to the response format
  provided above.
34
35 Caption: {caption}
36 Image:
```

Question-Answering Agent

The Question-Answering agent is involved in the third phase of our pipeline, returning five pairs of questions and answers per image. As mentioned in the paper, this agent provides a question-choices-answer set and the corresponding composition of interest.

```
1 Your Role: Questions and Answers
  Generator
2
3 Objective:
4 Given a sample that includes a prompt
  and textual reasoning that explains
  whether the given image is normal (
  reasonable and factual) or weird (
  hallucination, incorrect, distorted,
  and contrary to common sense),
  generate multi-choice questions that
  verify if the image is normal or
  weird.
5 When a correctly or incorrectly
  generated image is provided for a
  given prompt, these QA(Question and
```


Answers) sets are designed to accurately assess and distinguish the factual information regarding the image.

6 The last choice should be "None of the above."

7

8 You should also match each QA with the most related compositions of interest (CoI).

9 The table of CoIs includes existence, size, color, shape, posture, relation, scene, and counting.

10 If the generated QA does not match any of the CoIs listed above, select "others."

11

12 Constraints:

13 - If the correct answer to a question in the QA set is given for a given image, the image should contain factual information. In other words, an incorrectly generated image should not have the correct answer.

14 - In reasoning, exclude context evidence (i.e., "The Scream of Nature" falls under which art movement?) that cannot be visually recognized. In other words, you should only generate QA sets based on visual evidence; only referring to context evidence for better understanding.

15 - Avoid including explicit expressions (i.e., "The Scream of Nature", "Edvard Munch", etc.) in your QA. Rather, replace them with general expressions like "the image".

16

17 You should follow the response format below:

18

19 [Response Format]

20 Set 1:

21 Compositions of Interest:

22 Question:

23 Choices:

24 A)

25 B)

26 C)

27 D)

28 E) None of the above

29 Answer:

30

31 Set 2:

32 Compositions of Interest:

33 Question:

34 Choices:

35 A)

36 B)

37 C)

38 D)

39 E) None of the above

40 Answer:

41

42 Set 3:

43 Compositions of Interest:

44 Question:

45 Choices:

46 A)

47 B)

48 C)

49 D)

50 E) None of the above

51 - Answer:

52

53 Set 4:

54 Compositions of Interest:

55 Question:

56 Choices:

57 A)

58 B)

59 C)

60 D)

61 E) None of the above

62 Answer:

63

64 Set 5:

65 Compositions of Interest:

66 Question:

67 Choices:

68 A)

69 B)

70 C)

71 D)

72 E) None of the above

73 Answer:

74

75 Your Current Task:

76 Given the reasoning for an image, generate five possible QA sets. Ensure adherence to the response format provided above.

77

78 Caption: {caption}

79 Reasoning: {reasoning}

Evaluation Agent

The Evaluation agent represents the final stage of our pipeline, scoring I-HallA for a given image. Specifically, when GPT-4o determines that none of the choices correctly answer the question, it can return "E) None of the above". We then calculate the I-HallA score for the image by comparing the predicted answers from the five questions with the ground-truth answers. For example, if 2 out of 5 questions are answered correctly, we assign a I-HallA score of 0.4 for that image.

1 You are an agent who answers questions based on the given image.

2 Here, the given image could be wrong.

3

4 So, your job is to either choose the best answer choice or output "None of the above" for ambiguous questions.

5

6 Your answer must choose one of the given choices to your best knowledge.

7 It must be a single character that indicates its answer (i.e., "A", "B",

"C", "D", "E").

8 As shown below [Example], your answer
should not include texts other than a
single character ("A", "B", "C", "D",
", and "E").

9

10 [Example]

11 Question: How many continents are shown
as connected in the image?

12 Choices:

13 A) Korean Peninsula

14 B) One large supercontinent

15 C) Two separate continents

16 D) Three continents joined together

17 E) None of the above

18 Answer: B

19

20 Question: How many atoms of the molecule
in the image?

21 Choices:

22 A) 1

23 B) 2

24 C) 3

25 D) 4

26 E) None of the above

27 Answer: E

28

29 DO NOT use your external knowledge
obtained by pre-training large-scale
data.

30 You must base your judgment on the image
input's visual evidence.

31

32 You should answer the following question
:

33

34 [Question]

35 Question: {question}

36 Choices: {choices}

37

38 [Response Format]

39 Answer:

T2I-FactualBench: Benchmarking the Factuality of Text-to-Image Models with Knowledge-Intensive Concepts

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Abstract

Most existing studies on evaluating text-to-image (T2I) models primarily focus on evaluating text-image alignment, image quality, and object composition capabilities, with comparatively fewer studies addressing the evaluation of the factuality of the synthesized images, particularly when the images involve knowledge-intensive concepts. In this work, we present T2I-FactualBench—the largest benchmark to date in terms of the number of concepts and prompts specifically designed to evaluate the factuality of knowledge-intensive concept generation. T2I-FactualBench consists of a three-tiered knowledge-intensive text-to-image generation framework, ranging from the basic memorization of individual knowledge concepts to the more complex composition of multiple knowledge concepts. We further introduce a multi-round visual question answering (VQA)-based evaluation framework to assesses the factuality of three-tiered knowledge-intensive text-to-image generation tasks. Experiments on T2I-FactualBench indicate that current state-of-the-art (SOTA) T2I models still leave significant room for improvement. We release our datasets and code at <https://github.com/Safeoffellow/T2I-FactualBench>.

1 Introduction

In recent years, text-to-image (T2I) generation have made significant advancements in synthesizing high-fidelity and diverse style images from input textual descriptions (Rombach et al., 2022; Podell et al., 2024; Li et al., 2024c; Chen et al., 2024a; Sun et al., 2024b,a; Wang et al., 2024c; He et al., 2024). T2I models, represented by diffusion models (Rombach et al., 2022; Podell et al., 2024; Li et al., 2024c; She et al., 2025) and autoregressive models (Sun et al., 2024b,a; He et al., 2024; Wang et al., 2025b; Liu et al., 2025), have been applied to

a wide range of scenarios, including e-commerce, art and games (Oppenlaender, 2022; Vashishtha et al., 2024; Vimpari et al., 2023).

A significant challenge accompanying the advancement of T2I generation lies in evaluating the generated images (Hartwig et al., 2024). Most existing efforts on this challenge primarily focus on evaluating text-image alignment (Hessel et al., 2021; Radford et al., 2021; Yu et al., 2022), image quality (Kirstain et al., 2023; Li et al., 2024b; Xu et al., 2024; Saharia et al., 2022; Wu et al., 2023) and object composition capability (Park et al., 2021; Saharia et al., 2022; Li et al., 2024a; Wu et al., 2024b; Huang et al., 2023) *inter alia*, using automated metrics such as Fréchet Inception Distance (FID) (Heusel et al., 2017), Inception Score (IS) (Salimans et al., 2016), and CLIPScore (Hessel et al., 2021). Recently, several efforts have been made to evaluate the reasoning capabilities of T2I models, as exemplified by Commonsense-T2I (Fu et al., 2024) and PhyBench (Meng et al., 2024).

However, despite the aforementioned efforts, a comprehensive benchmark for evaluating the factuality of T2I models in generating knowledge-intensive concepts and their compositions is still lacking. Knowledge-intensive concepts differ significantly from general concepts or objects because their visual features are often difficult—or even unnecessary—to explicitly describe in the input textual description. For example, as shown in Fig. 1, when given prompts with general concepts, the SOTA T2I model effectively generate images that fulfill the instructions. However, when presented with specific knowledge-intensive concepts, such as a *LV M43986 Cannes handbag*, the generated images often struggle to accurately represent the intended concepts. This characteristic sets the evaluation of such concept generation apart from traditional text-alignment evaluations.

To our best knowledge, the most closely related studies on the evaluation of knowledge-intensive

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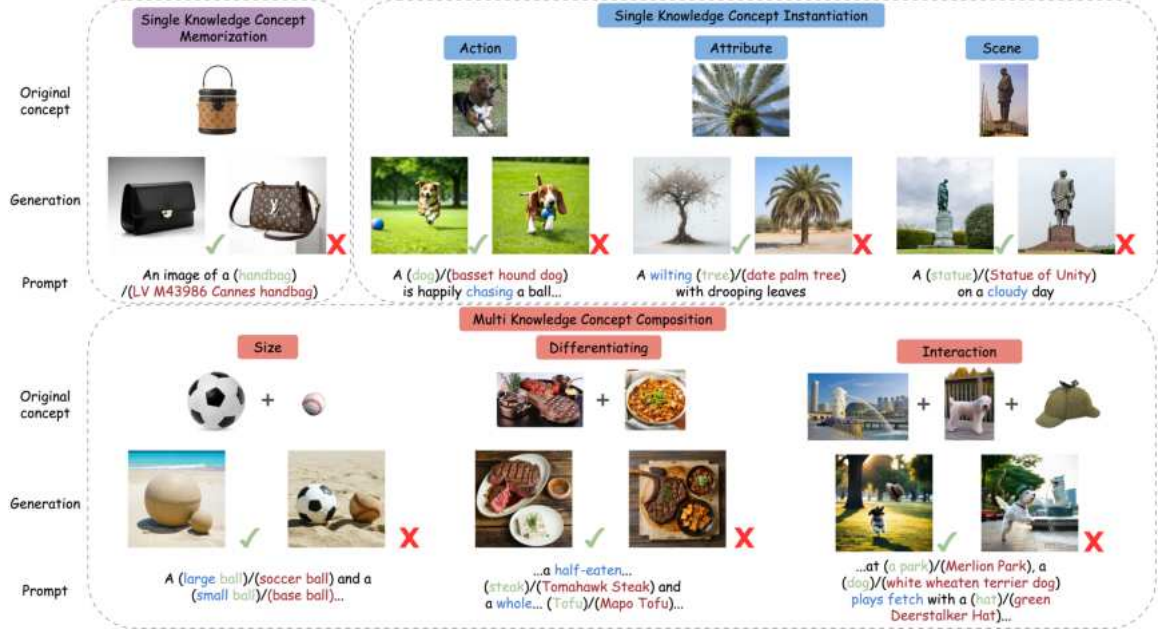


Figure 1: **General Concepts vs. Knowledge Concepts.** We use the SOTA T2I model Stable Diffusion 3.5 (SD 3.5; (Esser et al., 2024)) as an example to illustrate the challenges posed by knowledge-intensive concepts versus general concepts. When given prompts with general concepts (indicated in green), SD 3.5 effectively generates images (left in Generation) that fulfill the instructions. However, when presented with specific knowledge concepts (indicated in red), SD 3.5 (right in Generation) often struggles to meet the requirements or accurately represents the intended concepts. This issue is particularly pronounced when the images are required to compose multiple knowledge concepts. The blue text in the prompts highlights the specific tasks to be achieved.

concept generation are as follows: HEIM (Lee et al., 2024) conducts a holistic evaluation of T2I models across 12 different aspects, such as alignment, quality and aesthetic, etc. Among them, the knowledge dimension evaluate whether the model have knowledge about the world or domains. However, HEIM evaluates only a very limited set of real-world entities and employs superficial CLIP-Score to assess the factuality of entities. We also note the concurrent work KITTEN (Huang et al., 2024) explores the knowledge-intensive evaluation of image generation for real-world visual entities. However, KITTEN employs four pre-defined templates to generate input textual descriptions, restricting its flexibility to provide a comprehensive evaluation of concept under different scenarios.

In this work, we present T2I-FactualBench, the largest benchmark to date in terms of the number of concepts and prompts designed to evaluate the factuality of T2I models when generating images that involves knowledge-intensive concepts. The construction of T2I-FactualBench begins with collecting a set of **Knowledge Concepts, which are defined as concepts with a limited number of hyponyms in the knowledge base**. Knowledge concepts are specifically designed to challenge T2I models by requiring them to precisely generate

inherent visual details of each concept. Building upon the collection of knowledge concepts, we next propose a three-tiered knowledge-intensive text-to-image generation framework, spanning from the basic memorization of individual knowledge concepts to the more complex composition of multiple knowledge concepts.

To conduct an effective and efficient evaluation of existing T2I models’ performance on the proposed T2I-FactualBench, we also introduce a multi-round visual question answering (VQA)-based evaluation framework aided by advanced multi-modal LLMs. This multi-round VQA evaluation framework firstly assesses the factuality of the generated concept with respect to the reference image, and then evaluates the completeness of concept instantiation under different conditions, lastly examines the factuality of multiple concept compositions under varying scenarios.

We conduct a comprehensive evaluation of the performance of seven closed- and open-source T2I models on the proposed T2I-FactualBench, such as Stable Diffusion models (Rombach et al., 2022; Podell et al., 2024; Esser et al., 2024), Flux.1 (Andreas Blattmann, 2024) and DALL-E-3 (Betker et al., 2023). Furthermore, we explore two approaches for injecting external knowledge into the

models to facilitate the generation of knowledge concepts. The first approach is *Visual-Knowledge Injection*, where images of knowledge concepts are provided as references to guide the model in image generation. The second is *Text-Knowledge Injection*, where we provide textual descriptions of the visual features of knowledge concepts as external knowledge. Experiments on T2I-FactualBench indicate that current state-of-the-art (SOTA) T2I models still leave significant room for improvement.

2 Related work

2.1 Text-to-Image Generation Evaluation

Current evaluation metrics for text-to-image generation focus on key aspects such as the fidelity of generated images, assessed using FID and IS (Heusel et al., 2017; Salimans et al., 2016), and perceptual similarity evaluated by CLIP-I and DINO Score (Caron et al., 2021). Additionally, image-text alignment is measured using metrics such as CLIP-T, CLIP Score, and BLIP Score (Radford et al., 2021; Hessel et al., 2021; Li et al., 2022). However, these metrics often fall short in capturing the intricate nuances of text-image alignment. With the advancement of large language models (LLMs) and large multimodal language models (MLLMs) (Chen et al., 2024b; Wang et al., 2024a; OpenAI, 2024; Xiao et al., 2024b; Zhao et al., 2024; Xiao et al., 2024a; Liu et al., 2024; Lin et al., 2024; Bi et al., 2025, 2024), some approaches use MLLMs to employ VQA evaluation (Hu et al., 2023; Lu et al., 2024b; Huang et al., 2023; Kirstain et al., 2023; Xu et al., 2024). However, the binary yes-or-no answer format proves insufficient for detailed evaluations. Human evaluations (Ku et al., 2023b; Lu et al., 2024a; Huang et al., 2024) provide critical insights but are limited by significant costs and time-intensive processes. Some studies (Ku et al., 2023a; Zhang et al., 2023a; Wu et al., 2024a; Peng et al., 2024) highlight GPT-4V’s potential in image evaluation, indicating its effectiveness as a human-aligned evaluator for text-to-image generation.

2.2 Text-to-Image Generation Benchmarks

In terms of benchmarks, certain ones focus on assessing the images quality and alignment with human preferences (Kirstain et al., 2023; Li et al., 2024b; Xu et al., 2024; Yu et al., 2022), while others evaluate compositional generation by ana-

lyzing attributes like counting, color, and relationships (Huang et al., 2023; Park et al., 2021; Saharia et al., 2022). Recently, the focus has shifted to comprehensive evaluations (Lee et al., 2024; Li et al., 2024a; Wu et al., 2024b). For example, HEIM (Lee et al., 2024) comprehensively evaluates models across 12 distinct dimensions of capability. Furthermore, some benchmarks begin to emphasize the reasoning capability of T2I models. Commonsense-T2I (Fu et al., 2024) and Phy-Bench (Meng et al., 2024) focus on evaluating the multimodal commonsense understanding of generative models. However, the factuality of T2I models has not been sufficiently evaluated in the literature (Lee et al., 2024; Huang et al., 2024).

3 T2I-FactualBench Construction

In this section, we detail the construction process of T2I-FactualBench, a benchmark designed to evaluate the factuality of T2I models when generating images that rely on rich world knowledge.

3.1 Knowledge Concept Collection

The first step in constructing T2I-FactualBench involves collecting a set of knowledge-intensive concepts aimed to challenge T2I models by requiring generating precise visual details, rather than merely depicting general concepts. In this paper, we define a **Knowledge Concept** as a concept that has limited hyponyms in the knowledge base BabelNet (Navigli and Ponzetto, 2012).

Concept Category. We source to CNER (Martinelli et al., 2024) as the corpus to construct the knowledge concept set. CNER is a task designed for recognizing nominal concepts and named entities within a unified category space. It utilizes the completeness and broad semantic coverage of lexicographer files for nominal concepts in WordNet, along with the widely adopted semantic categories for named entities in OntoNotes, resulting in the establishment of 29 distinct categories. Among the 29 distinct categories, we focus on eight categories that can be grounded in real-world entities, such as *animal*, *artifact*, and *food*, while excluding abstract concept categories, such as *language*, *law*, and *discipline*.

Knowledge Concept Filtering. Given the training dataset of CNER, we begin by filtering out concepts in the eight categories and use SpaCy for lemmatization to obtain their standard lexical

Category	Subcategory	Examples	Num
ANIMAL	Mammals, Bird, Insect	<i>Bombay cat, Bombay cat, Keeshond dog, Aberdeen Angus, Damaliscus lunatus, Podilymbus, Crocodylus</i>	376
LOCATION	Landmark, Natural landform	<i>Kinderdijk, Leaning Tower of Pisa, Mont Saint-Michel, Oriental Pearl Tower, Butte, Danxia landform</i>	357
PLANT	Flower, Fruit, Tree	<i>Carnation, Balsam fir, Prunus armeniaca, Shumard oak, Syzygium, Coral bush</i>	312
ARTIFACT	Vehicle, Sports equipment, Musical instrument, Clothing, Tool	<i>Tesla Model 3, Ski pole, Kazoo, Pillbox hat, Chanel 2.55 flap bag, Sweater vest, DNA sequencer</i>	267
PERSON	Person	<i>Taylor Swift, Usain Bolt, Audrey Hepburn, Diane Keaton, Barack Obama, Mark Zuckerberg</i>	132
FOOD	Food	<i>Yakitori, Shrimp tempura, Macarons, Lasagna, Moon-cake, Mapo Tofu</i>	131
EVENT	Event	<i>COVID-19 pandemic, Ratha-Yatra, World Rally Championship, Tour de France</i>	40
CELESTIAL	Celestial	<i>Horsehead Nebula, Mars, Mercury, Uranus, Enceladus</i>	14

Table 1: Category, Subcategory, Examples, and Number of knowledge concepts in T2I-FactualBench

forms. We then utilize BabelNet to gather relevant information and determine whether it satisfies our definition as a knowledge concept. Each concept is queried in BabelNet to gather relevant synsets, synonyms, categories, hypernyms, hyponyms, and images. We aim to select knowledge concepts with fewer than four hyponyms, hypothesizing that such concepts are more likely to exhibit distinct visual attributes, making them well-suited for thoroughly evaluating the knowledge capabilities of T2I models.

We acknowledge that the popularity of knowledge concepts can significantly influence model performance. During the collection process, we find that the curated set of knowledge concepts includes both widely recognized concepts, such as "Husky dog," "British Shorthair cat," "bucket hat," "Macaron," and "Taylor Swift," as well as concepts that may be less commonly encountered in everyday contexts.

Ultimately, we curate a dataset of 1,600 knowledge concepts as a pool across eight domains, including *animals, artifacts, food, persons, plants, celestial bodies, events, and locations*. In Table 1, we detail the categorical distribution of knowledge concepts. For the detailed concepts collection, see Appendix A.

3.2 Text-to-Image Generation with Knowledge-Intensive Concept

Building upon the collection of knowledge-intensive concepts, we propose a three-tiered text-to-image generation task to comprehensively evaluate the factual accuracy of T2I models.

T1: Single Knowledge Concept Memorization.

We define the first level T2I generation task as Single Knowledge Concept Memorization (SKCM), which aims to assess whether T2I models can accurately generate a single knowledge concept, such as its specific visual attributes. Specifically, we utilize all concepts from the knowledge concept pool to construct task prompts using the following template: "An image of {Knowledge Concept}". Note that, in this task, the T2I model is permitted to determine the state or action of the knowledge concept.

T2: Single Knowledge Concept Instantiation.

Next, we introduce the second-level T2I generation task, referred to as Single Knowledge Concept Instantiation (SKCI). SKCI advances T2I generation evaluation by measuring the model’s ability to accurately instantiate knowledge concepts under reasonable conditions, such as depicting diverse actions for animals or varying attributes for objects. SKCI is designed to test the model’s understanding of the intrinsic properties and behaviors associated with knowledge concepts, which are often difficult to explicitly articulate in prompts.

Specifically, in SKCI, we define three types of instantiation: $T = \{Action, Attribute, Scene\}$. *Action* is designed for animal, artifact, person, plant and food categories, which instantiates knowledge concepts with different actions. *Attribute* is tailored for categories such as animal, artifact, person, plant and food, instantiating these concepts with diverse states. *Scene* is designed for the location category, instantiating knowledge concepts through various

environmental conditions, including weather and time of day. A knowledge concept $c \in C$ is randomly sampled, along with an instantiation type $t \in T$ selected according to the concept’s category. Given c and t , a powerful LLM M , such as GPT-4o, is prompted to sample one reasonable instantiation phrase p . For example, a phrase may be "chasing a ball" for knowledge concept "basset hound dog". Lastly, the reasonable phrase p_i is combined with the concept c to prompt the LLM M to produce a SKCI prompt S , which is used as the textual input for the T2I model.

T3: Multiple Knowledge Concept Composition with Interaction. Finally, we define the third-level T2I generation task as Multiple Knowledge Concept Composition with Interaction (MKCC), which is designed to evaluate a model’s ability to simultaneously compose multiple knowledge concepts within a single image. MKCC assesses not only the common challenges encountered in general concept composition (Huang et al., 2023), such as adherence to prompts and seamless integration, but also both the implicit and explicit semantic relationships between different knowledge concepts.

Specifically, given two randomly selected knowledge concepts c_1 and c_2 from the concept set C , we first prompt GPT-4o to determine whether a significant size disparity exists between c_1 and c_2 . If such a size discrepancy is identified, the LLM then proceeds to generate the prompt S . For example, given the prompt "A soccer ball and a baseball", the T2I model must correctly generate one image where the soccer ball is significantly larger than the baseball.

Next, if the two concepts do not show significant size discrepancy, we prompt GPT-4o to generate actions or attributes that can be used to instantiate each knowledge concept, following a process similar to that used in SKCI. This instantiated knowledge concept composition critically evaluates the ability of the T2I model to simultaneously represent the distinctive visual features of different knowledge concepts under various instantiations. Note that, in this task, the T2I model is granted the flexibility to determine the interaction between the instantiated knowledge concepts.

Lastly, we incorporate specific semantic relationships between knowledge concepts to further assess the model’s ability to composite multiple knowledge concepts that interact with one another. Specifically, given two foreground (animal, plant,

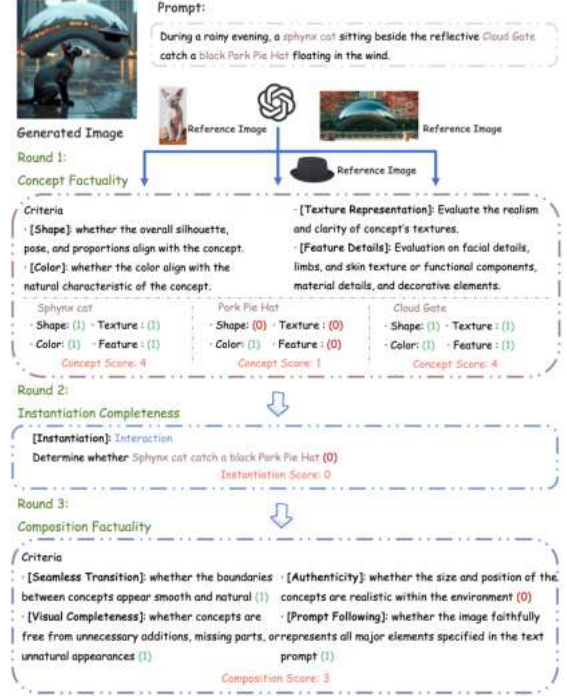


Figure 2: Multi-Round VQA based Factuality Evaluation Pipeline. We present an evaluation case in MKCC level.

food, person, artifact) concepts, c_1 and c_2 , and an optional background (location) concept, we first use GPT-4o to instantiate these concepts and then determine the interaction feasibility between the two foreground concepts, as well the suitability of these interactions occurring within the background concept. If appropriate, we use GPT-4o to generate one plausible interaction phrase, which is combined with the optional background concept to construct the prompt S .

Ultimately, we have constructed 3,000 prompts across all three levels. For detailed information of three-tiered framework, see Appendix A

4 Multi-Round VQA based Factuality Evaluation

To conduct an effective and efficient evaluation of the T2I model’s performance on the proposed T2I-FactualBench, we introduce a multi-round visual question answering (VQA)-based evaluation framework, aided by advanced multi-modal LLMs. This framework consists of three VQA tasks: (1) Concept Factuality Evaluation; (2) Instantiation Completeness Evaluation; and (3) Composition Factuality Evaluation. Figure 2 provides an overview of the multi-round VQA-based evaluation for T2I-FactualBench.

4.1 Concept Factuality Evaluation

At the core of the T2I-FactualBench evaluation is the precise assessment of the factuality of the generated knowledge-intensive concepts. To achieve this, in the first round of VQA, we employ an advanced multi-modal LLM combined with the reference image as an effective proxy for the human evaluator to assess the factuality of the generated image. The reference image is obtained in the knowledge concept collection process sourced from BabelNet.

Specifically, given the knowledge concept c_i , its model generated image I_i , and the reference image I_i^R , we design a dedicated evaluation prompt to instruct GPT-4o to assess the factuality of the knowledge concept in the generated image across four dimensions: *shape, color, texture, and feature details*, as outlined in DreamBench++ (Peng et al., 2024). For the detailed definitions of the four dimensions, see Appendix B.1. For each dimension, GPT-4o assigns a score of 1, accompanied by a rationale, if the generated image meets the defined criteria. Otherwise, a score of 0 is assigned.

At the SKCM and SKCI levels, the generated image contains only a single concept. In contrast, at the MKCC level, the generated image encompasses multiple concepts. Therefore, we define the concept factuality score of I_i as:

$$\text{Concept Factuality} = \frac{1}{N_i} \sum_{j=1}^{N_i} \left(\frac{S_{ij} + C_{ij} + T_{ij} + F_{ij}}{4} \right) \quad (1)$$

where N_i denotes the number of concepts within I_i . $S_{ij}, C_{ij}, T_{ij}, F_{ij}$ represent the scores for the respective dimensions for the j -th concept in I_i .

4.2 Instantiation Completeness Evaluation

In addition to evaluating the factuality of the concept, we next evaluate whether the T2I model can produce precise instantiation of knowledge concepts.

We prompt GPT-4o to determine if c exists and the instantiation phrase p is successfully completed. If both conditions are met, the concept instantiation completeness score is assigned to 1. Otherwise, it is assigned a value of 0. For the detailed evaluation instruction, see Appendix B.2.

4.3 Composition Factuality Evaluation

Finally, in the last evaluation round, we design a VQA task to comprehensively evaluate the factuality of composing multiple knowledge concepts within a single image.

Specifically, given the text prompt t_i and the model-generated image I_i , we instruct GPT-4o to evaluate the composition factuality of the knowledge concepts in I_i across four dimensions: *Seamless Transition, Visual Completeness, Authenticity, and Prompt Following*. For the detailed definitions of the four dimensions, please see Appendix B.2. For each dimension, we apply the scoring system outlined in Section 4.1 and the composition factuality score of I_i is defined as:

$$\text{Composition Factuality} = \frac{S_i + V_i + A_i + P_i}{4} \quad (2)$$

where S_i, V_i, A_i, P_i representing scores for the respective dimensions for the i -th prompt, with each score taking a value of either 0 or 1.

5 Experiment

5.1 Experimental Setup

Text-to-image models We comprehensively evaluate the performance of seven text-to-image models on the T2I-FactualBench, including three variants of Stable Diffusion: (1) Stable Diffusion v1.5 (Rombach et al., 2022), (2) Stable Diffusion XL (Podell et al., 2024), and (3) Stable Diffusion 3.5 (Esser et al., 2024). Other models evaluated include (4) PixArt-alpha (Chen et al., 2024a), (5) Playground v2.5 (Li et al., 2024c), and (6) Flux.1 (Andreas Blattmann, 2024). For API-based models, we evaluate (7) DALL-E 3 (Betker et al., 2023)¹.

In addition to the aforementioned T2I models, we develop a visual knowledge injection method based on two subject-driven generation models: (8) SSR-Encoder (Zhang et al., 2023b), based on Stable Diffusion v1.5 and (9) MS-Diffusion (Wang et al., 2024b), based on Stable Diffusion XL. These models are capable of referencing one or more images during the generation process to enhance the factuality of subject representation. We also introduce a text-based knowledge injection method utilizing (10) Stable Diffusion 3.5* and (11) Flux.1 dev* due to the robust semantic comprehension capabilities afforded by their Diffusion Transformer (DiT) architecture. For further details about the methods, please refer to Appendix C.2.

¹Note that due to policy restrictions on generating images of individuals, we did not evaluate DALL-E 3 for person knowledge concepts.

Model	SKCM	SKCI		MKCC		
	Concept	Concept	Instantiation	Concept	Instantiation	Composition
<i>Text-to-image Generation</i>						
SD v1.5	40.5	52.9	53.9	37.6	13.4	15.1
SD XL	45.8	59.9	65.8	51.7	28.0	35.4
Pixart	26.4	46.2	55.3	35.8	19.8	24.3
Playground	45.6	66.1	62.5	53.8	35.4	44.8
Flux.1 dev	35.6	54.9	58.0	56.9	54.1	63.8
SD 3.5	46.2	64.6	71.2	68.9	59.2	75.5
DALLE-3	55.5	72.4	88.5	71.3	70.2	85.6
<i>Visual-Knowledge Injection</i>						
SSR-Encoder	71.8 ↑ 31.3	69.0 ↑ 16.1	22.8 ↓ 31.1	43.1 ↑ 5.5	12.5 ↓ 0.9	9.5 ↓ 5.6
MS-Diffusion	84.8 ↑ 39.0	80.4 ↑ 30.5	50.8 ↓ 15.0	65.5 ↑ 13.8	32.9 ↑ 14.9	31.0 ↓ 4.4
<i>Text-Knowledge Injection</i>						
Flux.1 dev*	41.2 ↑ 5.6	60.3 ↑ 5.4	59.8 ↑ 1.8	64.2 ↑ 7.3	56.9 ↑ 2.8	72.6 ↑ 8.8
SD 3.5*	49.7 ↑ 3.5	66.7 ↑ 2.1	65.8 ↓ 5.4	67.9 ↓ 1.0	53.6 ↓ 5.6	64.7 ↓ 10.8

Table 2: **Main results** on T2I-FactualBench. We present the performance of text-to-image generation models and two distinct knowledge injection methods following Multi-Round VQA evaluation across three levels. We highlight the row of DALLE-3 in gray to denote the incompleteness of its evaluation data. **Model *** indicates that the model has undergone text-knowledge injection. ↑ and ↓ denote improvements and declines relative to their base models (SSR-Encoder→SD v1.5, MS-Diffusion → SD XL, Flux.1 dev* → Flux.1 dev, SD 3.5* → SD 3.5).

5.2 Evaluation Metrics

In addition to the proposed multi-round VQA-based factuality evaluation framework, we evaluate the T2I models using various metrics, including CLIP-T, CLIP-I (Radford et al., 2021), and DINO Score (Caron et al., 2021). For comparison, we also incorporate two MLLM-based evaluation methods, TIFA Score (Hu et al., 2023) and LLMscore (Lu et al., 2024b), which leverage MLLMs to provide a fine-grained assessment of text-image alignment.

5.3 Main Results

We first report the quantitative results of diverse text-to-image models on T2I-FactualBench.

Models Performance Across Three Levels. As shown in Table 2, the performance on T2I-FactualBench improves with the advancement of the backbone model. For example, Stable Diffusion 3.5 achieves higher concept factuality scores compared to previous models, such as SD v1.5 and SD XL. Furthermore, stronger models exhibit subtle changes in concept factuality scores when transitioning from SKCI to MKCC (e.g., SD 3.5: 64.6→68.9; Flux.1 dev: 54.9→56.9). They also perform better in composition evaluation (e.g., SD 3.5: 75.5; Flux.1 dev: 63.8). In contrast, weaker models experience significant declines in concept factuality scores (e.g., Playground: 66.1 → 53.8) and achieve lower composition factuality scores

(e.g., Playground: 44.8).

Moreover, as instantiation complexity increases from SKCI to MKCC, all models exhibit a decline in Instantiation Completeness scores. These more intricate tasks require not only the retention of multiple knowledge concepts but also the ability to distinctly instantiate and effectively compose these concepts during generation. This observation highlights the limitations of existing T2I models in generating images involving knowledge concepts and their complex interactions.

We also observe a counter-intuitive trend that concept factuality scores tend to increase as task complexity increasing, from SKCM to MKCC. We hypothesize that this is due to the number of concepts varies across tasks of different levels. In fact, SKCM utilizes all the collected concepts. However, as described in Section 3.2, the prompts for SKCI and MKCC are constructed by randomly sampling one or more concepts along with an instantiation type, which are then provided to the LLM to generate a coherent and reasonable prompt. If a concept fails to generate a coherent and reasonable prompt, it will be discarded and resampled. Furthermore, at the MKCC level, the LLM is more likely to select combinations of more prevalent concepts. For example, the LLM may prefer combining concepts like "Taylor Swift" and "Shiba Inu dog" over "Taylor Swift" and "flying lemur." Therefore, SKCI and MKCC show higher concept factuality scores than

Method	Spear.	Kend.
CLIP-T	0.256	0.196
CLIP-I	0.235	0.175
DINO	0.277	0.207
TIFA Score	0.354	0.311
LLMscore	0.262	0.233
Concept Factuality (Ours)	0.568	0.491
- w/o reference image	0.442	0.376

Table 3: Correlation Scores between previous metrics and human evaluation in **Concept Factuality**. Spear. and Kend. represents Spearman and Kendall correlations, respectively.

SKCM. In Appendix D.2, we conduct an ablation study to validate this hypothesis.

Effect of Visual-Knowledge Injection. Table 2 also shows that the visual knowledge injection method (i.e., using reference images as visual knowledge) can significantly improve concept factuality of base models. However, their performance on instantiation and composition declines. We hypothesize that while visual knowledge injection enhances the factuality of concept generation, it simultaneously impairs the model’s ability to follow instructions and accurately integrate multiple concepts.

Effect of Text-Knowledge Injection. In terms of the effect of text-knowledge injection, Flux.1 dev* shows significant improvements across various metrics with the additional textual descriptions of knowledge concepts. Conversely, SD 3.5* improves slightly in concept factuality but declines in instantiation completeness and composition factuality. This decline could potentially be attributed to the long prompts impairing instruction-following and concept composition capabilities. This finding suggests that while text-knowledge injection enhances concept generation accuracy, it may hinder instruction following with complex prompts.

5.4 Analyzes

Multi-Round VQA Metrics Better Aligning with Human Preference. We conduct experiments to investigate the effectiveness of the proposed multi-round VQA evaluation framework by comparing the VQA answers with human annotations. Specifically, we curate a validation set of 900 samples for all three level tasks and engaged three annotators on iTAG platform² to evaluate Concept Factuality,

²<https://www.alibabacloud.com/help/en/pai/user-guide/itag/>

MLLMs	Concept		Composition		Instantiation
	Spear.	Kend.	Spear.	Kend.	ACC
<i>Open-Source</i>					
Qwen2.5-VL-7B	0.351	0.293	0.325	0.284	0.42
Qwen2.5-VL-72B	0.488	0.405	0.490	0.434	0.61
Internvl-2.5-78B	0.413	0.345	0.406	0.361	0.57
<i>Closed-Source</i>					
GPT-4o mini	0.347	0.283	0.444	0.392	0.63
Qwen-VL-Max	0.512	0.433	0.434	0.387	0.59
Gemini-1.5-Flash	0.541	0.470	0.513	0.466	0.73
Gemini-2.0-Flash	0.478	0.413	0.578	0.502	0.74
GPT-4o	0.568	0.491	0.662	0.608	0.81

Table 4: Comparisons of using different multi-modal LLMs as the backbone model for multi-round VQA evaluation. Spear. and Kend. represents Spearman and Kendall correlations, respectively.

T2I Models	Concept	Instantiation	Composition
SD v1.5	34.8 / 46.9	10.0 / 13.0	10.5 / 16.3
SD XL	52.9 / 70.3	13.0 / 16.0	28.8 / 38.5
Pixart	38.2 / 71.9	11.0 / 18.0	16.3 / 42.0
Playground	55.7 / 79.7	13.0 / 23.0	40.0 / 59.0
Flux.1 dev	55.3 / 84.3	30.0 / 46.0	54.8 / 82.3
SD 3.5	65.0 / 85.3	31.0 / 38.0	69.5 / 75.5

Table 5: Comparison of T2I models’ performance on **Knowledge Concept** and **General Concept**.

Instantiation Completeness, and Composition Factuality. The questions presented to the annotators are consistent with the prompts for Multi-Round VQA. We employ Spearman and Kendall correlations to quantify the alignment between human ratings and scores generated by Concept and Composition Factuality. For binary Instantiation Completeness scores, we compute accuracy. Detailed information about human annotations can be found in Appendix C.4.

Table 3 shows that our Concept Factuality evaluation aligns more closely with human judgments than previous metrics, highlighting its superior accuracy as reliable evaluation methods. Furthermore, when removing reference images from Concept Factuality assessment, there is a significant reduction in correlation coefficients, underscoring the necessity of providing reference images for accurate evaluation. In Appendix D, we conduct an inter-human agreement analysis and an error analysis for the Multi-Round VQA.

Impact of MLLM Models on Multi-Round VQA

To explore how different MLLM models impact evaluation results, we test several closed- and open-source models. As shown in Table 4, GPT-4o aligns more closely with human preferences in Concept and Composition Factuality and achieves the highest **81%** accuracy in Instantiation Completeness. Therefore, we choose GPT-4o as our evaluation model to ensure more accurate assessments.

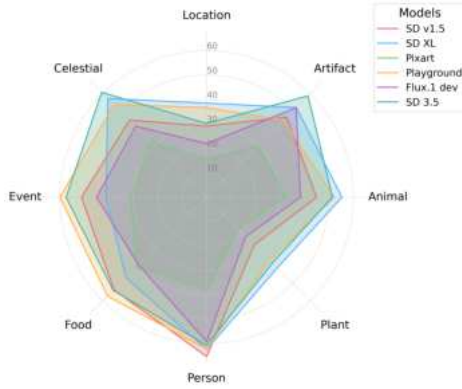


Figure 3: Concept Factuality Scores across 8 domains in the SKCM level for text-to-image models

While GPT-4o is chosen for its superior performance in T2I-FactualBench, we acknowledge that several open-source models, such as Qwen2.5-VL-72B (Bai et al., 2025) and Internvl-2.5-78B (Wang et al., 2025a), also demonstrate promising performance. These models present an opportunity to explore the trade-offs between performance and resource efficiency. We will release a trained open evaluation model in the future to enhance the reproducibility of our work.

Ablating Knowledge Concept with General Concepts. To determine if models’ poor performance on T2I-FactualBench is attributable to their limited factual generation capability related to knowledge concepts, we randomly select 100 prompts from the most challenging MKCC task and replace specific knowledge concepts with general ones (e.g., "Basset hound dog" to "dog"). Table 5 shows that models achieve significant improvements across three metrics when prompts including general concepts instead of knowledge concepts.

Models Face Challenge in Certain Domains. We analyze concept factuality scores across 8 domains within the SKCM level, as shown in Figure 3. Results indicate that models perform relatively well in animals, artifacts, and food, but poorly in plants and locations. We hypothesize that this discrepancy is due to the limited representation of plant and location concepts in the common training datasets (e.g., COCO and LAION). Further analysis of specific instances indicates that models struggle with generating concepts requiring detailed features. Specifically, when generating concepts related to plants and detailed elements of landmark buildings such as statues or architectural decorations, models fail to accurately capture

intricate characteristics and complex textures.

6 Qualitative Analysis

Through an extensive qualitative analysis of images generated by various models on our benchmark, we identify four key deficiencies in current generative models: **Concept Error, Instantiation Failures, Realism Error, and Feature Mixture Error**. Detailed presentation of these findings is provided in Appendix D.6

First, models often obscure fine details of knowledge concepts or to generate similar but incorrect concepts. such as missing food concepts "Mapo Tofu" and "Gyoza" texture details, producing overly smooth surfaces and generating similar animal "wild boar" instead of "Baird’s tapir". Second, models struggle with executing related attribute changes. For instance, they capture "ice skate" features but fail with variations like "broken ice skate". Third, when generating multiple objects, models often fail to integrate distinct features accurately, resulting in chaotic fusions that obscure individual characteristics. For instance, when generating both an "Egyptian Mau cat" and a "Basset Hound dog," the traits of these animals are often mixed inappropriately. Moreover, interactions among multiple concepts sometimes result in implausible scenarios, such as generating a "keeshond dog" embedded in a car window rather than positioned inside the car.

Additionally, with visual-knowledge injection, the output frequently lacks coherent integration of multiple knowledge concepts, appearing as simplistic concatenations. In text-knowledge injection, while improving factuality, it struggles with complex concepts like Mapo Tofu, failing to enhance instantiation performance or integration, indicating a need for refinement in handling complex tasks.

7 Conclusion

In this work, we introduce T2I-FactualBench, a benchmark to evaluate the factuality of knowledge-intensive concept generation as well as a multi-round VQA-based evaluation framework. Experiments on various T2I models show that current models still struggle with achieving high factuality in generating specific concepts and composing multiple concepts in one image.

Limitations

In this section, we discuss the limitations of this work: (1) **Knowledge concepts.** We only involve English knowledge concepts, which limits the comprehensiveness of evaluating the factual accuracy of text-to-image models on knowledge-intensive concepts. (2) **Generation task.** We propose a three-tiered generation task and design seven variations based on the intrinsic properties of knowledge concepts, including *memorization, action, attribute, scene, size, differentiating and interaction*. However, there are many more tasks relevant to real-world scenarios that we plan to explore in future work. (3) **Evaluation Scope.** The primary focus of our benchmark is to evaluate the factuality of text-to-image models when generating images that involve knowledge-intensive concepts. Consequently, we intentionally exclude other evaluation criteria, such as image fidelity and aesthetic quality, which, while significant, are beyond the specific scope of our study. We will provide a more comprehensive assessment of model performance in the future. (4) **Knowledge injection.** We explore two approaches for injecting external knowledge. However, both approaches come with their own limitations. We believe T2I-FactualBench will inspire further research on how to effectively inject external knowledge into T2I models.

Ethics Statement

During the collection of knowledge concepts, we rigorously eliminate any form of geographical or racial bias. Specifically, we employ random sampling from data sources and conduct strict manual curation to ensure a balanced concepts across different regions and ethnic groups. In the prompt generation phase, we implement keyword constraints to guide the model, ensuring that the content is reasonable and devoid of harmful elements. Additionally, we conduct a comprehensive manual review to further ensure the appropriateness and fairness of the content.

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A Dataset Details

A.1 Detailed Knowledge Concepts Collection Process

We elaborate on the meticulous process of collection and filtration employed for knowledge concepts pool.

The filtering process is outlined as follows: First, each concept is used as a query to retrieve relevant information from BabelNet, including the synset collection, categories for each synonym, hypernyms, hyponyms, and associated images. Because a single concept can have multiple synonyms (e.g., "Taylor Swift" may refer to the American singer or her eponymous album), we eliminate synsets that do not belong to the given category through keyword matching. To differentiate knowledge concepts from general concepts, we focus on selecting concepts with fewer than four hyponyms (Jumaeva, 2024). We hypothesize that such concepts are more likely to exhibit distinct visual attributes, making them well-suited for thoroughly evaluating the knowledge capabilities of T2I models.

A.2 Detailed Three-tiered Framework

We present detailed information on the three-tiered framework of T2I-FactualBench in this section. In Table 6, we present our proposed three-tiered structure and seven tasks, including the number of prompts for each task and the evaluation method. In Table 9, we compare T2I-FactualBench to the knowledge domains of existing text-to-image benchmarks. Our T2I-FactualBench is the largest benchmark to date in terms of the number of concepts and prompts specifically designed to evaluate the factuality of knowledge-intensive concept generation.

B Multi-Round VQA Details

B.1 Concept Factuality Evaluation

In Concept Factuality Evaluation, we assess the factual accuracy of model’s generated knowledge concept across four critical dimensions: *shape, color, texture representation, and feature details*. We provide a specific definition for each dimension in Table 10. Given that each category prioritizes distinct feature details, we design specific evaluation criteria for each category, as illustrated in Figure 5. The detailed Concept Factuality Evaluation prompt is provided in Figure 6.

Level	Task	Number	Evaluation
SKCM	Memorization	1600	Concept
SKCI	Action	200	Concept, Instantiation
	Attribute	200	
	Scene	200	
MKCC	Size	225	Concept, Instantiation, Composition
	Differentiating	225	
	Interaction	350	

Table 6: The three-level tasks in T2I-FactualBench

B.2 Instantiation Completeness and Composition Factuality Evaluation

In both the Instantiation Completeness Evaluation and Composition Factuality Evaluation, we first conduct a confirmation of presence, predicated on the existence of knowledge concepts within the generated images. If confirmed, the evaluation proceeds to the subsequent assessment stage; otherwise, the score is assigned 0. For the Instantiation Completeness Evaluation, we have tailored different evaluation prompts for each task, exemplified by the Size task prompt depicted in Figure 7.

In the Composition Factuality Evaluation at the MKCC level, we assess the composition accuracy of model’s generated knowledge concepts across four critical dimensions: *Seamless Transition, Visual Completeness, Authenticity, and Prompt Following*. We provide a specific definition for each dimension in Table 10. Furthermore, we have crafted distinct prompts based on the number of knowledge concepts. The interaction variation necessitates an additional assessment of the background’s composition factuality due to the presence of background knowledge concepts. Detailed prompts are illustrated in Figure 8 and Figure 9.

C Implementation Details

C.1 Model Details

We comprehensively evaluate the performance of 7 text-to-image models on T2I-FactualBench, including three variants of Stable Diffusion: (1) Stable Diffusion v1.5 (stable-diffusion-v1-5), (2) Stable Diffusion XL (stable-diffusion-xl-base-1.0), and the latest (3) Stable Diffusion 3.5 (stable-diffusion-3.5-large) which incorporates the Diffusion Transformer (DiT) to enhance its capability in processing complex textual inputs. The (4) PixArt-alpha (PixArt-XL-2-1024-MS), leveraging Diffusion-Transformer technology, stands out for its minimal parameter footprint and expedited training process. Additionally, We include (5) Playground v2.5 (playground-v2.5-1024px-aesthetic), an advanced model evolving from Stable Diffu-

sion XL, which is fine-tuned to generate visually superior images that better resonate with human aesthetic preferences.. Furthermore, we evaluate (6) Flux.1 (FLUX.1-dev) , a successor to Stable Diffusion, featuring a novel hybrid architecture that merges multimodal processing proficiency with the parallelized functionality of the Diffusion Transformer. For API-based models, we evaluate the performance of (7) DALL-E 3 (Betker et al., 2023). When generating images, we default to leveraging the GPT model to enrich the input text prompts with additional details.

We adopted the default hyperparameters specified for each model by their respective authors.

C.2 Two Knowledge Injection Methods

Visual knowledge injection models We use the knowledge concept images from Section 3.1 as reference visuals to assist the models in image generation. We select two subject-driven generation models: (8) SSR-Encoder (Zhang et al., 2023b), based on Stable Diffusion v1.5, and (9) MS-Diffusion (Wang et al., 2024b), built upon Stable Diffusion XL.

Text knowledge injection models We augment the prompts by appending the definitions of the knowledge concepts, acquired in Section 3.1. This textual augmentation aims to enable the models to generate more precise representations of the concepts. We select (10) Stable Diffusion 3.5* and (11) Flux.1 dev* model due to the robust semantic comprehension capabilities afforded by their Diffusion Transformer (DiT) architecture.

C.3 Evaluation Details

In our Multi-Round VQA evaluation, we utilized the GPT-4o-0513 model. The final score for each VQA task is computed by evaluating the accuracy of generated image against established criteria.

We also compute established metrics. Specifically, we derived the CLIP-T metric by calculating the cosine similarity between the textual features and visual features extracted from the generated images utilizing CLIP (ViT-L/14). Moreover, for the CLIP-I and DINO metrics, we computed the cosine similarity between the feature sets of reference images and those of generated images by CLIP (ViT-L/14) and DINO (Dinov2-small), respectively.

For TIFA score, we employed GPT-4o-0513 to generate question-answer pairs based on text

prompts, utilizing BLIP (blip-vqa-capfilt-large) VQA model to answer the questions with the generated image. The TIFA score was calculated as the accuracy of the answers produced by the VQA system. For LLMscore, we used QwenVL-2.5-72B model to generate image descriptions and employed GPT-4o-0513 to evaluate the alignment between the image description and the text prompt as LLMscore.

Notably, when evaluating the composition of multiple knowledge concepts in the MKCC, we calculate the CLIP-I and DINO scores by determining the cosine similarity for each individual knowledge concept and then averaging these values to obtain a composite score.

C.4 Human Evaluation Details

We conducted human evaluations on iTAG platform. Specifically, we engaged three annotators to evaluate Concept Factuality, Instantiation Completeness, and Composition Factuality to ensure the robustness of our assessments. The questions presented to the annotators were consistent with the prompts for GPT-4o, as depicted in Figures 6, 7, 8, and 9, to minimize bias introduced by question formulation. Figure 10, show the interface for human evaluation on Concept Factuality.

To assess the alignment between all evaluation metrics and human experts, we curate a balanced dataset comprising 300 concept validation samples, 300 composition validation samples, and 300 Instantiation validation samples. Each validation sample is evaluated by three annotators and we calculate the mean score from the three annotators' evaluations to ensure the reliability of the manual annotations.

D Additional Results

D.1 Results of Previous Metric Scores

In Table 7, we detail the CLIP-T, CLIP-I, and DINO scores on the T2I-FactualBench across three levels. Some results in the table are similar to those in Section 5.3.

However, our analysis reveals several critical limitations inherent in the current metrics:

- **Inadequate Assessment of Complex Instructions.** As a bag-of-words model, CLIP-T fails to accurately assess a model's ability to follow complex instructions. Specially, we observe that model performance appears to improve as instruction complexity increases,

Model	SKCM			SKCI			MKCC		
	CLIP-T	CLIP-I	DINO	CLIP-T	CLIP-I	DINO	CLIP-T	CLIP-I	DINO
<i>Text-to-image Generation</i>									
SD v1.5	31.0	75.2	38.4	31.3	76.0	44.7	33.0	67.0	26.6
SD XL	31.3	74.6	42.3	31.9	75.7	47.5	34.8	64.2	27.2
Pixart	27.8	68.0	30.2	29.8	71.5	40.1	32.9	64.1	26.3
Playground	30.8	73.6	42.4	31.6	73.8	47.1	35.0	64.9	29.3
Flux.1 dev	29.4	73.8	38.2	30.6	75.1	45.7	34.1	65.5	27.0
SD 3.5	31.4	76.9	43.4	32.1	76.6	47.7	35.6	67.3	29.8
DALLE-3	30.3	73.6	47.9	30.9	70.9	45.7	34.4	64.2	28.3
<i>Visual-Knowledge Injection</i>									
SSR-Encoder	30.4 ↓ 0.6	86.6 ↑ 11.4	63.9 ↑ 25.5	28.9 ↓ 2.4	87.2 ↑ 11.2	68.8 ↑ 24.1	30.4 ↓ 2.6	73.0 ↑ 6.0	36.5 ↑ 10.5
MS-Diffusion	31.0 ↓ 0.3	83.4 ↑ 8.8	65.3 ↑ 23.0	31.2 ↓ 0.7	81.6 ↑ 5.9	64.1 ↑ 16.6	33.6 ↓ 0.8	69.9 ↑ 5.7	37.1 ↑ 9.9
<i>Text-Knowledge Injection</i>									
Flux.1 dev*	29.8 ↑ 0.4	77.9 ↑ 6.1	44.7 ↑ 6.5	30.7 ↑ 0.1	77.1 ↑ 2.0	49.4 ↑ 3.7	34.3 ↑ 0.2	66.5 ↑ 1.0	29.0 ↑ 2.0
SD 3.5*	31.3 ↓ 0.1	79.6 ↑ 2.7	48.0 ↑ 4.6	31.7 ↓ 0.4	78.6 ↑ 2.0	51.0 ↑ 3.3	34.9 ↓ 0.7	68.3 ↑ 1.0	31.3 ↑ 1.5

Table 7: **Additional results** on T2I-FactualBench. We present the previous metric evaluation of text-to-image generation models and two distinct knowledge injection methods across three levels. We highlight the row of DALLE-3 in gray to denote the incompleteness of its evaluation data. **Model *** indicates that the model has undergone text-knowledge injection. ↑ and ↓ denote improvements and declines relative to their base models.

transitioning from SKCM to MKCC, which misrepresents the model’s true capability in handling intricate instructions.

- **Inability to Distinguish Different Models.** The previous metrics do not effectively distinguish between models. Notably, the performance discrepancies between strong backbone models and weak backbone models are minimal. For instance, the transition from SD v1.5 to SD 3.5 on the SKCM level reveals negligible changes in scores (CLIP-T: 31.0 → 31.4; CLIP-I: 75.2 → 76.9; DINO: 38.4 → 43.4), underscoring the metrics’ inadequacy in capturing model advancements.
- **Lack of Fine-Grained Evaluation.** The current metrics are insufficient for fine-grained evaluation of model capabilities. They fail to assess effectively the models’ ability to integrate multiple knowledge concepts, thereby providing an incomplete metric of the models’ performance in complex compositional tasks.

While TIFA Score and LLMscore leverage MLLM assessments to provide a fine-grained reflection of the alignment between text and image, they are **not ideally suited for evaluating the factuality of T2I models in generating knowledge-intensive concepts and their compositions**. TIFA primarily focuses on the presence of objects, the correctness of general objects, and the accuracy of object attributes. Without a reference concept image as input, it cannot accurately assess the factuality of knowledge-intensive concept generation.

For example, TIFA may recognize the presence of a "dog" in an image but cannot determine if it is specifically a "basset hound dog."

Similarly, LLMscore relies on VLMs to generate visual descriptions, which tend to use classified concepts instead of knowledge-intensive descriptions. For example, a VLM might describe a generated image as containing a "dog" rather than a "basset hound dog." This limitation affects the thorough assessment of factuality for generated knowledge-intensive concepts.

D.2 Ablation Study on Concept Factuality Across Levels

In Table 2, we noticed that concept factuality scores tend to increase as task complexity increasing from SKCM to MKCC. We believe this counter-intuitive trend is due to the number of concepts varies across tasks of different levels. While SKCM features a diverse range of **1600 concepts**, SKCI and MKCC use only **400 and 550 common concepts**, respectively, leading to higher factuality scores.

To validate this hypothesis, we collect the concepts that appear in MKCC, denoted as C_M . For SKCM, we calculate the Concept Factuality scores only for those concepts also found in C_M . As shown in the Table 8, the results reveal that, when considering the same set of concepts, the scores tend to be higher at the less complex SKCM compared to the MKCC.

Model	SKCM*	MKCC
SD v1.5	60.4	37.6
SD XL	65.7	51.7
Pixart	49.3	35.8
Playground	70.2	53.8
Flux.1 dev	68.9	56.9
SD 3.5	76.2	68.9
DALLE-3	82.3	71.3
SSR-Encoder	88.1	43.1
MS-Diffusion	92.9	65.5
Flux.1 dev*	74.3	64.2
SD 3.5*	80.8	67.9

Table 8: Ablation Results of diversity models performance in Concept Factuality. SKCM* indicates the subset where knowledge concepts are also present in MKCC.

D.3 Results of Model Performance in Different Domains and Dimensions

In Figure 4, we provide a comprehensive analysis of the concept factuality scores for 11 distinct models across eight knowledge concept domains at the SKCM level.

In Table 11 and Table 12, we present the performance of models across various dimensions of Concept Factuality and Composition Factuality on T2I-FactualBench.

Our comparative analysis reveals that even SOTA models exhibit stronger capabilities in representing overall Color, Shape, and Texture, while fine-grained **Feature Details remain challenging** for generative models. For composing multiple concepts, models achieve higher scores in Seamless Transition and Visual Completeness but **struggle with Authenticity and Prompt Following**. We suppose this is due to the inherent complexity involved in maintaining realistic spatial arrangements and precisely interpreting and executing detailed textual instructions.

D.4 Results of Error Analysis For Multi-Round VQA

We each collect 50 error cases for Concept Factuality, Task Completeness, and Composition Factuality evaluation, where GPT-4o assessment differed from human annotations. In Table 13, we present a comprehensive breakdown of these error cases, categorized by the specific dimensions of discrepancies observed in each evaluation metric. Our statistical analysis reveals: In Concept Factuality

evaluations, discrepancies often arise in Texture Representation (32%) and Feature Details (57%), because of the need for advanced visual feature capture and analysis capabilities. For Task Completeness, errors frequently occur with complex Size (24%), Differentiating (28%), and Interaction (30%) instantiations in MKCC, as the model must accurately distinguish between multiple concepts and assess their correct instantiation and interaction. In Composition Factuality, errors frequently occur in the Authenticity (56%), requiring strong spatial recognition skills and common world knowledge.

D.5 Results of Inter-Human Annotators Agreement Rates

We observe substantial agreement among the annotators on the validation set. Specifically, for binary evaluations (Instantiation Completeness), consensus was achieved between at least two annotators in **87%** of the cases, with all three annotators agreeing in **74%** of the cases. For Likert-scale evaluations (Concept Factuality and Composition Factuality), we calculated a Krippendorff’s Alpha (Krippendorff, 2018) of **0.72**, indicating a good level of agreement for a subjective task of this complexity. These metrics underscore the reliability of our human evaluations.

D.6 Qualitative results

We present additional qualitative cases in Figure 11. We identify four key deficiencies in current generative models: Concept Error, Instantiation Failures, Realism Error, and Feature Mixture Error.

E Cost of T2I-FactualBench and Multi-Round VQA Evaluation

We provide the necessary cost details as follows: We used the GPT-4o-513 API to filter knowledge concepts and generate different phrases for each task, and then created prompts based on the knowledge concepts and phrases. This process required 5,600 API calls, costing approximately \$30. In the multi-round VQA evaluation, we used the GPT-4o-0513 model for all three levels. The evaluation of each model required 6,350 API calls, costing approximately \$45.

Benchmarks	Tasks In Prompt Construction									Knowledge Concepts		Evaluation
	Basic	Action	Attribute	Scene	Size	Differentiating	Interaction	Back-Foreground	Multi-Concepts(3)	Domain	Num	
PartiPrompt	✓	✓	✓	✓	✗	✗	✗	✓	✓	6	212	Human and CLIP Human and CLIP Human and Automatic Metrics
HEIM	✓	✓	✓	✓	✗	✗	✗	✓	✓	6	342	
KITTEN	✓	✗	✗	✓	✗	✗	✗	✓	✗	6	322	
T2I-FactualBench	✓	✓	✓	✓	✓	✓	✓	✓	✓	8	1600	Multi-Round VQA

Table 9: **Comparing T2I-FactualBench to the knowledge domain of existing text-to-image benchmarks.** T2I-FactualBench covers more essential tasks in prompt construction, including single knowledge concepts understanding (marked blue) and multi-knowledge concepts composition (marked purple). Moreover, the knowledge concepts in our benchmark across 8 diverse domains (animals, artifacts, food, persons, plants, locations, celestial, events). With a total of 1,600 knowledge concepts, it stands as the most extensive benchmark in knowledge field.

Dimension	Definition
Shape	Assess whether the overall silhouette, pose, and proportions align with the common shapes associated with the concept.
Color	Assess whether the concept’s color scheme and lighting conditions align with the natural or expected hues, saturation, and brightness characteristic of the concept.
Texture	Evaluate the realism and clarity of concept’s textures, ensuring authentic representation in key areas, free from blurriness, pixelation, or artificial effects, to uphold realistic integrity
Feature Details	Evaluate the accuracy, completeness, and logical placement of the concept’s features. Focus on facial details, limbs, and skin texture to ensure they align with the natural or expected representation of the concept.
Seamless Transition	Assesse whether the boundaries between concepts appear smooth and natural.
Visual Completeness	Evaluate if concepts are visually consistent and free from unnecessary additions, missing elements, or unnatural appearances
Authenticity	Assess whether the size and position of the concepts are realistic within the environment. For example, a car should be much larger than a husky, and neither should be in nonsensical positions, like floating unsupported.
Prompt Following	Evaluate the extent to which the image faithfully represents all major elements specified in the text prompt.

Table 10: Definition of various dimensions in Concept Factuality and Composition Factuality evaluation.

Model	SKCM				SKCI			
	Shape	Color	Texture	Feature Details	Shape	Color	Texture	Feature Details
Text-to-image Generation								
SD v1.5	39.5	38.0	34.5	21.6	54.9	48.9	49.4	30.5
SD XL	50.7	49.4	47.5	35.6	57.7	58.0	54.8	38.3
Pixart	32.9	36.5	35.0	28.3	43.2	46.5	44.2	25.3
Playground	50.4	53.3	51.6	36.4	63.0	62.8	65.3	45.7
Flux.1 Dev	61.5	63.2	59.0	42.7	59.0	62.8	59.3	38.0
SD 3.5	66.4	66.5	65.2	55.3	62.5	60.8	59.9	42.5
DALLE-3	78.1	75.8	75.5	56.1	78.9	69.2	82.1	58.3
Visual-Knowledge Injection								
SSR-Encoder	48.0	49.0	45.5	29.5	81.8	72.7	73.3	48.2
MS-Diffusion	64.5	63.3	59.6	47.0	81.9	82.8	74.2	69.2
Text-Knowledge Injection								
Flux.1 Dev*	69.2	70.2	68.8	48.1	64.5	68.3	65.3	42.7
SD 3.5*	72.1	73.7	72.3	53.2	70.7	74.2	73.0	48.8

Table 11: Performance of models across various dimensions of **Concept Factuality** on SKCM and SKCI.

Model	MKCC–Concept Factualty				MKCC–Composition Factualty			
	Shape	Color	Texture	Feature Details	Seamless	Visual	Authenticity	Prompt Following
<i>Text-to-image Generation</i>								
SD v1.5	39.5	38.0	34.5	21.6	16.1	18.2	12.8	13.2
SD XL	50.7	49.4	47.5	35.6	37.2	38.8	34.3	31.4
Pixart	32.9	36.5	35.0	28.3	25.8	27.5	22.0	21.7
Playground	50.4	53.3	51.6	36.4	46.1	48.1	43.1	41.9
Flux.1 Dev	61.5	63.2	59.0	42.7	67.2	69.3	61.5	60.1
SD 3.5	66.4	66.5	65.2	55.3	77.0	80.1	74.3	70.6
DALLE-3	78.1	75.8	75.5	56.1	85.6	87.2	88.8	82.5
<i>Visual-Knowledge Injection</i>								
SSR-Encoder	48.0	49.0	45.5	29.5	9.4	12.0	7.4	9.0
MS-Diffusion	64.5	63.3	59.6	47.0	31.0	35.0	27.9	30.7
<i>Text-Knowledge Injection</i>								
Flux.1 Dev*	69.2	70.2	68.8	48.1	75.3	77.1	70.6	67.4
SD 3.5*	72.1	73.7	72.3	53.2	66.4	68.6	61.0	62.2

Table 12: Performance of models across various dimensions of **Concept Factualty** and **Composition Factualty** on MKCC.

Concept Factualty		Instantiation Completeness		Composition Factualty	
Error Type	Percentage	Error Type	Percentage	Error Type	Percentage
Shape	12%	Action	6%	Seamless Transition	12%
Color	8%	Attribute	10%	Visual Completeness	10%
Texture Representation	32%	Scene	2%	Authenticity	56%
Feature Details	57%	Size	24%	Prompt Following	22%
-	-	Differentiating	28%	-	-
-	-	Interaction	30%	-	-

Table 13: **Error Summary**. Breakdown of Error Cases in Concept Factualty, Task Completeness, and Composition Factualty.

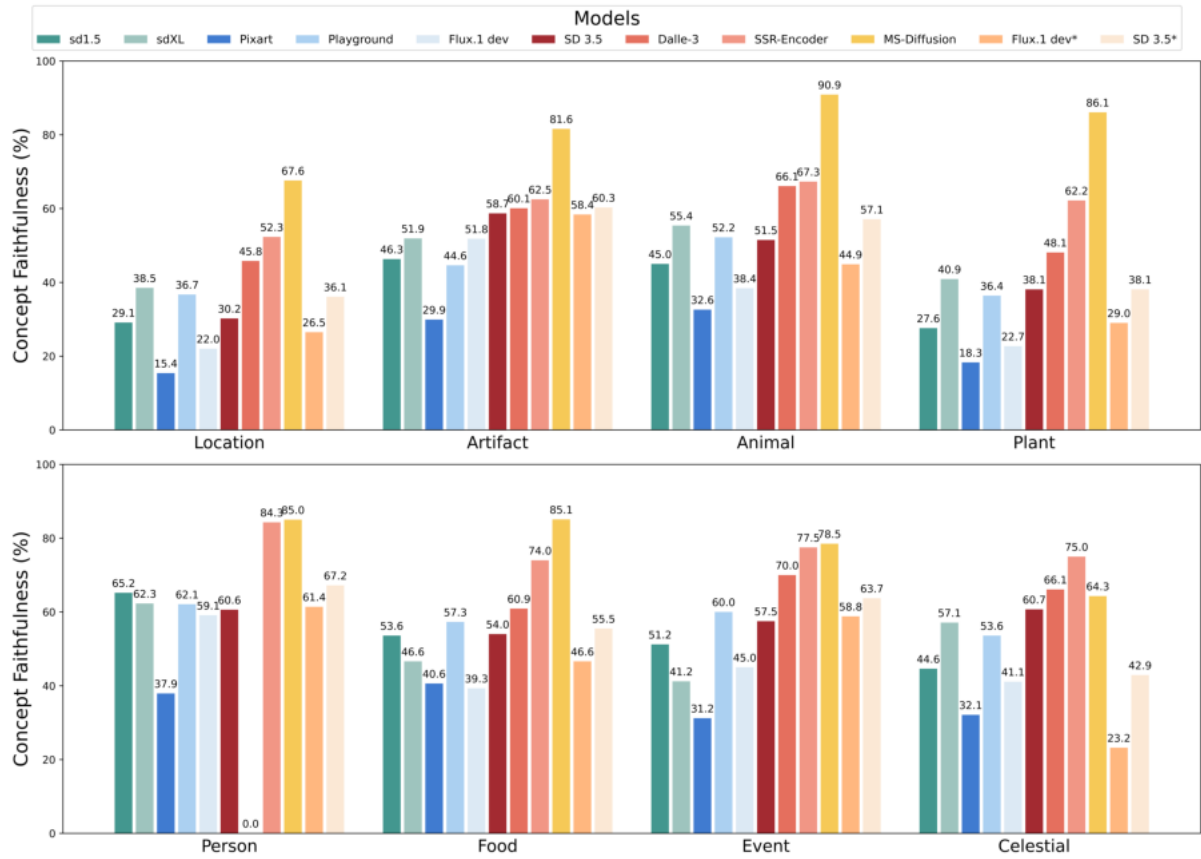


Figure 4: Concept Factuality Scores across 8 domains in the SKCM level for 11 Models.

Feature Details:

Animal: Evaluate the accuracy, completeness, and logical placement of the {concept}'s features. Focus on facial details, limbs, and skin texture to ensure they align with the natural or expected representation of the {concept}.

Person: Evaluate the accuracy, completeness, and logical placement of the {concept}'s features. Focus on facial details, limbs, and skin texture to ensure they align with the natural or expected representation of the {concept}.

Plant: Evaluate the accuracy, completeness, and logical placement of the {concept}'s features. Focus on facial details, limbs, and skin texture to ensure they align with the natural or expected representation of the {concept}.

Artifact: Evaluate the accuracy, completeness, and logical placement of the {concept}'s features. Focus on functional components, material details, and decorative elements to ensure they align with the natural or expected representation of the {concept}.

Food: Evaluate the accuracy, completeness, and logical placement of the {concept}'s features. Focus on functional components, material details, and decorative elements to ensure they align with the natural or expected representation of the {concept}.

Celestial: Examine the {concept}'s architectural details, unique decorations, and structural features, ensuring key elements and iconic details are precisely represented with correct proportions, symmetry, spatial layout and accurately reflect the original design.

Location: Examine the {concept}'s architectural details, unique decorations, and structural features, ensuring key elements and iconic details are precisely represented with correct proportions, symmetry, spatial layout and accurately reflect the original design.

Event: Evaluate the {concept}'s portrayal by examining the historical accuracy and representation of key figures, attire, and iconic scenes within the context of the event. Ensure that the visual narrative accurately depicts significant moments and the overall atmosphere, maintaining fidelity to documented historical accounts and cultural settings.

Figure 5: Feature details for eight knowledge concept categories.

Task Instructions:

You are provided with a concept image and a generated image. First, learn the concept of {concept} from the concept image. Then evaluate the {concept} in the generated image based on your own knowledge of {concept} and the concept of {concept} you learned from the concept image.

Concept Evaluation:

Evaluate whether the {concept} in the generated image matches the conceptual characteristics of the {concept} based on:

1. **Shape Accuracy:** Focus on the overall outline and structure of the generated {concept}. Assess whether the overall silhouette, pose, and proportions align with the common shapes associated with the concept.

2. **Color Accuracy:** Assess whether the generated {concept}'s color scheme and lighting conditions align with the natural or expected hues, saturation, and brightness characteristic of the concept.

3. **Texture Representation:** Evaluate the realism and clarity of the {concept}'s textures, ensuring authentic representation in key areas, free from blurriness, pixelation, or artificial effects, to uphold realistic integrity.

4. **Feature Details:** {feature_details}

Notice that: You should only consider the concept of {concept} in the generated image.

Evaluation Criteria:

Evaluate based on the criteria above. Score from 0 to 4 based on how many criteria are fully met:

- 0: None met
- 1: One met
- 2: Two met
- 3: Three met
- 4: All met

Explain your rating by providing a very brief core reason. Keep your explanation concise and to the point.

Input:

The generated image <image>. Caption of the generated image is {text}.

The concept image <image>. Caption of the concept image is {concept}.

Output Format(Notice that output must be without any bold formatting):

Total Rating: Your Rating

Shape Accuracy: 0/1 Reason:

Color Accuracy: 0/1 Reason:

Texture Representation: 0/1 Reason:

Feature Details: 0/1 Reason:

Figure 6: The prompt we used for Concept Factuality Evaluation with GPT-4o.

Task Instructions:

You need to analyze a generated image that contains two knowledge concepts: {type_1} {concept_1} and {type_2} {concept_2}. The task details are as follows:

1. **Existence Check:** First, confirm whether both concepts are present in the generated image. If either concept is missing, the Task Score should be 0.

2. **Size Restoration:** If both concepts are present, assess whether their size proportions in the image match their real-world counterparts.

3. **Evaluation Criteria:** Estimate the relative volume, height, or recognizable size ratio of each concept in the image to ensure they reflect the real-world size differences. If {concept_1} is indeed larger than {concept_2} as expected, the Task Score should be 1; otherwise, it should be 0.

Input Information:

The generated image: <image>.

Concepts: {type_1} {concept_1} , {type_2} {concept_2}

Output Format (Note: the output must be without any bold formatting):

Task Score:

Reason:

Figure 7: The prompt we used for Instantiation Completeness Evaluation with GPT-4o. A case of Size variation.

Task Instructions:

You are provided with a generated image, the text prompt, and two knowledge concepts: {type_1} {concept_1} and {type_2} {concept_2}. Your task is to evaluate the generated image based on these knowledge concepts and the text prompt.

1. Confirmation of Presence: determine whether both {concept_1} and {concept_2} are present in the generated image. If either is not present, the Total Rating should be 0. If both are present:

2. Integration Evaluation:

- Seamless Transition: Assess whether the boundaries between {concept_1} and {concept_2} are smooth, and ensure they integrate harmoniously with the surrounding environment.

- Visual Completeness: Evaluate whether {concept_1} and {concept_2} exhibit visual consistency without unnecessary additions, missing parts, or unnatural appearances.

- Authenticity: Assess if the size and position of {concept_1} and {concept_2} are realistic for the environment. For instance, a car should be much larger than a husky, and neither should appear in illogical positions, such as floating without support.

- Prompt Following: Evaluate whether the image faithfully represents all major elements specified in the text prompt and ensures that specific details, such as colors and shapes, are accurately depicted.

Evaluation Order: Please first confirm the presence of the concepts, then proceed to evaluate the integration.

Evaluation Criteria: Evaluate based on the criteria above. Total Rate from 0 to 4 based on how many criteria are fully met:

0: None met

1: One met

2: Two met

3: Three met

4: All met

Provide a concise explanation for the rating. Keep your feedback specific and to the point.

Input:

Generated image: <image>

Text prompt: "{text}"

Output Format (Notice that output must be without any bold formatting):

Total Rating: Your Rating

Seamless Transition: 0/1 Reason:

Visual Completeness: 0/1 Reason:

Authenticity: 0/1 Reason:

Prompt Following: 0/1 Reason:

Figure 8: The prompt we used for Composition Factuality Evaluation of **Two** knowledge concepts with GPT-4o.

Task Instructions:

You are provided with a generated image, the text prompt, and three knowledge concepts: {type_1} {concept_1}, {type_2} {concept_2} and background {type_3} {concept_3}. Your task is to evaluate the generated image based on these knowledge concepts and the text prompt.

1. Confirmation of Presence: determine whether {concept_1}, {concept_2} and {concept_3} are present in the generated image. If any of them is not present, the Total Rating should be 0. If all are present:

2. Integration Evaluation:

- Seamless Transition: Assess whether the boundaries between {concept_1}, {concept_2}, and background {concept_3} are smooth.

- Visual Completeness: Evaluate whether {concept_1}, {concept_2} and the background {concept_3} exhibit visual consistency without unnecessary additions, missing parts, or unnatural appearances.

- Authenticity: Assess if the size and position of {concept_1}, {concept_2}, {concept_3} are realistic for the environment. For instance, a car should be much larger than a husky, and neither should appear in illogical positions, such as floating without support.

- Prompt Following: Evaluate whether the image faithfully represents all major elements specified in the text prompt and ensures that specific details, such as colors and shapes, are accurately depicted.

Evaluation Order: Please first confirm the presence of the concepts, then proceed to evaluate the integration.

Evaluation Criteria: Evaluate based on the criteria above. Total Rate from 0 to 4 based on how many criteria are fully met:

0: None met

1: One met

2: Two met

3: Three met

4: All met

Provide a concise explanation for the rating. Keep your feedback specific and to the point.

Input:

Generated image: <image>

Text prompt: "{text}"

Output Format (Notice that output must be without any bold formatting):

Total Rating: Your Rating

Seamless Transition: 0/1 Reason:

Visual Completeness: 0/1 Reason:

Authenticity: 0/1 Reason:

Prompt Following: 0/1 Reason:

Figure 9: The prompt we used for Composition Factuality Evaluation of **Three** knowledge concepts with GPT-4o.



Figure 10: iTAG Interface for Concept Factuality Evaluation.

	Concept Error		Instantiation Failure		Realism Error	Feature Mixture Error
Concept Images						
Dalle-3						
Flux.1 dev						
Playground v2.5						
SD 3.5						
SD 3.5*						
MS-Diffusion						
Prompt	Mapo Tofu and ...gyoza ..., A Baird's tapir standing with Mount Fuji visible in the background ... by a riverbank at sunset. A broken ice skate a worn sackbut rests on a wooden chair, ... beside it, a brand new clarinet behind the Golden Gate Bridge, a Tesla Model 3 ... with a keeshond dog looking out the window... ..., an Egyptian Mau cat playfully chases a basset hound dog near the Lotus Temple ...					

Figure 11: **Qualitative results.** Error cases of diversity models in T2I-FactualBench.